The Peace of Fairness



Stephen Takowsky

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Reference Edition

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Publishing Details

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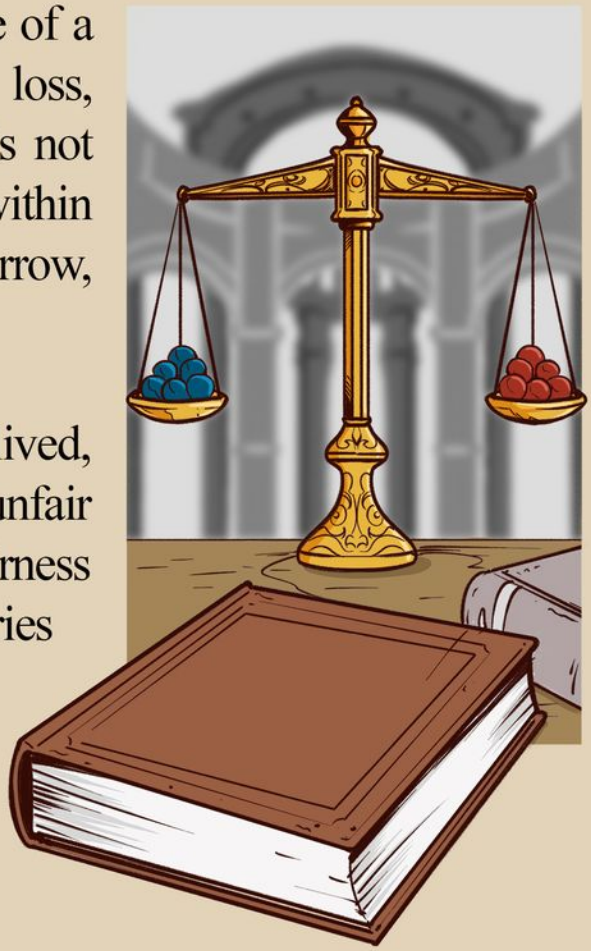
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Series Map

THE LAW OF FAIRNESS

A new way of seeing every life.

We judge fairness by the visible shape of a life: health and hardship, love and loss, opportunity and misfortune. Yet life is not lived from the outside. It is lived within consciousness, through joy and sorrow, pleasure and pain, peace and fear.



Could life be fair where it is actually lived, even when it appears profoundly unfair from the outside? The Law of Fairness proposes that every felt experience carries weight according to its intensity and duration. Across the completed conscious life, total positive and negative feeling exactly balance.

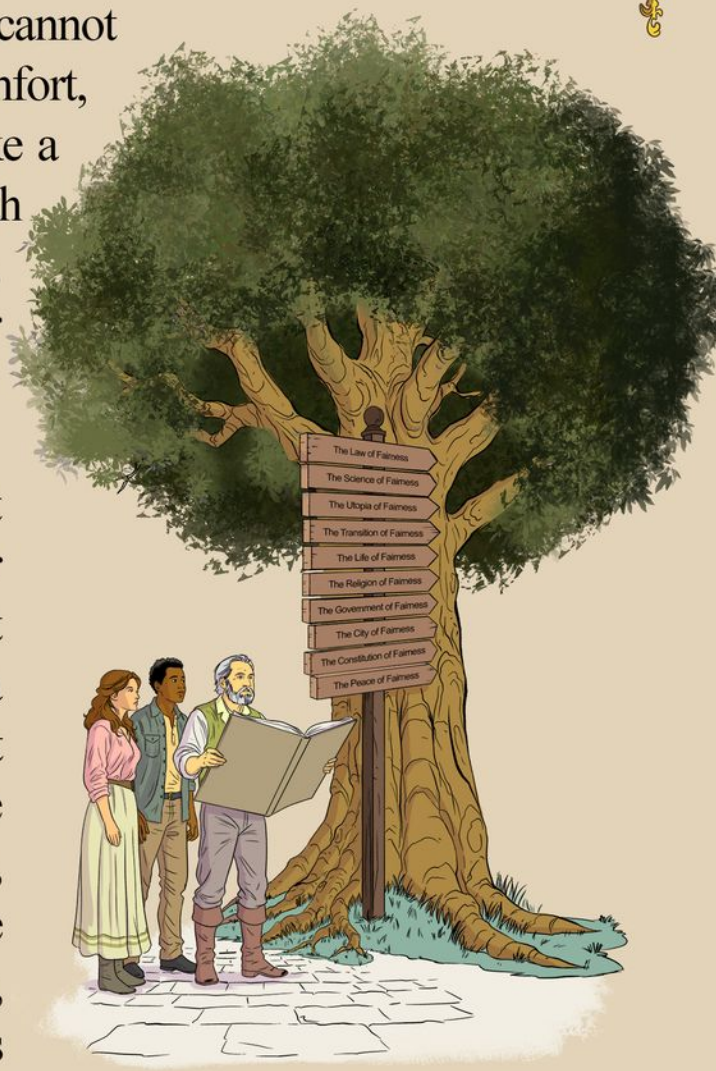
Picture a scale open for a lifetime. Joy and relief add weight to one side; pain and grief to the other. It may tip heavily for years, but when the conscious life is complete, the two sides balance. Each life carries its own scale. No life balances another.

Lives may differ enormously in length, choices, pleasures, wounds, and circumstances. The scale weighs neither wealth nor virtue, status nor faith. It weighs felt experience alone. Circumstances tell the visible story of a life, but feeling is how that story is lived. The Law asks us not to mistake the visible story for the whole.

WHAT IF THE LAW IS REAL?

A new meaning of utopia.

If the Law of Fairness is real, utopia cannot mean endless pleasure. More comfort, luxury, or consumption does not make a society ideal. If conscious lives end with no lasting surplus of positive feeling, utopia must mean something deeper and steadier than limitless happiness.



"There is no free happiness" does not mean joy is wrong, pain is payment, or suffering is holy. Pleasure alone cannot define a good life, nor appearances a good society. The goal is joy without excess. A fairer world would reduce needless suffering, poverty, disease, cruelty, corruption, and abuse while protecting love, beauty, rest, play, choice, and joy. Moderation restrains excess, work builds worthy goals, and stewardship preserves the world so future generations inherit something worth protecting.

If the Law is real, its deepest equality lies beneath wealth, status, power, culture, and belief. Lives may differ while reaching the same final balance of felt experience. These ten books follow that question from science into life, spirituality, government, cities, law, transition, and peace, exploring how humanity might build a lasting utopia.

THE TEN BOOKS OF FAIRNESS

A complete path from truth to practice.

THE LAW OF FAIRNESS



Explores the core question, defines key terms, and sets the ethical bounds of the series.

THE SCIENCE OF FAIRNESS



Evaluates the Law through evidence, measurement, rival models, protocols, and falsification.

THE UTOPIA OF FAIRNESS



Brings the series into a mature view of what a peaceful world could become.

THE TRANSITION OF FAIRNESS



Asks how a move toward utopia could unfold with care, step by step, under real limits.

THE LIFE OF FAIRNESS



Brings the Law into daily life and asks what it means for how we live.

THE RELIGION OF FAIRNESS



Explores faith, world religions, and the optional path of Fairness.

THE GOVERNMENT OF FAIRNESS



Examines civic structure and the public institutions that give it form in practice.

THE CITY OF FAIRNESS



Builds a civic space for learning, service, and peace with privacy kept safe.

THE CONSTITUTION OF FAIRNESS



Sets out the rights, rules, paths, and checks a lasting utopia would need.

THE PEACE OF FAIRNESS



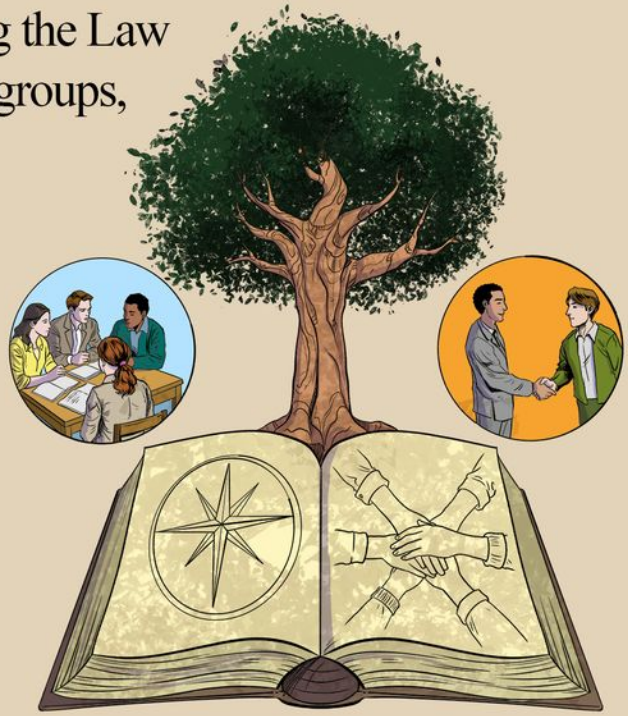
Builds the base for lasting peace and a fairer future for all humanity.

The series moves from the deepest question about reality to the structures that make a more peaceful world.

THE WORK AHEAD

A shared mission to build a better world.

Read the books, share them, and bring the Law of Fairness into classrooms, research groups, reading circles, and community conversations. Important ideas grow stronger when people examine them together, challenge assumptions, compare perspectives, and turn understanding into shared effort.



Scientists, researchers, clinicians, philosophers, and other experts can help test the Law, refine its measurements, improve its methods, and compare it with strong rivals. Serious criticism is essential. It exposes weaknesses, sharpens the questions, and helps reveal what should endure, what needs revision, and what should be rejected.

Beyond the hypothesis lies a practical challenge: building a world with less corruption and needless suffering, and more room for dignity, freedom, meaning, beauty, and opportunity. If this project speaks to you, bring your knowledge, experience, curiosity, and determination. Progress begins when people from different paths meet, share what they know, and build what survives scrutiny.

Help us make this generation
the one that brought peace to our world.

How to Read This Book

This Reference Edition explores the book's ideas in depth and develops the reasoning behind them. The companion Foundation Edition distills the essential ideas into a shorter, accessible introduction that stands on its own and requires no specialized background.

Peace is more than the absence of war. A household without food security, a worker trapped by debt, a city living through failing infrastructure, or a nation spending its future on permanent fear may be formally at peace while ordinary life remains unstable. The Peace of Fairness asks what material foundations make peace durable enough to be felt in homes, workplaces, public services, markets, borders, and the long decisions of nations.

Read the book as economics made human. A wage matters partly through the time and choices it makes possible; housing matters through the safety and continuity it provides. Debt places claims on future resources and can constrain later choices, while a queue can impose unequal burdens on people with different reserves. Maintenance can prevent a small fault from becoming an emergency. Corruption adds cost beyond the money diverted, and climate risk can intensify pressures across institutions.

The book is organized as a widening chain rather than a list of isolated policies. Parts I and II establish enoughness, hidden cost, public finance, strategic fear, and debt repair. Parts III and IV move through work, care, housing, education, maintenance, money, markets, production, trade, and essential systems. Parts V through VII widen the lens to sovereignty, diplomacy, arms reduction, treaties, development, migration, crisis, climate, and anti-capture. Parts VIII and IX then ask whether the whole architecture is buildable, affordable, revisable, and strong enough to survive serious objections. Reading in order makes those

dependencies visible, but the Table of Contents also supports a focused route into a field you already know.

Keep the scientific and practical claims separate as you read. Per-Stream Terminal Valence Closure (PTVC) proposes that positive and negative felt experience balance exactly within one qualifying conscious stream at its true endpoint; Section 1.6 gives the formal statement. The economic and institutional proposals require their own evidence and ethical warrant. Test them through staffing, maintenance, privacy, appeal, transition, and failure. The final red-team chapters are part of that method: a criticism that reveals a hidden cost, weak causal link, missing right, or better alternative should change the design. The practical horizon is secure homes, usable care, dignified work, maintained infrastructure, answerable institutions, and international relationships stable enough for cooperation to become less costly than permanent fear.

This volume is self-contained. The Law of Fairness introduces the series' central proposal, and The Science of Fairness considers how it might be tested. The Transition of Fairness examines the practical work of change. Readers moving between the Foundation and Reference Editions should follow subjects rather than chapter numbers. Let one question guide what you read next: where does an institution place a burden, and what would make a fairer arrangement durable?

Part I — Peace, Prosperity, and Enoughness

Conflict can be far from home when its cost reaches a household. A shipping route closes, a replacement supplier charges more, an employer cuts shifts, and a bill comes due before wages recover. The household may never see the original disruption, yet it must decide which necessary expense can wait. Economic instability travels through such chains, turning events at a border, port, public office, or power station into choices made at a kitchen table (Galtung, 1969; World Bank, 2011; United Nations and World Bank, 2018).

Stopping organized violence is an indispensable achievement. Material peace asks what happens afterward and around it: whether people can live with dependable access to work, care, shelter, public services, and the time needed to make ordinary plans. These material conditions do not exhaust justice, political freedom, cultural survival, or ecological responsibility. They form one essential dimension of a peace that people must be able to use. A public order can appear stable at a podium and precarious at dinner. The table knows whether the future is usable. A household that can plan meals, welcome guests, care for children, speak without panic, and rest has a kind of peace that statistics alone cannot express. A household that must turn every conversation into triage has lost more than money. It has lost some of the social room in which love, learning, patience, and humor can live.

Section 1.5.3 develops the method used throughout this Part: identify who carries a burden, how it reached them, and which feasible alternative could reduce it. The place where a burden can be interrupted need not be the place where it began. A town cannot reopen a distant shipping route, but it may be able to prevent a delivery delay from becoming a household eviction. Following the chain helps distinguish the cause of a problem from the nearest effective point of repair.

Chapter 1 establishes that method through the everyday passage of risk from institutions to households. Its subject is the difference between necessary protection and the recurring private labor of compensating for unreliable systems. Chapter 2 asks what dependable material life can make possible. Enoughness is not an instruction to diminish ambition. It is a context-sensitive floor from which care, beauty, invention, learning, and different forms of flourishing can become realistic choices (Sen, 1999; Nussbaum, 2011). Chapter 3 then examines corruption as a distortion of the routes people rely on: not only stolen money, but delay, guesswork, captured access, and the work imposed on those least able to avoid it (Olken and Pande, 2012).

Across these inquiries, averages cannot stand in for access and monetary totals cannot stand in for health, time, rights, or freedom. The aim is to make consequences traceable without pretending they are interchangeable. If an institution claims to create security or prosperity, the people carrying its costs should be visible in the account, and the route to correction should be as real as the promise.

Chapter 1 — Why Peace and Economics Belong Together

Consider a household whose annual income seems adequate but whose cash is exhausted this week. A work shift is canceled, rent comes due, and a prescription must be collected before the next payment arrives. No single charge explains the emergency. The collision between timing, access, and limited reserve turns a manageable interruption into a risk of debt or untreated illness. Peace and economics meet at precisely that point.

The chain may begin with war, damaged infrastructure, unreliable administration, or a credible fear that commitments will fail. Those causes are not interchangeable. A conflict can interrupt supply directly; a port rule may hold goods without physical shortage; market power or an ordinary equipment failure can produce a similar price increase. A serious account separates initiating events from the decisions that transmit them and tests rival explanations before calling any one factor responsible (Blattman and Miguel, 2010; World Bank, 2011).

The resulting burden is not always a line in a budget. It can be the hours spent finding another route, the training forgone because a job schedule is unstable, the private generator that substitutes for public reliability, or the appointment abandoned after repeated travel. Some protective expense is indispensable: insurers need reserves, hospitals need backup power, and states need credible defense. The question is whether each layer addresses a real risk at a proportionate cost, or whether mistrust has become a permanent and unexamined condition of ordinary life.

This chapter calls usable room for planning and recovery *material breathability*. Its detailed meaning is developed in Section 1.1. The household test then asks where distant systems meet, while the later sections distinguish justified protection from antagonistic overhead, examine how institutions shape expectations, and trace the interaction of conflict, corruption, and

scarcity. Public administration, health, infrastructure, and conflict studies measure different parts of this architecture. These fields can work together by tracing which function failed, who was affected, which dependency broke, and which institution can restore it, without forcing their findings into a universal score.

Tracing who bears the consequences makes the inquiry actionable. Suppose missed clinic visits rise after a bus service is withdrawn. An investigator should check whether the route change affected the relevant patients, whether clinical staffing or scheduling changed, and whether restored transport improves completed care. If the expected result does not appear, the account must change. The same standard applies to claims about international supply, public finance, and fear: specify a mechanism, a comparison, the people affected, and an observation that could overturn the explanation. The object of this comparison is completed care, not simply a restored line on a transport map.

Section 1.6 treats PTVC as a separate hypothesis on which this chapter’s material conclusions do not depend. Economic reliability cannot replace rights, democratic authority, cultural life, or independent consciousness research.

1.1 — Peace Beyond Non-War

Negative peace refers to the absence of direct violence, including war. Ending organized violence is a vital achievement. The material inquiry asks whether the systems governing shelter, work, care, movement, and public access also reduce the fear carried into ordinary life (Galtung, 1969). Reconstruction requires coordinated work: stabilize safety, restore damaged capacity, support displaced people, repair titles and records, reopen services, retain workers, and move public finance toward dependable provision.

1.1.1 — Fear That Survives the Battlefield

Civil war destroys far more than the assets visible in battle damage. It interrupts schooling, reallocates labor, fragments markets, raises transport and insurance costs, weakens health systems, displaces skilled workers, and leaves institutions operating under exceptional rules long after shooting subsides. The World Bank's January 2025 Global Economic Prospects identifies armed conflict among the major downside risks facing developing economies. Aggregate loss is only part of the transmission. Fear is also carried in credit terms, medicine availability, travel distance, title security, and willingness to invest where rules may not survive the next crisis (Blattman and Miguel, 2010; World Bank, 2011; World Bank, 2025b).

Fear changes collective time as well as individual feeling. Municipalities shorten planning horizons, firms postpone fixed investment, skilled workers leave, and households avoid commitments that would be rational under stable rules. These choices can preserve survival while weakening the very tax base, supplier network, professional workforce, and neighborhood confidence needed for recovery. The resulting trap is not simply psychological. Expectations alter investment, maintenance,

migration, hiring, and public revenue, which then alter the evidence on which expectations are formed (World Bank, 2011; United Nations and World Bank, 2018).

1.1.2 — Peace as Material Breathability

Material breathability is the room to plan, work, care, rest, heal, and begin again without every basic need becoming a crisis. It includes resources beyond unavoidable costs, dependable access to food and care, time outside emergency, and a credible route back after failure. These dimensions interact without becoming interchangeable: money cannot by itself replace a safe care route or stable housing.

Risk remains part of every human life. Illness, aging, accident, weather, conflict, uncertainty, and scarcity cannot be abolished. A mature peace distinguishes unavoidable burden from preventable burden, names who carries each, and refuses to hide avoidable pressure inside ordinary life for someone else's convenience.

Private households cannot substitute indefinitely for public repair. A family can patch a leak, find a ride, pay a fee, appeal a denial, care through a crisis, or borrow money for treatment, but each workaround spends time and future room. When official repair fails, households become unpaid maintenance departments for civilization. The cost is then hidden because it is scattered across kitchens, cars, phones, bodies, and sleepless nights.

Repair has a time dimension that prevention alone cannot cover. Even a well-designed system will produce error, injury, omission, and strain. A peaceful order is therefore judged not only by what it prevents, but by how quickly it detects damage and how fairly it repairs it. The route back matters: people need a usable reporting route and a fair hearing, clear responsibility for payment and repair, and a way to obtain apology and restoration. The response

should also change whatever would otherwise pass the same burden to the next person.

1.1.3 — Public Systems as Peace Infrastructure

Public systems become peace infrastructure when their reliability lowers the amount of private defense required for ordinary life. Safe water reduces the need to buy and store backup supplies, boil water, or seek treatment for illnesses caused by unsafe supply. Reliable transit lowers the private vehicle, time, and location costs of reaching work or care. A competent clinic reduces the pressure to save for catastrophe or rely on personal connections. The peace dividend appears first as released household capacity, not as a ceremonial announcement.

Nominal coverage does not establish the staff, supplies, access, maintenance, and reserve capacity needed for reliable service. Report coverage separately from outage duration, restoration time, surge performance, and the share of intended users who can obtain service. Those measures distinguish capacity on paper from capacity usable under strain.

Average reliability is not enough. A system can perform well in aggregate while concentrating failure on remote communities, disabled people, migrants, low-income households, caregivers, or those without digital access. Material peace requires distributional reliability: the people least able to purchase substitutes should not be assigned the most fragile public routes. Prevention studies and conflict research both become stronger when they ask not only whether a system works, but for whom it repeatedly fails (United Nations and World Bank, 2018).

Stress-test the recovery route from a household with no savings, car, or spare daytime hours. If help is reachable only with a credit card or flexible employer, the public route is not yet dependable.

1.2 — The Household as the First Test of Peace

A household total can conceal unequal access within the home. Duflo's (2003) analysis of South Africa's pension expansion estimated improvements in girls' weight-for-height and height-for-age when women received pensions; it found no comparable effect when men received them. Who receives a transfer can therefore matter alongside its amount. The burden account should ask who controls a payment, whose unpaid work increases, and who can use the time or access restored. Collect only the information needed for the decision, with a safe way to report unequal access.

1.2.1 — Household Panic as Public Evidence

Household panic is a public signal that deserves causal investigation. Similar distress may arise from low income, volatile scheduling, disability-related costs, caregiving, housing shortage, debt terms, discrimination, illness, family violence, or administrative failure. Treating the household as a sensor means following the route carefully enough to avoid blaming the person or diagnosing the wrong system (Mani et al., 2013; Ridley et al., 2020).

Average conditions can conceal a household's risk of severe loss. A household may manage ordinary months yet remain one repair, illness, school interruption, or caregiving emergency away from losing housing or work. The relevant buffer combines resources, reliable services, time, support, and enforceable rights sufficient to prevent a shock from becoming irreversible. Research on poverty, debt, and mental health shows why limited reserve can intensify both cognitive load and distress without implying that every outcome has one cause (Richardson, Elliott, and Roberts, 2013; Mani et al., 2013; Ridley et al., 2020).

Systemic strain becomes visible when shocks move together. A transport failure that affects one commuter may be manageable; a transport failure during a heat emergency, school closure, rent increase, and clinic backlog can overwhelm households and institutions at once. This co-movement is a public-capacity problem because the private substitutions people normally use may fail simultaneously. Material peace therefore depends on how essential systems behave under compound pressure, especially for people with disabilities, care obligations, insecure work, or no private alternative.

A useful signal must have a defined population and denominator. An agency studying abandoned applications should distinguish eligible people who never started, applicants who stopped after a particular requirement, applicants who completed elsewhere, and genuine withdrawals. It should compare the route before and after a change or with an appropriate alternative service, while recording staffing, seasonal demand, and outreach that could explain the difference. Publish aggregate completion and correction times, with uncertainty estimates and comparisons between affected groups that preserve privacy. Do not retain identifiable household histories merely to enrich a dashboard. If the signal cannot distinguish service failure from a change in who attempted access, it cannot establish that reform improved material peace.

1.2.2 — Time Horizons and Daily Security

The ability to plan ahead depends on economic conditions. A person who can predict next month’s rent, work schedule, medication supply, transport, and childcare can compare options and tolerate a temporary setback. A person facing repeated shocks must allocate attention to immediate survival and may rationally reject opportunities whose delayed payoff is too uncertain. Short

horizons are often an adaptation to the environment, not evidence of defective character (Haushofer and Fehr, 2014; Mani et al., 2013).

Daily security therefore includes temporal protections: notice before a schedule changes, continuity before a benefit stops, grace periods before an outstanding debt becomes due in full ahead of schedule, response times for urgent services, and predictable repair when a system fails. These arrangements do not promise an eventless life. They prevent avoidable surprise from becoming a routine method of governing people with little reserve.

Payment timing can change what workers are able to do. Kaur and colleagues (2025) staggered wage payments among low-income piece-rate manufacturing workers in India. After an earlier payment, output rose by about 7 percent and costly unintentional mistakes declined, with gains concentrated among more financially constrained workers. The result makes the collision between household obligations and the payment calendar a concrete object of study: a timing change can help existing skills translate into better performance.

A separate randomized survey-timing study of low-income households in the United States did not detect a before-versus-after-payday difference in measured cognitive performance, although respondents surveyed before payday favored earlier monetary rewards more strongly (Carvalho, Meier, and Wang, 2016). Different settings need not produce the same response. The productive research question is which constraint a timing change releases: access to cash, attention at work, or the ability to keep an appointment. Those pathways require their own measures rather than a single story about scarcity.

If a school schedule assumes a parent with flexible time, a clinic assumes someone can wait all day, a disability system assumes unpaid navigation, or elder care assumes endless family

availability, the burden has not disappeared. It has moved into care work, often onto women, low-income households, extended families, and workers with the least bargaining power (Addati et al., 2018).

Counting care work neither puts love up for sale nor makes it commandable or replaceable. It makes visible the caregiver's time, health, income, and future that institutions would otherwise leave out.

1.2.3 — Rent, Debt, and Medical Delay

Rent, debt, and medical delay form a recurrent feedback loop. Housing may take first claim on income; debt carries unmet costs into the future; delayed care can worsen health when money or time is short. Each system can be studied separately, but the household experiences them as one sequence. A payment missed to preserve housing may raise borrowing cost. A clinic visit postponed to preserve wages may later require more expensive care.

The strongest analysis keeps remedies attached to mechanisms. More housing supply may reduce some rent pressure but not disability exclusion, displacement from care, discriminatory access, or predatory financing. Debt relief may restore liquidity but not correct an unstable schedule or unaffordable treatment. Medical coverage may exist while transport, paid leave, language access, or appointment capacity remains missing.

1.3 — The Material Overhead of Antagonism

Antagonism has a balance sheet, but the entries must be separated before they are judged. Baseline protection, prudent redundancy, transaction friction, distributional burden, and capture are not the same thing. Protective spending may address genuine danger, compensate for unreliable institutions, or preserve an arrangement that profits from fear. The economic task is to identify which function each layer actually performs. Before comparing defensive expenditures, identify the specific threatened function, probability and severity of failure, expected protection, distribution of burden, and an observable condition for retiring the safeguard.

1.3.1 — Defensive Spending as Social Structure

Defensive spending becomes social structure when precaution is no longer an occasional response but a permanent condition of participation. Households may buy private transport because public routes are unsafe; firms may hold excess inventory because customs and supply are unpredictable. Each decision may be sensible in isolation while the aggregate system remains expensive and brittle.

Defensive systems also create constituencies for their continuation. Vendors, departments, professions, insurers, and political actors may become dependent on the spending that fear generated. This does not make their work illegitimate. It means exit has distributional consequences. Section 8.3 addresses the worker and regional conversion obligations that prevent avoided expenditure from becoming immediate free cash.

An additional defensive layer has an opportunity cost: the value of the best feasible alternative use of its resources. That cost exists even when the layer is the better choice. The allocation becomes inefficient at the margin when its additional benefit is lower than

that opportunity cost. Compare feasible alternatives over the same horizon, keeping fiscal cost, labor, privacy, delay, and protection distinct where no defensible common unit exists. Necessary protection can support care and maintenance rather than merely crowding them out (HM Treasury, 2026b).

1.3.2 — Antagonism Inside Public Administration

Antagonism inside public administration often appears as defensive procedure. Staff expect fraud, applicants expect hostility, managers expect blame, and oversight bodies expect concealment. Repeated verification, fragmented records, narrow discretion, and inaccessible appeal can then become policy choices rather than accidental inconvenience. Administrative-burden research treats the costs of public access as design responsibilities rather than private coping failures (Moynihan, Herd, and Harvey, 2015; Herd and Moynihan, 2018).

Administrative burden includes the learning cost of finding and understanding a route, the compliance cost of meeting its requirements, and the psychological cost of stigma or stressful treatment. Money, literacy, health, language, digital access, and flexible time affect who can carry those demands. A formally equal procedure can therefore produce unequal access (Moynihan, Herd, and Harvey, 2015; Herd and Moynihan, 2018; Halling and Bækgaard, 2024).

The legitimacy loss is cumulative. When lawful access repeatedly feels adversarial, people may withdraw, rely on brokers, avoid benefits, withhold information, or treat every office as a place where personal influence matters more than the written route. Agencies then encounter more incomplete cases and defensive behavior, which can confirm their suspicion. Public administration becomes more expensive because antagonism has become mutually reinforcing.

Administrative design should be evaluated through a lifecycle account. An agency may save staffing while users lose shifts, travel farther, purchase assistance, or abandon claims; a strict verification rule may prevent some improper payments while increasing delay, appeal, health deterioration, and emergency spending elsewhere. Evaluation must include fiscal cost, user time, distributional burden, error severity, correction time, and the value of access that the procedure either preserves or destroys.

1.3.3 — Opportunity Cost of Suspicion

Suspicion can close options that require durable commitments. Apprenticeships, supplier development and research partnerships depend on credible continuity. Delay may preserve flexibility while driving away skilled workers, worsening illness or leaving maintenance to become an emergency. Appraisal must consider both the protective value of caution and the capability lost while decisions remain unresolved.

Visible responses to fear can win more public honor than work that prevents it. That incentive shapes budgets and careers, making quieter alternatives easier to overlook.

Transaction-cost analysis examines the resources spent arranging, monitoring, and enforcing exchange. Reliable institutions can reduce repeated checking and bargaining, while search and enforcement remain necessary even when opacity is reduced (Coase, 1937; North, 1990).

1.4 — Economies as Political Psychology Made Concrete

Economic systems allocate goods and claims, but they also shape the expectations and habits people develop through reliable payment, debt, arbitrary access, status pressure, and institutional repair. Those experiences influence which strategies seem prudent, although no economic rule explains personality or culture in full (Bowles, 1998). Prices, production, scarcity, credit, investment, and public finance have distinct causal structures, while people can also contest and redesign the rules that shape their behavior (North, 1990). Rules that reward connections or costly signals may make hoarding, gatekeeping, concealment, or rapid exit rational responses rather than fixed personal traits (Bowles, 1998; North, 1990). Public offices carry a similar lesson: eligibility checks can be necessary, but requiring people to display desperation or tolerate humiliation provides no additional evidence. The analysis must therefore avoid both economic determinism and the fiction that institutions leave human development untouched.

1.4.1 — Status Pressure and Defensive Consumption

Status pressure becomes economic when visible consumption operates as insurance against exclusion. Housing in a costly district may purchase school access. Credentials may protect employability. Clothing, transport, and technology may signal reliability in markets that ration opportunity through appearance. Some consumption remains expressive or pleasurable. The system becomes coercive when dignity, safety, and access depend on winning a costly display contest (Bertrand and Morse, 2016).

A household may prefer lower consumption and still face inferior access, stigma or reduced mobility. Reducing that

pressure requires changing the rules that make costly display rational while preserving room for taste and cultural difference.

Defensive consumption can conceal insecurity: a household may look comfortable while depending on debt and overwork to maintain the appearance that protects its access.

Status pressure also shapes public politics. People who feel socially threatened may resist repair for others if they believe their own rank is fragile. Scarcity and status can combine into resentment: why should someone else receive help when I have exhausted myself to remain respectable?

1.4.2 — Debt as Future Psychology

Debt can narrow future choices through repayment risk or widen them by financing useful investment; its effect depends on terms, purpose, alternatives, and who bears risk (Richardson, Elliott, and Roberts, 2013; Haushofer and Fehr, 2014). Chapter 7 examines these mechanisms.

At public scale, debt service can crowd out maintenance spending and narrow policy choice. In both settings, the financial claim matters because it alters the options still open when conditions change.

Debt can also carry undeserved shame. Household survival, medical need or public recovery can create obligations without personal wrongdoing. Treating the debt itself as evidence of failure adds stigma to the financial constraint.

1.5 — War, Corruption, and Scarcity

War, corruption, and scarcity are distinct domains with different evidence, actors, and remedies, yet their effects often converge in the same household calendar. Conflict can disrupt supply and public capacity. Corruption can distort allocation, maintenance, and credibility. Scarcity can raise the value of controlling an essential route. The analytical task is to map the interfaces without turning the three into synonyms (World Bank, 2011; United Nations and World Bank, 2018; Olken and Pande, 2012).

1.5.1 — Why These Cannot Stay Separate

The common mechanism is institutional stress. Emergency procurement can concentrate discretion; damaged logistics intensify shortage; shortage raises the return to gatekeeping; and weak oversight makes diversion harder to detect. The sequence is contingent rather than inevitable. Robust records, distributed capacity, credible review, and protected local knowledge can interrupt it, so analysis must examine both amplification pathways and resistance capacity (World Bank, 2011; United Nations and World Bank, 2018; Olken and Pande, 2012).

Choke points may be as mundane as a port, payment channel, maintenance depot, or understaffed permit office. Control of such nodes can transmit a geopolitical or governance failure far beyond the original event. Mapping them reveals where modest repair may prevent a much larger cascade.

Fiscal translation is one route through which the domains meet. Conflict risk raises borrowing and readiness cost; corruption reduces value obtained from expenditure; scarcity increases emergency spending and political pressure. A budget can therefore deteriorate even when each ministry describes its own choice as necessary. The combined account must distinguish

unavoidable cost from cost amplified by weak institutions or concentrated power.

A material-incidence map records a countermeasure and a failure signal beside each point where war, corruption, and scarcity amplify one another.

Scarcity can accelerate conflict by narrowing acceptable options and raising the political value of controlling an essential route. Food, water, energy, housing, medicine, land, work, and public attention do not produce conflict mechanically, but instability around them can make scapegoating, patronage, coercion, and exclusion easier to organize. The effect depends on institutions, distribution, expectations, and whether people can see a credible route to fair allocation (World Bank, 2011; United Nations and World Bank, 2018).

1.5.2 — Strategic Fear as Economic Actor

Strategic fear acts through risk premiums, inventory policy, capital location, insurance, defense procurement, research priorities, and the willingness to enter long contracts. Its effects can arise without an order from the center because firms and households respond independently to perceived exposure. The burden is therefore distributed through markets before any public account of the threat is complete.

Incomplete information can justify readiness, yet reciprocal precautions may escalate when each actor treats the other’s defense as proof of danger (Jervis, 1978). Section 5.5 examines bounded verification and confidence-building responses.

Strategic fear also changes innovation. Research and industrial policy may prioritize military or protective uses, sometimes producing valuable spillovers and sometimes crowding out civilian repair. Evaluation should count security value,

technological learning, workforce capability, secrecy cost, procurement risk, and the civilian alternatives forgone.

The second-round costs of strategic fear appear when reciprocal precautions harden into material structure. Parallel standards, separated supplier networks, larger inventories, restricted finance, duplicated research, and long-lived procurement commitments can persist after the original signal has changed. The security dilemma is therefore economic as well as military: interaction can fragment capacity even when no participant initially sought escalation (Jervis, 1978).

A verified communication channel, inspection regime, reserve-sharing agreement, or compatible technical standard can create economic value before deep trust exists by reducing uncertainty each actor would otherwise price alone. Cooperation begins materially when fewer resources are required to protect against misread intentions.

A proposed review record would preserve each forecast's event definition, deadline, probability, evidence date, and subsequent updates. Compare performance with a stated baseline on the same questions and forecast horizons, not with an easier set chosen afterward. Keep classified inputs protected while allowing authorized independent assessment. Record protective actions taken after a warning: a forecast of events under an expected response differs from a forecast of danger without that response. Nonoccurrence alone establishes neither a mistaken warning nor successful prevention. Review should investigate that distinction and retain the costs of error alongside accuracy.

Forecasting skill can be developed rather than inferred from confidence or rank. In a two-year geopolitical forecasting tournament, Mellers and colleagues (2014) reported that probability training and team collaboration improved calibration and resolution: probabilities better matched observed frequencies,

and forecasts better distinguished events that occurred from those that did not.

1.5.3 — Burden Transfer Across Scales

A local service failure may shift care to families, then raise health costs, reduce labor supply, and eventually require national spending. Migration may carry the consequences of war or climate pressure into municipalities whose budgets were never designed for the arrival pattern.

Tracing transfer requires a chain of custody for cost. The account should name the initiating event, the institution that could absorb or redirect it, the group that ultimately carried it, and the time at which repair became more expensive. This will not settle every dispute about responsibility. It prevents each level from presenting the burden as if it arrived from nowhere and belonged naturally to the next weaker actor.

Burden transfer also has a temporal form. Deferred maintenance can improve a current budget while increasing failure probability, emergency cost, and future service interruption. Cheap debt can widen present room while narrowing future policy choices. Temporary security measures can become permanent contracts and institutional habits. A material peace analysis therefore records the time horizon of every claimed saving and distinguishes a genuine efficiency from a bill moved beyond the reporting period.

When legitimate burdens conflict, begin with the lawful decision, the feasible alternatives, and the consequences of continuing the present arrangement. Identify binding rights and minimum service conditions before comparing discretionary gains. Where no option meets those conditions, reconsider scope, resources, and interim protection rather than silently treating current failure as an acceptable default. For an initial dominance

screen among compliant options, set aside an option when another performs at least as well on every decision-relevant dimension and better on at least one across the credible assumptions being considered. An uncertain estimate alone does not establish that ordering. Where genuine tradeoffs remain, the authorized decision maker gives public reasons, identifies who still carries cost, and states what evidence would reverse the choice. A reviewer tests those reasons; a dashboard does not acquire authority to decide. Section 28.7.2 follows this process through a contested service redesign and an appeal.

Effective interfaces assign each scale a distinct job. Local institutions contribute knowledge and immediate service; national systems supply fiscal capacity, standards, and redistribution; international coordination addresses cross-border risks; households retain protection, privacy, and voice. Burdens fall through the system when authority, resources, and correction do not travel together, so interface design is itself a form of peace infrastructure.

1.6 — What the Hypothesis Adds to Material Peace

The terminal hypothesis and the material policy argument must remain distinguishable even when they inform the same inquiry. A formal claim about the endpoint of experience does not settle which present institutions are justified or how public budgets should be allocated. Material proposals can be tested through their effects on rights, access, service reliability, and avoidable burden without assuming the hypothesis is true. The relevant question is what additional reasoning the hypothesis contributes while leaving independently warranted care and repair intact. This section separates its formal implications from the cost interfaces that public institutions must still make workable.

1.6.1 — The Terminal Hypothesis

Per-Stream Terminal Valence Closure, or PTVC, is the hypothesis that within one separately routed maximal qualifying conscious stream, finite positive and negative phenomenal magnitudes are exactly equal at that stream's true endpoint under a fixed representation calibrated to a neutral origin. The claim is intrastream and nontransferable: it does not balance one person against another, one generation against another, or a population against an aggregate.

If the accumulated signed balance is nonzero, exact closure constrains the net signed remainder of that same stream. A negative balance requires a net positive remainder and a positive balance a net negative remainder. This fixes the remainder's sign and net magnitude under the declared representation, not the gross amount, form, timing, or cause of future experience. Required for closure does not mean morally deserved, and no particular reward, illness, injury, or event is prescribed.

Equal terminal balances leave a rich space of possible histories. Consider two hypothetical complete histories for a stream under

the same fixed, valid representation. One contains 10 units of positive magnitude and 10 of negative magnitude; another contains 100 of each. Both end at zero, although their gross phenomenal magnitudes differ. Sequence, duration, and opportunities for agency are further questions that the terminal total alone does not settle. These invented quantities are neither measurements of people nor a claim of interpersonal comparability. They show why the scientific question about terminal balance can coexist with a substantive material inquiry into which institutions expand usable choice and reduce preventable suffering.

The claim being tested must remain distinct from the observation used to test it. Exact zero, a finite tolerance band, and positive probability concentrated at zero are not equivalent targets. Probability-one closure permits exceptions with probability zero under a declared law. Limit-sure attainability allows policies whose success probabilities approach one without any single policy necessarily attaining it. Set-universal closure permits no qualifying exception. A model can satisfy more than one of these descriptions, but support for a weaker statement does not establish a stronger one. A true endpoint is not automatically biological death, a protocol endpoint, or the last observation. A study must specify its representation and neutral origin, the routing of experience to a stream, and its treatment of endpoints and missing observations. It must then justify the connection between measured indicators and phenomenal valence, while separating the quantity sought, the estimator used, and the resulting verdict.

PTVC is mechanism-neutral. Adaptation, opponent regulation, allostasis, learning, selection, endpoint conditioning, hidden-state dynamics, and other models may explain observed patterns without establishing the terminal target. Evidence for a route is

not automatically evidence for exact closure, and an action-support or policy-support result does not establish conscious ledger awareness, psychological motivation, moral instruction, or personal obligation.

Past experience also remains fixed in character. Later recovery, compensation, interpretation, or prognosis can improve the future and acknowledge wrong, but it does not make prior pain unreal. That boundary keeps present care and repair independently warranted rather than postponing them until a terminal theory is settled.

Finally, PTVC is not a consciousness detector or a public balancing rule. Even strong evidence within a declared reference class would not by itself identify which systems are conscious, where experience is located, or who has personhood. Material policy must therefore continue to rely on independent evidence, rights, ethics, consent, dignity, and human judgment.

1.6.2 — Independent Warrant and Cost Interfaces

The added value of cross-institution burden tracing can be tested against competent existing appraisal. In one proposed study, independent teams trained to a common standard would each review one case. Within each case, teams would be randomly assigned to standard appraisal or that appraisal plus explicit cross-institution burden tracing, with the same evidence and time budget. Case packets would be frozen to information available at the original decision date; later outcomes would be held separately rather than leaked into the appraisal. An independent group would establish the reference case record and adjudication rules from the same available evidence before seeing team reports. Reports with assignment labels removed would be scored for consequential omissions detected and feasible remedies, alongside missed burdens, unsupported additions, and time used.

Disputed findings would remain disputed where the record cannot settle them. Held-out cases would test whether the method travels beyond the examples used in training. This study would evaluate the review method; following recommendations into delivery would be a further test.

The method joins capability analysis, public finance, and administrative-burden research around the same decision, tracing an apparent saving beyond the originating budget to its eventual bearer and a funded correction route (Sen, 1999; Herd and Moynihan, 2018; GAO, 2020).

A cost interface identifies the resources an adjacent proposal needs to work. A legal remedy, for example, depends on accessible filing, evidence handling, adjudication and enforcement. Its design is incomplete until responsibility and funding for those functions are assigned. Chapter 29 applies this discipline to the project’s own institutions.

Costing must remain dimensional. A cost profile keeps money, time, risk, ecological effects, and lost options distinct unless a defensible conversion is supplied. It states the baseline and horizon, marks uncertainty, and avoids double counting. This permits comparison of projects under explicit objectives; it does not turn the result into a ranking of people’s worth or a complete measure of a good life.

Chapter 2 — Enoughness and Ordinary Abundance

Two households may have identical incomes and entirely different freedom to use them. One depends on accessible transport, temperature control, paid assistance, and frequent medical travel. Another lives beside reliable transit and public services and has fewer care obligations. A single threshold may classify their resources equally while overlooking the costs that determine what each person can actually do. The capability approach begins with this distinction between resources and usable opportunities (Sen, 1999; Nussbaum, 2011).

Enoughness combines private means and shared systems into a secure material floor, responsive to disability, geography, climate, care duties, and local infrastructure. Section 2.1 sets out the adequacy and review test. The comparison also changes with the question being asked. Equal cash grants can be a useful starting rule, yet they do not settle whether both households can reach the same essential service. Conversely, different support packages need not express different human standing. The chapter asks how a public promise can remain intelligible while allowing those differences to be examined rather than concealed.

The idea has a sharp distributional boundary. A call for restraint cannot become a lecture to people who lack safe housing, reliable care, adequate food, or discretionary time. Where basic capacity is missing, building and repair take priority. Scrutiny of avoidable excess should begin where the strongest actors can change course without sacrificing necessities. This is not a universal ceiling or an aesthetic preference for a modest life; it is an attempt to distinguish secure choice from deprivation dressed as virtue. Excellence may produce unequal outcomes through talent, discipline, risk, teamwork, or discovery. Fairness does not promise identical achievement.

Once that floor exists, ordinary abundance becomes a positive project. A maintained park, a library open when people can use it,

a comfortable home, an accessible clinic, and time for rest are not merely consolation prizes for those who cannot purchase prestige. Shared services are not always pure public goods; their value does not depend on that classification. They make social participation, pleasure, creative work, and exceptional achievement easier to choose without fear. Public beauty belongs here when the people who endured neglect can remain to enjoy the improvement.

Ecological comparison should guide how essential capacity is built without freezing deprived communities' aspirations (O'Neill et al., 2018; Fanning et al., 2022).

The chapter moves from an adequate floor toward stable, varied, and inventive lives, then asks how prosperity should be measured. National accounts remain valuable. Housing access, health, time, distribution, ecological condition, and repair must be examined beside them without pretending one composite figure can decide whose life matters. Enoughness succeeds only if the people who depend on the floor can test and improve it.

2.1 — Enoughness as Room to Live

The practical question is not whether one official can define enough for everyone, but whether an accountable institution can justify a minimum it is able to deliver. It must identify the protected function and the evidence used to estimate local need. It must also state the capacity and cost of meeting that need and provide a route for people whose disability, household duties, geography, or circumstances make the standard inadequate. If resources fall short, the authority must disclose the shortage and its allocation rule rather than relabel deprivation as maturity.

2.1.1 — Austerity and Unequal Sacrifice

A call for restraint deserves scrutiny when people already carrying scarcity are asked to absorb costs that insulated actors can avoid. A useful diagnostic is to compare the stated purpose of restraint with its actual incidence. A proposal may claim to protect fiscal stability while shifting travel time, care work, health risk, or administrative burden onto people who have no private substitute. Another may reduce a visible budget line while increasing arrears, homelessness, emergency care, staff burnout, or deferred maintenance elsewhere.

People can share the same basic need while requiring different goods or services to meet it; one administratively convenient basket will not fit every context (Rao and Min, 2018).

The same nominal reduction can be minor for a secure actor and catastrophic near the floor. An incidence analysis therefore reports starting reserve, access to substitutes, reversibility, exposure to cascade, and control over the decision instead of treating equal percentage cuts as equal sacrifice. Include lost time, privacy, unpaid leave, injury risk, and dependence on an unsafe relationship as well as cash.

Restraint might mean lower luxury energy use, reduced speculative land holding, longer product lives or fewer prestige subsidies.

Before cutting frontline capacity, accessibility, maintenance, or care, scrutinize executive security costs, ceremonial spending, dependence on consultants, vendor rents, and avoidable duplication. Include costs shifted to workers and users in that review.

2.1.2 — Enoughness Without Imposed Scarcity

Accessibility determines whether a resource can be used; differentiated support therefore belongs inside the adequacy standard. WHO evidence shows that avoidable barriers inside health systems themselves contribute to inequity for persons with disabilities (World Health Organization and World Bank, 2011; World Health Organization, 2022a).

Imposed scarcity appears when avoidable design choices make necessary goods harder to reach. It can arise through chronic underinvestment, rent extraction, artificial queues, monopoly control, exclusionary geography, procedural overload, or a refusal to maintain capacity that public authorities already know people depend on. These routes differ and require different remedies, but each shifts a preventable shortage onto ordinary life.

People need to know where to report failure, what evidence is required, who owns the review, what happens while it is pending, and how quickly a lost option can be restored. An appeal resolved only after eviction, worsening untreated illness, utility shutoff, or lost schooling may arrive too late to prevent avoidable harm. Indicators should locate where nominal provision fails in practice.

People should be able to change work, household, region, faith, community, or life plan without losing essential support.

Portability of benefits, records, and care helps make that freedom real.

2.2 — Enoughness as Civilizational Maturity

Enoughness begins as a personal judgment about what permits a life to function. At civilizational scale, it becomes a stewardship question about how shared systems use limited resources. Public maturity involves identifying whom a decision burdens and who has the power to alter it. It also requires comparing proposed restraint with feasible alternatives instead of praising sacrifice in the abstract. The measure is not austerity for its own sake but durable capability to protect ordinary life.

2.2.1 — Maturity as Burden Awareness

Once a burden is located, maturity assigns responsibility through causal contribution, benefit, control, and capacity. This approach is more useful than generalized guilt because it identifies who can change the route, who needs protection during adjustment, and which repair can be verified.

Compare the proposed activity with feasible alternatives, including inaction; each path can move risk to different people. A visible burden alone does not establish avoidability or bad faith.

Consider an illustrative drainage problem beside a public road. Water from several properties reaches an old culvert, while a recent construction project has changed the flow. The road authority can clear the blockage immediately. Establishing how much each property contributed may take longer. Repair and the final allocation of cost are therefore two connected decisions with different clocks.

A workable agreement can authorize the road crew to restore drainage while preserving the evidence needed to settle payment later. The authority to act follows the ability to restore the shared function. The eventual contribution follows the agreed account of cause and benefit. A household does not need to wait for the entire dispute to end before the road becomes usable.

This makes responsibility more precise. The person able to interrupt a problem is not necessarily the person who should carry its final cost. Separating those roles can make cooperation possible sooner. People can agree on the next useful action while continuing to examine a disagreement about the past.

2.2.2 — The End of Panic Growth

Panic growth is expansion driven partly by the fear that any pause will produce exclusion, insolvency, loss of rank, or strategic defeat. The concept describes a mechanism, not a diagnosis of every growing household, firm, or state. Growth may answer real need, diffuse useful knowledge, and build capability.

At household scale, the mechanism may involve debt service, unstable work, credential competition, housing inflation, or status signals that protect access. At institutional scale, it may involve refinancing, acquisition, attention extraction, planned obsolescence, market dominance, or budget growth tied to permanent threat.

A mature growth regime can move at several speeds at once. Experimental sectors may advance quickly, mature high-throughput sectors may consolidate, essential systems may expand steadily, and destructive practices may contract with worker and regional protection. No single growth rate describes that movement. The governing questions are whether capability rises, brittleness falls, and institutions can slow a harmful activity without collapsing the people and places built around it.

2.3 — Ordinary Abundance

Ordinary abundance is the positive form of enoughness. It exceeds a bare minimum by naming dependable access to the material and civic conditions that make life usable: shelter, food, care, time, energy, mobility, learning, safety, beauty, association, and a future that can still be planned. Finite abundance is not a contradiction but a design problem. Many forms of value can expand without proportional material throughput: knowledge, care quality, durable design, repair skill, art, public space, software, prevention, trust, and time recovered from friction. Material goods still matter, especially where scarcity is real, but prosperity need not depend on treating every resource flow as if it could grow without ecological or maintenance consequence.

2.3.1 — Housing, Food, Care, and Time

Secure, habitable, accessible housing anchors care, school, work, privacy, and rest; Section 10.1 develops how its instability transmits costs across those systems.

Food security is not only sufficient calories or a stocked market. It includes affordability, nutritional adequacy, cultural usability, safe storage, time to prepare food, and resilience when income or supply changes. A household that can eat only by sacrificing medicine, heat, transport, or dignity has not reached ordinary abundance.

Care and time determine whether housing and food can become a life rather than a survival schedule. Childcare, disability support, elder care, health care, rest, and discretionary time distribute freedom across households.

Research on poverty suggests that financial preoccupation and stress can affect attention and economic decisions (Mani et al., 2013; Haushofer and Fehr, 2014). Reliable housing, food, care, and time may therefore widen the room people have to compare

choices and participate in public life. That extension is a plausible hypothesis for collective settings and can be tested directly through participation, deliberation, and decision outcomes rather than assumed.

2.3.2 — Beauty as Public Repair

Public beauty is not a luxury when it arrives through green, shaded, walkable, safe, and accessible public space. Trees, water, quiet, light, durable materials, and places to sit can affect heat exposure, movement, social contact, restoration, and the ease of using essential infrastructure. Design cannot cure deprivation, but it can reduce or intensify the burden of moving through a city (Twohig-Bennett and Jones, 2018; World Health Organization Regional Office for Europe, 2016).

A place also communicates who is expected to belong. A shaded bench, clean public bathroom, dignified clinic, maintained school entrance, calm transit platform, or safe park tells ordinary people that income has not exiled them from beauty.

The displacement test keeps that repair honest. If beautification raises rents, removes informal users, polices harmless presence, or turns a neighborhood into a display for outsiders, it shifts burden under the name of improvement. Public beauty is fair when the people who carried neglect can remain to receive the repair.

Maintenance adds a temporal message. A cared-for public place says that someone expected the community to exist tomorrow. Neglect teaches the opposite by making lives seem temporary in the public imagination. Restoring beauty can therefore lengthen a community’s practical horizon.

2.4 — Stability Over Stimulation

Stability concerns the pattern of burden through time: average resources do not reveal whether a person can form reliable plans (Haushofer and Fehr, 2014; OECD, 2024a).

2.4.1 — Useful Speed and Dependable Time

Artificial urgency can benefit actors who gain from shortened judgment. A limited window, unstable schedule, rolling crisis, or performance alert can reduce bargaining time and make refusal costly. The proper audit asks what objective deadline exists, who designed it, whose option expires first, and whether the same purpose could be achieved without transferring planning risk downward.

Urgency should be classified before it is moralized. Some is protective, as in emergency medicine or disaster response. Some is logistical, as in a closing transport window. Some is contractual and negotiable. Some is manufactured to extract attention, labor, money, or compliance. The burden account names the class, evidence, beneficiary, and consequences rather than praising speed or slowness in the abstract.

A useful deadline closes a decision that somebody else genuinely needs in order to proceed. Suppose a supplier must know on Thursday whether a workshop wants a delivery on Monday. The deadline coordinates a vehicle and a receiving crew. Moving it to Wednesday merely to make the customer decide sooner gives the workshop less time without creating the same logistical benefit. Asking what depends on the answer reveals whether speed is doing productive work.

This distinction can improve a whole chain of decisions. The workshop might agree to a provisional quantity on Wednesday and a final quantity on Thursday, within a range the supplier can accommodate. Each side then receives the information it needs at

the moment it becomes useful. The arrangement replaces a single abrupt demand with commitments matched to actual dependencies. Predictable rules can make coordination faster because people spend less time renegotiating emergencies that the calendar itself created.

2.4.2 — Stability With Adaptation

Healthy stability is adaptive because it separates what must remain continuous from what should change. Water, care, identity records, income protection, and emergency access may require continuity; procedures, technologies, and organizational forms may be revised. Designing those layers explicitly prevents reform from destroying the essential service it was meant to improve.

One way to design that separation is to describe the promise independently of the machinery that currently delivers it. A public payroll service promises that an authorized payment will reach the right person on its agreed date. Its software can change while that promise remains fixed. During a replacement, the team can compare calculations before changing the route that actually sends money. The old route remains available until the new one has demonstrated the required function.

The same principle gives innovation a clear object. The team is free to improve the calculation process, simplify correction, or reduce repeated entry. It is not required to preserve an awkward screen merely because people know it. What carries forward is the service commitment, together with the information needed to verify it. Stability belongs to the dependable relationship; adaptation belongs to the means by which that relationship is kept.

This also makes success easier to describe. Staff should be able to explain which task became simpler while users can still recognize their entitlement and payment date. A change that

produces that combination has improved the system without asking every household to absorb its internal reorganization. The institution has carried the transition rather than exporting the work of understanding it.

2.4.3 — Lower Volatility and Ordinary Calm

The cost of volatility should be measured through behavior and process rather than inferred from a mood. The number of contacts about each case, schedule changes, payment volatility, missed appointments, emergency borrowing, repeated proof, abandonment, and correction time can reveal where instability is consuming attention. Qualitative evidence remains necessary to explain why the same indicator carries different consequences across lives.

Calm has distributional conditions. It is easier to praise serenity when another person is absorbing the night shift, dangerous commute, unpaid care, or emergency phone call. Ordinary calm counts as shared wealth only when the systems producing it do not export alarm to hidden workers, distant communities, or future repair budgets.

The shape of free time matters alongside its total. Consider two arrangements that each return three hours during a week. One gives a person six unpredictable half-hour gaps. The other gives a known three-hour interval. The total is identical, but the second arrangement may support a visit that requires travel or a task that needs uninterrupted concentration. A time account becomes more informative when it records notice and continuity as well as duration.

People will value those arrangements differently. Someone providing care may prefer predictable shorter intervals. Another person may need a longer block in order to leave the neighborhood. The practical improvement is to make the pattern

visible and give people some influence over it. An institution can then learn whether it has returned time in a form that its users can actually spend.

Ordinary calm also has a collective rhythm. When a household knows which evening is free, other people can make compatible plans. A single dependable interval can become a shared meal or a recurring rehearsal because it is available to more than one calendar. Stability creates room for these relationships through coordination, not by prescribing how the recovered time should be used.

2.5 — Growth That Builds and Growth That Depletes

Growth requires classification: higher output can build durable capability, or deplete the conditions on which future production depends. The phrase growth that must stop carries a high burden of proof. It cannot be aimed first at subsistence, basic development, or communities still building essential capacity. It applies most clearly where an activity depends on severe nonconsensual harm, fraudulent value, irreversible destruction without necessity, or cost shifting that cannot be repaired. Even then, stopping an activity does not authorize abandoning the people whose livelihoods were organized around it.

2.5.1 — Productive Growth

Productive growth expands the capacity to meet human needs. The first test is what lasts. After the announcement, financing cycle, or construction rush, does society retain useful skill, better tools, stronger care, safer infrastructure, cleaner energy, more resilient food systems, dignified work, or greater repair capacity? Growth that leaves no capability behind may have increased motion without increasing wealth.

The second test is the full lifecycle. Patient investment in education, health, infrastructure, research, and maintenance may create more value even when its early numbers look modest. Timing and persistence therefore matter as much as initial scale.

The third test is what actually improved. Higher prices, transactions, leverage, or measured output are not themselves proof of productive growth. Identify the physical, institutional, or human capability that increased and how it will be maintained.

Reparative growth is also a correction of accounting. Past systems often called themselves cheap by deferring maintenance, neglecting places, extracting from ecosystems, underpaying care, or letting trauma become private burden. Repair makes

yesterday's hidden cost visible in today's budget. The higher recorded cost may reflect obligations that the earlier budget left out.

2.5.2 — Extractive Growth

Extractive growth expands numbers while shifting burden elsewhere. It may raise revenue, valuation, output, or consumption while lowering wages, weakening care, degrading land, increasing debt, worsening housing pressure, or sending repair costs into the future.

Extractive growth can be legal. A system may permit rent shifting, labor precarity, ecological damage, predatory lending, or speculative price escalation.

The difficult cases are those in which extractive growth delivers a visible good. A platform may be convenient, a factory may employ people, and a financial product may provide temporary liquidity. The visible benefit should be acknowledged, then tested against the hidden burden on which it depends.

When refusing a transaction would mean losing housing, care, work or access to an essential service, a signature tells little about the freedom to choose. Assess bargaining power, substitutes, information and mobility. Include people unable to participate in the bargain, especially future people who will inherit its costs.

Speculation can finance useful risk-taking. To identify extraction, examine whether returns depend on controlling essential access or shifting costs onto others. Section 13.6 develops these financial mechanisms.

Speculation also becomes politically distorting when public authority begins to protect asset values more fiercely than ordinary access. If housing policy, tax policy, infrastructure policy, and financial policy quietly organize around protecting escalation, then the beneficiaries of rising prices have become a

governing constituency. People without those assets may have to devote more of their future income to securing access.

2.5.3 — Ecological and Social Limits to Growth

Ecological analysis must distinguish local scarcity, global boundary, reversible pressure, cumulative harm, and uncertain tipping risk. It should examine full life cycles, rebound effects, land and water use, biodiversity, pollution, and distribution across places and generations. No single footprint number can answer every question, but uncertainty is not a license to ignore potentially irreversible burden.

Meeting human needs with low energy use depends on how effectively energy is turned into housing, mobility, health, education, and public services. The comparison is not with a universal consumption quota, but with the social organization that allows a given level of resource use to produce broad capability. Inequality, weak public systems, poor settlement patterns, and extractive infrastructure can require much more throughput for the same lived result (Vogel et al., 2021; Millward-Hopkins et al., 2020).

Planetary boundaries identify constraints that accounting totals alone can conceal: climate stability, biosphere integrity, freshwater availability, and land-system change. An appraisal should state which limits a proposed expansion affects, how the effect is measured, and what alternative would meet the same need with less damage (O'Neill et al., 2018; Fanning et al., 2022).

Minimum access is not the main driver of many global ecological pressures. Research on inequality and affluence indicates that meeting basic needs for people who remain deprived is materially different from preserving high-consumption patterns at the top. Enoughness therefore directs restraint first toward avoidable excess, structural waste, and

burdens displaced onto poorer communities, rather than converting ecological limits into another demand that the least secure accept less (Rammelt et al., 2023; Wiedmann et al., 2020).

Decision under uncertainty should use robust and adaptive designs. Sensitivity analysis, scenario ranges, reversible stages, early-warning indicators, and predeclared review points show which choices remain defensible across plausible futures. This approach permits timely action while preserving correction when assumptions, prices, technologies, or ecological conditions change.

2.6 — Aspiration Without Defensive Excess

A first home, a first degree, or a period of financial comfort can represent liberation. Judgments about excess must account for what such gains make possible from a person's starting position. The relevant distinction is between developing capability and purchasing a defensive status that others must then match. Public policy should preserve meaningful ambition while reducing the pressures that make costly display a condition of dignity.

2.6.1 — The Right to Build and Excel

The right to build includes the right to fail. Ambition becomes coercive when people are allowed only the image of success. A fair culture lets people try, learn, stop, change paths, and begin again without permanent economic or civic exclusion.

Vocation should be distinguished from status anxiety. Vocation draws a person toward craft, inquiry, care, service, creation, or mastery because the work itself matters. Status anxiety seeks rank because dignity or access feels conditional. The outward effort may look similar, but the burden pattern differs.

Wealth can provide security, independence, productive investment, family care, philanthropy, and freedom to attempt valuable work. Domination appears when wealth controls essential access, captures public rules, suppresses competition, purchases impunity, or leaves others dependent on arbitrary favor. This distinction preserves ownership and achievement while requiring essential systems and public authority to remain contestable.

2.6.2 — Innovation That Lowers Burden

Novelty alone is not enough. A tool, process, firm, technology, or public system should be asked what hidden cost it reduces and what new cost it creates.

Large grids, transport networks and research facilities may be necessary. Their scale should be justified by the service required and assessed against dependence on the provider, public oversight, maintenance and the cost of replacement.

Adoption needs support and safeguards proportionate to the tool's risks. A successful demonstration is therefore not the same as successful ordinary use: staffing, support, incentives, and user conditions often change as adoption widens. Evaluation should ask whether the benefit survives that transition rather than assuming that performance in ideal conditions will automatically scale.

Apply the procurement and exit tests in Section 6.4. Credible alternatives also change bargaining while a system is working: a provider has stronger incentives to remain useful when users can leave.

Burden can move even when innovation delivers a real gain. Automation may reduce dangerous work while concentrating income; digital access may save travel while weakening privacy; efficient logistics may lower prices while intensifying labor or fragility. Honest adoption states the tradeoffs, funds worker and user transition, and establishes correction before celebration hardens into inevitability.

2.7 — The New Meaning of Prosperity

Production matters because a society must create the goods and services on which people depend. Yet a larger total does not establish who can use those gains or what it costs to maintain them. A complete prosperity account connects output with distribution, household time, health, and the condition of shared assets. It also asks whether today's choices preserve capacity for those who come later. The next measures separate those questions rather than asking GDP to answer all of them.

2.7.1 — Prosperity Beyond GDP Alone

Prosperity beyond GDP begins with a division of labor among measures. National accounts record production, income, consumption, investment, assets, and liabilities. Those records also support estimates of productivity and assessments of public finances. Well-being, distributional, environmental, and balance-sheet accounts ask whether those gains reached households, preserved time and health, strengthened public systems, and left future capability intact (HLEG, 2026; OECD, 2024a; Liu et al., 2024).

Accounts of output should also identify the unpaid labor and environmental conditions that sustain it. Public maintenance and inherited infrastructure require attention to their condition and service life as well as recorded expenditure.

Output can rise as household conditions worsen, while health or infrastructure gains may be only partly reflected in market transactions. Investigate those divergences rather than averaging them away (Stiglitz, Sen, and Fitoussi, 2009; OECD, 2024a; HLEG, 2026).

A mature framework distinguishes stocks from flows, averages from distributions, levels from volatility, and threshold failure from gradual change. Income, health, time, housing,

environmental integrity, trust, and repair capacity answer different questions. Keeping these distinctions explicit prevents a composite result from hiding a severe failure in one indispensable domain.

A housing measure must identify what deterioration calls for an investigation; a care measure must distinguish rising need from falling throughput; a maintenance measure must report the backlog that available crews and funding can actually clear. When a headline measure improves while conditions worsen for an affected group, the discrepancy triggers a public review rather than disappearing into an average. Decision makers should publish the affected population, the plausible causes, the options considered, the chosen response, and a date for checking whether access improved. These are separately reported quantities: no composite improvement may automatically cancel an unmet legal duty or essential-service failure.

2.7.2 — Prosperity Without Hidden Victims

A prosperity claim should include people whose contribution or loss falls outside its initial boundary, whether through distance, subcontracting, unpaid care, or delayed effects. Suppressed dissent and silenced household members cannot count as reduced burden.

The practical discipline is due diligence linked to remedy. Supply chains, procurement, labor conditions, ecological effects, debt exposure, and community impacts should be reviewable where feasible; workers and affected communities need protected voice; discovered harm needs correction, compensation, or redesign.

Residual uncertainty must be reported as part of the prosperity claim. An honest account separates known burden, plausible but unresolved burden, and effects outside current observation.

2.7.3 — Prosperity as Repair Capacity

Prosperity includes what a society can repair. A brittle rich society may have high output and low repair capacity; a more modest society with strong maintenance, trust, care, and public competence may be better equipped to preserve ordinary life through shock.

Measure the interval from a reported defect to restored service, identifying households still waiting after the system is declared operational. For people with little reserve, that delay can turn a repairable fault into irreversible loss.

Restoring an asset does not settle the claims of people harmed by its failure. Compensation and complaints must remain accessible after engineers declare a bridge safe and the emergency team departs; their findings should inform the next maintenance decision.

A repair schedule can show whether the longest-exposed neighborhoods are receiving priority or another postponed completion date.

Chapter 3 — Corruption Drag and Guesswork

A permit applicant receives one instruction at the counter, another by telephone, and a third from an intermediary who promises to expedite the file. No bribe has necessarily changed hands. Yet the applicant has already lost time, delayed investment, and perhaps borrowed money to survive an uncertain wait. A supplier facing unpredictable customs clearance may pay a similar price in storage, inventory, and lost contracts. Evidence from transport and port settings shows why corruption can alter decisions far beyond the payment itself (Olken and Barron, 2009; Sequeira and Djankov, 2014).

The first obligation is causal precision. Bribery, embezzlement, favoritism, selective enforcement, administrative incapacity, and ordinary error are different phenomena. Similar symptoms do not establish a common cause. A long queue may reflect understaffing; a suddenly accelerated case may warrant investigation but does not by itself prove a corrupt arrangement. Perceptions, reported experiences, audit discrepancies, and prosecutions also measure different things. The inquiry needs comparable cases, documented rules, decision times, records of exceptions, and a credible account of who possessed discretion (Olken and Pande, 2012).

That discipline makes the invisible cost more serious, not less. When a route cannot be trusted, citizens keep redundant records, firms purchase advice, public servants demand extra signatures, and reporters reconstruct decisions that should have been recorded. Actors with money can buy some protection from confusion; others carry the delay directly. The resulting loss is not one financial amount. Time, service quality, trust, safety, and foregone opportunities have related but distinct measures.

This chapter follows those costs from the initial irregularity into its wider consequences. It asks when verification is a necessary defense and when it has become a second barrier; when backups

are prudent resilience and when they reveal enduring public failure; and when a real shortage has been made worse by rules that sell access to essential capacity. An informal workaround may protect access when an official route fails, but becomes a private toll when relief depends on purchased connections. Adding or removing regulation does not settle the question. Controls need a defensible public purpose and must withstand scrutiny of their burden, beneficiaries, and alternatives.

Transparency is a possible remedy, but disclosure has costs and boundaries. Personal histories, security-sensitive details, and legitimate confidences are not public property. Consequential decisions nevertheless need owners, reasons, correction, and review. Useful visibility allows ordinary access to become less dependent on private interpretation while retaining confidentiality where it is warranted.

The destination of the chapter is practical rather than ceremonial. A cleaner procedure should reduce unnecessary steps, protect people from arbitrary delay, and leave a defensible record of what changed. Public confidence is earned through that performance. It cannot be manufactured by a dashboard, by demanding loyalty, or by calling every slow office corrupt.

3.1 — The Full Cost of Corruption

Corruption begins when entrusted power serves private advantage, but the amount diverted rarely captures the harm. A defensible cost account separates stolen public resources, private illicit transfers, and downstream injuries such as exclusion, inflated prices, delay, or inferior service (Olken and Pande, 2012). Embezzlement, false invoices, ghost workers, and diverted supplies may directly shrink public capacity, whereas a bribe paid from private money is not automatically an equal fiscal loss. Each claim needs an independently justified baseline, such as a comparable bid or engineering estimate adjusted for scope, timing, quality, and risk, without treating under-specified bids or inflated benchmarks as neutral. Propagation should be traced before costs are summed because overpayment, later repair, user delay, and lost service may overlap. The account identifies who carries each burden and states how uncertainty in the counterfactual changes the conclusion.

3.1.1 — Delay as Corruption Cost

Delay becomes a corruption cost when entrusted power is misused to make waiting a source of private advantage. The person who can wait has one kind of power. The person whose rent, medicine, business, shelter, safety, or legal status depends on the decision may have far less power to refuse. A corrupt system need not demand payment openly. It can make the lawful wait unbearable.

Delay attribution begins with a lawful or technically credible service horizon. Published standards, comparable cases, queue position, staffing levels, emergency conditions, and the actual sequence of decisions help distinguish ordinary delay from delay used as leverage. The inference strengthens when hidden intervention, unexplained reordering, selective acceleration,

retaliation, or repeated deviation from comparable cases appears in the record.

Record when the wait began and ended, and the period over which its consequences are tracked. Preserve the distribution and consequences of waits rather than treating all waiting minutes as interchangeable.

3.1.2 — Trust Destruction as Economic Loss

Loss of warranted trust can change behavior before a legal case is resolved. Firms may shorten contracts and citizens avoid programs, imposing costs on institutions that may themselves be clean. Distinguish documented corruption from credible risk, rumor and low confidence caused by other failures (Knack and Keefer, 1997; Algan and Cahuc, 2010).

Falling bid participation, greater use of intermediaries, rising compliance cost, low take-up of public services, survey evidence, staff exit, and repeated complaint patterns may all reveal coordination loss. None is specific to corruption. Economic shock, polarization, poor performance, discrimination, war, or technological failure can produce similar signals. A claim that corruption contributed becomes credible only when these indicators are read beside evidence about the hidden route that plausibly generated them.

The burden extends beyond the person unfairly excluded. Filling a post through favoritism can reduce service quality, narrow the supplier pool, drive skilled workers away, weaken professional norms, and teach younger people that connection outranks preparation. Some losses appear in vacancy, turnover, rework, error, or lower participation. Others remain visible in testimony and institutional decline. Keeping those consequences separate prevents one career injury from being counted repeatedly while preserving the wider public loss.

3.1.3 — Maintenance Failure

Corruption can become physical when poor materials fail early, inspections are softened, spare parts are missing, warranties cannot be used, or repair is unfunded.

Repeated failure can lower public expectations until broken elevators or unreliable water are treated as ordinary conditions. That resignation can protect poor performance by making complaints and repairs seem futile.

Maintenance failure has many possible causes, including fiscal shortage, climate stress, aging infrastructure, technical error, weak planning, abnormal use, and disaster. Attributing a failure to corruption requires evidence that entrusted power was misused for private advantage and that this misuse made the failure more likely, more severe, or harder to repair. Concealed conflicts, knowingly false reports, compromised inspections, captured budgets, or protected nonperformance can establish that connection when the evidence links them to the failure. When that connection is not supported, the accurate finding is maintenance failure with the causal question still open.

Frontline maintenance records can reveal missing parts, pressured inspections, repeat faults, and contractor defects before a headline audit detects them. Preserve those warnings in the procurement history.

The asset history should distinguish design, construction, inspection, wear, abnormal use, climate exposure, scheduled maintenance, budget release, contractor performance, complaints, and repair response.

Compare bids for equivalent service life and a common maintenance obligation, including delivery, repair, correction, and exit, before interpreting their initial price difference (Olken, 2007; Lewis-Faupel et al., 2016).

3.2 — Mistrust as Overhead

Mistrust adds recurring work to ordinary transactions, from checking records to buying protection against an unreliable service. Some verification is legitimate because error and fraud are real risks. The avoidable cost arises when uncertainty makes every participant repeat a check that a dependable common process could perform once. A useful account separates necessary safeguards from the fees, delays, and lost options imposed by a failing route.

3.2.1 — The Cost of Checking Everything

Checking becomes a second economy when brokers, certification services, legal advisers, premium support channels, and trusted intermediaries monetize route uncertainty. Some expertise is genuinely valuable. Rent appears when the official route is kept confusing or unreliable enough that ordinary access depends on purchasing interpretation, personal connection, or acceleration.

Contract-cost claims also require a proportionate baseline. Longer contracts can reflect genuine complexity, safety, labor protections, environmental duties, or cybersecurity needs. The benchmark is the safeguard burden appropriate to the same risk under a credible enforcement environment. Fewer bidders, tailored criteria, repeated change orders, concealed ownership, unexplained waivers, and tolerated nonperformance strengthen a capture inference; complexity alone does not.

3.2.2 — Public Services Under Suspicion

Trust repair must work on both sides of the counter. Clear eligibility, accessible language, human help, non-digital routes, proportional verification, appeal, privacy protection, and relief-first pathways reduce burden for users. Adequate staffing, usable tools, discretion with reasons, and protection from impossible

caseloads prevent frontline workers from becoming unofficial interpreters of a broken system.

Evaluate integrity controls against both leakage prevented and legitimate access lost. Report false-positive flags, appeal reversals, abandonment, delay, and staff burden across relevant groups. These measure route performance, not user honesty.

Market suspicion can be studied through bidder entry, supplier exit, price dispersion, warranty disputes, payment delay, product failure, intermediary dependence, and the premium paid for trusted brands or networks. Read these patterns against market scale, quality, regulation, geography, credit conditions, and technological change before attributing them to corruption. The claim becomes stronger when opaque ownership, selective enforcement, insider information, or unexplained access connects the market outcome to hidden power.

Policy disagreement, coalition bargaining, technical uncertainty, lawful confidentiality, and ordinary administrative error are not the same as hidden influence. Records of meetings, donations, conflicts, amendments, timing, enforcement, and reasons can help test the route.

3.2.3 — Trust Repair as Economic Repair

Trust is not simply a forecast that nothing will go wrong. Bohnet and Zeckhauser (2004) compared a laboratory trust decision with a risky choice offering the same material outcomes but determined by chance. Participants required, on average, a higher probability of the favorable outcome when another person controlled it. The comparison makes the source of uncertainty relevant alongside the possible loss. The authors interpreted the pattern as a cost of potential betrayal; it does not make every reluctance to trust a mistake.

The institutional extension is a research question: with service reliability measured separately, do accessible reasons and credible correction make people more willing to rely on a public office? Evaluation should examine whether that reliance is justified by performance and recourse, not merely whether reported confidence rises.

Early repair may produce more visible complaints because people finally believe reporting is safe or useful. A rise in recorded problems may therefore reflect worsening conditions, improved detection, or both. Time horizon, reporting access, retaliation protection, and resolution quality must be read together.

3.3 — The Hidden Tax on Attention and Public Time

An institutional burden may be costly without appearing on a bill or payroll. People spend scarce attention discovering rules, reconciling contradictory instructions, and correcting decisions that should have been intelligible the first time. Those demands can displace work, care, and civic participation even when the formal service carries no fee. Measuring them requires distinguishing the time spent searching for an answer from the mental effort of judging whether an answer is dependable. The following cases examine how navigation and public reconstruction duties shift those costs between institutions and the people they serve.

3.3.1 — Attention as Material Capacity

Attention is a scarce coordinating resource with an opportunity cost distinct from clock time. Two people can spend the same hour on a process while one also carries pain, caregiving, language translation, fear of status loss, or repeated context switching. A fair account therefore observes time and cognitive demand separately and records the conditions that make the burden harder to recover from.

Attention burden overlaps with delay, paperwork, and navigation, so the same hour should not be counted several times. One event may be described along several dimensions, but any estimate should identify the primary unit and note secondary effects. A two-hour office visit may be recorded as time lost, with childcare expense and missed wages listed separately when evidenced. Anxiety and lost concentration may remain qualitative rather than receiving invented prices.

The benchmark for paperwork is not zero documentation. It is the least burdensome process that still protects the legitimate

purpose. Reuse of verified information and risk-based review can reduce burden without abandoning integrity.

3.3.2 — The Navigation Tax

Finding the responsible office, current rule, required evidence, deadline, and correction route is unpaid work that grows when responsibility and case status are unclear.

The navigation tax is now frequently encoded in digital systems. Portals can reduce travel and speed routine cases, yet they can also shift clerical work onto users, hide the reason for failure, and penalize people whose lives do not fit standard fields. Human-centered digital government therefore requires accessible alternatives, clear status information, accountable data use, and a human route for exceptional cases (Hegemann and Ritz, 2026; Halling and Bækgaard, 2024; OECD, 2026a).

A static route map is insufficient when a live case has stalled. Status information should identify who is responsible, where the case stands, the reason codes and their explanations, the deadlines and expected next steps, and how to escalate a case that departs from the standard path.

3.3.3 — Journalistic Reconstruction Burden

Journalists, researchers, whistleblowers, auditors, and civil-society investigators reconstruct decisions from contracts, timelines, interviews, ownership records, leaked documents, and official claims. Independent investigation remains necessary even when institutions preserve good records. The avoidable burden arises when basic decision facts that public power should have recorded must first be rediscovered at private cost or personal risk.

The newsroom bears reporting time, verification and editorial costs, emotional strain, digital security, and the risk of error under deadline pressure.

Reconstruction also reveals a temporal asymmetry. Harm can accumulate for years before a hidden route is exposed, while the public record begins only when the scandal becomes visible. Disclosure should therefore preserve the timeline of prior warnings, missed interventions, and delayed correction so revelation is not mistaken for repair.

Source and bystander protection are part of the reconstruction account. Reporters, whistleblowers, affected people, and bystanders may face retaliation, doxxing, litigation, employment loss, or renewed exposure. Public accountability should not require vulnerable people to surrender private life or repeatedly perform their harm in order for a record to be believed.

Public records can lower reconstruction burden without becoming a single official story. Provenance, version history, disclosed changes, lawful protection of sources and private persons, and a right of correction allow outsiders to test the record.

Independent investigators also need lawful access and sufficient resources, including funding and staff time, to challenge official accounts.

3.4 — Defensive Duplication and Strategic Overbuild

Defensive duplication should be analyzed by function rather than appearance. A second route may provide independent review, continuity, or protection against capture. It may instead exist because records are unreliable, organizations do not trust one another, or no actor accepts responsibility for shared infrastructure. Each layer must identify the distinct failure it addresses and show whether that function can be preserved at lower total burden.

3.4.1 — Institutional Redundancy Under Mistrust

Redundancy can also slow repair when every actor protects itself against blame. More signatures, approvals, exceptions, and reviews may look responsible while making responsibility harder to locate. A useful audit distinguishes independent challenge from sequential avoidance: does the additional layer test a different risk, or does it merely delay the moment when someone must decide?

Fragmented redundancy often leaves agencies with conflicting records of the same case. Cases are reopened, staff reconcile fields by hand, and users repeat evidence because no system is responsible for keeping the authoritative record current. The resulting burden is a coordination failure, even when every individual office follows its own procedure.

A redundancy register can make that failure reviewable. Give each layer a named owner and a review date. Record the risk it addresses, its trigger, data source, protective value, and dependencies alongside the burden on users and staff. State when the layer can be safely retired.

Interoperability can reduce re-entry while preserving independent challenge. A shared verified core may support continuity, while separate oversight records, privacy boundaries,

and human review prevent one database from becoming unquestionable.

3.4.2 — Security Overbuild

Security overbuild begins when protective measures expand beyond a defensible risk model or remain after the environment has changed. Gates, guards, cameras, credentials, background checks, barriers, monitoring, and emergency posture may be necessary. They become overbuild when additional layers add little protection, impose large false-positive burdens, or survive mainly through procurement lock-in and institutional fear.

Lifecycle cost includes more than purchase. Security systems require staffing, training, calibration, data governance, cybersecurity, maintenance, appeal, error correction, and eventual removal. They also consume time, privacy, public space, and dignity.

Assess security false positives across affected groups. An average error rate can conceal the loss of work, movement, or essential access for people unable to resolve a mistaken flag quickly. Proportionality requires both the likelihood and the consequence of error.

3.5 — Manufactured Scarcity and Rent-Shifting

Physical and ecological constraints limit water, land, energy, materials, and safe use of ecosystems. Trained labor, time, infrastructure, and public finance impose further capacity constraints, each with different causes and remedies. Administrative scarcity appears when available capacity cannot be reached or coordinated. Scarcity can also be manufactured or preserved because restricted access creates leverage. Rent-shifting occurs when revenue depends on preserving control of a bottleneck rather than adding capacity. Those paying the charge may have no practical alternative while the restriction remains.

3.5.1 — Scarcity by Design

A licensing rule, zoning restriction or platform requirement can protect a legitimate public interest while also limiting access. The question is whether the restriction is necessary for that purpose or has been preserved to protect an incumbent's advantage.

Assess who benefits, who pays, the rule's public justification, supporting evidence, available review, and whether a less burdensome design preserves the legitimate protection.

Scarcity can also be preserved through chronic noninvestment. When demand is predictable and feasible capacity is repeatedly deferred while incumbents earn scarcity rents, the absence of supply becomes a governance outcome even without an explicit barrier. The analysis asks whether undercapacity reflects real fiscal, labor, ecological, or technical limits, or whether delay itself protects leverage.

The boundary between productive return and scarcity rent is not drawn by the existence of profit. It depends on what capacity was added, what risk was genuinely borne, whether rivals and users can exit, how bargaining power is distributed, whether the good is

essential, and whether revenue depends on repairing or preserving the bottleneck.

3.5.2 — Artificial Queue Power

A queue can allocate genuinely scarce capacity under clear rules. Artificial queue power arises when the queue is created, prolonged, manipulated, or monetized so that someone gains leverage over those waiting (Halling and Bækgaard, 2024; Hegemann and Ritz, 2026).

Artificial queue power is not solved simply by publishing lists. Public release can harm privacy, stigmatize vulnerable people, or expose sensitive status. The proper visibility is route visibility: how the queue works, what criteria govern it, what delays mean, what remedies exist, and how officials answer for the delays the system imposes.

Fear pricing turns insecurity into leverage over the buyer. People may pay urgency fees, accept predatory credit or purchase privileged access because losing an essential service would be worse.

Fear pricing must also be distinguished from the real cost of serving danger. Emergency logistics, insurance reserves, volatile inputs, and rapid deployment can justify higher prices. Exploitation becomes more plausible when cost and risk are concealed, market power blocks alternatives, urgency prevents meaningful refusal, or the seller benefits from preserving the insecurity that sustains demand. Evidence should therefore follow input cost, contractual terms, competition, availability, public subsidy, and the actor's role in reducing or prolonging the threat.

3.5.3 — Real Scarcity Still Exists

The existence of real scarcity does not make avoidable bottlenecks less consequential. When capacity is already constrained, corruption, patronage, exclusionary bottlenecks, or

other preventable frictions can impose larger opportunity costs because there is less slack to absorb them. The analysis should therefore separate the underlying physical constraint from the additional burden created by allocation, governance, or access design.

Local conditions can overturn a generic prescription. A housing rule may protect water in one place and create exclusion in another; a reserve excessive nationally may be essential in an isolated region. Claims should state geography, capacity, time horizon, and revision evidence. Where a bottleneck is administrative rather than physical, a bounded pilot can compare access before and after a rule change without pretending the result generalizes everywhere.

3.6 — Why Visible Power Releases Capacity

Making public power legible can release capacity before any money changes hands. Clear responsibility lets people direct claims to the institution able to act and reduces the need for repeated checking across offices. Visibility must still be paired with protection against retaliation, selective disclosure, and the false reassurance of an incomplete record. The question is not whether every detail should be public but whether affected people can verify the decisions that shape their options. The following tests distinguish useful transparency from trust based on appearance alone.

3.6.1 — Less Guesswork, More Action

One concrete test comes from Olken's (2007) randomized study of 608 Indonesian village road projects. Raising the announced probability of government audit from about 4 percent to essentially 100 percent reduced missing expenditure by about eight percentage points, measured by comparing reported spending with independent engineering estimates. The participation interventions had little average effect on that overall measure, despite increasing involvement.

Electronic procurement shows how a reform can improve delivery even without lowering the tender price. Studying public works in India and Indonesia through variation in adoption, Lewis-Faupel et al. (2016) found improved quality in India and fewer completion delays in Indonesia, alongside a greater likelihood that contracts went to contractors from outside the region. They found no evidence of lower procurement prices. These results show why a procurement reform should be judged by delivery as well as price.

Public expenditure tracking can reveal whether money reaches schools, clinics, local governments, or infrastructure sites,

especially when administrative records are compared with direct observation. Its value depends on a chain from discrepancy to inquiry, correction, recovery, and changed incentives. Visibility without an accountable response can document failure without reducing it (Reinikka and Svensson, 2004).

Measure the interval from the earliest point at which a defect could reasonably have been detected to the point at which the burden was actually remedied. Report detection delay separately from the time spent correcting it. If the starting point is reconstructed, state the evidence and uncertainty. A promised remedy is not a completed one. Temporary mitigation, restored service, compensation, contract correction, and prevention of recurrence are different achievements.

3.6.2 — Better Trust Without Naivety

Disclosure becomes more powerful when people can act on it. Ferraz and Finan (2008) studied Brazil's municipal audit program, comparing public disclosures before and after elections. Evidence of corruption affected incumbent electoral performance, with stronger effects where local radio could spread the findings. The result identifies a consequential link between information, its local distribution, and political accountability. The institutional question is whether people receive intelligible findings while they still have an effective opportunity to respond.

The World Development Report 2017 places commitment, coordination and cooperation within unequal distributions of power. Its framework directs attention to whether institutions can keep promises, coordinate fragmented actors and prevent powerful participants from rewriting a route after others have relied on it (World Bank, 2017).

Blind trust allows hidden power to escape scrutiny; total cynicism persuades people that exposure changes nothing and

makes repair seem impossible. Trust without naivety relies on records, plural review, correction, and an honest account of past consequences.

Part II — Hidden Cost and Debt Repair

Part II compares the visible price of repair with costs carried through neglect, delay, unpaid work, and narrowed future choice. Its accounts keep money, time, safety, privacy, and care distinct (North, 1990; World Bank, 2017). Visibility itself changes the apparent comparison. The cost of a proposed repair can be itemized at a meeting, while the continuing cost of doing nothing is scattered across people who never attend it. Putting those accounts beside one another does not predetermine which proposal should win; it establishes what the meeting must actually decide.

The Part follows each burden beyond the first transaction. A contract may look inexpensive because maintenance failure will be paid years later. A public record system may look costly because trained staff, cybersecurity, translation, privacy review, correction, and appeal are visible now. A debt settlement may appear generous while replacing one chain with another. A military reduction may be announced as immediate savings while worker transition, regional revenue loss, pensions, supplier obligations, verification, and residual security remain uncounted. The governing questions are where each burden lands across the complete lifecycle and whether a proposed saving releases usable capacity or merely shifts a bill.

Corruption and transparency open the Part because public argument often counts them asymmetrically. Chapter 4 places the hidden costs of failure beside the real costs of accountability, then asks when proportionate legibility creates more capacity than it consumes.

Strategic fear then carries the inquiry across states, markets, organizations, and households. Uncertainty can become a costly standing posture, even where readiness answers real danger and a reduction in that posture releases no immediately usable cash. Chapter 5 therefore follows a proposed peace dividend from its

baseline through conversion to its destination. It asks which gross costs are genuinely avoidable, when net savings become usable, who bears and pays for the transition, and what residual security, maintenance, and oversight still need funding. Each estimate must state its uncertainty.

Public finance receives the conversion question. Budgets, taxes, procurement, maintenance, gifts, reserves, deficits, guarantees, and off-budget obligations carry different kinds of authority and risk. The material account asks who supplies the money, who performs the labor, who receives the service, who waits when funding fails, and who gains leverage through dependence. Donor and vendor firewalls must prevent a gift from buying influence beyond its stated public purpose (Sen, 1999; Nussbaum, 2011).

Debt completes the Part because debt is a claim on future room. Household obligations can organize every decision around survival, while municipal and sovereign debt can narrow policy choices, delay maintenance, weaken bargaining power, and expose public life to soft capture. Repair distinguishes liquidity from insolvency, legitimate obligation from predatory terms, productive investment from unsustainable borrowing, and honest restructuring from disguised control. Its positive aim is to reopen options while naming the burdens carried by creditors, taxpayers, workers, residents, public services, and future borrowers.

Chapter 4 — Corruption and Transparency Cost

Corruption and transparency are often pictured as darkness and light. The moral image is understandable, but a field-level comparison needs more precision. Both hidden failure and trustworthy public visibility consume resources over time. Public institutions choose well only when the two are costed with equal seriousness (World Bank, 2020b; OECD, 2020; NIST, 2020).

Chapter 3 distinguishes corruption from incapacity, error, and other causes of poor performance. The accounting task here is to trace a supported mechanism from the decision and its beneficiary to the resulting burden, without treating every failure as evidence of corruption.

Capturing a record, granting access to it, and releasing it publicly are separate decisions: preservation alone authorizes neither inspection nor disclosure. Section 4.2 costs the staffing, safeguards, interpretation, review, correction, appeal, and retirement needed to make records useful.

This precision strengthens the constructive case. Well-designed visibility can detect failure earlier, preserve sequence, reduce procurement leakage, clarify responsibility, improve maintenance, protect whistleblowers, and make correction possible before crisis. It can also reduce the public's need to infer motives from rumor because institutions carry more of the burden of explanation. A record earns value when it is connected to a decision, a challenge route, and a repair mechanism, not merely because it has been published.

Section 4.4 directs visibility toward consequential public roles and decisions while protecting private life. A system that exposes ordinary need while leaving decisive power obscure has reversed that duty.

Section 4.5.1 distinguishes trust surveys from operational evidence: neither a change in public confidence nor an audit discrepancy alone measures the other. Consider an illustrative

purchasing dispute. The published total has not changed, yet the delivered specification differs from the one bidders priced. A reader of the total alone cannot tell whether the change reflects a legitimate adjustment, an error, or a concealed advantage. The useful record is the connection between versions: what changed, when approval occurred, and which obligations changed with it. Even that record does not settle the dispute by itself. It makes a question answerable that the aggregate concealed. This is the chapter’s practical scale: enough visibility to distinguish competing explanations and identify a consequential correction, without assuming that publishing more material necessarily produces more understanding. The account must include the work needed to make the record usable.

Scrutiny follows role and authority, not unequal human worth. A corrected public record should identify the decisions that relied on its superseded version, so an amended file can lead to correction of the consequences.

4.1 — What Corruption Actually Costs

A corruption cost map should connect diversion to procurement quality and service delay. It should also trace deferred maintenance and the loss of public capacity that may follow. A fiscal transfer is not itself proof of every downstream effect, so keep those effects in separate fields with a defined comparator. That distinction supports budgeting, investigation, and repair without turning a causal chain into an untestable total (World Bank, 2020b; Olken and Pande, 2012).

4.1.1 — Stolen Funds

For stolen funds, reconcile authorized expenditure with delivered work and defensible asset value. Record grants diverted, false invoices, or corruptly underpriced public transfers as fiscal losses; use the payment/loss/harm distinction established in Section 3.1 when a private bribe is also involved (World Bank, 2020b; Olken and Pande, 2012).

Direct theft can weaken voluntary compliance and support for collective provision even when individual tax liabilities remain unchanged. Expected diversion can undermine confidence in shared revenue, making later repair harder to finance. Fiscal harm can therefore extend beyond the amount taken (Alm, 2019; Slemrod, 2019; Knack and Keefer, 1997).

Recovery must also close the route that enabled diversion; returning the funds does not prevent the next loss.

4.1.2 — Broken Maintenance

Broken maintenance shifts burden onto workers: crews inherit unsafe backlogs, teachers improvise around failing buildings, nurses work in understaffed facilities, and transit workers manage equipment that should have been replaced. Public employees may then be blamed for maintenance debt they did not create.

A maintenance-exposure register should link each deferral to its authorizing decision and stated reason. Section 4.5.2 specifies the condition, criticality, uncertainty, and resource fields needed to assess the resulting risk. Postponement may be justified; an unrecorded decision prevents later reviewers from distinguishing that judgment from neglect.

4.1.3 — Opportunity Cost of Corruption

Consider an illustrative case in which a municipality proves that an inflated contract cost $100,000 more than equivalent delivery. Its approved alternative was a $100,000 clinic repair. The financial loss and the clinic repair forgone describe the use of the same resources; adding them as separate monetary losses would double count. Additional patient travel, service interruption, or later repair costs require separate causal evidence and an overlap check. If the clinic plan was only one of several possibilities, the account should show those alternatives rather than claim that one unrealized future was certain.

4.1.4 — Intergenerational Cost of Corruption

Intergenerational cost is the burden hidden power sends to people who were not present when the decision was made. A child inherits a school system weakened by old contracts. A future taxpayer pays for infrastructure that should have been maintained. A young worker enters an economy where corruption has hollowed out trust and opportunity (World Bank, 2017; North, 1990).

Future burden is not adequately represented by discounting a liability into a present price. Some effects are irreversible, path dependent, or distributed through lost capability rather than future expenditure. A child educated in an unsafe building, a region deprived of infrastructure, or an ecosystem damaged by captured permitting may carry consequences that no later budget fully

repairs. Monetary estimates may assist without exhausting the claim.

4.2 — What Transparency Actually Costs

Visibility requires funded institutional work; otherwise its cost falls on exhausted staff, affected people, advocates, vendors, or future budgets (OECD, 2020; NIST, 2020; W3C, 2023). A transparent system has distinct lifecycle stages, including capture, access, release, interpretation, review, correction, retention, deletion, migration, and exit. The disclosure rule can change at each stage because audit needs, confidentiality, security, privilege, and accessibility impose different duties. Records-management principles cover integrity across formats and technologies (ISO, 2016), while accessible publication determines whether people can use what is released (W3C, 2023). Indexing, version history, interoperable formats, correction routes, and tested portability therefore belong in the design and budget.

4.2.1 — Staffing and Training

Training matters because visibility without judgment can harm. Staff must understand privacy, redaction, correction, and accessibility across disabilities and other access needs. Training also needs to cover cultural context, trauma-sensitive communication, data limits, the greater visibility duty attached to public power, and the difference between system burden and personal exposure. A poorly trained system may disclose too much, hide too much, misclassify records, or treat ordinary people as objects of suspicion.

A credible visibility budget names the people who make public records trustworthy. Records officers, auditors, privacy reviewers, accessibility specialists, translators, technologists, security staff, legal reviewers, frontline personnel, and independent complaint reviewers all perform different work. The

budget also states caseloads, supervision, escalation, continuing education, professional independence, and recovery time.

Staffing estimates should distinguish permanent operation from launch work and surge demand. Training is not a one-time event because rules, technologies, threats, and public needs change. Caseload ratios, review complexity, protected-category handling, error correction, leave coverage, and staff safety all affect capacity.

4.2.2 — Cybersecurity and Tamper Resistance

Cybersecurity protects record integrity, the people named in a record, and the institution's ability to correct public power (NIST, 2020; NIST, 2024).

Public records must remain examinable, recoverable, and usable after a supplier leaves. Apply the exit and portability requirements in Section 6.4 before procurement, so a vendor cannot acquire a private veto over public answerability (NIST, 2024).

Security requires recurring expenditure on governance, threat modeling, access controls, provenance, key management, incident response, backups, vendor review, and independent testing. Tamper resistance must distinguish malicious alteration from lawful amendment. An append-only history can preserve an erroneous original while annotations, appeal decisions, access limits, and downstream notices identify the correction and repair its consequences.

Digital burden should be compared by scale and public purpose. A durable archive, a high-volume sensor network, a public analytics platform, and an artificial intelligence service impose different demands for energy, cooling, hardware, networks, replacement, minerals, and specialized labor. The relevant inquiry is whether the chosen computation is necessary for the

accountability function and whether a lower-burden design can provide materially equivalent protection. Environmental cost belongs in system design before prestige or convenience makes expansion feel inevitable.

4.2.3 — Translation, Format, and Accessibility Costs

Translation and accessibility turn formal visibility into usable visibility for readers with different language, literacy, technical, financial, and physical circumstances (W3C, 2023). Budget the production, testing, and maintenance of these routes.

Usability should be tested in practice, not inferred from the existence of a translated file or accessibility statement. People need to find the record, understand the decision, navigate the interface, request help, and challenge error across language, disability, literacy, device, and connectivity conditions.

Imagine that an office changes the closing date for a public application. Its original announcement exists in several languages, and a telephone service reads from a prepared script. Updating the principal webpage completes only one part of the work. Each route needs the same operative date and a clear indication of which version is current.

The useful unit of maintenance is the decision being communicated. A correction record can identify every version derived from that decision and assign its update to someone with the relevant language or access expertise. The office can then test whether a caller and a reader using an accessible document receive the same answer. The comparison concerns what each person can do with the information, not whether the pages look alike.

A publication budget becomes more realistic when it funds this continuing relationship among versions. Initial translation prepares a route; coordinated updates keep it dependable. Good

preparation can also reduce later work. A stable source document and a clear change note allow a translator to review the affected passage instead of reconstructing the whole announcement.

4.2.4 — Redaction and Privacy Review

Redaction is the art of making public power visible without making ordinary people vulnerable. A public record may contain names, addresses, medical details, family information, financial hardship, migration status, witness statements, worker complaints, sexual or intimate information, child information, or safety-sensitive details. These cannot be treated as debris in the path of transparency (NIST, 2020; Nissenbaum, 2010).

Officials may use privacy language to conceal misconduct, procurement capture, delay, discrimination, or abuse. A defensible review process must protect ordinary people without allowing privacy to become a shield for power.

Appeal must also be independent enough to matter. If the same office that created the error controls all correction without external review, trust remains fragile. Ombuds, independent auditors, administrative judges, public advocates, or review panels may be needed depending on the domain.

Costing correction and appeal requires intake, notice, temporary protection against ongoing harm, evidence review, independent escalation, decision time, version correction, downstream notification, and remedy. A cheap publication system with no practical challenge route externalizes its error cost onto the people misdescribed by it.

4.3 — Visible Burden Can Be Cheaper Than Hidden Failure

Distribution can reverse an apparently favorable lifecycle comparison (Fung, Graham, and Weil, 2007; World Bank, 2020b). A digital portal may lower agency processing cost while increasing navigation work for people with limited access. A public database may reduce search cost for journalists while creating privacy and reidentification risk for affected families. An audit regime may deter leakage while overwhelming frontline staff. A distributional comparison shows gains and burdens by group, not only as a net institutional total.

4.3.1 — Up-Front Cost Versus Downstream Harm

Compare launch, recurring operation, renewal, migration, incident, and decommissioning costs with the downstream harms the design may prevent. The record should also state which avoided harms remain speculative and which have direct evidence from comparable failures or near misses.

Early detection has its own failure modes. Weak thresholds can create false alarms, alarm fatigue, stigma, or unnecessary intervention. High thresholds can miss the very pattern the system was built to reveal. Detection design therefore needs escalation rules, human review, protected reporting, feedback from frontline workers, and a record of whether alerts led to correction rather than merely producing more alerts.

4.3.2 — Reduced Litigation and Dispute Cost

Many disputes concern what happened. A worker recalls a promise, a contractor disputes authorization, a citizen challenges a denial, or a community says a warning was ignored. Clear records may narrow factual disagreement and thereby reduce some dispute costs. This is a plausible mechanism, not a

quantified effect established here (Fung, Graham, and Weil, 2007).

A useful dispute record preserves the decision, its authority, evidence available at the time, notice to affected people, appeal route, and subsequent corrections so neither party can treat an obsolete version as conclusive. It may ease factual verification, leaving independent review to address remaining disagreement. Measure claimed savings through time, expense, and resolution quality.

Check whether fewer cases reflect short deadlines, obstructed challenges, or conclusive treatment of the institution's own record. Preserve counsel, appeal, affected-person access, and independent review.

Procurement disputes require the lifecycle record in Section 6.4, including beneficial ownership and evidence of exclusion.

Procurement gains should be separated into price change, quality change, avoided change orders, lower dispute cost, better maintenance terms, increased competition, and reduced dependence. An overlap rule prevents the same gain from appearing in several fields.

4.3.3 — Visibility Can Also Overbuild

The choice is not secrecy or total exposure, but bounded legibility under a stated purpose and threat model. Excessive visibility wastes staff time, exposes private life, rewards compliance theater, chills deliberation, and creates unusable archives.

Overbuild appears when documentation is duplicated without need, bystanders are exposed unnecessarily, security overhead or new gatekeepers add little protection, or activity is driven into less visible channels. Correctives include necessity review, role-proportionate burdens, sampling where continuous capture adds little, and authority to narrow or retire controls.

4.4 — Legibility Without Public Surveillance

System-level burden data describe budgets, queues, contract performance, service delays, error rates, and correction times. The object is institutional performance; service users are not being ranked for worth or deservingness (OECD, 2020; World Bank, 2017). Aggregate numbers can hide unequal burden. A service may look average while disabled users, language minorities, rural communities, migrants, debtors, or low-income households are being failed. The analysis must distinguish power from proximity. A person mentioned in a record is not necessarily the one carrying public power.

4.4.1 — Aggregate Before Identifiable

Aggregate before identifiable is a practical rule for privacy-preserving measurement. It asks whether a burden can be understood at the level of system, neighborhood, agency, queue, contract, program, or category before personal details are used. If aggregate data can answer the question, identifiable data should usually stay protected (NIST, 2020; Solove, 2006).

Apply that rule to collection, category design, linkage, access classes, retention, and deletion before data exist. Justify each personal identifier at collection rather than relying on later redaction (Nissenbaum, 2010; NIST, 2020).

If the issue is queue delay, the public may need distributions by service type, region, urgency, and appeal outcome. It may not need names. If the issue is procurement capture, the public may need beneficial ownership and contract routes, not private details about ordinary service users.

Combining records from unrelated settings can turn past vulnerability into a general judgment about a person. A person who sought debt relief might later be judged as a civic risk. A service applicant might become a research subject. A public vote

might be interpreted as moral seriousness. Voluntary community work might become an employment rank. A separate lawful purpose, a narrowly bounded connection, and meaningful consent or other valid authority may justify a specific use of records. Unrelated participation or vulnerability cannot establish guilt.

A credible separation regime combines purpose limits with technical and organizational barriers. Distinct stores or schemas may be required, along with role-based access, access logging, limits on model training, retention and deletion rules, independent audit, and consequences for informal merger (Nissenbaum, 2010; Solove, 2006; NIST, 2020).

4.4.2 — Public Power Carries More Visibility

Public power carries the greater visibility burden. Officials, agencies, contractors, donors, regulated monopolies, public platforms, and institutional decision-makers owe more explanation than private persons because visibility should follow consequence (OECD, 2017b; Nissenbaum, 2010).

Accountability does not expose the whole person in a public role. Life safety, family and medical privacy, lawfully bounded secrecy, and nonparticipant protection remain legitimate. The stronger duty attaches to decisions, funds, contracts, authorizations, conflicts, outcomes, and correction systems affecting people who cannot readily refuse.

This asymmetry prevents scapegoating. Frontline workers are visible, while decisive power may sit in policy, budgets, procurement, staffing, software, or leadership. Records should trace responsibility upward rather than leave the most reachable employee to absorb public anger.

Redress begins with notice: people should know when measurement affects them, which data and assumptions are used,

how to report error, who reviews disputes, and what happens during review.

Aggregate metrics also need challenge because categories can misdescribe communities without exposing individuals.

4.5 — Measuring Trust, Time, and Reduced Drag

Time, maintenance exposure, and institutional trust require different measures. Time accounting captures delay and repeated effort. Maintenance evidence distinguishes working capacity from exposure to failure, while trust surveys must specify whom respondents are asked to trust. Each measure supports a different decision, but competing explanations must remain open (OECD, 2021c; Moynihan, Herd, and Harvey, 2015).

4.5.1 — Trust Indicators and Their Limits

Trust is not uniform across institutions: a person may trust a local clinic while distrusting national government. Measurement must specify the institution and relationship being assessed (Knack and Keefer, 1997; OECD, 2021c).

Earned trust can reduce unnecessary defensive effort, but a fall in appeals is not sufficient evidence. Appeals may decline because decisions improved or because challenge became harder; they may rise when a newly accessible remedy reveals old errors. Read appeal volume beside adverse decisions, successful corrections, abandonment, and access to representation. Trust becomes informative when its behavioral indicators can be distinguished from silence or exclusion.

Changes in sample composition, subgroup experience, media exposure or question wording can move a trust indicator without an equivalent change in institutional performance. Compare administrative and lived evidence, and test whether repair changed experience, before interpreting a headline movement (Guiso, Sapienza, and Zingales, 2009; OECD, 2017a).

Trust evidence is strongest when it separates competence, reliability, fairness, openness, and care. Repeated measures, event-linked analysis, subgroup distributions, and open-ended

explanations can show why trust changes without pretending it is a stable substance owned by an institution.

4.5.2 — Maintenance Debt Measures

Maintenance debt is the accumulated repair that has been postponed, across physical assets, digital systems, public facilities, and environmental protections (ISO, 2017; World Bank, 2020b). A maintenance-debt register records condition, criticality, deferred-work age, safety and accessibility exposure, escalation risk, service dependency, responsible office, and review date (Flyvbjerg, 2014; World Bank, 2020b).

Priority should reflect consequence and reversibility as well as age. A newer defect capable of cascading into safety or service failure may deserve action before an older cosmetic backlog. Each entry also needs inspection confidence, climate exposure, workforce requirements, and a cost range. These details help distinguish prudent sequencing from hidden abandonment. Where condition evidence is weak, uncertainty should shorten the next inspection interval or increase contingency planning rather than disappear inside a precise estimate.

4.5.3 — Corruption Drag Measures

A corruption-drag portfolio should distinguish leading signals from later consequences: evidence of unusual procurement can justify inquiry before backlogs, maintenance deterioration or declining participation appear (Olken, 2009; Olken and Pande, 2012).

Price anomalies, concentrated bids, emergency procurement, change-order growth, unresolved complaints, audit exceptions, and persistent delay open review routes. Before attributing an anomaly to diversion, test relevant alternatives such as terrain, emergencies, supply limitations, labor conditions, market conditions, and quality constraints. The review protocol should

state in advance which signals, combinations, or thresholds justify deeper examination. Predetermined escalation rules make scrutiny more consistent and reduce the risk that unusual performance is investigated selectively because of politics, reputation, or convenience.

Connect a financial anomaly to a specific service consequence: state the suspected mechanism, affected population, plausible alternatives, expected lag, attribution confidence, and office responsible for repair. Compare both process and subsequent service outcomes with credible alternatives. If the financial anomaly falls but service performance does not recover, test whether the attribution was incomplete or another constraint still blocks access.

4.6 — What Material Records Can and Cannot Show

Dashboards, ledgers, and indices compress reality for different public jobs. A dashboard brings current signals into view. A ledger preserves sequence, obligation, status, and correction across time. An index combines selected indicators to locate patterns that merit closer review. Governance fails when one display is asked to perform another tool's work and becomes a target, ritual archive, or verdict it was never designed to issue (Fung, Graham, and Weil, 2007; NIST, 2020).

4.6.1 — Dashboards as Bounded Attention Tools

Every dashboard needs a charter. The charter names the purpose, intended users, decision supported, components, update rhythm, source owners, protected exclusions, interpretation responsibility, correction route, review date, and uses that are forbidden. A display without a charter can migrate from inquiry to management target without public notice.

A dashboard earns its place when a signal produces a responsible next action: investigate, contact affected people, add capacity, repair an asset, review a contract, test a threshold, or explain a change. Log the response and the reasons for declining action, so the public can distinguish an unattended warning from an examined one.

A ledger preserves sequence across decisions that would otherwise be scattered among budgets, contracts, minutes, incident reports, appeals, and public explanations. It records what was promised, decided, funded, delayed, changed, challenged, corrected, repaired, or left unresolved, together with the source and authority for each entry. This institutional memory prevents every budget cycle or leadership change from resetting the story of burden and allows affected people to test whether a stated repair reached the obligation it was meant to answer.

Ledgers are especially valuable for obligations that mature over time, including maintenance, debt, donor restrictions, contract performance, environmental restoration, incident response, appeals, and promised community repair. A durable record can distinguish a future liability from a present saving, reveal an exception becoming routine, and identify a promise carried forward without funding or authority.

An index compresses several indicators to identify possible risk or priority, such as procurement concentration, integrity weakness, maintenance vulnerability, service exclusion, climate exposure, or unusual delay. Before consequential action, readers must be able to inspect its components, sources, assumptions, missing data, distributions, uncertainty, and rival explanations without privileged access.

4.6.2 — Metric Correction and Retirement

Every metric should receive an exit condition at creation. Review dates, complaint routes, misuse triggers, privacy thresholds, decision-relevance tests, and named retirement authority prevent a provisional tool from becoming permanent merely because contracts, spreadsheets, automated systems, or public habits depend on it. Retirement includes downstream notice and removal from consequential use, not merely a warning label in the originating office (NIST, 2020; OECD, 2020).

Review should combine affected-person and frontline experience with independent technical, privacy, security, and distributional analysis. The institution that built the measure cannot be the only judge of whether it still deserves authority. A renewal decision should state the measure’s present use, known blind spots, complaint history, error burden, and the less intrusive alternatives considered.

A metric should be suspended, narrowed, or retired when it is shown to incentivize denial or misclassification, exposes personal information beyond its declared purpose, lacks a credible data denominator, or no longer guides an authorized decision. The responsible office records the defect, preserves a challenge route, publishes a corrected interpretation where safe, and tests the replacement before reinstating high-stakes use. An independent reviewer must be able to recommend retirement even when the score appears to reflect well on the sponsoring institution.

International indicators can support peer learning, aid allocation, risk monitoring, and research when their purpose, components, and limits are explicit. Consequential use also requires affected-country participation, access to source data, local interpretation, a route to challenge, and protection against punitive misuse or conditionality that deepens the burden being measured. The indicator should direct accountable support rather than become a remote authority over communities unable to correct its assumptions.

Chapter 5 — Strategic Fear and Peace Dividend

Fear is one of civilization's oldest expenditures, and modern institutions have given it budgets, supply chains, insurance premiums, legal departments, screening systems, hardened infrastructure, redundant networks, and regional economies. Under consequence, fear becomes a durable arrangement of labor, land, technology, secrecy, and public money. Chapter 5 asks which parts of that arrangement protect against real danger, which persist because uncertainty is expensive, and which capacities could be converted without moving risk onto people with less power (Jervis, 1978; SIPRI, 2026b; Knight, Loayza, and Villanueva, 1996).

Strategic fear can harden uncertainty into standing systems whose original rationale becomes vague, politicized, commercially useful, or inaccessible to review. A serious account begins by respecting the protective case: some threats are real, some secrecy is justified, some deterrence prevents harm, and some reserve capacity saves lives precisely because it exists before crisis. The task is to preserve effective protection while recovering the margin where readiness has become self-renewing cost.

The difficult work is attribution. A conflict that did not occur may have been deterred, prevented by diplomacy, made unlikely by geography, reduced by interdependence, or avoided through luck. A system that remains safe after a reduction may have been overbuilt, or it may be consuming reserves that will fail later.

Section 5.7 keeps fiscal savings, released resources, returned time, and lower risk in distinct accounts before any aggregation is justified.

Earlier peace-dividend debates offer a durable lesson: reductions can free resources while also creating industrial shocks, regional revenue loss, contract unwind, environmental liabilities, pension burdens, remobilization risk, and timing

problems. The useful question is what became safely unnecessary, what still has to remain, what needs conversion, and who carries the adjustment before any remainder is called usable (Markusen and Yudken, 1992).

Section 5.5 examines how reciprocal verification and bounded disclosure can reduce uncertainty while preserving justified protection. A proposed reduction can therefore be staged around what becomes knowable. The first step may test an agreement without committing the entire later release. That sequence matters to the economic account: capacity available after a successful test is not capacity available today, and delaying an irreversible commitment can preserve an option while the relevant uncertainty is resolved.

The account of a claimed dividend is complete only when its source, net amount or kind of noncash benefit, lawful destination, custody record, and revision rule are explicit.

A proposed civilian reuse should identify demand as well as spare equipment, since a machine without buyers, replacement parts, or an operator remains an idle asset. Training and capital transfer must be timed so workers and public services receive usable capacity before the old spending stream disappears.

5.1 — Opacity Feeds Strategic Fear

Opacity becomes material when missing or disputed information changes another actor's safety, options, or costs. Its consequences depend both on the threat and on the risks created by disclosure (Jervis, 1978; Fearon, 1995). A credible test asks whether history, geography, alliance obligations, political incentives, or independent threat evidence would have generated the same defensive action. Only the remaining difference can reasonably be attributed to the unresolved signal.

5.1.1 — Unknown Intentions

Separate declared purpose, demonstrated conduct, institutional route, legal authority, capability, historical pattern, and unresolved inference. A peaceful statement can coexist with threatening behavior, while ambiguous conduct can support more than one explanation. Identify which public decisions depend on evidence and which depend on inference.

Strategic fear is especially acute where actors doubt one another's ability or willingness to keep commitments. A promise can fail because circumstances change, because authority is fragmented, because domestic opposition overturns it, or because the promise was never sincere. Conflict research treats these commitment and information problems as central rather than incidental (Fearon, 1995; Kydd, 2005).

A rumor-to-budget route should be traceable. The record should identify the source of the claim, the evidence independently checked, the office that translated it into action, the spending or restriction that followed, the interests that benefited, and whether later correction changed the posture. This chain protects genuine early warning while making fear inflation harder to monetize.

5.1.2 — Domestic Opacity and Foreign Fear

Domestic opacity can raise external cost without making foreign access universally legitimate. A sudden change in command, emergency authority, procurement ownership, financial exposure, or military route may cause neighbors and markets to plan for instability.

A state can reduce external fear by making budgets, procurement routes, emergency authorities, debt obligations, and security exceptions lawfully legible to its own people and, where appropriate, to treaty partners. Domestic citizens should not be exposed for foreign reassurance, and weaker states should not be forced to surrender sovereignty as the price of being considered trustworthy.

5.1.3 — Market Fear and Capital Flight

Markets can translate fear into motion faster than public repair, creating currency pressure, debt stress, import difficulty, unemployment, budget strain, and household price shocks. Hidden debt, banking weakness, war, sanctions risk, corruption, inflation, rumor, or a global risk shock may each contribute. The mix has to be identified because the burden often falls on people who did not move the money (IMF, 2026b; Guiso, Sapienza, and Zingales, 2009).

Protective financial measures require instrument-specific review. Liquidity support, currency defense, bank support, capital-flow measures, and sanctions management may reduce systemic harm while concentrating discretion or shifting loss to households. Review must state function, incidence, duration, authority, exit condition, and alternatives (IMF, 2019; IMF, 2026b).

Capital flight must be separated from ordinary diversification. Hedging differs from insider exit after private warning, and

protecting suppliers differs from extracting public rescue while abandoning workers. Legitimate risk management, predatory exit, insider exit, and panic exit leave different burdens.

5.2 — Overreadiness as a Budget Line

Overreadiness is a durable mismatch between preparation and the marginal risk it addresses. Necessary readiness remains maintained, deployable, proportionate, reviewable, and connected to a live function (SIPRI, 2026b; Jervis, 1978). The next increment of capability should be tested against the additional harm it plausibly prevents, not against a desire for perfect assurance. Staffing, maintenance, interoperability, disposal, escalation, and opportunity cost all belong in that calculation (SIPRI, 2026b; Jervis, 1978).

5.2.1 — Emergency Procurement Overreadiness

Emergency procurement exists because crisis can arrive faster than ordinary purchasing systems (OECD, 2015b).

Emergency procurement can also conceal overreadiness, waste, and capture. Urgency can weaken review and give specialized vendors greater influence. Stockpiles may expire and temporary contracts may persist. Fear can raise prices, while crisis language can discourage questioning and leave public explanations thinner.

A procurement record should begin at authorization and continue through after-action review. It should identify the normal rule bypassed, life-safety reason, supplier ownership, price basis, delivery, quality, stock condition, maintenance, expiry, disposal, worker safety, use, and return-to-ordinary date. Review clocks, supplier diversity, stock rotation, and post-crisis audit allow proven capacity to remain deliberately while narrowing arrangements that survived only through emergency fog.

Name the authority permitted to bypass ordinary procurement and the evidence required.

5.2.2 — Private Security Overreadiness

Private security may protect its purchaser while weakening the constituency for public repair. Evaluation should ask which public failure produced the substitute and who remains exposed.

A gated road may displace risk onto others; private schooling can weaken support for public schools; and private generators can diminish pressure to repair the grid. Firms with private legal teams may navigate rules that smaller competitors cannot. Private defense can therefore amplify a shared-system weakness even when it offers legitimate protection.

5.2.3 — Legal and Compliance Overreadiness

Compliance overreadiness is paperwork or verification that exceeds its justified protective function (North, 1990; Williamson, 1985).

Compliance should be separated into prevention, evidence, coordination, and institutional self-protection. Rules preventing fraud, discrimination, injury, privacy harm, or public waste may justify their cost; duplicate forms created by interagency mistrust may not.

Compliance becomes excessive when an additional requirement imposes burdens disproportionate to its protective value, compared with feasible alternatives. The relevant comparison is not whether checking consumes more effort than direct prevention: some checks are indispensable to prevention itself. Each added form, signature, or approval must identify the distinct risk it controls and the evidence that would justify simplifying it.

5.3 — When Peace Releases Capacity

Give each claim an explicit status, such as projected, authorized, verified, disputed or unavailable for reallocation, so a promised benefit cannot be spent before it exists. Claim savings from reduced waste only when a real burden falls without an equal or greater burden appearing elsewhere, over a horizon long enough to reveal displaced costs (Knight, Loayza, and Villanueva, 1996; SIPRI, 2026b). Time changes the result. A cancellation may lower this year's spending while creating litigation next year. A smaller stockpile may look efficient during calm and fail during crisis. A workforce reduction may save salary and produce longer queues, broken maintenance, or costly rehiring.

5.3.1 — Transition Costs First

Before declaring savings, account for the cost of supporting affected people and meeting existing obligations during conversion (Markusen and Yudken, 1992; Knight, Loayza, and Villanueva, 1996).

Fund the income, skills, mobility, pension, and placement support required by affected workers, using the transition framework in Section 8.3.

Regional costs matter because a budget line may support a local economy. A defense facility, emergency procurement hub, border installation, compliance industry, or large vendor contract may sustain schools, housing, small businesses, and tax revenue. Abrupt savings can hollow a place. Transition must count the region, not only the agency.

Supplier contracts, pensions, health coverage, wage insurance, environmental remediation, equipment storage or disposal, land reuse, cybersecurity migration, regional revenue bridges, school and service stabilization, skill translation, infrastructure conversion, and administrative coordination may all be required.

Transition plans must also fund mental health support and small-business continuity. People do not only lose income when an old industry contracts. They can lose identity, community rhythm, pride, intergenerational skill, and trust in public promises.

Residual security preserves necessary deterrence, intelligence boundaries, cyber defense, emergency reserves, inspection capacity, disaster response, and lawful force as a last resort after excess is removed.

5.3.2 — Shared Costs and Reusable Capacity

Consider a planning example in which two depots share one maintenance contract. Closing the smaller depot removes its electricity bill and some transport journeys, but the contract price remains unchanged until the renewal date. An internal account may have assigned half the contract to each depot. That allocation helped describe their costs; it did not mean that half the contract could disappear when one depot closed.

The decision therefore has two different horizons. Before renewal, the saving consists of costs that actually cease, less the work of moving stores and maintaining service. At renewal, a different contract may become possible. The purchasing team needs to show what smaller service can be bought and at what price. A later saving becomes a supported estimate when the required service and available terms are known.

Capacity can still improve before the contract price falls. The remaining depot may serve the same area with a better stock arrangement, or a vacated building may become available for another purpose. Those changes deserve their own account. A usable building is a resource even while its conversion requires money; fewer journeys can release working time even when no employee leaves the payroll.

This way of reading a closure protects the promising part of the proposal. It identifies the precise decision that releases each resource. Some benefits follow from changing the operation, some from revising a contract, and some from preparing an asset for reuse. A credible conversion plan brings those decisions together so that a reduction in one activity can become capacity for another.

5.4 — Trust and Coordination Dividends

Reliable coordination can release capacity across households, firms and institutions without producing a corresponding public-budget saving (Knack and Keefer, 1997; Zak and Knack, 2001). Attribution is difficult because trust and performance influence one another. Reliable institutions may build trust, while preexisting trust may make coordination easier and failure more tolerable. Compare performance, user and staff experience, and credible alternatives; where causes remain entangled, report a contribution to lower friction rather than sole causation.

5.4.1 — Verification and Transaction Costs

As Section 1.3.3 explains, verification and enforcement costs can deter a small firm from pursuing an otherwise useful contract (Coase, 1937; Williamson, 1985). A verification demand can become avoidable coordination cost when an institution may lawfully reuse a relevant, reliable fact it already holds but instead requires the person to reproduce it (Knack and Keefer, 1997; Zak and Knack, 2001). Report the documents, staff and applicant time, correction loops, and access failures removed, alongside any new risk.

Speed should be attributed carefully. A process may move faster because trust improved, staffing increased, eligibility narrowed, rights were bypassed, or demand temporarily fell. The chapter counts a time-saving coordination dividend only when time falls without shifting burden into exclusion, error, privacy loss, or later rework.

5.4.2 — Better Crisis Cooperation

Crisis cooperation can produce avoided loss rather than visible saving (United Nations and World Bank, 2018; Organization for Security and Co-operation in Europe [OSCE], 2011).

An early warning, interoperable record, common protocol, or trusted communication channel may reduce deaths, downtime, panic buying, duplicated deployment, and recovery delay. Because the counterfactual crisis cannot be replayed, claims should remain modest and use drills, response times, after-action evidence, and comparison with prior failures. Preparedness remains funded even when cooperation improves.

Test whether the highest-burden groups can use the system under crisis conditions.

A joint exercise can reveal that compatible radios are useless when nobody is authorized to request the neighboring crew.

5.5 — Reciprocal Transparency as Peace Technology

Reciprocal transparency can lower strategic fear when bounded, credible, proportionate, and safe to maintain. One-sided exposure may instead increase vulnerability, especially when a weaker actor depends on aid, debt relief, recognition, alliance support, or market access. A confidence-building design must therefore first specify parity of purpose and burden. Requiring identical quantities of disclosure is not the same as creating comparable security (OSCE, 2011; Schelling, 1966).

5.5.1 — Parity and Capacity Matching

Capacity matching asks what each party can safely and competently carry given its exposure and resources (OSCE, 2011; Jervis, 1978). A state without strong cybersecurity may need support before sensitive reporting. A city with weak archives may need records infrastructure. A post-conflict society may need delayed disclosure. A community facing threat may need protected channels.

Capacity matching may require funded translation, secure systems, trained inspectors, legal assistance, data standards, cyber defense, and protected custody. These supports should build autonomous ability rather than create dependence on the stronger party’s vendors, intelligence, or interpretation.

One party may reveal budget categories while another opens inspection corridors; one may provide technical verification while another supplies protected data under custody. A compact can therefore be unequal in form while remaining reciprocal in burden, provided the asymmetry is explicit, contestable, and not a route for domination.

5.5.2 — Protected Security Domains

Some domains must remain protected because exposure itself can create danger. Military vulnerabilities, cyber weaknesses, intelligence sources, emergency response gaps, life-safety routes, sensitive diplomatic channels, and certain critical infrastructure details cannot always be public or fully reciprocal in real time (OSCE, 2011; Kydd, 2005).

Protected categories need a stated reason, custodian, review, and delayed explanation where possible. Protection must be justified by harm prevention, rather than embarrassment, prestige, or avoidance of accountability.

Protected domains require cybersecurity as well as secrecy discipline. A disclosure architecture can create new attack surfaces, allow sensitive capabilities to be inferred from aggregated data, or expose personnel and infrastructure indirectly. Protection should therefore include data minimization, segmented access, secure transmission, provenance, breach response, retention limits, and review of inference risk. The public may know the category and oversight route without receiving the dangerous detail.

The hardest cases are dual-use systems. A port, satellite, data center, laboratory, energy grid, logistics route, or communication network may serve ordinary life and security at once. Full exposure can create danger, while full secrecy can create mistrust. The fair method is layered review: public purpose, protected technical inspection, redacted findings, breach consequences, and sunset review where the security reason can be reconsidered.

Public humiliation can trigger political backlash, harden identity, feed propaganda, and make future cooperation harder. Verification that treats a people, agency, or state as inherently suspect may produce compliance on paper while increasing resentment in practice.

5.5.3 — Confidence-Building Steps

Confidence-building steps are small acts that reduce fear enough for larger cooperation to become possible. They can include shared incident records, joint inspections, limited data exchange, notification protocols, demilitarized practices, public budget explanations, third-party verification, emergency hotlines, or agreed review windows (OSCE, 2011; Schelling, 1966).

To build cumulative confidence, each step needs both a verification rule and a recovery rule. A failed notification, inspection dispute, data mismatch, or security incident should trigger diagnosis, proportional response, and a defined route back, rather than force an immediate choice between unqualified trust and hostility (OSCE, 2011; Kydd, 2005).

Each step should state what fear it addresses and what burden it leaves untouched.

Conditions attached to aid, debt relief, trade access, security support, or technical assistance should protect the undertaking rather than transfer continuing control. A reciprocal arrangement discloses who sets the standard, owns the data, chooses the vendor, interprets results, and controls the public narrative. It also preserves local authorship, refuses unnecessary exposure, and provides an exit route that does not punish independence.

5.6 — Where Savings Should Go

Allocate any verified remainder under Section 5.7 through a lawful destination and accountable custodian (World Bank, 2011; United Nations and World Bank, 2018). The relevant amount must exclude transition obligations that remain unpaid. No category is automatically first in every place because urgency, readiness, and reversibility differ. A public rule should explain which eligible repairs or protections can produce near-term relief and which require longer preparation.

5.6.1 — Repair Backlogs First

After conversion obligations and worker transition are funded, verified savings should carry a strong presumption in favor of urgent repair (OECD, 2019; World Bank, 2011).

Repair-first allocation needs a prioritization method that does not simply follow asset value or political visibility. Severity, safety, service dependence, duration of neglect, lack of private fallback, unequal exposure, and cost of further delay all matter. A modest repair in a long-neglected place may reduce more burden than a prestigious project with a larger price. The route should explain why one backlog moved before another.

5.6.2 — Debt Relief and Household Stability

Debt relief can restore options constrained by payment obligations (Richardson, Elliott, and Roberts, 2013; Dobbie and Song, 2015). Apply the relief-without-new-control safeguards in Section 7.5 when allocating verified savings to debt relief. Household stability may include rent relief, medical debt relief, utility stabilization, childcare support, emergency grants, food security, transportation support, legal aid, and help with predatory debt.

A debt-relief allocation should not treat the face value of an obligation canceled as an equal amount of new household purchasing power or available public revenue. The balance recorded, the amount a creditor could realistically collect, the sum the household would otherwise pay, and the public cost of intervention may all differ. Appraisal should identify actual payment relief, collection and litigation exposure, credit-record consequences, creditor incidence, and the new fiscal obligation before announcing a peace dividend. Section 7.4.1 examines evidence from specific medical-debt interventions; the principle here is to count the capacity actually released, not the largest printed balance.

5.6.3 — Public Capacity and Reserves

Public capacity and reserves are legitimate destinations for verified savings because a peaceful society still needs buffers, trained institutions, and room to act before panic (IMF, 2019; United Nations and World Bank, 2018).

A reserve needs a declared purpose: disaster response, emergency care, liquidity support, worker transition, cyber recovery, climate adaptation, critical imports, contract exit, or temporary reconstitution of security capacity. Its target range, custodian, eligible uses, audit, release trigger, replenishment rule, and review date should be known before political pressure arrives.

Reserves must not be built by starving urgent care or known repair. The public explanation should show why resources are being held, what immediate burdens remain unfunded, how the tradeoff was judged, and when the balance will be reconsidered.

Climate and disaster adaptation can also use released capacity (IPCC, 2023). A cooling facility, for example, needs electricity, staff, transport access, and an opening schedule that follows the heat.

Capital set aside for flood protection should not be reported as completed protection before the works can function.

5.7 — When Savings Become Usable Capacity

Usable capacity is the demonstrated remainder available for a new purpose (Knight, Loayza, and Villanueva, 1996; IMF, 2019). Authorization reduction, outlay saving, avoided future liability, released staff or equipment, lower household cost, returned time, and trust-based coordination are different kinds of change. A fiscal remainder may be appropriated; staff may need support and retraining, while assets may need adaptation. Returned household time is a welfare and capacity gain; lower uncertainty may appear through smaller inventories or lower risk premiums. The public record should show source, baseline, authorization, commitment, actual outlay, conversion and transition cost, residual obligation, maintenance and oversight, uncertainty reserve, destination, outcome, dispute, correction, and retirement.

5.7.1 — Timing, Conversion, and Double Counting

A gross reduction becomes usable capacity only after timing, conversion, and overlap are resolved. An appropriation can fall before a contract ends; a contract can end before cash is released; cash can be released before workers, facilities, equipment, or suppliers can move to another purpose. Treating those moments as one event makes a dividend arrive earlier and larger than the underlying capacity (IMF, 2019).

Net usable fiscal capacity is the saving left for new commitments after the costs of making it usable have been covered. Begin with savings that can actually be realized during the chosen period. Deduct conversion costs, worker and regional transition support, residual security and reversal capacity, maintenance, oversight and administration. Set aside a prudential reserve before treating the remainder as available for allocation. Use the same currency, price basis, and time horizon throughout. Deduct each incremental cost only once; an expense already

removed from the saving estimate must not be subtracted again. A negative remainder identifies a funding gap, not spendable savings. The reserve is money withheld against uncertainty, not necessarily an expense already incurred. Where a component cannot be estimated reliably as one figure, report a justified range.

Include verification and privacy in the oversight estimate, and justify the reserve through the declared sensitivity range. Establish the baseline and gross avoidable cost before estimating realizable savings. Then identify who bears the remaining costs, where the available funds will go, and what benefit they are expected to deliver.

Report returned household time, lower private precaution, released attention, improved coordination and reduced exposure in their own units, with a baseline, distribution and decision relevance, rather than treating them as cash. Across all claimed gains, identify shared causes, remove double counting before aggregation, and keep gains to different recipients distinct.

For illustration, suppose realizable savings over the chosen period amount to 100 units. Deduct 20 for conversion, 15 for worker and regional transition, 20 for residual security and reversal capacity, 10 for maintenance, 5 for oversight and verification, and 10 for uncertainty. Twenty units remain available. This illustration is not a forecast; who benefits and how the funds are used remain separate questions.

Time has to match the underlying risk. One-time proceeds cannot fund an indefinite obligation without a financing plan. A recurring saving should not be declared after one favorable quarter. An avoided liability needs a probability model, discounting assumptions, and evidence that the liability truly became less likely. Released capacity is dated when it can be deployed, not when a reduction is announced.

5.7.2 — Public Tracking and Revision

Verified savings can be redirected into vendor margins, executive rewards, prestige projects, creditor advantage, or general accounts without repairing the original burden. This need not involve theft: the decisive question is who controls allocation and what reaches those meant to benefit (United Nations and World Bank, 2018; World Bank, 2011).

Track each claimed reduction through delivery to its intended destination, recording both completed and failed transfers (OECD, 2015a; IMF, 2019).

Publish distributional results only where lawful and safe, without exposing individual records.

Apply Section 6.5's donor firewall to external funding for verification or repair. The tracker must name the sponsor, any control risk, and how the record can be exported when funding ends.

A replacement indicator should preserve a documented bridge to earlier reports, so a change in measurement cannot be mistaken for a change in the service itself.

The final comparison should distinguish a delayed transfer from a transfer completed on time that failed to deliver the expected service. Date every assumption change so readers can distinguish forecast error, delivery delay and incomplete implementation.

Chapter 6 — Public Finance and Donor Firewalls

Public finance is where social promises acquire dates, staff, contracts, maintenance duties, and consequences. A government may praise care while clinics remain understaffed, celebrate education while school buildings decay, or announce resilience while postponing the work that would make communities safer. A budget cannot reveal every moral fact, but it shows which needs have been made institutionally real enough to receive money, authority, labor, and time (IMF, 2019; OECD, 2015a; Addati et al., 2018).

Public finance requires several kinds of account. Annual cash flow records what enters and leaves during a period. Accrual reporting recognizes obligations incurred even when payment comes later. Capital expenditure acquires or improves long-lived assets, while operating cost keeps services functioning now. Maintenance backlogs, reserves, guarantees, and off-budget liabilities each describe a different claim on present or future capacity. Keeping those categories distinct prevents apparent balance from hiding the obligations that make public life work.

Taxation introduces a different mechanism. Simplicity can improve legibility and reduce compliance cost, while crude simplicity can erase ability-to-pay differences, invite avoidance, weaken revenue stability, or shift enforcement toward people with the least professional help. Complexity may protect equity or close loopholes, yet it can also tax time, literacy, privacy, and calm. Tax design is therefore assessed across legibility, distribution, administrability, enforcement capacity, avoidance control, certainty, privacy, correction, and durable public capacity.

Procurement determines what the public receives, what it can maintain and change, and who bears lifecycle risk. Section 6.4 examines how specifications and contract terms can preserve those capabilities or create lasting vendor dependence.

Donor firewalls govern the point at which private support can acquire public authority. A gift can fund valuable work while granting influence through conditions, access, data, or withdrawal leverage. Section 6.5 tests which rights remain legitimate and whether refusal is materially possible. Imagine a useful grant whose first payment arrives before the recipient can fund its recurring commitments. Acceptance changes the next negotiation: refusing a later condition may now threaten a service already in use. The relevant choice is therefore larger than accepting or rejecting today’s gift. It includes the obligations that acceptance would place between the institution and its future independence.

Reserves, deficits, and borrowing move burden across time. A reserve can protect services from panic, while an excessive or undefined reserve can postpone present repair. A deficit can stabilize a downturn, answer an emergency, or finance durable capacity; persistent mismatch, high debt service, currency exposure, interest-rate risk, and opaque guarantees can narrow later choice. Fiscal calm begins by naming these functions and risks precisely rather than treating every deficit as shame or every loan as progress.

6.1 — What Budgets Reveal About Priorities

Read a budget against inherited obligations, burdens displaced beyond its accounts, and the capacity future residents will inherit (Musgrave, 1959; OECD, 2015a). Reading a budget requires connected records. Appropriations show authorization; cash reports show payment; accrual statements show obligations; asset records show condition; and performance records show delivery. Guarantees, debt reports, and public-private arrangements reveal additional future exposure. Household and frontline evidence then identifies work, delay, or risk displaced beyond the accounts. Together these records show whether a promise became usable service while there was still time to correct it (IMF, 2018; International Budget Partnership, 2026).

6.1.1 — Recognition Through Funding

Funding makes a public obligation operational only when money, authority and delivery capacity are provided together (Musgrave, 1959; OECD, 2015a). A usable line item therefore states the service pathway, eligible population, staffing or unit assumptions, delivery constraint, review date, and office accountable for shortfall.

Underfunding can create false evidence against public systems. A service is starved, performs poorly, and is then accused of being inherently ineffective.

A budget becomes especially revealing at the point where money changes into an authorized action. Imagine a library funded to open later in the evening. The building budget covers light and heating, while a separate staffing account controls the hours of the people who must remain. An increase to the first account alone does not purchase a later opening. The promise is funded only when the two decisions can operate together.

The budget explanation should therefore follow one complete service rather than one convenient ledger. It can show the additional opening hours, the staff arrangement that makes them possible, and the date from which the public can use them. This gives residents something definite to compare with the original proposal. It also gives staff a way to identify a missing dependency before the opening is advertised.

Following that conversion makes public recognition more concrete. The relevant achievement is not that an important need received a larger number beside its name. It is that resources and authority were joined into a service people can reach. A budget document earns its place in ordinary life when a reader can follow that route without needing private knowledge of the institution.

6.1.2 — Prestige and the Repair Alternative

Prestige bias appears when spending raises institutional image more reliably than it lowers burden. Compare an easily announced new project with the maintenance or service improvement it displaces. Leaders seeking a visible legacy and donors seeking recognition can reinforce this bias.

Suppose a town is choosing between a new performance hall and repairs that would return an existing hall to regular use. Both proposals promise a place for shared cultural life. Comparing their construction prices alone leaves the decision unfinished. The town can first ask how many usable evenings each proposal would provide and who could reach them.

The existing hall might need a roof and accessible entrances. The new hall might support activities the old one could never accommodate. Those are reasons to compare the actual programs each building could sustain. A repair is not automatically the better choice, and a new building is not automatically an

indulgence. The stronger proposal is the one whose particular benefits justify the resources committed to obtaining them.

Recognition can then follow the completed public use. A restored hall filled with rehearsals and local gatherings gives a civic achievement a visible life after the announcement. The people who designed and maintained that improvement have something durable to point to. Prestige becomes more useful when it attaches to a place that keeps serving its community.

6.1.3 — Budget Silences and Institutional Memory

Budget silences can conceal obligations borne by groups or systems with little influence over allocation (IMF, 2019; Addati et al., 2018). An intelligible public account investigates omissions. A missing line for disability access, translation, or care may conceal an unfunded obligation, but the cost may also sit in another department, contract, or classification. Reconcile those accounts before concluding that the function is absent.

A budget process should therefore include distributional review and participatory evidence strong enough to reveal burdens that organized influence would otherwise omit.

Budget memory can begin with the decisions that have not yet produced an invoice. Suppose a school delays replacing a roof after an inspection finds that another year of service is feasible with specified repairs. The next budget should carry the inspection date and the repair that made the delay reasonable. A successor can then distinguish a planned interval from a forgotten obligation.

That record is most useful when it preserves the decision point. What condition justified postponement? What observation would bring the work forward? Which preparatory task can proceed while the larger expenditure waits? Design work or an updated survey may leave the next team better prepared to act.

Remembering a postponement can therefore create capacity before the final construction begins.

This is the constructive purpose of tracing a budget silence. The account makes unfinished work inheritable. It allows a new administration to continue a sound plan, revise a weak one, and explain the difference. Public memory becomes a tool for completing obligations rather than merely a catalogue of what earlier officials failed to do.

6.2 — Honest Costing Versus Buried Liabilities

Publishing a fiscal document does not guarantee that a legislature or citizen can understand the obligations behind it. The Open Budget Survey 2025 reported that half of 82 assessed governments omitted debt-composition data from budget proposals and that 82 percent omitted information needed to assess sustainability over the following decade (International Budget Partnership, 2026). Credible scrutiny distinguishes cash and accrual, follows commitments beyond annual statements, and identifies contingent and off-budget risks while decisions are still changeable (IMF, 2018). Fiscal risk concerns divergence from a projected path through shocks, guarantees, litigation, disasters, interest, currency movements, public enterprises, or contracts, whereas social burden also includes care, displacement, delay, illness, and lost access that may never appear as accounting liabilities. Failure-cost disclosure identifies the supported consequences of inaction, separates them from scenarios, and specifies what evidence would revise the decision (IMF, 2019; IPCC, 2023). Vendor dependencies also require explicit exit costs and portability tests, developed in Section 6.4 (NIST, 2024).

6.2.1 — Unpaid and Underpaid Care

Care burden does not fall evenly, and care itself can consume the time caregivers need to advocate for support. Budgets should identify who carries unpaid or underpaid hours, which services could share or replace that work, and whether respite, income support, leave, training, or formal provision would reduce excessive or unfairly imposed demands (Addati et al., 2018).

Record the need for and availability of respite and training. Also record emotional load, schedule predictability, continuity, safety, and the dignity of both the person receiving care and the person providing it.

6.2.2 — Pension and Workforce Liabilities

Public pension and workforce commitments should state who was promised what, how the promise is funded, which assumptions govern it, and how correction can occur without treating workers themselves as the liability (OECD, 2023c; IMF, 2019).

Workforce accounts should distinguish pension and healthcare obligations from operational risks such as staffing shortages, training gaps, overtime reliance, burnout, injury exposure, recruitment difficulty, and loss of institutional memory. These risks can create future costs without already constituting accounting liabilities. A budget can therefore meet its reporting rules while leaving the capacity needed to deliver services underfunded.

Underfunded obligations can create conflict between retirees, current workers, taxpayers, and service users. A burden-aware approach requires early visibility, public explanation, and burden-sharing rather than surprise austerity or promise-breaking.

6.3 — Tax Simplicity, Compliance, and Trust

Tax systems occupy a psychologically delicate place because they ask people, households, firms, and communities to release private resources into public trust. Confusing, humiliating, arbitrary, or visibly captured administration can weaken revenue and teach contributors that cooperation is naive (Luttmer and Singhal, 2014; OECD, 2025e). A simpler form can reduce time and error while a simpler tax base can shift burden regressively; complexity may close avoidance routes while also rewarding those who can purchase advice. The relevant question is therefore not whether a system is simple in the abstract. It is simple for whom, in relation to which obligation, with what privacy cost, and under which enforcement capacity.

6.3.1 — Tax as Shared Capacity

Taxation converts private resources into shared capacity (Luttmer and Singhal, 2014; OECD, 2025e). Tax is a burden-sharing instrument, not moral punishment for earning. Shared capacity includes readiness, prevention and insurance whose value may become apparent only in a crisis; it is not limited to services received by the same person in the same year. Report filing, assistance, correction, and appeal costs alongside statutory incidence, distinguishing taxpayers' access to accountants, reliable records, language support, and digital services (Alm, 2019; Slemrod, 2019; OECD, 2025e).

6.3.2 — Evasion and Trust Collapse

When people believe others are evading tax, their own willingness to contribute can weaken, threatening legitimacy as well as revenue (Alm, 2019; Slemrod, 2019). The analysis distinguishes desperation, confusion, avoidance, aggressive planning, and deliberate evasion. Not every error is moral

corruption. But large-scale evasion by powerful actors is a direct attack on shared capacity, especially when complexity, secrecy, and professional access let them shift enforcement and service losses onto people with fewer options.

Low trust can raise the cost of audits, penalties, investigations and litigation. Trust can fail from both directions. Visible impunity for powerful evaders tells ordinary contributors that the rule is optional for those with influence, while indiscriminate enforcement tells them that the system cannot distinguish organized avoidance from a correctable mistake. A credible response therefore combines proportionate detection, beneficial-ownership visibility, professional-enabler scrutiny, usable correction routes, and public evidence that enforcement reaches complex as well as simple cases. Provide reasonable deadlines, correction before punishment where appropriate, and human review for cases that do not fit the form.

6.3.3 — Administrative Capacity

Tax design must fund staff competence, service standards, secure records, reliable assistance and independent review (OECD, 2025e; IMF, 2019). Fiscal plans should disclose the cost and readiness needed to operate their rules. Before a filing rule takes effect, estimate the calls and correction requests that its first reporting cycle will generate.

Digital filing and matching must preserve non-digital access where needed and purpose limits that prevent tax records becoming dossiers spanning unrelated areas of life. A disputed automated match should have a named resolution path, rather than force the filer to reconstruct the matching system.

Visibility should follow revenue into authorized purpose, delivery, maintenance, distribution, and correction without pretending each payment funds a favored service. Public

reporting must show delay, failure, tradeoffs, and revision as well as success.

6.4 — Procurement, Maintenance, and Anti-Waste Design

Procurement creates distinct opportunities for value or failure before tender, at award, and throughout contract performance and exit (OECD, 2015b; ISO, 2017). Public dependence often begins in a specification that no ordinary reader will ever see. Proprietary interfaces, unavailable source code, weak documentation, missing training, nonportable data, short warranties, and unpriced recurrent costs can make a competitive purchase function like a long monopoly after award. A burden-aware review therefore asks not only who won, but what the public will still own, understand, maintain, test, and change when the original vendor is gone.

6.4.1 — Lifecycle Cost

Lifecycle review separates stages before it announces a total. Acquisition, installation, integration, training, staffing, operation, maintenance, energy, supplies, security, accessibility, correction, downtime, renewal, replacement, disposal, and exit each need assumptions and a responsible party (ISO, 2017; OECD, 2015b).

Users, maintainers, repair workers, accessibility specialists, and exit teams should participate in lifecycle review before award. Executive or vendor evaluation alone often misses the burdens that appear during daily operation, especially when knowledge, spare parts, documentation, or data portability are weak.

An audit purchase can be designed to change what evidence reaches the regulator. In a two-year field experiment in Gujarat, Duflo et al. (2013) changed the market arrangements governing environmental auditors' appointment, payment, and incentives for accurate reporting. Independent measurements provided a check on their reports. Auditors reported more truthfully, fewer plants were falsely classified as compliant, and treated plants

reduced pollution. The experiment evaluated a reform package rather than isolating each component. Its constructive lesson is that independence can be an operating design: who selects and pays the reviewer, how accuracy is checked, and whether truthful findings can change the regulated activity.

Procurement also shapes markets. If public contracts favor scale, opacity, political access, or incumbency, the public may unintentionally strengthen the very market power it later tries to regulate.

6.4.2 — Change-Order Risk

Change orders are a normal feature of complex work and a recurrent route through which approved cost expands. Construction, technology, infrastructure, health systems, and emergency services may reveal conditions that could not reasonably have been known when the contract was signed. Public review must distinguish legitimate adaptation to uncertainty from cost increases caused by strategic underbidding, weak scope, political pressure, or opaque negotiation. In those cases, the terms offered at tender may be substantially rewritten after competition has ended (Flyvbjerg, 2014).

A defensible change-order record should show cause, approving authority, price basis, schedule effect, accumulated amendments, vendor pattern, conflicts, alternatives, and the threshold at which the modified contract returns to competition or public reconsideration. It should also count the staff time and technical capacity required to review each amendment. Understaffed agencies can be outmatched by vendors that specialize in post-award expansion.

Necessary adaptation should remain visible, justified, and learnable rather than punished. Repeated small amendments may amount to a new project even when no single change appears

decisive, while a safety correction or genuinely altered public need may require prompt action. The review should state which facts were unknowable, which were foreseeable, and what future procurement practice must change.

Require the contract to preserve practical competition after installation through audit access, data return, transition support, continuity obligations, and source or escrow arrangements where appropriate (NIST, 2024).

Outsourcing expertise can help, but it must not leave the public institution permanently unable to understand or operate its own system. Before award, procurement should price knowledge transfer and retained public expertise alongside the supplier obligations above.

Pre-award red-team review should identify the affected users, uncertainty, mitigation owner, stop condition, and residual risk accepted by the approving authority. The team needs authority to pause, narrow, redesign, reject, or require public explanation.

The final record should include a public burden summary that explains which risks were tested and which safeguards became conditions of approval without exposing protected security detail. It should also preserve the risks named before signature as later performance criteria.

6.4.3 — Maintenance Before New Purchase

Maintenance review compares the marginal value of repair, upgrade, replacement, and expansion while identifying who will staff and finance each option across its useful life (OECD, 2015b; World Bank, 2020b).

The comparison also applies to digital infrastructure. Replacing a public platform before repairing data quality, staff training, accessibility, and support systems may simply digitize the old failure.

The choice between repair and replacement should include condition, remaining service life, safety, energy and operating cost, compatibility, accessibility, spare parts, staff skill, environmental effect, downtime, and the public’s ability to maintain the asset after acquisition.

6.5 — Donor Firewalls and Restricted Gifts

Before accepting support, an institution should identify the donor and lawfully reviewable source, every restriction and promised right, related intermediaries, operating burden, concentration risk, and the authority able to refuse or unwind the arrangement. Philanthropy can fund repair, research, care, culture, and public goods before routine finance is ready, but it can also transfer influence over access, governance, data, procurement, narrative, or renewal (Reich, 2018; Frumkin, 2006; OECD, 2026e). Boundaries protect donors as well as recipients by keeping public decisions with accountable institutions rather than shadow governors. Influence can also arise through roles outside the gift, because a funder may simultaneously be a contractor, creditor, landlord, platform provider, board member, employer, or source of emergency finance. Each role may be lawful in isolation while their combination creates leverage invisible in the donation agreement. Connected interests, discounted services, naming rights, employment offers, cross-contract defaults, and withdrawal threats therefore require joint review.

6.5.1 — Donation Is Not Ownership

Donation is not ownership of mission, doctrine, public meaning, staff judgment, participant access, research conclusions, record release, political timing, or institutional criticism. A donor may receive acknowledgment, a lawful receipt, audited use-of-funds reporting, and evidence that accepted restrictions were honored. Authority beyond that stewardship relationship requires a separate lawful basis and must never be implied by the size or urgency of the gift (Reich, 2018; Frumkin, 2006).

The decisive test occurs when donor preference conflicts with duty. Suppose a major sponsor threatens not to renew funding unless an unfavorable evaluation is delayed or a recipient appears

in a promotional story. Before acceptance, the recipient institution should have identified who can refuse the demand, what independent publication authority protects the finding, whether essential services can continue through the notice period, and which financial reserve or replacement plan covers withdrawal. An agreement that prohibits editorial control but leaves the sponsor able to shut down care within a week is not an effective firewall.

Inventory governance seats, vetoes, consultation promises, early drafts, branding, naming, staffing conditions, procurement preferences and renewal discretion. Identify prohibited rights, approval authority, conflicts requiring recusal, and remedies for violations: correction, suspension, repayment or exit.

Narrative capture occurs when a donor controls how the funded work and its participants are represented. Examine communication, branding, evaluation, photography, and naming clauses for approval power over reports, criticism, participant stories, research conclusions, crisis statements, and disclosure timing.

People receiving services retain consent, privacy, and the right not to perform gratitude. The institution retains authority to report failure, revise its interpretation, publish adverse findings, and end the relationship.

6.5.2 — Access Capture

Access capture occurs when funding purchases proximity that changes decisions without appearing in the formal governance chart. Private meetings, early drafts, advisory invitations, event placement, privileged briefings, naming privileges, and personal relationships can alter staff behavior even when no veto exists (Ostrower, 1995; Callahan, 2017).

Stewardship, audit, and lawful donor communication should be formal, role-limited, recorded where appropriate, and separated from program judgment, governance, research, and participant access.

Donors should receive no privileged access to people or private records connected with the funded work. Necessary contact must follow the same purpose, consent, privacy, safeguarding, and oversight rules that apply to comparable actors.

6.5.3 — Dependency Risk

Donor dependency puts essential functions at risk when replacement support cannot be secured in time. OECD's 2026 review of development philanthropy reinforces the need to examine concentration and operating models before private support becomes structural dependence (INCITE!, 2007; Reich, 2018; OECD, 2026e).

Ordinary concentration becomes coercive leverage when renewal, payroll, bridge finance, or vendor support is tied to a disputed judgment, delivered through informal pressure, or made impossible to replace. The review should identify whether refusal remains materially possible and whether the institution has already begun changing its conduct to preserve the relationship.

For functions whose sudden loss would harm people, acceptance should include a continuity plan with a reserve floor, diversified funding, staged commitments, and portability. Record each donor's share of the function, renewal timing, months of reserve coverage, the lead time before withdrawal would interrupt service, replacement options, exit costs, and payroll and contract obligations after termination.

Identify connected vendors, unfunded continuing costs, the lawful custodian of records and equipment, and who would carry

a shortfall. Specify the concentration threshold requiring redesign or refusal.

Test how services would continue if a donor withdrew, changed conditions, became unsafe, faced financial difficulty, or could no longer fund the work. Also test whether the institution could refuse renewal to protect its mission. Where one donor could interrupt a protected service before replacement finance or safe transfer is available, reduce the grant's scope, build the reserve first, or decline the terms.

Treat donor dependency in budget planning as a funding and continuity risk. Assess any contractual or other qualifying obligation separately under the applicable accounting rules.

Funding labels should travel with reports, dashboards, websites, platforms, and civic tools so readers do not mistake support for ownership. A readable label need not reproduce every legal clause, but it should identify the funder and supported function, summarize the main obligations attached to the funding, and state the limits on donor control.

6.5.4 — Donor Exit Plan

An exit plan should name the final funded period, continuing obligations to staff and participants, public notice, asset and data custodians, replacement financing, and any shortfall. Identify an independent custodian for records after organizational closure. State what can transfer, shrink safely, or must end.

Designate transition authority before dependence hardens. Notice periods should reflect the time needed to replace payroll or transfer care, not merely the grant's accounting year.

A smaller first cohort can be preferable when its full support period is funded rather than dependent on renewal. Clear exit terms also prevent the donor from becoming the inevitable

emergency rescuer, allowing support to remain finite, generous, and noncontrolling.

6.6 — Reserves, Deficits, and Fiscal Calm

The IMF estimated global public debt just below 94 percent of GDP in 2025 and projected 100 percent by 2029. That outlook warrants planning but cannot by itself judge every deficit (Blanchard, 2019; IMF, 2026a). Borrowing that stabilizes a downturn has a different time profile from a recurring operating gap or an unmanaged liability. A serious account therefore connects fiscal choices to the assets, services, risks, and obligations they create over time.

6.6.1 — Deficits as Tool, Not Shame

A deficit created by automatic stabilization during a downturn differs from a structural gap that would remain after temporary weakness passes. Emergency borrowing differs from recurring operating imbalance. Borrowing for a durable asset or productive capacity differs from borrowing that repeatedly finances ordinary commitments without a credible revenue path. Assess purpose, financing conditions, distribution, alternatives, and the route by which costs are carried (Blanchard, 2019; Furman and Summers, 2020).

Public explanation should identify which condition is present, how long it is expected to last, what would change the judgment, and what future capacity is created or consumed. It should also distinguish a planned fiscal choice from a forecasting error, an off-budget obligation, or a contingent liability that has suddenly materialized.

6.6.2 — Debt-Service Burden

A debt-service statement should show sensitivity to interest rates, revenue shocks, currency exposure where relevant, rollover risk, and the services most vulnerable if projections fail (IMF, 2026a; World Bank, 2025c).

A government may have a sustainable long-run path yet face near-term refinancing pressure, or meet current payments while accumulating unsustainable obligations. Maturity concentration, variable rates, foreign-currency exposure, creditor structure, and guarantees can change the burden even when the headline debt stock is unchanged.

For emergency borrowing, an after-action review should reconcile authorized borrowing with actual use, beneficiaries, the asset or service created, sunset decisions, the debt-service path, and any emergency mechanism that became permanent.

Separate financing that ends with the emergency from liabilities that survive it. Record debt-service dates, outstanding guarantees, contracts requiring exit or renewal, assets entering ordinary maintenance budgets, and temporary facilities whose recurrent costs became structural.

Emergency dependency should be separated into recurring categories: single-supplier dependence, temporary staffing dependence, crisis warehousing, emergency legal authority, foreign-currency exposure, humanitarian supply concentration, and systems that cannot operate outside exceptional procurement. Each category has a different exit path. A register should state the trigger that created the exception, the cost of remaining, the cost of leaving, the capability that must be rebuilt, and the authority responsible for returning the system to ordinary conditions.

6.6.3 — Fiscal Fear Rhetoric

Fiscal fear rhetoric can make cuts appear inevitable without testing the claimed mechanism, timing, distribution and alternatives. Some warnings identify real constraints; others turn uncertainty into political leverage (Musgrave, 1959; OECD, 2015a).

Consistency is the first safeguard against selective alarm. The same standards applied to social programs should also examine prestige projects, tax privileges, military excess, guarantees, captured contracts, and deferred maintenance.

A proposed cut, tax, borrowing decision, reserve target, or new expenditure should state its cost, incidence, purpose, hidden assumptions, and performance under a strong plausible alternative. Where uncertainty remains, the public should see the range, the contingency, and the event that would trigger reconsideration.

6.7 — Public Finance for Repair Rather Than Prestige

Repair, access, regional restoration, and capacity that protects future people deserve priority when ordinary systems fail (Mazzucato, 2018; OECD, 2015a). That presumption is not a universal numerical ranking: preventable harm, readiness, distribution, irreversibility, and long-term capability can point toward different choices. Some urgent repairs are not yet executable, while a larger investment may be safer than repeated patching, and a beautiful public project may also serve access, research, identity, or renewal. Decision makers should disclose competing burdens and explain the conditions that would change their allocation. Equity of attention matters because neglected regions may normalize decay that influential districts can quickly publicize, and the people using funded offices, forms, portals, queues, and appeals should help test whether the promised route works.

6.7.1 — Regional Repair Priority

Regional repair should identify historical extraction and neglect, remaining local capabilities, and the role of local authorship (OECD, 2014; World Bank, 2017). Identify regional differences in schools, water, hospitals, transport, public offices and climate exposure. Use place-specific baselines, local authorship, employment and supplier effects, service outcomes, maintenance funding, and long-horizon review rather than one uniform distribution formula.

Future-person review should trace asset condition, ecological exposure, debt service, maintenance, and foregone options across the relevant horizon while avoiding the fiction that one discount rate settles every obligation. Show the maintenance backlog alongside debt service so a postponed roof repair cannot masquerade as an unencumbered future budget. A prestige project

should retain an auditable counterfactual: which repair alternatives were rejected, what assumptions justified the choice, what threshold would trigger redesign or cancellation, and how maintenance, access, displacement, and regional incidence will be reviewed after opening.

A library, square, or cultural venue can earn public pride through sustained use as well as its opening ceremony.

6.7.2 — Turning a Repair Priority into a Work Program

A repair portfolio needs both an order of importance and a route into execution. Consider three proposed works: an urgent water repair whose design is complete, a bridge renewal awaiting a site investigation, and a school improvement that can be built during a short holiday. Their need and readiness are different facts. A useful funding plan assigns money to the next necessary task for each rather than treating every project as ready for construction.

For the bridge, an investigation can be valuable progress even though no new structure appears. It may establish which design is appropriate and allow a credible construction price. For the school, reserving a suitable crew for the holiday may determine whether the project can be completed without disrupting a term. For the water repair, waiting for the other projects to become equally ready could prolong a burden that can already be relieved.

This approach also creates a place for smaller local suppliers. Work can be offered in packages that fit real capabilities while preserving responsibility for how the packages connect. A project manager needs to know which completion unlocks the next stage and which tasks can proceed independently. The budget then supports a sequence of achievable work instead of a single impressive commitment whose dependencies remain unfunded.

Public reporting can distinguish preparation, construction, and usable service. Each is an accomplishment with a different

meaning. Preparation should leave a decision or design ready for the next team. Construction should produce the specified asset. Usable service arrives when the asset is staffed and connected to the system around it. The portfolio succeeds by moving worthwhile projects through those stages at a pace the available workforce can sustain.

Chapter 7 — Debt, Precarity, and Restoring Future Choice

Debt can bring housing, treatment, education, equipment, infrastructure, or emergency relief into the present when needed capacity cannot be funded at once. Used well, it spreads the cost across the period that benefits from the investment. The same instrument becomes restrictive when repayment consumes the options the financing was meant to create (Richardson, Elliott, and Roberts, 2013; World Bank, 2025c; Sen, 1999).

The chapter begins before any formal debt instrument appears because a contract never lands on neutral ground. The same amount owed has different force when one household has savings, paid leave, stable housing, legal help, transport, health, documents, family support, and time, while another is already negotiating survival one deadline at a time. Poverty is therefore treated as option collapse before debt is examined as one mechanism that can prolong it.

Household debt constrains choices through payments, collection pressure, disrupted care, and privacy exposure as well as principal. Section 7.2 traces those carrying burdens through daily life. For an illustrative household, an affordable monthly installment can still fall on the wrong day. Moving its due date may solve a timing problem without reducing principal; reducing principal may leave a payment-timing problem in place. Beginning with that distinction helps the reader ask what relief is meant to change before choosing an instrument or judging its generosity.

Municipal and sovereign debt raise related questions, although their law, scale, and remedies differ. Like household debt, they can become claims on future room and shift burden onto people who lacked a full voice in the original decision. They can force maintenance, care, education, housing, food, water, health, public safety, or climate adaptation to wait behind obligations whose full cost was not publicly understood when the promise was made.

The chapter asks how debt repair can restore choice without purchasing control. Section 7.5 sets out the enforceable limits on demands attached to assistance.

Institutions providing relief must also examine their own power. A credible provider states what the recipient must supply, which records are created, who can see them, how errors are corrected, whether support survives criticism, and how exit works. Because material need can make a formally optional request coercive, refusal, privacy, appeal, and access to unrelated essentials have to remain real.

Relief necessarily distributes burden. Creditors, guarantors, depositors, taxpayers, workers, pensioners, suppliers, public agencies, and future borrowers may all be affected. A legitimate repair therefore includes an incidence account: who absorbs loss, who receives restored capacity, which larger harms are avoided, how the route affects future access, and which new risks require monitoring. The debtor’s burden and the creditor’s claim both matter, but neither can settle the design alone.

7.1 — Poverty as Option Collapse

Poverty is commonly measured through income, assets, consumption, or access to services. Those measures are essential, yet its deepest material effect is often the contraction of possible futures. Poverty reduces what can safely be refused, where a person can go, how long a decision can wait, which risks can be absorbed, and whether a legal option is practically executable (Mani et al., 2013; Mullainathan and Shafir, 2013). Scarce liquidity turns ordinary setbacks into cascades, survival administration consumes time, and immediate problems claim attention that could have supported learning, care or long-range planning.

7.1.1 — Time Scarcity

Money scarcity can force people to spend more time securing necessities: a cheaper grocery may require a longer journey, while an affordable clinic may require missed work. The relevant measure is whether the complete route fits the person's available time and resources (Mani et al., 2013; Mullainathan and Shafir, 2013).

Administrative cliffs convert ordinary instability into denial. A portal closes at midnight, one missing document resets a claim, an interview is offered only while the applicant must be at work, or a notice arrives after the response period has begun. Trace each cliff to the step that risks denial or abandonment, and test simpler evidence reuse, alternative proof, good-cause correction, and a human route before requiring the process to start over.

7.1.2 — Risk Compression

Without a buffer, an ordinary setback can trigger a cascade of losses (Haushofer and Fehr, 2014; Ridley et al., 2020).

A car repair may be an inconvenience for one household and, for another, lead to lost work, missed rent, late fees, food reduction, debt, conflict, and possible eviction.

Risk compression means that the interval between problem and disaster shrinks. There is less time to compare options, seek advice, negotiate, appeal, wait for benefits, or recover from illness.

Risk compression can also punish truth. A person may hide a problem from a landlord, employer, creditor, partner, or public office because disclosure could trigger faster collapse. Late disclosure may then be treated as unreliability, even though the penalties attached to early honesty helped produce it.

7.1.3 — Option Restoration

Option restoration expands the usable capabilities a person retains after intervention (Sen, 1999; Nussbaum, 2011).

Restoration should be tested in the decisions that organize ordinary life. Compare emergency cash, safe exit from work or housing, access to care, time to appeal, mobility, data correction, and exposure to one missed payment before and after intervention. Record distribution, duration, and reversibility, because a short grant that expires before stability forms is different from durable room. The audit also identifies any new dependence on an official, donor, platform, creditor, or landlord.

An unused option may reflect a barrier rather than a settled preference. In Seattle and King County, Bergman and colleagues (2024) tested support for families receiving housing vouchers. A package combining tailored search assistance, landlord engagement, and short-term financial help increased moves to neighborhoods with high historical upward mobility from about 15 to 53 percent. A second randomized phase found substantially larger effects from the full services than from information and

financial assistance alone. The intervention helped turn an existing entitlement into a choice families could exercise.

This is a finding about neighborhood choice, not a completed observation of the children’s lifetime earnings. Its practical implication is to follow the steps between an offer and a usable option, then finance the help that removes the binding barrier. Assistance should preserve the family’s decision about where to live. Local evaluation can examine sustained access and satisfaction after a move, alongside the reasons other families choose to remain.

7.2 — Household Debt as Horizon Compression

The New York Fed reported $18.8 trillion in United States household debt in Q2 2026; the $13 billion reported quarterly decline included a temporary mortgage-reporting gap caused by a servicing transfer. That aggregate does not identify distress, but it shows the scale at which future income is pledged through household credit (Richardson, Elliott, and Roberts, 2013; Sweet et al., 2013; Federal Reserve Bank of New York, 2026). A debt account should show payment share, duration, variability, enforcement route, and the options left after payment, as well as the balance due. Assess why the debt arose, its terms and currency, any collateral or other security, the borrower's capacity to repay, and how the debt can be enforced. Compare the resulting burden with feasible alternatives and routes to repair.

7.2.1 — Debt and Family Stress

Debt can change a household's emotional climate as people ration disclosure, hope, patience, sleep and tenderness (Richardson, Elliott, and Roberts, 2013; Sweet et al., 2013).

Partners, children and relatives who never signed a debt may absorb added care, lost time, emergency cash needs or housing strain (Richardson, Elliott, and Roberts, 2013; Sweet et al., 2013).

Household review should therefore distinguish the legal debtor from those carrying secondary burdens. Collection contact, disclosure, co-signing pressure, asset risk, and service interruption should not make partners, children, elders, or other dependents invisible collateral.

7.2.2 — Debt Collection Humiliation

Debt collection can recover money while also imposing costs on privacy, calm, attention, work, and social standing. Calls, letters, lawsuits, garnishment, public records, workplace contact, and

repeated demands can make an obligation follow the debtor through settings far beyond the original transaction. The enforcement route therefore belongs inside the full cost of credit (CFPB, 2025; Board of Governors of the Federal Reserve System, 2026).

Some collection is legitimate. Creditors need ways to recover lawful debts, distinguish inability from refusal, and prevent abuse. The problem begins when collection practices convert enforcement into humiliation or use fear as leverage beyond the true claim.

A financial record can migrate into housing, employment, insurance, services or civic standing and become a general certificate of reliability. Its purpose and downstream use therefore matter as well as its accuracy (Solove, 2006; Nissenbaum, 2010; CFPB, 2025).

Keep enforcement tied to the lawful claim and the record's purpose. Provide notice, validation, dispute, proportional contact, correction, retention limits, and restrictions on unrelated reuse.

7.2.3 — Debt Data Privacy

Debt records can continue narrowing options after the underlying crisis has changed, so relief must address their lifecycle as well as the payment schedule (NIST, 2020; CFPB, 2025).

Apply Section 4.4's purpose, access, retention, and correction limits to relief records, and explain what is kept and when it will be deleted.

Relief records become especially sensitive when donors seek stories, platforms seek training data, administrators request intimate proof, or evaluators demand disclosure that a person in crisis cannot freely refuse. Funding and evaluation should rely on aggregate or de-identified evidence where possible and keep

service eligibility separate from publicity, research, and model development.

A relief program may still need to confirm identity, relevant need, priority, fraud risk, and repayment capacity.

Notify people of access to their debt information and repair the downstream consequences of inaccurate records.

7.3 — Municipal and Sovereign Debt as Soft Capture

Cities, regions, and sovereign states borrow against future capacity to finance water systems, schools, hospitals, energy grids, housing, transport, climate resilience, and emergency response. Debt service, creditor concentration, opaque terms, currency exposure, short maturities, or crisis conditionality can narrow policy space and weaken the capabilities the borrowing was meant to support. Soft capture arises when that dependence gives a creditor or other actor continuing control over public choices beyond what the legitimate claim requires (World Bank, 2025d; IMF, 2026a). Municipal and sovereign debt require specialist review because bankruptcy law, bond covenants, collective action clauses, sovereign immunity, fiscal federalism, currency exposure, and development finance cannot be resolved by a general material framework (World Bank, 2021a; IMF, 2026a). A rule that assists one borrower may alter risk or cost for another. Review must also face corruption, elite consumption, weak revenue, waste, and failed projects while keeping essential services, climate resilience, public capacity, and ordinary residents inside the burden account.

7.3.1 — Debt Service as Policy Narrowing

Under municipal or sovereign borrowing pressure, the budget sensitivities in Section 6.6 become bargaining leverage. The additional risk here is that refinancing deadlines or creditor conditions may constrain public choices about maintaining essential services (World Bank, 2025c; IMF, 2026a).

The timing of a principal repayment can matter more than the interest due in the same year. In an invented municipal account, a loan of 100 units carries annual interest of 4 units and requires all 100 units of principal at the end of year five. The annual interest line looks modest. The fifth-year obligation is 104 units unless an

agreed arrangement changes the payment path. An authority that plans only for the interest has left its largest financing decision outside the ordinary budget.

A second arrangement might repay 20 units of principal each year and charge interest at the same rate on the opening balance. Its payments would be 24, 23.2, 22.4, 21.6, and 20.8 units. The total interest would be 12 units rather than the first arrangement's 20. The second route requires substantially more cash in the early years. Comparing the two therefore means comparing both the total cost and the dates on which public resources must be available.

That comparison gives refinancing its proper place. Replacing the first loan when it matures may spread the principal over another period, but the new terms must actually be obtainable. The public decision becomes clearer when officials identify what they would do if renewal were more expensive or arrived later than expected. Preparing that choice before the deadline gives elected institutions more room to select among workable arrangements.

Borrowing can still be a means of sharing the cost of a durable asset across its useful life. The important connection is between the asset's continuing service and the payment calendar. A water system that remains productive for many years may support a different financing pattern from equipment that will need replacement soon. The account should make both lives visible: the life of the obligation and the life of what it helped build.

7.3.2 — Municipal Austerity Pressure

Municipal adjustment should connect debt service and fiscal rules to service levels, maintenance, workforce changes, neighborhood effects, and feasible alternatives before presenting cuts as inevitable (World Bank, 2025c; IMF, 2026a).

Small municipal cuts can accumulate into a larger loss of livability even when no single reduction appears catastrophic.

Fiscal adjustment should protect minimum service capacity and give special attention to residents with the least ability to move, buy private substitutes, or wait. The plan should include distributional review, public explanation, correction, and a timetable for restoring essential capacity.

A service reduction can be compared through the weekly timetable it would create. Suppose a municipal counter is open on five mornings and an adjustment would remove one morning. The saving depends on what paid work actually ceases or moves elsewhere. Its effect on access depends on which users could attend during the remaining hours. Four mornings with the same total staffing may provide a different service from five shorter sessions, even when both cost the same.

Residents can help make those alternatives concrete. A worker may need one late session rather than another morning. A resident who depends on a particular bus may need the counter open when that route arrives. Staff may know that concentrating appointments produces a manageable queue, or that a shared office can keep an essential function reachable on the closed day. These are different designs, not interchangeable descriptions of a percentage cut.

The comparison becomes productive when each option names the service it preserves and the work it releases. An authority may find that moving a session improves access without requiring a larger appropriation. It may instead find that the proposed saving cannot be achieved while retaining the required function. In either case, the public debate has advanced from defending a line item to choosing among observable arrangements.

7.3.3 — Sovereign Bargaining Weakness

Sovereign bargaining weakness appears when financing or restructuring is needed under conditions that make refusal unusually costly. Currency pressure, import dependence, food and energy exposure, capital flight, disaster, war, commodity shocks, weak reserves, and fragmented creditors can each narrow negotiating room. The analysis should identify the mechanism and counterparty (Guzman, Ocampo, and Stiglitz, 2016; World Bank, 2025d).

Delay in sovereign restructuring can interrupt medicine supplies, salaries, social protection and schooling, leaving ordinary residents to carry costs far from the negotiation table.

Credible public-debt records link each instrument's debtor entity, creditor, currency, maturity, interest structure, guarantees, governing law, priority, contingent exposure, and payment profile to constrained services and investment. The World Bank's 2025 Radical Debt Transparency report found loan-level disclosure of newly contracted debt in only 25 percent of low-income countries; its 2026 Debt Reporting Heat Map evaluates instrument and sector coverage, recently contracted external loans, and contingent liabilities. Borrowers and creditors share responsibility for accurate, reconcilable terms (IMF and World Bank, 2025; World Bank, 2025d; World Bank, 2026).

7.4 — Restructuring, Relief, and Restored Options

A humane reset begins with a comparison of futures. Continued enforcement may destroy more repayment capacity, stability, and human capability than it protects, while indiscriminate cancellation can hide legitimate creditor and public interests. Relief is justified when transparent restructuring offers a less destructive and more durable route after enforcement cost, relief cost, incidence, public trust, and future stability are compared (Dobbie and Song, 2015; World Bank, 2021a). When evidence is incomplete or contested, the process should identify what remains unresolved and preserve temporary protection where delay itself could cause irreversible harm. Measure debt repair through restored stability and agency as well as account status, while keeping creditor incidence, later access to finance, and administrative legitimacy visible (Dobbie and Song, 2015; Dobbie, Goldsmith-Pinkham, and Yang, 2017).

7.4.1 — Partial, Full, and Structured Relief

Debt relief is a family of instruments: deferral, interest or fee reduction, refinancing, settlement, principal reduction, moratorium, bankruptcy discharge, municipal restructuring, sovereign reprofiling, or fuller reset. Consumer, municipal, and sovereign processes operate under different law. International sovereign-debt frameworks have nevertheless emphasized good faith, transparency, impartiality, equitable treatment, sustainability, and workable majority processes as relevant disciplines (World Bank, 2021a; IMF, 2026a; UNCTAD, 2012; United Nations General Assembly, 2015).

Before committing relief funds, identify the constraint the chosen instrument is expected to change. Distinguish the face-value balance from payments the debtor would otherwise make, potential creditor recovery, collection pressure, credit-report

consequences where applicable, and continuing access to essential care. Predeclare when each channel should respond and how new arrears will be tracked. Stopping collection notices may produce different outcomes from reducing a bill before treatment, even if both remove the same nominal balance. Compare usable relief, administrative cost, and burdens shifted to creditors or future users without adding overlapping estimates.

Partial relief may work when the debt is valid and the borrower can repay under fairer terms. A lower interest rate, longer term, fee removal, or temporary pause can convert crisis into manageable coordination.

Full relief may be necessary when repayment would consume the future without realistic benefit. Medical debt, fraudulently imposed debt, disaster-related debt, debt attached to predatory terms, or debts beyond any credible repayment path can require stronger reset.

Medical-debt studies test different forms of relief at different points in the care and collection process. In two randomized experiments, Kluender et al. (2025) relieved medical debt with a face value of $169 million for 83,401 people. One experiment acquired unpaid hospital debt as it was about to enter third-party collections; the other acquired older debt already held by collectors. Credit-report outcomes were examined across both experiments. The health, healthcare-use, and financial-wellness survey reached 2,888 respondents from the hospital-debt experiment, a 19.4 percent response rate among those invited. Relief modestly improved credit access where reporting would otherwise have continued. In the hospital-debt experiment, it also reduced repayment of other existing medical bills and increased the likelihood of another bill entering collections. The survey did not detect average improvements in its health, utilization, or financial-wellness outcomes. In a separate study using an income-

eligibility threshold at Kaiser Permanente, Adams et al. (2022) estimated that financial assistance increased care use and the detection and management of treatment-sensitive conditions near that threshold; utilization effects dissipated after three quarters. These were different interventions and populations, not a head-to-head timing experiment. Together they motivate a sharper question: which combination of debt repair and continuing access addresses the constraint still active in the household's life?

7.4.2 — Creditor and Public Burden

Relief shifts losses as well as reopening options. Compare that incidence with continued enforcement, including effects on financial stability, essential services and future access to finance (Dobbie and Song, 2015; World Bank, 2021a; UNCTAD, 2012; United Nations General Assembly, 2015).

A predatory lender, a local supplier, a pension fund, a hospital, a public agency, a landlord, a foreign bondholder, and a family member occupy different ethical and institutional positions.

The public incidence of relief depends partly on where the claim sits. A municipality might owe a large bondholder and a small contractor who has already completed a repair. A payment delay of the same length could affect them differently. The contractor may need the receipt to pay the crew that did the work; the bondholder may hold the claim as one part of a much larger portfolio. Understanding that difference improves the design of an adjustment without deciding legal priority by sympathy.

A negotiated arrangement can distinguish the claim being changed from the service that still needs to continue. A reduced obligation on completed work and a new contract for future maintenance are separate decisions. Combining them in one unexplained payment can make it hard to tell whether the creditor accepted a loss or was compensated through later terms.

Recording each decision separately makes the agreement easier for all parties to evaluate.

The same clarity helps when public money supports relief. The account can show which obligation was reduced, who supplied the resources, and what continuing service was purchased. Residents can then see the public contribution without treating every payment to a former creditor as evidence that no relief occurred. A settlement is more useful when its parts remain intelligible after the negotiating team has left.

7.4.3 — Fraud and Abuse Safeguards

Relief safeguards should distinguish error, misunderstanding, changed circumstances, opportunistic misuse and organized fraud (World Bank, 2021a; CFPB, 2025). Use narrow screening to flag anomalies, human review to distinguish incomplete records from changed circumstances, and specialized investigation for evidence of organized fraud. Apply Section 9.3.2's notice, correction, and appeal safeguards, with defined authority, evidence thresholds, and response clocks at each tier; keep risk flags out of unrelated decisions.

Frontline service staff should not become investigators in every case, and applicants should not carry an anti-fraud campaign through repeated forms or delays. When a debt, eligibility decision, or fraud flag is wrong, harmful action should pause while evidence is disclosed, reviewed, corrected, and repaired. The institution should then find similar cases and examine why its controls failed.

7.5 — Financing Without New Chains

Fresh-start financing needs a noncontrol covenant as concrete as its financial terms (Sen, 1999; Nussbaum, 2011). Chains often survive under softened names. Loyalty becomes community, unpaid labor becomes service, residence restriction becomes continuity, public confession becomes storytelling, religious pressure becomes alignment, and data extraction becomes evaluation. The practical test asks what must be said, done, disclosed, believed, or surrendered to keep support, then ensures that refusal does not trigger retaliation, public shaming, or loss of unrelated essentials. Each possible chain needs a written prohibition, confidential reporting, independent review, and a remedy that does not depend on the benefactor.

7.5.1 — Relief That Restores Choice

A labor chain forms when relief is exchanged for work under conditions the person would not freely choose. Work connected to assistance must be voluntary, fairly compensated, safe, portable, protected by ordinary labor standards, and separable from relief already granted, identity, membership, belief and public gratitude. A person who would lose housing, food, legal status, or debt relief by refusing a placement may not have meaningful bargaining power.

In a hypothetical fresh-start program, a person receives a written decision settling a specified debt. The same organization also offers a paid apprenticeship. Treating those as separate decisions makes the opportunity easier to understand: the debt settlement has its own terms, and the apprenticeship has a wage and a defined course of work.

The person can compare the apprenticeship with a job elsewhere. Declining the offer leaves the completed relief decision in place. Accepting it creates an ordinary employment

relationship whose conditions can be examined on their own merits. The program can take pride in a participant choosing a different employer because that choice shows the additional room the relief was intended to create.

Separation also clarifies the program's account. The cost of settling debt belongs to the relief activity. Wages and supervision belong to the training activity. Keeping those costs visible allows reviewers to ask whether the apprenticeship teaches a useful skill without treating the participant's earlier need as a source of cheap labor.

7.5.2 — Relief Without Loyalty Purchase

A loyalty chain forms when relief creates political, institutional, personal, or social obedience. The recipient may learn that criticism endangers future support, disagreement is disloyal, or the helper now deserves a privileged claim on later choices (Reich, 2018; Nussbaum, 2011).

The correct substitute for loyalty is accountability. The recipient may have ordinary obligations attached to the relief: truthful application, lawful conduct, repayment if structured, reporting limited to the program's purpose, and correction of errors. None of these requires personal or ideological allegiance.

A practical test is whether two otherwise eligible recipients receive the same financial terms when one agrees to publicity and the other declines. The test should also cover a recipient who criticizes the program. Any difference must have a stated, relevant basis that can be challenged without going through the donor.

Before support begins, protect recipients' freedom of association, political independence and exit.

7.5.3 — Exit With Dignity

Exit with dignity is the final anti-capture test. A program may begin well and still fail if leaving requires shame, lost records, retaliation, public explanation, spiritual blame, disproportionate financial penalty, social exile, or collapse of unrelated support (Sen, 1999; Nussbaum, 2011).

Dignified exit requires portable records, clear closing procedures, no retaliation, privacy protection, and continuity planning where possible. A person should know what obligations remain, what support ends, what records are kept, how to correct errors, and how to complain after exit.

Lawful repayment obligations, misuse of funds, or fraud remain subject to proportionate, fair process after exit.

7.6 — Trauma, Crime, and Social Breakdown

Debt and precarity can compress response time, remove safe exits, and increase exposure to immediate loss. This section asks how those pressures affect safety, treatment, accountability, and recovery, and which supports reduce exposure before and after harm while keeping individual responsibility specific (Mani et al., 2013; Haushofer and Fehr, 2014; Ridley et al., 2020). A credible analysis holds three truths together: poverty is not criminality, victims and harms remain real, and conduct occurs within housing, debt, trauma, addiction, school, neighborhood, market, and service conditions. Context can identify risk pathways without assigning collective blame or dissolving individual responsibility. Prevention, treatment, protection, accountability, and repair then become distinct tasks rather than one moral verdict. Population-level risk pathways identify prevention and support needs; they do not identify which individual will harm another person (Haushofer and Fehr, 2014; Ridley et al., 2020).

7.6.1 — Trauma and Decision Narrowing

Trauma can narrow perceived options even when formal options remain. Violence, neglect, coercion, sudden loss, repeated humiliation, or chronic instability may make danger more salient and routes requiring trust, calm, or belief in the future harder to see or use (Shonkoff et al., 2012; World Health Organization, 2013).

The effects of trauma on attention, risk perception, emotional regulation, trust and decision horizons vary across people and situations; assessing their relevance requires appropriate clinical judgment and empirical evidence.

Review what was foreseeable, what control remained, what support or treatment was absent, who was endangered, and which response can reduce future harm.

Victims remain central throughout that inquiry. Explanation should never demand that they minimize harm, forgive, wait, or accept another person's history as compensation. Context improves prevention and proportional response only when safety, recognition, care, and remedy remain fully visible.

7.6.2 — What Clinical and Criminological Evidence Can Establish

Clinical, criminological, legal, and victim-services review is essential for this terrain. Trauma, addiction, youth development, domestic violence, poverty, deterrence, restorative practice, courts, policing, incarceration, reentry, and victim support each carry specialized evidence and disputed causal claims. The chapter maps material burden and institutional responsibility; diagnosis, adjudication, treatment planning, and jurisdiction-specific design remain with qualified specialists (Sampson, 2012; Western, 2018).

Long follow-up can reveal possibilities that a short trial misses. Blattman et al. (2023) report outcomes a decade after 999 high-risk men in Liberia were randomized to eight weeks of nonspecialist-led cognitive behavioral therapy, $200 cash, both, or neither. Antisocial behavior was about 0.20 standard deviations lower with therapy alone and 0.25 lower with therapy plus cash than in the control group. These standardized index effects are not percentage reductions in crime, and their difference does not by itself establish that adding cash outperformed therapy. A new application should specify whom it serves, how trained practitioners deliver it, and how outcomes are followed without substituting a group average for individual assessment.

Victim-advocacy review tests whether harmed people remain subjects of the argument rather than evidence for someone else's theory. Safety, privacy, recognition, compensation, voice, and

freedom from renewed contact may point in different directions across cases.

Poverty and social-work reviews test for stigma, class suspicion, savior language, and policy confidence unsupported by operational reality. Benefits administration, housing instability, disability, domestic safety, caregiving, immigration status, and family complexity can change what an apparently simple intervention does. Service design must therefore preserve local knowledge, refusal, appeal, and practitioner judgment rather than treating a general material model as complete.

7.7 — Repair, Accountability, and Domestic Order

Some harms require restraint, some restitution, some separation, and some long work. Consequences should be judged by the protection, truth, prevention, rehabilitation, and repair they make possible, as well as by the burdens they impose and the risks they leave behind. Repair before revenge begins with the burdens that have to be carried after harm: safety, medical care, housing, income replacement, legal support, trauma support, child stability, community trust, and future risk reduction. Accountability enters through truthful responsibility and effective protection rather than through the amount of suffering imposed (UNODC, 2020; Sherman and Strang, 2007).

7.7.1 — Community Repair Cost

Community repair cost is the work required after harm spreads beyond the direct parties. Violence, theft, fraud, exploitation, public corruption, and repeated disorder can change how neighbors move, how children play, how businesses operate, how elders feel, and whether people trust institutions (UNODC, 2020; Western, 2018).

The burden includes unpaid work and costs that fall outside a formal repair budget. Families comfort children, neighbors organize watches, teachers manage fear, local businesses add security, tenants move, public workers absorb anger, and community leaders try to restore trust. The burden does not vanish because the court file closes.

A repair-centered order brings those costs into the public account. Lighting, transit, youth services, victim support, housing repair, safe public space, addiction treatment, mediation, economic stabilization, and trusted complaint routes can each restore part of the community’s practical freedom.

Community repair needs funds, staff, safety, and services. Assigning that work to residents without resources transfers the burden again; participation must be supported.

Compare confinement, supervision, treatment, restitution, reentry, and community repair by their functions and costs, including victim safety, due process, family disruption, lost work, public expenditure, and the risk of an ineffective response. Ask which arrangement protects others, makes responsibility effective, and reduces future harm.

7.7.2 — Institutional Responsibility

Institutional responsibility asks what public systems contributed to preventable breakdown. Did schools miss distress, housing systems tolerate dangerous instability, benefits arrive too late, courts delay correction, mental-health services disappear, debt collection intensify crisis, or policing ignore early harm and appear only after escalation? The questions turn recurrent failure into specific duties of prevention and repair (Sampson, 2012; UNODC, 2020).

Accepting institutional responsibility creates budget, staffing, training, data, legal, and leadership obligations. Those duties can give institutions an incentive to locate failure entirely in the individual. But if the same harms recur in the same neglected routes, attributing each failure solely to the individual leaves that recurrence unexplained.

Sanctioning an actor does not repair the policy or service failure that made repetition more likely. Accountability should close each preventable route.

7.8 — Keeping Futures Open

Follow debt relief beyond formal account closure. Record whether the intended payment burden actually changed. Check whether new fees, lost services, or replacement borrowing absorbed the gain before calling the program successful. Keep creditor incidence and correction routes visible without turning a past debt into a permanent label for the borrower.

7.8.1 — Youth Debt and Early Constraint

Youth debt deserves special attention because it can shape a life before adult stability and bargaining power have formed. Student loans, family debt, juvenile fines, medical bills, housing instability, care responsibilities, and high-cost credit may claim future earnings and mobility before a young person has had time to build a buffer (Board of Governors of the Federal Reserve System, 2026; Mezza et al., 2020).

Early debt can make the safest path look like the only path, discouraging experimentation.

Early constraint compounds through sequence. A delayed credential narrows work; unstable work delays housing; unstable housing weakens health, care, mobility, and savings; damaged credit raises the price of the next necessity.

Youth-facing finance should provide clear total-cost disclosure, affordable payment, hardship and discharge routes, limits on high-cost fees, data correction, and education or training paths that do not require untenable borrowing.

7.8.2 — City and State Future Room

Public debt policy should be judged by the options left after service obligations are met and by the reliability of the capacity financed (World Bank, 2025c; IMF, 2026a).

Distinguish the time horizons of pension stress, deferred maintenance, tax-base erosion and refinancing pressure rather than treating every high debt ratio as capture.

At public scale, refinancing should be traced through to the maintenance and services it is intended to support, before later liabilities absorb the released capacity. Where arrears, service interruptions, or deferred maintenance motivated restructuring, monitor whether they actually fall.

Part III — The Domestic Floor

Peace becomes credible inside the ordinary systems on which households rely. Growth, investment, productivity, trade, institutional trust, and national security matter, but their value remains incomplete until they reach work, wages, time, care, access, housing, health, education, maintenance, and reliable public service. The domestic floor is where broad economic order becomes dinner, sleep, shelter, medicine, learning, recovery, and a future people can plan.

Work begins the Part because labor converts finite lifetime into shared production and household security. Wages and schedules determine whether people can plan meals, arrange childcare, respond to illness, sleep, study, care for elders, recover from injury, refuse abuse, and imagine a future. Employment supports peace when contribution is matched by enough income, predictability, safety, care, and recovery to leave a life beyond the shift (Addati et al., 2018; International Labour Organization, 2026). Work can carry dignity without making every work condition dignified. Unsafe, degrading, coerced, underpaid, surveillance-intensive, or family-destroying arrangements may still produce goods and services. Respect belongs to the worker; scrutiny belongs to the system.

Access follows work because income alone cannot overcome a system that makes people wait beyond their margin. Queues, documentation, digital gates, absent human routes, appeal delay, and fragmented life-event services can turn help into a second job. Present-care design asks whether the first responsible door opens early enough to preserve health, housing, income, legal position, and trust before damage becomes harder and more expensive to repair.

Housing, health, and care are the systems through which calm becomes durable. Shelter provides the place from which people sleep, recover, store medicine, maintain records, and remain

connected. Healthcare brings suffering into contact with competent relief. Disability, aging, mental health, addiction, and family care determine whether dependence is supported or hidden in unpaid exhaustion. Together they form an interdependent infrastructure for planning, participation, and trust (World Health Organization and World Bank, 2025; World Health Organization, 2025b).

Education and generational stability widen the horizon. Children inherit income and debt, but also time, safety, attention, institutional treatment, public repair, stress, language, expectation, and the reliability of the systems around them. Education becomes peace policy when it develops durable capability, craft, judgment, discovery, and useful skill across different routes. A society that sorts successfully while childhood conditions deteriorate has measured position more carefully than possibility.

Maintenance and public capability close the Part because ordinary freedom rests on systems that keep working after attention moves elsewhere. Roads, water, sanitation, transit, energy, records, response times, sidewalks, parks, digital access, public counters, and repair crews form the quiet architecture of trust. Reliability gives households room to plan; erratic failure forces them to become private emergency managers (Hallegatte, Rentschler, and Rozenberg, 2019; ISO, 2024a).

Chapter 8 — Work, Wages, and Household Time

Work is where economic abstraction becomes finite human time. Every job draws on hours that could otherwise hold sleep, care, study, recovery, civic life, friendship, worship, exercise, imagination, or silence. A serious account therefore asks what work produces, what it pays, which risks it carries, and what remains after the workday has made its claim. The household is where those answers become visible.

Praise for essential work should be tested against pay, safety, predictable time, care support, and voice. A celebrated workforce can still lack household security. A worker offered a higher hourly rate with no assurance of next week’s hours faces a different choice from one offered a smaller rate increase on a dependable schedule. The chapter follows such differences through the household account. A nominal improvement should be read alongside the commitments the worker can now make, and the risks the worker must still carry.

A wage is a planning instrument. It determines whether rent, food, transport, utilities, medicine, childcare, debt, rest, and a modest emergency reserve can coexist without recurring panic. Legal compliance alone cannot show whether compensation supports material peace. The anti-despair test asks whether pay, realized hours, benefits, essential costs, and risk leave a household able to form a future.

Care shortages and underpaid care work constrain employment and shift burdens into households, often concentrating them among women, migrants, racialized workers, disabled people, and low-wage families (Addati et al., 2018).

Retraining is valuable when it connects a worker to a real destination. A course cannot carry the transition if nearby jobs are absent, credentials are ignored, time and childcare are unaffordable, disability access is missing, age discrimination persists, or relocation destroys the support on which the

household depends. Effective programs connect curriculum, employers, income support, placement, regional strategy, and follow-up. Enrollment is the beginning of a route, not evidence that the route works.

Worker voice supplies operational knowledge that management data often miss. Schedules, tools, routes, customer conflict, physical risk, fatigue, care conflict, informal workarounds, and retaliation fear are known most clearly by the people who encounter them. Unions, councils, safety committees, surveys, grievance systems, and direct participation in transition planning can convert that knowledge into correction. Consultation becomes meaningful when it can change the design.

Headline employment can remain resilient while work is materially insecure. The International Labour Organization projected global unemployment at 4.9 percent in 2026, about 186 million people, alongside a broader jobs gap of roughly 408 million and approximately 2.1 billion workers in informal employment. These aggregates describe neither every labor market nor every worker’s condition, but they show why job counts should be read beside protection, predictability, bargaining power, regional opportunity, and work quality (International Labour Organization, 2026; OECD, 2026c).

8.1 — Work, Dignity, and Extraction

Work can bring skill, care, strength, patience, imagination, and service into shared life. Assess those contributions alongside the demands imposed on workers (Kalleberg, 2011; Sen, 1999). Dignity does not depend on employment or output. People who cannot work, should not work, or contribute outside employment retain full worth and must be included when judging a labor system.

8.1.1 — Contribution, Pay, and Material Recognition

Material recognition appears in pay, safety, staffing, voice and time to live outside work (Addati et al., 2018; Nussbaum, 2011).

Contribution also includes work that markets often underprice. Care, maintenance, sanitation, translation, accessibility support, administration, local repair, and emotional labor can hold public systems together while prestige and compensation flow elsewhere.

Extraction often arrives with hopeful language. A worker is told the job offers flexibility, exposure, entrepreneurship, passion, mission, training, family culture, or future opportunity. Sometimes those words describe real benefits. Sometimes they conceal risk transfer, underpayment, unpaid labor, unstable hours, missing benefits, unsafe conditions, or no credible path upward.

If an employer or platform keeps the upside while the worker carries demand risk, equipment cost, injury risk, downtime, customer conflict, algorithmic change, or legal uncertainty, the arrangement has shifted cost under the name of opportunity.

Immigration status can intensify this extraction. A worker who fears deportation, blacklisting, document exposure, wage retaliation, or language-based exclusion may carry the cost of silence.

8.1.2 — Meaningful Work Without Coercion

Paid work can offer something that an equal cash transfer does not. Hussam et al. (2022) studied 745 Rohingya adults in Bangladesh, randomly assigning residential blocks to an eight-week employment opportunity, cash payments of equal average value, or a comparison condition. Employment improved measured psychosocial well-being more than cash alone. The comparison identifies a benefit of the offered work package beyond its income component in that setting; it does not separate the contributions of routine, social contact, recognition, and activity. A useful next study could vary worker choice or social contact while holding pay constant. The aim would be voluntary opportunities that people value, with adequate pay and continuing support for those who do not participate.

Meaning can also be used as a chain. Workers in care, education, public service, religious institutions, public-interest work, the arts, startups, research, and mission-driven organizations are often asked to accept overwork, low pay, unclear boundaries, or emotional exhaustion because the cause is important (Kalleberg, 2011; Nussbaum, 2011). Meaningful work still requires fair wages, safety, predictable time, care support, and the freedom to leave.

8.1.3 — Bad Work as Public Cost

Workplace injury, instability, stress and care burdens can disappear from an employer’s accounts while falling on workers and public systems (Pega et al., 2021; Kalleberg, 2011). A public cost account can record injury rates, turnover, sickness absence, staffing gaps, schedule volatility, compensation claims, retraining need, family care disruption, public-benefit use, and service failures. Trace how work design moves these costs beyond the firm.

Voice must include workers who are easiest to silence. Temporary staff, care workers, migrant workers, disabled workers, domestic workers, warehouse workers, platform workers, retail workers, public-facing service workers, and workers on visas or informal terms often see the real burden but have the least safe access to speak (Addati et al., 2018; Kalleberg, 2011).

8.2 — Wage Floors and Anti-Despair Economics

Wage floors operate within labor markets as well as household budgets (International Labour Organization, 2024b; Cengiz et al., 2019). Wage floors interact with regional costs, firm size, productivity, prices, public support, taxes, benefits, enforcement, migration, sector structure, and informal work. Detailed policy therefore requires current expert review. The material-peace contribution is a clear adequacy test: compensation should be judged against the real costs and risks a household must carry, with adjustment evidence reported across workers, firms, sectors, and places.

8.2.1 — Below-Floor Work

Below-floor work is employment that keeps people active in the labor market while leaving them materially unstable. Employment counts alone cannot establish whether realized earnings sustain housing, food, transport, care, health, and emergency resilience (International Labour Organization, 2024b; Neumark and Shirley, 2022).

Below-floor work can arise through involuntary part-time work, seasonal instability, unpredictable commissions, on-call work without sufficient guarantee, low-paid subcontracting, platform work with hidden costs, and jobs that require workers to buy tools, vehicles, uniforms, insurance, or digital access from wages already too small to carry them.

A worker may receive too few hours, or hours so irregular that a second job, childcare plan, training course, or medical appointment becomes impossible.

8.2.2 — Regional Cost Differences

A wage cannot be fairly judged apart from place. Housing, transport, food, utilities, childcare, healthcare, climate, taxes,

insurance, and local service access vary widely. The same nominal wage can produce breathing room in one region and despair in another.

Regional cost analysis should also avoid humiliating poorer places by treating low prices as proof that low wages are harmless. Low rent may coexist with poor transit, fragile health access, weak schools, dangerous housing, food deserts, low bargaining power, or few alternative employers. Cost analysis must ask what life requires, not only what the cheapest local basket appears to cost.

Regional wage comparisons must account for enforcement as well as prices. Fear of retaliation, immigration vulnerability, legal cost, lack of documentation, arbitration clauses, confusing rules, or weak labor agencies can make violations rational for bad actors.

Domestic and care work make this especially visible. When the workplace is a home, the boundaries of time, task, emotion, and privacy can blur. An economy organized around material dignity cannot rely on intimacy, gratitude, family language, or undocumented vulnerability to make labor cheaper than it is.

8.2.3 — What Wage Evidence Can and Cannot Show

Studies can estimate how a defined wage change affected pay, employment, hours, turnover, prices, firm entry, or hiring within a specified place and period. Those results do not identify one timeless wage adequate for every household, sector, region, or stage of life.

The first discipline is to separate the statutory rate from realized compensation. A posted hourly wage may differ from what a worker receives after unpaid preparation, unstable hours, illegal deductions, unpaid waiting, equipment costs, travel requirements,

benefit loss, or wage theft. Evidence that measures only the legal rate may miss burden carried through the design of the job.

Wage-floor evaluation should specify the counterfactual and the adjustment margins before results are known. Alongside hourly pay, report total earnings, hours, hiring and separations, compliance, prices in affected markets, and the ability of workers with irregular schedules to retain care and transport. Separate short-run adjustment from medium-run changes in productivity or firm organization, and report uncertainty for low-wage subgroups rather than allowing one average to settle the debate. Compare a wage change with plausible complements, such as enforcement against wage theft and better access to childcare, rather than assuming that the minimum wage must carry every burden alone. Evidence of reduced employment in a subgroup should trigger examination and mitigation, not automatic denial of other groups' gains (Cengiz et al., 2019; Neumark and Shirley, 2022).

A simple illustration shows why several margins matter. A worker paid $15 for 30 hours receives $450 a week before taxes. At $16 for the same hours, gross pay rises to $480; at $16 for 28 hours, it falls to $448. These are arithmetic scenarios, not estimates of how employers respond to wage policy (International Labour Organization, 2024b; Cengiz et al., 2019; Neumark and Shirley, 2022).

8.3 — Salaries, Severance, and Transition Income

A society committed to material peace cannot ask workers to carry reform alone. Defense conversion and fossil transition may be necessary for reasons that reach beyond one workplace. Automation, public-sector redesign, climate adaptation, institutional repair, and technological change can also disrupt livelihoods. Necessity does not erase the household costs of transition or the duty to make those costs visible (OECD, 2024b; Causa et al., 2024).

8.3.1 — Income Bridges

An income bridge protects the interval between an old livelihood and a viable next one. Its purpose is to prevent a transition driven by firms, governments, technological change, or market shifts from becoming an immediate household emergency.

Design must distinguish temporary displacement from long-term structural loss. Short interruptions may require rapid replacement income and return-to-work coordination. Plant closure, public-sector redesign, defense conversion, climate transition, or automation may require wage insurance, severance, recognized training, relocation support, local investment, or a staged path into another occupation (International Labour Organization, 2015; Causa et al., 2024).

Evaluation should follow outcomes at the system level: time to stable reemployment, earnings recovery, benefit continuity, training completion, regional job creation, involuntary relocation, and household arrears (Causa et al., 2024; OECD, 2026c).

A well-built bridge ends when the next platform can bear weight. That may be a stable job, a viable enterprise, a supported retirement, a recognized caregiving role, or another lawful arrangement that restores planning capacity.

8.3.2 — Defense and Fossil Transition Income

Defense and fossil transitions affect regional tax bases, infrastructure, identities, and skill networks as well as individual jobs.

Workers should not be assigned moral responsibility for sectoral histories they did not design. A machinist, miner, refinery worker, logistics specialist, engineer, maintenance worker, or base employee can be tied to a system whose future must change without having authored its public burden (Causa et al., 2024; Sen, 1999).

Defense conversion and fossil transition should also map skills carefully. Technical maintenance, energy systems, logistics, safety culture, heavy manufacturing, project management, environmental remediation, infrastructure repair, grid work, transit, housing construction, and climate adaptation may all use existing competence if planning is serious.

The same worker protections matter in transitions involving artificial intelligence, which can move decision power away from workers while leaving them accountable for errors, customers, pace, and emotional labor. Review whether workers have training, authority, time, appeal routes, and protection when automated systems change the work beneath them.

Automation can remove dangerous, repetitive, exhausting, or inefficient work. It can also remove income, skill pathways, social status, and bargaining power.

If productivity rises because machines assist workers and the gains support wages, safety, shorter hours, better service, and shared prosperity, the burden may be reduced. If those gains come with greater worker risk, intrusive monitoring, deskilling, or unsupported displacement, efficiency alone does not make the technology fair.

8.3.3 — Public-Sector Redesign Support

Agencies may need better technology, clearer access, shorter queues, stronger review, improved records, fewer duplicated processes, and more humane service design. But public redesign is labor, and its burden often falls on staff before it benefits the public.

A public employee may be asked to learn new systems, handle old and new workflows at once, absorb citizen frustration, correct software errors, protect privacy, explain policy changes, and maintain service while staffing remains thin. If this burden is not counted, reform becomes another hidden pressure (OECD, 2026a; Herd and Moynihan, 2025).

Public-sector redesign also requires support for roles that disappear or change. Tasks can become unnecessary as services improve, but affected workers need funded income bridges, redeployment, training, severance, or retirement support rather than quiet discard.

Reform budgets must include training, staffing, transition time, complaint routes, manual fallback, non-digital access, privacy review, and recovery after implementation failure.

Workers and regions deserve truthful explanation of why change is needed, what harm old systems produced, what support will arrive, and how their skill can help build the replacement (Causa et al., 2024; Sen, 1999).

8.4 — Hours, Schedules, and the Politics of Time

Time is one of the most commonly hidden labor burdens. A wage statement may show hours paid, but it rarely shows last-minute schedule changes, commuting, unpaid availability, recovery after exhaustion, coordination with care, or the household uncertainty created by irregular work (Schneider and Harknett, 2019; Pega et al., 2021). Schedule predictability should include notice, stability, minimum hours where appropriate, real ability to trade shifts, protection from retaliation, and accessible ways to request schedule adjustments for disability, care, school, religious practice, and medical appointments. Algorithmic scheduling deserves close scrutiny. A system can optimize labor costs while scattering worker lives into fragments that no manager has to see. When software makes last-minute changes, cancels hours, clusters shifts badly, or rewards constant availability, the hidden burden is a design choice whose cost has moved into households.

8.4.1 — Last-Minute Work Burden

Last-minute work burden arises when workers are called in, sent home, held late, rescheduled, or made available with little warning. The hours may appear as labor flexibility, but the burden enters family systems, care arrangements, transport, rest, meals, and emotional stability.

A canceled shift can be as harmful as an added shift. The worker may lose expected pay after arranging childcare, transport, or other commitments. An added shift can produce overtime, exhaustion, missed care, or family strain. Both forms of instability should be counted (Schneider and Harknett, 2019; Pega et al., 2021).

Responsible scheduling can also improve an employer's operations. In a nine-month randomized field experiment in 28 Gap stores, Kesavan et al. (2022) tested a package designed to

improve schedule consistency, predictability, adequacy, and employee control. Store productivity rose by an estimated 5.1 percent, with sales up 3.3 percent and labor use down 1.8 percent. These are results for the tested package, not isolated effects of advance notice. They challenge the assumption that predictable work must be purchased entirely through lower performance. The next comparison should follow weekly earnings and household coordination alongside store results: lower labor use and better productivity do not, by themselves, establish how workers share the gain.

Last-minute work also redistributes burden to people who are not on the payroll. Grandparents, partners, older siblings, neighbors, informal caregivers, and children may absorb the employer's scheduling uncertainty.

8.4.2 — Overwork and Health Cost

Excessive hours, insufficient rest, chronic stress, understaffing, and permanent availability affect sleep, injury risk, mental health, caregiving, judgment, and long-term capacity.

Accountability begins by asking who benefits from overwork and who later pays. A firm, agency, hospital, platform, school, or public institution may save staffing cost by relying on overwork, while the worker, family, healthcare system, and public services absorb the consequences (Schneider and Harknett, 2019; Nussbaum, 2011).

Overwork can be voluntary in language and compulsory in structure. A worker may choose overtime because base wages are too low, staffing is too thin, professional culture rewards self-erasure, debt requires extra hours, or a visa, promotion path, or contract renewal depends on compliance.

Workload and support offer distinct opportunities for improvement. Across three occupational groups, Demerouti et al.

(2001) found job demands principally associated with exhaustion and lack of job resources associated with disengagement. Their model used reported and observed working conditions. The result guides a testable design distinction rather than establishing the causal effect of a staffing reform: reducing excessive demands and improving support may address different parts of the problem. A workplace can examine both, then measure whether employees gain sustainable capacity alongside better service.

Remote and hybrid work can reduce commute burden for some workers, but not for all. Care workers, drivers, cleaners, manufacturing workers, health staff, public-facing service workers, construction workers, and many others cannot move their labor into the home.

Rest should be designed into work through staffing, leave, predictable shifts, protected breaks, reasonable overtime limits, realistic allowance for commuting, and the right to disconnect where appropriate. The detailed law varies by jurisdiction and sector.

8.4.3 — Time Sovereignty and Dignity

Time sovereignty is enough control over time to plan a life amid the demands of work, care, public systems, debt, and other institutions.

Time sovereignty also protects civic life. A worker may be unable to attend a union meeting or testify at a hearing despite having a formal right to participate.

Coordination is unavoidable in hospitals, transport, care, farms, emergency services, logistics, manufacturing, restaurants, security, and public works. The relevant distinction is whether coordination is jointly designed, foreseeable, and compensated, or whether uncertainty is transferred almost entirely to those with the least bargaining power. Workers need not control every hour

to retain meaningful authority over sleep, care, study, recovery, and family life (Schneider and Harknett, 2019; Nussbaum, 2011).

8.5 — Childcare, Eldercare, and Disability Support

Care determines whether a work schedule is actually usable. Childcare, eldercare, disability support, family caregiving, and the worker's own health needs shape attendance, hours, mobility, training, and recovery. This section therefore treats care capacity as part of employment design and asks when employers, public systems, and households are each carrying costs that should be made explicit (Addati et al., 2018; Nussbaum, 2011). A material account of care must follow both sides of the relationship. It asks whether the person receiving care has safety, continuity, agency, privacy, access, and genuine choice. It also asks whether paid and unpaid caregivers have wages, training, rest, support, and a life beyond constant readiness.

8.5.1 — Childcare Access and Labor Access

Childcare functions as public capacity as well as private family choice. When it is unavailable, unreliable, unaffordable, physically or developmentally unsafe, or incompatible with work schedules, the burden appears as lost income, reduced hours, career interruption, family stress, child instability, and gendered inequality. Those consequences travel beyond the household into labor supply, public revenue, health, and long-term opportunity (Addati et al., 2018; Sen, 1999).

The test also needs schedule fit. A parent working evenings, weekends, rotating shifts, emergency shifts, or multiple jobs may be excluded by childcare systems designed around standard office time.

8.5.2 — Eldercare and Disability Support

Dependence is an ordinary condition of life. Disability support should remove avoidable barriers to participation while

preserving agency and privacy (World Health Organization, 2022a; Nussbaum, 2011).

Aging households often require coordination among paid work, medical appointments, bathing, food, medication, mobility, paperwork, transport, supervision, and emotional presence. When eldercare is unsupported, adult children, spouses, neighbors, and low-paid care workers become the hidden infrastructure of the economy.

Caregiver support requires respite, predictable services, workplace protection, income support where appropriate, accessible information, training, transport, and a reliable route for urgent help.

Care should never be assigned by gender destiny or treated as lesser ambition. Women, men, nonbinary people, grandparents, siblings, disabled people, adult children, and chosen family members carry care in different settings. The material test is whether care is supported, shared, chosen where possible, and compatible with the caregiver's own health, income, rest, and future.

8.5.3 — Care Worker Wages

Care workers perform skilled, relational, physically and emotionally demanding labor. Low pay can make it difficult to attract and retain skilled caregivers. Unstable hours, understaffing, injury risk, migration dependence, and weak progression can further destabilize the workforce and the people who rely on it. Care quality and care worker conditions therefore belong in the same account (Addati et al., 2018; World Health Organization, 2022a; Sen, 1999).

Workload, insurance, training, and dependence on multiple jobs also shape continuity (Addati et al., 2018; Sen, 1999).

A paid care hour and an hour spent directly with a recipient are not always the same unit. In a simple planning example, a worker has six paid hours available for a set of home visits. Necessary journeys take one hour and preparation and handover take another. Four hours remain for visits. A budget that prices all six hours as direct contact has promised more contact than the stated working day can supply.

Travel can sometimes be reduced through a better route. Preparation may become easier when reliable information is available before a visit. Those improvements should preserve the recipient’s agreed time and the worker’s opportunity to make a sound handover. Removing a line from a schedule is a real efficiency only when the task it represented is no longer needed or has genuinely become easier.

This distinction creates a clearer conversation about pay. The worker’s contribution includes the work that makes a visit possible and continuous with the next one. A provider can explain the relationship between paid hours and contact hours rather than presenting coordination as an unexplained surcharge. Recipients gain a more dependable promise because the whole working day, not only its visible encounters, has been funded.

The example also shows why comparing providers requires a common service definition. Four hours of planned contact with supported handovers is different from six hours that can be achieved only by leaving travel or preparation outside the account. The comparison should follow the care being delivered rather than reward the account that leaves the most work unnamed.

Count care needs without requiring intimate proof of love or family conflict, protecting the privacy of both caregiver and recipient under Section 4.4.

8.6 — Retraining and Mobility

Retraining only improves a transition when the new skill has a realistic path into useful work. Course completion, a certificate, and a job with stable hours are different outcomes, and public programs should report them separately. Geographic constraints can defeat a good curriculum when employers, housing, transport, or caregiving arrangements do not support the move. Workers also need a voice in identifying which skills are transferable and which promises are not credible. This section connects the design of training to regional capacity and the terms on which affected people can participate.

8.6.1 — Retraining That Leads Somewhere

A stronger model of retraining connects learning to employers that use and reward the acquired skill. Katz et al. (2022) examine randomized evaluations of sector-focused programs that combine occupational preparation with screening and practical support. Their analysis of WorkAdvance points toward access to higher-wage jobs and sectors, rather than employment alone, as an important source of earnings gains. The evidence concerns a program package, not the independent effect of a certificate. That gives transition planning a concrete question: which employer-recognized skills and supports open a better job pathway? Evaluation should retain the assigned comparison group and follow everyone offered the program, including noncompleters, so successful graduates do not stand in for all those asked to retrain.

Telling a worker to retrain for a distant occupation may amount to telling a family to leave its home, support network, school, or caregiving arrangement. Mobility support can expand choice, but regional investment and local job creation are equally important.

Evaluation should distinguish initial placement from durable placement. A job obtained for several weeks is not equivalent to a stable role with usable hours, adequate pay, safety, progression, and benefits. Follow-up should examine whether the new path restored household planning capacity and whether the occupation remains viable after the program's publicity has ended (Causa et al., 2024; OECD, 2026c).

A training contract also needs a serious alternative at the same cost. McIntosh and Zeitlin (2022) randomized workforce training and cash transfers among poor, underemployed young adults in Rwanda. Varying the transfer amounts made cost-equivalent comparisons possible. Training improved several outcomes, but cash performed better on a number of economic measures, while training performed better on business knowledge. The comparison asks what the program adds beyond the choices participants could make with equivalent resources.

A follow-up about three and a half years after treatment found continuing gains from both approaches in different outcomes, including productive time from training and assets or savings from cash (McIntosh and Zeitlin, 2026).

8.6.2 — Regional Transition Capacity

Regional transition depends on coordinated training, employers, public services, local finance, infrastructure, and trusted institutions. The 2026 OECD employment outlook emphasizes how strongly geography still shapes access to jobs and disposable income. Transition policy therefore has to build opportunity in place as well as support people who choose to move (Causa et al., 2024; OECD, 2026c; Sen, 1999).

Imagine a region whose largest employer is reducing production. A local college begins by examining the work people already know how to do. Precision measurement may matter in

several industries. Equipment maintenance may travel farther than familiarity with one particular product. Mapping those tasks with workers and prospective employers gives the college a better starting point than naming an entirely new occupation and treating everyone as a beginner.

Several smaller employers could share a training workshop that none could sustain alone. Each would identify the practices required for its own work, while the common facility would teach the skills they actually share. The region could preserve useful equipment and experienced instructors even as the mix of employers changed.

The outcome would be a network of destinations rather than a single replacement employer. A worker who learns a transferable maintenance task could compare several roles without leaving the region. Employers would gain a place to develop further skills. Regional renewal becomes more plausible when the next opportunity grows from knowledge already present and leaves room for several different futures.

8.6.3 — Worker Voice in Transition

Worker voice is planning intelligence, not ceremonial consultation. People closest to the work can expose training for nonexistent jobs, unsuitable equipment, overlooked safety hazards, unrealistic timelines, and household burdens that remote models miss. Their knowledge does not replace broader evidence, but plans that exclude it are more likely to fail after public money and private lives have already been committed (Causa et al., 2024; OECD, 2026c).

Voice also protects against elite imagination. Designers may picture mobile, credentialed, healthy workers with savings and flexible households. Actual workers may be caregivers, disabled, older, undocumented, regionally rooted, facing language barriers,

traumatized by prior layoffs, or tied to housing that cannot be sold.

New industry requires land, finance, utilities, skills, procurement, and time. Worker support requires budgets, responsible institutions, eligibility, appeal, and continuity.

8.7 — Stabilization as Freedom Infrastructure

Stabilization matters because shocks often arrive in clusters rather than isolation. A lost shift can collide with care, transport, rent, or illness, turning a manageable disturbance into forced choices. Freedom infrastructure gives households enough continuity that one disruption does not close several future paths at once (Sen, 1999; Nussbaum, 2011). Institutional margin appears when systems tolerate ordinary disruption. Benefits do not vanish after one missed notice; treatment does not collapse after one administrative error; schools can respond to a family crisis; employers can accommodate a temporary care need; and appeals can correct mistakes before damage compounds.

8.7.1 — Reduced Panic as a Shared Benefit

Household stability may improve school, work, care, and service outcomes; the scale and causal routes of these spillovers require local testing.

To test whether the domestic floor works jointly, follow a worker through an involuntary schedule change. Does the employer supply notice; can the worker keep a medical appointment; is childcare still available; does income support bridge an actual pay gap; and can a failed handoff be corrected before rent is due? Compare the total time and monetary burden across these systems with the previous route, rather than celebrating each program separately. A wage rise that coincides with loss of predictable hours or childcare access may leave usable freedom unchanged. Linked outcomes, recorded without unnecessary household surveillance, are a stronger test of work-centered peace than employer payroll averages alone.

The public value of reduced panic can also be assessed through fewer emergency escalations, shorter recovery, lower staff rework and greater use of preventive services. No single measure

captures inner life, but several system indicators can show whether institutions are producing fewer avoidable emergencies.

8.7.2 — Making a Better Opportunity Reachable

A job offer can become a genuine option only when the interval between jobs is workable. Imagine a worker whose new role pays more but whose first payment arrives three weeks after the old role ends. The annual comparison favors the move. The household still needs to meet obligations during the interval. A clearly dated income bridge can make the improvement reachable without changing the wage attached to either job.

The bridge should match the actual sequence. An advance on earnings creates a later repayment obligation; a grant does not. Continued support may depend on the date the new earnings arrive rather than the date the offer is accepted. Making those distinctions visible lets the worker compare the move with staying put. It also lets the provider budget for the interval it is actually supporting.

The same reasoning applies to a care arrangement that changes with the job. A new shift may require an introductory visit before care starts. That visit consumes time before the improved schedule becomes available. Supporting the handover can be more useful than adding another promise about the eventual arrangement. The transition becomes feasible when the old support lasts until the new support is usable.

This is how stabilization can expand choice rather than simply preserve the current position. A household gains room to accept a worthwhile change because its essential arrangements can cross the interval with it. The institution has helped make a better future selectable in the present.

Chapter 9 — Queues, Capacity, and Care Now

A queue may appear orderly from the office. From the household, life continues while the request waits: rent is due, pain changes, school continues, and legal or financial clocks may expire. Imagine two requests that enter on the same morning. One concerns a routine document; the other needs that document before a fixed housing deadline. Their elapsed waiting time may be identical while the consequences diverge. The chapter examines what a queue must know to respond to that difference without making the person's ability to insist the deciding factor.

Some queues protect safety, expertise, lawful review, or scarce capacity. Their legitimacy depends on necessity, proportionality, explanation, support, revision, and distribution. A defensible system can show why sequence exists, how urgency is recognized, what interim protection is available, and how people challenge error. Waiting earns legitimacy only when it performs a real public function rather than concealing undercapacity or friction.

Hidden rationing occurs when help is formally available but practically unreachable. Repeated documents, uncertain status, missed notices, narrow office hours, digital-only access, unaffordable travel, and unresolved errors turn persistence into the admission price. People with time, money, devices, fluency, confidence, and professional help are then more likely to complete the route, while exhaustion can be misread as indifference or fraud.

Present-care access offers a bounded first promise: recognize urgency, preserve the person's place, reduce foreseeable deterioration, explain the next step, and connect the case to a responsible human pathway. Full adjudication may still require evidence and expertise. The gain is that uncertainty no longer means abandonment while health, housing, income, or safety continues to deteriorate.

Frontline workers often carry the other side of the same failure. Clerks, nurses, social workers, call-center staff, interpreters, eligibility reviewers, housing officers, and case managers may be trapped inside fragmented systems with impossible caseloads and little authority to repair obvious problems. A humane access design protects the public without turning workers into shock absorbers for underfunding, broken software, contradictory rules, or leadership decisions they did not make.

Digital, telephone, in-person, and assisted routes can share a common case without requiring identical interactions. Each must provide lawful, comprehensible, timely, humane access, without penalties for the channel the person's circumstances require. A queue's entry timestamp should survive a change of channel, so someone moving from the portal to telephone assistance does not silently lose their place.

Interface repair cannot substitute for real capacity. A clean portal attached to too few clinicians, caseworkers, inspectors, judges, or repair crews makes scarcity easier to view without making service more available. The chapter therefore joins process simplification to staffing, training, scheduling, maintenance, and human escalation. Digital tools expand reach when they remain accessible, correctable, and answerable (OECD, 2026a; W3C, 2023).

9.1 — Queue Burden as Hidden Suffering

The burden of a given wait also depends on the resources available to absorb it. One person has savings, flexible work, stable housing, private transport, and a professional advocate. Another has food for three days, an expiring prescription, no paid leave, and a landlord who will not wait. Queue burden is examined across six dimensions without combining them into a score. Time, health, income, dignity, inequality, and material repair remain analytically distinct because each has different evidence, mechanisms, and remedies. Their interaction matters, but interaction does not create a natural common unit.

9.1.1 — Waiting as Time Loss

On average across ten surveyed countries, 18 percent of respondents waited more than a week for primary care and 52 percent waited at least a month for a specialist appointment. These averages describe access conditions, not any individual's burden (OECD, 2025c; Little, 1961; Green, 2006).

Queue analysis should distinguish at least three forms of loss. Passive waiting is elapsed time before service. Active waiting is the labor of calling, traveling, uploading, gathering documents, checking status, and correcting errors. Uncertain waiting adds vigilance: remaining available for a call, avoiding travel, postponing work, retaining temporary papers, or repeatedly testing whether a case still exists. These burdens can accumulate even when the official record contains only an opening date and a completion date. The analysis should also distinguish one uninterrupted wait from recurrent episodes. Ten contacts of forty minutes each may impose more planning damage than one scheduled appointment of the same total duration because each contact breaks concentration, work, care, or sleep. A complete queue account therefore records elapsed time, active user-hours,

contact frequency, notice quality, channel changes, and the share of people whose cases exceed consequential deadlines.

Waiting becomes dangerous when the underlying condition deteriorates. Delayed care can allow symptoms to worsen; delayed housing repair can prolong exposure to heat, mold, injury, or inaccessibility; delayed income support can interrupt food, medication, or utilities. Timeliness is therefore part of the service itself, not a convenience added afterward (Institute of Medicine, 2015; OECD, 2025c; Green, 2006).

Basic queueing logic separates the size of a queue from the capacity needed to reduce it. In a stable system, Little's Law states that the average number of cases in the system equals the average arrival rate multiplied by the average time each case remains. If ten cases arrive per working day and the average time in the system is five working days, the average number present is fifty. This illustration assumes consistent units and includes service time as well as waiting. It is an accounting relation, not a prescription for faster service. Separately, if admissions persistently exceed completions and other exits, the number of open cases grows. With variable arrivals and service times, waits can rise sharply as utilization approaches full capacity. Reform therefore needs evidence about demand, available staff, case complexity, urgency, and exits such as abandonment; a disappearing case is not necessarily a completed service (Little, 1961; Green, 2006).

9.1.2 — Waiting as Dignity Cost

Being suspended by a system can make a person feel invisible, doubted, dependent, childish, or powerless.

Dignity cost grows when the system gives no explanation. A person may not know whether they are forgotten, rejected, under

review, missing a document, waiting for a supervisor, or trapped in a backlog (Herd and Moynihan, 2025).

Queue design can expose vulnerability. Waiting in a public office or food line, sitting in a crowded benefits room, or repeating private facts within earshot may reveal needs that someone with money can address privately. Impersonal treatment, such as being addressed only by a case number, can add a separate dignity cost.

Queue inequality can also arise from geography. A person far from an office, clinic, court, school support center, or service hub may pay more in travel, time, and weather exposure. Rural, remote, island, border, and under-served urban communities often carry queue burdens that central systems underestimate.

9.1.3 — Queue Reduction as Peace Work

Earlier service can prevent one deadline from becoming several problems at once: a resolved case may preserve a shift, a prescription, a rent payment, or an appeal that would otherwise be lost. Returned time can therefore protect options as well as hours.

Better staffing, clearer rules, service coordination, triage, digital support, human review, and early relief can all reduce avoidable delay (Institute of Medicine, 2015; Green, 2006). Choose among them by locating the bottleneck. Simplification can release existing capacity; a shortage of trained staff or functioning equipment requires an increase in usable capacity.

Measure queue changes against a common service standard, with a usable appeal and assisted route. Include people who never entered the queue, and identify barriers affecting disabled or otherwise excluded applicants.

Follow the person from first attempt through eligibility, service, correction, appeal and closure. Record shadow queues in telephone lines, document collection, referral systems, unpaid

advocacy, provider search and repeated restart. A referral remains unfinished until the receiving service accepts it.

A useful report includes percentile waits, abandonment, repeat contact, missed deadlines, urgent escalation, assisted completion, correction time, accessibility failures, case complexity, and staff workload. It should identify whether throughput improved because capacity increased, process steps were removed, demand fell, eligibility narrowed, or unresolved cases were transferred elsewhere. Staff turnover, overtime, error, and burnout belong beside output because a short queue produced by exhausting the workforce is not durable (Herd and Moynihan, 2025; Little, 1961). Publish the denominator used for each waiting-time measure so changes in the counted population remain visible.

9.2 — Administrative Burden and Distress Gatekeeping

Measure the attrition that occurs before a formal eligibility decision. A process that reports only grants and denials can hide people who left when instructions conflicted, correction routes failed, or repeated evidence requests became unaffordable. Pre-decision withdrawal is an access outcome, not evidence that no eligible applicant existed (Moynihan, Herd, and Harvey, 2015; Herd and Moynihan, 2018). Distress-gatekeeping occurs when the system makes visible suffering part of the admission test. People may be asked to prove desperation repeatedly, disclose financial, medical, family, or migration details unrelated to the decision, display emotional collapse, or accept suspicion as the price of assistance. The result can be perverse: people who communicate calmly may be judged insufficiently needy, while people in acute distress may be treated as unreliable.

9.2.1 — Documentation Burden

Documentation is necessary when a service must establish identity, need, authority, safety, or lawful eligibility. The problem begins when proof requirements become detached from the decision they are meant to support. A form can be administratively ordinary while consuming days of travel, calls, fees, translation, and repeated explanation.

Section 1.3.2 establishes the learning, compliance, and psychological-cost framework. Under distress, the additional question is whether an evidence request protects against a live risk or merely requires the applicant to bear the institution’s uncertainty (Herd and Moynihan, 2025).

A burden budget can make proof requirements visible before a program launches. Designers can estimate documents, visits, signatures, identity checks, expected corrections, and user hours

for one decision, then distinguish institutional processing time from labor shifted to the applicant. The budget should include travel, fees, translation, childcare, lost work, device and data needs, accessibility support, and the probability that a common error forces restart. It should identify which facts can be verified once and lawfully reused, which must be refreshed, and which should never be collected because they do not change the decision. Pilot testing should include people with limited time, unstable records, disability, language difference, fluctuating conditions, weak connectivity, or no reliable digital access. Proportionality is judged across the whole pathway and against the value and risk of the decision.

A randomized study makes the access question concrete. Finkelstein and Notowidigdo (2019) studied 30,000 older people likely eligible for, but not enrolled in, the Supplemental Nutrition Assistance Program. Over nine months, enrollment was 6 percent in the control group, 11 percent with information alone, and 18 percent with information plus application assistance. The additional enrollees were generally less sick and had higher net income than control-group enrollees. Assistance therefore widened access without automatically reaching those facing the greatest barriers. The next design question is specific: what additional support lets people with heavier needs use the same doorway? Measure the distribution of successful access alongside the headline enrollment gain.

9.2.2 — Trauma Repetition

Trauma repetition occurs when a person must recount harm to different offices, staff members, platforms, forms, reviewers, or appeal bodies. Ordinary information collection can feel like reopening a wound as the price of care. A better design uses a tell-once principle where lawful and safe: a concise trauma summary

remains under the person’s control, records only what the service needs, can be corrected, and moves through a warm handoff rather than unrestricted copying. Later staff should distinguish verified facts, the person’s account, professional interpretation, and unresolved uncertainty. No summary replaces a conversation the person chooses or a genuinely required specialist assessment, but every institutional boundary need not reset the story to zero.

Before intake, decide which facts are necessary, which can be checked or self-attested, and which require later review. Preserve an exception route when the required proof is inaccessible.

People fleeing violence may lack papers. Disabled people may be unable to attend repeated appointments. Migrants may have documents awaiting translation. Disaster survivors may have lost records.

Dignified verification ends with a reasoned record. The decision should show what facts were accepted, what remained uncertain, how uncertainty was handled, and how correction can occur (Herd and Moynihan, 2018).

9.3 — First Access, Fast Relief, and Present-Care Priority

An institution must identify what cannot safely wait. It should state which evidence suffices for provisional action and who can authorize review before the person disappears into a queue. A first response can reduce immediate downside without pretending the full case has been resolved. Urgency, reversibility, proportionality, and accountability should govern each such decision.

9.3.1 — The First Door Opens

A capable entry point provides orientation, assigns a temporary case identifier, starts the relevant clock, names the office responsible for the next action, and records immediate risk that cannot safely wait. Where full resolution is impossible, it should create a warm transfer in which the receiving unit accepts the handoff and the person does not reconstruct the case from the beginning. First-contact resolution is valuable for simple matters, but first-contact ownership is equally important for complex ones. Someone remains answerable until the transfer is complete, the person knows what will happen next, and an escalation route exists if the promised action does not occur.

Urgency must be defined by preventable deterioration, not by who speaks most forcefully. Health change, medication interruption, eviction risk, loss of heat or water, disability access, child safety, and imminent income collapse may require accelerated review. Professional thresholds should be designed by the relevant field and updated as evidence changes (Institute of Medicine, 2015; Herd and Moynihan, 2025).

9.3.2 — Fraud Control Without Cruelty

Poverty, unstable addresses, missing documents, inconsistent spelling, informal work, fluctuating disability, family complexity, or language difference can create irregular records without dishonest intent. A sound integrity system distinguishes intentional misrepresentation from clerical error, stale data, ambiguous rules, incomplete records, and changes in circumstance. For each control, name the targeted defect, such as fraud or eligibility error, and report both false negatives that miss it and false positives that flag it when it is absent. It also measures delay, denial, investigation, and the burden and error introduced by the control itself. Use the least intrusive reliable evidence, separate investigators from frontline care where possible, and preserve notice, explanation, appeal, and correction.

The trigger for an integrity review should be a specific and relevant inconsistency, corroborated information, duplicate claim pattern, conflict of interest, or evidence of material misrepresentation. Broad demographic proxies, emotional presentation, neighborhood, immigration background, disability, or prior use of services should not serve as hidden substitutes for evidence (Herd and Moynihan, 2025).

Emergency human review is the pathway for cases that do not fit ordinary processing and cannot safely wait for ordinary appeal. It is triggered by imminent harm, an automated or routine denial with serious consequences, conflicting evidence, a disability or language barrier, a sudden change in condition, or a case that frontline staff reasonably believe the standard pathway is mishandling.

The reviewer can record uncertainty, limit the temporary measure, require follow-up, and refer a separate integrity concern. Immediate safety should not depend on resolving every suspicion

first, particularly when the temporary action is bounded and reviewable.

The relevant burden test is whether exceptional power prevented harm without becoming hidden favoritism, unreviewable discretion, or a substitute for repairing ordinary access.

9.3.3 — Fast Relief as Cost Prevention

Fast relief becomes cost prevention when a credible causal chain connects delay to escalation. Arrears aid may avert eviction; a medication bridge may reduce the need for emergency care; repairs may prevent injury or property loss; and childcare or transport support may prevent job loss. Evaluation needs a defined trigger, realistic comparison, full delivery cost, and enough time to observe avoided harm and displaced burden. It should ask what was plausibly prevented, who benefited, who might have stabilized anyway, and whether staff, providers, or caregivers absorbed unrecorded work. Savings matter, but so do reduced pain, regained time, preserved tenancy, and avoided family disruption when they fall outside the same budget (Institute of Medicine, 2015; Kruk et al., 2018).

Include delivery capacity, compensation, training and maintenance in the cost of rapid relief.

9.4 — Service Design Around Life Events

A birth can change household time, needs, risk, and responsibility across several public systems, but fragmented administration turns that one event into repeated proof and delay (Heinrich et al., 2022). Life-event design should instead begin with the event's actual clock. Scheduled births and retirements can be prepared for in advance, while aging and progressive disability emerge gradually. Job loss, displacement, and disaster may demand action before a complete record can be assembled. Timing therefore determines what support must begin at once, what can be coordinated later, and where delay could cause irreversible harm.

9.4.1 — Birth and Childhood Events

Birth and early childhood create a compressed period in which health, identity, income, housing, nutrition, leave, childcare, disability, and family support may change together. The clock begins before birth for prenatal care and preparation, accelerates at delivery, and continues through recovery, registration, feeding, sleep disruption, and the child's first months.

The event trigger may be pregnancy, expected delivery, birth, adoption, guardianship, or placement. Services should not wait for every downstream document before offering information and support that can safely begin earlier (Moynihan, Herd, and Harvey, 2015).

Coordinating registration, care, benefits, and safeguarding does not by itself authorize a unified family file. Use a high sharing threshold, applicable consent, lawful purpose, role-based access, and protected summaries under Section 4.4.

9.4.2 — Illness and Disability Events

Illness and disability do not follow one clock. Some conditions arrive suddenly, some progress, some fluctuate, and some are lifelong. A person may move between diagnosis, treatment, recovery, relapse, accommodation, income loss, transport needs, caregiving, and changed housing without a clear moment when the event begins or ends.

The trigger may be a clinical event, reported functional change, hospitalization, loss of capacity, new diagnosis, injury, treatment side effect, or failure of an existing accommodation. Access systems should recognize that a person can need support before diagnostic certainty is complete and can continue to need support after a crisis appears resolved on paper (Institute of Medicine, 2015; World Health Organization, 2022a).

Appeal and correction should preserve essential care and access where delay creates serious risk. The person needs a route to challenge inaccurate clinical summaries, misunderstood function, inaccessible assessments, and administrative errors. Corrected information should reach every downstream system that relied on the mistake.

A functioning life-event route needs a primary service owner, not a new authority over the person. For an illness-related transition, the health provider remains responsible for clinical judgment; benefit and housing offices retain lawful eligibility decisions; and a designated coordinator is answerable for the handoff timeline, unresolved tasks, and safe escalation. The person can view or correct the minimal shared status record, decline unrelated data reuse, and reach a human alternative when digital identity fails. In a trial, measure the time to uninterrupted treatment and the number of separate explanations a person must give. A coordinator who merely forwards messages without

authority or capacity to resolve a failed handoff is another layer of burden.

Support should not force a person to choose between proving incapacity and attempting recovery.

9.4.3 — Job Loss and Debt Events

Income can stop on a Friday while rent, utilities, food, transport, debt payments, childcare, insurance, and medication continue on schedule. The trigger may be dismissal, redundancy, contract loss, reduced hours, business failure, unpaid wages, injury-related separation, or a sudden caregiving demand. Systems should recognize partial loss as well as total unemployment. A person whose hours collapse can face the same debt spiral while remaining technically employed (Moynihan, Herd, and Harvey, 2015; OECD, 2024b).

Provisional action may include temporary income support, a pause on utility disconnection, rent or mortgage mediation, food access, benefit continuity, childcare preservation, transport, or a short debt standstill. Assess retraining and mobility against labor-market evidence without making immediate income support wait for that decision.

9.4.4 — Migration and Displacement Events

Migration and displacement may interrupt documents, residence, income, care, and family contact at once. Evidence minimums should anticipate document loss, transliteration differences, disrupted registries, family separation, and trauma. Identity and relationship claims may require staged verification. The absence of perfect documents should not erase observable need, while the institution should record uncertainty honestly and provide a route for later correction.

A migration service pathway neither grants universal status nor turns access into covert adjudication. Its task is to prevent

administrative disappearance and support lawful decisions without avoidable exposure, hunger, illness, or family fracture.

9.4.5 — Aging and Care Events

Aging is rarely a single event. It is a sequence of changes in mobility, memory, health, income, housing, bereavement, driving, caregiving, and daily function. The burden may grow slowly, arrive after a fall or hospitalization, or become visible only when an unpaid caregiver can no longer sustain the work.

For older people, especially relevant warning signs include caregiver exhaustion, repeated falls, missed medication, unsafe housing, and the loss of a partner. Age alone should never imply incapacity; the threshold for review is whether timely support can prevent foreseeable harm without overriding the older person’s agency (Institute of Medicine, 2015; World Health Organization, 2024).

An integrity review should distinguish suspected fraud by the applicant from exploitation of the applicant. The main risk may be abuse by a caregiver, vendor, relative, or institution. Controls should examine billing, coercion, undue influence, and service quality while avoiding suspicion based solely on changes in memory or communication.

Privacy requires care because aging records can expose health, finances, family conflict, cognition, and vulnerability to exploitation. Shared information should be limited to the action required. Family involvement should follow consent and lawful authority rather than an assumption that relatives automatically own access to the older person’s life.

Appeal and correction need accessible routes, advocates where appropriate, and timelines that reflect declining function. A person should be able to challenge an inaccurate capacity note, care assessment, financial record, or service termination.

9.4.6 — Disaster and Crisis Events

Disaster compresses many service needs into the same hours while damaging the institutions expected to respond. First access should distinguish immediate survival from the longer work of recovery, with a named authority responsible for each handoff.

Emergency authority should identify who can provide food, shelter, health support, transport, cash assistance, replacement documents, and protection, and how those roles coordinate. Local institutions, community organizations, and trusted networks may be essential, but delegation should include accountability, worker safety, compensation, and a route for public complaint.

Fraud control should focus on organized diversion, vendor abuse, duplicate large claims, and corruption without withholding basic relief from people who cannot document loss. Emergency distribution can use limits, receipts, audits, and later reconciliation. The cost of some over-inclusion may be lower than the cost of leaving people without water, food, medicine, or shelter.

Recovery introduces a second queue. Insurance, rebuilding, housing, schools, health, debt, permits, and infrastructure repair can remain delayed long after media attention leaves. A responsible disaster pathway follows burden through recovery and does not declare success when the acute shelter closes. The material test is whether the least-resourced households are being asked to carry the longest aftermath (Hallegatte et al., 2017; IPCC, 2023).

9.5 — Digital Access, Human Access, and Rights Boundaries

Public service may involve a digital tool, a remote interpreter, a community worker, and staff sharing a case record. Each channel can remove a practical barrier while also introducing its own point of failure. The person seeking help should not have to reconcile conflicting systems or diagnose a broken portal. Technology extends human capability only when the institution remains responsible for making the whole route work.

9.5.1 — Digital Convenience

Digital service can return time when it is designed as a genuine option. A person may apply after work, upload a document without travel, receive a status update, schedule an appointment, or correct a simple error without waiting on a telephone line. These gains should be measured in user-hours returned, journeys avoided, successful completion, correction speed, and continuity across devices and channels, not simply in the share of transactions moved online. The OECD's 2026 digital-government review found broad adoption but weak evaluation: only one in four surveyed OECD countries conducted cost-benefit analysis of completed digital projects. A strong service therefore saves progress, confirms receipt, explains required formats, permits authorized assistance, and allows a shift to telephone or in-person help without losing work already completed (OECD, 2026a; W3C, 2023).

The payment route matters as much as the identification device. Muralidharan, Niehaus, and Sukhtankar (2016) studied a randomized rollout of biometric smartcards in Andhra Pradesh's employment and pension programs. The incompletely implemented system made employment payments faster, more predictable, and less corrupt without a detected reduction in

access. This was evidence for a delivery package, not for every possible biometric requirement.

Their later study of stricter biometric identification in India's food-distribution system found reduced corruption alongside interrupted access for legitimate beneficiaries; the transition rules were central to the adverse effects (Muralidharan, Niehaus, and Sukhtankar, 2025). These were different programs, not a direct comparison of two devices. Together they identify a practical design question: does verification support payment or become a new condition that can block it? A digital reform should specify what happens when authentication fails, who can authorize a lawful alternative, and how missed benefits are restored.

Measure digital access through abandonment, assisted completion, recovery, accessibility defects, correction time, channel switching, and unequal failure across groups (OECD, 2026a; W3C, 2023).

9.5.2 — Disability and Language Access

Provide accessible forms, plain language, screen-reader compatibility, captions, sign-language interpretation, translation, easy-read formats, supported decision-making, quiet spaces, physical access, flexible proof routes, and human assistance across intake, service, correction, and appeal (W3C, 2023; World Health Organization, 2022a).

Disability is not one access problem. Visual, hearing, mobility, cognitive, intellectual, neurological, communication, psychiatric, and fluctuating conditions can create different barriers. Design should therefore offer multiple routes and allow individual accommodation rather than assuming one accessible version fits everyone.

Accessibility should be tested under realistic conditions. A portal that passes a technical checklist may still fail on a shared

phone, slow connection, screen reader, low-cost device, or interrupted session. Testing should include people who use the relevant access routes and should examine whether a task can be completed, corrected, and appealed without expert help (W3C, 2023; World Health Organization, 2022a).

9.5.3 — Data Minimization in Service Access

Data minimization begins with purpose discipline. A service should not collect information because it might someday be useful, because a vendor template asks for it, or because suspicion has become habit. Each field needs a named decision purpose, legal basis, source, access class, retention period, correction route, and deletion or sealing rule. Intake should not automatically receive audit data; audit should not automatically receive clinical detail; research should not automatically inherit service records; and a model-training environment should not be treated as an ordinary extension of administration. Derived attributes and inferences require the same scrutiny as submitted facts because they can expose health, family, location, finances, or vulnerability without the person supplying that label. Purpose-bound design protects privacy and improves data quality: fewer irrelevant fields mean fewer stale records, fewer contradictions, and a clearer explanation of what information changed the decision.

Purpose-bound records limit what a breach can expose and make correction easier. Protect them through proportionate access controls and tested recovery throughout their lifecycle, including vendor processing and deletion. Digital delivery still needs connected identity, payment, grievance, and human-support routes; connecting those functions does not justify unrestricted access to their underlying records (Karippacheril et al., 2024; NIST, 2020; OECD, 2026a).

Correction after digital-service failure should preserve queue position, prevent deadline expiry, and maintain accessibility, language support, and human review (OECD, 2026a; W3C, 2023; World Health Organization, 2022a).

9.6 — The Cost of Delay in a Fair Society

Delay is not a uniform inconvenience, because its consequences depend on what is waiting and what alternatives remain available. A postponed appointment may worsen illness, while a late benefit can trigger debt or eviction before the payment arrives. Institutions can also pass costs to other agencies when slow decisions create avoidable emergencies. A careful account records the starting condition, delay duration, available substitutes, and the harm that follows rather than charging every subsequent problem to one queue. The next distinctions make health, income, and administrative delay comparable without treating them as identical.

9.6.1 — Health Delay Cost

Health delay cost counts harm from postponed care, diagnosis, treatment, medication, therapy, disability support, health-related housing repair, or crisis response. A complete account distinguishes the delay before first contact, the interval between referral and treatment, interruptions inside a care course, and the time required to correct a mistaken decision. It then traces plausible consequences: symptom progression, avoidable pain, emergency use, hospitalization, loss of function, long-term disability, family stress, lost work, and additional caregiver labor. The analysis should identify the condition, baseline risk, effective-action window, and uncertainty about causation instead of assigning every later harm to the wait. It should also record unequal exposure, because the same delay is more dangerous for a person without medication reserves, transport, flexible work, stable housing, or a caregiver.

For caregivers, record changes in work hours, symptom-monitoring time, travel and stopgap supplies during the wait.

For delayed housing repairs involving heat, water, ventilation, accessibility, electrical systems, locks, or structural hazards, assess health and safety risks across seasons and household composition. A repair delayed in mild weather may become dangerous during heat, cold, storms, or illness. Children, older people, disabled people, and people working from home may carry consequences that a property-level average does not reveal.

9.6.2 — Benefit Delay Cost

Benefit delay cost counts hunger, debt, utility shutoff, missed rent, medical interruption, family stress, and crisis produced when income, food, childcare, disability, unemployment, housing, or other support arrives late. The relevant record follows who carried the delay: the applicant, children, caregivers, landlords, utilities, charities, clinics, emergency services, or future public budgets. It distinguishes elapsed days from consequential thresholds, such as an eviction date, prescription lapse, school absence, or disconnection notice (Herd and Moynihan, 2018; Heinrich et al., 2022).

Count the costs added by waiting for support. Late fees, payday loans, overdrafts, utility reconnection fees, transport, document fees, missed work, childcare, appeal time, and food substitution can consume the value of support before it arrives. The system may pay the benefit while the delay has already spent it.

Legal delay creates a related burden through unresolved status, postponed hearings, slow appeals, missed protection, and uncertainty that gives stronger parties leverage. The record does not rewrite law, but it makes time inside legal process visible as a material condition.

Interim legal protection is part of delay analysis because final remedy may come too late. The existence, accessibility, and duration of stays, temporary orders, representation, emergency

hearings, and status preservation should be recorded separately from the final outcome.

9.6.3 — Institutional Delay Cost

Distinguish scarce capacity, fragmented authority, preventable friction and a protected review function as causes of institutional delay (Herd and Moynihan, 2025; Institute of Medicine, 2015).

A delay-cost record should begin with the causal pathway. What decision or service was postponed? What condition existed at the start? What changed during the interval? Which later intervention became necessary, and what other explanations could have produced the same outcome? The record should state the baseline, exposure window, mechanism, affected population, comparison, uncertainty, and time horizon. Where identification is weak, the honest result may be a bounded estimate, a set of plausible consequences, or no adequate causal estimate.

Each stage needs a named owner, expected completion window, escalation rule, and handoff confirmation so delay cannot vanish between institutions. Evaluation should distinguish elapsed time from active user labor and follow deterioration, downstream expenditure, correction, and recovery after service arrives (Heinrich et al., 2022; Herd and Moynihan, 2025). Use the distribution, denominator, access, and staff-workload checks in Section 9.1.3.

Named cases can illuminate the mechanism, but they must not become spectacle. Housing loss, untreated pain, disability exclusion, or addiction crisis should be described only to the level needed to understand the institutional sequence and the repair (Desmond, 2016; World Health Organization and UNODC, 2020).

Chapter 10 — Housing, Health, and Care Infrastructure

Rent pressure can interrupt treatment; a medication gap can destabilize housing; and a broken care schedule can end paid work. Housing, health, and care therefore meet inside the same household even when institutions divide them into separate programs.

Healthcare becomes peace infrastructure when suffering can reach competent relief in time. Advanced equipment, specialist excellence, and research capacity are valuable, but their social meaning depends on an accessible route through cost, distance, waiting, language, disability, records, and continuity. A system earns trust when people can seek help before pain, illness, or uncertainty becomes catastrophe.

Mental health and addiction require material seriousness and clinical humility. The chapter supports early, rights-respecting, person-centered care while treating housing, work, debt, isolation, trauma, and administrative conditions as part of the environment in which recovery occurs (World Health Organization, 2025c; World Health Organization, 2025b).

Disability support, aging, and care systems reveal the labor beneath ordinary independence. When dependence is treated as a private inconvenience, the resulting system rests on unpaid family labor, inaccessible places, underpaid care work, and the exclusion of people whose bodies or minds do not fit administrative defaults.

Continuity is the operating principle that connects these systems. A service can be excellent at one moment and still fail the household if eligibility changes, a provider disappears, medication is interrupted, records do not travel, an appeal stalls, or a benefit cliff removes support. Each break can undo earlier gains because the person experiences one life even when institutions experience separate cases.

Stable housing supports continuous care; unsafe conditions can intensify illness, injury, and stress. Global monitoring shows substantial long-run gains in health-service coverage and financial protection, but progress remains uneven and millions of households still face material hardship from care. Availability therefore has to be read beside affordability, distance, waiting, discrimination, and continuity (World Health Organization and World Bank, 2025; Nussbaum, 2011). For a person changing address, continuity planning should establish where medicines, correspondence, and follow-up arrangements will go. Only the information needed to preserve continuity should follow the person across that move; unrelated household details need not travel with the service record.

Section 10.5 specifies accepted handoffs for discharge, relocation, renewal, caregiver change, and transitions between child and adult services. Consider a person whose treatment plan is clinically appropriate but cannot be followed in the place where they sleep. The gap is not necessarily a request for a different treatment. It may be a housing condition that prevents the existing plan from working. Starting from that distinction keeps the discussion close to the person’s actual difficulty while preserving the different responsibilities of clinical and housing services.

10.1 — Housing as Civilizational Calm

Stable housing supports sleep, care, food storage, secure records, recovery, privacy, and planning. It gives other services a reliable place to reach the household (World Health Organization, 2018; Desmond, 2016). Housing insecurity has no single cause or universal solution. Land, finance, zoning, construction cost, infrastructure, local opposition, wages, speculation, public investment, tenant protections, property rights, regional growth, migration, and climate interact. The burden discipline is more stable: housing policy should be judged by whether it lowers hidden strain without moving people, communities, or future costs out of view. Track affordability, stability, habitability, displacement risk, utility burden, access to services, repair, privacy, and downstream public cost separately.

10.1.1 — Shelter and the Loss of Ordinary Options

Baseline security has several dimensions. Tenure must be sufficiently stable that a household can plan. The dwelling must be physically safe, weather-protective, and healthy. It must also be usable for the people who live there, including children, older people, and people with disabilities.

Quality cannot be inferred from occupancy alone. Heat, ventilation, water, sanitation, damp, pests, electrical safety, crowding, noise, and structural condition shape health and recovery. A household may be formally housed while living in conditions that continually create illness or stress (World Health Organization, 2018).

Timely assistance can preserve a home before disruption spreads. Evans, Sullivan, and Wallskog (2016) studied 4,448 callers seeking emergency financial help in Chicago during 2010 to 2012. Fluctuating funding availability created a plausibly exogenous comparison: callers reaching the service when funds

were available were 76 percent less likely to enter a homeless shelter. This is a relative reduction in the studied outcome, not a 76-percentage-point change or a result for every housing program. Local evaluations can ask which impending losses can be prevented through prompt help and which households require continuing housing or care support.

Climate resilience is now part of baseline shelter. Heat, flood, fire, storm, smoke, drought, and insurance withdrawal can change whether a dwelling remains safe or affordable. Repair and adaptation programs should prioritize preventable harm while avoiding forced displacement disguised as resilience (IPCC, 2023).

Homelessness removes the stable base that help systems often assume. People may be required to maintain contact, documents and appointments while lacking safe storage, rest or reliable hygiene.

People may need housing first, treatment continuity, protection from violence, income, disability support, family reunification, or several forms of support at once.

10.1.2 — Housing Supply and Anti-Speculation

Supply, finance, land policy, and speculation interact through price, timing, and ownership. New construction can be absorbed by investors, credit can widen access or bid up scarce stock, infrastructure can unlock homes while raising nearby land values, and anti-speculation measures can reduce pressure without solving a physical shortage. The analysis should identify which link is locally binding before choosing the remedy.

New housing can benefit people who never occupy the new building. Using population-wide register data from the Helsinki metropolitan area, Bratu, Harjunen, and Saarimaa (2023) traced moves initiated by new centrally located market-rate homes. The

chains quickly reached middle- and low-income households and neighborhoods. A household moving into a new unit may free an older unit, whose next occupant frees another. The relevant reach is therefore wider than the income profile of the first occupants.

Tracing a moving chain is not the same as estimating how much rents would have changed without construction. A useful local account would follow units actually vacated, distinguish vacancies from household splits that leave a dwelling occupied, and check where a chain ends. That would show how supply becomes usable capacity rather than treating every move as another home created. Direct help remains relevant where discrimination or unaffordable rents block access; construction and assistance can address different links in the same housing system.

Displacement is not only removal from a unit. It is also removal from a school route, a clinic, a bus line, a grandmother who watches children, a trusted shop, a language community, a faith or secular gathering place, and the informal care map that makes daily life possible. Housing repair fails when it improves buildings while dissolving the social infrastructure that allowed residents to survive neglect.

A displacement review should ask not only who is forced out immediately, but who becomes unable to return. Repair can exclude through rent, taxes, insurance, parking, fees, cultural loss, business displacement, school disruption, or changed social networks. The burden may be gradual enough to avoid headlines.

This principle also applies to climate adaptation, transit investment, school improvement, public safety repair, and health access. Public investment can raise land values, and rising values can expel the people the investment was meant to serve. Housing without displacement therefore requires early safeguards (Sen, 1999; Nussbaum, 2011).

10.2 — Healthcare as Relief Infrastructure

Health-system design and claims about clinical outcomes require qualified review. A material burden account should separately track delayed access, financial exposure, preventable deterioration, emergency overload, workforce strain, and lost continuity. Those dimensions may interact, but one count should not be used as evidence for all of them. Public decisions must therefore pair competent clinical judgment with transparent measures of service and household cost.

10.2.1 — Care Access as Peace Infrastructure

The health-system question is whether a person can move from first contact through diagnosis, treatment, follow-up, and recovery without cost, transport, staffing, or administrative gaps breaking the route. Track access, quality, timeliness, continuity, and financial protection across that course (OECD, 2025c; World Health Organization and World Bank, 2025).

Care access can improve lives through more than one route. Using Oregon's lottery for the opportunity to apply for Medicaid, Baicker et al. (2013) estimated that coverage reduced positive depression screens by 9.15 percentage points after about two years, with a 95 percent confidence interval of 1.60 to 16.70 percentage points lower. It also reduced financial strain. The study did not find significant improvements in the measured blood-pressure, cholesterol, or glycated-hemoglobin outcomes over that period. This pattern opens a more useful evaluation than one undifferentiated success score: track financial protection, mental health, treatment, and physical outcomes on their appropriate timelines.

Care access also requires trust that the system will not punish need. People may delay care because they fear bills, stigma, immigration consequences, employment consequences, child-

welfare involvement, racial bias, disability disbelief, or being treated as exaggerating pain. Access is therefore geographic, financial, relational, and institutional.

Medical debt is especially corrosive because it often follows a moment when refusal was not realistic. A household can choose many purchases, but illness, injury, birth, disability, crisis, and medication need do not wait for financial readiness.

10.2.2 — Prevention, Health Gains, and Cost

Some preventive services reduce later costs, some mainly reduce suffering, and some require long horizons before the benefit appears. A serious analysis does not need every humane intervention to prove short-term fiscal savings; it asks whether avoidable deterioration is being prevented before crisis becomes the expensive default.

Prevention also includes the nonclinical conditions that make care possible. Housing stability, food security, safe workplaces, clean water, accessible transport, and reduced pollution may prevent suffering before a medical appointment is needed. A narrow clinical record can miss these upstream burdens.

Prevention can improve when early warnings prompt action across service boundaries. Housing services can flag health-threatening conditions, schools can connect children to care, workplaces can reduce exposure before injury, and transport systems can preserve access to routine treatment. Each interface needs a named owner and response path so prevention is not merely acknowledged upstream but acted on before crisis.

Repeated emergency presentations involving hunger, exposure, untreated addiction or mental distress, medication gaps, or preventable injuries can signal gaps in earlier support. Review those patterns with the services able to respond before an emergency.

10.2.3 — Health Workforce Burden

Workforce burden should be understood as a continuity risk. When staffing is thin, the system may still open its doors but lose memory, patience, explanation, follow-up, and the ability to notice deterioration early. Patients experience this as rushed appointments, repeated stories, delayed calls, preventable mistakes, and a sense that the institution is present in name but absent in care.

Health workers need safe schedules, adequate staffing, equipment, professional autonomy and interpreter support without exemption from clinical accountability (World Health Organization and World Bank, 2025; Sen, 1999).

The workforce record should include violence risk, documentation, supervision, occupational exposure and the work of explaining institutional failures to patients.

Technology should be judged by whether it relieves or increases this burden. Digital records, automated scheduling, decision support, billing systems, and remote care can help, but they can also create documentation pressure and screen-mediated frustration. Assess a tool’s claimed time saving alongside any additional worker and patient attention it requires to establish whether burden fell, shifted, or both.

The workforce record should distinguish moral distress when institutional constraints block care a professional judges necessary from moral injury associated with serious violations of deeply held values. Constraints such as staffing, policy, cost, time, or bed availability can create the first condition; exposure alone does not establish the second in every worker. Both deserve attention in workforce support and retention planning (British Medical Association, 2021).

Pair staffing and continuity indicators with patient access, while retaining clinical review of prevalence, treatment effects, crisis

thresholds and service standards (Institute of Medicine, 2015; Kruk et al., 2018).

10.3 — Mental Health, Addiction, and Anti-Spiral Design

Mental health and addiction belong within the domestic floor because distress can narrow housing, work, relationships, judgment, and future choice before an emergency becomes visible. The first duty is to make help reachable before a preventable crisis is treated as the gateway to care (World Health Organization, 2022b; World Health Organization and UNODC, 2020). Anti-spiral design joins timely clinical care with housing, social support, and continuity to interrupt reinforcing burdens. Identify the reinforcing loop and assign responsibility for breaking it.

10.3.1 — Mental Health as Material Stability

Mental health cannot be reduced to income, housing, or public services. Distress and mental illness have biological, psychological, relational, developmental, cultural, and social dimensions, and clinical assessment belongs to qualified professionals. Material systems matter because they can intensify risk, interrupt care, or create conditions in which recovery is harder to sustain.

Housing threat, food insecurity, untreated pain, violence, discrimination, debt pressure, unstable work, isolation, and exhausted caregiving can keep the body and mind under persistent strain. Repairing those conditions is not therapy, but it can remove burdens that repeatedly reactivate crisis.

A capable mental health system needs more than emergency response. It requires prevention, primary-care connection, community services, specialist care, crisis support, rehabilitation, peer support, social care, and continuity across settings. The mix will differ by place, but fragmentation can leave people and families to coordinate care themselves. The WHO’s Mental

Health Atlas 2024, based on data from 144 countries, provides a current picture of large gaps in policy, financing, workforce, and community-based service capacity (World Health Organization, 2022a; World Health Organization, 2025b). Workforce planning should distinguish the functions of clinical specialists, primary-care teams, peer specialists, community workers and support staff (World Health Organization, 2025b; World Health Organization, 2025c).

Apply Section 4.4's privacy rules to keep diagnoses separate from civic rank, employment worth, political participation, and general surveillance. The World Mental Health Report supports rights-based, community-oriented reform while preserving qualified clinical judgment (World Health Organization, 2022b).

10.3.2 — Crisis Response and Continuity of Support

Crisis response must protect safety without making jail, force, or public humiliation the ordinary doorway to care. Some crises involve real danger and require skilled coordination, but danger does not erase the need for clinical competence, de-escalation and privacy. A trustworthy system asks what response reduces the next crisis rather than merely clearing the scene.

The post-crisis record should include transport to care, safe handoff, family communication where appropriate, short-term stabilization, privacy protection, and a review of whether the same person has been cycling through crisis systems. Repeated crisis contact should trigger review of the pathway. It may reveal a failed handoff, unmet housing or care needs, or an illness whose course includes recurrence despite appropriate support. The review should identify the cause and the next useful action without treating recurrence alone as proof that care failed.

Crisis-response design must protect the person in crisis, responders, bystanders, families, and the surrounding

community. First identify the need and risk: a medical, psychiatric, or substance-related crisis, intoxication, domestic danger, homelessness, cognitive impairment, or a threat to public safety. Match the response to that assessment.

After discharge from a hospital, clinic, detox setting, shelter, jail or emergency room, follow whether transport, medication, housing, appointments, insurance and trust allow the recovery plan to operate. Recovery planning should connect treatment with food, housing, work, benefits, legal support and the person's chosen relationships. Continuity should also accommodate uneven recovery and preserve access to peer support. Clinical professionals and people with relevant lived expertise must shape the treatment and recovery questions that material analysis alone cannot answer.

10.4 — Care Systems, Disability Support, and Aging

Care systems must distribute support work without taking agency from the person receiving care (World Health Organization, 2022a; World Health Organization, 2024). Fragmented service responsibilities can turn a formally available benefit into an unusable route. The person's preferences and consent should guide how support is delivered, not only whether a file records eligibility. Reliability across settings matters as much as any individual appointment or service.

10.4.1 — Accessible Routes Across Services and Places

The issue here is continuity across housing, healthcare, work, and public space. Fragmented accommodations may leave each individual service formally open while the journey between them remains unusable; the access-infrastructure principle is developed in Section 8.5 (World Health Organization, 2024).

Access also means reliability. A blocked ramp, broken lift, absent interpreter, screen-reader failure, or unpredictable paratransit service can make a right unusable. Disability support should therefore include maintenance and response time.

Flexible schedules and administrative routes that avoid repeated proof also belong in the design.

Accessibility should be treated as a design condition rather than a late accommodation. Retrofitting after exclusion is often more expensive, slower, and less complete than designing with diverse bodies, senses, cognition, and communication from the beginning (World Health Organization, 2022a; Nussbaum, 2011).

The built environment also determines whether help-seeking itself is dignified. A clinic that is physically accessible but impossible to reach by transit remains partly inaccessible. A service office with stairs, confusing signage, fluorescent overload, no seating, no language support, or hostile security

procedures can turn assistance into ordeal. Access begins before the appointment and continues until the person returns home safely.

Accessible environments can also prevent costly downstream care. A home that allows safe movement, bathing, cooking, and entry can reduce falls, isolation, and caregiver injury while helping people remain at home (World Health Organization, 2018).

10.4.2 — Aging and Long-Term Care

Home can preserve familiarity and dignity when support, accessibility, respite, safety, and medical coordination exist. Home can also become isolation and hidden family exhaustion when those supports are absent. Facilities can provide skilled care and social structure when they are humane, staffed, transparent, and accountable. Facilities can also become warehouses when budgets, oversight, and culture fail.

A long-term-care burden map should follow the elder, caregivers, workers, payers, the household, and the community. Record where financial constraints limit choice, how risk is shifted, whether loneliness develops, and whether the person retains a voice in care decisions (World Health Organization, 2024; Nussbaum, 2011).

Aging also raises the question of memory and identity. A humane care system should not treat an elder only as a safety problem or bed assignment. Relationships, language, faith or nonfaith, food, music, routines, privacy, sexuality, friendship, grief, and autonomy remain part of personhood.

Long-term care should be evaluated by continuity, autonomy, safety, social connection, caregiver support, and protection from neglect or abuse. Family care cannot be assumed without asking what happens to the caregiver’s body, income, sleep, and future.

10.4.3 — Caregiver Support

Caregiver support must include unpaid family care and paid care work because each can conceal shortages in the other. When paid provision is weak, families may have to provide the missing care. When families are exhausted, services may encounter needs only after they have become crises.

Caregiver support also needs a continuity plan for the moment the primary caregiver is ill, exhausted, unavailable, or no longer able to continue. The plan should identify who can step in, how routines and medication information are transferred, how paid leave or emergency support is activated, and how the caregiver returns without penalty (Addati et al., 2018; World Health Organization, 2024).

Peer support and protected time also help preserve the caregiver's own life.

10.5 — Continuity and Access

Continuity allows gains from treatment and support to accumulate across administrative changes (Kruk et al., 2018; World Health Organization and World Bank, 2025). Interruption can erase those gains even when each institution reports formal coverage. An accountable service records duration, clinical risk, household shock, recovery cost, staff time, caregiver strain, and trust damage as distinct dimensions. Those records should trigger correction before a predictable gap becomes an emergency.

10.5.1 — Care Interruption as Harm Multiplier

A missed appointment may also produce a medication gap, a lost shift, a renewed crisis, an anxious child, a caregiver's missed sleep, and another round of paperwork.

Name high-risk interruptions before they occur. Gaps in medication, home care, housing support, therapy, or disability equipment should trigger review, as should interruptions during a change of provider.

A person may ration medication, delay appointments, stop disclosure, avoid work changes, or remain in unsafe arrangements when care may disappear. The cost is behavioral and emotional as well as clinical.

Eligibility transitions often arrive during life changes that already create stress. A birthday, job change, move, disability determination, income shift, pregnancy, release from custody, divorce, migration event, or aging threshold may change the formal category while the underlying need continues. The stronger design treats each threshold as a foreseeable risk event, with advance notice, temporary continuity, clear responsibility, human contact, and a route to correction.

10.5.2 — Medication and Treatment Continuity

Changes in insurer, prescriber, pharmacy, employer, residence, hospital, school, correctional setting, or disaster zone can interrupt care even when clinical need is unchanged.

Medication continuity begins with a current record the person can review. Names, doses, allergies, prescribers, refill dates, and adverse reactions need reconciliation after hospitalization or emergency care. The record supports but never replaces clinical judgment.

Continuity includes non-medication treatment. Therapy, rehabilitation, addiction care, home health, disability support, and long-term care can fail through canceled contracts, workforce turnover, relocation, or lost referrals. Relationships and routines may be clinically important even when they are not visible in a billing record.

Systems should track interruption, delayed refill, failed handoff, repeated assessment, adverse event, and restoration time (Institute of Medicine, 2015; World Health Organization and World Bank, 2025).

Addiction treatment requires particular attention to continuity because interruption can raise relapse, withdrawal, overdose, and criminalization risk. Clinically governed standards should maintain care across settings without turning treatment history into a permanent civic label (World Health Organization and UNODC, 2020).

At any transfer, the sending provider identifies outstanding clinical risks, the receiving provider confirms acceptance, and the patient receives a plain-language explanation and an urgent contact route. No agency should presume that a referral or electronic transmission is a completed handoff until acceptance or a documented safe alternative exists. Where an authorization dispute threatens indispensable treatment, a lawful interim

pathway and expedited review should prevent avoidable interruption. Measure not only how many referrals were sent, but the interval until usable care was restored, the number that failed, and who carried the consequences of failure.

10.5.3 — The Full Cost of Interrupted Care

An interrupted-care record should identify recurring interruption points, such as authorization delays, provider churn, eligibility cliffs, inaccessible notices, medication shortages, transportation breakdowns, or disputed services. The record becomes useful when it helps institutions repair their own handoffs instead of forcing each affected person to litigate the same failure alone.

Bills, readmissions, transport, staff time, lost wages, and appeals may support numeric estimates. Testimony can reveal family strain or declining confidence that must be reported separately. Distinguish costs caused by an interruption from events that merely followed it (World Health Organization, 2021; World Health Organization, 2025b; Institute of Medicine, 2015; Kruk et al., 2018).

10.6 — The Household Floor of Dignity

The household floor becomes visible when several ordinary systems must work at the same time. A safe dwelling cannot provide stability if its utilities fail, essential food cannot be stored, or caregiving depends on inaccessible transport. Each service may be supplied by a different institution, but people experience the combined result in daily routines. A credible standard should therefore assess continuity under foreseeable disruption as well as nominal eligibility. The following analysis connects basic provision to the family stress produced when those dependencies break.

10.6.1 — Food, Utilities, and Basic Stability

Refrigerated medication needs electricity. Cooking requires water and power. Heat, cooling, light, internet, washing, charging, and safe storage all shape the practical meaning of shelter.

Buffers should keep a utility interruption, spoiled food, or broken phone from becoming a household crisis.

A household food plan can be tested by following one meal all the way to use. During a planned water interruption, delivering ingredients may leave the household with food it cannot prepare. A different arrangement might provide a ready-to-eat meal at a reachable place until normal preparation is possible again. The provision changes because the missing function has changed.

This way of tracing a need is especially useful across seasons. An arrangement that works on an ordinary afternoon may need different transport or opening hours during extreme weather. The household can explain which part of its routine depends on the affected service. Providers can then organize support around that particular dependency.

The practical gain is continuity with less improvisation. People know where the temporary service is, when they can use it, and

what marks the return to the normal arrangement. A small interruption remains a limited change to the day rather than a demand to reconstruct the whole household routine.

10.6.2 — Family Stress and Material Systems

Family stress can reflect pressures from wages, housing, care, and inaccessible services as well as private relationships.

Children absorb instability without choosing it. They may experience school moves, caregiver absence, food insecurity, adult anxiety, crowded housing, or interrupted health care. Support should strengthen the household without turning children into evidence for institutional success or failure (Cunha and Heckman, 2007; World Bank, 2018).

Education systems also affect the domestic floor. Access and equity depend on location, disability, wealth, language, and the resources available to families and schools. Public support should reduce the need for private households to repair educational inequality through money, time, or social connections (UNESCO, 2024; Global Education Monitoring Report Team, 2026).

Information sharing inside a family requires care. Adults may want support without disclosing every clinical or financial detail to relatives, and young people may have rights and needs that differ from those of caregivers. Coordination should use consent and minimum necessary information rather than assuming the household is one data subject.

Chapter 11 — Education and Generational Stability

Education carries peace across generations by expanding what people can understand, make, repair, question, and become through academic, technical, artistic, civic, practical, and relational routes.

Childhood conditions belong inside educational analysis before achievement is interpreted. Housing, food, sleep, health, disability access, safety, trauma, parental time, transport, school resources, language, and digital conditions shape attention and opportunity. High expectations remain possible, but they are paired with an obligation to repair the circumstances that make learning needlessly difficult (Global Education Monitoring Report Team, 2026; Heckman, 2006).

Vocational strength matters because a peaceful civilization must know how to build, maintain, heal, feed, move, wire, program, teach, and repair. Craft and technical education connect intelligence to material capability and provide routes into contribution that status hierarchies often undervalue. A durable system links training to real equipment, apprenticeships, wages, safety, progression, and regional demand.

Universities and research institutions carry a different but equally important responsibility. They preserve inquiry, archives, method, professional formation, fundamental research, and the ability to challenge what public life takes for granted. Their public usefulness is weakened when debt, credential inflation, prestige competition, donor influence, staff precarity, or closed access displace learning and truth-seeking.

Children and students need age-appropriate consent or assent where applicable, protection from humiliation, and correction rights in research, school evaluation, discipline, and public communication. Section 11.3.3 applies record protections to children’s information and images.

Education data should distinguish learning, access, safety, progression, debt, and placement, with interpretation tied to the measure, baseline, language, geography, and household conditions. Section 11.1.3 shows how assessment can guide instruction.

The generational test looks beyond current enrollment. A prosperous society can still pass forward debt, climate exposure, degraded schools, broken care systems, unpaid maintenance, weak archives, and exhausted trust. Stability means handing children systems they can use and repair after adults are gone. Peace is expressed in the liabilities one generation refuses to hide in the life of the next.

Durable capabilities reduce dependence on a single gatekeeper and help people adapt when industries or communities change (Global Education Monitoring Report Team, 2026; National Academies of Sciences, Engineering, and Medicine, 2017). A qualification also has a life after the classroom. Imagine a learner who completes a demanding course just as the local employer changes its equipment. The certificate records a real achievement, but its practical use now depends on whether the learner can update a specific skill without beginning the entire route again. A durable system preserves earlier learning while providing a practical route for updating skills as requirements change.

11.1 — Education as Capability and Discovery

UNESCO's 2026 Global Education Monitoring Report estimated that 273 million school-age children and youth were out of school in 2024, roughly one in six worldwide (Global Education Monitoring Report Team, 2026). The scale matters, but access alone does not establish what a learner can do after attending. Assessment can confirm safety-critical competence and course prerequisites or show whether instruction is working. Ranking becomes harmful when the measure displaces the development and discovery it was meant to support.

11.1.1 — Learning as Future Room

Learning opens future room because it increases the number of paths a person can perceive and use. A child who can read deeply enters more worlds. A young person who understands number, evidence, tools, history, law, ecology, and care has more ways to interpret life before crisis narrows it.

Education can widen future room without guaranteeing unlimited mobility. A person may learn well and still face labor-market exclusion, discrimination, disability barriers, family duties, debt, local decline, migration pressure, or climate instability. Treating education as a private escape hatch allows public failure to remain intact while blaming those who cannot escape it (OECD, 2025a; Nussbaum, 2011).

Learning should also help communities solve problems that no isolated person can solve and recognize their own places as worth repairing.

Assess whether lessons remain useful in work, care, household judgment, civic participation, technical pathways, public reasoning, or later study.

Civic judgment requires practice. Students should learn to deliberate, revise, ask good questions, read budgets and evidence,

understand basic law, interpret public systems, and listen across difference. They should also learn that disagreement is not failure.

Civic judgment also includes media and information literacy. Students need to know how images, statistics, rankings, viral clips, machine summaries, rumors, and institutional statements can compress complex burden into a misleading signal. That knowledge helps them ask what evidence is missing, who was not heard, and what harm may be hidden by the cleanest story.

11.1.2 — Basic Skills and Advanced Paths

Foundational literacy, numeracy, communication, digital access, practical reasoning, and life competence give learners usable agency. They require support alongside advanced paths.

Complex medicine, engineering, law, science, climate adaptation, public finance, translation, arts, archives, infrastructure, agriculture, care systems, and governance require deep specialization. A trustworthy system opens these advanced paths widely (OECD, 2025a; World Bank, 2018).

Demanding programs, universities, and apprenticeships can deserve honor for the capability they cultivate; prestige distorts education when it follows scarcity of access instead.

Prestige distortion teaches families to buy security through signals rather than capability. Private tutoring, unpaid internships, elite networks, résumé inflation, legacy advantage, donor influence, and strategic extracurriculars can turn education into a costly arms race. Ambition is not the problem; the question is whether competition produces knowledge and competence or mainly converts parental resources into public status.

11.1.3 — Learning Outcomes and Their Limits

Disaggregating learning outcomes can locate a gap, but it cannot by itself identify the cause. A transport barrier, interrupted instruction, inaccessible materials, and household instability may

produce similar patterns. Use attendance and service records alongside appropriate comparison and local inquiry to test the proposed explanation. Fund a response to the identified mechanism, then examine whether both the barrier and learning outcome changed (OECD, 2025a; World Bank, 2018).

A learning measure becomes useful when it changes what a child is taught next. Pratham's Teaching at the Right Level approach reorganizes instruction around children's current learning levels rather than assuming that a grade-level syllabus matches everyone's preparation. Banerjee and colleagues (2017) followed a sequence of randomized evaluations as the approach entered government schools. Some early designs did not yield detectable learning gains, but their results informed two scalable versions that did. Evaluation became part of developing the institution, not merely a verdict delivered after it was built.

A local adaptation should specify how a demonstrated gap changes the next lesson and when progress will be checked again. Grouping should remain revisable rather than becoming a permanent judgment of ability. Learning records need privacy and an explanation accessible to families.

11.2 — Vocational Strength, Craft, and Useful Skill

Vocational dignity requires safe equipment, fair wages, sound supervision, and a credible path for progression, not just a course labeled practical. A program that exposes learners to danger, unpaid labor, discrimination, poor tools, or a dead-end credential shifts the burden onto those least able to refuse it (International Labour Organization, 2024a; Global Education Monitoring Report Team, 2026). Local builders, repairers, growers, caregivers, technicians, and maintainers can return capacity to places otherwise dependent on distant systems. Craft intelligence includes sensory and situational knowledge, such as noticing a machine's change in sound or knowing how a patient can transfer safely, that abstract descriptions may miss. Public support should recognize practical judgment as innovation while verifying that training leads to usable work.

11.2.1 — Technical Education Pathways

Apprenticeships, trade schools, community colleges, professional institutes, certification programs, work-based learning, union training, adult retraining, and technical high schools provide routes into the skills public systems need.

Before a learner commits, disclose wages, safety, transfer options, credential value, employer accountability, and routes into later study or leadership. Judge the pathway by the usable skill and bargaining power it develops.

Occupational expectations form early in education. Children hear which jobs are praised, mocked, feared, or treated as failure. They learn whether cleaning, caregiving, farming, repairing, cooking, driving, building, teaching, coding, nursing, researching, and managing carry different degrees of human respect (Kalleberg, 2011; Sen, 1999).

11.2.2 — Maintenance Skill

Maintenance training covers inspection, repair, cleaning, calibration, replacement, monitoring, documentation, adaptation, and preventive care across physical and digital systems.

Maintenance skill is learned through practice, supervision, repetition, failure memory, and responsibility for real systems. It deserves stable institutions, equipment, instructors, apprenticeships, and careers rather than emergency praise after breakdown (ISO, 2024a; Andrews, Pritchett, and Woolcock, 2017).

Training for maintenance must include both technical and institutional knowledge. Workers need tools, parts, documentation, authority to report risk, protection against retaliation, fair wages, and enough staffing to act before failure. A maintenance education that gives skill without institutional power may leave workers knowing what is wrong but unable to repair it.

Care skill should be taught, certified, paid, and respected without turning human tenderness into a checklist. Safe lifting, medication support, dementia communication, disability accommodation, child development, trauma awareness, end-of-life presence, and family coordination all require disciplined judgment. These skills help prevent dependence from becoming hidden suffering.

11.3 — Childhood Conditions and Fair Starting Stability

Starting stability begins before school, when nutrition, sleep, caregiver attachment, health, language, play, safety, and disability recognition shape later learning (Heckman, 2006; Cunha and Heckman, 2007). Children should not have to prove they deserve conditions adults failed to provide. Hunger or exhaustion may be mistaken for defiance or low motivation even in a child showing effort and ability. Predictable meals, stable housing, sleep-supportive schedules, and family security are therefore part of learning infrastructure. Education should respond to these needs without turning them into stigma.

11.3.1 — Housing and Neighborhood Stability

Neighborhood stability also shapes learning. A child may live with noise, violence, pollution, poor transit, limited parks, unsafe routes, weak services, or constant displacement pressure. The cost-and-capacity question asks whether educational burden is being misread as individual deficit when part of it belongs to place (World Bank, 2018; OECD, 2025a).

Housing and neighborhood movement should be recognized as instructional disruption. A child who changes homes, schools, routes, caregivers, or neighborhoods may lose records, friendships, quiet space, sleep, and continuity of attention. Portable support, careful records, privacy, transportation awareness, and stable handoffs reduce that loss.

Chronic exposure to violence, threat, instability, abuse, community danger, police fear, family crisis, disaster, displacement, or public humiliation can make immediate safety dominate attention and leave less room for learning.

Trauma-aware education should recognize that behaviors labeled defiance, distraction, or withdrawal can have different

causes, including fear and self-protection, and restore access to learning with structure, care, and consistent expectations.

11.3.2 — Parental Time and Care

Parental time is an educational condition because children develop inside relationships. A caregiver who has time to talk, read, cook, attend appointments, notice changes, help with homework, comfort distress, advocate at school, and rest is part of a child's learning environment. When labor systems consume parental time, education feels the loss.

Some caregivers have flexible work, paid leave, transport, health, language access, and confidence with institutions. Others have unstable schedules, multiple jobs, night shifts, caregiving responsibilities, disability, migration stress, debt, or past institutional harm. Schools that demand involvement without counting these differences may convert burden into blame (Addati et al., 2018; Sen, 1999).

11.3.3 — School Resource Inequality

Unequal staffing, facilities, learning materials, support services, transport, meals, and enrichment shape the opportunities behind a common educational standard.

A standardized expectation is credible only when the conditions for meeting it are reasonably supported. If one school has stable staffing, advanced courses, safe facilities, and rich support while another has turnover, crowding, deferred maintenance, and unmet need, comparison requires context (World Bank, 2018; OECD, 2025a).

Resource inequality is not only funding. It includes institutional stability, leadership capacity, teacher experience, community trust, administrative support, access to internships, cultural recognition, disability services, and extracurricular life. A well-funded school can still misuse resources. An underfunded school

can still hold extraordinary dedication. But dedication should not be used to justify deprivation.

Children's learning records, images, disability information, trauma histories, family circumstances, migration status, and religious participation need heightened protection against identification, coercion, lasting exposure, and future constraint. School dashboards, research reports, and donor accounts should show aggregate outcomes, methods, uncertainty, and correction without making individual children promotional proof (Nussbaum, 2011; Sen, 1999).

11.4 — Universities, Research, and Public Usefulness

Public usefulness does not mean political obedience. Universities must be useful partly because they can resist immediate popularity, donor pressure, government pressure, market fashion, and public impatience. Their usefulness depends on method, archive, critique, expert formation, and the courage to preserve difficult inquiry even when the result complicates a preferred story (National Academies of Sciences, Engineering, and Medicine, 2017; UNESCO, 2023). At the same time, autonomy is not immunity from burden review. If a university shifts cost to adjunct labor, students, neighborhoods, patients, public grants, or indebted families while speaking only in the language of excellence, it has hidden part of its social price.

11.4.1 — Inquiry and Truth-Seeking

Research should account for costs borne by participants, communities, students, staff, animals, ecosystems, data subjects, and future users. Producing knowledge does not justify exploiting vulnerable people.

A university's public-value claim should be separated into research validity, educational capability, and the conditions of its own labor. An expensive laboratory may be justified by long-horizon discovery that cannot be ranked by immediate commercial return; a low-cost program may still fail if students lack supervision or its essential staff are precarious. Evaluation should specify the knowledge produced, its reproducibility and access conditions, training and career outcomes, operating and maintenance obligations, and conflicts introduced by donors or vendors. Negative findings are contributions when methods are sound.

11.4.2 — Debt and Credential Pressure

Higher education can widen futures while placing debt at the doorway. Borrowing may finance valuable learning and still compress housing, family formation, occupational choice, entrepreneurship, or public-service work for years. A responsible account therefore follows both the capability gained and the obligation created, including completion risk, earnings uncertainty, interest, repayment design, and alternatives to debt-intensive credentialing.

Credential pressure grows when employers, professions, families, and institutions treat degrees as filters for seriousness even where actual work does not require them. This can create a credential treadmill: more schooling is required to stand still, while costs rise and risk shifts to students (Sen, 1999; National Academies of Sciences, Engineering, and Medicine, 2017).

The same educational price creates unequal burdens: savings and professional networks let one household absorb uncertainty that another must finance at interest (Global Education Monitoring Report Team, 2026; Sen, 1999). Alongside those debt and credential risks, follow completion, credential recognition, stable work, and debt serviceability under realistic earnings (World Bank, 2018; OECD, 2025a).

11.4.3 — Knowledge Access

Public health guidance, legal information, climate knowledge, education materials, technical manuals, archival records, research summaries, and civic explanations should be accessible enough to support public judgment. A readable account should let a resident identify which finding bears on a proposed local decision and which does not.

Knowledge access and expertise belong together. Specialized fields require training, method, and careful interpretation, while

public funding and public consequence create reasons for wider access, clear explanation, and accountable use. Wider access can broaden participation and open a path into expertise, but complex competence still takes time to develop (OECD, 2025a; World Bank, 2018). Supervised projects and public questions that researchers answer in their own terms can support that path.

Universities can support knowledge access through libraries, public lectures, open resources, repositories, accessible formats, community partnerships, translation, technical assistance, and adult learning. Fund staff time for both release and updating; preserve a correction history and visible supersession notices so obsolete summaries do not outlive their evidence.

11.5 — Youth, Status Pressure, and Enoughness Formation

Digital life can connect young people to learning, collaboration, and creativity while intensifying status comparison, exposure, compulsion, commercial attention capture, and surveillance. The relevant household question is which designs widen belonging and which monetize insecurity (UNESCO, 2023; OECD, 2018a). Protection cannot mean denying children the tools they need to understand and navigate modern life. Enoughness offers another measure of ambition, allowing chosen excellence without requiring constant display or credential escalation, as Chapter 2 develops.

11.5.1 — Status Anxiety

Status anxiety begins when dignity feels conditional on visible position. Young people may feel they must prove intelligence, attractiveness, productivity, popularity, moral identity, athletic ability, wealth, activism, creativity, or future success before they are allowed to rest inside themselves.

Accountability begins by asking who benefits from status anxiety. Schools may compete for rankings. Platforms may profit from attention. Brands may profit from insecurity. Universities may profit from credential pressure. Employers may benefit from overachievement. Families may transmit fear because they see the future narrowing (UNESCO, 2023; OECD, 2018a).

Status anxiety becomes a material issue when it changes sleep, spending, course choices, friendships, body image, family pressure, and willingness to take intellectual risks. A young person who lives inside permanent comparison may choose the safer-looking signal over the truer path.

Examine how scholarship rules, housing costs, debt and fear of borrowing, social-media pressure, and labor-market barriers

contribute to anxiety, rather than asking students to absorb institutional scarcity as private distress.

11.5.2 — Youth Future Confidence

Youth future confidence is a grounded expectation that adulthood offers enough stability, meaning, repair, and possibility to justify preparation. Without it, education becomes harder to believe.

A material-burden account treats future confidence as a social outcome. It is shaped by climate anxiety, debt pressure, housing prices, labor instability, war, public distrust, digital comparison, family stress, and visible institutional failure (UNESCO, 2023; OECD, 2018a).

Future confidence should also include agency. Young people need to see ways they can contribute without being asked to save everything alone. Climate, care, technology, public service, local repair, art, research, and craft can all become routes into meaningful authorship when the burden is shared across adults and institutions. When young people contribute, institutions should show what changed because of their work; visible consequence helps turn participation into credible agency.

Hope becomes credible when a young person can identify a real beginning: an apprenticeship, a supported service role, or a local repair project. The responsible institution should provide a mentor and bounded duties, then remain answerable when an opportunity falls through.

11.6 — Generational Stability as Peace Policy

Generational stability cannot depend on one successful official or funding cycle. Records must be maintained, and successors must be trained to use them. Stable funding, repairable buildings, and usable digital infrastructure keep commitments available after sponsors leave. Responsibility for renewal and correction must transfer with leadership (World Bank, 2018; ISO, 2024a; OECD, 2026a).

11.6.1 — Debt and Climate Inheritance

Debt and climate inheritance are among the clearest ways adults transfer burden to future people. Borrowing can finance repair that future generations benefit from. Climate investment can protect future life. Borrowing for current consumption can leave repayment obligations without a corresponding asset; deferred climate action can leave physical risks that later spending cannot fully reverse.

Across generations, the question is what future people receive in exchange for obligations imposed on them. If they inherit debt, did they also inherit infrastructure, health, education, climate resilience, housing, water systems, public capacity, and knowledge? Or did they inherit repayment obligations without usable assets (Sen, 1999; Nussbaum, 2011)?

Education should teach debt and climate as connected forms of time ethics. Students should understand budgets, borrowing, maintenance, emissions, adaptation, risk, and intergenerational cost without being made to carry guilt for costs imposed by earlier generations.

Underfunded schools, polluted land, broken housing systems, debt traps, racial or caste exclusion, disability neglect, destroyed local economies, climate exposure, lost language, and institutional distrust may carry the effects of earlier decisions.

Identify the relevant history and the repair needed, so symbolic acknowledgment does not substitute for an effective response.

Climate inheritance can be especially difficult to reverse because future people cannot renegotiate physics. Heat, water stress, disaster risk, insurance collapse, food instability, migration pressure, and ecological loss may arrive because present adults delayed. A future generation may be asked to adapt to conditions it did not choose and cannot cheaply reverse.

Because future needs are uncertain, build flexible, repairable, transparent systems that later generations can revise.

To protect future capabilities (Sen, 1999; Nussbaum, 2011), this book proposes a named person or office responsible for renewal, a funding plan, and triggers based on age or condition that allow time for procurement and training. Section 12.1 develops the maintenance account.

11.6.2 — Carrying a Cohort Through Institutional Change

An educational promise also extends across a cohort. Suppose a technical program changes its equipment halfway through a two-year course. New learners can begin with the revised sequence. Existing learners need a route that recognizes what they have already mastered and identifies the additional practice the new equipment requires. Treating both groups as if they were starting from the same place would waste learning already achieved.

A cohort transition plan can preserve completed work while making the change intelligible. Teachers map the old sequence to the new one, identify the skills that carry across, and provide a short bridge where practice differs. Learners can see what remains to be done and how the qualification will be completed. The record follows competence rather than merely preserving the name of a discontinued module.

The principle matters when a provider changes as well as when a course changes. A partnership between institutions can assign responsibility for completing the current cohort before recruitment moves elsewhere. Staff time and access to equipment must follow that responsibility. A receiving institution can then build on the learners’ work instead of making them repeat a history whose original administrator has departed.

Generational stability includes these modest continuities. Knowledge survives not only because it is stored but because a new teacher can recognize it and help the next person use it. The public inheritance is a functioning route from effort to competence, carried across institutional change.

Chapter 12 — Maintenance and Public Capability

Civilization depends on maintenance long before it notices maintenance. Water, bridges, public buildings, records, software, transit, drainage, clinics, and energy become visible mainly when they fail. By then, the failure may already have consumed school days, work hours, health, household savings, public trust, and the attention of emergency staff. Maintenance is the practice of keeping those losses from becoming ordinary.

New construction and public ambition can be beautiful, useful, and necessary. Their credibility depends on the operating, renewal, and retirement duties they create and the older systems they may displace; Section 12.1 develops that account. A completed repair creates knowledge as well as a functioning asset. The next crew benefits from knowing which part failed, which replacement fitted, and which apparent fix did not last. When that knowledge leaves with a contractor, the same physical fault can become a new investigation. Maintenance therefore raises a question about continuity of competence as well as the condition of things.

State capacity is judged by performance under limits. Can public systems deliver help, give effect to rights, repair infrastructure, protect privacy, resist corruption, train staff, respond in time, and explain error? Size and data volume cannot answer those questions. A powerful institution may be incompetent, and a small one may still be coercive. Capacity without surveillance directs knowledge toward service rather than control over people.

Land is the shared condition beneath housing, food, work, transport, ecology, memory, public facilities, and the ability to remain. Prices and property rights matter, but so do rent extraction, speculation, displacement, public stewardship, customary and Indigenous claims, taxation, ecology, infrastructure value, and local voice. The chapter examines who

benefits from land decisions and who bears their costs, drawing on legal, historical, ecological, and community expertise.

Transit, water, sanitation, waste, and energy are dignity infrastructure. Transit returns time; safe water protects the body; sanitation protects health and privacy; reliable energy preserves food, medicine, communication, care, and work. Neighborhood repair signals whether public systems can see and answer ordinary deterioration. These services form the material grammar through which peace becomes routine.

Regional balance asks whether places supplying labor, resources, and productive capacity receive durable services and investment in return; Section 12.5 follows the effects of extraction, industrial loss, and neglect (Andrews, Pritchett, and Woolcock, 2017; Rodríguez-Pose, 2018).

The Flint water crisis shows how technical failure, governance, structural inequality, records, and public voice can interact. The lesson is broader than intent: danger grows when warnings cannot travel, exposed communities are not believed, and correction is slower than harm. Public capability therefore includes testing, listening, maintenance, transparent escalation, independent review, and authority to stop a failing decision (EPA Office of Inspector General, 2018; Michigan Civil Rights Commission, 2017).

12.1 — Maintenance Before Spectacle

Maintenance is easier to postpone than new construction because an opening ceremony gives a project a visible champion. The benefits of renewal are often the failures that do not occur. Budget discipline therefore requires lifecycle reserves and condition assessments before systems visibly break. Public backlog records and clear responsibility for renewal make postponed obligations harder to conceal (ISO, 2024a; Flyvbjerg and Gardner, 2023).

12.1.1 — Maintenance as Civilization Work

The first requirement is an asset register that reflects reality. Institutions need to know what they own or operate, where it is, what condition it is in, who is responsible, which components are critical, and what replacement horizon applies. An incomplete register turns maintenance into emergency discovery.

Training and career ladders matter because maintenance skill takes time to build. Apprenticeships, certification, supervision, safe staffing, and knowledge transfer should be financed as part of the asset's lifecycle plan rather than treated as optional overhead (ISO, 2024a; International Labour Organization, 2026).

Physical and digital systems are increasingly connected. Sensors, software, authentication, remote control, and data storage can improve maintenance while adding cyber and vendor risk. Updates, access control, backups, incident response, and manual fallback belong in the same lifecycle plan (NIST, 2020; ISO, 2024a).

Decision records should preserve assumptions so later reviewers can see whether a lifecycle forecast was wrong or the maintenance promise was broken (Flyvbjerg and Gardner, 2023; ISO, 2024a).

12.1.2 — Maintenance Labor Dignity

Maintenance labor is knowledge labor because repair workers hold the memory of failure. They know which parts break, which forms confuse users, which vendors disappear, which workarounds have become unofficial policy, and which public promises depend on unpaid improvisation. Give that knowledge standing before design, procurement, staffing, and budget choices are made.

Maintenance labor also carries safety burden. Workers enter flooded tunnels, climb poles, repair sewers, patch roads in heat, manage hazardous waste, troubleshoot electrical systems, calm angry users, and keep aging buildings usable with equipment that may itself need repair or replacement (International Labour Organization, 2024a; Andrews, Pritchett, and Woolcock, 2017).

Fund maintenance pay, safety, equipment, staffing, advancement and worker voice.

A competent maintenance culture should include formal channels for workers to report risk without retaliation. A technician who sees a bridge risk, a clerk who sees a benefits failure, or a sanitation worker who sees a recurring hazard should not need personal courage to be heard.

Administrative maintenance includes record correction, cybersecurity patches, translation and accessibility coordination.

Publish backlog priorities, funding status, responsible teams, review dates, and reasons for deferral, with a route for residents to challenge priorities that normalize their neighborhood’s burden.

For each deferred repair, state the likely service disruption, safety risk, accessibility burden, equity effect and emergency cost of continued delay.

12.1.3 — Repair as Peace Practice

Repair can reduce daily confrontations produced by broken buses, unsafe crossings, flooded streets, lost records, unusable ramps, or repeated outages. It can remove unnecessary occasions for fear, blame, suspicion, and defensive private substitution.

Failure memory belongs in maintenance practice. Work orders, inspections, near misses, repeated patches, vendor changes, parts shortages, and staff warnings should remain traceable so the next team does not rediscover the same weakness after every turnover.

During staged repair, publish the condition and work sequence, protect access, and deliver promised updates.

12.2 — Service Capacity, Competence, and Response Time

Public capacity must be assessed together with rights, correction routes, and the burdens a system imposes. Weak delivery and coercive enforcement can both harm the people an agency serves (Andrews, Pritchett, and Woolcock, 2017; Fukuyama, 2013). A capacity register should distinguish service delivery, competence, response time, staffing, corruption resistance, privacy boundaries, and recovery from error. Each dimension needs its own evidence, followed by a review of how failures in one dimension undermine the others.

12.2.1 — Capacity and Response Time

Capacity analysis should distinguish authorized positions, funded positions, filled positions, available shifts, productive time, and the skill mix required for the work. A service may appear fully staffed on an organization chart while leave, vacancies, training gaps, turnover, broken equipment, or administrative load reduce usable capacity (Andrews, Pritchett, and Woolcock, 2017; World Bank, 2017).

Debates about institutional size become misleading when they treat all public functions as if they were identical. A flood response, a building inspection, a benefits appeal, a park repair, a water test, a procurement review, a disability accommodation, and a cyber incident do not need the same structure. They need competence appropriate to risk, proximity, expertise, privacy, and speed.

Competence also means knowing which tasks should not be centralized and which tasks cannot be safely outsourced. Some functions need local judgment; some need national standards; some need independent review; some need public ownership of data; some need private expertise under strict exit terms.

Start the response clock when the institution first receives enough information to act, rather than when it later classifies the case as urgent.

Support public-facing staff with escalation, supervision, recovery time and authority to resolve problems caused upstream.

12.2.2 — Corruption-Resistant Capacity

Corruption-resistant capacity requires transparent hiring, procurement, inventory, maintenance, scheduling and complaint routes, with beneficial-ownership review where lawful, conflict rules, worker protection, audit trails, separation of duties, appeal and appropriate role rotation. These controls need adequate staffing and enough lawful discretion to solve real problems without turning exceptions into informal favors or emergency shortcuts into a permanent system.

Alongside enforcement, timely and intelligible access, usable review, and protection against retaliation make lawful routes more workable and reduce opportunities for corruption created by institutional failure.

12.3 — Land, Rent, and Anti-Speculative Capture

A parcel can be a home, farm, watershed, sacred site, school site, transit corridor, inheritance, tax base, speculative instrument, habitat, or field of shared memory (Ostrom, 1990; Sen, 1999). Its uses carry overlapping claims rather than a single market value. Every land proposal should identify who gains stability, who loses place, who captures appreciation, and who bears ecological cost. Consultation, applicable rights and histories, and a correction route are necessary if actual effects differ from promises.

12.3.1 — Land as Shared Condition

Land concentrates power when access to location determines who can work, learn, receive care or remain in a community.

Property rights can support security, investment, stewardship, and independence. They are also exercised inside a larger system of access, nuisance, taxation, planning, infrastructure, ecology, and neighboring rights. Treating either private ownership or public purpose as absolute hides the burdens placed on others.

Ecological function is part of the material account. Floodplain, watershed, soil, heat, habitat, fire risk, and coastal change affect what can safely be built and maintained. Conservation decisions should state who bears restriction, who benefits, and how transition or compensation is handled (IPCC, 2023).

Public land requires stewardship rather than automatic disposal or indefinite retention. The governing body should state the public purpose, maintenance duty, access rule, review date, and conditions for reuse. Credible rights, responsibilities, monitoring, and conflict resolution support the governance of common resources (Ostrom, 1990).

12.3.2 — Public Land Stewardship

A parcel may look vacant in a spreadsheet while carrying ecological value, cultural memory, informal use, drainage function, future school capacity, emergency use, food potential, or affordable housing possibility. Selling such land for short-term revenue can appear efficient while destroying options that the public will later need to buy back at higher cost.

A public land register should identify parcels, current use, restrictions, ecological value, infrastructure connection, potential public uses, maintenance cost, revenue agreements, community input, and long-term burden. The register should be transparent enough for public review without exposing sensitive sites where protection is needed.

Public land stewardship should also include conditions on partners. A public-private project using public land should state affordability, access, maintenance, labor standards, environmental safeguards, data rules, anti-displacement protections, and public return before the agreement becomes hard to reverse.

Land decisions must hear tenants, workers, disabled residents, young families, elders, Indigenous communities, migrants, small firms, and service users alongside owners and developers, with participation support for those unable to afford the meeting.

12.3.3 — Competing Claims on Land

Expert disagreement should change the option comparison rather than expand the attendance list. Record the strongest housing, ecological, accessibility, and affected-community arguments against the preferred use. Identify which objections concern a binding right, which concern uncertain consequences, and which concern a contestable priority. The authorized body must answer each kind on its own terms; a favorable cost estimate cannot settle

a title or consent question. Section 28.7.2 illustrates a decision and appeal sequence, but land-specific rights and authorities must be established independently (OECD, 2023a; Rodríguez-Pose, 2018).

Identify who is harmed first if a land policy increases displacement, fails to produce housing, damages ecology, or traps public capacity (Nussbaum, 2011; Ostrom, 1990).

12.4 — Transit, Water, Waste, and Energy

Failure in transit, water, waste, or energy can interrupt access to housing, schools, clinics, and digital systems (World Health Organization and UNICEF, 2025; International Energy Agency, 2026a). Energy outages can disable water, transit, communication, cooling, and medical or waste systems; flooded streets can block transit and emergency services, and waste failure can create health hazards. Section 17.7.2 tests a combined failure. Service-specific records should reveal failure, dependency, repair time, and responsible authority.

12.4.1 — Transit as Time Return

Assess transit by the essential destinations people can reach within predictable time and cost, rather than vehicle or route totals alone.

Transit as time return depends on operational accessibility. Routes, stops, vehicles, information, payment, transfers, weather protection, lighting, and assistance determine whether mobility is real for disabled people, elders, children, caregivers, and people carrying work or care equipment (World Health Organization, 2022a; World Bank, 2024).

Land use and transit reinforce one another. Housing, jobs, care, and schools located far from service increase car dependence and travel time. Transit investment that triggers displacement can also remove the very households it was meant to serve. Planning should therefore connect service, affordability, and development.

Transit evaluation should include access to opportunity, total travel time, frequency, reliability, affordability, accessibility, safety, emissions, and distribution across neighborhoods. One average travel time cannot show whether the longest and most consequential journeys are improving (OECD, 2023a; Hallegatte, Rentschler, and Rozenberg, 2019).

12.4.2 — Energy Access

Energy access requires reliable delivered power, not merely installed capacity. Section 17.2.2 examines the network investments needed to make generation usable (International Energy Agency, 2026a; ISO, 2024a).

Energy access now includes reliability under climate stress. Energy transition without reliability can create backlash; reliability without transition can pass climate burden forward (International Energy Agency, 2026a; IPCC, 2023).

Energy access also requires attention to billing design. A household may have physical connection but still live under shutoff fear, arrears, predatory fees, confusing bills, or seasonal spikes.

The energy register must also cover demand peaks, vegetation management, cybersecurity, interconnection queues and the siting burdens of generation and transmission.

12.4.3 — Neighborhood Repair

Sidewalks, shaded bus stops, safe crossings, lighting, drainage, playgrounds, clean public restrooms, accessible entrances, and parks affect everyday safety and use of a neighborhood.

Neighborhood repair should include residents because they know the daily routes of burden. They know which crossing frightens parents, which bus stop floods, which stair blocks elders, which vacant lot becomes danger, which alley is dark, and which repair would return the most life.

Apply Section 12.1.2’s backlog record to residents’ reports so they can see which repairs followed and need not restart their case after staff changes.

A neighborhood route is usable only when its necessary links work together. A repaired pavement can lead to a crossing that someone still cannot use. An accessible bus stop can remain

unreachable from the homes it is meant to serve. The next repair may therefore be valuable because it completes a route, even when another isolated project would improve more square meters of surface.

Residents' journeys can provide a practical unit of planning. Follow a route from a home to a shop or public service and identify where continuity breaks. Then examine which small set of works would make the complete journey possible. The result may join responsibilities held by separate departments: one repairs the surface while another changes a crossing or removes a recurring obstruction.

This approach gives modest works a clear public purpose. Instead of reporting several disconnected completions, the authority can show that a particular journey has become usable. It can also identify the remaining link when most of the work is done. A route-based account turns local knowledge into an explanation for sequence and helps residents see what the next repair will unlock.

12.4.4 — Reliability Metrics Without Ranking People

The first step is to name the object precisely: a water system, bus route, clinic, lift, benefit pathway, power feeder, school facility, or digital service. A vague score called community reliability invites interpretation beyond what the data can support.

Every measure needs a definition, unit, period, geography, and data source. Uptime, interruption frequency, mean time to restore, repeat failure, missed trips, repair backlog, and coverage can each be useful, but they answer different questions.

Water quality, waste overflow and energy affordability require measures alongside reliability. Complaint frequency should be read as evidence about the route, not as a profile of the

complainant, and reliability data should never become an eligibility shortcut (OECD, 2023a; ISO, 2024a).

Public dashboards should show trends, definitions, data dates, and planned corrective work. They should avoid league-table design that invites moral comparison among neighborhoods or agencies with different duties and inherited conditions.

Averages should travel with distributions. A network can report acceptable average performance while a small group of users experiences repeated or extreme failure. Tail duration, recurrence, neighborhood variation, accessibility failure, and essential-service impact reveal burdens hidden by the mean.

Coverage matters as much as performance. A system may be highly reliable for the people it reaches while leaving others outside the network. Reporting should distinguish service quality from service availability so exclusion cannot appear as success.

Service-level objectives can make commitments testable. A target may specify restoration time, maximum interruption, accessibility response, or communication frequency; it should also state exclusions, uncertainty, and the repair required when the target is missed.

A mature reliability record separates leading from lagging indicators. Asset condition, overdue inspection, vacancy, parts availability, backup testing, and unresolved defects can reveal rising risk before an outage occurs. Outage duration, injury, service interruption, and emergency cost describe the consequences once a failure occurs.

12.5 — Regional Balance and the Repair of Abandoned Places

A region can carry closed mines, empty mills, lost ports, abandoned rail, degraded land, weakened schools, or healthcare gaps long after an economic shift (OECD, 2023a; Rodríguez-Pose, 2018). A national assessment should compare what the region contributes through taxes, royalties, labor, procurement, or land with the durable capacity and public investment it receives in return. That record must also track environmental liabilities, service staffing, and infrastructure siting rather than interpreting abandonment as local failure. Repair should protect the skills, apprenticeship networks, work traditions, and identity of declining, rural, remote, border, island, postindustrial, or overlooked urban areas without using heritage as a reason to forbid necessary change (OECD, 2026c; Rodríguez-Pose, 2018). Lost industry can weaken school funding and local enterprise, so a credible remedy must address the capabilities that disappeared as well as visible assets.

12.5.1 — Infrastructure Inequality

Infrastructure inequality appears when some regions receive reliable transport, broadband, water, energy, schools, clinics, and maintenance while others live with chronic delay and breakdown. Compare those differences with the geographic gaps in jobs and incomes discussed in Section 8.6.2, without treating any association as proof of one policy cause. Regional opportunity needs to be assessed across work, services, mobility, and maintained assets (OECD, 2026c; Rodríguez-Pose, 2018).

Infrastructure inequality can persist through maintenance schedules. A wealthy district may receive resurfacing, drainage, broadband, clinic renovation, and transit upgrades before failure

becomes severe, while a neglected region waits until breakdown is undeniable (IPCC, 2023; Rodríguez-Pose, 2018).

Regional repair should let communities shape priorities, reject or negotiate extraction, prestige projects, environmental risk, displacement, and donor conditions, and preserve memory without being accused of ingratitude.

Regional repair must ask who can speak with authority for a place. The anti-displacement safeguards in Section 12.3 apply when established owners or employers claim to represent residents who cannot afford to challenge them.

Regional repair should fund local institutions with authority to continue correction after outside attention moves on.

12.5.2 — Anti-Resentment Repair

Repeated neglect, humiliation, extraction, broken promises, and cultural contempt can make distrust reasonable. Repair earns trust through sustained, visible improvement.

Treat anger as information about burden, not as proof of every attributed cause (OECD, 2026c; Andrews, Pritchett, and Woolcock, 2017).

Repair should then separate the grievance into claims that can be tested: lost income or services, unfair treatment, status or cultural injury, perceived demographic change, and attributed causes. Each claim can require different evidence and a different remedy.

Terms such as backward, unskilled, resistant, low-information, or culturally deficient can obscure extraction and neglect. Repair plans should draw on local knowledge and fund communities' ability to shape and maintain the work.

Repair without patronizing also means telling the truth about harmful local patterns. Respect does not require romanticizing corruption, exclusion, racism, violence, environmental harm, or

failed leadership where they exist. It means criticism must be tied to repair rather than superiority.

12.6 — The Maintained Floor of a Fair Civilization

The ordinary floor must be reviewed as technology, climate, cost, health, work, housing, and public expectations change. A standard that once secured access may become unusable without updated maintenance or staffing. New ambitious projects cannot be evaluated independently of the service obligations they may displace. Periodic review should identify who bears the gap while a system is adapted.

12.6.1 — Floor Before Horizon

Ordinary service reliability makes ambitious projects less dependent on private rescue. When a horizon project competes with a failing floor, the decision record should state which essential obligation would be deferred, for how long, who would carry the interim burden, and what evidence justifies the trade. Ambition remains credible when it can explain why a new initiative outranks a known maintenance, care, housing, or access failure and when that priority will be reviewed (Sen, 1999; Nussbaum, 2011).

The floor-before-horizon rule is a constraint on preventable sacrifice, not a ban on every transition cost. Some maintenance, adaptation, or public-health measures must inconvenience people now to avert larger and less reversible loss. Such a proposal may proceed only through lawful authority and a public comparison that identifies the threatened floor, the people asked to carry interim cost, realistic less-burdensome alternatives, and the assistance required to keep essential access intact. A claimed future benefit does not by itself authorize present deprivation. The decision should make the affected group’s refusal or appeal meaningful, fund mitigation before disruption, and state the event that would require stopping or redesigning the project.

12.6.2 — Reliability Across Domains

Map the dependencies, propagation times, backups, and gaps that can turn one service failure into a household crisis; Section 17.7.2 provides a combined-system test.

Backup concentrated in wealthy areas leaves other regions exposed. Mutual aid, shared procurement, mobile teams, technical assistance, and investment in neglected places can shorten that repair gap (Andrews, Pritchett, and Woolcock, 2017; Rodríguez-Pose, 2018).

A visible lead, record, response clock, escalation route, and correction process spare residents the work of coordinating repair across agencies, vendors, levels of government, or service categories. A broken lift connecting a transit stop to a clinic needs one coordinating contact even when several organizations own the components.

Check whether restoration in one network depends on another network excluded from the first operator’s uptime figure.

Part IV — Money, Markets, and Essentials

Part IV begins where the domestic floor meets the institutions that keep it usable across time. The preceding Part established the practical conditions of work, care, housing, health, education, maintenance, and public capability. This Part asks whether money, markets, production, trade, and essential networks convert those conditions into durable room, or take that room back through instability, gatekeeping, and deferred cost.

Exchange, enterprise, investment, and price discovery can widen capability. This Part tests whether their costs and risks are visible, refusal is usable, and errors remain correctable.

The central question is whether production can sustain essentials without shifting its costs onto workers, ecosystems, public budgets, or future maintainers. An order for replacement equipment illustrates the junction. Financing can approve the purchase before a manufacturer has a production slot; the finished item can arrive before compatible components or installation skills are available. Each organization may have met its own commitment while the intended service still waits. The chapters follow those different constraints without treating money, productive capability, and delivery as substitutes for one another.

Money comes first because it is the language through which much public and private coordination becomes possible. It can widen future room, allow strangers to cooperate, and help households respond to change without pleading before a gatekeeper. It also carries a dangerous falsehood: the suggestion that an account balance, wage, credit score, or liquidity position reveals the worth of the person.

Production deserves the dignity usually reserved for invention because peace depends on the competence that keeps systems usable. Engineers, machinists, nurses, farmers, builders, coders, drivers, technicians, line workers, caregivers, inspectors, planners, and maintenance crews convert designs into shelter,

medicine, transport, food, energy, and repair. Redistribution without buildability eventually redistributes shortage; production without worker dignity manufactures a quieter form of insecurity (Hidalgo and Hausmann, 2009; Graham and Thrift, 2007).

Trade and supply chains widen the frame beyond the border. Trade can reduce isolation, stabilize access, share capacity, and create repeated cooperation that makes conflict less attractive. It can also hide forced labor, ecological damage, chokepoint vulnerability, dependency, sanctions spillover, shipping risk, and extraction from communities too distant to appear in the final price (Baldwin and Freeman, 2022; International Labour Organization, Walk Free, and IOM, 2022).

Every argument in Part IV eventually reaches the systems that keep people alive. Energy, food, water, shelter, medicine, sanitation, mobility, communication, waste systems, compute, storage, cybersecurity, and digital access are the material grammar of peace. Failure in any one can turn monetary stability or market abundance into an abstraction. The Part therefore closes where economic claims become physically testable in ordinary life.

Chapter 13 — Money, Credit, and Macro Stability

Money is one of the oldest public languages of trust. It lets strangers cooperate without becoming kin, lets households carry tomorrow in a usable form, lets firms coordinate labor and materials, and lets public institutions turn collective obligation into schools, clinics, roads, water systems, emergency support, pensions, wages, repair, and resilience. When money works well, it is quiet infrastructure for freedom.

This chapter traces the incidence, tradeoffs, and evidence in inflation, credit, banking, reserves, capital flows, exchange rates, and macroeconomic policy. Technical judgments remain answerable to economists, central bankers, financial supervisors, and affected publics.

Credit receives special attention because credit is future time made available now. It can build a house, start a firm, stabilize a farm, fund a clinic, repair a water system, bridge a household emergency, or finance productive transition. It can also seize tomorrow by fees, opacity, refinancing pressure, coercive collateral, variable-rate shock, or debt service that grows faster than the capacity it was supposed to build.

Banking visibility then becomes a difficult balance. Hidden ownership, shell routes, opaque leverage, illicit finance, political capture, and too-big-to-fail structures create public burden. But financial transparency without privacy discipline can expose ordinary people, overburden low-risk households, exclude migrants and poor customers, or convert financial systems into broad surveillance tools.

Inflation and currency instability bring the macro level down to the body. They change what a household can afford to eat, whether it can pay the rent, and whether care, medicine, schooling, or travel to work remains within reach. They can also strain confidence in public institutions. The same headline rate

can be an inconvenience for one household and a crisis for another.

Reserves, capital mobility, and macro defense appear later because societies also need buffers. A household without savings is exposed to emergency; a public system without reserves is exposed to shock; a country without adequate buffers may become vulnerable to currency pressure, import strain, capital flight, creditor dependence, or crisis bargaining. Yet buffers are not virtue medals, and vulnerability is not national shame.

Because credit creates purchasing power and leverage, banking stability depends on balance sheets, liquidity, capital, collateral, payment systems, supervision, and public confidence. The 2008 financial crisis showed how apparently dispersed instruments could concentrate risk through correlated exposures, short-term funding, and shared assumptions about housing and liquidity (Bank for International Settlements, 2026; Financial Crisis Inquiry Commission, 2011). A bank can hold long-lived assets while owing money payable today. If depositors or other short-term creditors demand payment before those assets can be sold, the timing mismatch can threaten its ability to pay.

13.1 — Money as Coordination and Social Trust

Money coordinates settlements across time and distance and preserves usable claims without assigning every exchange through a central authority, but that usefulness does not make it a measure of character or contribution (Bank for International Settlements, 2026; McLeay, Radia, and Thomas, 2014). Payment systems need timely settlement, accessible agents, and protection from arbitrary delay and hidden fees. In Kenyan household panel data, Jack and Suri (2014) estimated that negative shocks reduced consumption by about 7 percent among mobile-money nonusers while they did not detect a corresponding decline among users, who received more remittances from a wider network. Fees, the burden on senders, cash-out reliability, and functioning during a shared disruption still require attention because faster transfers move resources rather than creating them. Income and wealth can buy time, transport, privacy, education, healthcare, legal help, safer housing, and room to refuse a bad bargain without proving personal worth. Benefits offices, tax systems, and lenders must keep need, capacity to contribute, and repayment estimates tied to their distinct purposes; any use of creditworthiness beyond lending requires separate authority, relevance, correction, and appeal.

13.1.1 — Prices and Moral Judgment

Prices are compressed signals, not moral verdicts. A number may reflect scarcity, demand, quality, risk, public subsidy, market power, regulation, speculation, fear, bargaining weakness, or costs pushed outside the transaction. The same price can therefore carry several causes at once, and no invoice explains by itself whether the exchange is productive, coercive, or merely unavoidable.

Quality change can hide inside an apparently stable price. Smaller packages, reduced durability, slower service, narrower insurance networks, lower staffing, more advertising, and increased user labor can preserve the sticker price while transferring work and risk to the purchaser. A serious price inquiry therefore asks what quantity, quality, reliability, warranty, privacy, maintenance, and human support the buyer receives for the same nominal amount (Stiglitz, Sen, and Fitoussi, 2009).

A low price can result from genuine efficiency, better logistics, learning, and competition. It can also depend on suppressed wages, unsafe work, unpaid care, ecological depletion, public infrastructure, tax preference, forced migration, or maintenance deferred to another generation. The buyer sees affordability while the worker, community, treasury, or future maintainer carries part of the cost that the price leaves unnamed (Polanyi, 1944; Sandel, 2012).

Rising markups may reflect weakened competition, but cost structure, innovation, quality, risk, and measurement choices complicate the diagnosis. Chapter 14 develops the market-power inquiry (De Loecker, Eeckhout, and Unger, 2020).

Price transmission also has a sequence. A commodity shock can move through wholesale contracts, transport, inventories, retailer margins, wages, interest rates, public benefits, and household expectations at different speeds. Policy that responds to the final consumer price without locating that route may suppress demand where the original bottleneck cannot be repaired. Mechanism-specific evidence matters because the same headline inflation rate can arise from very different structures.

Subsidies and price caps can relieve immediate burden while creating other risks. Poorly designed support may favor high users, be captured by producers, weaken supply, conceal fiscal cost, delay conservation, or become difficult to withdraw.

Unbuffered liberalization can impose the opposite error by transferring transition cost to households least able to adapt. Policy must therefore state the objective, incidence, duration, financing, supply response, exit rule, and expected distributional consequence rather than treating any instrument as intrinsically fair.

13.1.2 — What Economic Indicators Can Show

Economic indicators are instruments of attention. They can reveal inflation, unemployment, output, wages, productivity, debt service, inequality, household cost, trade exposure, financial stress, and access to essentials. Their value begins with disciplined definition: what is counted, in what unit, for which population, over what period, and at what price basis.

Read average income alongside the distribution of gains (Stiglitz, Sen, and Fitoussi, 2009; OECD, 2024a).

Stocks and flows must also remain separate. Income is a flow, wealth is a stock, debt service is a flow, debt is a stock, and liquidity is the ability to meet obligations when due. A household can own an illiquid asset and still be unable to pay for medicine. A state can have productive assets and still face foreign-currency pressure. Conflating the categories turns analysis into false reassurance.

Nominal and real values create another common error. A wage can rise in currency terms while purchasing power falls. An asset can appreciate because supply is constrained, not because useful capacity improved. A budget can grow while maintenance need grows faster. Price adjustment, basket choice, base year, and local cost belong in the interpretation, not in technical fine print.

The release date and version of economic data matter. Revisions, lags, seasonal adjustment, informal activity, missing populations, and rapid shocks can change the picture.

Contemporary macroeconomic and financial claims should identify the release used and remain open to update rather than allowing a polished sentence to outlive its evidence (Bank for International Settlements, 2026; International Monetary Fund, 2026c; International Monetary Fund, 2026d). When later revisions materially alter the picture, the record should preserve both what decision makers could know at the time and what became visible afterward. That distinction permits learning without rewriting the evidentiary conditions under which the original judgment was made.

13.2 — Credit Creation and Productive Lending

Credit can turn future productive capacity into present purchasing power, but the promise carries an obligation that must remain serviceable. Its value depends on the use of funds, the borrower's position, and the terms on which uncertainty is assigned. Access rules can widen opportunity or lock people out of activity they could sustain. Data collection may help judge repayment while creating separate privacy and discrimination risks. These questions need different tests, even when they arise in the same lending decision.

13.2.1 — Credit as Future Capacity

A lender or banking system supplies purchasing power now in exchange for a future claim on income, production, tax capacity, collateral, or another source of repayment (Minsky, 1986; McLeay, Radia, and Thomas, 2014).

The decisive question is what the borrowing leaves behind. Credit that finances a durable home, productive equipment, preventive care, water repair, training linked to real work, or an energy upgrade may create capacity that helps carry its own repayment. Credit that merely bridges low wages, inflated rent, medical exclusion, speculation, or a recurring budget hole may offer temporary relief while tightening the next period’s constraint.

Loan size or interest alone cannot establish productive value: a small vehicle-repair loan may preserve employment, while a large low-rate loan may inflate land values or fund prestige without maintenance. Compare the capacity remaining after fees, risk, displacement, upkeep, and repayment.

When a bank makes a loan, it generally creates a matching deposit; it does not simply lend a fixed stock of prior deposits. This expands financial claims, not the stock of workers, materials,

or equipment. Lending remains constrained by profitability, capital and liquidity requirements, settlement needs, borrower demand, and monetary conditions. Borrowers gain purchasing power and an obligation; the material test is what that purchasing power finances and who carries the risk (McLeay, Radia, and Thomas, 2014).

Credit cycles create collective risk even when each contract appears defensible in isolation. Rising asset prices can increase collateral, invite more lending, and validate the optimism that produced the rise, while a reversal destroys collateral and credit together. Macroprudential tools, stress tests, countercyclical buffers, underwriting standards, and sector-specific limits try to interrupt this feedback before private leverage becomes a public emergency. Their use should remain proportionate, transparent, and open to revision.

Small firms illustrate the information problem. They often lack the standardized records, marketable collateral, and long credit history available to larger borrowers, even when they possess local knowledge and viable customers. Relationship lending, public guarantees, cooperative finance, cash-flow underwriting, and patient equity can reduce that gap, but each creates governance and capture risks that must be disclosed (Berger and Udell, 1998).

Working capital makes the interval between doing useful work and receiving payment visible. Imagine a workshop with an order priced at 120 units. Materials require 40 units before production, and wages and other operating payments require another 50 before the buyer pays. The order leaves 30 units before financing and any further costs, yet the workshop must first find 90 units. A positive eventual margin does not supply the cash needed to reach it.

A buyer’s deposit of 30 units reduces that funding gap to 60. A supplier’s agreement to receive 20 units after delivery would reduce the workshop’s early cash requirement again, while moving a financing obligation to the supplier. A loan can bridge part of the interval, but its repayment date needs to follow a realistic receipt date. Each arrangement changes who carries the wait. Following those dates reveals how a viable order can become financeable.

The calculation also distinguishes financing from demand. Borrowing can help the workshop complete an agreed order. It cannot establish that an additional product will find a buyer. If the order’s specification is unresolved or acceptance depends on an untested feature, the uncertainty concerns the sale itself. A lender and a producer need to understand which part of the plan is a timing gap and which part remains a commercial question.

Good coordination can reduce the financing required without increasing anyone’s total revenue. A clear acceptance procedure may shorten the interval before payment. Staged delivery may allow part of the order to be paid before the rest is complete. These are productive contractual choices because they align obligations with completed work. The capacity gained comes from making a viable sequence easier to carry, not from treating credit as a substitute for production.

Climate and infrastructure risks complicate the credit horizon. Flood, fire, heat, insurance withdrawal, energy transition, and changing regulation can alter the value or operating cost of an asset long before the loan matures. Lenders, borrowers, insurers, and public authorities should not hide these exposures in optimistic appraisal. Nor should risk models become a pretext for abandoning whole communities without financing adaptation and repair.

13.2.2 — Credit Access and Exclusion

Exclusion from credit is also a burden. Communities, small firms, farmers, students, households, cities, and regions can be denied useful capital because of discrimination, weak collateral, poor infrastructure, prior extraction, data gaps, local decline, or risk models that reproduce past injustice.

Access also requires guarding against overcorrection. Opening credit to excluded communities without preventing predation can repeat historical harm under the banner of inclusion. The fair question is not how many loans are approved, but whether the approved credit leaves people and places with more future room.

Underwriting is legitimate because lending requires risk assessment. A lender, public bank, cooperative, agency, or community fund must ask whether a borrower can repay, whether the financed project is sound, and how loss will be carried. Risk review is not inherently unjust.

Risk review should explain its criteria, account for hardship, and avoid unnecessary exposure or judgments about character (Demirgüç-Kunt et al., 2022).

Underwriting also needs to distinguish inability from exploitation. A loan that cannot be sustained should not be approved merely because fees, collateral, or refinancing make failure profitable to the lender. Honest refusal can protect dignity when reasons are explained and safer alternatives remain available.

13.2.3 — Credit Data and Privacy

Credit data is powerful because it travels. Payment history, debt balances, income, employment, location, assets, court records, banking behavior, utility payments, and purchasing patterns can be used to evaluate risk. They can also become a total profile of a person.

Borrowers should be able to inspect the records used in a lending decision, challenge inaccuracies, and learn when stale information expires.

Alternative data can widen access for thin-file borrowers, but rent, utility, cash-flow, platform, device, location, education, and transaction records may reproduce inequality, freeze temporary hardship into a durable label, or reveal intimate behavior unrelated to repayment. State the purpose, test predictive value and bias, minimize collection, and bar unrelated reuse (Nissenbaum, 2010; Solove, 2006).

A person denied credit needs more than the statement that a model made the decision. Adverse-action explanations should identify the principal factors, distinguish inaccurate data from legitimate risk, and provide a realistic path to correction or human review. Explanation should not be a technical recital that no ordinary applicant can use. Its function is to make the decision contestable enough that error, drift, and misuse can be repaired.

Model governance continues after deployment. Lenders need to monitor data quality, performance across relevant groups, changing economic conditions, security incidents, override patterns, complaint outcomes, and the consequences of updates. A model that behaved acceptably in one credit cycle may change after inflation, unemployment, disaster, migration, or a shift in the applicant pool. Periodic review and retirement are therefore part of responsible underwriting, not evidence that automation has failed.

Small-business finance needs an additional boundary because enterprise and household life often overlap. Owners may be asked to expose family assets, household accounts, platform activity, tax records, or personal guarantees to obtain working capital. Some linkage can be relevant to risk, but relevance does not authorize unlimited access. Information collection should be

limited to the lending decision, retained only as long as needed, protected from unrelated reuse, and subject to correction when a business request would otherwise become an open-ended household search.

13.3 — Banking Visibility and Anti-Capture Finance

Financial visibility is useful only when it connects formal transactions to the people and institutions exercising power. An account name may identify the legal holder without revealing control of the underlying activity. Investigations must also respect the rights of people who are not involved in wrongdoing and the limits of available evidence. A complete design therefore asks what can be verified, which burdens a reporting rule imposes, and how mistaken inferences can be corrected. The section considers ownership, anti-money-laundering duties, and the strength of the resulting evidence as separate questions.

13.3.1 — Beneficial Ownership and Financial Route

Beneficial ownership asks who actually controls or benefits from an entity, asset, account, contract, property, or financial route. The formal owner may not be the real power. Shell companies, nominees, trusts, affiliates, layered ownership, and offshore structures can obscure control.

Hidden ownership becomes a public problem when it controls public consequence. Procurement, land, housing, infrastructure, political funding, public contracts, energy systems, media, platforms, and essential goods require enough beneficial-ownership visibility to identify who exercises control and who ultimately benefits (Financial Action Task Force, 2023; van der Does de Willebois et al., 2011).

Banking instability reaches beyond shareholders when payment systems, deposits, payroll, credit and public services depend on continued access (Financial Crisis Inquiry Commission, 2011).

Losses are rarely carried evenly. Large actors may diversify, hedge, negotiate bespoke terms, or exit before a danger becomes public; depositors, workers, suppliers, and small firms often learn of the risk after their choices have narrowed. Capital, liquidity,

resolution planning, deposit protection, operational continuity, and clear public communication help protect people who cannot monitor a bank like a professional.

Stability language also deserves scrutiny. A genuine run risk may require careful sequencing, but an institution can invoke confidence to resist disclosure or reform that threatens its position. Independent supervisory analysis and scenario testing should explain why a safeguard protects payment and credit continuity rather than merely an institution's prestige.

13.3.2 — Anti-Money-Laundering Burden

Anti-money-laundering systems respond to real harms. Hidden financial routes can carry corruption, organized crime, trafficking, sanctions evasion, terrorist finance, tax crime, and theft from public institutions. Financial secrecy is not victimless merely because the transfer appears as data on a screen, and capable investigation is part of protecting the material floor (Basel Committee on Banking Supervision, 2020; World Bank, 2016).

De-risking is a warning sign. When a bank exits an entire region, correspondent relationship, remittance corridor, nonprofit class, or customer category because compliance appears expensive, the apparent reduction in institutional risk may transfer burden to people who still need lawful payment routes. Activity can move to cash, informal intermediaries, or less supervised systems, making both inclusion and enforcement harder (World Bank, 2016).

Sanctions and conflict contexts create a special danger of overcompliance. Banks and payment providers may block lawful humanitarian, family, or commercial transfers because the penalty risk is easier to avoid than to interpret. Clear licenses, authoritative guidance, escalation channels, and protected

humanitarian routes are therefore part of credible financial integrity rather than exceptions to it.

Evaluate whether fees, distance, digital and language barriers, identity rules, or sudden closure make accounts, payments, remittances, savings, insurance, credit, cash, and human assistance unusable (Demirgüç-Kunt et al., 2022).

Assess both the burdens of anti-money-laundering controls and the harms they aim to prevent. Count institutional compliance costs, customer delay or denial, privacy exposure, disrupted remittances and aid, correction time, and displacement into less visible channels. Also assess how weak enforcement exposes people to stolen assets, captured procurement, organized violence, and lost public revenue.

13.3.3 — What Finance Evidence Can Establish

Financial evidence can describe claims, flows, leverage, maturity, liquidity, capital, concentration, ownership, losses, and exposure. It may show that a balance sheet is fragile, funding is short-term, a route is opaque, or a shock is concentrated. It cannot establish motive by itself, show that every actor understood the structure, or select the fairest remedy without additional legal, institutional, and distributional analysis.

Credit growth may finance investment or speculation; a rescue may prevent collapse while preserving weak governance; inflows may expand capacity or create reversal risk. A causal claim must distinguish those mechanisms against a stated counterfactual, timing, and identification strategy.

Stress tests use specified scenarios to examine how a financial system could respond to adverse conditions. They do not predict which conditions will occur. Their value lies in exposing sensitivity to interest rates, exchange rates, defaults, margin calls, deposit flight, asset-price declines, cyber events, or funding

interruption. Their limits lie in model structure, behavioral response, network effects, and shocks that were not imagined (Bank for International Settlements, 2026; International Monetary Fund, 2026b).

13.4 — Inflation, Currency Stability, and Public Trust

The IMF's April and July 2026 outlooks show that global price pressures do not reach every economy in the same way (International Monetary Fund, 2026c; International Monetary Fund, 2026d). Commodity shocks and war can raise the cost of essentials abruptly. Technology demand, expectations, and financial conditions can shift relative prices and borrowing costs through different routes. Households experience those changes through their own consumption patterns and exposure, which national averages may hide (Jaravel, 2021).

13.4.1 — Inflation as Household Pressure

Inflation becomes most tangible when prices reach essentials. Food, rent, fuel, electricity, medicine, transport, childcare, insurance, school supplies, and basic repairs are not luxuries. When their prices rise faster than income, households lose room, and the composition of spending becomes as important as the headline rate. A moderate average can coexist with severe pressure when essential categories rise faster than the basket, while exposure differs with wages, benefits, debt structure, homeownership, geography, and available buffers (Jaravel, 2021; Auclert, 2019).

National averages can hide a geography of inflation. Rural households may face transport and fuel pressure; cities may face rent and childcare pressure; islands and remote regions may face freight and import exposure; drought or crop loss may reshape local food prices. Regional baskets and access conditions are therefore part of the evidence, not optional color.

Price change is not itself unfair. The burden becomes unfair when essential shocks move through systems that leave some people without buffers, substitutes, bargaining power, timely support, or a protected route to relief. The material question is

where the shock is stored, how long it remains there, and who finances the adjustment with health, time, or future choice.

13.4.2 — Currency Trust

Currency trust is the belief that money will remain usable across time. People accept wages, hold savings, lend, borrow, price goods, sign contracts, and plan budgets because they expect the currency to retain enough meaning for tomorrow.

When currency trust weakens, people may spend quickly, shift into foreign currency, buy physical goods, demand shorter contracts, raise prices preemptively, move capital, or avoid long-term commitments. Risk premiums may rise, and people may retreat into less fair or less safe informal arrangements. These responses can reinforce the loss of trust (Bank for International Settlements, 2025; Sargent, 1982).

Anti-inflation tools carry burden of their own. Higher interest rates may reduce demand or anchor expectations, but they can also raise mortgage costs, strain borrowers, slow investment, increase unemployment risk, pressure public budgets, and affect countries through exchange rates and capital flows.

13.4.3 — Wage and Price Dynamics

Wage and price dynamics are complex because wages are income for workers and cost for firms at the same time.

Workers should not be blamed reflexively for inflation. When wages lag behind food, housing, transport, energy, or care, attempts to restore purchasing power are often defenses of the material floor rather than evidence of irresponsibility. Wage growth may interact with prices, productivity, bargaining power, markups, and expectations, but its moral meaning cannot be inferred from a headline rate (International Labour Organization, 2024b).

Wage changes may also reflect labor scarcity, minimum standards, collective action, monopsony or delayed correction after years of suppressed pay.

Measures that stabilize the headline index but abruptly raise housing insecurity or local-government financing pressure need explicit mitigation and independent review. A central bank, fiscal authority, wage-setting institution, and public-service provider have different mandates; coordinated analysis does not erase their independence. Report the dated assumptions and the conditions under which the preferred mix would change, rather than presenting one macroeconomic tool as universally fair.

13.5 — Reserves, Capital Mobility, and Macro Defense

Macro defense requires real buffers and explicit tradeoffs because reserves, capital-flow tools, exchange-rate choices, emergency financing, and controls can protect continuity while also altering prices, debt, freedom, migration, and public trust. These measures need defined risks, oversight, and exit conditions. Buffers can take many forms, from foreign exchange and fiscal reserves to emergency credit, food stocks, deposit insurance, grid capacity, and trained response teams; each covers a different interruption. Holding them diverts resources from housing, care, education, infrastructure, or debt relief, with the tradeoff shaped by how each buffer is financed and maintained. Adequacy therefore depends on the risk insured, the carrying cost, the reliability of alternatives, deployability before hardship deepens, and a public explanation of the chosen level.

13.5.1 — Capital Flight Risk

Capital-flow measures should distinguish residents moving funds abroad from nonresidents withdrawing funding or reducing new inflows. Forbes and Warnock (2012) classify unusually large changes in gross outflows as flight or retrenchment and changes in gross inflows as surges or stops. These episodes can interact, but they are not one transaction.

Capital flight exposes an asymmetry of escape. Mobile wealth can protect itself from uncertainty, while departure may weaken the currency, raise import costs, reduce investment, strain banks, pressure public budgets, and intensify debt burdens for people who cannot move their assets or lives. The issue is whether those costs are privately borne or shifted onto those with the least exit capacity. Distinguish lawful protective transfers from tax evasion and coercive exit threats (Calvo, 1998; Ndikumana and Boyce, 2011; Ostry et al., 2010).

Capital-flight prevention begins with the conditions that make flight rational: arbitrary policy, corruption, unsafe financial routes, weak legal protection, unstable currency expectations, opaque liabilities, and poor explanation. Protective instruments may still be necessary, but they should be reviewed for effectiveness, duration, evasion incentives, and the burdens they place on ordinary transactions.

A useful exposure record separates the amount leaving from the obligations that remain. A firm earning local currency but owing foreign-currency debt may face a larger repayment burden after depreciation, while a supplier waiting for imported parts may face delay and higher cost. The same outflow can therefore reach production and households through different routes.

For a hypothetical purchase on otherwise identical terms, one supplier quotes US$100,000 and another 1.05 million local currency units. At 10 local units per dollar, the dollar quote costs 1 million; at 11, it costs 1.1 million. The quotes are equal at 10.5. This is a sensitivity test, not a forecast: conversion timing, who bears the exchange-rate risk, and hedge terms still matter.

Currency defense uses imperfect instruments: liquidity support, reserve use, debt management, prudential measures, communication, interest-rate changes, fiscal adjustment and narrowly designed flow measures. Officials should identify the channel each tool addresses, who may be harmed, the alternatives, review date and correction route.

13.5.2 — Controls and Freedom Risks

Emergency controls on capital, withdrawals, foreign exchange, prices, or transactions require unusually strict safeguards. Each restriction needs a stated risk, defined scope, duration, evidence for continuation, appeal route, and accountable reviewer. Exceptions must protect medicine, food, remittances, family

support, small firms, and essential imports. A temporary tool becomes unfair when opacity, indefinite renewal, or selective enforcement turns stabilization into unreviewable control (International Monetary Fund, 2022; Ostry et al., 2010).

Controls can trap savings, block lawful trade and remittances, privilege insiders or shift activity into black markets.

Dependence on foreign creditors, reserve currencies, imported essentials, payment systems, technology, or security support can become leverage when an outside actor can interrupt an essential without bearing comparable loss (Ostry et al., 2010). Trace the specific route: narrowed policy space, terminated platform access, withheld parts, amplified foreign-currency debt, or obstructed medicine.

The risk increases when dependency is concentrated, opaque or unbuffered, or when creditors, suppliers or insurers can change terms faster than local people can adjust.

13.5.3 — Macro Defense Without National Ranking

Macro indicators describe operating conditions and exposure; they are not rankings of a population's worth or maturity.

The same number can carry different meanings in different systems. A reserve stock adequate for an economy with limited foreign-currency debt may be fragile where refinancing is concentrated, import dependence is high, capital flows are volatile, or contingent liabilities are large. Context is not an excuse added after the fact; it is part of the object being measured (Bank for International Settlements, 2026; International Monetary Fund, 2026c; International Monetary Fund, 2026d).

The diagnostic also separates liquidity, solvency, and real-resource shortage. A temporary loss of market funding calls for different tools than an insolvent balance sheet, and neither is repaired by treating scarce fuel, food, transformers, medical

supplies, or skilled workers as purely financial abstractions. Interest rates, swap lines, guarantees, controls, fiscal support, and reserve use reach different channels. Policy becomes more credible when it names the mechanism under repair and the burden each instrument moves.

A policy that appeared successful under a favorable commodity cycle, reserve-currency privilege, abundant external finance, or an unusually mild shock cannot be exported as a universal recipe. Analysts should name the starting balance sheet, exchange-rate regime, debt currency, institutional capacity, distributional structure, external conditions, and time horizon before drawing lessons.

Tie macro indicators to bounded, revisable decisions: a foreign-exchange shortage may warrant import financing, with a stated threshold, affected groups, review date, and evidence that would reverse the response (Stiglitz, Sen, and Fitoussi, 2009; Sen, 1999).

13.6 — Finance in Service of the Real Economy

Finance should be judged by whether it expands productive capacity or crowds it out, how failure costs are shared, and whether borrowers and communities retain meaningful alternatives. This standard does not require every investment to be immediately tangible. Research, insurance, liquidity, hedging, software, training, and long-horizon risk pooling can all support real capacity. Their contribution and the burdens they create should be traceable and open to review.

13.6.1 — Speculation Versus Capacity

Speculation can perform real functions through price discovery, liquidity, risk-bearing, and investment in uncertain futures. The problem begins when price play detaches from capacity and burdens essentials. For housing, commodities, land, and platforms, assess how speculation affects useful capacity, access costs, and dependence. The relevant distinction is between risk-taking that finances useful possibility and extraction that profits from dependence (Arcand, Berkes, and Panizza, 2015; Cecchetti and Kharroubi, 2015).

Material peace requires patient capital. Infrastructure, housing, climate adaptation, care systems, public health, education, regional repair, water, energy, and maintenance often produce benefits over long horizons. In infrastructure, short-termism in capital markets can restrict institutional investors' ability to provide long-term finance (Della Croce and Yermo, 2013).

Patience must be governed as carefully as speed. Long horizons can conceal cost overruns, favoritism, greenwashing, weak maintenance, or burdens imposed on communities that lack bargaining power. Patient capital becomes capacity-building when project records include lifecycle cost, labor standards, local

consequence, maintenance finance, milestones, correction, and evidence that the asset can carry the claim placed upon it.

13.6.2 — Care and Social Infrastructure Finance

The financing problem is distinctive because many returns do not appear as a direct cash flow. Good care may prevent hospitalization, reduce caregiver exit from work, support childhood development, stabilize a household, shorten crisis, or preserve independence. These benefits are real but distributed across families, employers, health systems, schools, public budgets, and future years, which makes private capture of the full return difficult (Addati et al., 2018).

Different care needs require different funding logics. Universal services can spread risk and simplify access; social insurance can pool predictable life-course needs; targeted support can direct resources where burden is greatest; community and cooperative models can preserve local authorship. Each arrangement has tradeoffs in taxation, eligibility, administration, choice, continuity, and political durability. The financing architecture should be judged by the care people can actually receive, not by institutional label.

Geography matters. Dense urban markets may support several providers while rural and remote regions face distance, small volumes, workforce shortage, and high travel cost. Equal formal eligibility can therefore produce unequal actual access. Financing should include transport, mobile provision, regional workforce strategies, housing for staff, and cross-provider backup where ordinary market density is absent.

The capital stack should match the kind of care being financed. Land and buildings may support long debt; equipment may require leases or renewal reserves; operating care needs recurring revenue; workforce development may need grants or public

investment; emergency capacity may require standby funding. Combining unlike needs in one financing instrument can make the service appear viable while leaving its most important cost unfunded.

Insurance can pool rare and expensive risks, but only when coverage remains intelligible and usable. Exclusions, narrow networks, prior authorization, delayed reimbursement, and complex appeals can transfer risk back to households and providers. The financing record should therefore distinguish nominal coverage from timely access to the care the contract claims to support.

Care’s value includes independence and participation even when it does not raise measured output (Sen, 1999; Nussbaum, 2011).

Chapter 14 — Market Power and Rent Extraction

Market power can impose rent pressure, unavoidable fees, one-sided terms, platform lock-in, employer dominance, or exclusion without visible violence. Its burden is sharpest where refusal threatens shelter, income, access, or participation.

Market analysis becomes useful when it follows power through observable mechanisms. Can either side refuse? Are alternatives technically present and economically usable? Who waits because access is controlled? Which costs are shifted into labor, suppliers, public infrastructure, or future maintenance? When does scale become a gate over shelter, credit, land, platforms, data, or essential services? These questions locate extraction without turning the whole market into a moral caricature.

Rent extraction is return tied chiefly to control over a gate rather than to new capacity. The chapter tests how that mechanism appears in monopoly, land, housing, finance, platforms, and inherited power while preserving legitimate profit, risk-bearing, labor, and stewardship.

Market power becomes especially dangerous around essentials. A person can refuse a luxury more easily than shelter, medicine, water, energy, food, transport, communication, banking access, legal access, or work. When refusal is weak, price and contract terms no longer express ordinary choice. They reveal dependency. The comparison with discretionary goods guides remedy design: essential access must remain usable while the specific source of market power is addressed.

Inheritance begins in a legitimate desire to protect descendants from instability. Public concern arises when inherited advantage hardens into durable control over land, firms, education, politics, media, platforms, or institutional access. Family continuity can be respected without treating hereditary command as politically neutral. The task is to preserve care and stewardship while

keeping entry, mobility, and public power open to those born outside established networks.

Market power changes the meaning of a price. Under robust competition, price can carry information about scarcity, quality, and preference. Under durable dominance, it can also reflect exclusion, switching costs, control of a platform, ownership of land or data, or the ability to wait out smaller rivals. Research on rising markups and superstar firms suggests that concentration can alter labor shares, investment, innovation, and the distribution of bargaining power even when consumers still see abundant products (De Loecker, Eeckhout, and Unger, 2020; Autor et al., 2020).

Scale can create genuine efficiencies, fund research, lower unit costs, and coordinate complex networks. Anti-extraction policy therefore cannot punish success merely for being large. It examines contestability, interoperability, entry, labor conditions, acquisition patterns, essential facilities, political influence, and whether wealth can convert itself into rules that prevent challenge. Contestability should be checked at renewal and exit, where switching costs reveal whether apparent choice remains usable after dependence has formed (Piketty, 2014; Besley and Persson, 2011).

14.1 — Productive Wealth and the Limits of Domination

Wealth can become stewardship when owners train workers, maintain assets, support research, accept patient returns, and build durable institutions, so blanket suspicion would destroy initiative while unaccountable prestige could conceal harm (Mazzucato, 2018; Piketty, 2014). Private success also rests on roads, schools, law, stable money, public research, worker health, care, and trust without reducing the value of enterprise or risk. Building a firm, maintaining a reserve, or saving for children does not turn an owner into a public ruler. The accountability question changes when concentrated assets govern access to housing, employment, politics, contracts, information channels, platforms, or essential infrastructure. As the number of people unable to refuse a decision grows, responsibility must become more visible and enforceable.

14.1.1 — Productive Wealth and Control Rights

Productive wealth can be assessed through useful output, reliable pay, safety, skill development, and repaired defects. A capacity register should also identify the public conditions supporting production, genuine investment risks, labor and ecological costs, future maintenance obligations, and whether the firm depends on coercive relationships (Mazzucato, 2018; Sen, 1999).

Business categories can hide public power. A vendor becomes infrastructure when no practical substitute exists; a landlord becomes a regional gate when housing alternatives disappear; a platform becomes a regulator when it controls discovery, payment, reputation, and access. The legal label may remain private while the material consequence becomes public.

The distinction turns on the rights bundled with an asset. Ownership may include use, income, exclusion, transfer, voting,

data access, platform governance, patent control, or the ability to set terms for dependent users. A return can be proportionate while one of these control rights becomes excessive because people cannot reasonably exit.

14.1.2 — Wealth Visibility Where Consequence Is Public

Differentiated scrutiny is more useful than theatrical exposure. Beneficial ownership, conflicts of interest, procurement trails, lobbying routes, major political spending, dominant platform rules, and essential-service control may require reviewable public information. Intimate spending, family life, medical choices, private grief, and ordinary household use do not become public property because a person is wealthy (Piketty, 2014; Financial Action Task Force, 2023).

A useful comparison is between two successful suppliers. One earns more by improving quality, training workers, and reducing repair costs. The other earns more because dependent customers cannot transfer their records or obtain compatible parts. Both may report higher profit, but the second case requires scrutiny of the barrier to exit. Mixed cases require separating the useful service from the avoidable toll.

14.2 — Monopoly, Oligopoly, and Civilizational Drag

Concentration can be hidden by a crowded storefront when common ownership, suppliers, data bottlenecks, payments, or standards still restrict entry and switching (De Loecker, Eeckhout, and Unger, 2020; Autor et al., 2020). A credible market definition identifies the product, geography, time horizon, substitutes, entry barriers, data rights, network effects, regulations, and the real cost of exit. National competition figures may conceal local monopoly, and apparently broad digital markets may depend on a single gate. Civilizational drag includes time lost by workers retraining under imposed changes, small firms paying to reach customers, agencies locked into vendors, and households navigating extra fees that no price index captures. Dynamic review also asks whether acquisitions preempt future rivals or conduct lowers quality, squeezes suppliers, captures data, blocks interoperability, or prevents entry before prices change. Remedies should depend on evidence that a feasible excluded alternative would reduce burden, not on novelty, firm count, or political appearance alone.

14.2.1 — Pricing Power Over Essentials

Market power over an essential good is not proved by the good's necessity alone. A regulator must identify the relevant alternatives, barriers to entry, cost and quality differences, consumer switching constraints, and the source of the seller's markup. A high price may finance safe production or reflect genuine scarcity; it may instead exploit control of land, a network, intellectual property, information, or an emergency bottleneck. The remedy follows the mechanism. Where entry can work, interoperability or competition may help; where the service is a durable natural monopoly, obligations on price and continuity may be more appropriate; where access depends on subsidy, the

funding and delivery conditions require public scrutiny. The central test is whether essential use can remain reachable without making investment and maintenance impossible.

Refusal strength should determine the intensity of scrutiny, not predetermine the remedy. Emergency patients who cannot switch hospitals need enforceable clinical and billing duties at the point of use. Where tenants cannot move because housing supply is fixed, examine construction, legal protections, mobility, and land rules. If a digital platform locks in business users, test portable data and nondiscriminatory access before assuming a breakup will solve the problem. For every remedy, estimate enforcement capacity, investment incentives, service continuity, and the cost of mistaken intervention. A rule that weakens existing market power but creates an unaccountable public gate is not a successful correction.

A useful essential-price register should include direct price, hidden fees, time cost, quality decline, distance cost, debt created by purchase, rationing effects, and health or safety consequences of nonpurchase. This prevents an essential market from appearing stable because only the sticker price is measured.

14.2.2 — Labor Market Power

Employer power can arise through local dominance, credential control, noncompetes, immigration leverage, scheduling, platform ratings or informal blacklisting. Wages capture only part of that power.

Formal job mobility can coexist with practical entrapment. Leaving may mean losing health coverage, housing stability, immigration status, transport access, childcare arrangements, professional references, local community, or a credible path into the next job. Labor-market power is therefore assessed through

usable exit, not vacancy counts alone (Manning, 2003; Azar, Marinescu, and Steinbaum, 2022).

Labor market power can also hide inside subcontracting and platforms. The visible firm may disclaim employment responsibility while controlling price, access, ratings, discipline, data, dispute resolution, and termination. When control is real but responsibility is routed away, burden has been shifted downward. The fair analysis follows control, not labels alone.

Political influence through scale is not only campaign donation. It includes lobbying, revolving-door expertise, regulatory consultation, model legislation, trade-association pressure, litigation threats, standards-setting, philanthropic agenda-shaping, research funding, procurement dependence, and media access. These routes can shape policy and institutional agendas before any statute changes.

Concentrated economic resources can make public rules vulnerable to capture (OECD, 2017b). The risk to examine here is dependence on one firm for regional employment or essential services: threats of relocation, job loss, price increases, service disruption, data withdrawal, or investment slowdown can become bargaining tools over the rules meant to constrain that firm. Such threats may be credible, exaggerated, or strategically timed. Counterweight requires technical competence, independent evidence, diversified capacity, and public institutions able to distinguish warning from leverage.

14.2.3 — Antitrust Evidence and Competing Risks

Antitrust review remains expert and jurisdiction-specific. It depends on market definition, competitive effects, entry barriers, buyer power, monopsony, platform self-preference, vertical foreclosure, merger retrospectives, innovation effects, data control, remedy design, and administrability.

A disciplined antitrust posture should be able to distinguish at least four cases: scale that builds real capacity, scale that creates fragile dependency, scale that suppresses rivals while seeming efficient, and scale that has become private governance over essential access. Different cases require different remedies. A remedy may break up a firm, regulate prices or conduct, open access, strengthen supervision, require interoperability, or fund alternatives; in other cases, no intervention may be warranted (U.S. Department of Justice and Federal Trade Commission, 2023; Baker, 2019).

Expert review should include economics, law, labor evidence, consumer evidence, data governance, public procurement, small-firm experience, regional effects, and technical capacity. A market can look efficient in price terms while degrading labor, innovation, resilience, or public autonomy.

Regions tied to a dominant employer need credible worker and community transition support when a competition remedy threatens local collapse.

Remedy review should test public-platform gatekeeping, regulated-utility complacency, the costs of maintaining interoperability after a breakup, and privacy risks introduced by interoperability mandates.

14.3 — Land, Housing, Finance, and Platform Rent Capture

Land, housing, finance, and platforms require remedies matched to the particular gate creating dependency (Glaeser and Gyourko, 2018; Greenwood and Scharfstein, 2013). Rent-capture risk grows where essentiality and concentrated control meet scarcity or high switching costs. People may face weak refusal even when a legal option to leave exists. These conditions guide investigation but do not alone prove extraction, which requires evidence about conduct, counterfactual alternatives, and resulting burden.

14.3.1 — Land Rent

Land differs from produced capital because location cannot be manufactured. Buildings, irrigation, soil improvement, drainage, utilities, and care can add value, but the site itself is shaped by geography, law, public infrastructure, neighborhood life, ecological condition, and expectations about future development.

Several tools can return location value to shared purposes: land-value taxation, betterment levies, infrastructure charges, community land trusts, public acquisition, long leases, anti-vacancy rules, or negotiated development obligations. Each has legal, administrative, and distributional risks.

Agricultural and Indigenous land require additional care. Market price may not capture food security, customary use, water access, cultural continuity, biodiversity, or obligations across generations. A transfer that appears voluntary on paper can still occur under debt, legal asymmetry, displacement pressure, or incomplete recognition of collective rights.

Evidence should separate land from structures, current use from permitted use, transaction price from assessed value, and private improvement from publicly created location gain (Glaeser and Gyourko, 2018; Piketty, 2014).

14.3.2 — Housing Rent Pressure

Housing rent is a recurring claim on household income. When it absorbs the margin needed for food, care, transport, savings, or education, insecurity spreads beyond shelter.

The causes differ by place. Population growth, job concentration, household formation, construction cost, interest rates, insurance, land scarcity, zoning, permitting, infrastructure limits, vacancy, short-term rental demand, ownership concentration, and income distribution can all affect rent. A serious analysis begins by identifying which mechanisms are binding rather than importing one city's explanation into another.

Measurement must extend beyond rent-to-income ratios. Overcrowding, eviction filings, involuntary moves, commute expansion, school disruption, deferred care, housing quality, energy cost, homelessness risk, and proximity to support all reveal how shelter pressure travels through a household (Glaeser and Gyourko, 2018; Hsieh and Moretti, 2019).

Remedies should be assessed as a portfolio. Construction, land reform, social and cooperative housing, vouchers, tenant protection, infrastructure, taxation, vacancy rules, zoning change, and finance regulation solve different problems and can create different side effects. The relevant question is which combination expands secure, maintained, well-located housing without transferring the cost elsewhere.

14.3.3 — Financial Rent

Financial rent begins where control of an essential route permits fees, spreads, complexity, or asset gains that exceed the service and risk actually provided. The boundary is empirical, not rhetorical.

The cost of finance includes more than the stated interest rate. Origination fees, account charges, advisory layers, insurance add-

ons, trading spreads, penalties, refinancing cost, platform commissions, payment fees, and the time required to understand or correct a transaction can all shape the effective burden. Complexity makes comparison difficult and can turn opacity into a source of revenue.

The cost of moving payroll, recurring payments, credit history, identity records, or trusted relationships can keep a customer with an expensive provider even when competitors exist.

A defensible inquiry compares financial cost with service, risk, and alternatives. What information was produced? What uncertainty was carried? What capital remained at risk? What customer support, fraud protection, and correction were provided? Could another provider enter? Could the customer leave? Did the activity create real capacity or mainly rearrange claims on existing assets (Greenwood and Scharfstein, 2013; Philippon, 2015)?

Assess financial depth by allocation, risk, and institutional purpose. Expansion may support innovation, insurance, homes, trade, and public investment. Beyond a point, however, it may draw credit and talent into asset trading, regulatory arbitrage, and intermediation that crowds out productive uses (Arcand, Berkes, and Panizza, 2015; Cecchetti and Kharroubi, 2015).

14.3.4 — Rent Capture Versus Useful Return

One payment can contain both a useful return and a rent; an accounting statement alone may not separate them.

A landlord may provide maintained housing and also benefit from location scarcity. A platform may coordinate exchange and also collect a toll created by network lock-in. A lender may evaluate and fund risk while also exploiting emergency need. A patent may reward invention while extending control beyond what continued innovation requires.

Evidence should separate price from markup, markup from market power, and market power from harm. High margins can reflect innovation, temporary scarcity, or accounting choices. Low margins can coexist with labor extraction or supplier domination. The analysis needs mechanism, duration, alternatives, and incidence rather than one ratio (De Loecker, Eeckhout, and Unger, 2020; Tirole, 1988).

14.4 — Inheritance, Power Persistence, and Fair Circulation

Expert review matters because poorly designed circulation tools can harm small family firms, farms, ordinary homes, and community continuity while sophisticated wealth escapes. The public objective is not symbolic punishment. It is to keep opportunity, leadership, capital, land, and institutional access from freezing permanently in the same hands. Inherited advantages can reinforce one another across domains. Property can finance education; education can open professional networks; networks can improve legal and political access; those advantages can then protect or enlarge the original assets. A persistence register should therefore track reinforcing pathways, not merely the size of a bequest.

14.4.1 — Family Stability Value

Circulation policy should distinguish ordinary family continuity from concentrated control and from inherited advantages that close access for people outside the family line (Beckert, 2008; Nussbaum, 2011). Merit language often hides unequal starting conditions. One person may work hard with strong schools, safe neighborhoods, family capital, professional introductions, legal help, unpaid internships, emotional margin, and the expectation of rescue; another may work as hard under debt, danger, care burden, weak networks, and no buffer. Effort remains real in both lives, but it does not erase inherited differences in risk and opportunity. Fair circulation requires seeing those conditions without denying individual agency.

Entrenched advantage becomes especially hard to see when it presents itself as culture. Confidence, language, professional familiarity, institutional ease, and inherited expectations can all become invisible capital. People outside those settings may be

judged as less prepared when they were simply not raised inside the code.

14.4.2 — Political Dynasty Risk

Political dynasty risk arises when family wealth, name recognition, donor networks, media access, institutional connections, and inherited legitimacy make public leadership circulate inside narrow lineages. The problem is not that a child of a leader can never serve. The problem is when public choice becomes structurally tilted toward familiar names and concentrated networks.

Dynasty risk is a market-power issue because politics and economics reinforce each other. Wealth can finance campaigns, shape think tanks, support media ecosystems, fund legal action, influence universities, and build philanthropic authority. Public office can then protect the economic structures that sustain the dynasty (Dal Bó, Dal Bó, and Snyder, 2009; Querubin, 2016).

Circulation should not be confused with churn for its own sake. Constant disruption can destroy memory, craft, trust, and regional identity. Fair circulation means open pathways for entry and renewal while preserving the productive continuity communities need (Querubin, 2016).

A practical circulation register can track entry into ownership, access to capital, regional investment, public education quality, housing mobility, procurement openness, and leadership pathways.

14.4.3 — Tax, Law, and the Limits of Circulation

Sophisticated legal structures can make public responsibility optional for those with enough advice while ordinary families face blunt rules. Trusts, holding companies, family offices, charitable vehicles, carried interests, valuation discounts, offshore routes, and corporate succession arrangements can serve

legitimate purposes or preserve control beyond ordinary accountability. Technical design must distinguish modest continuity from structures that entrench command.

Inheritance and estate rules, capital-gains taxation, land law, and wealth taxation require jurisdiction-specific expert review alongside the legal structures already named. Evaluate the incidence questions: what continuity is protected, what control persists, who can enter, and which public burdens remain unanswered?

Administration matters as much as rate design. Complex valuation, opaque exemptions, expensive advice, slow appeals, and intrusive proof can humiliate ordinary families while sophisticated holders route around the system. A legitimate circulation regime must be understandable, reviewable, proportionate, and capable of correcting its own errors.

14.5 — Competitive Order and Anti-Stagnation Discipline

Competition is valuable when it gives people and ideas a route into the future. It disciplines incumbents, tests quality, lowers complacency, and lets neglected needs become the basis for new enterprise. Competition without floors can also become a ritual of discard, asking workers and regions to prove market vitality by absorbing ruin (Baumol, Panzar, and Willig, 1982; U.S. Department of Justice and Federal Trade Commission, 2023). Local repair firms, specialist suppliers, cooperatives, midsize manufacturers, and large enterprises can serve different resilience functions; assess those roles using the scale and remedy distinctions in Section 14.2.3.

14.5.1 — Entry and Contestability

Contestability is more than legal permission to start a business. It requires access to capital, premises, suppliers, workers, customers, licenses, payments, data, distribution, insurance, broadband, professional networks, technical standards, and a realistic chance to be discovered. A market may be formally open while every practical route is blocked by incumbents.

The same question applies to workers and ideas. Can workers move without losing their future? Can a new product reach customers without a gatekeeper burying it? Can a local firm bid without navigating procurement designed for giants? Can a community enterprise survive standards written by the actors it must compete against? Entry is the beginning of contestability, not the whole of it (Baumol, Panzar, and Willig, 1982; Djankov et al., 2002).

An entry test should also examine who can survive early losses while meeting worker and consumer protections.

Anti-capture rules should cover the influence routes in Section 14.2.2 as well as permitting offices, procurement specifications, professional licensing, tax preferences, zoning boards, data-access terms, and emergency exceptions.

Anti-capture review asks who helped write the rule, who benefits from its complexity, who can comply cheaply, who is excluded by cost, who receives exceptions, who audits performance, and how the rule can be challenged.

14.5.2 — Fair Failure and Restart

A competitive order needs failure, but it does not need lifelong ruin. Firms must be able to exit when a model no longer works, and people must be able to begin again after an honest mistake. Without exit, resources remain trapped. Without restart, one failure becomes a permanent social sentence.

Failure has several claimants. Workers may lose wages, benefits, identity, and local opportunity; customers may lose deposits, warranties, data, or essential service; suppliers may remain unpaid; lenders and investors may lose capital; and communities may lose tax base. These losses cannot all be eliminated, but priority, notice, continuity, and review can prevent failure from becoming abandonment.

Bankruptcy and restructuring rules shape the balance. They can preserve viable operations, convert debt, sell assets, protect essential contracts, or close a firm in an orderly way. They can also reward insiders, exhaust small creditors, or destroy knowledge because procedure moved too slowly (Armour and Cumming, 2008).

Failure statistics require context. High entry and exit can reflect dynamism or desperation; low failure can reflect stability or protected incumbency. The relevant evidence includes

productivity, worker outcomes, creditor recovery, customer continuity, innovation, market entry, and the geography of loss.

14.5.3 — Local and Small-Firm Capacity

Local and small-firm capacity supports material peace because nearby firms often understand local needs, employ local workers, adapt quickly, maintain relationships, and keep economic life from depending entirely on distant command.

Small firms can exploit workers, local elites can close entry, and informal networks can exclude outsiders. Their public value depends on fair standards and remedies.

Local capacity should be treated as a resilience asset with maintenance needs. Small firms need credit, technical assistance, predictable payment, legal clarity, worker standards, and fair access to customers (Berger and Udell, 1998).

Late payment can reach a small firm’s household as well as its balance sheet. An owner may defer wages, draw down family savings, or add unpaid hours to keep a viable business operating. Prompt, predictable payment is therefore a concrete way for large buyers and public agencies to strengthen local capacity.

14.6 — Markets That Reward Contribution Without Entrenchment

Market success can create capabilities that outlast the transaction, or it can harden into rules that exclude the next contributor. The difference cannot be inferred from firm size alone. It requires examining how productivity gains are made, who can challenge exclusion, and what happens when a supplier or platform fails. Public institutions need enough capacity to enforce fair conditions without becoming another unaccountable gatekeeper. These tests connect legitimate reward to the continuing possibility of entry and improvement.

14.6.1 — Success That Builds Capacity

To make contribution assessable, company reporting should identify supply-chain and community impacts, data use, safety, and exit risk.

Fee aggregation makes part of that account visible: charges split across contract lines or billed outside a headline price should be shown together. Platform reporting should also explain changes to ranking rules that alter business users' access.

Plain terms, reviewable charges, accessible disputes, data portability, beneficial-ownership visibility, and enforceable essential-service standards let users act on that information without having to become specialists.

A market can reward an improvement more reliably when buyers can compare the complete offer. Consider two hypothetical service contracts. One quotes 80 units for the core service and charges 15 for required support plus 10 for a necessary data transfer. The other quotes 100 units for the same complete service. With those assumptions, the first offer costs 105 units. Its lower opening figure does not make it the cheaper route.

The useful comparison is not simply the total of every possible fee. Optional services should remain distinguishable from charges a customer must incur to obtain the advertised function. The account should also identify when the charge arises. A payment required at entry affects the decision to join; an unavoidable charge for retrieving records affects the decision to leave. Those dates shape the practical meaning of the price.

A clear offer helps a genuinely better supplier compete. It can show that its higher initial price includes work another provider bills separately. Buyers can then decide which level of service they need. Price clarity supports enterprise by making useful differences visible, rather than forcing competition to revolve around which part of the same obligation can be omitted from the headline.

14.6.2 — State Capacity as Counterweight

Public capacity can create a more useful market rather than merely constrain one. Greenstone and colleagues (2025) worked with the Gujarat Pollution Control Board to evaluate a particulate-matter emissions market. Randomly assigned industrial plants reduced emissions by 20–30 percent relative to plants under the existing regulatory regime. The authors separately estimated an approximately 11 percent reduction in abatement costs at the same emissions level, using plant cost curves inferred from permit bids. The emissions effect and the modeled cost comparison are different findings; neither is a direct estimate of household health or an automatic saving in a public budget.

The regulator defines the environmental obligation; the permit market lets firms respond to differences in the cost of meeting it. Measurement and enforceable rules make that mechanism work. Before extending the design, examine who gains from lower abatement costs and whether exposure improves where people

live: lower aggregate emissions alone do not specify their distribution.

Staffing is part of institutional independence. Chronic vacancies, short appointments, low pay, weak training, and dependence on contractors can make an agency vulnerable to delay or capture. Expertise should be rewarded without creating an unreviewable technocracy. Clear mandates, professional norms, conflict rules, public reasons, appeal, and external scrutiny keep competence answerable.

Capacity must also exist below the national level. Local and regional bodies often see land, housing, transport, labor, water, small-firm, and service consequences first, but they may lack bargaining power against large firms. Shared expertise, pooled procurement, national support, and interoperable records can strengthen local authorship without centralizing every decision.

Evaluate whether public institutions can correct error, coordinate remedies, distinguish urgent action from permanent exception, end failed programs, and preserve what they learned (Besley and Persson, 2011).

Chapter 15 — Production and Industrial Skill

A civilization unable to produce and maintain what it needs must borrow its future from luck, depletion, emergency, or another society's spare capacity. Moral clarity can name suffering, but it cannot build a clinic, manufacture medicine, repair a water main, maintain a grid, retrofit housing, move food, or secure digital systems. Production is not beneath peace. It is one of the ways peace becomes material.

Production should be neither worshiped nor treated as a morally empty background. A factory, laboratory, farm, construction site, care network, software team, or repair shop can create extraordinary capability and still impose unsafe pace, pollution, surveillance, fragility, or regional dependence. The task is to preserve buildability while making those consequences answerable.

Capacity is not spectacle. A society can produce prestige objects while depending on distant suppliers for basic medicines, transformers, tools, seeds, water-treatment inputs, or repair parts. Productive strength is revealed by what can be built, maintained, adapted, and restored when ordinary systems are under pressure. Section 15.2 examines the links among firms, suppliers, and repair trades, including the cases in which losing a specialized component or skill can disable an otherwise capable production system.

Artificial intelligence appears here in material and organizational form, not as scientific authority or a sovereign planner. Its productive promise becomes credible only when physical demand, labor, opportunity cost, failure modes, and human authority enter the account.

Productivity must be translated into shared breathing room. Higher output, faster throughput, and lower unit cost remain useful measures, but they are incomplete when achieved through injury, unpaid user labor, ecological depletion, delayed

maintenance, or intensified work. A gain becomes material progress when it improves safety, wages, time, quality, affordability, resilience, and the life remaining after work.

Industrial capacity grows through research and patient learning across an interdependent production network. It cannot be summoned instantly when crisis arrives. Automation may raise productivity and remove dangerous or repetitive work, but its gains depend on ownership, task redesign, worker transition, and the regional institutions available to absorb change. Evidence on robots and employment shows both productivity potential and concentrated local disruption (Hidalgo and Hausmann, 2009; Acemoglu and Restrepo, 2020).

Industrial policy needs a discipline stronger than slogans about picking winners. It identifies the capability gap, public purpose, time horizon, performance test, exit rule, tradeoff, and capture risk. Protection or subsidy cannot become permanent entitlement. The public record should state which learning, resilience, employment, decarbonization, or security capacity the intervention is expected to create and when that claim will be reviewed (Juhász, Lane, and Rodrik, 2024).

15.1 — Productive Capacity Beyond Prestige

Claims about industrial capacity need evidence because nominal output can hide dependency. Import content, supplier concentration, tooling, standards, workforce depth, energy, and logistics reveal important constraints. Inventories, lead times, maintenance, and the ability to scale safely show whether those constraints can be managed under stress. A headline production figure is therefore only the beginning of an audit.

15.1.1 — Capacity as Buildability

Buildability is the ability to make, maintain, repair, adapt and deliver needed goods or services where people need them.

A practical capacity register has six linked fields. Useful output asks what is made; maintenance ability asks whether it can keep working; skill continuity asks whether people know how to operate and repair it; supplier reliability asks whether parts and materials remain available; resilience margin asks what happens under stress; and hidden burden asks who pays the unseen cost.

Useful capacity can grow through better operation of existing equipment. Bloom and colleagues (2013) randomly offered intensive management consulting to large Indian textile plants. The intervention package raised productivity by about 17 percent in the first year through improvements in quality and efficiency and lower inventory. This was a change in how production was organized, not simply a larger investment announcement. It gives the capacity register an empirical question: what usable output could better routines release from resources already present?

Nine years later, a follow-up found that roughly half the adopted practices had been dropped, yet a substantial treatment-control gap in management practices remained. Practices had spread within firms much more than between them; respondents often cited managerial turnover and limited director time when

explaining losses (Bloom et al., 2020). The original productivity estimate is not a nine-year estimate. A practical next step is to build succession and routine handover into the improvement itself, then test which gains survive staff change.

Claims for prestige production should compare its jobs, pride, and spillovers with the repair, essentials, or regional capacity the same resources could provide; apply the priority rule in Section 12.6.1.

Repair production supplies replacement parts, maintenance tools, trained crews, spare equipment, software patches, building materials, inspection systems, manuals, safety gear, calibration equipment and the logistics needed to put them where failure occurs.

Regional production policy must also resist the temptation to promise every place the same future. Some regions may be strongest in repair production, others in food systems, energy components, care infrastructure, advanced manufacturing, logistics, climate adaptation, or technical education. Fair regional repair means matching capacity to place without assigning places to lower dignity categories.

15.1.2 — Evidence of Productive Capacity

Capacity evidence is strongest when the register's different indicators agree. Complexity measures and export sophistication can illuminate accumulated knowledge, but they remain proxies rather than a complete account (Hidalgo and Hausmann, 2009; Hausmann et al., 2014).

The ability to adapt production under stress is a sharper test than nameplate capacity. A medicine plant may produce one formulation at volume yet need months to validate a different line; a machine shop may own suitable equipment but lack tooling, drawings, metrology, certified material, or operators; a utility

may have nominal treatment capacity but no spare pump, chemical inventory, laboratory coverage, or electrical backup. Scenario evidence therefore records conversion time, yield loss, workforce, regulatory steps, and upstream inputs required for emergency or substitute production.

Supplier substitution requires more than a name in a database. An alternative must meet specifications, quality controls, cybersecurity requirements, tooling interfaces, safety rules, warranties, transport conditions, payment terms, and legal permissions. In medicines, energy equipment, aviation, semiconductors, water treatment, and medical devices, qualification may be the longest constraint. Capacity maps therefore distinguish identified, technically compatible, qualified, and scalable alternatives.

The strongest capacity evidence combines records with exercises. Tabletop stress tests, controlled production trials, supplier-failure simulations, maintenance drills, and cross-agency exercises reveal whether plans survive real lead times and human limits. Public reporting can show whether essential capacity exists and is improving while protecting details whose release would create security or commercial harm.

Asset age alone does not establish usable capacity. Maintenance history, failure frequency, software support, and replacement lead time show whether nominal capacity is decaying beneath reported output.

Physical plant becomes usable capacity only if working capital and payment systems keep it operating. A firm may have workers and machines yet be unable to buy inputs, insure shipments, carry receivables, absorb temporary yield loss, or survive delayed payment. Public purchasers can create the fragility they claim to solve when emergency orders are large but payment is slow or contract risk is shifted to smaller suppliers. Capacity evidence

therefore includes cash-conversion cycles, payment reliability, financing access, currency exposure, and the distribution of contract risk (Berger and Udell, 1998).

15.2 — Firm Ecology, Scale, and Technical Depth

Firm ecology concerns how enterprises, suppliers, workers, finance, standards, and infrastructure combine to sustain specialized capabilities and contestable power (Piore and Sabel, 1984; Berger, 2013). Technical depth includes tooling, process engineering, quality control, repair knowledge, trained judgment, safety practice, and supplier memory that may disappear even while familiar brands and buildings remain. Procurement can weaken that depth when compliance, insurance, and change-order terms shift disproportionate risk onto small suppliers. Many useful firms also depend on one owner, craftsperson, relationship, or generation of knowledge, making succession a regional resilience issue. Apprenticeships, cooperatives, employee buyouts, patient purchasers, and reliable documentation can preserve capacity without freezing every firm in place.

15.2.1 — Large Firms and Technical Capacity

Large firms can carry capacities that smaller actors often cannot: complex manufacturing, advanced research, high-volume production, global procurement, capital-intensive equipment, and system-level integration.

Large firms should be asked to carry continuity duties where their operations become essential. These may include maintenance transparency, contingency planning, worker transition support, interoperability, vendor-exit pathways, public contract accountability, and emergency cooperation.

Mid-sized firms can hold specialized skill while remaining connected to region, craft, and supplier relationships. They link local resilience with national capacity; losing them can leave production either too fragmented or too concentrated.

The economic value of scale becomes clearer when fixed preparation is separated from repeated production. Suppose a

specialized process requires 600 hours of design and setup, followed by two hours for each unit produced. At 100 units, the process requires 800 hours, or eight hours per unit. At 1,000 units, it requires 2,600 hours, or 2.6 hours per unit. Under these teaching assumptions, the preparation is spread across more useful output.

That relationship helps explain why a large producer and a small specialist can be complements. A specialist may develop the setup knowledge while a larger enterprise coordinates a long production run. Neither contribution disappears because the final product carries only one company’s name. The institutional question is how contracts recognize the preparation, the repeated work, and the knowledge needed when the process changes.

Scale has a different meaning when the required output is varied. A workshop that can switch efficiently between small batches may be more useful for local repair than a line optimized for one long run. The relevant comparison is the quantity and variety people actually need. A production ecology can therefore contain both high-volume facilities and adaptable workshops without treating one as an incomplete version of the other.

Technical depth lies partly in the connection between them. A repair workshop may discover a recurring defect that the high-volume producer can remove from the next design. The producer may supply reliable specifications that make local servicing easier. When that exchange works, scale and proximity contribute different knowledge to the same product’s useful life. Maintaining the relationship can be more valuable than making every firm resemble the largest one.

15.2.2 — Supply Networks

A supply network maps work, finance, deadlines, standards, inventory, and bargaining power. Its apparent efficiency may depend on risk pushed out of view (Gereffi, Humphrey, and

Sturgeon, 2005; Christopher and Peck, 2004). Section 16.3 examines how those transfers affect labor.

Supply-network mapping should expose only the detail needed for safety, continuity, labor protection, ecological accounting, and anti-capture review, with ordinary privacy and legitimate commercial confidentiality protected.

Restart tests ask whether a facility, supplier, water plant, clinic, archive server, grid component, or factory line could recover after a month-long stop. Check drawings, parts, manuals, vendor exits, apprenticeships, standards, institutional memory, and people able to recognize failure before alarms do. Include the years needed to rebuild lost workshops and maintenance skills in the cost of allowing them to disappear (Christopher and Peck, 2004).

15.2.3 — Firm Power and Public Burden

A firm's public obligations rise with its control of essential supply, employment or infrastructure. For example, a sole regional equipment supplier can create continuity risk even with a small national market share.

The relevant dependency may lie in a repair manual or proprietary connector rather than in ownership of the entire service.

Review should occur before crisis, while alternatives, continuity duties, transition plans, and public options can still be built. Waiting for failure or abuse leaves the public bargaining from panic and dependency. Testing an alternative supplier before renewal can reveal whether an exit clause describes a usable option or only a contractual promise.

15.3 — Innovation, Automation, and Shared Gains

Innovation should be evaluated by the work it makes possible and the burdens it changes, not by novelty alone. A new system can increase output while shifting risk, surveillance, or maintenance cost onto workers and users. Productivity gains also raise a distribution question when the people who enabled the change receive little of its benefit. Computing and automated systems have physical demands that belong in the same assessment as their promised efficiency. The subsections separate technical improvement, shared gain, and the resources required to sustain deployment.

15.3.1 — Innovation as Burden Reduction

Before deployment, document the burden reported by affected workers or users, its baseline measure, and the technical and nontechnical alternatives considered (Mazzucato, 2013; Acemoglu and Johnson, 2023).

A burden-reducing innovation should be tested in the place where burden is actually carried. If the burden is in a clinic, test with clinicians and patients. If it is in a warehouse, test with workers and families affected by scheduling. If it is in public access, test with disabled users, language minorities, and low-digital-access households.

Displacement should be assessed across four levels: the individual worker, the household, the region, and the skill ecosystem. A severance payment may help the individual briefly while leaving a region with fewer apprenticeships and a household with long-term insecurity. The burden cannot be counted at one level only.

15.3.2 — Shared Productivity Gains

Productivity measures output relative to specified inputs. Higher output per hour can become profit, lower prices, higher wages, shorter hours, better quality, tax capacity, reserves, or greater market power. Ownership, competition, bargaining, institutions, and task changes determine that distribution. Assess the gain alongside changes in safety, time, quality and usable capability.

Task analysis is more informative than job labels. Technology may automate part of a role, complement another part, create new supervisory work, or shift effort to customers and contractors. A job can survive while its autonomy, pace, skill, schedule, surveillance, or injury risk changes substantially. Exposure is therefore not the same as displacement, and displacement is not the only burden that matters (Autor, 2015; Gmyrek et al., 2025).

Workplace evidence makes the possibility of shared gains concrete. Studying the staggered introduction of a conversational assistant among 5,172 customer-support agents, Brynjolfsson, Li, and Raymond (2025) estimated a 15 percent average increase in issues resolved per hour. Less experienced and lower-skilled workers improved both speed and quality; the most experienced and highest-skilled workers saw small speed gains and small quality declines. The study also found evidence of learning. These results suggest a promising mechanism: a tool can help newer workers use practices that previously depended on accumulated experience. Whether that advantage becomes better pay, more time, improved service, or intensified workload is a further institutional question, not something the productivity estimate decides.

Where productivity gains depend on public support or have public consequences, the account should show what capacity returns through maintenance, climate adaptation, debt repair, worker transition or essential services, and who receives it.

15.3.3 — AI Energy and Labor Cost

Artificial intelligence depends on physical infrastructure and specialized labor; Section 17.6 examines that resource base. Here the industrial question is who gains productive capacity and who carries the work of operating and correcting the system (International Energy Agency, 2025; International Energy Agency, 2026b).

Location and timing determine much of the electricity burden. A data center drawing power during abundant low-carbon generation differs from one that worsens a local peak, delays household or industrial connections, or requires new fossil backup. Interconnection queues, transmission limits, substation capacity, backup generation, curtailment rules, and hourly demand therefore belong beside annual electricity totals. The relevant question is not only how much energy is used, but what system must be built, reinforced, or displaced to supply it (International Energy Agency, 2026b).

Opportunity cost belongs in the same ledger. Grid upgrades, water infrastructure, skilled electricians, construction labor, public financing, and scarce equipment devoted to one computing project are unavailable for other uses during that period. This does not make artificial intelligence illegitimate. It means that promised gains should be compared with lower-compute methods, conventional software, organizational repair, and alternative public investments.

Thermal and material design shapes the burden after electricity enters a facility. Waste heat may be reusable only where temperature, distance, season, and local infrastructure make recovery practical. Short replacement cycles can also turn servers and accelerators into electronic waste if repair, component reuse, resale, and secure decommissioning are not planned in advance. Lifecycle accounting therefore includes embodied materials,

cooling equipment, backup systems, hardware turnover, and the destination of retired assets.

Apply the digital custody and exit rules in Section 17.6 when retiring models, data, hardware, credentials, or vendor accounts: name shutdown authority, preserve essential service and review evidence, and notify affected people.

Before deployment, identify changing tasks, affected workers and suppliers, available training and redeployment, wage and schedule protections, and the share of verified gains reserved for adjustment. Apply Section 8.3's transition safeguards and review quality, safety, autonomy, accessibility, energy use, regional effects, and complaints.

Local capability is part of that agreement. A firm or public agency that adopts automation without retaining people who understand the work can become faster in routine conditions and more fragile during exceptions. Skilled operators, maintainers, auditors, trainers, and frontline workers preserve the knowledge needed to recognize error, override a bad output, restore service, and improve the tool.

15.4 — Labor, Skill, and the Quality of Work

The quality of work depends on more than wages and the number of jobs created. Tasks can cultivate competence and judgment, or leave workers with little control over pace and conditions. Monitoring may improve coordination in some settings while also creating pressure that is hard to contest. Repetitive work deserves attention to safety and dignity without assuming that every worker seeks the same career path. These distinctions help evaluate skill development and managerial power on their own terms.

15.4.1 — Skill Development

Skill development is the growth of human competence through work. It includes technical skill, judgment, craft, troubleshooting, communication, safety awareness, process understanding, maintenance ability, and the capacity to teach others.

Deskilling can preserve a job while reducing a worker's bargaining power and mobility by making competence dependent on a narrow process or proprietary system (OECD, 2023b; World Bank, 2019).

When technology, climate policy, trade, or restructuring changes required skills, employers and public institutions should share training costs, time, income support, and placement risk under the transition principle in Section 8.3.

Worker dissent can protect the production system. A person who slows an unsafe line, challenges an impossible target, reports a defect, or questions a misleading metric may be preserving the firm's future as well as public safety. Retaliation destroys an early-warning channel and teaches the institution to hide its own failure.

15.4.2 — Surveillance and Work Intensity

Sensors, keystroke logs, cameras, route trackers, biometric systems, algorithmic schedules, and performance scores can support safety, staffing, coordination, and quality. They can also intensify pace, punish normal variation, invade privacy, and suppress breaks. Review their effects alongside purpose, access, retention, worker voice, and appeal (Kellogg, Valentine, and Christin, 2020; Ajunwa, Crawford, and Schultz, 2017).

Surveillance review should ask whether the same safety or quality goal could be met with less intrusive design. Sometimes the answer will be yes: aggregate data, equipment-level monitoring, random audit, worker reporting, or environmental sensors may solve the problem without tracking every person continuously.

15.4.3 — Dignity of Repetitive Work

Repetitive work can be honorable without being romanticized. Many essential tasks involve routine, and routine can coexist with mastery, teamwork, and pride. It becomes degrading when pace, ergonomics, autonomy, respect, compensation, recovery, and opportunities to develop skill are poor (Sennett, 2008; Crawford, 2009).

Metrics can also be captured. A firm may improve the reported number while hiding pressure elsewhere. Safety may improve on paper because injuries are underreported. Worker voice may appear strong in identifiable surveys if respondents fear the consequences of criticism. Schedule dignity may look adequate in policy but fail in practice. Lived testimony must accompany data.

A work-quality dashboard should keep improvement data separate from individual disciplinary files unless a directly relevant, reviewable employment purpose requires otherwise. Published results should use role- or system-level aggregation

where feasible, protect small groups from re-identification, and preserve a worker route to correct inaccurate records. Data then support repair without becoming a portable personnel history.

15.5 — Productivity, Skill, and Human Depth

Productivity should be evaluated by what it releases for people, not only the output produced per hour (Syverson, 2011). Freed time can support recovery, care, creativity, and life outside paid work. Its value depends partly on whether work design preserves attention, judgment, learning, discretion, and cooperation. A gain that intensifies harmful hours or transfers fatigue elsewhere requires a fuller health and labor account (Pega et al., 2021).

15.5.1 — Exhaustion as Hidden Cost

Work design creates much of this burden. Staffing, quotas, schedule volatility, heat, noise, surveillance, mandatory overtime, commuting distance, emotional labor, and fear of discipline can make fatigue structural rather than personal. Advice about resilience cannot repair a production system that consumes recovery time as an input.

Long-hours evidence supplies a concrete occupational-health reason to examine working time. WHO/ILO estimates attribute ischemic heart disease and stroke burdens to exposure to long working hours (Pega et al., 2021).

Fatigue also propagates through interfaces. A tired worker may hand incomplete information to the next shift, miss a weak signal, postpone a small repair, or take a shortcut that increases the burden on someone later in the process. In tightly coupled systems, these small losses can accumulate into quality failure, outage, injury, or emergency maintenance. Recovery is therefore not only a private health benefit. It is part of system reliability and the preservation of institutional knowledge.

Schedule timing matters alongside total hours. Night work, rotating or split shifts, short notice, long commutes, and repeated transitions between day and night impose different recovery demands even when weekly totals match.

Fatigue-risk management should remain a system tool rather than an individual suspicion score. Useful practice includes staffing floors, maximum duty periods appropriate to the field, protected breaks, predictable rotations, transport after extreme shifts, confidential reporting, incident learning, and worker participation in schedule design. Wearables or performance monitoring may sometimes assist safety, but they should not become continuous surveillance, a proxy for character, or a way to make workers responsible for exhaustion created by the production plan.

15.5.2 — Technology That Supports Workers

The effects of worker-support technology may not be obvious at launch. A scheduling tool can make shifts fairer or less humane. A warehouse system can reduce injury or intensify pace. A diagnostic aid can support clinicians or undermine judgment. A maintenance sensor can prevent failure or create constant alert burden. The material-incidence question asks how the tool changes the worker's actual day (Autor, 2015; Acemoglu and Restrepo, 2020).

Worker-support technology should be co-designed with workers where possible. The people who use a tool know the friction points, safety risks, exception cases, and dignity harms that designers may miss. A tool imposed from above can solve a managerial problem while creating a shop-floor problem.

Automation leaves people training models, maintaining machines, handling exceptions, cleaning data, moderating content, repairing errors, supporting users, and managing failures; count that continuing labor.

15.6 — Industrial Policy, Resilience, and Public Purpose

A public strategy for productive capacity must distinguish capability-building from subsidy capture, incumbent protection, donor influence and national prestige (Juhász, Lane, and Rodrik, 2024; International Labour Organization, 2015). The market-state binary is too crude for industrial policy. Some capabilities emerge through private initiative; others require public coordination, public options, cooperatives, universities, unions, regional institutions, mission-driven nonprofits, or carefully governed partnerships. The appropriate tool depends on the technology, time horizon, externalities, security need, financing structure, and distribution of failure costs.

15.6.1 — Public Purpose in Industry

Learning and spillovers create a rationale for public support. A firm may not capture the full value of knowledge, workforce skill, supplier capability, or regional infrastructure created by its investment. Public support can help where social return exceeds private return. The support should be tied to learning, diffusion, access, and performance rather than becoming a permanent reward for political proximity (Mazzucato, 2013; Juhász, Lane, and Rodrik, 2024).

Conditionality connects public purpose to private execution. Depending on the project, conditions may concern investment, employment, wages, training, local suppliers, price, access, emissions, data, open standards, repayment, profit sharing, maintenance, or geographic commitment. The conditions need to be few enough to enforce and strong enough to prevent public money from becoming unrestricted private advantage.

Set milestones that can be verified without letting the subsidized firm define its own success. Verify additional capacity

and workforce learning against a realistic no-subsidy comparison, so firms are not rewarded for investment they would have made anyway. Track contract performance, maintenance, spillovers, regional employment, concentration, and public exposure on failure. Before support becomes irreversible, establish dated review, modification or clawback powers, lawful asset recovery where appropriate, and worker and supplier protection on closure. State what evidence would end support, rather than changing the objective afterward (Juhász, Lane, and Rodrik, 2024; Andrews et al., 2017).

15.6.2 — Worker and Region Conditions

Before contracts are signed, industrial agreements must account for concentrated effects on workers, suppliers, housing, public revenue and local services. Negotiate wages, benefits, safety, schedules, training, worker voice, anti-retaliation, surveillance, subcontracting, local procurement, environmental duties, and closure terms before land assembly and signed contracts reduce worker and community leverage.

During implementation, map dependence beyond the main payroll, including suppliers, transport, repair, food, care, and professional services. Arrange finance, technical help, market access, or diversification before losses spread.

After award, evaluation should follow people and places beyond the ribbon cutting: wage progression, placement, benefit continuity, supplier survival, housing cost, public services, environmental repair, migration, local business formation, and regional options after subsidy (International Labour Organization, 2015; Juhász, Lane, and Rodrik, 2024).

Chapter 16 — Trade and Supply Chains

Trade is one of the oldest arguments about peace because it sits at the border between need and power. People have long hoped that exchange could make strangers useful to one another before they become enemies. They have also watched trade become a machine for hiding exploitation, ecological damage, debt dependence, strategic vulnerability, and humiliation behind a low price or a full shelf.

Distance can become moral cover. An affordable good may depend on a distant worker absorbing danger, a river receiving waste, a supplier carrying impossible deadlines, a farmer losing bargaining room, a port community bearing congestion, or a small country depending on one buyer, creditor, corridor, platform, or payment system. Fair trade analysis follows the good beyond the checkout line. The bargain is incomplete until remote burdens and repair duties are visible.

Buffers buy time before essential shortages cause harm. Section 16.2 examines which goods need reserves and how to control storage cost, decay, waste, and capture.

Interdependence can support restraint or create coercive leverage; Section 16.4 distinguishes those outcomes through usable alternatives and asymmetric replacement costs.

Resilience is not autarky. Producing everything domestically can raise costs, reduce quality, and leave a country exposed to its own disasters, monopolies, or politics. The practical task is to distinguish ordinary dependence from concentrated vulnerability, then combine supplier diversity, repair capability, buffers, reciprocal agreements, digital readiness, and transparent emergency rules without pretending every link is safe (OECD, 2021a; Bonadio et al., 2021).

For each essential route, assess time to harm, supply concentration, substitution, reversibility, and who bears adjustment; the map in Section 16.2 applies these questions to

actual chokepoints. Consider an illustrative buyer choosing between two suppliers. Their quoted prices are comparable, but their replacement routes differ: one product can be obtained elsewhere quickly, while the other requires a long qualification process. A larger stock may bridge the second interruption, yet it also ties up resources and must remain usable. The relevant comparison concerns the whole interval between failure and replacement. It cannot be read from the import share alone. Nor does a border tell the reader where the dependency ends: a domestic supplier may use the same imported input as its foreign competitor. Mapping that connection reveals why apparently separate alternatives can fail together, and why the chapter follows actual routes before treating diversification as protection.

Trade gains can be dispersed while workers and regions bear concentrated losses from imports, automation, sanctions, environmental rules, or changing demand. Apply Section 8.3’s transition protections and account for the time needed to rebuild local capacity.

16.1 — Why Trade Still Matters in a Fair World

The useful distinction is not open versus closed. It is breathable exchange versus burden-displacing exchange. Breathable exchange lets a society obtain what it cannot sensibly make alone while preserving enough local capacity, bargaining room, and source knowledge to refuse abuse. Burden-displacing exchange uses distance to hide costs borne beyond the buyer (Baldwin and Freeman, 2022; Goldberg and Pavcnik, 2007).

16.1.1 — Mutual Capacity and Distributional Burdens

Exchange can combine places' different resources, skills, technologies, and production strengths; labor conditions, ecological costs, and usable exit remain part of the assessment.

Trade must therefore be evaluated over time. A first contract may look beneficial because it brings goods or revenue; a decade later the same arrangement may have weakened local firms, trapped suppliers in low-margin work, exposed workers to dangerous conditions, or made an essential system dependent on one external gate. The relevant account follows capability, bargaining power, and repair across the life of the relationship.

Exchange can create channels that remain useful in crisis: suppliers, buyers, researchers, shippers, and repair partners offer contacts through which threats can be clarified. Repeated contracts, customs procedures, letters of credit, shared standards, inspections, port coordination, insurance, and arbitration also require predictable conduct. Those habits can support restraint, but remain fragile where dependency is one-sided or rules cannot be challenged.

Incidence begins with the worker. Imports may lower consumer prices while reducing employment or bargaining power in exposed industries; exports may create jobs while intensifying schedules, land pressure, or environmental harm. The distribution

depends on local labor markets, mobility, institutions, supplier networks, and whether affected workers receive credible routes into new capacity (Autor, Dorn, and Hanson, 2013; Goldberg and Pavcnik, 2007).

16.1.2 — Evidence for Trade Claims

Trade evidence begins by defining its object. Merchandise volume, customs value, services, value added, bilateral balance, tariff revenue, freight cost, and firm-level sourcing answer different questions. A defensible claim identifies the measure, route, period, and price basis that support it.

The March 2026 WTO outlook and 2025 maritime review support dated trade scenarios whose tariffs, sanctions, war risk, freight, insurance, energy, climate, and technology assumptions require updating (World Trade Organization, 2026; United Nations Conference on Trade and Development, 2025a).

Evaluate openness and restriction using the same questions about prices, wages, employment, resilience, environment, security, retaliation, and distribution, and compare each with a credible alternative.

16.2 — Supply Chains, Buffers, and Chokepoints

Supply-chain analysis must trace actual routes, handoffs, payments, inspections, permissions, data systems, workers, maintenance needs, failure thresholds, and usable substitutes rather than rely on ideological categories (Christopher and Peck, 2004; Carvalho et al., 2021). A chokepoint may be physical, financial, legal, digital, political, or ecological, as drought, sanctions, cyberattack, monopoly supply, or understaffed inspection each illustrate. Mapping must also include labor burden, ecological risk, security exposure, and how public consequences reach different communities. A public account can disclose the existence and nature of a critical dependency without revealing sensitive details that create a specific security danger. The test is whether the map supports repair and emergency access rather than merely describes uninterrupted operation.

16.2.1 — Buffer Capacity

Buffer duty belongs at the level with both capacity and consequence. Households cannot store what wages never allow, small firms cannot finance inventories that dominant buyers refuse to support, and public systems cannot assume private stocks will remain available under profit pressure. Firm, regional, national, and international reserves serve different functions. Fair resilience places financing and responsibility where they can be carried without turning scarcity into private panic.

Buffer capacity is the time and usable backup available between disruption and serious harm (Tang, 2006; Christopher and Peck, 2004).

Buffer governance matters because reserves can be captured. Insiders may receive access first. Firms may lobby for subsidized inventories that protect profit more than public need. Emergency stockpiles may expire from neglect. Public agencies may hide

shortages to avoid embarrassment. A buffer must therefore include ownership, access rules, rotation, transparency, exception procedures, and independent review.

16.2.2 — Just-in-Time Fragility

Just-in-time systems can save storage and working capital while retaining disruption risk that the accounts do not price fully. When a route breaks, shortage and emergency freight are realized losses and response costs. Carvalho et al. (2021), studying the 2011 Great East Japan Earthquake, traced disruption through direct and indirect suppliers and customers of affected firms. The result gives resilience appraisal a concrete object: the dependency path, not simply a supplier count. Compare the cost of buffers and genuinely independent routes with the losses they are designed to prevent; the existence of propagation alone does not select the best buffer (Barrot and Sauvagnat, 2016).

Trace environmental burdens through supply-chain tiers using proportionate chain-of-custody checks, risk tiering, independent review, and correction when hidden mining, water, pollution, emissions, or disposal costs emerge.

16.2.3 — Critical Goods Priority

Criticality is defined by consequence, not prestige. A humble chemical, spare valve, basic antibiotic, refrigeration part, translation service, or small electrical component may matter more than a glamorous technology. The question is what breaks if the item is missing, how soon harm begins, and who suffers first (OECD, 2021a; Bonadio et al., 2021).

Use criticality to set reserve, procurement, transport, and emergency-allocation priorities, including accelerated customs access and quality safeguards.

Priority classification must be revisable. A good that appears ordinary in calm conditions can become critical in a crisis.

Shortages of masks, oxygen, shipping containers, semiconductors, fertilizer, infant formula, fuel additives, data-center capacity, or spare parts can suddenly impose severe public burdens. The classification system should learn from near misses, not wait for catastrophe to prove dependence.

Resilience should be tested as a portfolio of substitutes that remain usable in the same crisis. Two suppliers in one floodplain or dependent on the same software, port, fuel source, or mineral are not two independent backups. A critical-goods review should examine correlated failure, lead time for qualification, spare-part compatibility, stock rotation, contractual priority in emergencies, and whether a second route can actually serve affected regions. Choose the lowest feasible lifecycle cost among strategies that meet the specified continuity threshold, while disclosing whose time and labor carry the safeguard. Periodic exercises should trigger correction or retirement of inventories and supplier contracts that fail a real substitution test.

16.3 — Fair Trade, Labor Standards, and Material Honesty

Contract compliance alone does not show that a supply chain's labor and ecological burdens were responsibly carried (International Labour Organization, Walk Free, and IOM, 2022; OECD, 2018b). The burden test is deliberately two-sided. Standards must protect people and ecosystems while remaining attainable for weaker producers without exclusion, humiliation, or dependency on costly certification controlled by outsiders. Fair trade is neither permission for exploitation nor a moral gate reserved for wealthy buyers.

16.3.1 — Labor Visibility

Labor visibility begins with a simple fact: the final seller often knows far less about working conditions deep in a supply chain than the workers who experience them. Production can pass through contractors, labor brokers, farms, mines, factories, warehouses, ships, platforms, and home-based work before reaching the buyer. A first-tier audit cannot by itself reveal that entire route.

Forced labor and severe exploitation often depend on recruitment debt, document retention, threats, migration status, isolation, withheld wages, or control of housing and transport. Product-level due diligence is important, but remedy must reach the mechanisms that make refusal unsafe (International Labour Organization, Walk Free, and IOM, 2022).

Audits have limits. Announced visits, coached interviews, falsified records, translator dependence, worker fear, and rapid supplier substitution can create a clean file while hidden burden continues. Audits are strongest when combined with worker organization, complaint channels, purchasing-practice review,

unannounced checks, remedy records, and independent follow-up (Locke, 2013).

16.3.2 — Supplier Power Imbalance

A dominant buyer can set terms that a dependent supplier cannot safely refuse; formal agreement alone does not establish fair bargaining.

Buyer efficiency can be built on supplier fragility. A supplier that must finance inventory, accept late payment, absorb cancellations, meet sudden deadlines, pay for compliance, carry liability, or cut labor costs to survive is functioning as the buyer's shock absorber (Gereffi, Humphrey, and Sturgeon, 2005; Distelhorst, Hainmueller, and Locke, 2017).

Imbalance can travel through tiers: a brand squeezes a first-tier supplier, the supplier squeezes a subcontractor, the subcontractor squeezes a labor broker, and the broker squeezes migrant workers. Burden analysis must follow how each tier passes pressure to those less able to refuse, instead of stopping at the brand visible to consumers.

Supplier power imbalance also affects ecological choices. A supplier under extreme price pressure may underinvest in waste treatment, worker safety, energy efficiency, or maintenance. The buyer may publicly demand standards while privately setting prices that make standards hard to meet. Material honesty requires aligning demands with the cost of meeting them.

A responsible supply relationship can still be demanding. Quality, safety, delivery, and environmental requirements matter. The difference lies in whether compliance is financed, timelines are realistic, price reflects upgraded standards, and disputes can be heard.

Standards can be made more attainable through transition periods, technical assistance, shared financing, local worker

participation, small-producer access, recognition of different legal systems, and independent review of exclusion effects.

16.3.3 — Enforcement Capacity

A labor rule, ecological promise, anticorruption clause, or supplier code means little if no one can inspect, translate, verify, protect witnesses, review documents, hear complaints, impose remedies, and correct failure. Enforcement capacity is part of the cost of fair trade.

Burden-route analysis asks who enforces and who watches the enforcer. Private audits may be captured by buyers, suppliers, or the certification market; state enforcement may be corrupt, underfunded, politicized, or inaccessible; worker organizations and civil society may reveal harm while facing retaliation. Credible oversight therefore needs multiple routes of evidence, correction, and appeal (OECD, 2018b; Locke, 2013).

Cross-border enforcement depends on jurisdictional cooperation. A violation may involve a recruiter in one country, a labor broker in another, production in a third, financing elsewhere, and a buyer headquartered far from the worksite. Customs agencies, labor inspectorates, environmental regulators, courts, procurement bodies, unions, and independent monitors need compatible evidence rules and lawful channels for cooperation.

Response should be proportionate to the harm and the possibility of repair. Imminent danger, forced labor, violence, document confiscation, or deliberate toxic release may require immediate suspension and protective intervention. Other failures may be better addressed through a time-bound corrective plan with worker participation, back pay, safety investment, changed purchasing terms, and independent follow-up. The design must

neither normalize serious abuse nor use automatic exit to protect a buyer's reputation at the expense of people left behind.

The cost of enforcement should follow power. Small suppliers and low-capacity jurisdictions cannot be expected to finance every audit, laboratory test, information system, training program, and remediation order demanded by wealthy buyers. Lead firms, public purchasers, financiers, and importing systems should contribute when their standards or commercial practices create the need. Cost sharing makes protection more credible and reduces the temptation to convert compliance into another extraction channel.

Evaluation focuses on conditions changed, remedies delivered, recurrence, and the distribution of enforcement burden. Public disclosure can report patterns, repeat violations, correction rates, buyer practices, and system weaknesses without publishing worker dossiers or reducing a country to one score. Debarment, penalties, or import restrictions may sometimes be necessary, but they need clear criteria, evidence, review, and a route to reconsideration when conditions genuinely change. A reopened facility should be judged by corrected conditions and workers' access to remedy, not the replacement of its certificate alone.

16.4 — Interdependence Without Domination

Interdependence can discourage disruption, but reliance becomes dangerous when one party can dictate the terms of exit. A trade relationship may create genuine mutual benefit even when the parties possess unequal bargaining power. Diversification can improve resilience without implying that every good must be produced domestically. Reciprocity matters most when commitments are testable and neither side can quietly shift the cost of adjustment to people with the fewest options. This section separates cooperative dependence from vulnerability that permits domination.

16.4.1 — Mutual Dependence

Mutual dependence exists when parties rely on one another, giving each an interest in the relationship's continuation. A buyer may need a supply while the producer needs a market. The relationship can discipline aggression because each party has something to lose from rupture.

Mutual dependence can be real and still highly asymmetric. A party with ready substitutes has more bargaining room than one facing large replacement costs. That asymmetry, not the mere fact of exchange, creates the possibility of leverage. The practical question is whether diversification, enforceable commitments, or other safeguards preserve meaningful options for the more vulnerable party (Keohane and Nye, 1977).

Asymmetric dependence can constrain policy without an explicit threat: a government may avoid environmental rules, labor protection, taxation, or diplomatic positions because a supplier, buyer, creditor, or investor might leave.

Fairness requires a distinction between ordinary bargaining and coercive leverage. Negotiation is legitimate when each side can protect its core dignity and essentials. Coercive leverage appears

when one side uses dependence to force concessions that a less trapped partner would reject. The remedy is not always exit.

16.4.2 — Diversification as Sovereignty Support

Diversification supports sovereignty by reducing any one actor's power to interrupt an essential function. It can involve suppliers, routes, technologies, currencies, energy sources, inventories, skills, standards, financing, and domestic repair capacity. Replacing one concentrated dependency with another is not diversification.

Technology diversity can provide options, but incompatible systems also create cost. Multiple standards, spare inventories, training paths, and maintenance regimes may reduce efficiency. The decision should compare the cost of redundancy with time-to-harm and the consequence of failure.

Counting suppliers is only the beginning of assessing diversity. Imagine that a service buys half its essential components from one firm and half from another. Both firms obtain the critical input from the same upstream plant. The service has diversified its immediate contracts while retaining one production dependency. A disruption at that plant can reach both orders at once.

A useful map follows the essential function far enough to identify that shared point. It may show a common processing facility, transport crossing, or source of a specialized part. The next decision then becomes specific: develop an alternative at that point, hold a suitable reserve, support repair capability, or choose a design that uses a more widely available input. The appropriate response depends on what can be changed before the service is needed.

Diversification can also be temporal. Two delivery routes that fail for different reasons may still be unavailable during the same season. An alternate source is useful only if it can deliver within

the interval the buyer can tolerate. The account should therefore ask when each alternative becomes available, not merely whether its name appears on an approved list.

The purpose is to build meaningful choice with proportionate effort. A second distributor may still improve service and bargaining even when it shares an upstream source. Recognizing its actual contribution allows the institution to value that improvement while directing its next investment toward the dependency that remains.

16.4.3 — Reciprocity and Fair Bargaining

Reciprocity does not require identical exchange. Countries, firms, workers, and communities begin with different resources, risks, and needs. A reciprocal arrangement gives each side real benefit, voice, information, and a route to challenge terms; it does not demand mirror-image concessions from unequal positions (Keohane, 1984; Rodrik, 2011).

Consider an illustrative one-off order expected to leave a supplier 20 units after its existing costs. A new buyer-required certification costing 30 units would turn that margin into a loss of 10 unless the terms change. The parties can negotiate a buyer contribution, a price adjustment, or a less costly equivalent that preserves the standard. If certification also serves later orders, allocate its cost over that expected use and test the result if those orders never arrive. Reciprocity requires an agreement about that risk, not merely the same formal obligation.

16.5 — Sanctions Exposure and Shipping Risk

Economic conflict tools deserve the same material honesty as trade agreements. Sanctions, export controls, shipping restrictions, insurance exclusions, banking de-risking, port closures, and vessel rerouting can be lawful or unlawful and necessary or reckless. Their stated targeting does not establish where the burden lands. The analysis follows pressure through food, medicine, energy, payment, transport, and everyday prices without substituting an economic account for a legal judgment (Hufbauer et al., 2007; Peksen, 2011).

16.5.1 — Sanctions as Burden Instruments

The trade-and-shipping inquiry follows how sanctions reach households through banking de-risking, insurance withdrawal, port delay, rerouting, import interruption, freight cost, and private overcompliance. Chapter 21 develops the full coercive-instrument test, including lawful purpose, civilian incidence, humanitarian channels, proportionality, review, and off-ramp.

The fairness discipline asks what sanction burden is intended and what burden is collateral. A targeted asset freeze differs from broad financial exclusion. A military-goods restriction differs from a measure that disrupts medicine or food. A sanction aimed at a decision-maker may be morally different from one that causes inflation for households (Peksen, 2011; Hufbauer et al., 2007).

Apply Chapter 21’s civilian-incidence and humanitarian-access tests to protect food, medicine, medical devices, water, survival energy, disability support, and basic communications wherever possible. At the trade interface, protect lawful banking and licensing routes, correct overcompliance, and prevent diversion to armed groups, corrupt officials, or sanctioned actors. If civilians bear the pressure while decision-makers remain insulated, urgently review both humanitarian cost and effectiveness;

unlawful collective punishment remains a separate legal determination (Hufbauer et al., 2007).

16.5.2 — Shipping and Maritime Risk

Public purchasers as well as private firms bear the cost of disrupted shipping. Rerouting changes fuel use, crew time, insurance premiums, port congestion, spoilage risk, inventory timing, and emissions. A cheaper purchase may become expensive after the route is counted.

Maritime comparisons should include port and fuel pollution, ballast water, noise, dredging, coastal damage, accidents, waste, and climate vulnerability alongside price and speed.

Resilience may require alternative routes, regional stocks, port capacity, crew protection, shared data, and crisis communication. Those measures carry cost and security tradeoffs that should be stated (Notteboom, Pallis, and Rodrigue, 2022).

16.5.3 — Evasion and Shadow Trade

Closed or costly lawful routes can encourage parallel markets under sanctions, tariffs, quotas, currency controls, shortages, or corruption. Predatory evasion and civilian survival needs require different assessments: medicine, food, fuel, and income may still be needed when formal routes fail. Trace brokers, shell firms, false labels, rerouting, hidden ownership, informal payments, and bribery without assuming that survival pressure makes every route lawful or safe (Early, 2015; Financial Action Task Force, 2023).

Rules should anticipate likely evasion, civilian need, enforcement capacity, corruption risk, and repair routes. The objective is not merely to punish shadow activity, but to reduce the conditions that make it profitable or necessary.

Sanctions, maritime restrictions, export controls, tariffs and humanitarian exceptions require jurisdiction-specific legal and

geopolitical review (Hufbauer et al., 2007; Early, 2015; International Committee of the Red Cross, 2015).

Geopolitical review should examine how a measure interacts with relative state power, historical grievances, alliances, colonial memory, security threats, and dependency.

16.6 — Trade as Peace Architecture

Trade's contribution to peace depends on the institutions and dependencies governing exchange, so neither commerce nor vulnerability should be treated as an automatic friend-or-enemy label (Martin, Mayer, and Thoenig, 2008; Copeland, 2015). The research is not unanimous: Martin, Mayer, and Thoenig (2008) report contrasting bilateral and multilateral openness relationships with conflict, while Lee and Pyun (2016) estimate peace-promoting effects for both, conditioned in part by proximity. Yakovlev and Spleen (2022) find that including trade–distance interactions can materially change diversification estimates. These disagreements show why a researcher should hold samples and conflict outcomes fixed while varying dependence measures and geographic interactions. The effect of a trade relationship must therefore be investigated through the structure of dependence, the cost of rupture, and the institutional alternatives available rather than a universal claim about trade volume.

16.6.1 — Trade That Builds Capacity

Foreign-exchange and financing terms can determine whether learning survives. A project may be technically sound yet collapse when imported inputs rise in price, payment arrives late, interest resets, or revenue is earned in a weaker currency than the debt. Contracts should identify these exposures, allocate them explicitly, and avoid placing every macroeconomic shock on the local producer. Capability is not durable if one currency movement can erase years of industrial learning.

Public procurement can function as a capability compact when it states the public need, quality and safety requirements, local learning objective, price discipline, maintenance duty, worker conditions, data and intellectual-property terms, and exit route.

The contract should purchase both a usable service and the capacity to sustain it after the original supplier leaves, while widening the field of qualified alternatives rather than creating a protected dependency (Rodrik, 2018; Juhász, Lane, and Rodrik, 2024).

16.6.2 — Trade That Shares Repair

Trade can move spare parts, emergency inputs, technical knowledge, finance, repair teams, standards, and productive capacity after disruption. These arrangements support mutual recovery when shocks exceed the capacity of any one place.

Inventories and spare capacity are forms of traded insurance. A region may maintain reserves of critical goods, while partners commit surge supply or reciprocal access. The arrangement has visible cost in storage, maintenance, and unused capacity, but it can prevent panic and coercive bidding when disruption arrives (Baldwin and Freeman, 2022).

Affected communities should set recovery priorities with outside partners; finance or equipment must not substitute donor-preferred projects or purchase long-term control through emergency access (United Nations and World Bank, 2018).

Match recovery finance to the loss and repayment capacity. Short debt maturities can outrun recovery, tied aid can restrict suppliers, and guarantees can socialize risk while privatizing return; grants, concessional finance, insurance, debt suspension, equity, or commercial credit should be chosen against those constraints.

After activation or a near miss, record delays, exclusion, unmet need, worker burden, and terms that became predatory under stress. Use the findings to revise the next agreement.

Chapter 17 — Energy, Food, Water, and Digital Infrastructure

Energy, food, water, mobility, waste, and digital systems share three features: people cannot readily refuse them, failures cascade across sectors, and dependable service rests on long chains of labor and maintenance. Each domain also has distinct engineering, ecological, financial, legal, and geographic constraints. The chapter therefore uses a common reliability frame without pretending that one policy fits every system.

Energy is one of the chapter's largest burden systems. Electricity, fuel, heat, cooling, grid stability, storage, transmission, affordability, energy poverty, regional dependence, clean-transition cost, climate risk, cybersecurity, and workforce capacity all shape whether daily life is secure. Reliability must be judged at the household, facility, network, and regional levels.

Water and land carry the chapter's deepest survival logic. Water is not optional infrastructure. It is health, agriculture, energy, sanitation, ecological stability, social trust, and peace. Land is not merely property or resource. It is housing, farming, watershed, memory, culture, biodiversity, conflict risk, and future room.

Mobility, logistics, waste, and physical networks show how essentials actually move. Hospitals need supply routes, schools need transit, farmers need storage, ports need labor protection, and communities need roads, resilient last-mile delivery, and safe waste systems. These dependencies make essentials chains of work, maintenance, movement, and trust as well as goods and services.

Digital infrastructure is often described as weightless, yet its resource demand, skilled labor, and failure risk belong in the same lifecycle account as other essential systems. Section 17.6 examines those obligations. A digital service can be essential at one hour and deferrable at another. A delayed entertainment download and a failed dispatch connection can use similar equipment while imposing very different consequences. The

chapter begins from the function being served, so technical similarity does not silently determine which interruption deserves priority or which backup is worth maintaining.

Energy, water, transport, communications, and storage form an interdependent service chain. Test peak demand, restoration and black-start capacity, cyber resilience, local repair, inventories, and protection of vulnerable users during combined disruption, as in Section 17.7.2 (International Energy Agency, 2026a; International Telecommunication Union, 2025).

The scale of unmet need remains severe. The 2026 edition of The State of Food Security and Nutrition in the World estimated that about 645 million people faced hunger in 2025, that 2.1 billion people, or 25.8 percent of the global population, experienced moderate or severe food insecurity, and that about 2.69 billion could not afford a healthy diet. Those figures do not select one production model. They require every efficiency claim to answer to access, nutrition, affordability, conflict, climate, land, water, price volatility, and the bargaining position of farmers and households (FAO et al., 2026).

17.1 — Essentials Before Luxury

The Essential Reliability Floor tests nine distinct dimensions of service: access, affordability, continuity, maintenance, redundancy, crisis protection, privacy, redress, and resistance to ranking people by worth (Sen, 1999; Hallegatte, Rentschler, and Rozenberg, 2019). It identifies what headline capacity conceals because success on one dimension cannot erase failure on another. Essentiality depends on the harm caused by lost access, so food, water, shelter, power, medicine, sanitation, mobility, communication, and some digital functions require context-sensitive protection. When employment, schooling, warnings, banking, benefits, or health appointments rely on a communication channel, institutions have made that channel essential even if it was once optional. The floor must be revised as technology and social organization change rather than fixed by prestige or market label.

17.1.1 — Essential Access and Price Shock

Price shock is especially severe when it arrives through several essentials at once. Higher fuel costs can raise food transport, water pumping, heating, cooling, manufacturing, school budgets, and clinic operations.

Responses may include temporary relief, reserve release, supply repair, anti-profiteering scrutiny, income support, or structural investment (FAO et al., 2026; International Monetary Fund, 2026a).

Assess exposure to essential-price shocks through the share of the household budget devoted to necessities, the size of price changes, available substitutes, household reserves, and time to relief.

Staffing, training, safety, rest and retention belong in the ordinary cost of reliability before a crisis exposes their absence

(Graham and Thrift, 2007; International Labour Organization, 2024b).

Essential work should be judged under surge conditions as well as calm ones. Storms, heat waves, holidays, cyber incidents, crop failures, outbreaks, strikes, and supply disruptions can turn ordinary schedules into overtime, sleep loss, family strain, injury risk, and emergency improvisation.

17.1.2 — Essential-System Reserves

Reserves also need moral priority rules. During shortage, the question is not who can pay fastest or who has the best private access. Hospitals, water systems, disability-related equipment, children, elders, emergency shelters, food preservation, and core communications may need priority because failure there creates wider harm. Those priorities have to be written before scarcity turns judgment into panic.

Reserves need rotation, compatibility tests, release authority, and replenishment as food and medicines expire, fuel and batteries degrade, parts become obsolete, backups lose compatibility, and staff move (Hallegatte, Rentschler, and Rozenberg, 2019; International Organization for Standardization, 2024a).

Two backups can share one point of failure. In a fictional example, each of two alternative delivery routes is unavailable in 5 percent of comparable time intervals. If their failures are independent, both fail in 0.25 percent of intervals. If the same disruption always disables them together, both fail in 5 percent. The separate failure rates alone permit joint unavailability anywhere from zero to 5 percent. Counting routes therefore cannot establish the protection they supply. Reserve planning should test shared power, access, staff, and suppliers, then measure whether the alternative still works when the primary

route fails. These figures illustrate probability, not an estimate for an actual network.

After shortages and near misses, record expired stock, distribution failures, vulnerable users, release delays and recovery time so the next budget reflects operational evidence (Hallegatte, Rentschler, and Rozenberg, 2019; International Organization for Standardization, 2024a). Stock rotation can connect ordinary purchasing with emergency readiness, provided replenishment keeps the reserve genuinely available.

Test reserve-access instructions with people who must use them under stress, including those who cannot complete a digital form.

17.2 — Energy Reliability and Long-Horizon Transition

Energy planning must reconcile reliability, affordability, emissions, supply dependence, and delivery constraints in permits, materials, labor, transmission, and trust. IEA analysis through 2030 identifies networks and flexibility as central bottlenecks amid industrial demand, electric transport, cooling, and data centers (International Energy Agency, 2026a; IPCC, 2023). A defensible regional portfolio considers generation, efficiency, storage, demand response, cybersecurity, vulnerable loads, workforce transition, and ecological repair under the actual grid and resource constraints. Compare options by lifecycle emissions, air pollution, water, land, waste, mining, biodiversity, decommissioning, and service reliability rather than a single technology label. Different mixes can be justified when costs, assumptions, sequencing, and burdens are disclosed.

17.2.1 — Reliability as Peace Infrastructure

Reliability also shapes institutional trust. When outages recur without explanation, when restoration favors wealthy districts, when bills rise while service declines, or when public authorities cannot say what loads will be protected, people begin to plan around abandonment (Hallegatte, Rentschler, and Rozenberg, 2019; International Energy Agency, 2026a).

Energy poverty can arise through disconnection, unaffordable use, unsafe substitutes, debt cycling, or fear of operating heating, cooling, refrigeration, cooking, lighting, or medical equipment despite a formal connection (International Energy Agency, 2026a).

Affordability failure can hide as demand reduction. A household may consume less energy because it is efficient, or because it is enduring unsafe temperatures, spoiled food, foregone

medical-device use, or other suppressed needs. Energy planning should therefore distinguish efficiency gains from deprivation-driven underconsumption before interpreting lower household demand as success (International Energy Agency, 2026a).

Operational review should track arrears, disconnection risk, indoor temperature, unsafe fuel substitution, medically necessary load, energy share after housing cost, seasonal exposure, and the time required to restore service. Results remain household- and climate-sensitive; they guide tariff, efficiency, housing, health, and protection design.

Disconnection policy is one of the sharpest tests of essential-service design. Utilities need workable payment discipline, but loss of power can create immediate health and safety harm. Arrears plans, targeted support, seasonal protections, medical exemptions, notice, and appeal should be considered before disconnection becomes punishment disguised as billing.

17.2.2 — Clean Transition Costs

Network investment is often the binding condition. New generation does not automatically deliver usable power where and when it is needed. Transmission, distribution, storage, interconnection, control systems, permitting, cybersecurity, and trained crews determine whether capacity becomes reliable service. Underinvestment in the network can make both clean supply and existing supply appear less dependable than they are (International Energy Agency, 2026a).

Households experience transition through bills and equipment. A heat pump, vehicle, insulation project, appliance, or connection upgrade may lower lifetime cost while requiring cash, credit, landlord cooperation, skilled installation, and confidence in future prices. Rebates that arrive after purchase can exclude people

without liquidity. Rental housing creates split incentives between owners and residents.

Date forecasts and state uncertainty, stress cases, and revision triggers for changes in demand, fuel prices, weather, technology cost, industrial location, or policy (International Energy Agency, 2026a; IPCC, 2023).

17.3 — Food Security and Agricultural Sanity

Food policy must connect nutrition and affordability with the systems that produce and deliver meals (FAO et al., 2026; HLPE, 2020). Those systems depend on storage, transport, trade, finance, labor, scientific knowledge, and climate adaptation. A country can have adequate aggregate supply while households still lack a healthy, reachable diet. Analysis must therefore follow the cost and reliability of actual access, not only the quantity grown or imported.

17.3.1 — Nutrition Before Commodity Abundance

Commodity abundance can hide nutritional fragility. A country may produce or import enough calories while healthy diets remain unaffordable, fresh food is inaccessible, school meals weaken, elders skip protein, and children grow up on inexpensive diets that supply calories without meeting their nutritional needs.

Nutrition also has time and equipment conditions. A household needs cooking space, storage, water, energy, schedule stability, disability accommodation, cultural fit, and safe travel. A food policy that offers ingredients without recognizing time, transport, and kitchen capacity may look generous while leaving the meal impossible (FAO et al., 2026).

Household knowledge, health, and money also affect the conversion of ingredients into nourishment.

School meals, elder meals, emergency food, and community food programs should test whether their lines, cards, forms, cafeterias, deliveries, and pantries provide nourishment without shaming people.

Farmer stability is also institutional stability. Credit terms, crop insurance, water rights, input prices, land tenure, market concentration, extreme weather, labor availability, and public procurement can determine whether a farm survives. A society

that wants stable food but leaves producers under permanent volatility is asking farmers to absorb the public's peace risk privately.

Severe financial pressure can leave farmers with less room for long-horizon land care, even when they value ecological stewardship.

17.3.2 — Supply Diversity and Local Capacity

Test concentration in crops, seeds, fertilizer, export markets, ports, buyers, irrigation, and storage against the time and alternatives available after disruption.

Supply diversity must include dietary and cultural diversity. Food security measured only by a few commodities can erase allergies, disability needs, religious diets, infant needs, elder needs, rural access, and local foodways. The dignity of food lies partly in whether nourishment arrives in a form people can actually use (HLPE, 2020).

Food security needs both local capability and wider exchange. Local production can shorten routes, preserve knowledge, support farmers, and create emergency options. Trade can smooth seasons, share ecological advantage, widen diets, and supply places that cannot produce enough of every needed food.

A seasonal purchasing plan can make local capacity and wider exchange work together. Imagine a community kitchen that can obtain a particular crop locally during its harvest months but relies on another region during the rest of the year. Its annual plan can specify what local growers are able to supply, what storage preserves, and when the remaining deliveries are needed. The question is how the seasons connect, not which source deserves a permanent moral preference.

A standing order may give growers a clearer basis for planning. The kitchen still needs agreement about quantity and quality, as

well as a way to adjust when a harvest differs from the forecast. A storage arrangement adds another choice: some of the crop can be carried forward, but space, handling, and preservation use resources. The comparison is between complete routes to usable meals.

Wider trade can then fill a defined gap rather than displace local knowledge by default. Local production can support freshness or rapid replenishment where it is well suited, while exchange draws on different growing conditions. The resulting resilience comes from the fit among those capacities. It is strongest when the plan can explain what each source contributes during the periods when the others contribute less.

Environmental fit matters. Producing a water-intensive crop locally may be less resilient than importing it, while dependence on imports for a staple may be dangerous where routes are fragile. The right comparison includes water, land, emissions, biodiversity, labor, price, and time-to-harm.

Soil health, pollinator protection, water efficiency, biodiversity, emissions reduction, animal welfare where relevant and reduced chemical harm may require transition finance, technical assistance, risk sharing, market access, research and fair pricing.

By the time reports confirm a food-price shock, households may already have changed diets, skipped meals, borrowed money, sold assets, delayed medicine, or withdrawn children from activities. Early signals should therefore come from schools, clinics, food banks, farmers, markets, and household surveys.

Read weather signals, commodity prices, fertilizer and energy costs, transport disruptions, currency movement, conflict risk, crop disease and household food insecurity together. The aim is to buy time before hunger becomes public panic, without treating warning signals as certain predictions.

During food-price shocks, publish clear information on supply, reserves, relief, timing, and uncertainty using the crisis-communication discipline in Section 19.5.

17.4 — Water, Land, and Ecological Stewardship

Water and land are where material peace becomes least abstract. Energy can sometimes switch fuels, transport can sometimes change routes, and digital systems can sometimes relocate. Water basins, soils, aquifers, watersheds, forests, wetlands, and land memories cannot move so easily. Their degradation narrows the future room of whole communities and can intensify pressure across food, health, energy, migration, and conflict systems (UNESCO World Water Assessment Programme, 2025; IPBES, 2018).

17.4.1 — Water as Non-Optional Peace Condition

Water access can improve life even without raising earnings. In Tangiers, Devoto et al. (2012) studied randomly offered help with obtaining a home water connection on credit. Households already had access to water from the grid. Connections returned time for leisure and social activity and reduced water-related conflict, with improved reported well-being rather than demonstrated health or income gains. The result makes an important distinction concrete: time made available for life is a benefit, not a failed productivity intervention. A local appraisal should still identify who collects water and who gains usable time, including differences associated with gender, age, disability, and distance. Connection cost and continuing service obligations belong beside those gains.

Sanitation belongs beside drinking water. Failed wastewater systems can spread disease, damage ecosystems, and prevent safe, dignified use of schools and public facilities.

Consider an accounting illustration, not field data. Irrigation withdraws 100 units in a season, consumes 60, and returns 40 in usable form within the same basin and period. After an upgrade it withdraws 75, consumes 60, and returns 15. Assume unchanged storage and other inflows. Withdrawal falls by 25, but net

depletion remains 60: the extra 25 left at the intake is offset by 25 fewer return-flow units. The farm may gain from the upgrade; the basin has not thereby gained 25 units for another consumptive allocation. Review should also test changes in planted area or crop use, while recording where and when return water is actually recoverable. This illustrates the field-efficiency and basin-availability distinction examined by Grafton et al. (2018).

17.4.2 — Land Degradation and Conflict

Land degradation amplifies pressure by narrowing livelihoods. Soil loss, desertification, deforestation, contamination, erosion, heat, flood risk, and declining yields can push households toward migration, debt, land disputes, resource conflict, or dependence on outside aid. The pathway is rarely direct, but the burden is real.

Land stress cannot be converted into blame against vulnerable communities. Conflict risk grows through interacting causes that may include governance failure, unequal rights, violent actors, market pressure, climate disruption, corruption, and exclusion. People already carrying ecological loss are not the moral cause of instability. Repair reduces pressure and secures rights without recasting harmed populations as threats (IPBES, 2018; von Uexkull et al., 2016).

Commons stewardship begins with the recognition that shared resources can be damaged by poorly governed extraction and neglect. Fisheries, aquifers, forests, wetlands, and grazing lands need rules for access, monitoring, maintenance, and resolving disputes that fit the resource and the community using it (Ostrom, 1990).

Community monitoring can be supported and tested rather than merely prescribed. Slough and colleagues (2021) coordinated randomized trials in six countries, assigning monitoring interventions to 400 communities and observing 347 comparison

communities. Most intervention communities adopted regular monitoring over roughly a year. Across the sites, the pooled results showed modest improvements in resource conservation, satisfaction, and knowledge. The promising mechanism is information that users can bring into decisions about a shared resource. The next local test should follow that information into an actual rule or response, record the work required, and examine whether practice continues after outside support ends. A pooled gain does not establish the same result in every setting.

Commons stewardship should also ask who does the stewardship labor. Monitoring, restoration, rule enforcement, meeting attendance, conflict mediation, and education take time. If that labor is unpaid, gendered, or carried by already burdened communities, a commons can look participatory while hiding extraction through civic duty.

17.4.3 — Indigenous and Local Knowledge

Indigenous and local knowledge can reveal features of a place that a distant plan overlooks. Seasonal patterns, species behavior, fire history, flood memory, soil change, sacred sites, migration routes, contamination histories, and informal use may be known before they are written in official systems.

Include Indigenous and local knowledge before outcomes are fixed, with translation, participation capacity, applicable rights, refusal, and a reasoned response explaining whether and how that knowledge affected the proposal (Tengö et al., 2014; Whyte, 2017).

Ecological accounting does not imply that every value can be priced. Relations to land, water, species, health, memory, and future can be distorted by monetary reduction. Yet refusing to price everything cannot become refusing to count anything. Money, physical indicators, rights, testimony, thresholds, and

uncertainty can work together without pretending to be interchangeable (Whyte, 2017; IPBES, 2019).

Water, energy, food, and land decisions should be reviewed together because each can move burden into the others. Power may require cooling water and corridors; irrigation may raise food output while depleting aquifers or increasing electricity use; desalination may widen supply while adding energy and brine costs. Separate permits can approve components whose combined demand exceeds the basin, grid, or community that must carry them.

Drought allocation makes priority visible because households, hospitals, sanitation, ecosystems, farms, industry, energy, and public safety may share one constrained supply. Rules should be prepared before crisis, state protected minimums, distribute conservation duties, and provide emergency review. Scarcity becomes more dangerous when allocation is improvised or powerful users preserve convenience while basic needs are rationed.

Restoration requires finance and maintenance beyond construction announcements. Wetlands, soils, forests, rivers, contaminated sites, groundwater, urban shade, and floodplains may take years to recover and may never return fully to an earlier condition. Budgets should identify responsible parties, milestones, monitoring periods, long-term stewards, reserves, and what happens if restoration fails.

Cross-boundary governance needs durable dispute routes. Watersheds, airsheds, species ranges, migration corridors, and coasts cross jurisdictions. Shared monitoring, data access, local participation, scientific review, emergency communication, negotiated allocation, and independent mediation can reduce the chance that ecological pressure becomes political humiliation

while keeping unequal geography and history visible (UNESCO World Water Assessment Programme, 2025).

17.5 — Mobility, Logistics, Waste, and Physical Networks

Apply lifecycle appraisal to infrastructure siting, construction, operation, maintenance, replacement, failure, and decommissioning, including the land, materials, energy, water, labor, and public authority each requires. Distributional analysis should follow both benefits and harms. Users may be far from the people living beside the asset, and future generations may inherit its repair. Public usefulness does not erase concentrated local burden; it creates a stronger duty to minimize, compensate, monitor, and restore (IPCC, 2022; Hallegatte, Rentschler, and Rozenberg, 2019).

17.5.1 — Mobility Networks as Access

Mobility is plural. Walking, rolling, cycling, buses, trains, roads, paratransit, ferries, emergency transport, rural routes, and freight access matter differently by place. Assess whether people can reach essential life without excessive time, risk, cost, exposure, or dependence (Lucas, 2012; Martens, 2016).

Logistics resilience should be tested by time to failure. What happens after one day, one week, one month, or one season of disruption? Some delays are inconvenient; others become hunger, treatment interruption, crop loss, unsafe waste, closed schools, or panic buying. Routes should be classified by the burden of delay, not merely by commercial value.

Measure whether essential supplies actually reach households in rural areas, islands, informal settlements, and disaster zones, including disabled people, elders, and language-minority communities; full warehouses do not establish successful last-mile delivery.

17.5.2 — Waste Systems

Waste systems are health infrastructure because every society produces what it must remove, treat, reuse, recycle, contain, or make safe. Garbage, sewage, medical waste, chemical waste, construction debris, e-waste, agricultural runoff, and industrial residue do not vanish when they leave the visible district.

Waste has to be followed to its destination. Landfills, transfer stations, incinerators, sewage plants, informal dumps, recycling centers, and export routes often concentrate near communities with less voice. Fair waste policy counts exposure, transport, labor, leakage, local consent, repair, and the places that absorb what others discard (UNEP and ISWA, 2024; Kaza et al., 2018).

This inquiry asks whether throughput creates hidden burden. Excess packaging, disposable goods, short product lives, poor repairability, high waste, long supply chains, and planned obsolescence can all shift cost into workers, land, water, waste systems, households, and future maintainers (Kaza et al., 2018).

Material throughput should be evaluated against use duration. A durable public bench, repairable phone, long-lived pipe, reusable medical container, or modular grid component carries a different burden from disposable equivalents. Longevity is not always best, but planned short life should justify itself against repair and waste costs.

17.6 — Digital, Compute, Storage, and Archive Infrastructure

A public benefits office may present cloud storage and artificial intelligence through a simple portal. The interface conceals dependencies in identity, accessibility, language support, human appeal, backups, archives, and vendor exit. Failure in any one component can make the service unusable even when the application appears to be running. Those functions belong to the material service account (International Telecommunication Union, 2025; Karippacheril et al., 2024; World Bank, 2025a; NIST, 2024).

17.6.1 — Digital Access as Essential Access

Affordability is recurring. A household may own a phone yet ration data, lose service after a bill, share one device among several people, or lack storage and updates required by a portal. Connection statistics should be joined to cost, quality, device, and usage evidence (International Telecommunication Union, 2025).

The World Bank's 2025 report on artificial-intelligence foundations treats connectivity, compute, context, and competency as interdependent foundations, not software alone (World Bank, 2025a).

Preserve safe, affordable access and a genuine human alternative when the digital interface fails.

17.6.2 — The Material Cost of Compute

Computing depends on physical infrastructure. Servers require chips, buildings, electricity, cooling systems, water where needed, network equipment, skilled staff, maintenance, security, and eventual replacement.

The IEA's 2025 base case projected about 945 terawatt-hours of global data-center electricity use in 2030; its 2026 update

projected about 950 terawatt-hours. Local grid effects can be much larger than the global share suggests. Include land and public incentives in the opportunity-cost account (International Energy Agency, 2025; International Energy Agency, 2026b; Masanet et al., 2020).

Include minerals and hardware supply alongside work on data, system evaluation, and content moderation in the local account. State where costs fall and which public capacities are displaced, reinforced, or newly required.

Retain records needed for memory, correction, and audit; let others expire. Keep work, public, Fairism, research, donor, vendor, and ordinary-life records separate, applying the purpose and access limits in Section 4.4.

17.6.3 — Cybersecurity as Ongoing Burden

Cybersecurity is maintenance under adversarial conditions. It requires governance, asset knowledge, access control, patching, monitoring, backups, training, incident response, recovery, vendor review, and communication. It is a continuing function, not a product installed at procurement.

Digital outages can delay care, interrupt supply, freeze income, or expose identity when essential services depend on the affected system (NIST, 2024; Anderson et al., 2013).

Asset and dependency inventories are foundational. Institutions need to know which systems they operate, which software they depend on, where their data sits, and which vendors have access. A usable inventory names owners, critical functions, update routes, dependencies, and recovery priorities.

Incident planning must fund public explanations, recovery assistance, and learning after failures instead of shifting repair work to ordinary users.

17.6.4 — Vendor Dependence and Exit Cost

Data rights are central to vendor exit. Contracts need explicit terms for ownership, permitted use, location, retention, deletion, training use, breach duties, and the format and timing of export. Personal and public records cannot become bargaining chips held inside a provider's platform. Migration has to preserve privacy, evidentiary integrity, and the ability of people to correct information about themselves.

Claims of digital savings should account for work transferred to local staff through self-service or systems described as automatically maintained, as well as the staff needed to verify, correct, secure and explain the system (Gmyrek et al., 2025; Kellogg, Valentine, and Christin, 2020).

Cybersecurity creates another dependency. A vulnerable supplier, update channel, library, identity service, or managed platform can transmit risk across many institutions at once. Contracts should allocate patching, disclosure, incident response, evidence preservation, and supply-chain duties, while public buyers retain independent monitoring and the ability to isolate critical functions (NIST, 2022; CISA et al., 2023). Test whether a compromised update can be stopped without disabling every institution that depends on the same supplier.

A usable exit includes a working replacement and intelligible records, not merely permission to receive a large data export.

17.7 — Essential Systems as Peace Infrastructure

Reliable essential systems can lower the pressure that turns ordinary shocks into public emergencies. Their peace value depends on continuity in places and moments where substitute services are scarce. Investment at the point of construction is not enough if staff, maintenance, and repair routes disappear later. The analysis must distinguish the social benefit of dependable access from the resources needed to keep it dependable. The final two subsections connect those functions without treating infrastructure as a substitute for political judgment.

17.7.1 — Essentials Reduce Conflict Pressure

Essential insecurity can intensify conflict pressure without making people under stress morally responsible for conflict.

Research on essential-service repair can test changes in household precautions and a specified conflict outcome. Even a randomized repair does not automatically show that reduced anxiety caused a change in conflict, because the intermediate response was not itself randomized. Causal mediation analysis must state its additional assumptions, including how common causes of that response and the outcome are addressed, and test sensitivity to plausible violations (Imai, Keele, and Tingley, 2010).

17.7.2 — Essentials Require Maintenance

Apply Chapter 12’s maintenance duties to aging digital components as well as physical assets (NIST, 2024). During planned work, preserve access through notice, alternatives, temporary support, language access, and rapid complaints, especially for disabled people, caregivers, hourly workers, rural residents, and households without backup.

Consider a heat emergency that reduces grid output just as water-treatment pumps and clinic refrigeration require more power. A regional plan cannot establish readiness through three separate reliability tests if the same backup generators, drivers, electrical technicians, and fuel deliveries have been promised to all three systems. Before declaring readiness, list shared dependencies, calculate usable capacity rather than nominal capacity during the combined event, identify who decides the lawful priority of loads, and reserve an alternative for patients whose treatment would otherwise stop. The emergency authority must be able to protect water safety and clinical continuity without exposing household health records or quietly transferring the outage to poorer neighborhoods. Exercise failure detection, activation of authority, fuel and staffing arrangements, accessible public instructions, and service restoration together, and record which threshold was missed.

Part V — Sovereignty and Peace Architecture

A credible domestic floor also needs protection against disruption across borders. Shelter, care, work, food, and energy can still be disrupted by sanctions, arms races, cyberattack, debt pressure, forced migration, and mistrust between states. Material peace therefore asks whether international systems extend ordinary stability or leave it exposed to distant decisions and strategic fear.

Sovereignty enters as local authorship: the capacity to preserve law, memory, public institutions, culture, infrastructure, and political judgment without absorption into a stronger actor's template. That capacity also carries an internal duty of answerability. Corruption, repression, violence, and administrative cruelty cannot become unreachable merely because they are carried out in sovereignty's name. Durable peace joins self-government to public accountability.

Transparency can reduce hidden harm while respecting safety, capacity and lawful self-government.

The same disclosure duty can impose unequal costs across administrative capacity, security exposure, and bargaining power. Chapter 18 tests how protection, timing, funding, and review can make cooperation reciprocal.

Security realism remains part of the moral argument. Diplomacy, deterrence, protected channels, intelligence, and emergency communication can preserve lives because premature disclosure may endanger people, installations, negotiations, witnesses, or crisis stability. The constructive standard is narrow protection with a stated purpose, accountable owner, defined scope, durable record, review clock, and a route back into public memory when the danger changes (Jervis, 1978; United Nations and World Bank, 2018).

Reciprocal safeguards can support durable arms reduction by protecting shared information, diplomatic commitments, verification, and the people carrying conversion. The following

chapters test whether those safeguards provide sufficient assurance under the actual threat; parity alone does not establish that a reduction is safe.

The resulting order is polycentric. States can share standards, facts, inspections, crisis channels, and common safeguards while preserving separate sovereignty and lawful authority. Smaller states, under-resourced administrations, vulnerable communities, and ordinary people remain participants with correction and refusal rights. Peace deepens when coordination removes hidden burden and enlarges local authorship.

The sequence develops that standard in three stages. Chapter 18 builds reciprocal parity as the non-imperial grammar of sovereignty. Chapter 19 protects diplomatic timing and narrow secrecy through record, review, and crisis communication. Chapter 20 tests arms reduction through verification, worker protection, regional repair, and failure memory. Together they show what international peace must carry before its gains become durable.

Chapter 18 — Sovereignty and Reciprocal Parity

Reciprocal parity begins with four practical questions: who becomes exposed, who carries implementation labor, who gains interpretive power, and who can refuse, contest, or correct the result? Sovereignty matters because it determines whether cooperation enlarges a people's authorship or turns that people into an object of administration.

Sovereignty carries centuries of struggle, dignity, abuse, aspiration, memory, violence, legal development, anticolonial resistance, and statecraft. It has protected peoples against invasion, forced ideology, outside command, resource extraction, and institutional humiliation. Rulers have also invoked it to shield corruption, predation, censorship, private violence, and official secrecy. The aim of reciprocal parity is to preserve that protection while making abuses answerable.

Reciprocal parity means comparable burden with preserved agency. Constitutions, records, technology stacks, administrative vocabulary, and implementation speed may differ. The governing requirement is that visibility, verification, protected exceptions, correction, and explanation leave each party able to author its future without giving one side convenience, leverage, or reputational reward at another's expense (Barnett and Duvall, 2005; Lake, 2009).

Equal disclosure duties can expose poorly protected infrastructure records, divert essential-service staff, or endanger witnesses and minorities. Sections 18.1 and 18.4 examine those differences in exposure and capacity.

Local authorship and reciprocal repair provide the answer to transparency imperialism. A standard loses legitimacy when one institutional model is treated as every society's destination, one donor class becomes examiner, one platform becomes default government, or one dashboard becomes the world's measuring rod. Cooperation gains legitimacy when it addresses the debt

pressure, extraction, climate exposure, security dependence, and capacity gaps that make visibility costly.

Polycentric cooperation is the constructive alternative. States and institutions can coordinate around climate, health, migration, security, finance, and digital infrastructure while authority remains distributed across legitimate levels. Common safeguards, reciprocal explanation, and interoperable records can connect these levels without sovereign merger or a global ruler (Ostrom, 2010; Slaughter, 2004).

Security context changes the burden of disclosure. Invasion risk, insurgency, organized crime, cyberattack, resource dependence, and institutional fragility can turn ordinary transparency into a route to harm. Burdened specificity requires the protected interest, plausible threat, scope, granularity, custody, review, and release condition to be stated and periodically retested.

Capacity is the labor beneath answerability. Records, audits, secure systems, archives, translation, accessibility, procurement skill, legal counsel, public explanation, and correction require trained people and stable budgets. Support succeeds when the receiving institution is more capable of governing itself after outside assistance ends; otherwise assistance has purchased compliance rather than strengthened sovereignty.

18.1 — Sovereign Equality and Material Asymmetry

The material-burden analysis asks who pays for the same rule under different conditions. Consider a hypothetical compact that requires monthly procurement reporting. A ministry with secure archives and trained auditors may absorb the work within its existing budget; another may have to divert the staff who purchase medicines. Equal deadlines would then conceal unequal service costs. Before either party commits, the compact should identify the staffing, translation, cybersecurity, and finance needed to meet the obligation while keeping essential services running.

18.1.1 — Asymmetric Exposure Risk

Payment, cloud, satellite, shipping-insurance, standards, and data networks can constrain sovereignty without crossing a border. Map chokepoints, fallback capacity, switching costs, and escape time to identify where an outside actor can exercise practical command (Keohane and Nye, 2012; Barnett and Duvall, 2005).

Interpretive power is another asymmetry. A powerful state, rating agency, platform, newsroom, donor consortium, or creditor group may frame disclosed information more loudly than the society that produced it. When one side controls the microphones, models, analysts, and markets, a record can become a narrative weapon. Context rights, reply rights, provenance, and correction are therefore part of material parity (Keohane and Nye, 2012; Barnett and Duvall, 2005; Nissenbaum, 2010).

18.1.2 — Security Context Differences

Distinguish a plausible disclosure-related threat to people, infrastructure, negotiations, witnesses, or defensive systems from protection of embarrassment, privilege, or convenience; Chapter 19 develops the review process.

For each proposed disclosure, ask how it could change exposure to harm. Could it reveal a port dependency, grid weakness, food reserve, procurement concentration, ministerial vulnerability, or protected identity? Could it make corruption easier to challenge, reformers easier to target, or both? Granularity, timing, custody, access, and review should follow the causal answer instead of a binary label of open or closed (Jervis, 1978; Fearon, 1995; United Nations and World Bank, 2018).

Status language should name the specific condition under review instead of summarizing an entire society. Terms such as low-capacity, unready, fragile, or lagging may hide distinct conditions: too few trained staff, an unreliable archive, unsafe disclosure channels, debt pressure, missing review authority, or no funded correction route. Separating those conditions makes criticism exact and turns a label into a repair agenda.

18.1.3 — Parity Before Deep Trust

The party asked to open records should receive reciprocal disclosure, security support, legal standing, and limits on the other party's leverage. Enforceable correction routes allow cooperation to continue when confidence is incomplete rather than depend on goodwill after exposure.

Timing is part of parity. One party may be ready for public release while another still needs administrative repair, cyber protection, or legal review. Staged disclosure can preserve reciprocity better than identical release dates when the reason, support plan, and review date are explicit (Kydd, 2005; Ostrom, 1990; Koremenos, Lipson, and Snidal, 2001). Further disclosure should be pausable after a breach, but information already made public may be impossible to recall. The pause rule should specify which material remains protected, which obligations continue, who may restart disclosure, and what evidence that decision

requires. Those conditions make caution reviewable without promising that exposure can be undone.

18.2 — Local Sovereignty Inside Global Visibility

Other volumes in the series examine legal architecture and the sequence of institutional rollout. Here the narrower question is whether partners have the material capacity to cooperate on usable terms. A reporting requirement can become de facto foreign control if one side supplies nearly all the funding, technical knowledge, and interpretation. Material sovereignty requires examining that imbalance alongside any formal treaty or consent.

18.2.1 — Sovereignty as Local Authorship

Institutions carry language, memory, legitimacy, law, and the scars of earlier intervention. A reporting form or office design that works in one setting may weaken practical authority in another. The durable test is whether the community that will live with the adopted form can revise it in a voice it recognizes after outside attention has moved on (Sen, 1999; United Nations, 2007; OECD, 2011).

An absorption audit asks whether shared visibility changes who effectively governs. If records are hosted, translated, summarized, funded, searched, and interpreted mainly by outside actors, public access can coexist with outside control over the records and their interpretation. The compact may need custody rules, context rights, and local summary capacity.

18.2.2 — Plural Legal Forms

Legal plurality is design reality. Common-law, civil-law, customary, religious, Indigenous, regional, federal, unitary, hybrid, and post-conflict systems carry different forms of legitimacy and risk. Shared safeguards may require notice, review, remedy, and non-domination while allowing jurisdictions to use distinct legal instruments. The Peace of Fairness treats that

plurality as material to cooperation rather than a temporary obstacle to uniformity (Merry, 1988; Griffiths, 1986; Berman, 2012).

Overlapping legal orders can shape the same dispute through customary authority, municipal law, religious norms, treaty duties, constitutional rules, professional standards, and transnational commercial practice. Peaceful coordination therefore needs conflict rules, translation, appeal, and protection for people vulnerable inside any one order. Respect for plurality includes a route to challenge unreviewable local authority (Merry, 1988; Griffiths, 1986; Berman, 2012).

Every legal form brings costs, staffing needs, language requirements, evidentiary rules, access barriers, appeal routes and institutional rhythms.

Cultural and historical context cannot excuse cruelty, yet ignoring it can create new injury. Communities remember occupation, forced reform, missionary pressure, debt conditionality, resource extraction, proxy conflict, sanctions, migration trauma, and elite capture. A proposal that appears neutral to its designers may carry old echoes to those asked to receive it. A fair process asks how the standard will sound, who will be trusted to explain it, which wounds it touches, and whether affected communities can reshape it before adoption.

18.2.3 — Global Coordination Limits

Mandate creep often begins with useful adjacent information. A body may seek wider data because it improves work on emissions, procurement, debt, migration, or security, but that benefit warrants review, not automatic authority. Expansion needs a lawful route, stated burden, public reason, bounded custody, and use limits.

Purpose is the first limit. Name the specific burden the compact addresses so it cannot become general governance by another name.

Authority is the second limit. Coordination must not silently gain power over domestic policy, lawful institutions, societies' worth, or public narrative. Any added authority requires an explicit legal route, public consent, constitutional review, and transition governance (Keohane, 1984; Slaughter, 2004; Zürn, 2018).

Jurisdiction supplies the third limit. A compact must identify which institutions authorize its duties, settle disputes, and amend or end the arrangement. The same allocation of responsibility should name the local institution that can correct shared records after the compact changes or ends.

18.3 — Reciprocal Transparency Compacts

A reciprocal transparency compact identifies its purpose, parties, affected groups, scope, safeguards, verification, review, and breach response. It must also specify who may access shared material, how long access lasts, what citizens will be told, and how changed conditions alter the arrangement, including review of protected exceptions (Abbott and Snidal, 2000; Koremenos, Lipson, and Snidal, 2001; United Nations, 2003). An emissions compact should not quietly reach all industrial data. A procurement compact should not become a political dossier. A border-incident compact should not become surveillance of entire communities.

18.3.1 — Mutual Burden Terms

Mutuality concerns comparable burden, not identical tasks. Inspection duties may be matched by limits on financial leverage, military posture or donor influence (Ostrom, 1990; OECD, 2011; Koremenos, 2016).

The mutual-burden register should assign each obligation to an owner, funder, deadline, and remedy. In a hypothetical procurement compact, the disclosing administration might prepare the records while a jointly governed fund pays for secure storage, translation, independent legal review, and participation time. The terms would also identify who restores service after a platform failure and who pays to move the records to a replacement system.

Fund recurring protection for the life of the compact; the full one-time and recurring cost distinction is set out in 18.4.3.

Mutual terms should also govern interpretation. Alongside the context, reply, provenance, and correction rights specified in 18.1.1, provide expiry for provisional findings so that an early

assessment cannot harden into permanent stigma (Ostrom, 1990; OECD, 2011; Koremenos, 2016).

The compact's ethical test is whether mutuality survives adversity. When scandal, shortage, cyberattack, or conflict hits, do both sides still carry comparable duties, or does one side retreat into protection while the other remains exposed?

18.3.2 — Verification Bodies

Verification bodies earn trust only when their own authority is visible, limited, and corrigible. A credible mandate identifies what may be inspected, what evidence may be requested, which conclusions may be issued, and which matters remain outside the verifier's jurisdiction. Appointment, funding, expertise, conflicts of interest, security practice, and correction procedures therefore belong to the compact itself, not to an unpublished operating custom (IAEA, 2022; OPCW, 1993; OSCE, 2011).

Reporting and verification perform different jobs. Reporting creates a declared account, reveals implementation gaps, supports coordination, and identifies assistance needs. Verification independently tests selected claims under a defined mandate and evidence rule. Conventional-arms reporting research shows that political will, national capacity, and interagency coordination shape the quality of reports, which is why reporting support and verification design must be planned separately (Nicolin, Holtom, and García García, 2025).

Reciprocity has to reach inspection practice: access, notice, observation, protected zones, travel, translation, evidence handling, and reply. In 2026, the OSCE reported that the South-Eastern European Article IV process had destroyed more than 10,000 weapons and completed more than 830 inspections, involving nearly 1,300 arms-control specialists and more than 280 observers from 30 participating States. The record shows how

declarations, visits, shared procedures, and locally built capacity can become a durable verification institution without turning operational success into proof of peace (OSCE, 2011; OSCE Secretariat, 2026).

Verification should not expose people to compensate for deficient institutional records. Where personal data are unavoidable, fund applicable consent, access, correction, and deletion safeguards under Section 4.4.

When parties disagree, the record should identify the precise question in dispute, the evidence each side relies on, and what further observation or review could narrow the disagreement.

18.3.3 — Exit and Breach Rules

Exit and breach rules keep a compact voluntary, governable, and repairable. Failure may follow bad faith, cyberattack, corruption, war, disaster, incapacity, or a changed threat, and the design needs to distinguish those causes before assigning consequence. Notice, reasons, review, interim protection, and public explanation where safe help separate accountable exit from evasion and prevent an unsafe arrangement from becoming captivity (United Nations, 1969; Helfer, 2005; Koremenos, 2016). The compact should define its principal breach thresholds before a crisis occurs, including the duty at issue, the evidence required, the urgency of response, and any realistic period for cure.

The compact should identify which duties survive breach. Personal-data protection, witness safety, archive custody, correction, emergency communication, and unfinished repair may remain necessary after cooperation pauses. Ending an agreement does not erase the people exposed by its records or the obligations created while it operated. Each surviving duty should therefore have a named custodian, the resources needed to carry it, and a condition for completion or lawful transfer.

Remedies may include technical correction, repair funding, delayed disclosure, third-party review, temporary pause, renegotiation, proportionate consequence or termination (United Nations, 1969; Helfer, 2005; Koremenos, 2016). Identify who keeps critical services running during a pause. Incapacity may call for support, narrower scope or more time; deliberate concealment or repeated evasion may require independent review, stronger consequence or termination.

18.4 — Material Parity Bands and Synchronization Costs

Closing cross-national capacity gaps takes funded synchronization rather than ranking countries by dignity or legal standing (Fukuyama, 2013; Andrews, Pritchett, and Woolcock, 2017; UNDP, 2009). Parity bands are purpose-bound categories for administrative ability, security exposure, economic cushion, and the cost of joining a specific compact. Economic cushion must be assessed through liquidity, solvency, and real-resource constraints for the function at issue, not national income alone, following Section 13.5.3. Each profile needs a named function, evidence and provenance, uncertainty, authorized users, responsible review, expiry, appeal, prohibited uses, and a support route that can retire the category when no longer necessary. Comparable measures may include patch time, audit backlog, language access, procurement risk, debt exposure, or service continuity, but unlike conditions should not be collapsed into a composite country rank. The test is whether synchronization can be financed and reviewed without shifting hidden risk onto the partner with less capacity.

18.4.1 — Administrative Capacity Bands

An administrative profile should distinguish existence from performance. A law may create an appeals office, but the material question is whether people can reach it, receive a decision, correct an error, and avoid retaliation within a usable time. Staffing, backlog, language access, and non-digital routes reveal more than formal institutional presence.

Capacity should be tested under combined stress rather than through a single compliance demonstration. A mistaken record, whistleblower disclosure, translation dispute, server outage, ministerial conflict, and citizen appeal may arrive in the same

month. A useful band therefore measures service continuity, correction time, staff load, and whether compliance displaces clinics, schools, procurement oversight, or local emergency work.

18.4.2 — Security Exposure Bands

The assessment asks how disclosure could change adversary, market, creditor, or internal intimidation behavior, and should reward candor by attaching protection to the risk (NIST, 2024; ENISA, 2024; Buzan, Wæver, and de Wilde, 1998).

Every band needs a narrowing path. Where exposure is driven by weak cyber protection, the repair is technical capacity; where activists or witnesses could be targeted, the repair is identity protection, lawful remedy, and safer custody; where markets could overreact, the repair may include staged explanation and stabilization.

Assign exposure by domain: safe publication of aggregate budgets does not imply safe release of threatened officials' identities or critical infrastructure vulnerabilities.

External actors can convert a technical phrase into leverage. A donor may attach conditions to an alleged lack of readiness, an opposition party may turn a repair need into a charge of national failure, and a market actor may price risk from a public band. That possibility is itself part of the publication burden and should inform whether the assessment is released in full, aggregated, delayed, or accompanied by contextual safeguards.

Public artifacts must keep the same discipline. Maps, summaries, captions, and diagrams should show the function assessed, the burden at issue, uncertainty, and the repair route. A substituted-purpose test is useful: if the same graphic can be detached from its declared task and used to justify an unrelated funding, eligibility, or prestige decision, the design needs more context or tighter limits.

18.4.3 — Synchronization Cost

Synchronization has a material price. Legal systems translate terms, agencies harmonize files, cyber teams secure interfaces, courts hear disputes, journalists interpret records, citizens learn new procedures, officials train, contractors are governed, and legacy systems are cleaned without erasing evidence. These tasks are the work that makes shared standards real. Fair architecture states who pays, who waits, who gains, and which essential domestic functions remain protected during conversion (Pritchett, Woolcock, and Andrews, 2013; UNDP, 2009; Andrews, Pritchett, and Woolcock, 2017).

Synchronization budgets should separate one-time conversion from recurring operation. Budget initial interface design, data cleaning, translation, training, vendor migration, and legal revision as conversion work. Fund their continuing upkeep or repetition alongside cybersecurity, audits, appeals, public explanation, archival custody, accessibility, and correction.

Before signing, check whether compact duties compete for specialists who maintain essential domestic services. If so, phase obligations, fund secure shared technical support, or narrow disclosure until capacity exists. Assistance must not require the weaker party to surrender its records, procure from the sponsor or accept external interpretive control. Both sides must be able to comply, contest findings and withdraw lawfully while preserving the domestic capabilities the compact is meant to protect.

18.5 — Local Agency Against Transparency Imperialism

Transparency requirements can become a route for stronger actors to acquire influence while avoiding burdens they helped create (Scott, 1998; Escobar, 1995; OECD, 2005). The risk is greatest when outsiders host, interpret, and control the evidence needed for local decisions. Meaningful participation therefore requires practical access to records and a way to contest their use. Cooperation should increase local agency rather than exchange one form of opacity for dependence on a more powerful observer.

18.5.1 — Local Consent and Participation

Local participation requires usable ways to understand and change a proposal. Consent is a distinct authorization: it depends on who holds the relevant right, what decision is being authorized, and whether the applicable conditions for giving or withholding consent are met. A consultation may influence a plan without obtaining consent; attendance, silence, or acceptance of needed services must not be treated as agreement. Where Indigenous peoples' free, prior and informed consent is required, the relevant rights and representative processes must be respected. Other affected people retain their own applicable participation, challenge, and remedy rights (UNECE, 1998; UN General Assembly, 2007; Sen, 1999).

Local participation also strengthens design. Communities know which disclosures could endanger them, which records are inaccurate, which authorities are trusted, which languages matter, which symbols insult, and which promised repairs have failed before (UNECE, 1998; United Nations, 2007; Sen, 1999).

Ask the people most likely to be misread to review the language and symbols before adoption.

18.5.2 — Repair Before Instruction

Before adding a transparency or verification duty, address the debt, infrastructure, climate, extraction, data, language and staffing burdens that prevent safe compliance (World Bank, 2017; Andrews, Pritchett, and Woolcock, 2017; Ferguson, 1990).

Stronger actors also need a reciprocal self-audit. Before asking another state to disclose corruption, has the stronger actor disclosed its own firms, banks, arms sales, tax havens, lobbying, extraction history, or emissions burden? Reciprocity begins when the instructor becomes answerable for the burden it helped create.

Urgency does not erase the principle. If a government is harming people now, care and protection may be immediate; even then, intervention should ask how to protect affected people without creating permanent external rule, dependency, or a narrative of superiority (World Bank, 2017; Andrews, Pritchett, and Woolcock, 2017; Ferguson, 1990).

18.6 — Plural Cooperation Beyond a World State

Shared standards can establish a humanitarian floor, reciprocal duty, review route, data boundary or minimum safeguard. Specify what must be common for protection and what must remain local for authorship (Ostrom, 2010; Abbott and Snidal, 2000; Slaughter, 2004). A treaty body, regional organization, professional network, local authority, and civil-society monitor may each carry part of a shared task while retaining distinct scopes, memberships, enforcement tools, and exit routes. No single institution is authorized to absorb the rest (Ostrom, 2010; Slaughter, 2004; Zürn, 2018).

18.6.1 — Local Institutions Remain Real

Local institutions need lawful authority, practical competence, resources, interoperability and discretion to adapt shared obligations to local conditions (Ostrom, 1990; Sen, 1999; World Bank, 2017).

Apply Section 17.6's custody, correction, and migration requirements so local institutions retain usable public memory after a vendor or donor exits; outside hosting must not become exclusive control over that memory.

Plural authority can add the cost of consultation, translation, appeal, and record reconciliation, as well as the time these require. That cost can be justified when it prevents concentrated error and preserves correction. Speed gained by bypassing the people who will carry the decision also has a cost.

Through regional networks, ports, hospitals, universities, laboratories, regulators, unions, municipalities, and professional associations can share training, procurement, evidence, and mutual aid while retaining their own mandates. Network redundancy also gives communities another route when a national office, donor, or global secretariat is captured.

Shared exercises make institutional reality visible before crisis. A cross-border grid drill, public-health simulation, archive-recovery test, or humanitarian logistics exercise can reveal language gaps, incompatible records, unclear authority, vendor dependence, and missing maintenance. The lesson should return to local budgets and staffing rather than remain in an international report.

18.6.2 — Cooperation in Modules

Cooperation can be organized around a task small enough for each participant to understand its own authority. Consider a proposed river-warning arrangement among neighboring states. One maintains upstream gauges, another relays observations, and each decides how to notify its own communities. The common undertaking concerns the information needed for a warning. Local evacuation routes and public-service decisions remain the responsibility of the institutions that know the affected places.

The arrangement needs agreement about what a reading means and when it was taken. It also needs a way to identify a missing reading. Those common elements make the information usable across borders. They do not require every participant to buy the same equipment or organize its administration in the same way. Compatibility can attach to the exchanged result while implementation remains plural.

Participation can then grow by adding functions rather than expanding an institution’s authority by implication. A later agreement might support joint maintenance training or a shared stock of replacement parts. Each addition has its own responsibilities and funding. A state can assess that specific proposal instead of choosing between total integration and no cooperation at all.

The value of this modular approach is practical. It gives cooperation a clear place to begin and makes its benefits inspectable. Communities can see whether warnings arrived sooner or whether a broken gauge was repaired more reliably. Success in one task supplies experience for the next while leaving the choice of further cooperation with the participants.

Chapter 19 — Diplomacy and Narrow Secrecy

Peace needs protected rooms because premature performance can destroy fragile openings. Diplomacy is strongest when protected discussion lasts long enough to reduce danger and is connected from the start to mandate, record, review, and eventual explanation.

Secrecy has sheltered lives and negotiations, but it has also sheltered embarrassment, corruption, failed policy, private bargains, illegal acts, and vendor capture. The operative distinction is protected purpose versus ownerless shadow, tested through authority, scope, record, clock, review, challenge, and return to public memory. The protected room is justified by the harm that would follow premature exposure, not by the status of those inside it.

Diplomacy and secrecy are material systems. They require staff, translation, secure communications, legal care, archival custody, incident response, oversight, and public trust. A protected interval also places burden on people who cannot yet see what is done in their name. The incidence test follows who carries risk during concealment, who controls the record, who can review it, and who is harmed when delay becomes permanent.

Unaccountable power and reckless exposure are separate failures. Public authority should remain answerable while negotiators, intelligence sources, cyber defenders, witnesses, and ordinary people are protected from disclosure that creates a concrete route to harm. The same architecture should prevent officials from hiding an ordinary mistake behind an undefined security label.

Section 19.4 treats deterrence as costly insurance, testing both inadequate protection and unnecessary readiness against credible alternatives.

Sections 19.2 and 19.3 develop the protection lifecycle: justified secrecy remains accountable through review and a

reasoned decision to release, summarize, redact, or continue protection. A protected negotiation also contains statements of different kinds. An exploratory suggestion is not yet an undertaking, and an undertaking is not the same as an authorized commitment. If a later account erases those distinctions, secrecy can leave the public unable to reconstruct how a choice was made even after the words are released. Attention to that sequence gives the chapter a practical question: what must be preserved now so later scrutiny can interpret the record fairly, without exposing the discussion before its protective purpose has ended? The answer concerns the character of the record as well as the date of disclosure.

The Cuban Missile Crisis remains instructive because public posture, private channels, intelligence, misperception, organizational routines, and time pressure shaped what leaders believed was possible (Schelling, 1966; Jervis, 1978).

Section 19.5 follows a warning through authentication, interpretation, and referral to someone able to act, asking how a protected channel can clarify danger before precaution becomes escalation. The accountability trail should also preserve access history while protected content remains unreleased.

19.1 — Diplomacy After Shadow Reduction

Diplomatic timing must remain tied to public authority and a record of who carries risks and costs (Berridge, 2022; United Nations, 2012; Putnam, 1988). Some negotiations require protected time before details can safely be released. The protection must have a defined purpose, custodian, and opportunity for later review rather than becoming a permanent exemption from accountability. A material assessment asks whose safety would change if disclosure came too early or too late.

19.1.1 — Diplomacy Still Needs Timing

The most fragile diplomatic moments often occur before a public agreement can be named. A prisoner exchange, humanitarian corridor, ceasefire phrase, cyber stand-down, border clarification, sanctions pause, or safe-passage arrangement may begin as a reversible test of whether the other side can move. Timing protects that test long enough to learn whether authority, coalition control, and reciprocal capacity are real (Zartman, 2000; Stedman, 1997; Fearon, 1995).

Premature exposure has an incidence that public debate often misses. The cost may fall on a negotiator’s reputation, but it can also reach refugees at a crossing, families waiting for detainees, towns under bombardment, ship crews in contested waters, or hospital systems dependent on a pause in violence. Timing is therefore a material protection question, not merely a diplomatic preference.

Without time to test reciprocal delivery, public performance can reward maximal statements and make exploratory concessions harder, particularly when every retreat is treated as humiliation (Zartman, 2000; Stedman, 1997; Fearon, 1995).

19.1.2 — Public Mandate and Diplomatic Route

A peace goal does not create a diplomatic mandate by itself. Private individuals, donors, corporations, platforms, factions, intelligence contacts, or informal envoys can produce large consequences by speaking as if they carry public authority. A fair route distinguishes who may converse, explore, propose, negotiate, commit, and ratify, and it records the passage from private contact to public consequence (Putnam, 1988; United Nations, 2012; Berridge, 2022).

A public mandate does not require a legislature or electorate to prewrite every negotiating sentence. It requires the negotiator's room to move to be connected to a legitimate purpose, an authorized route, and a later account. Domestic constituencies shape what can be offered, accepted, and ratified, which is why diplomacy often operates as a two-level negotiation rather than a single conversation across a table. The discipline is to preserve enough flexibility for peace while preventing the private channel from manufacturing authority it was never given (Putnam, 1988; United Nations, 2012).

The route may be an envoy, a mediator, a regional organization, a humanitarian intermediary, a technical cyber contact or a trusted backchannel. Each route remains bound by the mandate.

Diplomatic records must remain usable for later scrutiny. Notes, drafts, authorizations, dissent, translations, access logs, and decision points need durable custody, indexing, and context. A record scattered across private devices or unreadable after staff turnover cannot support review.

19.1.3 — Diplomatic Labor

Diplomatic capability depends on analysts, interpreters, envoys, cultural and legal advisers, security teams, archivists, and

administrative staff who keep fragile channels usable (Berridge, 2022; Sharp, 2009; Bjola and Kornprobst, 2013).

Fund crisis channels for the hours, authentication, translation, repair, exercises, and succession they require. Test connection success, verification time, escalation errors, contingencies, and documented incident resolution. These measures test operational readiness; they do not count wars prevented by a quiet year (Berridge, 2022; Sharp, 2009; Bjola and Kornprobst, 2013).

Budget childcare for staff in long negotiations alongside the staffing and support required to keep channels available.

Protected consultation should reach people who can identify where a bargain will fail: refugees, host communities, detainees' families, border and local officials, aid workers, minority representatives, and operators of schools, farms, ports, and health services. Provide feedback and support for their participation.

Those close to war also carry the emotional burden of ambiguity. They may know that every sentence can be misread, every delay can cost lives, and every public statement can close a path. That pressure should not be romanticized. It should be supported with staffing, rotation, mental-health care, and institutional memory so that exhaustion does not become policy (Berridge, 2022; Sharp, 2009; Bjola and Kornprobst, 2013). A documented handover between negotiating shifts can preserve commitments and unresolved questions without relying on one exhausted individual's memory.

19.2 — What May Remain Secret and Why

Secrecy categories should correspond to distinct protected interests. Life-safety, bargaining, military capability, intelligence-source, personal-privacy, and cybersecurity holds create different harms, scopes, custodians, and review clocks. Naming the category makes it possible to protect what is necessary without giving one undifferentiated label control over every hidden record (Bok, 1982; Pozen, 2010; NIST, 2024). This is a material-burden account, not a universal secrecy code. Legal and institutional design remain with law, government and security domains.

19.2.1 — Life-Safety Secrecy

Life safety provides the strongest case for confidentiality. The category covers information whose premature disclosure creates a credible route to death, violence, abduction, retaliation, targeted harassment, or severe vulnerability, including threatened witnesses, hostages, negotiators, intelligence sources, reformers, and civilians moving through danger. Protection should be as narrow as safety permits and as durable as the risk requires (Bok, 1982; Nissenbaum, 2010; Sagar, 2013).

Time belongs inside the life-safety reason. Some risks last hours, others years, and a few may persist indefinitely. The record should still state a review date, responsible owner, and event that could change the decision, such as evacuation, relocation, arrest, threat reduction, source consent, or completion of a protective operation. Continuing protection then remains an active judgment rather than forgotten inertia.

Life-safety protection requires staff and money. Secure communications, relocation, witness support, translation, legal counsel, identity protection, cyber defense, trauma-informed practice, and long-term case management are material

obligations. Hiding a name without funding safety can create the appearance of protection while leaving the person exposed.

Digital systems complicate the category. Metadata, access logs, geolocation, document history, facial images, and cross-linked databases can reveal a protected person even when visible text is redacted. Life-safety secrecy therefore needs data minimization, access control, retention limits, secure deletion, and breach notification designed around the actual threat rather than the document alone (NIST, 2024; Nissenbaum, 2010).

The life-safety category cannot shelter wrongdoing by the institution. Evidence of abuse, illegal detention, corruption, civilian harm, or unlawful surveillance may require protected disclosure to an authorized reviewer, court, inspector, or independent body. The identity of a vulnerable person can remain protected while the conduct of public power is examined (Pozen, 2010; Roberts, 2006).

19.2.2 — Cybersecurity Secrecy

Cybersecurity secrecy has a double timing problem. Early disclosure can guide attackers toward unpatched systems; delayed disclosure can leave hospitals, utilities, banks, schools, ports, households, and public agencies unknowingly exposed. Coordinated disclosure therefore requires triage, repair ownership, affected-party notice, protection for researchers, bounded delay, and a reasoned timetable for public explanation, including review when remediation is delayed or unavailable (NIST, 2024; ENISA, 2024; Nissenbaum, 2010).

Vulnerability details, credentials, defensive architecture, and incident-response tactics may need protected handling while remediation is underway (NIST, 2024; ENISA, 2024).

A defensible cyber exception begins with dependency. Reviewers need to know which systems rely on the vulnerable

component, how quickly they can patch, who lacks capacity, which services would fail if the information were exploited, and who receives notice and repair support. Cyber secrecy is justified only while disclosure would create a specific security risk, and it must allow authorized reviewers to examine avoidable delay, vendor negligence, or institutional failure. Embarrassment alone does not justify withholding (ENISA, 2024; NIST, 2024; ISO, 2016).

The burdens of people who cannot patch quickly must be counted. Rural hospitals, small municipalities, schools, small firms, humanitarian organizations, and poorer states may receive an urgent instruction while lacking staff, funds, or even reliable notice. A defensible cyber response pairs repair support with advisory language instead of treating unequal repair capacity as negligence.

19.2.3 — The Material Cost of Secrecy

Fund the full secrecy lifecycle, including classification, secure custody, vetted access, oversight, redaction, and eventual review or declassification (Roberts, 2006; Pozen, 2010; Bok, 1982).

Interoperability debt is another cost. Agencies may possess fragments of the same problem in incompatible networks, vocabularies, classifications, or retention systems. Lawful bridges, common metadata, and preauthorized emergency routes can preserve necessary protection without making the state blind to itself (ISO, 2016; Roberts, 2006; NIST, 2024).

Declassification backlogs create future work because later staff must reconstruct context after names, capabilities, and institutions have changed. Planning and funding review when a record is classified can preserve that context and reduce avoidable reconstruction; the actual cost and accuracy gains should be

assessed rather than assumed (Pozen, 2010; Roberts, 2006; ISO, 2016).

19.3 — Logged Secrecy and the Burden of Explanation

Logged secrecy preserves an accountable trace while protected content remains confidential (ISO, 2016; Bovens, 2007; Roberts, 2006). The public may not be entitled to the substance of a sensitive negotiation, but oversight still needs to know that a decision was authorized and recorded. Metadata about purpose, custody, review, and expiry can establish an audit route without revealing protected details. Such records make confidentiality a bounded responsibility rather than an unreviewable absence.

19.3.1 — A Traceable Secrecy Record

A traceable record should preserve more than the hidden content. It should preserve the authority invoked, the category of protection, the custodian, the date created, the access class, the review trigger, and the disposition. That structure lets later reviewers distinguish a lawful confidential negotiation from an unrecorded exercise of power. Records management standards are useful here because accountability depends not only on disclosure but on authenticity, integrity, context, and retrievability over time (ISO, 2016; Bovens, 2007).

The record benefits from a two-layer design: protected content under risk-appropriate access controls, and separate authorization and review metadata that do not reproduce dangerous detail. An inspector, court, archivist or other authorized reviewer can then test the secrecy system without receiving every operational fact (ISO, 2016; Bovens, 2007; Sagar, 2013).

A trace should show when protection began, which event or interval triggers review, whether the clock was paused, who approved an extension and what evidence justified continued protection (Pozen, 2010; Roberts, 2006).

A trace must also survive institutional boundaries. Diplomatic records may move between ministries, cyber records between vendors and public agencies, defense records between commands and contractors, and humanitarian records between international and local actors. Each transfer creates a risk of broken custody, altered context, duplicated copies, or incompatible retention rules. A durable trace therefore records transfer, receipt, version, access change, and eventual return or destruction (ISO, 2016; Bovens, 2007).

19.3.2 — Reason Codes

A reason code states the protected interest, plausible harm, authority, scope, and review condition in compact form. It does not replace judgment. It makes judgment inspectable by preventing an institution from relying on undefined labels such as security, diplomacy, privacy, or sensitivity. Good reason codes also make later comparison possible across similar decisions (Bovens, 2007; ISO, 2016; Roberts, 2006).

Review clocks should match the risk: a crisis decision may need review in hours, a negotiation in days or weeks, and source protection at defined milestones.

Public summaries can reveal the distribution of secrecy without revealing protected content. Institutions can report how many decisions used each category, average duration, review outcomes, correction rates, and confirmed misuse. Aggregate reporting helps the public see whether narrow secrecy is staying narrow while preserving the reasons for withholding individual details (Pozen, 2010; Roberts, 2006).

Apply the same reason codes and public classification authority to contractors, with the access, breach, custody, correction, migration, and audit duties in Section 17.6 specified in their contracts (ISO, 2016; NIST, 2024).

19.3.3 — Bad-Faith Secrecy Cost

Misused secrecy creates measurable drag. It hides corruption, delays correction, multiplies litigation, protects weak vendors, distorts budgets, burdens whistleblowers, injures victims, and forces future institutions to reconstruct truth from fragments. It can also weaken trust in legitimate confidentiality during later diplomacy, intelligence protection, cyber response, and crisis communication (Pozen, 2010; Sagar, 2013; Roberts, 2006).

A recurring institutional pattern can reveal bad faith: vague categories, missing records, serial extensions, punishment of internal challenge, selective leaks, and sudden claims of sensitivity when embarrassment becomes likely. No single sign proves abuse, but the pattern can trigger independent review. The goal is to make concealment contestable before a scandal or declassification decades later becomes the first moment of accountability (Pozen, 2010; Sagar, 2013).

Journalists, whistleblowers, courts, auditors, historians, and civil-society groups have legitimate oversight roles whether or not an institution has already failed. Officials who obstruct those roles and then condemn the resulting scrutiny deepen the loss of trust (Pozen, 2010; Sagar, 2013; Roberts, 2006).

Correction should not depend on a heroic leak or on the good character of a single inspector. A protected objection can be routed to more than one lawfully authorized reviewer, with preserved authorization logs and a secure route to challenge conflicts of interest or unexplained denial. Reviewers require sufficient access to test the asserted danger while remaining subject to security duty, record retention rules, and sanctions for disclosure abuse. A contested secrecy extension should disclose a non-sensitive reason and review status when lawful, while keeping operational details protected. If the normal reviewer is implicated, a preauthorized alternative forum must be able to

examine the record without allowing classified material to circulate publicly.

Fear of blame, budget protection, legal exposure, institutional pride, or habit can sustain concealment without an exceptional villain.

19.4 — Deterrence Without Mythology

Credible defense may be necessary against real threats, but misperception, prestige, alliances, and institutional incentives can enlarge capabilities beyond what protection requires. Review should compare each capability's civilian burden, cost, residual risk, and actual protection with alternatives such as resilience, diplomacy, cyber defense, or supply-chain repair (Schelling, 1966; Jervis, 1978; Freedman, 2004). The test must be updated when weapons become obsolete or signals invite mistaken escalation. Threats also include coercion through cyberattack, debt, energy cutoff, disinformation, maritime harassment, hostages, or manipulated migration routes. These pressures can reach hospitals, ports, households, schools, and currencies before resembling battlefield violence.

19.4.1 — Deterrence as Costly Insurance

Deterrence is costly insurance against severe harm. Its value depends on the threat model, credibility, resilience, communication, and the probability that defensive signals will be interpreted as intended (Schelling, 1966; Jervis, 1978; Kydd, 2005).

The deterrence register should include residual risk. Even a strong posture cannot make a population safe from all harm. It may reduce some dangers while increasing others, reassure allies while alarming neighbors, or protect borders while crowding out health, housing, climate adaptation, and maintenance.

Deterrence also needs maintained infrastructure, rumor correction, cyber protection, diplomacy and civilian recovery capacity (Schelling, 1966; Jervis, 1978; Kydd, 2005).

19.4.2 — Security Industry Influence

Security industries supply real capabilities: engineering, cyber defense, logistics, protective equipment, secure communications, maintenance, and analysis. Their influence requires special burden review because revenue can align with persistent threat posture. Procurement transparency, conflict rules, independent threat assessment, lifecycle costing, and competing expert review help keep public fear from becoming a private market (Stigler, 1971; Markusen and Yudken, 1992).

Capture arises when security-industry expertise controls the definition of threat, the options considered, or the evidence used to evaluate its own products (Stigler, 1971; Markusen and Yudken, 1992).

Vendor influence can shape what danger becomes visible. A system with a purchasable solution may receive attention while diplomacy, maintenance, care, training, language capacity, resilience, or de-escalation receives less. The public may be shown the cost of not buying a product more vividly than the cost of buying it forever.

Trace industry influence through advertising, think-tank funding, political donations, media commentary, technical language and prestige projects. Vendor support alone is not a reason to dismiss an argument.

Vendor and industry influence can also shape secrecy itself. Procurement data, testing results, cyber flaws, cost overruns, or performance failures may be withheld under security categories. Sometimes the withholding is legitimate. Sometimes it protects a business model. A fair review asks who gains from the hidden classification and whether independent technical assessment exists (Stigler, 1971; Markusen and Yudken, 1992).

19.4.3 — Civilian Cost of Deterrence

Trace deterrence's civilian effects through land use, proximity to installations, environmental exposure, defense-dependent employment, surveillance, fear, and future remediation, alongside its protective value (ICRC, 2022; Ireland, 2022; United Nations, 1977).

Civilian costs can arise before a weapon is used, through displacement planning, hardened infrastructure, emergency medicine, disrupted schools, inaccessible transport, and fear. The account must also include contamination risk and the long repair of housing and public services that may follow the use of force. In populated areas, blast and fragmentation effects can extend far beyond the nominal target. Civilian burden accounting therefore belongs inside deterrence policy, not only inside postwar relief (ICRC, 2022; Ireland, 2022).

Base closure, arms reduction or procurement shifts can impose concentrated losses on defense-dependent workers and towns; transition support belongs in the civilian-cost account.

Under common evidence rules, strategic, technical, fiscal, historical, regional, and civilian-impact reviewers, including dissenting analysts, should test the threat model, alternatives, cost distribution, and exit conditions.

19.5 — Crisis Communication and De-Escalation Channels

Facts are often incomplete during a crisis, and rumor or decentralized precaution can trigger mobilization, border incidents, market disruption, or violence before the picture clears. Communication channels can shorten that interval when they turn uncertainty into verified information rapidly enough to affect decisions. Readiness therefore requires more than a nominal contact list; staffing, authentication, language capability, and rehearsed procedures matter. Later public explanation should follow when safety permits (United States and Soviet Union, 1963; United States and Soviet Union, 1971; OSCE, 2011).

19.5.1 — Hotlines and Direct Channels

Direct channels among governments, militaries, border commands, cyber teams, emergency managers, or mediators require 24-hour staffing where appropriate, authentication, language capacity, exercises, redundancy, and a known escalation route (United States and Soviet Union, 1963; United States and Soviet Union, 1971; United States and Soviet Union, 1972).

Reliability has a human-factors dimension that equipment lists often miss. Operators must recognize accents, stress, uncertainty, coded language, and the difference between a political message and an operational warning. They need authority to seek clarification without improvising policy, and they need a handoff procedure when senior decision makers are unavailable. Fatigue, turnover, translation delay, and fear can disable a technically perfect connection. Direct-channel design should therefore include role continuity, cognitive load, interpreter practice, and clear escalation thresholds, not only encryption and backup power (Berridge, 2022; United States and Soviet Union, 1971).

Exercises should be adversarial enough to expose weakness without becoming theater. A useful drill can test authentication failure, partial network loss, conflicting translations, simultaneous incidents, leadership turnover, or a message arriving with incomplete evidence. The after-action review then separates technical failure, procedural ambiguity, authority confusion, and avoidable delay, assigns owners and repair dates, and requires a retest (OSCE, 2011; United States and Soviet Union, 1972).

A direct channel also needs a protocol for silence. No answer may indicate equipment failure, internal confusion, deliberate delay, coercion, or a decision to avoid contact. Agreed repeat attempts, alternative routes, neutral intermediaries, temporary restraint measures, and a threshold for public communication reduce the risk that silence is misread as a threat while preserving legitimate caution (United States and Soviet Union, 1963; OSCE, 2011).

Regional and subnational channels matter as well. Many dangerous incidents begin below the level of heads of government: a border unit, port authority, air-traffic controller, cyber-response team, coast guard, grid operator, local commander, or humanitarian coordinator encounters a problem before national leaders do. A peace architecture that centralizes every clarification can lose the minutes in which harm remains preventable. Local channels need bounded authority, clear referral routes, documentation, and protection from political retaliation when they make responsible contact (Berridge, 2022; OSCE, 2011).

Clarification should be connected to restraint authority. If frontline units receive clarification but command rhetoric has already demanded retaliation, the system may still fail. Channels,

rules of engagement, public messaging, and leadership incentives need to be aligned before ambiguity arrives.

19.5.2 — Shared Fact Protocols

Use the verification distinction in Section 18.3.2 to agree what crisis evidence can be checked quickly, through authenticated channels, with known provenance, limits, and challenge routes (OSCE, 2011; IAEA, 2022; OPCW, 1993).

The protocol should identify who may check the evidence, which evidence is admissible, how uncertainty is reported, and how a disputed result is escalated.

Chain of custody protects credibility. The record should identify who collected the material, which device or system generated it, how it was stored and transmitted, what transformations occurred, what was redacted, and which uncertainty remains. Evidence without custody can become another contested object.

Proportional disclosure can preserve both security and verification. Cyber defenders may share indicators while exploit details remain restricted. Border imagery may be summarized while source capability remains protected. Humanitarian access can be confirmed without publishing the location of people still at risk.

Synthetic media and manipulated digital evidence raise the verification burden because images, audio, video, documents, and telemetry may be generated, altered, stripped of provenance, or released through compromised accounts. Provenance tracking, authentication, corroboration, and clear uncertainty language can help, but no single technical technique guarantees truth or context. Crisis protocols should therefore combine technical evidence with trusted channels and human review (NIST, 2024; Chandra et al., 2024).

19.5.3 — Public Communication During Crisis

Public communication during crisis shapes material behavior. People decide whether to flee, shelter, buy supplies, send children to school, trust a bank, avoid a road, answer a rumor, or support restraint from the information public authority provides. Timely updates, uncertainty labels, correction, accessible formats, and separation of instruction from political performance can lower panic without manufacturing false certainty (United Nations, 2012; Berridge, 2022; Putnam, 1988).

Leaders can state what is known, what remains uncertain, what protective action is underway, and what channel is being used without humiliating the other side or locking themselves into maximal retaliation. Precision matters because ambiguity can inflame panic, while theatrical certainty can destroy a private route to correction (United Nations, 2012; Berridge, 2022).

Public communication should be prepared in advance for foreseeable categories. A state can draft templates for cyber disruption, border incident, public-health risk, evacuation, infrastructure failure, hostage rumor, and diplomatic clarification without knowing the exact future case. Preparation reduces the chance that panic writes the first statement (United Nations, 2012; Berridge, 2022; Putnam, 1988).

Fund protected time, qualified instructors, realistic scenarios, cultural and legal expertise, mental-health awareness, technical support, and review so frontline staff can practice restraint before crisis.

Public message authentication is part of crisis readiness. Publics need preannounced ways to verify which accounts, numbers, broadcast channels, and local authorities are genuine. A forged evacuation order, synthetic voice, or compromised official account can move people into danger even when the underlying response is competent. Redundant channels, provenance checks,

and rapid correction reduce exposure (NIST, 2024; Chandra et al., 2024).

After a crisis, document each message's reach, language, channel and delay, along with any observed misunderstandings; use the record to repair communication failures and resulting harm.

19.6 — When Prudence Decays Back Into Ordinary Shadow

Review can lapse as custodians retire and temporary crises become precedent. Apply Section 19.3's lifecycle controls to orphaned records and overdue decisions, with independent sampling and explicit responsibility for inherited files (Buzan, Wæver, and de Wilde, 1998; Pozen, 2010; Sagar, 2013). Review may move a record from full protection to partial summary, a redacted archive and fuller release as risk changes. Reentry can be gradual; it must not be forgotten.

19.6.1 — Security Language as Shield

Security language becomes a shield when it expands from a reason into an identity. Once an institution defines itself as guardian of an exceptional realm, ordinary questions about budget, evidence, error, and duration can be cast as disloyal. Securitization scholarship helps explain that move. The safeguard is to tie exceptional treatment to a specified threat, bounded authority, and reversible procedure (Buzan, Wæver, and de Wilde, 1998; Pozen, 2010).

A shield often reveals itself by refusing narrower alternatives. If partial release, independent review, delayed summary, redaction, or protected briefing would satisfy safety while allowing accountability, but officials still demand total silence, the protection claim deserves skepticism. The burden test requires officials to explain why narrower options would be insufficient (Buzan, Wæver, and de Wilde, 1998; Pozen, 2010; Roberts, 2006).

A no-review zone forms where secrecy, specialization, urgency, hierarchy, or technical complexity prevents meaningful challenge. Its decisions may then persist without correction: a flawed threat model guides spending, a failed contractor renews, a

harmful practice or diplomatic assumption hardens, or a cyber vulnerability remains untreated.

19.6.2 — Capture of Secrecy Classes

Capture risk rises when the people who create, classify, use, review, and extend secrecy belong to the same professional network with little external challenge. Expertise is necessary, but closed expertise can normalize overclassification and make correction socially costly. Rotating reviewers, judicial or legislative access where lawful, protected internal dissent, statistical audits, and retrospective sampling can widen accountability without broadcasting sensitive content (Sagar, 2013; Roberts, 2006).

Some exclusion is unavoidable in immediate crisis. Not everyone can be consulted while hostages are at risk, missiles are moving, systems are being patched, or negotiations are fragile. The fair review asks what happened before and after the emergency: whether civilian burdens were anticipated, affected communities had safe representation, harms were examined afterward, and explanation followed when possible.

Civilian exclusion is especially dangerous for communities that security systems already mistrust: border residents, minorities, migrants, dissidents, religious groups, workers in strategic industries, and people living near military or critical infrastructure sites. Security decisions may land on them as surveillance, displacement, suspicion, or silence. Their burden should not be treated as collateral inconvenience.

Before secrecy is needed, emergency planners can learn from disability groups, local officials, language communities, hospitals, schools, ports, workers, and border residents without revealing sensitive plans.

Chapter 20 — Arms Reduction and Conversion

A world organized around permanent threat spends resources before a shot is fired. Weapons, bases, doctrines, procurement networks, training systems, secrecy classes, regional dependencies, and public identities are built around anticipated violence. Some of that capacity protects people because danger is real. The chapter distinguishes necessary readiness from preventable overreadiness and asks how safely released capability can become repair.

Arms reduction is treated as verified and reciprocal material change. Protected information, inspection, residual security, expert challenge, worker and regional safeguards, and contingency plans for renewed danger all belong to the design. The central question is which capability remains necessary, which has become excess readiness, and which maintained public goods can be built from the verified net release.

Defense conversion is the chapter's human center because industries are made of people and places. Engineers, machinists, welders, software specialists, logistics workers, maintenance crews, administrators, unions, base communities, and suppliers carry transferable skill, household obligations, and local public finance. Peace becomes credible when that capability is redirected with workers and regions as authors of the transition.

Section 20.3 develops the worker non-abandonment rule: replacement demand and funded transition support must be ready before defense orders and local revenues disappear. The transition is already under way when orders begin to thin, even if the official closure is still distant.

Arms reduction is an industrial and regional transition, not a photograph of destroyed weapons. Defense systems contain laboratories, tooling, certifications, suppliers, pensions, municipal revenues, and relationships of deterrence that depend on perceived capability. Failed civilian qualification can revive

pressure to restore the old posture. The same discipline improves strategic judgment: residual risk, conversion cost, and the price of error remain visible beside the promised dividend. Where reconstitution costs are high, the plan can retain drawings, certifications, training pipelines, or reserve tooling without retaining the full production posture.

Historical conversion and threat-reduction efforts show that dismantling systems, securing materials, employing specialized workers, remediating sites, building verification, and sustaining cooperation require budgets, institutions, and time. The durable lesson is practical: worker protection, regional planning, social dialogue, credible civilian demand, verification, and contingency make conversion governable without imposing one universal template (Markusen and Yudken, 1992; ILO, 2015). Conversion plans should schedule evidence reviews before major procurement expirations and site closures, while alternatives remain affordable. Early review can distinguish a genuine capability need from an expiring procurement relationship, preserving useful skills without automatically preserving the old contract.

20.1 — The Cost of Militarized Overreadiness

Readiness answers a defined danger with proportionate capacity. Overreadiness begins when the posture exceeds what the risk warrants. Habit, prestige, industrial interests, identity, or political comfort can keep the excess in place after the marginal benefit declines. An honest comparison separates indispensable protection from the ongoing cost of maintaining a capability whose justification has changed.

20.1.1 — Defense Spending as Public Burden

Record direct outlays, long-term personnel obligations, veterans' care, classified support systems, environmental cleanup, procurement overruns, cybersecurity maintenance, base-community dependence and civilian repair displaced by the same funds. Even justified spending must identify where its burden lands and what would count as successful review (SIPRI, 2026a; SIPRI, 2026b; United Nations, 2018).

The Stockholm International Peace Research Institute estimated world military expenditure at US$2.887 trillion in 2025, equal to 2.5 percent of global gross domestic product and marking the eleventh consecutive annual increase (SIPRI, 2026a; SIPRI, 2026b).

A serious public register must also distinguish annual spending from committed spending. A weapons system bought this year can create decades of training, parts, software updates, storage, base use, contractor dependence, retirement, remediation, and replacement.

A threat economy can persist when employment, contracts, regional identity, campaign narratives, institutional prestige, and sunk procurement costs help determine readiness spending alongside the assessed threat.

20.1.2 — Opportunity Cost of Overreadiness

Opportunity cost applies to the resources committed to defense, including justified commitments. A proposal to change readiness should therefore compare the marginal costs and benefits of feasible alternatives, not assume that the whole budget can be redeployed. Aggregate spending establishes scale, not which capabilities are dispensable. Additions and replacements need a common horizon and an explicit account of uncertainty (SIPRI, 2026a; SIPRI, 2026b; HM Treasury, 2026b).

Count the costs of leaving a defense commitment as well as maintaining it. Without replacement demand, apparent savings may reappear as unemployment, migration, property decline, service loss and future political anger (Markusen and Yudken, 1992; Rodríguez-Pose, 2018).

When applying Section 5.7.1, include relevant site-remediation costs, pension obligations and debt service, counting each incremental cost only once (SIPRI, 2026b; Brauer and Dunne, 2012).

20.1.3 — What Defense Cost Evidence Can Establish

A defense-cost review should examine how secrecy affects civil liberties and require each participating discipline to disclose its evidentiary limits and institutional incentives (Dunne, Smith, and Willenbockel, 2005; Alptekin and Levine, 2012; SIPRI, 2026b).

Defense-cost evidence is especially vulnerable to counterfactual error. A lower budget does not reveal what would have happened under a different threat environment, and a higher budget does not reveal whether deterrence succeeded or waste accumulated behind necessity. Cross-country regressions can mix security exposure, regime type, growth, alliance structure, and conflict risk. Responsible conclusions therefore identify the spending category, time horizon, comparison, distribution, and

uncertainty before claiming a peace dividend or security loss (Dunne, Smith, and Willenbockel, 2005; Alptekin and Levine, 2012).

Record which premise drives each disagreement so the alternatives review can resolve substantive differences rather than merely place disciplines beside one another.

A defense-cost review should compare plausible security strategies rather than only current expenditure with a hoped-for peaceful alternative. One option might maintain existing readiness; another might substitute verified notifications, resilient infrastructure, and bounded reserve capacity; a third might alter procurement without changing deterrent force. For each, state the same threat scenarios, credible adversary responses, capability gaps, conversion costs, effects on workers and regions, and the danger of being wrong. Classified technical inputs may be independently reviewed without public exposure (Dunne, Smith, and Willenbockel, 2005; Alptekin and Levine, 2012; SIPRI, 2026b).

A claim may be current-source confirmed, contested, classified but independently reviewed, obsolete, unverifiable, or outside the available evidence. Where evidence remains classified, the public report can still state which assumption was tested, what alternatives were compared, and what conclusion the reviewer could not verify.

20.2 — Arms Ceilings, Inspection, and Confidence-Building

Arms ceilings need verified reciprocity rather than wishful trust. If one actor reduces capability while another conceals, cheats, or exploits the opening, the result is unilateral vulnerability. A credible ceiling therefore requires common definitions, protected data, declarations, inspections, and challenge procedures. It also needs correction and failure memory capable of distinguishing misunderstanding, incapacity, and bad faith (Schelling and Halperin, 1961; OSCE, 2011; IAEA, 2022).

20.2.1 — Ceilings as Burden Limits

A ceiling should be judged by the burden it actually prevents. A nominal limit may leave modernization untouched, shift capacity into another domain, or allow replacement systems that preserve the same strategic pressure under a different name. A defensible ceiling map specifies the category limited, baseline, loopholes, modernization rules, reserve allowances, verification method, review date, and expected burden reduction (Schelling and Halperin, 1961; OSCE, 2011; OPCW, 1993).

Define what the ceiling constrains: warheads, launchers, deployed systems, reserve systems, production capacity, delivery range, readiness, and geographic placement are different dimensions (Schelling and Halperin, 1961; OSCE, 2011).

Ceilings should be paired with reduction pathways, not merely final numbers. The pathway should show when systems are retired, how destruction or storage is verified, how personnel and regions are supported, what residual capacity remains, and what conditions trigger review.

Inspection needs access that can constrain cheating, independent judgment, technical skill, and procedural fairness. Fund the staff, laboratories, secure travel, evidence custody, legal

support, and dispute resolution needed for that mandate. Protect inspectors from risk and political pressure, alongside civilians and sensitive information; support lower-capacity parties without dependence on the inspected party or instrument supplier.

Inspection should have its own failure triggers. Denied access, inconsistent records, unexplained movement, tampered seals, missing serials, unsafe conditions, delayed reports, intimidation, or repeated anomalies should activate review.

20.2.2 — Mutuality and Parity

Apply the comparative-burden audit in Chapter 18 to each party's border exposure, cybersecurity, fiscal room, and inspection duties (OSCE, 2011; IAEA, 2022; Kydd, 2005).

Mutuality is credible only when a violation has a defined response short of automatic escalation. A staged compact should specify which observation counts as an anomaly, what independent verification is available, how quickly the affected side may request clarification, and which further reductions pause while the discrepancy is examined. Deliberate obstruction, ambiguous sensor failure, and minor reporting error require different procedures. Neither party should be forced to expose operational vulnerabilities to prove good faith, and neither may claim that an unverifiable assurance is equivalent to inspected compliance. The sequence should preserve defensive capability and communication while concentrating consequences on the disputed obligation (Kydd, 2005; OSCE, 2011).

The Article IV process discussed in Section 18.3.2 also illustrates gradual transfer of ownership. In 2014, Bosnia and Herzegovina, Croatia, Montenegro, and Serbia took over implementation of the arrangement, while the OSCE moved to a supporting role (OSCE Secretariat, 2026).

20.2.3 — Cheating and Evasion Risk

Verification can sometimes establish a narrowly defined fact without revealing the information a party must protect. Philippe et al. (2016) demonstrated physical zero-knowledge object comparison in a laboratory using simple test objects. Under the protocol's assumptions, matching items could be confirmed without revealing their geometry or composition. Tested substitutions could also be distinguished, but failed comparisons were not guaranteed to remain zero-knowledge. This was a proof of principle, not a deployed treaty-verification system. Its importance here is a new design possibility: disclosure and assurance need not always rise together. Arms-control research can ask which facts must be established, which information must remain protected, and which deception or implementation tests would make that separation reliable in practice.

The breach taxonomy should also distinguish ambiguous definitions, contractor error, disputed ownership, inherited stock and emergency deviation. These may require different responses from deliberate concealment, negligent reporting or technical malfunction.

Verification must combine declarations with independent checks. Data exchange, inspections, observation, imagery, tags, inventory records, production data, transport evidence, and challenge procedures constrain deception but have different blind spots. The Vienna Document illustrates a layered approach through annual information exchange, advance notification, observation, consultation, and on-site inspection. Its current limitations are evidence for maintaining and updating verification, not for pretending that either trust or surveillance can replace design (OSCE, 2011; OSCE Secretariat, 2026).

The reporting–verification distinction established in Section 18.3.2 becomes consequential during arms reduction: transfer

reports support coordination, while independently checked claims determine whether restraint is safe to extend (Gonçalves Costa and Holtom, 2026).

Where immediate danger allows, disputes should enter a defined clarification process before public escalation. Requests for explanation, supplementary evidence, repeat inspection, technical consultation, and neutral review can keep uncertainty from hardening into accusation. The process needs deadlines and escalation rules so correction remains possible without burying a serious breach. An inspection protocol can preserve a dated baseline for each disputed item, so a replacement, repair, or relocation is not mistaken for an unexplained change during the review. The parties can agree in advance which routine maintenance changes require an explanatory update to that baseline.

If verification cannot constrain evasions that would materially change security, narrow, delay or redesign the proposed reduction while retaining protective capacity.

20.3 — Defense Conversion With Worker and Regional Protection

Plan replacement demand, multiyear finance, and service stabilization with workers while defense contracts remain (ILO, 2015; OECD, 2019; OECD, 2023d). Regional conditions determine which transition routes can be credible. Engineering, logistics, manufacturing, quality control, cybersecurity, systems integration, materials, and maintenance skills may have valuable civilian uses. A tested transfer plan must establish feasibility and protect necessary defense capacity while workers and regions change course.

20.3.1 — Worker Non-Abandonment in Practice

Worker protection must be funded before closure is announced. Map occupations, wages, benefits, credentials, commuting patterns, suppliers, and workers in precarious roles. Then specify income bridges, health and pension continuity, paid training, credential recognition, placement partners, relocation choices, retirement options, and a dispute route. Each commitment needs a responsible institution, timetable, and budget (ILO, 2015; OECD, 2019).

Workers should help design and test those commitments. The test is usable support, not the existence of a plan: can affected people reach the new work without an unaffordable gap in income, care, or transport? A saving financed by lost livelihoods and collapsing local services is burden transfer. Credible conversion protects people and makes a reduction easier to sustain; it does not guarantee that every job or location can remain unchanged (Autor, Dorn, and Hanson, 2013).

20.3.2 — Regional Transition Plans

Conversion should honor a defense town's skill and service through credible new work, without treating its industrial identity as shame (OECD, 2019; Rodríguez-Pose, 2018).

Place is productive infrastructure: suppliers, test facilities, rail links, colleges, union halls, utilities, and a tax base may depend on one industrial system. Anchor loss can weaken demand far beyond the factory gate. Multiyear finance, local procurement, site reuse, service stabilization, and revisable governance are therefore core conversion tools (Bartik, 2020; Kline and Moretti, 2014).

The OECD's 2023 framework links strategic programming, stakeholder engagement, experimentation, skills, small-firm support, inclusion, and smart specialization rather than treating them as isolated projects (OECD, 2023d).

Regional plans must connect procurement demand to local capacity. When public funds buy grid equipment, transit parts, water systems, disaster supplies, housing components, or medical devices, affected regions should know whether they can compete.

20.3.3 — Skill Transfer Mapping

Skill transfer needs a feasibility grade. Some skills move directly, some need bridge training, some require new certification, and some may not transfer well. Disciplined translation names the skill, civilian demand, credential gap, wage risk, employer commitment, and support required for a dignified move, avoiding both automatic optimism and needless waste of capability (ILO, 2015; OECD, 2019; OECD, 2023d; Markusen and Yudken, 1992).

Machining, systems engineering, quality assurance, logistics, cybersecurity, materials science, project management, and safety

practice may transfer, but civilian sectors use different standards, customers, liability rules, volumes, and price pressures.

Review how long transition support lasts against the time needed for a usable transition, including debt, care, transport and relocation costs.

If the civilian order book or retraining placements fail to materialize at a declared checkpoint, the authority must revise the schedule and support rather than treating a plant's closure as evidence of successful conversion.

Age and life stage matter. A fifty-five-year-old machinist, young apprentice, single parent, disabled worker, and manager with transferable credentials need different routes.

20.3.4 — Worker and Union Voice in Conversion

Worker voice must enter before closure dates, product choices, and training programs are fixed. People closest to production know which machines can be adapted, which processes depend on tacit skill, which suppliers are fragile, which safety practices are nonnegotiable, and which promises will fail on the factory floor. Participation is both technical evidence and a democratic claim (ILO, 2015; Ostrom, 1990).

Unions are one route, not the whole workforce. Nonunion employees, subcontractors, temporary staff, logistics workers, cleaners, security staff, apprentices, disabled workers, and small suppliers may carry different risks. A conversion council must identify who is missing instead of treating formal representation as complete.

Regional councils should track losses beyond the prime contractor and connect them to housing, schools, healthcare, land use, environmental repair and municipal finance.

A conversion plan should also record which worker recommendations were accepted, rejected, or deferred and why.

That record turns consultation into evidence about whether voice changed design.

20.4 — Civilian Uses of Saved Expenditure

Use verified net conversion savings to repair civilian systems and fund their continuing maintenance (Hallegatte, Rentschler, and Rozenberg, 2019; European Union, United Nations, and World Bank, 2018). Include continuing diplomacy and de-escalation capacity when calculating net savings under Section 5.7.1, counting each cost only once. After funding residual security and conversion costs, compare candidate uses by avoided harm, distribution, readiness, lifecycle cost, maintenance and reversibility. Publish the allocation sequence so competing projects do not draw on the same saving. Use the public tracking record in Section 5.7.2 to follow the allocation through delivery, maintenance, and lived relief. Apply Chapter 7's debt-transparency and relief safeguards when directing conversion savings to debt stabilization (World Bank, 2025d; Guzman, Ocampo, and Stiglitz, 2016).

20.4.1 — Repair Backlogs

Prioritize repair backlogs by current burden and the rate at which delay raises cost, disruption, or irreversibility. Deterioration, disappearing spare parts, delayed inspections and temporary workarounds can turn a manageable defect into an outage (Graham and Thrift, 2007; Hallegatte, Rentschler, and Rozenberg, 2019).

Publish the backlog priorities and reasons for deferral, using the repair-priority method in Section 6.7. Allow local challenge where central data miss an existing burden, and keep deferred projects visible across budget cycles (World Bank, 2017; Hallegatte, Rentschler, and Rozenberg, 2019).

20.4.2 — Education and Skill Reinvestment

Direct skill reinvestment toward burden-reducing capabilities and communities long denied access, applying Section 6.7's priority discipline to guard against capture by institutions already adept at attracting funds (World Bank, 2018; Heckman, 2006; ILO, 2015).

Connect technical colleges, apprenticeships, research, adult education, language support, and portable credentials to labor demand and funded transition under Section 20.3. Combine early foundations, later retraining, and employer participation, with support for people unable to undertake years of unpaid study (Heckman, 2006; World Bank, 2018).

Include young people in defense-dependent regions, with locally viable pathways into grid and water systems, transit, health technology, robotics, resilient construction, cybersecurity, disaster logistics, or precision manufacturing.

Show reserve allocations alongside education and skill reinvestment. When verified savings are assigned to reserves, report the release rule, the shocks that could renew emergency overreadiness, and which urgent repairs or transition needs remain unfunded.

20.5 — Force, Restraint, and the Hidden Baseline

A serious peace architecture must protect people against aggression, atrocity, invasion, and immediate organized violence while holding force to rules of necessity, restraint, civilian protection, command accountability, and review (United Nations, 1945; ICRC, 1949; United Nations, 1977). It cannot allow military force to remain the unexamined default for budgets, prestige, diplomacy, policing, borders, or technology. Last-resort judgment needs a living record of lawful alternatives, including diplomacy, mediation, arbitration, crisis channels, confidence-building, defensive containment, humanitarian protection, and sanctions review. Those alternatives must be funded and rehearsed early enough to be usable, with reasons recorded for attempts, rejection, or urgency when delay itself would increase danger (United Nations, 1945; United Nations, 2012). The goal is competent protection and disciplined restraint rather than either automatic force or an automatic command to wait.

20.5.1 — Force Remains Sometimes Necessary

Necessity should never be used to make force feel pure. Within this material-peace framework, even justified force remains tragic public burden. It may be required to stop worse harm, yet it still produces injury, trauma, displacement, fiscal strain, ecological damage, veteran obligations, and future memory. The decision therefore demands a graver standard than ordinary policy: bounded authority, evidence, review, civilian protection, and repair (United Nations, 1945; ICRC, 1949; United Nations, 1977).

The discipline also requires evidence. Fear, humiliation, domestic politics, alliance pressure, prestige, retaliation, and media urgency can all mimic necessity. A fair process asks what is known, what is uncertain, what has been attempted, what

civilian harms are likely, what success means, what exit looks like, and who will review the claim afterward.

20.5.2 — Civilian Burden Accounting

The calculation has to begin before operational choices are fixed. It includes expected death and injury, displacement, damage to water, energy, health, communications, food systems, cultural property, and the indirect effects of interrupted care and transport. It also asks how uncertainty is handled when civilians are not fully visible to the decision-maker (ICRC, 2022; Ireland, 2022).

Uncertainty belongs inside the authorization. Where the full civilian burden cannot be known, the decision should state the missing information, plausible range of harm, available mitigation, and review trigger. Carrying that uncertainty forward prevents a bounded judgment from hardening into false confidence (ICRC, 2022; Ireland, 2022; United Nations and World Bank, 2018).

The account must also follow burdens within the society using force. Military families, veterans, reservists, care workers, taxpayers, dissenters, and communities near bases or factories carry burdens that public narratives may hide. Fair accounting follows the burden into both populations without flattening aggressor and victim.

Command accountability should also reach systems that make force easier to authorize. Automated targeting support, private contractors, intelligence vendors, cyber teams, logistics providers, allied partners, and political advisers may shape the conditions of command. Responsibility cannot disappear because the decision chain is complex. The distributional test asks where public power actually operated and whether each actor was subject to review proportionate to its influence over harm.

The record should preserve the chain of authorization, lawful basis, civilian-harm review, alternatives considered, expected cost and review route, while protecting operational details.

If public culture glorifies force while underfunding veteran care, moral injury support, disability support, family reintegration, housing, employment and truth-telling, glory has become a substitute for carrying the burden of service. Public language can honor those placed in danger, support veterans, and remember victims without romanticizing destruction or making later restraint harder.

20.5.3 — Peace Capacity Beside Defense Capacity

Peace capacity needs lifecycle funding across diplomacy, mediation, interpretation, disaster response, humanitarian logistics, cyber defense, public health, reconstruction, verification and de-escalation. If defense systems receive permanent maintenance while peace systems receive temporary attention, force can become the only capacity ready on demand (United Nations, 2018; United Nations, 2023; United Nations and World Bank, 2018; Brauer and Dunne, 2012).

Operational readiness can be specified without turning peace into militarized imitation. Measures may include channel uptime, mediator availability, language coverage, evacuation and relief capacity, verification staffing, public-health surge time, reconstruction inventories, exercise results, and the speed with which lessons become funded repairs. These are capability measures, not guarantees that conflict will be avoided.

Compare marginal preparedness across defense, diplomatic and civilian-protection tools. Equal spending is not required; hotlines, mediation, humanitarian logistics and reconstruction need reliable capacity for the tasks assigned to them.

The United Nations' 2023 peace agenda places renewed emphasis on prevention, diplomacy, regional frameworks, and nationally owned capacities (United Nations, 2023).

20.6 — Defense Transition Failure Memory

Conversion-failure review should identify the burden that was underestimated. Worker income, regional revenue, replacement demand, remaining security needs, contract exit cost, repair capacity, or administrative ability may each explain the shortfall. The finding must then name a correction owner, funding route, timetable, and restoration trigger. A failure category that cannot change the next decision is only a postmortem label.

20.6.1 — Worker Harm Records

Co-design worker-harm records with workers, keeping testimony, administrative data and public reporting distinct. Use them to identify household strain, disability, wage loss, deskilling and unsafe replacement work early enough to trigger repair (ILO, 2015; Autor, Dorn, and Hanson, 2013; OECD, 2019; OECD, 2023d).

Records must follow people and households over time rather than stop at program completion. Aggregate job counts can hide lower wages, unstable hours, benefit interruption, deskilling, involuntary migration, early retirement, repeated short contracts, and losses among contractors or suppliers. Workers need correction rights, and reporting should show distributions and attrition rather than a single success rate (ILO, 2015; OECD, 2019; OECD, 2023d).

The record becomes useful when it is connected to a response matrix. A wage-loss threshold may trigger income support, a placement delay may trigger training redesign, a benefit interruption may trigger administrative repair, and a cluster of supplier failures may trigger regional finance or procurement review.

The variable set should follow the transition question. Pay, hours, benefits, placement quality, and job stability may require

administrative records; health, debt, housing, and care effects need separate justification and appropriate safeguards. Do not collect every sensitive field for every worker merely because it appears on a checklist. Report who the evidence misses, including subcontractors and temporary workers, and use protected aggregate or voluntary evidence where it can answer the question with less exposure.

Worker-harm records require nonpunitive use. A worker who reports harm must not become a public failure statistic, employability score, or political weapon.

Independent review should compare the plan's promises with the lived record and publish bounded findings that workers can challenge. Reviewers should also identify support that arrived late and local services that weakened.

20.6.2 — Regional Decline Signals

Employment quality, wages, supplier orders, municipal revenue, service capacity, migration, vacancies, housing stress, training outcomes, and environmental liabilities can reveal burden before decline becomes entrenched. Place-specific evidence should trigger support matched to the mechanism rather than automatic restoration of military demand (Markusen and Yudken, 1992; Rodríguez-Pose, 2018; Bartik, 2020; OECD, 2023d).

Supplier signals include order books, payment delays, customer concentration, credit access, inventory, certification costs, machinery utilization, and the survival of sole-source capabilities. Small firms can fail before a prime contractor's closure appears in official employment data, and the resulting loss of tacit knowledge may be difficult to reverse (Markusen and Yudken, 1992; Kline and Moretti, 2014).

Municipal finance often reveals the next layer. Falling property values, sales or income tax receipts, utility revenue, and business

activity can weaken schools, transit, emergency response, libraries, parks, and maintenance.

Schools and youth outcomes deserve attention because regional decline travels across generations. Enrollment change, absenteeism, vocational pathways, teacher turnover, apprenticeships, youth migration, and confidence in local opportunity can reveal whether the region still offers a believable future (Autor, Dorn, and Hanson, 2013; Rodríguez-Pose, 2018).

Corrections may include income support, training redesign, public procurement, supplier finance, site remediation, infrastructure investment, housing protection, service grants, or a slower conversion timetable. The intervention should match the mechanism and be revisited after implementation (Bartik, 2020; Kline and Moretti, 2014; OECD, 2023d). A placement program should distinguish a training enrollment from paid work that uses the worker’s skills and supports the household.

Security reversal risk remains real. A threat can change, an agreement can fail, or a capability can prove harder to reconstitute than expected. Planners should identify preserved skills, dormant tooling, supplier lead times, dual-use capacity, reserves, and review conditions without using every imaginable danger to block reduction. The time needed to recall a specialist workforce may differ substantially from the time needed to restart preserved machinery.

Keep the initial employment, supplier, and regional service baselines so later recovery can be distinguished from a change in who is counted. If jobs return while municipal capacity or supplier diversity remains impaired, the burden account is still incomplete.

Part VI — Global Coordination Without Empire

A treaty may be signed in a capital, yet its first material test occurs elsewhere: at a port that must recognize an exemption, a clinic waiting for medicine, a school receiving displaced children, or a border post deciding whether a promise of refuge is real. A ceasefire without transport, interpreters, monitors, and secure communications remains fragile text. Global peace is built in the distance between decision and consequence (Keohane, 1984; Ostrom, 2010; Zürn, 2018).

Different shared problems require different architectures. Aggregate-effort problems depend on many cumulative contributions. Weakest-link problems can be undermined by one unprotected node. Best-shot problems may be solved by a small number of capable actors whose work benefits many others. Finance, monitoring, knowledge transfer, and remedy should follow the way each public good is actually produced, including who supplies the decisive contribution, who protects the vulnerable node, and who benefits from the result (Kaul, Grunberg, and Stern, 1999; Sandler, 2004).

Answerable asymmetry is the central design problem. A powerful state may demand disclosure while preserving its own strategic opacity. A donor may fund repair while acquiring narrative or procurement control. A tribunal may promise equal law while remaining unreachable to parties without counsel, records, money, language access, or time. Fair cooperation requires rule-makers to disclose their own influence, finance the burdens their rules create, and submit unequal effects to review and correction.

Durable solidarity also protects the receiving side of compassion. Refuge becomes stronger when host communities receive housing, schooling, healthcare, language, employment, infrastructure, and fiscal support. Development becomes more legitimate when recipient institutions retain authorship over

priorities, data, contractors, and public narrative. Sanctions become more defensible when humanitarian channels work in practice and civilian incidence is monitored. Financing these burdens gives compassion the institutional capacity to last (UNHCR, 2018; UNHCR, 2026; World Bank, 2023).

Development repair begins from unequal starting positions without converting them into judgments of national character. Extraction histories, climate exposure, debt structures, currency vulnerability, infrastructure gaps, trade position, conflict damage, and accumulated maintenance costs shape present options. Repair enlarges local capability and bargaining room; it does not replace local institutions with permanent donor management.

The sequence begins with law as a shared language into which conflict can move before it becomes force. It then follows finance and repair into local capability, and finally follows displacement into receiving institutions and long-horizon belonging. The constructive promise is practical: reciprocal rules, accessible channels, civilian-sparing pressure, locally authored repair, and financed refuge can reduce conflict without installing a sovereign above peoples.

Chapter 21 — Treaties, Sanctions, and Civilian Consequences

A ceasefire can be formally complete and materially unusable. Restraint and access clauses need funded implementation that weaker parties can use (Chayes and Chayes, 1995; Koremenos, 2016). The participants may agree on an obligation while disagreeing about what would count as fulfilling it. A promised route can be nominally open yet unusable at the times people need to cross. The chapter follows the movement from agreed language to observable performance, where disputes about interpretation become consequences for people who were not at the negotiating table.

Every legal route carries an incidence profile. Treaties can transfer cost to people absent from the negotiation. Arbitration can privilege parties with money, counsel, translation, and time. Shared facts can become permanent dossiers. Sanctions can interrupt medicine, food, banking, remittances, and humanitarian procurement. Peacekeeping can protect life or become dependency, misconduct, and indefinite presence. These outcomes are design questions through which lawful cooperation either earns or loses material credibility.

Arbitration offers a pre-force channel in which conflict can be heard without battlefield escalation or public humiliation. Its peace value depends on access and enforceability: a usable route for weaker parties cannot become a private corridor for power.

Shared fact-building can stabilize negotiation when evidence can be compared and challenged; Section 21.3 sets the protected, purpose-limited terms.

Sanctions are treated here as transmission systems rather than declarations. Section 21.4 follows how formal restrictions pass through finance and supply, and asks whether the resulting burdens reach intended targets or civilians.

Peacekeeping is treated here as temporary public capacity: its value lies in protecting civilians and political space while local institutions recover, not in substituting indefinitely for them.

Test each legal promise at the point of use. Consider a hypothetical exemption allowing a clinic to import medicine. Permission on paper is not enough: the clinic also needs a bank willing to process payment, a shipper and insurer able to act, clear customs instructions, and a contact empowered to resolve an error. If one link fails, the exemption can fail in practice.

Civilian incidence also changes over time. A measure that appears narrow at authorization can widen through overcompliance, supply-chain caution, exchange-rate effects, insurance withdrawal, administrative delay, or the disappearance of lawful intermediaries. A protective mission can become dependency when local authority, staffing, finance, and exit are never built. The chapter therefore asks for review dates, burden indicators, correction routes, humanitarian continuity, and off-ramps from the beginning. A review trigger should identify who receives evidence of civilian spillover and who can narrow the measure when those effects exceed its stated purpose.

21.1 — Treaties as Material Burden Maps

A treaty can create binding duties without supplying the staff or money to carry them. Translate each obligation into an operating map of tasks, financing, deadlines, and remedies (Koremenos, 2016; Chayes and Chayes, 1995). The map should distinguish an actor's unwillingness to comply from a genuine shortage of administrative or technical capacity. Protect essential domestic services while assigning a funded route to meet the commitment.

21.1.1 — Treaty as Burden Map

Implementation burdens depend on existing regulators, laboratories, courts and fiscal space. Periodic reporting should therefore distinguish lack of capacity from other causes of noncompliance, estimate transition finance, and disclose whether private actors can shift adjustment costs onto workers, consumers, or municipalities. Flexibility clauses can be legitimate when they preserve the agreement while acknowledging unequal implementation conditions, but they need transparent triggers and review so that flexibility does not become a permanent privilege for the strongest party (Koremenos, 2016; Chayes and Chayes, 1995).

21.1.2 — Public Explanation of Treaty Cost

Before approval, publish the treaty's expected costs, affected communities, implementation demands and uncertainty (Putnam, 1988; Chayes and Chayes, 1995; United Nations, 1969).

Public explanation should include a counterfactual because treaty debate is distorted when only the agreement is costed. Governments should compare implementation with plausible nonagreement paths, including trade friction, strategic duplication, environmental loss, security risk, and foregone cooperation. The comparison must state assumptions and ranges

rather than present a single net benefit as settled fact. Distribution matters as much as the aggregate: a treaty can be socially beneficial while imposing concentrated losses that deserve transition support (Putnam, 1988; Abbott and Snidal, 2000).

21.1.3 — Ratification Without Spectacle

The process leading to ratification should give the public a serious opportunity to examine burden, finance, implementation capacity, and unequal effects while meaningful revision remains possible within the relevant constitutional process (United Nations, 1969; Koremenos, 2016; Helfer, 2005).

Ratification review should test stability, revisability, and protection against unilateral rewrite. Sunset clauses, review conferences, amendment procedures, reservations, withdrawal rules, and emergency derogations determine whether an agreement can adapt without collapsing. A rigid instrument may generate evasion or exit, while unlimited discretion may never bind. Legislative scrutiny should therefore distinguish commitments that require durability from those that require controlled revision, then specify how new evidence, technological change, or unequal effects may alter implementation without allowing a powerful party to rewrite the bargain alone (Helfer, 2005; Koremenos, Lipson, and Snidal, 2001). Identify which joint procedures can resolve an implementation disagreement before either party must choose between acquiescence and formal withdrawal.

This material-peace volume assesses the costs, incidence, conflict effects and expert-review needs of legal instruments. Its burden maps and civilian-impact reviews do not supply binding doctrine. A reporting requirement, for example, has a staffing cost even when it creates no new substantive legal obligation.

The assessment should identify the authority able to amend that requirement when its cost exceeds its demonstrated practical value. The amendment record should name the evidence threshold, consultation route, and transition period for institutions already relying on the rule, including how service or reporting continuity will be preserved during change.

21.2 — Arbitration, Access, and Escalation

Access begins with legal availability. The relevant instrument must establish consent to arbitration, the disputes covered, and who may bring a claim; an affected community does not automatically gain access because it bears the consequences of the dispute. Where arbitration is available, the weakest plausible participant should be able to identify fees, translation and expert needs, interim-relief options, timelines, cost-sharing support, and the consequences of a missed procedural step. Where it is unavailable, the burden map should identify another authorized route to protection or remedy.

21.2.1 — Dispute Channels Before Force

Access has a temporal structure as well as a financial one. Some disputes can tolerate months of written exchange; others concern blocked medicine, an endangered ceasefire, a frozen payment channel, or an imminent environmental release. Effective systems need differentiated tracks, including emergency consultation, provisional protection, expedited fact finding, mediation, arbitration, and full adjudication. The route should match urgency, reversibility, evidentiary complexity, and the risk that delay will defeat the remedy. A channel that becomes usable only after a bridge is destroyed or a population displaced may still offer a later remedy, but it has missed the opportunity to prevent that harm (United Nations, 2012; Doyle and Sambanis, 2006).

Independence needs funded procedure as well as an impartial decision maker. Translation and technical advice should not depend entirely on one disputant’s resources; appointment and review must remain usable by the weaker party. These conditions matter in formal arbitration and deserve separate examination in informal mediation, which is a different process.

Hartman, Blair, and Blattman (2021) followed a randomized community rollout of dispute-resolution education in rural Liberia. Three years later, among respondents reporting a land dispute, the treatment group reported less escalation on a combined measure of threats, property damage, or violence. The incidence of disputes did not clearly fall, and property-rights and investment outcomes were mixed. The stronger statistical evidence concerned that combined escalation measure in the dispute-reporting subgroup, not a general reduction in physical violence alone. A new trial should report escalation across the full assigned population as well as among disputants, since an intervention can change which disputes occur or are reported. Safer disagreement and a rights-protecting settlement remain separate outcomes. A proposed forum should therefore pair accessible mediation with an independent route for contesting coercion or an unfair settlement; that pairing is a design implication to test, not an outcome already established by this study.

21.2.2 — Evidence-Sharing Standards

Evidence-sharing standards shape what a forum can assess. Without standards, the process may reward the party with better archives, more technical experts, more surveillance, or more capacity to produce documents. A fair dispute channel must ask not only what evidence exists, but who can gather, translate, verify, protect, and challenge it (OHCHR and Human Rights Center, 2022; ICRC, 2020; ISO, 2016).

Parties need to know how a dataset was produced, which categories were excluded, how uncertainty was handled, what transformations occurred, and whether the same term means the same thing across systems. Common schemas can reduce confusion, but they can also import one institution’s assumptions

into every comparison. A defensible standard therefore preserves provenance, version history, challenge procedures, and competing interpretations (OHCHR and Human Rights Center, 2022; ISO, 2016).

Purpose limits should be operational rather than rhetorical. The forum should define which records may be requested, which identifiers may be removed or masked, who may inspect sensitive material, how long it is retained, and when a party can challenge a request as unnecessary to the dispute. Evidence relevant to one claim should not automatically become available for unrelated enforcement, profiling, or public exposure (OHCHR and Human Rights Center, 2022; ICRC, 2020; ISO, 2016).

21.2.3 — Remedy and Compliance Pathways

Compliance pathways should distinguish restoration, compensation, assurance, institutional reform, and guarantees of nonrepetition. These remedies answer different injuries. Money cannot restore a destroyed cultural site, a public apology cannot finance reconstruction, and a monitoring mission cannot substitute for access to medicine. Orders should identify sequence, responsible actors, milestones, verification, and the consequence of partial performance. Where several institutions share responsibility, a coordination map should assign each task and prevent burden from disappearing between bodies (Chayes and Chayes, 1995; United Nations, 2012).

A compliance pathway distinguishes good-faith capacity limits from obstruction, supplies assistance where appropriate, preserves consequences when delay is strategic, and records recurrence so repeated renegotiation cannot reward nonperformance.

Arbitration can also displace political decisions that should remain public. Some disputes require legal judgment, but others

require public bargaining, budget, apology, reconstruction, or democratic ratification. A fair order does not force every wound into private procedure simply because procedure appears cleaner.

21.3 — Shared Fact-Bases and Evidence Interoperability

Agreement about evidence does not require agreement about political authority. Parties can share a bounded account of an event while retaining their own legal duties and rights to contest interpretation. The arrangement must identify who can submit, protect, inspect, and correct information, including sensitive material that cannot be published openly. Interoperability is valuable only when records can be compared without erasing provenance or exposing protected people. The following distinctions address joint fact-finding and the safeguards that keep cooperation from becoming control.

21.3.1 — Building a Shared Fact Base

A shared fact base is a bounded evidence environment for a specific dispute, protection need, compliance question, or repair task. Participating institutions retain their own authority and source custody while agreeing on the definitions, provenance, access, and challenge procedures needed for the common purpose. The bridge expires or contracts when that purpose ends (OHCHR and Human Rights Center, 2022; ISO, 2016; United Nations, 2012).

A shared fact base also needs clear labels for evidence status. Records should distinguish confirmed, disputed, unverified, inaccessible, and not measured, because absence of verification is different from evidence of absence. Confidence language must be tied to method, especially when one party controls territory, sensors, archives, or witness access (OHCHR and Human Rights Center, 2022; Nissenbaum, 2010).

21.3.2 — Protected Evidence Channels

Use protected channels for witnesses, survivors, humanitarian workers, inspectors, cyber analysts, intelligence-derived warnings, medical reports and documents from vulnerable institutions (ICRC, 2020; OHCHR and Human Rights Center, 2022; Nissenbaum, 2010).

Protection requires a threat model that follows evidence through its entire lifecycle. Risk can arise at collection, transmission, translation, storage, analysis, publication, archival transfer, or deletion. Encryption alone does not prevent coercive access, insider misuse, metadata leakage, or inference from small samples. Governance should therefore define lawful purpose, least privilege, dual control for sensitive release, breach notification, independent complaint, retention periods, and tested secure deletion. Evidence remains available long enough to support accountability only under a documented purpose and review rule (ICRC, 2020; Nissenbaum, 2010).

Verify claims by evidence and method, without treating poorer communities or weaker states as presumptively less credible.

21.3.3 — Fact Sharing Without Empire

The No Cross-Architecture Dossier rule is a firewall against administrative merger. A conflict evidence system should not become a permanent profile combining migration status, debt history, health records, biometric identifiers, humanitarian aid records, legal claims, school records, platform data, religious affiliation, political activity, and ordinary civic life (Nissenbaum, 2010; ICRC, 2020; Slaughter, 2004).

Polycentric fact sharing requires a credible exit right. Institutions should be able to end participation, retrieve or destroy data where law permits, preserve audit records of prior use, and continue essential functions without technical captivity. Open

standards and portable documentation can reduce dependence on a vendor, state or secretariat after an emergency ends (Slaughter, 2004; Ostrom, 2010).

21.4 — Sanctions and Noncooperation

Sanctions and noncooperation need a stated purpose and an accountable theory of effect (United Nations Security Council, 2022; United Nations Security Council, 2024; Peksen, 2019). Civilian spillovers and private overcompliance can diverge from the official target. Monitoring must therefore include how payment, shipping, medicine, and emergency access are affected over time. Changes in evidence or conditions should be able to trigger revision or termination.

21.4.1 — Sanctions as Public Burden

Civilian-incidence monitoring should begin before sanctions take effect. Baselines should cover food, medicine, energy, remittances, banking, shipping, humanitarian procurement, and sectors capable of cascading failure. Distinguish legal restrictions from private overcompliance and domestic countermeasures. Give banks, insurers, shippers, suppliers, and customs staff clear implementation guidance, responsible contacts, and a remedy when lawful transactions are refused (United Nations Security Council, 2022; United Nations Security Council, 2024).

Test the full payment and delivery route: whether a provider can obtain a written license or safe-harbor decision where authorized, make and receive payment, purchase insurance, clear goods, and resolve a blocked shipment within a declared emergency horizon. The test should expose uncertain compliance risk, customs officers' fear of liability, and licensing delays. Humanitarian organizations need a rapid route to clarification.

21.4.2 — Targeting and Overbreadth Review

Targeting and overbreadth review asks whether sanctions burden the intended actors and how far the effects spread to others. The review should examine financial networks, ownership structures,

supply chains, intermediaries, regional effects, household exposure, humanitarian access, and private over-compliance (Drezner, 2003; Early, 2015; Felbermayr et al., 2020).

The proportionality inquiry should be dynamic. Early effects may differ from long-run substitution, evasion, elite adaptation, and household exhaustion. A measure that appears targeted during the first months can become broader as inventories fail, insurers withdraw, or informal markets expand. Review should therefore specify intervals, leading indicators, and thresholds for modification. It should also compare who can adapt: officials with foreign assets, firms with political connections, and wealthy households often have more routes around pressure than people dependent on regulated banks or public services. Without distributional tracking, civilian pressure may persist while intended targets retain ways to avoid it (Peksen, 2009; Early, 2015).

The review record should distinguish intended pressure from incidental spillovers, flag consequences outside the measure's stated scope, and record uncertainty about attribution. Later decisions can then address both the burden observed and the evidence linking it to the measure (Drezner, 2003; Early, 2015; Felbermayr et al., 2020).

Monitoring evasion requires evidence, but evidence collection must not become indiscriminate surveillance. Trade records, customs data, ownership registries, shipping patterns, and financial flows may be relevant. Personal data from civilians should remain minimized. The target is evasion architecture, not every person trying to survive inside scarcity.

21.4.3 — Exit Conditions

A credible off-ramp links coercion to a remediable objective. Observable conditions, evidence sources, decision authority,

review deadlines, and procedures for contested facts should be stated in advance. Phased relief, escrow, probationary licenses, or snapback provisions may preserve leverage where reversal is feasible. The target must be able to know what conduct will change the measure, while civilians must be able to see that pressure will not harden into an inherited relationship between populations (Drezner, 2003; Hufbauer et al., 2007).

Civilian suffering does not establish that a sanction is achieving its stated purpose. Under the series' ethical firewall, suffering is never moral credit, proof of justice, or owed payment.

21.5 — Peacekeeping, Local Authority, and Exit

Peacekeeping can protect civilians, monitor agreements, and support political space while local institutions recover (United Nations, 2008; United Nations, 2025). These mandates require different skills and cannot be financed or evaluated as one task. Expectations should be matched to personnel, authority, logistics, and an intelligible public explanation. Exit conditions must connect security gains to local capability rather than simply the passage of time.

21.5.1 — Peacekeeping as Temporary Support

Mandate and means must be reconciled before deployment. A mission charged with civilian protection, ceasefire monitoring, electoral support, police mentoring, humanitarian access, and institutional reform may be assigned more tasks than personnel, mobility, intelligence, language capacity, or political backing can sustain. That gap is not only an efficiency problem. It can invite reliance by civilians and local partners while leaving the mission unable to respond. Planning should therefore publish priority functions, geographic limits, reserve capacity, escalation authority, and the conditions under which a task will be suspended rather than performed symbolically (United Nations, 2008; United Nations, 2023).

People need clear explanations of a mission's mandate, complaint routes, limits, rules on the use of force, data practices, and exit plan. Those explanations must be accessible, translated, and repeated as conditions change.

Formal consent does not by itself establish local legitimacy. A formal invitation from a desperate authority may not represent affected communities, minority groups, local workers, or displaced people. A fair peacekeeping frame seeks consent,

contestation, and participation without treating emergency permission as permanent legitimacy.

21.5.2 — Civilian Protection Priority

Assess mission success through changes in civilian violence, fear, access to essentials, usable complaint routes, and the ability to resume ordinary life (United Nations, 2023; United Nations, 2025; Hultman, Kathman, and Shannon, 2013). Civilian protection needs threat analysis, presence, mobility, early warning, community engagement, language competence, medical evacuation, secure reporting, disciplined use of force and accountability when personnel cause harm (United Nations, 2023; United Nations, 2025; Hultman, Kathman, and Shannon, 2013).

Protection strategy should be co-produced with communities without transferring risk onto them. Local residents often understand threat patterns, routes, armed actors, and warning signals that external missions cannot see, yet consultation can expose informants or create retaliation. Engagement therefore needs confidentiality, representation beyond convenient elites, feedback on what was done with information, and a refusal to promise protection that cannot be delivered. Civilian protection is strongest when community knowledge changes patrols, access, and contingency planning while the institution carrying coercive power remains answerable for the decision (United Nations, 2023; United Nations, 2025).

Count the risk and moral strain carried by peacekeepers, interpreters, local drivers, medical staff, logistics workers, investigators and community liaisons. Independent investigators, host-state coordination where appropriate, troop-contributing-state responsibility where applicable, public case-status reporting that preserves survivor confidentiality, and protected reporting

can help address allegations of mission misconduct. Keep misconduct findings, protected evidence, confidential survivor support and aggregate reporting distinct; public accountability does not require exposing a victim's identity or endangering testimony.

21.5.3 — Exit Without Abandonment

Remaining too long can create dependency, resentment, external rule, and political stagnation. Leaving too quickly can expose civilians, collapse services, embolden armed actors, and undo fragile repair (Ruggeri, Dorussen, and Gizelis, 2017; Donais, 2009; Autesserre, 2014).

Exit without abandonment requires a sequenced handover of the services people actually rely on. Police support, courts, transport security, explosive-hazard work, local mediation, payroll systems, detention monitoring, communications, and humanitarian access may not mature at the same speed. A single withdrawal date can therefore conceal multiple capability cliffs. An honest exit plan identifies which functions can transfer, which need regional support, which require continued financing, and which should end. A receiving institution should demonstrate an ordinary service cycle before the external provider withdraws from that particular function.

Local institutions may inherit vehicles, facilities, communications systems, forensic tools, or training programs whose recurrent costs exceed domestic budgets. The handover should therefore include total cost of ownership, domestic revenue prospects, regional support, simplified technology, and a decision about which assets should not be transferred. Sustainability is shown when the function remains reliable after external subsidies and contractors recede, not by a ceremony or signed inventory (Doyle and Sambanis, 2006; United Nations,

2025). A simpler maintainable vehicle fleet may preserve more local capability than equipment whose spare parts require permanent foreign support.

A handover plan should identify each function as ready, conditional, or not ready, with a responsible receiving institution, funding source, unresolved gap, and review date. Readiness may differ across records, payroll, transport, security, and survivor support. Residents need one clear contact for failures that fall between the outgoing and receiving bodies. A post-handover review should compare promised and actual service continuity before the remaining support ends.

21.6 — Civilian Protection and the Material Floor of Peace

Civilian protection cannot depend entirely on the outcome of negotiations or the speed of military decisions. The basic test is whether people retain access to life-preserving services and a credible route to safety when institutions are under strain. Protecting that floor requires restraint by armed actors as well as practical capacity among humanitarian and public services. The duties differ by context, but the burden of failure must remain visible. This section examines the minimum protections and the operating capacity that gives them effect.

21.6.1 — The Anti-Atrocity Material Floor

A minimum anti-atrocity floor begins with the refusal to treat severe harm as a local detail beyond common concern. Mass killing, enslavement, systematic torture, forced disappearance, deliberate starvation of civilians, mass sexual violence, ethnic cleansing, and similar grave harms cannot be hidden behind diplomatic politeness or sovereignty language without betraying peace itself (ICRC, 1949; United Nations, 1977; United Nations and World Bank, 2018).

The material floor should be organized around prevention as well as response. Civilian registration that protects rather than exposes, resilient communications, secure evacuation information, emergency health capacity, food and water continuity, documentation of missing persons, and rapid repair of essential services can reduce the scale of atrocity without waiting for a full coercive intervention. These capabilities need lawful authority and local trust, but they are often less escalatory than improvised action after mass harm is underway (ICRC, 2024; United Nations and World Bank, 2018).

21.6.2 — Civilian-Sparing Capacity

The civilian-sparing floor states that ordinary people should not be made instruments of strategic pressure. Civilians should not be targeted, starved, displaced, exposed, deprived of essentials, made into bargaining chips, or treated as acceptable scenery for coercion (United Nations, 1977; ICRC, 2022; Ireland, 2022).

Civilian-sparing capacity includes the ability to withhold force when lawful targets cannot be reliably distinguished from protected people or objects.

Audit evidence should record target verification, weapon and timing choices, feasible precautions, no-strike and protected-site processes, civilian-harm tracking, independent investigation, acknowledgment, remedy, and operational learning. New information must be able to stop an action and change future practice. Legal and contextual judgment remains necessary: a lower recorded count can reflect poor observation rather than safer conduct, and civilian lives must not be reduced to a performance score (ICRC, 2024).

Chapter 22 — Development, Debt, and Repair

A sovereign debt payment can clear on schedule while a clinic postpones repairs, a water system defers maintenance, or a government abandons adaptation it can no longer finance. Compliance in the financial account can therefore reduce public capacity elsewhere. Chapter 22 begins with that divided reality and treats development as repair under unequal histories, bargaining power, climate exposure, infrastructure, trade, and currency conditions (World Bank, 2025c; UNCTAD, 2025c).

Development here means the repair of accumulated material burden under local authorship. Its strongest evidence is locally governed capability that remains usable and revisable after outside attention recedes.

Sovereign debt is a claim on future public room and therefore on the sequence of choices a society can still make. Borrowing can finance infrastructure, stabilize emergencies, and distribute cost across time. It can also intensify currency exposure, crowd out care and maintenance, and move public choices into creditor negotiations before citizens are heard. The central distinctions are finance that enlarges future capacity versus finance that narrows it, and relief that restores autonomy versus relief that acquires indirect control (World Bank, 2025c; World Bank, 2025d; Guzman, Ocampo, and Stiglitz, 2016).

Local authorship must also be resourced. It cannot mean leaving communities alone with burdens created beyond them or pretending that debt contracts, climate shocks, commodity prices, security threats, disaster damage, and infrastructure gaps can be overcome by pride. That autonomy requires funded operating capacity and a planned exit. An outside partner’s departure is therefore an illuminating point from which to read a proposal backward. What would the receiving institution still be able to do the following month? Which decisions would remain within its reach? These questions help distinguish a completed transfer from

a continuing dependency before the appearance of completion makes that dependency harder to discuss.

Reconstruction shows the full meaning of repair. A bridge changes market access, school attendance, clinic reach, ambulance time, food prices, and the public sense that the future remains usable. A river carries health, land, labor, food, memory, and culture. An archive carries legal identity, title, pensions, claims, accountability, and mourning. Repair begins by understanding the system that was damaged, not merely by counting visible assets replaced.

Development projects are sequences of capability decisions, not single transfer events. Before money moves, institutions should identify who defined the problem, whether debt payments are already constraining services, what currency and refinancing risks remain, who will procure and supervise the work, and who owns maintenance knowledge. After opening, the test becomes whether local staff can operate the system, budgets can sustain it, records and software remain portable, and the external actor can leave without taking the function away (World Bank, 2025c; World Bank, 2025d).

22.1 — Structural Vulnerability and Unequal Starting Positions

History should be named with causal discipline. Extraction, climate exposure, debt, currency dependence, governance, security, local decisions, and international rules may interact but should not be treated as interchangeable explanations. A serious account traces how they combine through identifiable burden routes and where responsibility differs from capacity to repair. It then compares interventions by the options they could realistically restore to affected people.

22.1.1 — Debt and Bargaining Weakness

Debt burden is not captured by the amount owed alone. Repayment deadlines, reserve depletion, refinancing risk, emergency procurement, currency pressure, collateral or guarantee structures, and the possibility that refusal will interrupt essential imports all shape bargaining. Lending becomes predatory when repayment pressure is used to obtain unrelated concessions over resources, public assets, data, procurement, or policy beyond legitimate debt recovery (World Bank, 2025c; World Bank, 2025d; Guzman, Ocampo, and Stiglitz, 2016).

Extraction also shapes what counts as infrastructure. A port, rail line, or road may count as investment, but its route can reveal whom it serves. Does it connect people to services, producers to diverse markets, regions to each other, and public systems to maintenance capacity? Or does it move minerals, crops, or labor outward while leaving communities disconnected?

Extraction may involve colonial rule, slavery, forced labor, unequal concession contracts, resource enclaves, debt-financed infrastructure, trade dependency, or domestic elites who profited from outside power.

22.1.2 — Climate Exposure

Climate exposure is a fairness problem because vulnerability often follows a different map than responsibility. Drought, heat, flooding, cyclones, sea-level rise, crop stress, water contamination, and disease movement can strike publics whose historical emissions were small and whose fiscal room is already narrow. The burden extends through insurance cost, damaged roads, school interruption, health-system strain, migration pressure, food-price volatility, and adaptation borrowing, including the repeated administrative cost of seeking emergency finance for damage one did not mainly create (IPCC, 2022; IPCC, 2023; World Bank, 2021b).

22.1.3 — Currency and Trade Exposure

Currency and trade exposure transmit external price shocks into domestic budgets. Hard-currency import bills, volatile commodity-export receipts and limited reserves can constrain fuel, medicine, education and infrastructure spending (UNCTAD, 2025a; World Bank, 2025c; Rodrik, 2011).

Section 13.5.1 compares exchange-rate changes at the point of purchase. Development planning extends that comparison across the operating life of an investment. An imported machine may require consumables and replacement parts priced in the same currency over many years. Its purchase price is therefore the opening entry in a continuing relationship. Comparing the proposal through that relationship shows which resources must remain available and which local skills could reduce dependence on external servicing.

The equipment schedule can bring those obligations into view before a funding agreement is signed. A component replaced every few months creates a different recurring need from a specialist repair expected once in several years. Recording the

purchase currency alongside each renewal date connects the engineering plan to the financing plan. A team can then consider whether to hold suitable spares, develop local repair capability, or arrange continuing access to the original supplier.

This account supports a more capable conversation about the investment itself. The question becomes how a useful asset will keep contributing after the initial purchase has been celebrated. The community receives a service whose continuing requirements have been considered together, rather than separate technical and financial plans that meet for the first time when a part needs replacement.

22.2 — Sovereign Debt and Anti-Predatory Lending

Debt analysis should trace who receives cash, fees, interest, and guarantees over the life of an obligation (World Bank, 2025c; Guzman, Ocampo, and Stiglitz, 2016). Principal alone cannot show how repayment timing and enforcement terms affect a public budget. The relevant question is which services or options become less available when fiscal room narrows. Records should make those consequences visible without assuming that every borrowing decision was predatory.

22.2.1 — Debt-Service Crowding

Citizens may be told that a policy is inevitable when the actual constraint is a bond covenant, collateral clause, bilateral side agreement, commodity prepayment, arbitration venue, guarantee, or contractor claim. Visibility does not mean public exposure of every sensitive negotiating detail while negotiations are active. It means that the public can eventually know who held leverage, under what terms, with what remedies, and with what consequences for services, assets, taxation, and future borrowing (World Bank, 2025d; Guzman, Ocampo, and Stiglitz, 2016).

Debt-service crowding should be traced through financing terms and actual budget choices. The World Bank's International Debt Report 2025 and UNCTAD's A World of Debt 2025 show why the debt stock alone is insufficient: interest costs, external payment burdens, and available financing shape the resources left for development (World Bank, 2025c; UNCTAD, 2025c).

To identify crowding, compare debt-service changes with revenue, grants, other spending, and the timing of specific service decisions. A postponed clinic or unpaid teacher may reflect several causes; the analysis should show which part of the financing gap is attributable to debt service and which remains uncertain. Then translate the evidenced loss into lived

consequences, such as longer journeys for care, missed maintenance, or interrupted schooling. This preserves the human account without treating every missing service as proof of a debt effect.

22.2.2 — Predatory Restructuring Risk

Conditionality requires mechanism-by-mechanism review because some conditions protect public value while others transfer authority away from the public. Anti-corruption requirements, audit standards, procurement transparency, and maintenance obligations may be legitimate when they are proportionate and jointly governed. Conditions become unfair when they dictate social priorities without local consent, impose one model regardless of context, weaken essential services, or use crisis vulnerability to lock in ideology (UNCTAD, 2012; IMF and World Bank, 2018; Guzman, Ocampo, and Stiglitz, 2016).

Official creditors can also delay relief through geopolitical competition, coordination problems, slow common frameworks, unclear comparability of treatment, or contested debt-sustainability analysis. Track the resulting service losses and uncertainty whether delay originates with private, bilateral or multilateral creditors, or domestic leaders.

22.2.3 — Evidence and Law in Sovereign Debt

Apply the no-chain relief principle in Section 7.5 to sovereign restructuring: debt relief must not purchase unrelated political, military, resource, religious, data, procurement or narrative concessions. Transparent use and proportionate corruption controls remain legitimate (UNCTAD, 2012; IMF and World Bank, 2018; World Bank, 2025d).

The same debt stock can imply different repayment burdens under different growth paths, export structures, fiscal capacity, climate risk, and exchange-rate exposure. Debt-sustainability

review should state its assumptions, uncertainty, effects on essential services, and evidence that would change the conclusion.

Compare the risks emphasized by debt lawyers, macroeconomists, development economists, civil-society advocates, creditors and debtor governments, then show how the proposed restructuring distributes burden under their competing assumptions.

22.3 — Development Finance, Local Authorship, and Exit

Grants, loans, guarantees, and technical assistance should be assessed by the rights, risks, and operating authority they transfer. Section 6.5's donor firewall applies to project finance and delivery, including consultant dependence, parallel implementation offices, procurement or data control, and political scheduling. Recipient authorship is necessary but does not excuse local elites from scrutiny when they capture projects, exclude communities, conceal contracts, or punish critics. Boundaries therefore run both ways: donors cannot govern through money, and local intermediaries cannot invoke sovereignty to shield theft, coercion, or neglect. The standard is accountable repair to affected people rather than deference to whichever side already holds power.

22.3.1 — Recipient Authorship

Before a project proposal is written, identify the locally legitimate body defining the problem, affected groups with standing to object, budget and procurement authority, accessible technical evidence, and decisions the outside financier can lawfully block (OECD, 2011; Andrews, Pritchett, and Woolcock, 2017). Agree funding terms before the donor becomes indispensable, using Section 22.7's firewall. Local institutions must be able to revise the project without losing essential support, including promised humanitarian essentials, surrendering control of participants' information, or exposing critics to retaliation. During delivery, apply the operating and handover tests in Sections 22.4.2 and 22.5.4.

22.3.2 — Restricted-Gift Risk

Restricted gifts become dangerous when their terms serve the donor's self-image more than the recipient's needs. A restriction may favor a visible building over the recipient's more urgent operating needs. Apply Section 6.5's right to accept, renegotiate or refuse the gift (Ferguson, 1990; Escobar, 1995; OECD, 2011).

Restricted finance should pass two distinct tests. Additionality asks what finance, terms, or capability the arrangement provides beyond a credible account of what would otherwise occur; relabeling existing assistance does not establish a new contribution. The recipient-benefit test then counts tied procurement, local co-finance, currency risk, fees, outward payments, and lifetime operating costs alongside the services and skills gained. An additional commitment can still offer poor value or create unaffordable obligations.

22.3.3 — Local Procurement and Capacity

Local procurement can create jobs, skills, tax base, repair capacity, and institutional memory. Politically connected firms can also capture it when competition, quality, and audit are weak. Support feasible local supply under transparent rules, performance review, worker protection, and long-term maintenance (OECD, 2005; Pritchett, Woolcock, and Andrews, 2013; Andrews, Pritchett, and Woolcock, 2017).

Make the exit plan visible to workers and recipients, identifying which functions can be financed domestically and which still require external inputs. Identify the first external input whose loss would interrupt service, then reduce that dependency through local substitution, reserve capacity or a durable supply agreement before handoff.

22.4 — Technology, Knowledge, and Fair Access

Technology transfer must leave local institutions able to operate, adapt and replace the system (UNCTAD, 2025b; NIST, 2024). Transfer also requires decision authority. The receiving institution should be able to configure the system, control operational data, choose suppliers, approve updates, respond to incidents, budget recurring costs, and decommission or replace the tool without losing an essential service. Research, invention, software, equipment, medicine and industrial know-how require investment and stewardship. In public repair, assess how licensing, pricing, patents, standards, data and procurement terms affect urgent access and continuing dependency. Access secured through hidden control can itself become extraction.

22.4.1 — Knowledge Transfer and Digital Sovereignty

Knowledge transfer becomes durable when local evidence can change the imported method. Documentation should expose assumptions, test conditions, known limits, and revision points clearly enough that local researchers and operators can adapt or reject a method when it fails under local conditions (UNESCO, 2021; Hess and Ostrom, 2007; WTO, 1994).

AI transfer requires attention to hidden assumptions, bias, data extraction, energy and water demand, foreign compute, and updates controlled outside the receiving institution. Its local usefulness in translation, agriculture, education, diagnosis support, maintenance or disaster response must be tested under those conditions.

Digital sovereignty must be tested by protection from extraction, political targeting, commercial resale, foreign control and unappealable automated decisions, rather than by a server's national location alone. Apply the No Cross-Architecture Dossier

rule in Section 21.3.3 to development records (UNCTAD, 2025b; NIST, 2024; Nissenbaum, 2010).

Offline fallback and exportable records are practical tests of whether a digital system can fail without trapping the people who depend on it.

22.4.2 — Training and Maintenance

Training also requires time away from other work, language adaptation, supervision, and retention. If trained people leave because wages, safety, or respect are inadequate, the receiving institution can lose the skills it needs to sustain the technology (UNCTAD, 2025b; UNESCO, 2021; Andrews, Pritchett, and Woolcock, 2017).

The first serious fault tests whether technology transfer has created usable local capacity. A device or platform may work during a donor-supervised pilot, then stall when a component fails, a license expires, a password is lost, the network changes, or the vendor's help desk no longer answers. Lifecycle design should therefore include source documentation, open or portable standards where feasible, credential custody, license continuity, spare parts, local diagnostic authority, vendor-exit provisions, and a tested restoration path (UNCTAD, 2025b; UNESCO, 2021).

Resilience requires distributed competence rather than dependence on one expert, one contractor, one password, or one unusually committed employee. Shift coverage, written procedures, local trainers, succession plans, accessible documentation, and supervised practice protect an essential system from ordinary turnover.

22.5 — Reconstruction After War, Disaster, and Extraction

Reconstruction begins with a damage inventory broad enough to describe what was lost. Physical destruction may be joined by ecological harm, broken records, land disputes, institutional collapse, cultural loss, trauma, weakened trust, and interrupted care. Households may carry additional costs while public systems remain absent. The first authority question is who may define the injury before contracts determine the repair.

22.5.1 — Local Labor and Ownership

Ownership includes design decisions. A rebuilt market should fit local trade. A housing program should fit household structure, climate, land tenure, disability needs, and cultural practice. A school should fit language, safety, transport, and staffing. A clinic should fit referral routes, medicine supply, power, and privacy. Reconstruction that imposes an outsider's image of a finished place can leave these requirements unmet.

Local labor also protects dignity after catastrophe. When outsiders do all visible rebuilding, affected people may be cast as passive victims even when they hold knowledge of the terrain, materials, families, customs, and risks that outside teams lack. Fair reconstruction treats local skill as part of recovery, not as a courtesy added after contracts are signed.

22.5.2 — Anti-Corruption Reconstruction Controls

Reconstruction spending creates a compressed market in which urgency, scarcity, weak records, damaged institutions, and extraordinary sums meet at once. Those conditions can enable overbilling, concealed ownership, political favoritism, coercion by armed actors, land grabs, ghost workers, counterfeit supplies, and contractors who profit from weakened scrutiny. Corruption

control must therefore be part of emergency design rather than a later audit of money already lost (World Bank, 2020b; World Bank, 2020a; United Nations and World Bank, 2018; OECD, 2011).

Physical verification complements financial records. Inspectors, engineers, residents, workers, auditors, and service users may each see a different part of whether a road was built, a clinic is equipped, a water line works, or a housing allocation reached the intended household. Multiple forms of verification reduce capture without turning affected communities into unpaid surveillance labor.

Controls should preserve public memory as well as prevent immediate losses. Emergency exceptions, contract changes, complaints, sanctions, unresolved defects, and lessons learned need a record strong enough to guide the next crisis.

22.5.3 — Trauma and Community Repair

A road may reopen while fear keeps children from school, families from returning, witnesses from speaking, and neighbors from trusting one another. Reconstruction that repairs concrete but leaves public life organized around terror is physically visible and socially incomplete. Trauma belongs in the material account because it changes attendance, work, care, health, migration, cooperation, and the ability to use rebuilt institutions (World Bank, 2020a; WHO, 2022b; Autesserre, 2014).

Trauma care requires clinical expertise, local knowledge and cultural safeguards. Reconstruction budgets and timelines should allow for safety, trusted channels, family support, mourning and pathways back into public life.

Trauma repair should not be reduced to therapy programs alone. Public safety, housing stability, school continuity, memorial

practices, family tracing, livelihoods, land return, and trustworthy institutions all affect whether communities can breathe again.

22.5.4 — Infrastructure and Maintenance Future

Maintenance financing should survive the handoff from reconstruction to ordinary administration. Before commissioning, the plan should identify which recurrent costs move into local budgets, which remain externally supported, how spare parts and specialist skills will be replenished, and what service indicator triggers renewed capital support.

Before handover, apply the first-fault and restoration test in Section 22.4.2, alongside warranty and inspection duties (Graham and Thrift, 2007; Hallegatte, Rentschler, and Rozenberg, 2019; World Bank, 2020a).

Maintenance also has an incidence pattern. A network may look restored while rural districts, informal settlements, displaced communities, or low-income households still face unsafe water, unreliable power, inaccessible transport, or unaffordable tariffs. Acceptance should therefore test who receives continuity, who pays for it, whose land or labor carried the rebuild, and which failures remain concentrated.

22.6 — Repair, Independence, and the End of Dependency

Capacity transfer has a pace. Moving authority too fast can abandon fragile systems, while moving it too slowly can entrench dependency. Fair transfer uses staged responsibility, paired training, local budgeting, and public explanation. Review belongs with the people who will carry the system after outsiders leave.

22.6.1 — Reciprocal Learning

Repair should support learning in both directions: local knowledge of scarcity, weather and social cooperation can revise outside expertise without romanticizing poverty (OECD, 2011; UNESCO, 2021; Hess and Ostrom, 2007).

Development can deliver useful public goods while raising a separate question about institutional change. In a randomized community-driven development program in Sierra Leone, Casey, Glennerster, and Miguel (2012) found improved local public goods and economic outcomes, but no evidence of a sustained transformation of local institutions or collective action. Their preanalysis plan helped separate the results. That distinction invites a stronger partnership: evaluate what was built and who benefited, then test whether local decision authority, participation, and maintenance capability also endured.

Keep complaint channels available before enrollment, during delivery and after exit.

22.6.2 — The Ability to Prepare a Successor

The strongest transfer leaves a local institution able to prepare its own successor. A technician who can operate a water pump has gained a useful skill. A team that can diagnose faults, order suitable parts, and teach another technician has gained a more

durable capability. The second achievement keeps the service usable when the first trained person changes jobs.

A handover can be organized around that difference. At first, an experienced partner demonstrates a repair and explains the reasons for each step. Next, the local team performs the work while the partner observes. Later, the local team teaches a new colleague, using records it controls. Questions raised during that teaching reveal which parts of the knowledge have genuinely traveled and which still depend on someone outside the institution.

Purchasing authority is part of the same transfer. Knowing the correct component is of limited use when only an external project office can place the order. The handover needs to connect technical knowledge with a working budget and access to suppliers. A locally held stock record then becomes practical memory: it helps the next person understand what was fitted and which alternatives have already been tested.

Independence can grow through continued exchange. The local team may choose to consult its former partner on a difficult fault while handling ordinary work itself. That relationship is different from dependence on an outside signature for every decision. Assistance becomes expertise that can be invited, rather than an indispensable gate through which the service must always pass.

The lasting achievement is the ability to continue learning. When the institution can recognize a new problem, seek appropriate help, and incorporate the result into its own practice, capacity transfer has become more than the delivery of equipment or a completed training course. It has created a place where competence can renew itself.

22.7 — Donor Firewall and Restricted-Gift Discipline

The donor firewall applies to official, philanthropic, and platform-supported projects, not only direct grants (OECD, 2011; OECD, 2026b; Ferguson, 1990). Recipients need the freedom to describe a project's effects in their own terms. Support should not purchase private authority over their speech, participation, data, or future choices. This makes recipient independence a design requirement rather than a courtesy that can be withdrawn.

22.7.1 — Mission Independence From Donors

Keep mission ownership with the people who must live with the project (OECD, 2011; OECD, 2026b; Easterly, 2006).

No donor control of recipients extends the firewall from institutions to persons. People receiving assistance must not be pressured into labor, testimony, religious participation, political display, research consent, data exposure, migration choice, public gratitude, silence, or loyalty. Some programs require participation in repair work or training by design, but those requirements must be proportionate, transparent, appealable, and connected to the service itself. The person in need must never become the price of the donor's moral story.

Make this condition explicit in intake forms, consent forms, media releases, field visits, monitoring calls, and evaluation interviews.

22.7.2 — Narrative Independence and Capture

A donor may prefer stories of transformation over slow maintenance, profiles of successful women entrepreneurs over explanations of structural debt relief, innovation over salaries, or heroic recovery over trauma. Those preferences can guide future funding toward what flatters the donor rather than what repairs burden (Ferguson, 1990; Escobar, 1995; OECD, 2026b).

Use Section 6.5.4’s exit plan and preserve an archive of unfulfilled development commitments, so donor withdrawal does not erase accountability.

Consider a water project whose residents choose to report the hours during which a dependable supply is available. The donor may contribute a short account of its funding, while the operating team explains how repairs and staffing changed those hours. Residents can describe what the service has made possible in their own words. The report has several authors because the achievement has several contributors.

This division gives public recognition a more durable object. A photograph records one visit; a continuing service record lets later readers follow what happened after that visit. A donor can receive credit for helping create capacity without becoming the owner of the community’s account of its own life.

Local authorship also makes the next decision better informed. Residents may care most about a remaining interruption that an outside report would overlook. Keeping that observation in the record gives the operating team a concrete next task. Reporting then supports improvement instead of merely closing the story at its most flattering moment.

Chapter 23 — Migration, Refuge, and Shared Responsibility

At a school enrollment desk, migration stops being a headline. A parent explains a missing transcript. A child enters a classroom in an unfamiliar language. An administrator searches for a seat, an interpreter, transport, meals, and support the budget did not anticipate. The border may be far away, but this ordinary room is where a national promise of refuge becomes material and where receiving capacity becomes part of protection.

Displacement reaches clinics, housing offices, labor markets, legal-aid desks, transit systems, and neighborhoods already carrying their own shortages. By the end of 2025, forced displacement still affected roughly one person in seventy worldwide, and most refugees were hosted by low- and middle-income countries. The governing material question is how dignity and real receiving capacity can be strengthened together (UNHCR, 2026; World Bank, 2023).

Two obligations meet in every receiving place. Displaced people must not be turned into enemies, symbols of decline, or units of pressure. Host communities must not be left to absorb housing scarcity, school crowding, clinic load, language needs, legal complexity, fiscal strain, and trust pressure while distant authorities claim the moral credit. Durable protection joins human standing to financed reception, visible burden-sharing, and lasting public capacity.

Migration systems have both protective and administrative responsibilities. Borders should function without humiliation, and refuge should be joined to local investment. Screening should preserve due process, family safeguards, data restraint, and humanitarian access. Integration should allow plural belonging, and return should remain voluntary and safe. These responsibilities continue after arrival: no person should live indefinitely under suspicion simply because movement exposed need.

A host-community capacity and compensation framework makes housing pressure, school capacity, care systems, labor markets, transport, public trust, timing, and investment visible. It distinguishes preexisting deficits from added demand, identifies the institutions and workers carrying the response, and links observable service thresholds to timely finance, staffing, and maintenance before overload becomes the trigger for action.

Movement combines constraint with agency, and those dimensions can change during the same journey. Some people are expelled from ordinary life, some pursue opportunity, and many do both. A material account separates the reason for leaving from the resources and relationships that shape possible destinations (McAuliffe and Ceriani Cernadas, 2026; de Haas, 2021).

Plans should connect additional funding to forecast arrivals and actual service demand, with a continuation review before leases and staff contracts expire (UNHCR, 2018; UNHCR, 2026; World Bank, 2023).

Interpretation helps people use schools, clinics, housing offices, and legal services, but each institution must remain accountable for its own decisions (WHO, 2026b; WHO, 2026c; Ager and Strang, 2008).

23.1 — Why People Move When Systems Break

A movement-cause register replaces the undifferentiated category of migrant with distinct routes and decision needs. War, ecological stress, debt, persecution, labor demand, family networks, study, and aspiration create different legal and evidentiary questions. They also alter service needs and whether a safe return is plausible. Policy becomes more accurate when it begins upstream, where viable choices may already have narrowed and preventable pressures may still be reduced (McAuliffe and Ceriani Cernadas, 2026; IPCC, 2022; de Haas, 2021).

23.1.1 — Conflict, Climate, and Ecological Stress

Climate and ecological stress often interact with conflict rather than arriving as separate causes. Heat, drought, damaged water systems, crop loss, destroyed hospitals, mined roads, armed control, and fear can converge until ordinary predictability disappears. People may flee a battle, an ecological collapse, or the compound system in which each makes the other more dangerous (IPCC, 2022; IPCC, 2023; IDMC, 2026; World Bank, 2021b).

Conscription threats, sexual violence, targeted reprisals and lost documents create distinct evidence and reception needs. The burden map should record what forced movement, what a person could safely preserve, trauma and family separation, and which protection route fits the evidence without turning danger into spectacle (IDMC, 2026; IPCC, 2022; IPCC, 2023; World Bank, 2021b).

Climate movement often appears first as household adjustment rather than formal displacement. A family sells livestock, sends one worker away, withdraws a child from school, takes debt for water or seed, or leaves seasonally before anyone calls the pattern migration. The analysis therefore keeps climate language close to

food, water, heat, insurance, debt, land, and school continuity. Climate stress can deepen political and economic pressures until movement becomes the least impossible option.

Insurance withdrawal can restrict access to housing credit, and repeated rebuilding can turn attachment to place into a financial trap.

23.1.2 — Debt and Economic Collapse

Debt can move people without appearing as violence. A household under impossible repayment may send one member abroad; a farmer facing debt after crop loss may move to a city; and a worker may leave when wages collapse, prices rise, or local industry disappears. At public scale, debt pressure can also narrow services until ordinary life becomes less viable (McAuliffe and Ceriani Cernadas, 2026; World Bank, 2025c; de Haas, 2021).

Trace the roles of creditors, commodity buyers, sanctions systems, employers, remittance intermediaries, recruitment networks and climate shocks alongside local unemployment, services and family obligations. Those pressures can narrow choices without extinguishing agency.

Economic-collapse migration should include remittances without romanticizing separation. Money sent home may stabilize food, rent, medicine, schooling, or debt service, while the sender carries loneliness, dangerous work, recruitment fees, legal precarity, family distance, and pressure to keep sending. Receiving economies may benefit from labor while origin households carry the emotional and caregiving cost. Both the financial flow and the human load belong in the account (McAuliffe and Ceriani Cernadas, 2026; World Bank, 2025c; de Haas, 2021).

23.1.3 — Persecution and Legal Exclusion

People move when the law itself becomes danger. Persecution can be political, ethnic, religious, racial, gendered, sexual, ideological, tribal, linguistic, caste-based, disability-based, or directed at family status, journalism, labor organizing, minority identity, or public dissent. The relevant burden is not only private hatred. It is public vulnerability under systems that fail to protect or actively harm (United Nations, 1951; United Nations, 1967; UNHCR, 2018).

Persecution and legal exclusion often produce evidentiary difficulty by design. People may flee without documents, conceal identity to survive, lose records at borders, or fear that disclosure will endanger relatives. Rigorous screening should therefore combine credibility assessment, corroboration where possible, trauma-informed interviewing, interpretation, confidentiality, review, and appeal. It should not reserve protection for the rare person whose escape preserved perfect documentary continuity.

Evidence can itself be part of the danger. Identity files, police records, health systems, employers, informants, and family networks may be controlled by, visible to, or vulnerable to the persecuting authority, so verification is not neutral. Expert review should assess country conditions, source independence, and the risk that a request for corroboration will disclose location, relationships, or protected identity back into the threatening network (United Nations, 1951; United Nations, 1967; UNHCR, 2018).

23.1.4 — Labor Demand and Family Networks

Migration does not always begin with collapse. People also move to meet labor demand, join family, study, use needed skills, or sustain care and kinship. These ordinary movements belong to human life and should not be forced into crisis language merely

because they cross a border (Massey et al., 1993; de Haas, Castles, and Miller, 2020; de Haas, 2021).

The institutional question is whether status, wages, recruitment, credential review, housing, family unity, and public capacity make movement secure rather than exploitable (Massey et al., 1993; de Haas, Castles, and Miller, 2020; de Haas, 2021).

When an economy needs migrant labor but withholds lawful status, wage protection, housing, or complaint routes, it creates precarity and blames workers. Family networks can help people arrive but cannot replace public rules.

23.2 — Refuge, Receiving Capacity, and Shared Duty

Refuge requires receiving capacity as well as lawful protection. Housing, schools, healthcare, and municipal services need real staffing and funding, not merely an allocation on paper. Section 23.7 converts forecast demand into a compensation and delivery account for receiving communities (UNHCR, 2018; World Bank, 2023). The responsibility should be shared without making displaced people the scapegoat for preexisting service shortages.

23.2.1 — Local Trust and Social Cohesion

Migration systems that bypass local voice can create a sense of imposition even where residents want to help, weakening support for reception (World Bank, 2023; Ager and Strang, 2008; OECD and European Union, 2018).

Cooperation can grow through a shared task before it appears as a general change of attitude. Mousa (2020) randomly assigned displaced Iraqi Christians to all-Christian soccer teams or teams mixed with Muslim players. Mixed-team participants showed more cooperative behavior toward Muslim peers, including willingness to train together six months later. The effects did not consistently extend to other social settings or broad intergroup attitudes. The opening for local peacebuilding is therefore concrete: build repeatable opportunities for cooperation, then test whether those gains extend to other settings. Applying that idea to a shared neighborhood project would be a new intervention, requiring its own comparison and attention to residents who do not participate.

23.2.2 — Anti-Scapegoating Safeguards

Anti-scapegoating safeguards should include public correction of false claims, visible investment in shared services, labor enforcement against exploitation, protection of migrants who

report abuse, community liaison, transparent funding, and public explanation of who benefits from migrant labor (UNHCR, 2018; World Bank, 2023; Bansak, Hainmueller, and Hangartner, 2016).

Weak national transfers, slow permit systems, and insufficient legal aid can also intensify local strain. Public explanations should name the responsible institution, market rule, funding gap, or employer practice so residents can direct demands for correction to those responsible.

After each pressure episode, leaders should publish what strain was measured, what capacity was added, what claims proved false, and which problems remain unresolved.

23.3 — Borders, Screening, and Human Dignity

Borders can perform lawful functions while preserving dignity. Identity, status, trafficking risk, safety, and public order matter, but every procedure also carries bodily and family cost. Controls need a stated purpose, relevant evidence, and time limits. Review, health safeguards, data restraint, and protection from degrading treatment keep the exercise of authority tied to a legitimate function (United Nations, 1951; UNHCR, 2012; WHO, 2026b).

23.3.1 — Border Function Without Dehumanization

Border procedures need notice, interpretation, counsel or legal information, evidence review, reasons, appeal, correction of error, protection against retaliation and special handling for vulnerable people.

Family safeguards must see real dependency. A child may rely on a grandparent, aunt, older sibling, foster caregiver, or community member when war or flight has broken formal records. Rigid paperwork can sever protection; careless acceptance can expose children to trafficking or abuse. The fair route is urgent, child-centered review with documentation repair and independent oversight.

Fair screening also needs sequencing. Immediate triage should identify urgent protection, medical need, family vulnerability, and safety concerns before paperwork becomes an endurance test. More complex evidence can then be examined through the ordinary decision process, with access to review, appeal, and correction.

23.3.2 — Detention as High-Risk Burden

Detention is confinement and must be analyzed as a high-burden intervention rather than a neutral administrative pause. It can intensify uncertainty, family separation, trauma, dependency,

legal difficulty, and health risk while removing the ordinary choices through which people protect themselves. The World Health Organization's 2026 evidence brief synthesizes these health and well-being risks and calls for stronger safeguards, timely screening and care, improved conditions, and policies that treat detention as a last resort rather than routine case management (WHO, 2026b; UNHCR, 2012).

Because the burden is high, detention requires more than a general reference to identity, flight risk, public safety, or processing. The stated purpose must be lawful and individualized, the evidence reviewable, the duration as short as possible, and the person able to reach counsel, healthcare, interpretation, family contact, complaint channels, and independent review. Administrative convenience cannot supply the missing justification (WHO, 2026b; UNHCR, 2012).

Children should not be detained for immigration purposes. Use family-based or other appropriate community arrangements that protect their best interests and family unity (WHO, 2026b).

Alternatives require investment rather than wishful thinking. Community case management, stable housing, legal orientation, reporting arrangements, sponsorship, transport, interpretation, and risk-tailored supervision can support attendance and public order without reproducing the health and dignity costs of detention. The least restrictive effective route should be considered before confinement (WHO, 2026b; UNHCR, 2012).

Institutional incentives also require scrutiny. Beds can become budgets, contracts can reward occupancy, and administrative delay can become normalized confinement. Spending, private provision, legal access, health outcomes, family contact, complaints, alternatives, and release routes should be visible enough to prevent detention from becoming an industry hidden inside procedure (WHO, 2026b).

23.3.3 — Humanitarian Access Channels

Migration records can expose people and relatives to persecuting authorities, employers, traffickers or abusive partners. Apply Section 21.3's purpose, retention and protected-access rules at collection, with independent audit, breach response and human appeal, including school and interpreter records (United Nations, 1951; UNHCR, 2018; ICRC, 2020; Nissenbaum, 2010).

Humanitarian channels should not depend on public pity or media attention. A fair system needs predesigned routes for medical urgency, trafficking risk, torture survivors, unaccompanied children, family reunification, statelessness, disability, pregnancy, and sudden conflict or disaster. These channels do not erase ordinary law. They prevent the ordinary system from forcing people in acute danger to prove eligibility through delay.

Activation rules should be defined before crisis and tested during use. Each channel should name the authority that can open it, the evidence threshold, maximum decision time, responsible receiving service, funding source, and review after activation.

23.4 — Shared Responsibility Across States and Cities

Migration is often legislated nationally but administered by cities, towns, and community institutions. This split in authority must be explicit before a receiving locality is judged for service delays (UNHCR, 2018; UNHCR, 2026; World Bank, 2023). Responsibility for displacement extends beyond the border where arrival becomes visible. It includes admission, finance, family unity, resettlement, labor access, municipal capacity, and the conditions for voluntary safe return, while recognizing how origin, transit, destination, and distant states contribute through conflict, climate, trade, and finance. The architecture should allocate burden across that full chain (UNHCR, 2026; McAuliffe and Ceriani Cernadas, 2026). Use Section 23.7's timely funding formula so national admission commitments do not consume municipal service reserves (World Bank, 2023; UNHCR, 2018).

23.4.1 — Language and Legal Support

Provide interpretation and translated notices without making people depend on a child, employer, trafficker or interested broker. Keep translation distinct from advocacy, with confidentiality appropriate to each role (Ager and Strang, 2008; UNHCR, 2018; OECD and European Union, 2018).

Legal support should identify who provides orientation, document help, counsel, appeal assistance, case tracking, family-reunification guidance, labor-rights information and protection from fraud, and how to reach that help before deadlines pass.

Agencies should examine comprehension, appointment completion, document errors, appeal access, interpreter quality, confidentiality, and waiting time for counsel (Ager and Strang, 2008; UNHCR, 2018; OECD and European Union, 2018). When failures occur, identify the specific step, such as notice,

interpretation, scheduling, document review, counsel, or appeal, so recurring failures at that interface can be corrected.

Support thresholds should be operational. When interpreter demand, legal case volume, document backlog, or unresolved-status time crosses a declared level, the framework should identify the staffing, funding, or service response that follows and when it scales back. A threshold crossed briefly may call for temporary support, while repeated or prolonged excess can show that ordinary capacity needs to expand.

Use the compensation matrix in Section 23.7 to keep language and legal-access needs visible alongside housing and other receiving costs. Donor preference for visible newcomer services must not erase interpretation, counsel or existing local pressure from the account. Distinguish measured strain from unsupported claims, and keep an identified need visible even when no funder will support it.

For interpretation and legal support, distinguish existing service gaps and backlogs from added arrival-related work. Where one expansion serves both, apportion its costs transparently and count the added capacity once.

23.4.2 — A Working Handoff Between Levels of Government

A shared responsibility becomes easier to use when the receiving city knows which decision belongs to which institution. Consider a proposed reception arrangement in which a national office determines status while a municipality arranges school enrollment and ordinary local services. The city needs dependable information about the documents it may accept. The national office needs to know when an unresolved record is blocking a service that can otherwise begin.

A named liaison can help resolve that interface without turning every local worker into a specialist in national procedure. The liaison receives questions that affect several cases, obtains an authoritative answer, and returns it in a form that local offices can use. A recurring question about the same document may reveal an instruction that should be clarified once rather than interpreted separately at every counter.

Responsibility also travels with a person who moves between municipalities. The first locality should be able to transfer the information necessary for a continuing service through an agreed route, with the person’s involvement and applicable privacy protections. The next locality needs to acknowledge what it has received and explain which local arrangements must still be made. A completed transfer is a new institution ready to act, not merely a file sent elsewhere.

National coordination adds value when it solves these recurring interfaces. It can provide a common answer where a common rule exists and direct a local question to the authority able to resolve it. Municipal experience can then improve the national instruction. The relationship becomes reciprocal: local delivery identifies the ambiguity, and national clarification reduces the work required at the next point of arrival.

23.5 — Integration, Return, and Long-Horizon Belonging

Arrival begins an institutional journey whose success cannot be judged at the reception desk. Durable belonging depends on work, skill recognition, language, housing, schooling, health, safety, legal clarity, social connection, and time. Integration can support participation without erasing plural identity, while voluntary return requires safety, dignity, reliable information, and genuine choice (Ager and Strang, 2008; Hainmueller, Hangartner, and Lawrence, 2016; OECD and European Union, 2018). Governments, donors, and receiving communities may each have incentives to close a case before the conditions for return are met. Destitution, benefit withdrawal, confusion, threats, misinformation, or family separation cannot be allowed to convert apparent agreement into coerced return.

23.5.1 — Work Pathways and Housing Security

The time spent waiting for legal clarity can shape work after the decision. Hainmueller, Hangartner, and Lawrence (2016) studied 17,360 people who applied for asylum in Switzerland in 1994–2004 and received subsidiary protection within five years. Using variation in administrative processing times, their two main specifications estimated that an additional year of waiting reduced employment in the year after protection by roughly 3–5 percentage points. This was a quasi-experimental comparison, not randomized delay. Its causal interpretation depends on comparable applicants not sorting into shorter waits because of better employment prospects.

The practical opportunity is to remove avoidable idle time while preserving accurate decisions and accessible appeal. A redesigned route should follow the period before the decision as well as the first year afterward, pairing employment with job quality and

continued protection. An earlier decision alone does not remove credential barriers, unsafe recruitment, or missing childcare (Betts et al., 2017; Dustmann et al., 2017). Protection remains grounded in need and rights, not in a person's economic return.

Safe mobility expands when qualifications are recognized across borders, eligible social benefits remain accessible, and lawful pathways are available. Portable social protection, transparent recognition of credentials, ethical recruitment, circular or return pathways, and diaspora investment can reduce deskilling and exploitation. These mechanisms must remain supplemental to asylum and non-refoulement, not substitutes for protection. Their value lies in widening real options so that migration is less often clandestine, coerced, or governed by the actor with the most bargaining power.

Civic orientation should prepare both newcomers and receiving institutions. Newcomers need to understand law, rights, obligations, schools, healthcare, work norms, emergency systems, and public expectations. Receiving institutions also need to understand who has arrived, what languages are spoken, what trauma may be present, what skills exist, and what misunderstandings are likely.

Housing stability is also protection against exploitation. People without stable shelter are easier to underpay, threaten, traffic, shame, or force into unsafe arrangements. Housing therefore belongs beside labor law and child protection, not in a separate social-service silo (World Bank, 2023; Ager and Strang, 2008).

23.5.2 — Children's Future Belonging

Children experience migration on a different clock from adults. A year of uncertain status, disrupted schooling, repeated moves, or family fear occupies a substantial part of childhood. Protection should therefore support continuity early: stable housing,

learning, healthcare, language access, family relationships, play, and friendships (UNICEF, 2016; UNHCR, 2019; Ager and Strang, 2008).

Children's school, health, disability, family, and legal records require privacy. Their experiences should not be displayed as proof that a policy is compassionate or blamed as proof that it is failing. The practical test is whether a child can learn, heal, form relationships, and imagine adulthood with greater security.

Belonging without erasure is also a security principle. People who are forced to hide language, faith, memory, or grief may withdraw from public systems and become easier for exploiters to isolate. Participation becomes stronger when dignity is not conditional on sameness. A plural public order can require lawfulness and mutual respect without asking people to amputate the lives they brought with them.

23.6 — Capacity With Dignity

Numbers and images stripped of method can fuel fear, while compassionate stories can also conceal pressure on local services. Both distortions weaken the coalition needed for durable migration policy (UNHCR, 2018; UNHCR, 2026; World Bank, 2023). Public evidence should distinguish preexisting gaps from new demand and identify the institutions responsible for each. Accurate accounts protect displaced people and host communities against narratives that erase either burden.

23.6.1 — Migration as Shared Repair Test

Examine who benefits when migrants become public targets: employers who underpay, landlords who overcrowd, politicians who deflect from underfunding, smugglers who exploit closed routes, and media systems that profit from outrage. Pair public correction with enforcement against wage theft, predatory housing, fraudulent brokerage, trafficking and exploitation tied to insecure status (UNHCR, 2018; World Bank, 2023; McAuliffe and Ceriani Cernadas, 2026).

Agencies should also keep dates, definitions, service changes, and responsibility lines consistent across public notices so administrative disagreement does not become fuel for rumor. When information changes, the corrected version should be easy to identify and trace back to the office responsible for it.

At each stage, update Section 23.7's record of transferred burdens, added capacity and expiring emergency authority. The receiving institution should acknowledge each referral, and the sending institution should know whether it was accepted.

23.6.2 — Shared Services That Make Room

Capacity with dignity can be visible in the way a service makes room. Imagine a library that adds language support because more

residents need help reading official notices. Existing users also include people who struggle with the same documents. The institution can offer a clearly described service to everyone who needs that function while providing additional specialist assistance where the subject requires it.

The distinction between a common service and specialist support makes the expansion easier to understand. A general help session can improve access to ordinary information. A confidential appointment can address a person’s particular case. Staff need time for both, and the budget can show how the arrangement increases usable service rather than treating one group’s presence as another group’s subtraction.

Shared use also creates practical opportunities for contribution. A resident who has learned the local system may later help another person find the right office, while professional staff retain responsibility for matters requiring expertise. Voluntary help can complement a funded service without becoming its hidden staffing assumption. The institution gains a wider network of ordinary familiarity while the core promise remains dependable.

The public story can then be about a place becoming more capable. A newcomer has found a usable route into the community; an established resident has gained a clearer service; staff have better tools for a recurring task. These outcomes should remain separately visible so that the shared benefit is explained through actual work. Dignity becomes a feature of how the institution serves people rather than a label attached to a publicity campaign.

23.7 — Host-Community Capacity and Compensation

Receiving capacity should be calculated separately for each service as forecast demand minus staffed usable supply, with justified readiness and timing buffers, and a distinction between preexisting deficits and new arrivals. Housing, schools, clinics, transport, legal access, language support, and municipal administration have different constraints and must not be collapsed into a single count. For illustration, a clinic with 100 existing appointment requests, capacity for 90, and 20 additional requests faces a gap of 30 weekly slots: 10 preexisting and 20 added. A separately justified readiness allowance of 5 raises the planned requirement to 35 slots, but a hiring-lag buffer must not count that same shortage again. Those slots are a capacity requirement, not a cash request or a judgment about newcomers; staff, rooms, interpretation, and lead time need independent costing. Assign funding for the inherited and incremental gaps, protect temporary service until capacity arrives, and update the calculation with actual load, uncertainty, appeal, and correction.

23.7.1 — Housing Capacity

The housing-capacity field begins with the actual stock and the rate at which that stock can change. It tracks safe shelter, vacancy, rent, overcrowding, homelessness, public-housing waitlists, maintenance, tenant protection, utility access, land, construction lead time, and infrastructure (UNHCR, 2018; World Bank, 2023).

Stock and flow answer different questions. Vacancy and habitable units describe what can be used now; permits, rehabilitation, utilities, construction labor, finance, and land describe how quickly capacity can expand. A national housing total can hide a neighborhood emergency, while a temporary bed can hide the absence of durable housing. Funding decisions must

distinguish capacity available now from capacity that still needs to be built or restored.

Geographic concentration also matters. A region may have aggregate space while a small number of neighborhoods, schools, clinics, or transit corridors absorb most of the arrival. The field should therefore map distribution, season, local vacancy, disaster recovery, student cycles, eviction pressure, and construction lead time rather than relying on one average (UNHCR, 2018; World Bank, 2023).

Scarcity can create profit as well as hardship. Landlords, speculators, hotel contracts, emergency vendors, informal brokers, and overcrowded employers may gain while residents and newcomers blame each other. The housing field should distinguish payment for actual shelter and capacity from arrangements that merely monetize desperation.

Housing support must be designed to avoid rivalry. If newcomers are housed only by competing for the same scarce units as low-income residents, the arrangement can intensify resentment. Investment should expand supply, repair vacant units, support shelters, protect tenants, improve maintenance, and avoid concentrating poverty.

23.7.2 — School Capacity

The school-capacity field is a compensation instrument, not merely a count of children. It costs placement and support duties, including meals, transport and facilities, and assigns them to the school or district carrying the work (UNHCR, 2019; UNICEF, 2016; World Bank, 2023).

Staff time is a primary unit of school capacity. Intake, assessment, planning, translation, family outreach, individualized instruction, special-education coordination, counseling,

safeguarding, and attendance follow-up all require staff time in addition to physical classroom space.

Distribution within a district matters. One school may receive most newcomer students because of housing placement, transit routes, language programs, or informal family networks. Compensation should follow the actual concentration of work while avoiding segregation and preserving choice.

Reception begins with placement, but placement is not only the assignment of a seat. Staff must understand prior schooling, literacy, age, language, disability, safeguarding, health, transportation, and family circumstances without turning intake into an interrogation. Flexible assessment helps a child enter the right learning pathway while recognizing that records may be missing or education may have been interrupted.

Use enrollment forecasts to arrange contingency staffing, portable funding, rapid credentialing, interpreters, and temporary classroom capacity before teachers are left to absorb the full arrival burden through unpaid effort and emergency improvisation (UNHCR, 2019; UNICEF, 2016; World Bank, 2023).

Success should be reviewed through attendance, continuity, learning access, language progress, safeguarding, family communication, staff workload, special-education access, and the experience of resident and newcomer students.

23.7.3 — Care-System Capacity

Care needs often require attention before a person enters routine local healthcare. People may arrive after violence, hunger, dangerous travel, interrupted medication, pregnancy risk, disability, poor sanitation, detention, family separation, or prolonged uncertainty. Health support is therefore part of arrival

safety and system continuity rather than a benefit reserved for later settlement (WHO, 2025a; WHO, 2026b; World Bank, 2023).

A receiving health system needs an integrated route from first contact to ordinary care. Triage, primary care, vaccination where indicated, maternal care, chronic-disease management, disability support, mental-health services, interpretation, medication continuity, and referral should connect rather than force people through repeated intake. Many risks arise from conditions before and during displacement and from barriers to care, not from migrant status itself (WHO, 2026b; WHO, 2026c).

Delayed care can increase suffering and shift demands into emergency services. Earlier access may prevent some escalation, but whether it reduces total public spending depends on the service, population, cost, and horizon. Humane reception should therefore distinguish improved health and continuity from demonstrated budget savings. A worthwhile care program may require more funding rather than finance itself.

The care-system field counts added capacity for eldercare, childcare, emergency response, casework, and referral networks. Migration can bring urgent care needs into systems that were already stretched (WHO, 2025a; WHO, 2026c; World Bank, 2023).

The care-system field should treat interpreters and community health workers as core capacity. When people lack needed language or community support, medical visits can take longer, diagnoses can become less reliable, consent less informed, privacy harder to protect, and staff workload heavier. Mental-health services need similar attention because trauma does not always announce itself in the language expected by local systems (WHO, 2025a; WHO, 2026c; World Bank, 2023).

The care-system field must count migrant workers as providers as well as people who may need care. New workers can relieve

shortages, while insecure status can expose them to underpayment and exploitation. Fund enforcement, worker education, employer compliance, credential pathways and care staffing together.

Include safe housing, union access, inspection and anti-retaliation routes in that workforce support.

23.7.4 — Compensation and Investment

Community anxiety is not one thing. It may reflect prejudice, underfunding, rumor, poor communication, sudden arrival, or the accumulated memory of being neglected by public institutions. Each cause calls for a different response. The matrix preserves dignity by refusing to collapse these conditions into one number or accusation (UNHCR, 2018; World Bank, 2023).

The matrix should be used in a joint operating exercise rather than merely in a budget annex. Suppose an arrival forecast raises clinic demand before extra staff can be hired and school demand before classrooms can open. The receiving authority, relevant national funder, health and education providers, and affected residents should identify which capacity can be delivered immediately, which needs a funded bridge, and which cannot safely be promised by the announced date. Money promised later does not replace a nurse or interpreter needed tomorrow. Reporting should distinguish arrivals from service demand, and actual capacity from contracts or positions not yet filled. Support for established residents and newcomers must be transparent about the shared improvements delivered; neither group should be asked to subsidize the other through hidden queues.

Capacity finance should activate from observable service pressure rather than visible breakdown. Arrival volume, shelter occupancy, school enrollment, clinic caseloads, interpreter demand, and sanitation load can trigger staged transfers or

temporary staffing authority, with review as demand changes. The rule should be known before crisis so local governments can expand services without bargaining for emergency capacity.

Part VII — Shock and Climate Resilience

Calm conditions flatter institutions. Shelves are full, payments clear, weather ordinary, and help is not urgently contested, so budgets appear prudent, supply chains efficient, services humane, and trust secure. The harder test begins when time compresses. Months become hours, inconvenience becomes danger, and hidden dependencies become public facts. Under pressure, every system reveals who has reserves, who can wait, who can leave, and who must act immediately to keep ordinary life from collapsing.

Part VII examines what material peace can preserve under pressure. Financial panic reveals where liquidity, information, and confidence were concentrated. Shortage reveals how thin essential supply was. Climate and disaster expose accumulated weakness in maintenance, insurance, land use, health, housing, and public finance. Shadow economies show where lawful routes became slow, humiliating, corrupt, costly, or absent. Organized crime can establish coercive systems of private protection that compete with public authority. Crisis is therefore an exposure test of capacity, not a verdict on the worth of a society or its people.

A shock can destroy real capacity; a weak response can multiply the loss. Flood, disease, bank runs, cyberattack, port closure, drought, war, sanctions, and crop failure can damage systems directly. Unclear eligibility, hidden rescue, inaccessible portals, humiliating proof, language failure, unusable shelter, rumor, price manipulation, corruption, hoarding, and abandonment then add avoidable burden. This Part follows both layers: what the event damaged and how preparation, access, maintenance, and accountability can keep response failures from becoming a second disaster.

Climate converts episodic shock into recurring material conditions. Repeated heat, flood, drought, fire, insurance withdrawal, crop stress, displacement, and price pressure reshape

the environment in which ordinary choices are made. Panic prevention then becomes long-horizon infrastructure. Maintenance, early warning, adaptation finance, relocation support, regional investment, water systems, grid resilience, and insurance design preserve options before households are forced into debt, flight, exposure, or humiliating dependence.

The shadow-economy chapter examines lawful failure from the ground up. When formal systems cannot deliver essentials, or punish survival more efficiently than they repair scarcity, informal and illegal routes acquire demand. Participants occupy different positions: some improvise around exclusion, some are coerced, and some organize profit through violence, unsafe goods, laundering, or political protection. Material peace requires lawful channels fast and dignified enough to reduce dependence on predatory exchange while preserving answerability for organized harm (ILO, 2018; Schneider and Enste, 2000).

Four disciplines organize the Part: prepare before fear arrives, keep humane channels open under stress, review every exceptional tool after use, and retire what was justified only by rupture. Crisis does not suspend fairness. It reveals whether fairness was built deeply enough to operate at speed, under scarcity, and with incomplete information.

Chapter 24 — Crisis, Shortage, and Anti-Panic

Panic can misread conditions while also revealing thin reserves and remembered failure: people may expect supplies to run out, help to arrive late, or large actors to be rescued first.

The chapter begins with cascades because crisis rarely remains inside the system where it starts. A banking scare becomes a household cash scramble. A shelf shortage becomes hoarding, price pressure, and accusation. A platform rumor becomes a clinic queue or fuel run. A procurement shortcut becomes corruption, mistrust, and poor quality. Capital outflow becomes currency pressure, import shock, refinancing trouble, and cuts to public service. A credible peace architecture must follow transmission fast enough to interrupt it.

Shortage turns the same discipline toward essentials. Food, water, medicine, energy, shelter, sanitation, disability support, communication, transport, and care continuity can all become life-shaping under stress. Scarcity may require rationing, price tools, anti-hoarding rules, distribution controls, fraud review, exceptions, and appeals. Any allocation system will have to choose among competing urgent claims. Its legitimacy comes from protecting necessity openly, accommodating medical and disability needs, preserving dignity, and correcting rules that prove too blunt.

Crisis can close the human, digital and accessible routes through which help normally arrives. Section 24.5 tests continuity under those failures. Communication must say what remains open, what is uncertain and when the next update will arrive. Consider an illustrative payment interruption. A household cannot withdraw money and a shop cannot accept it, but the goods are still on the shelves. Additional deliveries would leave the transaction blocked. Now change one fact: payments work, but deliveries have stopped. The same visible queue calls for a different response. If both failures occur together, repairing either

alone leaves the other constraint in place. In this illustration, a test transaction through an independent payment route and a stock check upstream provide different evidence. Neither substitutes for the other. Reporting both prevents a visible queue from becoming the sole basis for action. Anti-panic design begins with that diagnosis, not with a promise that every frightened response is mistaken. The first useful update identifies the failed link and the transaction expected to become possible next. That is more informative than an assurance about the system as a whole.

Current financial-stability evidence sharpens this taxonomy. Cross-border portfolio flows, nonbank leverage, bank-nonbank linkages, sovereign rollover risk, and sudden shifts in risk appetite can transmit stress before a local institution appears insolvent. A sound crisis architecture monitors funding structure and transmission channels, then separates tools that restore payment and market functioning from tools that absorb private solvency losses. The distinction is technical, and its human consequence is immediate: a wrong diagnosis can preserve asset positions while wages, working capital, and essential imports still fail (IMF, 2026b).

24.1 — How Panic Cascades Through Systems

The operational question in a crisis is what people had to guess and how quickly that uncertainty became costly action. Unclear deposits, empty shelves, disrupted benefits, medicine supply, or a silent emergency office can rationally trigger withdrawal, hoarding, duplicate ordering, or unsafe substitution. Individual precautions become a cascade when everyone relies on the same fragile signals. The analysis asks which verified message, payment channel, or service continuity could have slowed that relay.

24.1.1 — Financial Contagion

A useful diagnostic separates channels that the word contagion often hides. Solvency contagion transmits losses through exposures that weaken other institutions' ability to meet their obligations. Liquidity contagion spreads because cash or funding cannot be obtained against claims that may remain serviceable. Information contagion spreads because outsiders cannot distinguish strong institutions from weak ones. Operational contagion spreads when payment, settlement, payroll, custody, or access systems fail. Market-functioning stress can then force sales and widen losses even before underlying value is known. Different channels call for different combinations of instruments, and several may be active at once (Diamond and Dybvig, 1983; Allen and Gale, 2000; IMF, 2026b).

Stress tests become materially credible when they extend beyond institutional balance sheets to household and municipal transmission. A payment interruption can become missed rent, delayed wages, frozen benefits, unpaid suppliers, utility arrears, or emergency borrowing within days. Those effects may not threaten the financial system immediately, but they are the route by which instability enters ordinary life. A complete scenario

therefore records the time required to restore critical payments, the groups without substitutes, the fees and penalties paused, and whether temporary support arrives before a brief interruption becomes a lasting debt spiral (IMF, 2025a; Herd and Moynihan, 2018).

Contagion unfolds on overlapping clocks. Payment continuity, verified access to deposits, liquidity against sound collateral, and clear communication may all be urgent in the first hours and days. Recapitalization, resolution, restructuring, or closure may take longer, but their assessment must begin while temporary support operates. Treating these tasks as one undifferentiated emergency allows temporary instruments to become permanent subsidies. A staged response names the function protected at each horizon, the evidence that licenses continuation, and the losses that additional liquidity cannot repair (Bernanke, 1983; IMF, 2025a; IMF, 2026b).

24.1.2 — Firm and Supply Panic

Firms' defensive actions can preserve survival or transfer fear through the network. A large buyer can secure scarce stock while hospitals, farms, utilities, and smaller competitors wait. A platform can change terms and move working-capital risk to sellers overnight.

Crisis visibility therefore needs concentration, dependency, and substitutability maps, not only firm-level solvency. The same defensive act carries different public consequences when performed at a chokepoint (Perrow, 1984; National Research Council, 2012).

Supply panic often begins as a forecast and becomes real through synchronized defense. Firms lengthen lead times, place duplicate orders, refuse unfamiliar customers, and reserve stock for the buyers most likely to pay. Each action can be

understandable alone, yet together they create false demand, distorted inventories, and scarcity in the wrong places. Allocation data should distinguish verified final use from speculative ordering, protect commercially sensitive information, and reward truthful inventory rather than strategic exaggeration.

Recovery from supply panic requires more than restocking. Firms need repaired contracts, working capital, credible public demand schedules, prompt payment, alternative suppliers, logistics access, and a way to address debts accumulated during interruption. Public purchasers can stabilize the network by paying on time and publishing realistic schedules; they can destabilize it through abrupt cancellation and opaque change orders (OECD, 2025d; IMF, 2025a; World Bank, 2017).

24.1.3 — Media and Platform Amplification

Rumor travels through whatever channels a society has built, from public speech, radio, newspapers, and word of mouth to group chats, short-form video, automated accounts, algorithmic outrage, viral screenshots, search manipulation, and targeted advertising. Platforms do not create every fear, but they can amplify it, change how it is framed, and spread it faster than its basis can be checked. A local shelf photo can be mistaken for evidence of national collapse before the store has restocked.

Neither public silence nor a centralized claim to truth defeats rumor. Institutions become more trustworthy than speculation by communicating quickly, separating observation, forecast, warning, and instruction, and visibly correcting errors rather than silently replacing them. Translation, disability access, local messengers, time stamps, and supersession notices make messages usable (Slovic, 1987; Sunstein, 2002).

People with cars, cash, storage, flexible work or digital fluency can act on warnings first; others may arrive after shelves empty,

prices rise or portals crash. Compare when reliable information reaches each group with the point at which the relevant resource, service or route becomes constrained.

Crisis platform governance must distinguish harmful conduct from inconvenient criticism. Fraud, impersonation, coordinated manipulation, incitement, and fabricated emergency instructions may justify rapid action; minority testimony, legitimate warnings, and criticism of official failure remain protected. Every intervention needs an identified behavior, evidentiary basis, disclosed category, accessible appeal and correction routes, and a time limit on emergency labels. Treat comparable conduct consistently for supporters and critics of the institution managing the emergency. Record the rule, evidence, and later review outcome so that consistency can be inspected.

Rumor control improves when institutions treat the public as a source, not merely an audience. Reports from pharmacists, shopkeepers, transport workers, disability organizations, clinics, and neighborhoods may reveal shortages early. A structured listening function should verify reports, publish corrections, and explain when local events are not system-wide (WHO, 2026a; World Bank, 2017). Verification should distinguish independent observations, which can strengthen a signal, from repetitions of the same underlying report, image, message, or claim.

Attention systems can sustain yesterday's emergency. Platforms and public communicators should date instructions, mark resolved claims, retire obsolete guidance, and show recovery. Because rumor also moves through private groups, workplaces, and families, accurate messages should be easy to forward and verify without universal monitoring (Slovic, 1987; Sunstein, 2002; WHO, 2026a; World Bank, 2017). Recovery communication should travel through the same channels used for the warning and state clearly which restrictions, shortages,

closures, or protective instructions have ended and which remain in force.

24.2 — Liquidity, Backstops, and Emergency Stabilization

Using Section 24.1.1's channel diagnosis, test which payment, credit, settlement, payroll, utility or supply function the backstop protects, who receives access, who waits and what public risk is assumed. Recapitalization, restructuring, resolution or closure still require their own authority and loss-allocation rules. Support must reach households and productive relationships beyond the intermediary (Bagehot, 1873; IMF, 2025a; IMF, 2026b). Before an emergency facility opens, record its protected function, eligibility, public risk, downstream duties and exit test. Specify consequences for later profit, ownership change, concentration or failure to deliver the protected service.

24.2.1 — Liquidity Backstop Purpose

A backstop should not protect owners merely because they dislike private funding terms or want public protection from an ordinary loss. Collateral, pricing, cash-flow evidence, protected-service obligations, and a resolution path should make eligibility visible (Bagehot, 1873; Holmström and Tirole, 1998; IMF, 2026b).

A backstop is judged by both the failure it prevents and the structure it leaves behind. The immediate result may be fewer collapses, while the longer result includes concentration, fiscal exposure, management incentives, public trust, and the willingness of actors to self-insure before the next shock. A successful facility therefore carries a post-event reform path (Diamond and Dybvig, 1983; IMF, 2025a).

Financial recovery is incomplete without a service test. An institution receiving exceptional support reports whether deposits remained accessible, payroll cleared, essential suppliers were paid, settlement continued, and viable small firms obtained working capital on stated terms. Aggregate service results can be

disclosed without exposing private accounts. When the balance sheet improves but the protected function does not, the design has reached the wrong outcome and pricing, governance, loss allocation, or continued eligibility must change (Holmström and Tirole, 1998; IMF, 2025b).

Pass-through design keeps public liquidity from ending at dominant intermediaries. Standardized downstream terms, aggregate reporting by borrower size and region, multiple application channels, and prompt reasons for denial make the protected bridge visible beyond its first recipient. Smaller banks, community lenders, municipal utilities, and viable firms should not require political access or a preexisting relationship with the largest beneficiary (IMF, 2025b; World Bank, 2017).

Access procedures should fit applicants' capacity and follow a clear, public timetable. The route states when applications open, which facts are essential, when decisions are due, how provisional support will be verified, and how an urgent mistake can be corrected. Simplified forms, staffed assistance, and delegated review preserve speed for small firms, community lenders, and local utilities without weakening later audit (World Bank, 2017; Andrews et al., 2017).

24.2.2 — Moral Hazard Review

Moral hazard concerns the lesson an intervention teaches about control, gain, and loss. It is strongest when actors choose the risk, retain the upside, and expect the public to absorb failure. The position differs when support protects depositors, workers, patients, suppliers, or households that did not control the underlying decision. A serious review therefore separates the function worth preserving from the owners and managers whose incentives require correction, rather than treating every beneficiary of continuity as the author of the risk.

The relevant history includes risk management, compensation, dividends, buybacks, leverage, concentration, insurance, regulatory compliance, and contingency planning. Public conduct belongs in the same record. Policymakers may have encouraged fragile structures while celebrating efficiency; regulators may have ignored warnings; procurement systems may have rewarded thin inventory; underfunded agencies may have made private workarounds unavoidable. Moral hazard can be produced by institutions, markets, and governments as well as firms (Minsky, 1986; Holmström and Tirole, 1998).

The moral-hazard review should also examine nonrescue. Refusing support can impose discipline on one institution while causing unemployment, supplier failure, service interruption, or a generalized run. Those spillovers are not arguments for unconditional rescue. They are reasons to compare the full incidence of intervention, resolution, and failure rather than treating visible punishment as evidence of prudence (Gorton and Metrick, 2012; IMF, 2025b).

An effective review reaches forward as well as backward. Institutions should know before the next shock which capital, liquidity, insurance, living-will, collateral, governance, and data obligations follow from using an emergency facility, and repeated users should face escalating scrutiny and conditions. Predictable ex ante terms reward self-protection and reduce the temptation to bargain for exceptions after losses appear (Holmström and Tirole, 1998; IMF, 2025a).

Compensation rules reveal whether stabilization is protecting an essential function or preserving status. Extraordinary bonuses, dividends, and buybacks are incompatible with reliance on public liquidity intended to keep that function operating. Restrictions remain tailored and time-limited, yet they also close obvious evasion through deferred awards or affiliates. Visible rules let the

public judge whether users of the system come before those who managed the risk (IMF, 2025a; World Bank, 2017).

24.2.3 — Exit and Repayment Rules

Publish each emergency instrument's renewal and recovery-based taper tests early enough for supported institutions to plan. The tests should lead to ordinary finance, restructuring, resolution, or closure while preserving the function the facility was created to protect. Every extension should explain what remains impossible without support and which costs continuation shifts to taxpayers, services, competitors, or households (IMF, 2025a; Andrews et al., 2017).

Household relief may need to be a grant when repayment would recreate the emergency; business aid may become forgivable after payroll, service, or public-purpose conditions are met (IMF, 2025a; IMF, 2025b).

Closure of a national support facility can conceal local nonrecovery. Lending may stop while suppliers remain unpaid, workers lose hours, revenues weaken, and viable firms remain trapped in emergency debt. Exit review should track arrears, employment, supplier health, service continuity, ordinary-credit access, and regional concentration after markets stabilize.

Recovery terms should reflect the instrument and its public purpose. A secured loan may use a fixed schedule; an equity stake may yield dividends or sale proceeds; attached warrants may provide an additional contingent return; guarantees may require fees and capital changes; grants preserving public functions may be unsuitable to recover from protected users. One model for unlike instruments can weaken recovery or hide subsidy. The record should state principal, pricing, maturity, contingent return, loss allocation, and public purpose (IMF, 2025b; Reinhart and Rogoff, 2009).

Large repayment demands falling due together can drain working capital, reduce lending, force service cuts, and destabilize recovery. A fair schedule uses transparent triggers, prevents opportunistic delay, and accounts for interacting public claims. The public should recover justified value without recreating crisis through an unrealistic timetable (Reinhart and Rogoff, 2009; IMF, 2025b).

24.3 — Capital Flight, Sanctions Shock, and Contagion

Financial panic can spread across borders before the institutions responsible for relief agree on a diagnosis. A flight from one asset or currency may defend an individual position while weakening the systems on which others depend. Sanctions, debt maturities, and emergency controls can transmit separate shocks through the same payment channels. Responses therefore need a credible timeline of obligations and a way to distinguish temporary liquidity strain from deeper insolvency. The policy question includes rights risks created by the protective measures themselves.

24.3.1 — Capital Flight Dynamics

Capital flight begins with expectations about loss, convertibility, inflation, confiscation, payment failure, political danger, or future rules. The same transfer can protect one household, weaken a bank, pressure a currency, raise refinancing costs, and deprive firms of working capital. Analysis should therefore separate motive from system effect, identify who possesses a safe exit and who does not, and show how any protective measure allocates the cost of remaining (Forbes and Warnock, 2012; Rey, 2015).

Wealth, foreign accounts, legal advice, currency options, and advance information allow some actors to protect themselves before depreciation or controls reach ordinary users. Households paid locally, small importers, public services, and workers may face the same risk without a comparable route. A fair response therefore asks whether policy uncertainty, corruption, weak reserves, or elite impunity made flight rational, and whether stabilization prevents the least mobile from financing the exit of the most mobile (Forbes and Warnock, 2012; Rey, 2015).

Funding can contract through residents' outward investment, nonresident withdrawal, deposit runs, refusal to refinance, or a decline in new investment and trade credit. Not all are recorded as gross capital outflows. Each has a different timing and institutional response. A blunt restriction can interrupt lawful remittances or essential imports while leaving the underlying financing problem unresolved. Diagnosis should identify the transaction and the balance sheet affected before choosing a control (Forbes and Warnock, 2012; Rey, 2015).

Global financial conditions can prompt capital withdrawal even when domestic policy is stable. Diagnosis therefore requires both local and external variables rather than attributing every outflow to national policy (IMF, 2022).

Protective financial measures can slow destabilizing flows, but their administrative burden requires separate rights review. Section 24.3.3 examines how documentation, exemptions, and compliance can exclude smaller lawful actors even when the control itself is justified (IMF, 2022; Herd and Moynihan, 2018).

The distributional consequence often arrives through the exchange rate. Depreciation raises the local cost of imported food, fuel, medicine, machinery, and foreign-currency debt. Exporters may benefit while households and public services absorb higher prices before wages adjust. Measures aimed at restoring confidence should therefore be judged not only by whether capital returns, but by whether essential imports, credit, and employment remain accessible during the adjustment (IMF, 2022; Reinhart and Rogoff, 2009).

Durable stabilization therefore combines temporary safeguards with credible institutions, clear macroeconomic policy, and lawful treatment of capital. Coercion without repair may slow the visible outflow while intensifying hidden channels and long-term distrust (IMF, 2022; Forbes and Warnock, 2012).

24.3.2 — Debt Refinancing Risk

Refinancing risk concerns both when debt falls due and who bears the cost if funding fails. A government, municipality, bank, utility, or viable firm may face a payment wall when market access contracts or rates surge. The maturity map should show currency, rate structure, holders, collateral, guarantees, rollover options, and the wages, maintenance, health, education, climate adaptation, or supplier payments that would be displaced (Reinhart and Rogoff, 2009; IMF, 2026b).

The distinctive danger is timing mismatch. Assets, future revenue, or productive capacity may remain valuable while a concentrated obligation falls due during closed markets or adverse rates. That calendar can turn a manageable long-run position into immediate service cuts or disorderly sale. Early negotiation, maturity diversification, and contingent financing protect room for repair; they do not erase the underlying debt or the need to allocate loss (Reinhart and Rogoff, 2009; IMF, 2025b).

A credible refinancing framework reduces surprise. It publishes debt data, limits hidden guarantees, clarifies legal priorities, protects essential functions, and creates a route for early negotiation before all parties are forced into a disorderly contest (IMF, 2025b; Drezner, 1999).

Alongside the maturity map, show floating-rate exposure, callable terms, contingent liabilities, and sensitivity to higher rates (IMF, 2025b; Reinhart and Rogoff, 2009).

24.3.3 — Protective Measures and Rights Risks

Cross-border stress can justify capital-flow tools, import priorities, emergency currency rules, sanctions compliance, export restrictions, banking safeguards, or accelerated procurement. Before use, assess each instrument’s effects on

lawful access, prices, data and exemptions; review those effects against its purpose.

Protective measures remain legitimate through a continuation test rather than a permanent presumption. The test names the harm addressed, the evidence that the measure reduces it, less burdensome alternatives, affected groups, review interval, correction route, and sunset condition. A measure that no longer reduces the initiating burden, or creates greater avoidable exclusion than the harm it prevents, is narrowed, replaced, or retired.

Emergency economic controls can create administrative exclusion even when their purpose is legitimate. A small importer, charity, remittance provider, or family business may be unable to afford the documentation demanded by banks, insurers, and shippers. Large firms can hire compliance teams and secure interpretations; smaller actors disappear from formal channels. Proportionality therefore includes the cost of proving eligibility. Standard documents, advisory support, published decisions, and rapid correction can preserve lawful access without weakening the underlying control (Herd and Moynihan, 2018; United Nations Security Council, 2022). Guidance should be updated fast enough that private intermediaries are not still enforcing a superseded restriction after the public rule has narrowed.

Emergency risk data require an end as definite as their purpose. Transaction histories, travel, nationality, family links, and business relationships are sensitive and easily misread. Apply Section 21.3’s purpose, access and retention limits, with correction rights and a ban on turning temporary risk categories into broad measures of character. A cleared person must not remain excluded because a vendor or institution retains an obsolete flag (World Bank, 2017; IMF, 2022). The review should

record any lawful actors still excluded after the change, because aggregate stabilization can conceal a persistent access failure.

Apply the private-overcompliance review in Section 21.4 to emergency financial controls, including refusal patterns, safe-harbor guidance and contractors' correction routes (United Nations Security Council, 2022; Herd and Moynihan, 2018).

24.4 — Shortage, Rationing, and Fair Emergency Allocation

Shortage exposes allocation rules that ordinary conditions conceal behind price, geography, time, information, and administrative advantage. When essential supply is limited, a legitimate authority must explain who receives service first, on what evidence, and how an excluded person can challenge the decision. Rationing may protect against panic purchasing, price explosion, corruption, and informal power, but it can also discriminate, humiliate, disregard disability or household size, and reward political connections (Sen, 1981; Sphere Association, 2018; National Research Council, 2012). The starting quantity must be staffed, safe, usable stock within the relevant time window, not nominal inventory. A triage body should honor binding legal and clinical duties, compare substitutes and urgency, explain unavoidable waiting or reduced service, mitigate disproportionate loss, and set a time for reconsideration. Protected review must test for hidden priority channels and geographic exclusion, while paper or human access remains available if records or networks fail.

24.4.1 — Essential Goods Priority

Priority criteria should be public and revisable. Immediate risk to life or health, irreversibility of interruption, number and vulnerability of people affected, absence of substitutes, cascade effects, and restoration time provide reasons that can be explained. The register should also record exceptions and changed local facts (Sen, 1981; Sphere Association, 2018).

Anti-hoarding design distinguishes household prudence from predatory accumulation. A family building a reasonable buffer is not equivalent to a reseller clearing shelves, a firm stockpiling critical inputs beyond operational need, or an insider diverting

goods before public distribution. Effective rules identify essential goods, emergency windows, quantity limits, resale restrictions, business necessity, and special household circumstances. They avoid criminalizing survival by people whose distrust was created by unreliable systems.

Essentiality changes with context. A heating fuel is critical during a cold snap; cooling, hydration, and transport become critical during extreme heat. Communication may be essential when warnings, benefits, or medical appointments depend on it. Menstrual supplies, infant formula, oxygen, mobility devices, and culturally usable food can be overlooked by systems designed around an imagined average household. Priority review should therefore include people whose needs are least visible.

Essential designation should attach to a dependency, a minimum service, and a public obligation, not to organizational size or political access. A firm or facility seeking priority should identify the function it protects, minimum quantity, duration, downstream recipients, pass-through duties, available substitutes, and evidence needed for renewal. Publishing designations and outcomes allows competitors, workers, and communities to challenge hidden favoritism while preserving urgent supply (Sphere Association, 2018; World Bank, 2017).

Priority also requires attention to geographic isolation. Remote, rural, island, mountain, and conflict-affected communities often receive supply later and at greater cost, while national averages conceal the delay. Reserve placement, transport contracts, local repair capacity, and decentralized decision authority can reduce that disadvantage (National Research Council, 2012; Hallegatte et al., 2017).

24.4.2 — Price Controls and Supply Risks

Price intervention should be judged as a complete access package. Unchecked price spikes can exclude low-income households and reward hoarding; rigid ceilings can reduce formal supply, quality, or investment and shift exchange into queues or hidden payment. Evaluation therefore includes posted price, quantity, quality, queue time, search cost, informal fees, supplier response, and the household's ability to obtain the good.

Affordability and supply rarely fit inside one price rule. Temporary ceilings may work only with procurement, imports, public stock release, transport support, working-capital finance, targeted subsidies, rationing, benchmark-cost disclosure, or additional suppliers. For each instrument, publish the quantities, beneficiaries, duration, performance, and exit conditions (Rockoff, 1984; Amaglobeli et al., 2023).

Exit matters here as well. Controls that remain after supply normalizes can create hidden privileges, discourage investment, or preserve arbitrary exceptions. Controls removed too early can expose households before wages and distribution have recovered. Review should therefore use indicators of inventory, delivery time, competition, household access, and informal-market spread (Sen, 1981; IMF, 2025a).

Whenever a price rule creates excess demand, queue design becomes allocation policy. First-come service can reward time, transport, information, physical endurance, and digital fluency rather than need. Digital reservations can exclude people without devices; identification rules can exclude migrants or people with unstable housing. A credible package limits repeat purchasing, preserves human access, protects care and disability priorities, and publishes how errors are corrected (Herd and Moynihan, 2018; Sphere Association, 2018).

24.4.3 — Distribution Integrity

Scarcity can create rents at each handoff. Procurement, warehousing, transport, eligibility lists, digital booking, local distribution, resale, and exception approval can be captured by vendors, officials, armed actors, platform operators, or informal gatekeepers. Distribution integrity therefore begins with a route map that names custody, discretion, value, and complaint access at each stage.

Follow goods, custody changes, contracts, exceptions and complaint patterns at supply chokepoints. Apply Section 21.3's data limits so ordinary recipients do not become the main surveillance object (Sphere Association, 2018; WHO, 2026a).

Local knowledge is an early detection system for diversion. Residents, frontline workers, civil-society groups, shopkeepers, and local officials often see manipulated lists, missing stock, unsafe goods, or unreachable delivery sites before a central dashboard does. Protected complaints, whistleblower routes, independent monitoring, and public summaries should convert that knowledge into correction without exposing the speaker.

Distribution integrity means that the good reaches the intended person in usable condition. Procurement success, warehouse inventory, and dispatch records are insufficient if the final route fails. Temperature control, packaging, local transport, opening hours, identity requirements, physical accessibility, and safe collection all determine whether nominal supply becomes real access (Sphere Association, 2018).

A household receiving spoiled food or the wrong medical supply cannot be asked to restart the entire application process. Replacement, refund, escalation, and urgent exception routes need named owners and response times. Aggregated complaint data can reveal recurring vendor or location failures while

individual complaints remain protected (WHO, 2026a; Herd and Moynihan, 2018).

Physical security and independent oversight must protect distribution without deterring lawful recipients. Route design should separate protection from eligibility, train personnel in de-escalation, and preserve civilian complaint channels. Auditors or community monitors may verify stock, queues, storage, and complaint handling under privacy and safety rules, while operators retain a chance to correct factual errors (Sphere Association, 2018; WHO, 2026a; World Bank, 2017).

24.5 — Keeping Humane Channels Open Under Stress

Emergency plans fail when the protective channel exists on paper but cannot be reached by the people who need it. Care depends on supplies, personnel, power, records, and transport that may be managed by different authorities. Language and disability access are operating requirements during a crisis, not optional additions after normal service resumes. Complaint and correction routes must remain usable when decisions are made quickly and imperfectly. The subsections test continuity from the affected person's point of view.

24.5.1 — Care Continuity

Care continuity is designed around dependency chains, not facility doors. Insulin depends on reliable supply and appropriate temperature control; dialysis on transport, power, staff, water, and records; home care on workers, access, protective equipment, and communication; mobility on devices, charging, repair, and assistance. A clinic may remain open while the route to care has failed. Planning should therefore trace each time-critical need through the staff, power, transport, information, and material inputs that make treatment usable (WHO, 2026a; Sphere Association, 2018).

A continuity inventory separates functions that cannot pause, functions that can change mode, and households requiring proactive contact. Health care, food assistance, housing support, disability services, eldercare, childcare, mental-health care, addiction treatment, medical devices, transport, and records access each need a contingency owner and an alternate route. The inventory also identifies people who cannot safely present themselves after failure (Sphere Association, 2018; WHO, 2026a).

Advance registration remains voluntary, minimal, and correctable. A person with a time-critical dependency may choose a protected continuity record containing only what responders need, with role-based access, expiration, correction, and withdrawal. Emergency care cannot be conditioned on surrendering a broad medical or family dossier.

Recovery requires restoration of relationships as well as appointments. Patients may lose contact with a trusted clinician, care coordinator, interpreter, or support worker during displacement. Reconnecting records without reconnecting responsibility can leave people technically registered but practically abandoned. A continuity plan should name who follows the person across settings, how outstanding tests and prescriptions are reconciled, and how missed care is found rather than waiting for the patient to navigate a fragmented system (Herd and Moynihan, 2018; World Bank, 2017).

Care reserves should reflect replacement time as well as average use: a rarely imported device part or a fragile cold chain may need more protection than a readily substitutable consumable (National Research Council, 2012; Hallegatte et al., 2019).

24.5.2 — Language and Accessibility Support

Apply Section 9.5.2's disability and language access standards to emergency warnings, eligibility, shelters, health guidance, appeals and payments. Budget interpreters, accessible formats, staffed assistance and backup channels for use under stress (WHO and World Bank, 2011; WHO, 2026a).

Emergency help should not require sensitive medical, migration, disability, or family facts to be disclosed in public because private support was never built.

Warnings and guidance should combine text, audio, pictograms, and human explanation as needed. Physical access

also matters: transport, ramps, seating, sensory conditions, queue design, and the ability to bring equipment or a support person.

Test the completed path under adverse conditions, including a noisy shelter or failed network. Involve people using different communication and access supports, and identify the fallback when the first channel fails (WHO and World Bank, 2011; Sphere Association, 2018).

Public testing also measures the time required to reach a human being. An accessible website cannot resolve a complex eligibility question when a caller needs interpretation, clarification, or an exception. Agencies can compare completed access, abandonment, delay, and correction by language and accommodation need rather than counting translated documents alone (Herd and Moynihan, 2018; World Bank, 2017).

24.5.3 — Complaint and Correction Routes

Complaint routes are emergency infrastructure because a small error can become a large deprivation when time is short. A wrong name, missing shipment, inaccessible shelter, rejected payment, or unsafe queue needs more than a future grievance form. A staffed urgent channel, a response target, authority for provisional relief, and a later evidentiary review make correction fast enough to change the lived outcome.

Independent escalation is necessary when the office causing the burden controls the answer. An ombuds function, external reviewer, court, regulator, or protected community monitor may be required depending on the decision and legal system. The route should state who can pause harm, obtain records, order correction, and protect against retaliation.

Different emergencies require different complaint routes. An immediate threat to health or safety needs operational escalation; a payment or eligibility error needs rapid correction; abuse,

discrimination, or corruption may require independent investigation. Sending every problem into one queue obscures severity and delays repair. The public needs to know which channel applies, what information is essential, and when silence itself triggers escalation (WHO, 2026a; World Bank, 2017).

A systemic error requires a systemic correction. When the same flawed rule denies many people, repairing only the cases of those who complain rewards persistence and leaves other affected people burdened. Institutions can identify the affected class, reopen decisions, notify people, and correct downstream records. Administrative data serve fairness here by locating and repairing the institution’s mistake (Herd and Moynihan, 2018; World Bank, 2017).

24.6 — Learning After Rupture

UNDRR's July 2026 After-Event Review guidance connects observed failures to changes in the next plan, budget, exercise, and public record (UNDRR, 2026a). An investigation should reconstruct how warnings reached or failed to reach people who could act. It should also identify whether resources, authority, or accessible alternatives were missing at the decisive moment. The result is useful only when someone is responsible for correcting the next response, not merely describing the last one.

24.6.1 — After-Action Burden Audit

An after-action burden audit records costs, exclusion, exposure, exceptional work, gains and people missing from the response record.

An after-event review reconstructs the full chain from detection and warning through receipt, interpretation, decision, action, and protection. Breaks in governance, risk knowledge, communication, preparedness, and response are then connected to people who never received a warning, could not act on it, or reached safety only by carrying hidden cost (UNDRR, 2026a).

Carry accepted recommendations into budgets, contracts, exercises and operating plans. Public reporting should state which actions were accepted, rejected, deferred or superseded and why (National Research Council, 2012; UNDRR, 2015; UNDRR, 2026a).

Decision reconstruction follows turning points. At each one, the record states the information then available, alternatives considered, authority used, expected consequences, people affected, dissent raised, and the later evidence that changed the assessment. This protects against hindsight certainty while distinguishing bad luck from weak design, ignored warnings,

negligence, capture, or concealment (Vaughan, 1996; Reason, 1990).

Recovery labor is a finite public resource. Staff, utility crews, health workers, volunteers, caregivers, local organizations, and small firms may contribute exceptional hours, accept injury risk, postpone ordinary work, and need months to recover. The audit should count hours, compensation, injury, turnover, lost income, delayed services and rest requirements, then carry those costs into future staffing and reserve plans (Aldrich, 2012; ILO, 2024a).

A corrective action is complete only after its operating condition has been tested. Replacing an alarm system is not enough if the night-shift team has never used it; signing a mutual-aid agreement is not enough if transport, contacts, and payment authority fail during an exercise. Each action should have a funded owner, a due date, an observed test, an unresolved-risk statement, and a route for frontline staff or affected people to challenge a premature completion label. Include a combined-failure exercise using the dependency method in Section 17.7.2.

24.6.2 — Public Explanation of Mistakes

The public explanation should draw on the decision reconstruction in 24.6.1, identify who was harmed, and show how correction can be tracked (Vaughan, 1996; Reason, 1990).

Name the next verification point, the remaining disagreement and the observation or result that would change each competing interpretation. Later review should show whether the disagreement narrowed, persisted or widened.

Recurrence should be tracked by mechanism as well as by event name. If the same weakness reappears in another office, vendor, location, or stage of the process, moving the failure should not be mistaken for eliminating it (Vaughan, 1996; Reason, 1990; Tetlock, 2005).

Responsibility should follow evidence. If the outcome arose from reasonable uncertainty, the remedy may be better monitoring or contingency planning. If officials ignored contrary evidence, the remedy includes decision-process reform. If concealment, corruption, retaliation, or a conflict of interest shaped the response, independent investigation and stronger accountability are required. Public language should preserve these distinctions so correction is neither diluted into a generic lesson nor inflated into indiscriminate blame (Vaughan, 1996; Reason, 1990). The record should also distinguish responsibility for causing the failure from responsibility for detecting, escalating, correcting, and explaining it. A person or office that inherits the repair should not automatically inherit blame for the decision that created the problem.

Explanation remains available after attention moves on. A stable archive preserves the decision, later corrections, evidence, implementation status, and supersession notices. Personal data remain minimized (World Bank, 2017; Vaughan, 1996).

Chapter 25 — Climate, Disaster, and Displacement

Climate disruption is a peace concern because it changes whether people can remain, work, eat, insure, borrow, care, move, govern, and trust under stable material conditions. A heat wave can become a lost shift, cooling bill, school closure, medical emergency, grid constraint, debt decision, and migration pressure within the same week. The chapter follows those conversions from physical hazard to lived burden.

Climate pressure becomes strategic when it travels across systems. Heat enters labor, health, housing, schools, prisons, farms, grids, warehouses, and care homes. Water stress enters food, sanitation, agriculture, industry, migration, and trust. Insurance retreat enters household wealth, mortgages, rents, municipal revenue, and relocation. The chapter gathers those routes into one material system while keeping hazard, exposure, vulnerability, and policy failure analytically distinct.

Preparedness is peace infrastructure when warning connects to usable action. In 2025, 119 countries, about 60 percent of the global total, reported multi-hazard early-warning systems, but coverage and comprehensiveness gaps remained. The relevant test is whether warnings reach people in time and connect to protective options they can actually use (UNDRR and WMO, 2025).

Displacement often begins before it becomes visible as a public event. A household may send one member away for work, sell assets, leave school, abandon a fishing ground, relocate after repeated floods, or remain trapped because moving would destroy care, culture, land memory, language, status, or livelihood. A serious peace analysis therefore sees both compelled movement and compelled staying, along with the narrowing choices that precede either.

Prevention becomes more credible when its evidence matches place, design, hazard, and time horizon. Some measures have

strong cost-saving evidence; others depend sharply on context; still others are warranted because avoidable death, displacement, cultural loss, or loss of agency cannot be reduced to one monetary return. Claims should name the counterfactual, uncertainty, distribution, maintenance burden, and horizon. That rigor strengthens prevention by showing exactly what it protects and under which conditions.

Disaster is hazard filtered through exposure, institutions, and social conditions. The 1995 Chicago heat wave became catastrophic when extreme heat interacted with social isolation, limited mobility, inadequate access to cooling, and dangerously hot indoor conditions (Centers for Disease Control and Prevention, 1995). Adaptation therefore spans cooling, drainage, fire management, buildings, grids, water, shelters, evacuation, health surveillance, worker protection, relocation, and finance for places unable to carry the cost. The objective is not merely survival of the event, but recovery of ordinary life without permanent debt or abandonment. That recovery is stronger when responsibility for restoring housing, health, utilities, records, local revenue, and other essential services is assigned before emergency attention moves elsewhere.

25.1 — Climate Disruption as Strategic Instability

The World Meteorological Organization reported that 2015–2025 were the eleven warmest years on record. For 2025, the global mean near-surface temperature was about 1.43 degrees Celsius above the 1850–1900 average, with reported uncertainty of plus or minus 0.13 degrees (WMO, 2026). Separately, UNDRR's 2025 global disaster assessment estimated annual direct costs near US$202 billion and total annual costs above US$2.3 trillion when cascading and ecosystem effects are included (UNDRR, 2025). These disaster estimates cover multiple hazards; they do not measure losses attributable to climate change alone. Trace each climate hazard to exposed activity, protective capacity, the bearer of lost income or service, delayed effects and the actor able to change the route.

25.1.1 — Heat, Water, and Food Systems

Heat changes the protective value of nighttime. The effects of a hot day depend partly on whether bodies, buildings, transit, clinics, and care homes can cool after sunset. Hot nights can interrupt sleep, worsen chronic illness, leave workers depleted at the start of the next day, and reduce caregivers' capacity to protect others. A heat register should therefore include recovery time, not only peak temperature.

Heat changes the meaning of place and work. A dwelling that is legally shelter may become unsafe if it traps heat. A job that is formally available may become dangerous outdoors, in agriculture, construction, delivery, sanitation, kitchens, warehouses, transit, repair, or poorly cooled care facilities. Worker protection therefore includes water, shade, rest, scheduling, acclimatization, emergency response, and enforceable occupational standards (WMO, 2026; ILO, 2024a).

Heat tests the whole relief route. A cooling center fails when transport is unavailable, language access is weak, disabled people cannot enter or travel, families are excluded, hours are too short, or people fear immigration enforcement, policing, or data exposure. Energy subsidies fail when the grid itself cannot deliver. Warnings fail when workers cannot refuse dangerous heat. The distributional test follows the path from alert to actual relief.

Water stress separates physical scarcity from governed scarcity. Rainfall and aquifer conditions set real limits, while leakage, pricing, irrigation priorities, industrial claims, land rights, upstream use, corruption, monitoring, and emergency reserves determine how those limits reach households. A useful water register keeps the hydrological constraint distinct from the institutional route. The first may require adaptation and allocation; the second calls for repair, accountability, and a better distribution of power.

For food systems, carry Section 17.3's access analysis into compound climate shocks: production losses can coincide with damaged transport, failed cold storage and reduced household earnings. Identify which link makes food unavailable or unaffordable before choosing a supply, income or distribution response (IPCC, 2022).

25.1.2 — Insurance and Housing Shock

Insurance retreat narrows the household menu before formal displacement begins. People may defer repair, accept exclusions, borrow for premiums, move without support, rent informally, or remain uninsured because every alternative is worse. Those actions can look voluntary in administrative records while the range of survivable choices is collapsing.

Insurance translates modeled risk into premiums, deductibles, mortgage access, rents, property values, local tax bases, and public expectations. Withdrawal can leave a physically standing home financially unstable, alter landlord behavior, constrain lending, and weaken municipal revenue before a disaster occurs. Risk pricing is therefore a form of private climate governance whose assumptions, concentration, and distributional effects deserve public review (Kousky, 2019; Kunreuther and Michel-Kerjan, 2011).

Housing shock is broader than ownership loss. It travels through mortgage underwriting, rent, public-housing maintenance, shelter demand, property taxation, repair labor, and speculative acquisition after distress. A high-risk designation can become a trap if residents receive information without adaptation support, tenant protection, anti-predatory safeguards, compensation where warranted, or a dignified route to move. Disclosure must be paired with support that widens residents' options, rather than merely announcing that the market is leaving.

Renters and informal residents may never see the insurance notice that changed their future. The signal reaches them through a rent increase, neglected maintenance, refusal to renew, a landlord sale, reduced public investment, or an unsafe repair. A complete housing register therefore includes owners, tenants, public housing, informal settlements, lenders, insurers, local government, and the workers rebuilding after loss.

25.1.3 — Fiscal Stress and Security Pressure

Disaster debt often enters ordinary accounting without a climate label. A household loan, municipal bond, reserve draw, delayed school repair, reduced clinic contract, utility surcharge, or supplier arrear can all be later stages of the same hazard. A climate-debt register keeps the initiating event, borrower,

instrument, interest, repayment horizon, displaced service, and distribution of loss connected so the burden is not normalized into unrelated fiscal stress.

Disaster borrowing also risks crowding out prevention. If public finance is consumed by recovery, less remains for adaptation, maintenance, schools, health, care, and local capacity. This creates a dangerous loop: underprepared systems suffer more damage, borrow more for recovery, and then have less capacity to prevent the next harm (Hallegatte et al., 2017; UNEP, 2025).

Compare the recurring cost of replacing an exposed asset with the distinct cost and consequences of protecting or relocating it.

Assess a climate-affected household's need for assistance separately from security forecasts about its region of origin. Border, logistics and public-order planning must preserve the person's standing as a rights bearer.

Compare the protective effects of food reserves, water agreements, disaster response, transport routes, public health, communication and receiving-region support with reliance on border enforcement alone.

25.2 — Disaster Readiness, Mutual Aid, and Recovery

The WHO's 2026 framework treats law, evidence, flexible finance, workforce, surveillance, communication, and clinical guidance as connected preparedness capabilities (WHO, 2026a; UNDRR and WMO, 2025). Warnings will not protect people who lack time, transport, shelter, or income security to act on them. Those enabling conditions need funding and testing before a disaster. The after-event chain in Section 24.6.1 helps determine whether workers could leave and residents could seek help safely (UNDRR and WMO, 2025; UNDRR, 2026b).

25.2.1 — Mutual Aid Infrastructure

Community groups, faith communities, unions, disability organizations, neighborhood networks, local businesses and volunteer teams can identify urgent needs, translate information, deliver supplies, check on isolated people and notice failures before formal systems do. Before emergencies, public agencies should establish contact routes and fund training and equipment while preserving these groups' autonomy (Aldrich, 2012; Sphere Association, 2018).

Mutual aid cannot become an excuse to budget permanent undercapacity. Volunteers and neighbors may carry exceptional work, but food, transport, shelter, health care, child protection, disability access, translation, safety, and recovery administration require durable institutions and paid labor. The partnership rule is complementary capacity: community knowledge improves reach, while public systems remain responsible for scale, continuity, protection, and the burdens that cannot be shifted onto goodwill.

Mutual aid also requires safeguarding. Informal networks can reproduce local hierarchies, exclude outsiders, pressure vulnerable people, or lack privacy protections. Funds can be misused. Volunteers can be exposed to danger. A fair mutual aid

architecture supports local trust while providing training, transparency, boundaries, and access routes for those who are not already socially connected (National Research Council, 2012; UNDRR, 2015).

Public support can make community capacity safer and more durable. Small grants, preapproved reimbursement, insurance, storage, transport, communication tools, protective equipment, training, legal guidance, and recovery time keep local groups from financing public response through private sacrifice. The route remains simple enough to use and transparent enough to resist capture. Prepared partnership strengthens public capacity without turning every community relationship into a contract.

25.2.2 — Emergency Shelter and Care

Shelter capacity should be measured as usable protection, not bed count. Alongside Section 24.5.2's access standards, shelters need food, water, sanitation, medication, sleep, privacy, family unity, child and elder protection, trauma care, relevant pet accommodation, legal information and a route back to stable housing. Fear of policing, immigration consequences or data misuse can still make an open building unusable (Sphere Association, 2018; UNDRR, 2015).

Family and child safeguards matter because crisis can separate people quickly. Evacuation, shelter admission, hospital transfer, immigration fear, custody disputes, school closure, and communication failure can all fracture family systems. Emergency care should preserve family unity where safe, protect children from exposure, and avoid making parental need look like parental failure (WHO, 2025a; Sphere Association, 2018; National Research Council, 2012).

Shelter audits should compare design with actual use. Who did not come, who left early, which accommodations failed, which

rules separated families, where privacy broke down, and which groups relied on informal alternatives? The audit should produce a corrective plan for transport, accessibility, staffing, safeguarding, records, and communication before the next event. Empty beds can indicate success, fear, exclusion, or a better option elsewhere; the reason matters.

25.2.3 — Local Response Capacity

Before outside help arrives, local responders need defined roles, contacts, authority, equipment and relief staffing.

The first operational window is measured in minutes and hours. Roads may close, power may fail, communications may fragment, and regional or national resources may not arrive immediately. Local plans therefore need a minimum independent capability for warning, triage, shelter opening, backup power, water, transport, accessible communication, and protection of critical records. The duration of that capability should be explicit rather than implied by optimism.

No locality can retain every specialist or asset needed for rare disasters. Regional mutual aid, shared technical teams, interoperable communication, preapproved contracts, reserve inventories, and clear reimbursement rules can widen capacity without stripping local authorship. The design question is which function must be local, which can be pooled, and how pooled help arrives without displacing local knowledge or delaying urgent decisions.

Capacity also includes endurance. Count rotations, relief crews, mental-health support, safe lodging, childcare, compensation, protective equipment and recovery time as conditions of available staffing.

Local capacity includes the authority to refuse unsafe conditions. Plans should identify who can reject a dangerous

vendor offer, resist political pressure, request national support, and disclose a capacity gap before failure.

25.2.4 — Recovery Time Horizon

Recovery begins while response is still underway, but it operates on a different clock. Restoring electricity, reopening a clinic, repairing a road, resolving insurance claims, rebuilding housing, replacing documents, recovering schooling, and restoring livelihoods may take radically different periods. A credible plan names those clocks separately and provides interim support.

Recovery time is unequal. Wealthier households can hire contractors, move, pursue claims, and absorb delays. Poorer households may wait for aid, lose work, face fraud or debt, or accept unsafe repairs. Tenants may be displaced; migrants may avoid assistance because of data fears. Systems must account for unequal margins.

A recovery register should track physical repair, housing and service continuity, mental health, schooling, small firms, debt, public revenue, staffing, disputes, insurance delays, and community trust. Visible infrastructure alone cannot establish complete recovery.

25.3 — Ecological Stress and Displacement Pressure

Ecological displacement analysis must include people who cannot leave an increasingly unsafe place. Money, transport, legal status, caregiving duties, and a destination that can receive them determine the real options available. Remaining in place may be a constrained choice rather than evidence that conditions are tolerable. Protection plans should therefore measure immobility as well as the number who move.

25.3.1 — Land Loss and Livelihood Loss

Land loss and livelihood loss reinforce one another; neither is only a change in area or income. Such losses can erase a home, farm, fishing ground, burial place, sacred site, public revenue base, local network, and the practical knowledge through which a community lives. Livelihood loss can occur before land becomes physically uninhabitable because insurance, credit, transport, water, or markets withdraw.

The compensation record should identify which livelihood and continuity functions must be restored, relocated, or supported over time rather than treating market value as the whole loss (IPCC, 2022; World Bank, 2021b).

Livelihood loss can produce debt before migration. Farmers borrow for inputs after a bad season. Fishers invest in longer trips. Households sell assets. Young people leave. Local businesses shrink. Public revenue falls. Schools lose students. Health systems lose workers. A land burden becomes a regional burden through many small exits and debts (IDMC, 2026; McAuliffe and Ceriani Cernadas, 2026).

Water and sanitation collapse can produce hidden indignities that statistics understate. People may reduce bathing, lose access to safe menstrual hygiene, miss school, risk assault while fetching

water, use contaminated sources, hide illness, or pay informal providers. These burdens are intimate and often politically quiet.

The practical safeguard is early support without forced exit. If assistance arrives only after movement is unavoidable, the public system has waited too long. If assistance arrives as pressure to leave before communities have agency, the system becomes coercive. Fairness requires support that widens choices before both staying and moving become desperate.

25.3.2 — Internal Migration Pressure

Internal migration pressure often appears before formal displacement categories capture it. Families may split, seasonal movement may become permanent, rents may rise in receiving cities, schools and clinics may absorb new demand, and informal settlements may grow near work. The 2026 global report counted more than 82.2 million people living in internal displacement at the end of 2025, including 68.6 million displaced by conflict and violence and almost 13.6 million by disasters. Those stocks do not attribute every movement to climate; they show the scale of durable mobility and service obligations that public systems must address (IDMC, 2026).

Apply the shared housing and receiving-service approach in Sections 23.4 and 23.7 to internal destinations, including protection for established residents and newcomers (McAuliffe and Ceriani Cernadas, 2026; IPCC, 2022).

Policy should recognize people moving within the country as holders of continuing rights and receiving residents as people entitled to honest planning. Support can include portable benefits, documentation, school continuity, health access, job matching, housing assistance, municipal grants, and affected-person review.

The World Health Organization’s review of 95 interventions found important service innovations alongside gaps in the

evidence on long-term, migrant-inclusive, climate-resilient planning (WHO, 2025a).

25.3.3 — Cross-Border Displacement

Cross-border climate movement sits uneasily beside older legal categories. People may move because war, persecution, food collapse, disaster, debt, family networks, labor demand, and climate stress are tangled together. Climate or disaster exposure alone does not establish refugee status under the 1951 Convention; eligibility depends on its criteria, including a well-founded fear of persecution on one of its protected grounds. Regional refugee rules and complementary protection may also apply. Legal uncertainty must not become a reason to deny care. Humanitarian access, due process, family unity, child protection, data restraint, and receiving-region support remain relevant even when categories lag behind reality (United Nations, 1951; United Nations, 1967; UNHCR, 2020).

Cross-border displacement also raises responsibility questions. Who contributed to the risk? Who has capacity to support? Who benefits from fossil systems? Who owes adaptation finance? Who owes loss and damage support? This chapter does not settle those legal and diplomatic questions, but it insists that they be asked without converting vulnerable countries into petitioners for pity (IDMC, 2026; UNDRR, 2025).

25.4 — Burden-Sharing Across Regions and States

Climate burdens cross administrative borders even when legal responsibility and public budgets remain local. A region exposed to disaster may lack the fiscal capacity to fund the protection that its risk profile warrants. Burden sharing has to distinguish physical exposure, past contributions to harm, present ability to respond, and the interests of people not yet born. None of those measures is a ranking of the worth of populations. The following approaches examine what can be mapped, compensated, and maintained under reviewable commitments.

25.4.1 — Exposure Mapping Without Ranking Worth

Connect each exposure classification to funded protection and maintenance, receiving capacity and a reviewable action pathway; disclosure without support can accelerate lending withdrawal, property stigma or political abandonment.

A risk map earns trust when context travels with the color. Model assumptions, uncertainty, time horizon, data gaps, adaptation already in place, exposed populations, and the difference between present and projected risk remain visible. High risk is not destiny, and weak data are not evidence of safety. Readers need to know what the map supports, what it cannot decide, and when new evidence will revise it.

Exposure classifications do not measure the worth of a country or community.

25.4.2 — Loss and Damage Questions

Any serious treatment of loss and damage must hold grief without pretending to price every grief fully. Economic loss matters: homes, crops, roads, boats, clinics, wells, and businesses. Non-economic loss matters too: graves, language landscapes, sacred places, ecological memory, family separation, health, dignity, and

trust. A material-burden book can name these categories and require expert review without claiming to settle legal responsibility or the final method of compensation.

Compensation can also create risks. Money can be delayed, captured, politicized, tied to conditions, routed through corrupt actors, or used to silence claims. A community receiving compensation may still lose land, memory, and future. The relevant test is whether loss-and-damage support reaches affected people, respects local authorship, and preserves the right to speak after payment (UNDRR, 2025; Gardiner, 2011).

25.4.3 — Intergenerational Climate Cost

Intergenerational climate cost is often hidden through discounting, deferred maintenance, debt, ecosystem loss, and infrastructure that transfers risk to future users. Future people cannot consent to today's underinvestment, yet they also cannot be represented by a single assumed preference. The discipline is to make long-lived liabilities visible, test sensitivity to discount rates and scenarios, preserve options, and avoid using distant uncertainty to justify near-term exposure that is already preventable (Stern, 2007; Gardiner, 2011).

Record the adaptation deferred, the alternatives rejected and the burden passed to future budgets, so later reviewers can reconstruct and challenge the decision.

The review structure should also ask who gets to challenge the expert frame. A community may know that a proposed seawall blocks fishing access, that a relocation site breaks kinship support, that a data collection plan creates surveillance fear, or that a resilience project benefits a contractor more than residents. Expert seriousness includes the humility to hear burden evidence from people without credentials.

25.5 — Prevention, Preparedness, and the Cost of Delay

Prevention earns support through the harm it can avert, not only through a claim that it will save money. The comparison should name the hazard, the feasible protective action, its maintenance, and the consequences of waiting. Health, safety, and continuity can justify an intervention whose monetary benefits do not exceed its fiscal cost; that judgment should be stated rather than described as a financial saving. Keeping these reasons distinct allows a costly but worthwhile protection to be defended on its actual merits.

25.5.1 — Avoided Disaster Cost

Estimate avoided losses as ranges, including distributional effects and harms that cannot or should not be reduced to one price. Show sensitivity to rival assumptions rather than presenting a single benefit-cost ratio as certainty (National Institute of Building Sciences, 2019; UNDRR, 2025).

After a hazard occurs, the avoided-cost model should be calibrated against observed intensity, asset performance, evacuation time, service continuity, and actual losses in comparable exposed areas. That update can narrow earlier ranges, reveal which protective mechanism worked, and show where the model credited prevention that did not perform.

A disaster's fiscal aftermath can continue in accounts that do not carry its name. Deryugina (2017) found that U.S. hurricanes increased ordinary government transfers, including unemployment insurance and public medical payments, for years after the event. The estimated present value of the additional nondisaster transfers exceeded direct disaster aid. That finding changes where an appraisal looks: a repair decision can affect social-insurance budgets long after emergency appropriations

close. These transfers can provide valuable protection; expenditure is not itself a measure of social waste. Keep the fiscal cost, the support received, and the underlying loss separate so the same event is not counted repeatedly.

25.5.2 — Maintenance and Prevention Link

Maintenance priorities should follow consequence and fallback. A failed drain, substation, cooling system, bridge joint, water pump, school roof, or clinic generator matters differently depending on who depends on it and what substitute exists. Inspection data, failure history, climate projections, and community reports can guide sequencing, while equity review prevents politically visible assets from displacing the quiet systems on which low-margin communities rely (Hallegatte et al., 2019; UNDRR, 2015).

A prevention portfolio can combine physical protection with maintenance, warnings, social support, and recovery finance. Layers matter because a single barrier can fail. Redundancy is justified by the distinct risks it covers and its performance under stress, compared with feasible alternatives and lifecycle cost. Transparent duplication can still be wasteful; poorly documented capacity can still be protective. Visibility enables that judgment but does not determine it.

Prevention equity asks whether safety investment follows power or exposure. Wealthy neighborhoods may receive tree canopy, drainage, fire protection, backup power, and insurance support earlier because they have political voice and property value. Poorer areas may receive evacuation instructions instead of adaptation. A fair prevention system should counter that bias by directing investment toward places where burden is greatest, options are fewest, and delay would be most destructive.

The prevention register should show the order of investment, the reasons for it and when the next exposed community receives protection. A later promise is not protection already delivered.

25.6 — Climate Finance, Insurance, and Adaptation Burden

Climate finance turns promises into instruments, conditions, repayment paths, maintenance duties, and lived protection. Categories such as mitigation, adaptation, preparedness, loss and damage, insurance, relocation, resilience, and recovery do not reveal who actually benefits. Grants, loans, guarantees, subsidies, and public allocations should be compared by currency, cost of capital, conditions, recipients, and distributional effects. Local operating obligations and the risk of project failure also belong in that assessment.

25.6.1 — Adaptation Budget Categories

Organize adaptation budgets around household and municipal services. The 2025 Adaptation Gap Report estimated annual developing-country adaptation needs by 2035 at US$310 billion using modelled costs and US$365 billion using extrapolated nationally stated needs, both in 2023 prices. International public adaptation finance was US$26 billion in 2023 (UNEP, 2025).

Separate capital, recurring operating, contingency and replacement costs in adaptation budgets, and match financing terms to those time profiles. One-time climate finance must not leave local institutions with an unfunded continuing obligation (UNEP, 2025; IPCC, 2022).

A replacement reserve translates an asset's useful life into a continuing budget question. In an invented adaptation project, a pump costs 60 units and is expected to need replacement after six years. Setting aside ten units at the end of each year would accumulate 60 by the replacement date, assuming a stable price and no interest. The reserve funds a future purchase. It does not pay for the pump's electricity, routine servicing, or the people who operate it.

Now add a control unit costing 12 units that must be replaced after three years. An additional four units a year would fund each three-year replacement under the same assumptions. The combined reserve contribution is fourteen units a year. Different components therefore give one project several renewal clocks. A single asset total can conceal those clocks unless the budget keeps them visible.

Inspecting the equipment and revisiting replacement prices lets the institution adjust the contribution as knowledge improves. A grant may fund the initial purchase while local revenue supports annual contributions. Another arrangement may assign part of renewal to the funding partner. What matters is that the continuing commitment has an owner before the first equipment is installed.

This gives adaptation finance a constructive measure of durability. The project is not only ready to begin operating. It has a stated route for continuing after its first components wear out. The community inherits both the equipment and an account of how the next generation of that equipment can be obtained.

25.6.2 — Insurance Access and Retreat

Insurance access depends on more than disclosure and price. Insurer solvency, reinsurance capacity, claims performance, public backstops, last-resort markets, affordability support, and the availability of adaptation finance determine whether risk can still be pooled rather than simply abandoned. The public question is whether the insurance system preserves a workable route through rising risk while signaling where prevention and relocation are becoming necessary.

Insurance retreat should trigger public scrutiny of the underlying risk model and the options it leaves residents. Regulators should distinguish changing hazard from model

uncertainty or concentrated market withdrawal, then connect disclosure to adaptation and lawful relocation before coverage disappears (Kousky, 2019; Kunreuther and Michel-Kerjan, 2011; UNDRR, 2025).

Risk protection can change what people attempt before a disaster. In northern Ghana, Karlan et al. (2014) randomized cash support and access to rainfall insurance. Agricultural investment responded strongly to insurance even without a cash grant. The result identifies uninsured risk as an important constraint in that setting: protection could mobilize resources that were otherwise withheld from farming. It does not establish a universal increase in farm profits. An adaptation program can examine that productive channel alongside later payouts. Its record should also distinguish a loss not captured by the rainfall index from a payment owed but not delivered. Those failures call for different repairs, and both affect whether people can plan around the protection.

25.6.3 — Public Backstop Design

The need for a backstop can arise from market withdrawal, correlated loss, unaffordable reinsurance, thin local tax bases, or an essential service that cannot simply exit. The relevant comparison is among adaptation, risk pooling, targeted subsidy, public provision, managed retreat, and abandonment, with the burden of each route made visible (UNEP, 2025; UNDRR, 2025).

A public climate backstop should insure continuity of essential life, not guarantee the appreciation of exposed assets. Eligibility ought to distinguish critical public facilities, households with weak substitutes, small employers, and discretionary high-risk investment. Support can be conditioned on feasible adaptation and maintenance, with attention to whether low-income residents had the resources or authority to comply. The contract should

specify the public loss ceiling, reserve and replenishment rules, the trigger for reducing coverage, and the route for assisted relocation when the risk can no longer be borne safely. If a backstop repeatedly rebuilds the same avoidable exposure while shifting premiums and fiscal costs to others, its design requires revision.

When a backstop funds relocation, include receiving-place capacity using the framework in Section 23.7.

25.6.4 — Local Government Burden

Fund the operating capacity required by local adaptation awards (UNDRR, 2025; Hallegatte et al., 2019). Local fiscal room is often narrowest where responsibility is most immediate. Repeated shocks weaken tax bases while increasing demand for water, roads, shelters, health coordination, schools, waste removal, and housing support. Intergovernmental formulas can reflect exposure, service population, poverty, displacement, infrastructure condition, and recurring readiness costs. Shared grant writers, engineers, procurement support, legal assistance, data systems, and reimbursement advances keep exposed jurisdictions from being excluded by the very application capacity they lack.

These shared services, together with mutual-aid agreements, reserve inventories and emergency transfers, need named owners, activation triggers, payment clocks and local decision rights. The same budgets protect officials, responders, utility workers and community organizations through staffing, training, rest, retention and succession.

A no-abandonment rule rejects quiet triage through market withdrawal, inaccessible insurance, unfunded local mandates, or delayed disclosure without promising that every asset can remain or every loss can be prevented. Protection, adaptation, managed

retreat, or relocation may still be necessary. Affected people need open and reviewable decisions, compensation, attention to cultural ties, receiving capacity, and honesty about irreparable loss.

Chapter 26 — Black Markets and Anti-Capture

The inquiry into a shadow economy begins before the illegal act: which lawful route failed, who improvised or was coerced, who organized profit, and what repair would reduce dependence without excusing predation?

The chapter's governing distinction is among survival, coercion, and predation. Survival participation arises when lawful access is absent, unaffordable, slow, humiliating, discriminatory, dangerous, or captured. Coercion operates through threat, debt bondage, migration insecurity, trafficking, gang rule, addiction, captivity, or fear. Predation organizes profit from scarcity, unsafe goods, corruption, violence, laundering, and dependency. Because one person may occupy more than one status over time, classification must follow actual power, knowledge, harm, and room to refuse.

Following burden through enforcement changes which actors, institutions, and remedies become visible. An illegality-only lens can punish people at the bottom while leaving scarcity and protection networks intact; a social-causes-only lens can excuse deliberate harm; crime spectacle can turn poor neighborhoods, migrants, informal workers, street vendors, border regions, or sanctioned populations into suspect categories. A stronger discipline combines lawful-access repair, targeted accountability, due process, community protection, anti-corruption review, and transparent revision when rules help create the shadows they condemn.

A medicine shortage makes the stakes concrete. When the public supply chain fails, informal routes may keep some patients alive while exposing others to counterfeit products, unsafe storage, ruinous prices, or protected intermediaries. A credible response repairs lawful supply, warns and treats affected patients, targets organizers and corrupt protectors, reviews procurement and quality control, and avoids displaying desperate buyers as

evidence of enforcement strength. Human need diagnoses system failure; it does not license predation (OECD and EUIPO, 2025).

Anti-capture analysis follows hidden money upward. Contract rigging, beneficial-ownership opacity, property laundering, professional facilitation, customs protection, selective enforcement, political donations, and procurement influence can convert shadow profit into respectable power. Arresting low-level participants while ignoring those interfaces displaces blame downward. The account is incomplete until routes from illicit gain to land, status, law, silence, and office become inspectable and answerable.

Institutional failure requires more than punishment. Rules that reliably create dangerous shadows deserve review; lawful systems that exclude people from essentials need access repair; captured enforcement requires audit; communities controlled by violent actors need protection before public blame; and carefully bounded amnesty may sometimes reveal systems and reduce harm without erasing accountability. The objective is a public order that stops manufacturing hidden markets, weakens predatory networks, and refuses to make the desperate carry the moral theater of failure.

26.1 — Why Shadow Economies Form

A shadow-economy map must distinguish demand for usable goods from the power exercised over the route. It should identify who obtains access, who accepts unsafe quality, who is coerced, and who earns income. It must then follow payments for protection, information control, and the conversion of proceeds into legitimate-seeming assets or influence. Those distinctions prevent a survival purchase from being equated with the organized control profiting from it.

26.1.1 — Scarcity Channels

When formal supply or lawful access fails, people may buy medicine, fuel, housing or documents through unsafe informal routes (Schneider and Enste, 2000; ILO, 2018).

Scarcity-driven markets often weaken quality assurance along with supply. Section 26.3.2 develops that access dilemma; here the key point is that a scarcity premium can buy access while shifting product risk to the buyer.

A serious scarcity register asks which lawful route failed, how long people waited, what proof and price were demanded, who possessed alternatives, who was endangered, and who profited. Without that register, the public sees only the final prohibited transaction and misses the system that made the shadow useful (de Soto, 1989; Sen, 1981).

The burden register should distinguish episodic shortage from engineered scarcity. A storm, strike, port closure, disease outbreak, or conflict may interrupt supply. A captured distributor, corrupt procurement chain, speculative hoarding, or political favoritism may create scarcity as a profit strategy. The remedies differ, and so must the language.

26.1.2 — Administrative Exclusion

Administrative exclusion creates demand for brokers when lawful access requires documents, time, language, transport, digital skill, fees, or discretion that people cannot supply. The broker may translate a complex system, exploit vulnerability, or control a scarce appointment. Simplifying lawful processes, providing human help, publishing decisions, and creating urgent correction can remove the premium without criminalizing the person who needed assistance.

Blanket refusal by lawful providers can push legitimate users into cash, informal credit, forged documents or protected intermediaries. Apply the proportional-access and private-overcompliance reviews in Sections 21.4 and 24.3.3 to this displacement risk (FATF, 2025b; Herd and Moynihan, 2018).

26.1.3 — Sanctions and Trade Disruption

The question is who is constrained, who adapts, and who profits when restrictions change access. Follow medicine and remittance channels alongside enforcement costs, rather than treating either a legitimate purpose or a civilian hardship as a complete verdict on the measure.

Parallel imports of genuine goods are not automatically illicit; their legal status depends on the importing jurisdiction (WTO, n.d.). The informal supply routes considered here may deliver essentials when official channels fail, sustain predatory actors, or combine needed goods with corrupt fees, counterfeits and protection payments. Assessment must distinguish the goods, route and conduct rather than treating all trade outside authorized distribution channels as one category.

Other disruptions, including war, blockade, currency collapse, border closure and damaged transport, can create the same scarcity premium. Review why predictable civilian and

commercial needs lacked a resilient lawful channel before shadow routes became the de facto logistics system (Andreas, 2005; Early, 2015).

Neighboring states and border regions often carry sanctions burden without designing the sanctions. They may absorb diverted trade, refugee movement, smuggling pressure, price shifts, corruption risk, and enforcement obligations. Fair sanctions analysis should therefore ask who carries the spillover and whether they receive support, review, and lawful channels.

A review should track when lawful channels recover, because persistence after access normalizes can show that the shadow route has acquired its own constituency.

26.2 — Organized Crime and State Opacity

Organized crime can become governance under threat when a network supplies protection, credit, employment, transport, dispute resolution, punishment, and access where public authority is weak, absent, complicit, or feared. People may depend on the same network that exploits them. Anti-capture therefore has to replace functions as well as remove actors; otherwise the vacuum is filled by the next organization with money, information, and force (Gambetta, 1993; Varese, 2011). State opacity allows a protected market to become an informal architecture of public power. When enforcement can be rented, contracts can be rigged, ownership concealed, proceeds washed, land acquired through opaque structures, and objection silenced by fear, the market is no longer marginal. It governs access, price, work, and safety while the formal state retains the visible name (FATF, 2026; Varese, 2011; Rose-Ackerman and Palifka, 2016).

26.2.1 — Protection Networks

A protection network is the hidden infrastructure that lowers the operating risk of an illicit market. It may include corrupt officials, armed groups, business fronts, local brokers, lawyers, accountants, landlords, transport operators, customs contacts, police contacts, political patrons, or community enforcers. The public sees the market; the network sees the permissions behind the market.

Protection can be violent, bureaucratic, reputational, or financial. A payment may buy relief from harassment, avoided inspection, a shaped bid or political protection. Honest competitors, public funds, frightened residents and exploited workers carry the cost (Gambetta, 1993; Varese, 2011).

Protection networks gain power from fragmented agencies, weak records and dangerous complaint routes. Dismantling them

therefore requires community protection, witness safety, reliable services and integrity inside public institutions, so lawful security remains available after a raid.

Capture does not require every official to be corrupt. A few compromised nodes can distort a whole system if information, discretion, and fear travel through them. Control of a border post, procurement office, licensing desk, police unit, inspection team, tax office, or local court may be enough. Honest officials may then carry increased danger, moral injury, and isolation because the institution around them no longer reliably protects truth.

26.2.2 — Money Laundering and Real Estate

Money laundering in real estate converts hidden proceeds into durable control over land, housing, rents, construction, and political relationships. The risk is not limited to spectacular luxury purchases. Shell ownership, professional enablers, cash-intensive transactions, opaque trusts, manipulated values, and captured development decisions can reshape ordinary housing markets. Beneficial-control review should target high-risk structures and professional routes while protecting legitimate household privacy and avoiding a public dossier of ordinary buyers and tenants (Levi and Reuter, 2006; van der Does de Willebois et al., 2011; FATF, 2026).

Laundered property wealth can raise prices, distort development, displace residents, reward vacant speculation and influence municipal planning. Follow those effects into rents and neighborhood stability (FATF, 2026; van der Does de Willebois et al., 2011).

26.2.3 — Political Influence and Fear

Organized crime can narrow public choice without formally taking office. Influence may occur through donations, threats, patronage, local employment, media intimidation, judicial

pressure, control of public contracts, capture of unions or associations, financing of campaigns, or informal rule over territories. The public may still vote, attend meetings, and read laws, but meaningful choice is narrowed by fear and hidden money.

Fear reallocates daily life. It changes where people work, shop, speak, gather, report, organize, worship, testify, and raise children; it changes whether journalists investigate, officials resist, judges act, witnesses appear, and businesses refuse extortion. In captured systems, fear is not background atmosphere. It is a material institution that raises the price of lawful refusal and directs opportunity toward hidden power (Rose-Ackerman and Palifka, 2016; Varese, 2011).

Reducing that price requires survivable dissent. Witness protection, secure reporting, independent journalism, public-service integrity, safe relocation where necessary, family protection, and credible lawful security give people room to refuse without private martyrdom. Appeals to courage are inadequate when retaliation is predictable. Anti-capture capacity is measured partly by whether an ordinary person can tell the truth and continue living (Varese, 2011; Rose-Ackerman and Palifka, 2016).

Criminal political influence often hides inside respectability. A donation, development project, security service, media purchase, charity event, or local employment program can carry social approval while also buying protection. Scrutiny should follow any dependence, silence, or exemption from public review that the apparent generosity buys.

Community review matters as much as expert review. People living near shadow economies often understand coercion, dependency, fear, and informal survival with more precision than distant commentators. Field expertise should therefore include

community organizers, service providers, local journalists, harm reduction workers, migrant advocates, public defenders, and survivor advocates.

When those sources converge on a burden route, the record should identify which protective or oversight institution can act without exposing the people who supplied the evidence.

26.3 — Smuggling, Sanctions Leakage, and Parallel Trade

A parallel-trade route is a logistical and political system built from transport, storage, payment, documents, labor, protection, and legitimate commerce used as cover or carrier. The relevant questions are which demand sustains it and which chokepoints confer power. Investigators also need to identify the workers and communities carrying danger and the officials or firms supplying protection. That network account can reveal lawful channels that reduce dependence more effectively than a map of movement alone.

26.3.1 — Smuggling Routes

Smuggling often recruits ordinary infrastructure. A port, trucking company, warehouse, payment provider, freight forwarder, customs broker, or small shop may carry both lawful and illicit flows. Blanket closure can destroy legitimate livelihoods and essential supply; superficial compliance can leave the network untouched. Risk-based intervention needs transaction evidence, independent review, targeted inspection, protected reporting, and a continuity plan for lawful trade.

Public smuggling-route analysis should explain the mechanism without publishing actionable route detail. Use an abstract diagram with a clear evidence-status label, lawful alternatives and repair options. Images of raids, contraband, cash, migrants, neighborhoods or grieving families require the same dignity review as the text; they should clarify the account rather than create stigma or spectacle.

Disruption must be followed beyond the seizure. Goods may move through more dangerous paths, migrants may use riskier brokers, prices may rise, the quality of counterfeit products may deteriorate, or violence may concentrate. An enforcement action

is not the same as burden reduction. Its human and market consequences remain part of the result after the visible operation ends.

Corruption at this interface can take many forms: informal payments to move goods, selective inspection, false classification, document manipulation, protection of favored brokers, harassment of vulnerable traders, or leakage of enforcement information.

Border integrity also depends on institutional support for officials who resist corruption. When corruption is normal, the person who refuses needs colleagues, salary, supervision, complaint channels, and legal backing.

26.3.2 — Civilian Access Dilemmas

The civilian-access test is whether the lawful system can restore an essential good or status soon enough to change the person's decision to use a shadow route.

Institutions should forecast where unmet need is likely to turn when sanctions, border closure, rationing, licensing, procurement, or migration rules leave a known essential gap.

Fake medicine, contaminated fuel, unsafe food, exploitative transport, and false documents can deepen the original harm. Lawful repair therefore joins quality, price, availability, information, and protection. Access without safety is inadequate, and safety without access leaves the need unresolved (OECD and EUIPO, 2025).

26.3.3 — Financial and Shipping Intermediaries

Banks, insurers, brokers, freight forwarders, registries, flag states, ports, and payment systems are control points and possible exclusion points. Their decisions can interrupt illicit finance or freeze lawful humanitarian and commercial activity. Risk-based due diligence should therefore include usable guidance,

proportional verification, protected information, correction, and monitoring for whether charities, remittance providers, small firms, and frontier regions are excluded more readily than sophisticated clients with counsel (FATF, 2025b; FATF, 2026).

Beneficial-ownership information, suspicious-transaction reporting, trade documentation, safeguarded information sharing, sanctions guidance, auditability, and appeal all have legitimate anti-capture roles. Each also creates privacy, error, and exclusion risks. Controls should focus on high-risk structures and professional routes, preserve correction, and avoid turning nationality, poverty, migration status, or ordinary cash use into a proxy for guilt (FATF, 2021; FATF, 2026; van der Does de Willebois et al., 2011).

Before enforcement against an intermediary, identify how lawful payments, medicines, food, remittances and shipping will continue during investigation and transition. When no safe channel can be maintained, record the expected burden, mitigation, review date and evidence that could change course.

26.4 — Visible Enforcement and Anti-Capture Safeguards

Visible enforcement means that the public can inspect the authorization, priorities, evidence, data use, civilian protections, error review, and repair attached to coercive power. It does not mean publicizing sensitive operations or exposing ordinary people. The burden of visibility rises with the authority to search, seize, detain, freeze, exclude, or use force. Priorities should follow harm, power, and replaceability rather than ease of arrest. Organizers, violent actors, corrupt officials, professional enablers, procurement cartels, money launderers, and high-level protectors deserve different attention from coerced workers, survival sellers, or users of a failed lawful channel. A priority framework should state the harm targeted, evidence threshold, expected burden reduction, civilian risk, displacement risk, and lawful alternative that will remain after disruption.

26.4.1 — Enforcement Route Visibility

Operational metrics should reveal both action and effect. Arrests, seizures, inspections, accounts frozen, and cases opened show activity; they do not establish that the underlying predatory route weakened.

Read enforcement results with lawful access, safety, coercion, worker protection, corruption, price, quality, complaint patterns, and evidence that the route moved elsewhere. These observations test whether dependence and predatory control declined rather than treating activity alone as success.

Periodic review compares each tactic’s expected and observed effects, lawful access and complaint patterns. Evidence of more dangerous substitution or concentration in a rival network must be able to narrow or retire the tactic.

26.4.2 — Evidence and Due Process

Recovering illicit gains can finance restitution, but seizure does not itself prove wrongdoing. Decisions to seize must be insulated from a government's interest in the proceeds. Applicable law should protect innocent owners and tenants, provide notice and an accessible challenge, preserve records of chain of custody, and require proportionate judicial or independently authorized review. The agency recommending seizure should not be the sole institution determining either liability or disposition. Recovered assets should first be evaluated against lawful claims of victims and third parties; sale proceeds require transparent management and audit so enforcement does not develop a budgetary incentive to seize rather than to prevent harm. The burden test includes wrongful deprivation and correction time as well as value recovered.

Hidden networks, vulnerable witnesses and intense public pressure make evidence harder to examine. Those conditions call for protected testimony and competent review, not a reduction of due-process standards (United Nations, 2000; United Nations, 2003).

Protection must last longer than the press conference. Networks often retaliate after attention moves on, so the burden register extends into the following weeks and years. Residents, witnesses, small businesses, and families discover during that quieter period whether public protection was durable or merely visible.

26.4.3 — Data and Surveillance Limits

Apply the purpose and secondary-use limits in Section 21.3.3 to enforcement data. Keep customs, banking, migration, welfare, health, education, housing and policing records separate unless a narrow, lawful and reviewable need justifies a connection.

An alert system becomes more useful when reviewers understand its expected mix of signals. In an illustrative set of 10,000 records, suppose 100 contain the target condition. A tool detecting 90 percent of those cases produces 90 true alerts; a 5 percent false-positive rate among the other 9,900 produces 495 false alerts. Only about 15.4 percent of the 585 alerts therefore identify the target. These are invented inputs, not a measured model. The calculation tells reviewers what corroboration and staffing the alert queue needs. It also explains why model performance must be reported by intended use and relevant population rather than reduced to an impressive detection percentage.

Artificial intelligence can assist anomaly review, document comparison, and prioritization under accountable human judgment; it cannot establish guilt or justify coercion on its own. A model should not replace evidence, deny services without human review, infer guilt from poverty or association, or rank people by an opaque risk score (Herd and Moynihan, 2018; World Bank, 2017).

Surveillance effectiveness should be tested against less intrusive alternatives and against adaptation by sophisticated actors. Broad collection can burden ordinary people while professional enablers, shell structures, or protected networks change routes. Narrow targeting, financial and procurement evidence, beneficial-control analysis, integrity work, protected testimony, and audited access may produce better anti-capture value with less collateral exposure.

The data tool itself remains auditable. Version, purpose, training or rule basis, access history, error rate, affected groups, overrides, complaints, and retirement decisions form the minimum record. Independent reviewers need enough information to determine whether the system detects high-harm

structures or merely reproduces easy targets. When adequate evaluation is impossible, reporting “no adequate estimate of effectiveness” is more honest than expansion by assumption.

Legal authority, admissibility, warrants, detention, punishment, and formal remedy remain with the legal and governmental volumes and applicable jurisdictions. This Section retains the material questions: what the tool costs, who it exposes, which lawful access it blocks, whether it reaches control rather than visibility alone, and how error is repaired.

26.5 — Reintegration and Amnesty

A credible lawful exit can reduce recruitment, recover testimony, weaken coercive control, and restore ordinary work. It must offer a realistic alternative to the income and protection an illicit network supplied. Safety, accountability, and victim interests limit which pathways are appropriate for each participant. Reintegration should therefore be judged by whether people can leave without being forced immediately back into the same dependence.

26.5.1 — Survival Participant Pathways

A survival-participant pathway makes lawful life materially possible through documents, safety, treatment, housing, income, debt relief, childcare, transport and access to ordinary services. Assess what the participant would lose through lawful exit, including income, housing, protection, medicine, fuel, or credit, so legality does not require immediate collapse (Braithwaite, 1989; Maruna, 2001).

Surface activity can conceal captivity: a worker may be debt-bound, a child exploited, or a person in a scam compound compelled to commit the visible offense. Classification turns on actual room to refuse.

Coerced-participant protection also serves public safety. People trapped in networks often know routes, officials, payments, threats, and vulnerabilities. But information should never be purchased by abandoning the person afterward. Cooperation must be voluntary, safe, legally represented, and proportional. Otherwise the state becomes another extractor of vulnerable testimony.

Coercion often works through administrative fear. A trafficker, boss, broker, or gang may claim that police, immigration authorities, courts, creditors, or public agencies will punish the

person who seeks help. Lawful systems break that control only when protection is credible in practice: confidential access, independent advice, temporary status where authorized, safe housing, income, family protection, and rules that prevent a plea for help from becoming automatic self-incrimination or abandonment.

26.5.2 — Accountability for Predation

Apply Section 26.4's harm, power, and replaceability framework when setting investigative priorities for predatory actors and their facilitators (FATF, 2025a).

Accountability also reaches enabling institutions. A bank that ignored clear warning signs, a contractor that rigged bids, a real-estate gatekeeper that laundered proceeds, an official who sold protection, or a regulator who looked away belongs inside the causal chain (Levi and Reuter, 2006; United Nations, 2003).

Predatory accountability includes economic harm that lacks a dramatic image. Counterfeit medicine, fraudulent debt, wage theft, procurement cartels, laundering, forced labor, and exploitative brokerage can destroy health, livelihood, housing, and public capacity without visible street violence. A 2026 OECD and EUIPO study reports a robust cross-country association between labor exploitation and counterfeit-export intensity (OECD and EUIPO, 2026). Correlation does not establish a single causal route. The finding nonetheless supports connecting customs, labor protection, victim identification, and financial investigation rather than counting seizures alone.

Conditional amnesty begins with narrow, public, evidence-based eligibility. Role, coercion, harm, truthfulness, restitution, cooperation, continuing risk, and the ability to disengage safely may all matter. Exclusions for grave organized harm remain explicit. Independent review and a correction route prevent

amnesty from becoming a private bargain reserved for powerful insiders.

Victims and affected communities need a protected role in the design without being forced to decide every individual outcome or perform forgiveness. Their knowledge can reveal hidden harm, intimidation, unpaid loss, and risks that a cooperation agreement may miss. Participation should be supported, optional, trauma-informed, and separated from retaliation. The process remains public in principle while protecting people whose safety depends on confidentiality.

26.5.3 — Restitution and Repair

Restitution begins with the harmed person or community, the loss, the responsible route, what can be returned, and what requires care, compensation, service repair, memory, or prevention. The amount seized from an offender is only one possible resource and may bear little relation to the injury. A complete register preserves both recoverable property and harms that call for another form of repair.

Avoid selling recovered property into the same speculative market or routing funds through the same compromised office; asset management should reduce the chance that the laundering route recurs.

Where unsafe or predatory markets grew around inaccessible supply, credit or licensing, repair includes a workable lawful route. Section 26.6.2 develops that redesign (UNODC, 2020; Braithwaite, 1989).

Keep fear, lost schooling, delayed care, family separation, cultural loss, stigma and institutional distrust visible in the restitution register (FATF, 2025a; UNODC, 2020).

Restitution remains specific enough to test. Procurement crime may require rebuilding the missing service; wage theft may

require back pay and labor protection. Repair funds need independent audit, conflict controls, affected-community input, and safeguards against purchasing loyalty (FATF, 2025a; United Nations, 2003).

26.6 — Transparent Revision After Failure

A failed intervention can change the illicit market it intended to disrupt. New routes, substitute actors, and displaced risks may emerge even when enforcement statistics initially improve. Revision therefore needs evidence about both the original abuse and the effects of the remedy, including burdens transferred to lawful users. A market redesign should specify what failure would trigger reconsideration rather than treating the first plan as irreversible. The two subsections distinguish diagnosis from the authority to change course.

26.6.1 — Failure Diagnosis

Failure diagnosis reconstructs both the illicit market and the response to it. It identifies the lawful path that failed, the demand or need that remained, the actor who gained control, the people who lost options, the official or professional interface that reduced risk, and the harm created by both the market and the response. It also asks which public metric made activity look like success while dependence, fear, or unsafe substitution continued (Reason, 1990; Vaughan, 1996).

Use the protected reporting and witness safeguards in Section 26.2.3 to obtain evidence of coercion, bribery, inaccessible services and reliance on informal routes that formal records may miss (World Bank, 2017; Schneider and Enste, 2000).

26.6.2 — Market Redesign

Market redesign requires a lawful alternative that addresses the diagnosed access failure, reduces dependence on the shadow route, and makes predatory control less profitable. Cost its staffing, enforcement, and effects on workers and users.

Market redesign is not automatic liberalization. Some markets should remain prohibited because the goods or services are

inherently coercive, dangerous, or destructive. Even then, redesign may be needed around prevention, treatment, victim support, inspection, financial tracing, and lawful alternatives (ILO, 2018; Schneider and Enste, 2000).

Where harmful demand persists, redesign may include care. Addiction treatment, debt counseling, safe reporting, labor inspection, youth support, mental health care, violence prevention, and social stabilization can reduce the pressure that feeds hidden markets. Demand is not only consumer preference. It can be pain, dependency, fear, or narrowed choice. Material peace reduces the conditions that make predatory supply profitable.

Hidden exchange adapts, official data are incomplete, and some lawful restrictions will continue to sustain demand for shadow routes.

Part VIII — Material Peace and Buildability

Part VIII gathers the book's earlier material arguments into a question readers can picture: what becomes possible when ordinary systems keep their promises? Corruption, debt, queues, work, housing, care, markets, trade, climate, migration, diplomacy, and shadow economies now converge on the texture of a day no longer organized around preventable emergency.

Hope becomes credible when it carries invoices. A peaceful horizon names who builds, maintains, funds, and repairs it; who pays during transition; who may be displaced by improvement; and who has standing to contest an unfair allocation. Capability and human flourishing depend on resources and on the real freedom to use, refuse, and revise them (Sen, 1999; Nussbaum, 2011).

Enoughness comes first because a peace economy is judged by the life it releases, not only by the damage it avoids. Calmer time, longer horizons, reliable essentials, usable places, protected attention, trusted childhood, and ordinary beauty are material achievements in their own right.

Buildability follows enoughness because the ordinary good has to survive contact with labor, finance, procurement, regional dependence, resistance, maintenance, and public trust. Sequence, pilots, pause, and rollback belong to The Transition of Fairness. This volume identifies the material conditions that any sequence must carry, including full cost, workforce duties, maintenance capacity, uncertainty, refusal, and financed exit (ISO, 2017; GAO, 2020).

The burden lens also turns inward toward the Law of Fairness ecosystem. Interfaces, archives, public artifacts, moderation, compute, research, legal review, accessibility, staff, volunteers, donors, events, security, correction, and shutdown all consume money, time, skill, authority, and care. Trust begins with showing those invoices and the people who carry them instead of allowing

moral language to conceal them (Pfeffer and Salancik, 1978; Ebrahim, 2003).

Consider a repair crew handing a restored public space back to its users. Reopening can mark real progress even while unpaid hours, missing equipment, or an unfunded maintenance task remain out of view. Following that repair from the user's benefit through the crew's working conditions to the institution's continuing obligations shows what a claim of success must include.

Beauty gives accounting a destination, and accounting gives beauty an honest foundation. Public imagination with no cost discipline can hide labor and displacement. Cost discipline with no beauty can make a humane future feel unworthy of effort. A credible plan should show both the life it could make possible and the work required to sustain it.

Three chapters complete the buildability test. Chapter 27 makes the destination tangible in ordinary life. Chapter 28 asks whether institutions can finance, construct, maintain, correct, and leave it without coercion or self-exemption. Chapter 29 costs the project's own institutional ecosystem. The final red-team can then test whether those recorded obligations are credible and enforceable.

Chapter 27 — The Ordinary Good of Material Peace

Peace becomes materially credible when the analysis reaches the life that repair is meant to protect. The earlier chapters traced how avoidable burdens reach households and communities; this chapter asks how an ordinary day changes when institutions carry burdens within their capacity.

Chapter 2's enoughness standard becomes tangible here in an ordinary day with usable time, reliable essentials and room for different aspirations (Shah et al., 2012; Mani et al., 2013).

Repeated institutional failure can make short planning horizons an adaptation, not evidence of deficient discipline, financial literacy or culture (Haushofer and Fehr, 2014; Mullainathan and Shafir, 2013).

Public beauty makes care visible in the places people use each day. Section 27.3 asks whether those places remain usable and welcoming as needs and conditions change (Klinenberg, 2018; Gehl, 2010).

Improvement is public when the people who endured neglect can remain and belong after repair arrives. Affordability, accessibility, local authorship, rent pressure, and maintenance therefore sit inside the beauty question. Repair deepens home only when it does not prepare displacement.

Section 27.5 asks what attention returns when institutions stop demanding unnecessary vigilance. An ordinary day is also a way to test the limits of a success story. Imagine a repaired service that restores an hour to one household. The hour might become conversation, sleep, a postponed errand, or nothing that an evaluator can readily observe. Its value need not be exhausted by the next measurable activity. The repair has changed the person's range of possible evenings, and choosing among them remains theirs. This makes the chapter's perspective different from an account of organizational output. It asks what becomes possible once a system has done its work, including the possibility of

leaving public purposes behind for a while. Timing still matters: an hour available while a friend can visit differs from an hour returned after the visit must end. The calendar contains opportunities that a total of saved minutes alone cannot describe. Sections 27.6 and 27.7 carry this inquiry into childhood and local belonging, where reliable institutions become part of what people expect from one another.

Picture an ordinary day after these systems begin carrying their share. A child eats before school without anyone calculating whether breakfast means a late bill. A worker reaches a job on transit that arrives, returns with enough time to help a parent, and does not spend the evening decoding a threatening notice from an office that already holds the needed record.

On the same ordinary day, a clinic answers, a park remains safe without becoming exclusionary, a sidewalk is repaired before someone falls, a public office explains a decision in usable language, and a neighborhood welcomes newcomers without hiding the cost. Nothing miraculous has occurred. Workers, budgets, records, maintenance, and correction have carried burden in the right places often enough for daily life to become less defensive.

27.1 — Calmer Time and Longer Planning Horizons

Emergency thinking is rational when essential risks reach households faster than available buffers can absorb them (Shah et al., 2012; Mani et al., 2013). An audit should distinguish failures requiring legitimate vigilance from burdens that a higher-level institution could carry. Track unplanned changes involving rent, health, food, transport, paperwork, safety, or work, along with crisis frequency, response time, available help, and recovery costs across populations. Restored time appears in fewer deadline-driven borrowings, rushed moves, or purchases made to replace failed service. It also makes civic deliberation possible before manufactured urgency narrows options and turns fear into policy (Sen, 1999; Nussbaum, 2011).

27.1.1 — Institutional Time Returned

Count the time institutions shift to people through repeated proof, status checks and opaque handoffs. Lawful record retention, coordinated decisions, usable explanations and prompt error repair can return that time (Herd and Moynihan, 2018; Moynihan et al., 2015).

An illustrative time account makes the distinction concrete. Suppose 1,000 applicants each spend four hours completing a service request. A redesign reduces that average to one hour while requiring 400 additional staff hours. Applicants regain 3,000 hours; after counting the added staff time, the measured reduction is 2,600 hours. These are hypothetical figures, not an observed result. A real evaluation would also test errors, abandoned applications, accessibility, privacy, and whether another office inherited work.

The applicant time and the staff time should remain separately visible because they fall on different people. Neither number is automatically a cash saving. The human gain is the room returned

to care, rest, learning, and ordinary life; the institutional test is whether that gain survives complete accounting.

27.1.2 — The Value of Being Able to Return

A longer horizon changes which undertakings can begin. Some activities need repeated participation before their value appears. A neighborhood group restoring a garden needs to know that water will be available next season. A person learning an instrument needs occasions for practice across months. A workshop teaching a demanding skill needs confidence that instruction and equipment will still be there when the learner returns.

The relevant gain is continuity across attempts. A single free afternoon may permit a beginning. Dependable arrangements make returning possible. Repeated participation allows people to use what they learned last time instead of spending each meeting reconstructing the conditions for work. The material system supports a cumulative process whose value cannot be inferred from one isolated visit.

This gives a time account another useful question: which recurring commitment became feasible? The answer can remain in ordinary language. A group met through a season. A learner completed a sequence. A family kept a regular visit. Such descriptions identify what the returned time carried without assigning a price to the relationship or asking the institution to decide which pursuit matters most.

Longer planning horizons are therefore visible in promises people can reasonably make to one another. The promise may be small, but keeping it repeatedly changes what the next promise can contain. Peace acquires depth through that accumulation of dependable occasions.

27.2 — Security as Shared Reliability

Households purchase private substitutes when shared systems fail, although some backups are prudent and others reveal unequal access (Foster and Steinbuks, 2009; Carroll, 1997). An audit should identify which failure each substitute addresses and who cannot afford it. Reliability must be tested at the moments of greatest need, such as heat, illness, peak demand, or predictable seasonal stress, not only through annual averages. Shared continuity also makes long commitments safer, including training, caregiving, local investment, and public projects. The benefit is a reduction in catastrophic dependence on private rescue rather than a promise that no disruption will occur (Tyler, 2006; OECD, 2017a).

27.2.1 — Institutional Suspicion and Household Risk

Institutional suspicion often forms through repeated contact with opacity, unequal treatment, unexplained delay, insider advantage, or promises that fail at the office door. Suspicion can outrun evidence, but it is not irrational merely because officials dislike it. Reliable service, usable reasons, correction, and fair treatment must replace the experiences that taught people to expect concealment.

Frontline staff should not have to defend decisions the institution refuses to explain. Give them usable reasons, correction routes and escalation authority.

Cross-national trust evidence reinforces the importance of responsiveness and fair process, but no survey can substitute for local histories of harm or protection (OECD, 2026d).

Motive-guessing also corrodes trust between neighbors. When institutions fail to explain, people are left to guess who received help, who cut the line, who was favored, who lied, and who belongs. Public opacity becomes social suspicion; usable

explanation helps prevent hidden power from poisoning relationships among the governed.

27.2.2 — Confidence Without Knowing a Neighbor’s Private Life

Shared reliability can also protect relationships from unnecessary investigation. Consider a housing association that allocates a limited number of repair visits each week. Residents can be told the priority criteria and their own expected next step without being given another household’s private circumstances. A clear explanation lets a resident ask whether the rule was applied correctly instead of having to guess which neighbor had influence.

The explanation has to be useful at the point of disagreement. It should identify the relevant feature of the person’s own case and the route for correcting it. A resident who reports that a leak now affects electrical equipment has supplied a possible reason to reassess urgency. Staff can respond to that changed condition rather than defending an old position because it has already been entered in the schedule.

Over time, this creates a different form of confidence. People need not agree that their own request should always come first. They can rely on a process in which relevant new information can change the answer. The institution carries the burden of explaining its decision, and neighbors are less often required to explain themselves to one another.

That division of responsibility is a material benefit of a trustworthy public route. It keeps a service dispute attached to the organization able to resolve it while leaving ordinary social relationships more room to remain ordinary. Fair process supports cooperation partly by reducing the occasions on which people must compete through speculation about someone else’s life.

27.3 — Repair, Maintenance, and Public Beauty

Ordinary places reveal what a society expects people to endure. Broken sidewalks, leaking schools, dark transit stops, unsafe parks, inaccessible offices, unreliable lifts, and neglected libraries communicate abandonment more clearly than official slogans. Repair is public speech made physical, and its credibility rests on the maintained systems beneath the visible surface. Maintenance, social infrastructure and public-space design support use, belonging and health through different mechanisms. Assess whether improvements remain usable by the people whose daily life gives the place meaning (ISO, 2017; Klinenberg, 2018; Twohig-Bennett and Jones, 2018).

27.3.1 — Repair Before Monument

Apply the repair-before-prestige rule in Section 6.7 and count lifecycle duties before approving a monument (ISO, 2017; ISO, 2024a).

The people who walk, clean, operate, repair, and depend on a place often encounter failure before a distant planner can see it. They know where water pools, where an elder cannot cross, where a child feels unsafe, which lift fails, which route is avoided after dark, and where beauty already belongs. Use that knowledge as design evidence.

Local authorship also protects communities from consultant extraction. Residents are often asked to narrate problems without compensation while outside firms convert their knowledge into reports, renderings, and contracts. Fair practice pays expertise wherever it lives, records contributors, returns findings in usable form, and explains which recommendations were accepted or refused.

A public maintenance register makes routine care legible without turning workers into performers. Residents can see what

is inspected, scheduled, delayed, repaired, funded, and still backlogged. Workers need enough authority to report latent failure without punishment.

27.3.2 — Beauty, Belonging, and Public Life

Public beauty requires continued access for longtime residents, disabled users and children, continuity for local businesses, regular upkeep, local authorship and tenant protection before nearby values rise.

A public-space audit should observe use across time rather than inspect features only at opening. Heat, rain, darkness, crowding, broken lifts, closed toilets, missing signs, security practice, and maintenance cycles can change who can remain from one hour or season to another. Repeated observation with disabled users, caregivers, elders, children, workers, and nearby residents turns design intention into evidence about actual usability.

Beauty becomes belonging when ordinary people encounter care in the physical world without needing wealth, status, purchase, or special access. Trees, light, color, cleanliness, quiet, art, shade, safety, music, play, and welcome are material because they shape who can remain and how a place supports health and connection (Mitchell and Popham, 2008; Twohig-Bennett and Jones, 2018).

Experimental evidence complements the observational work. South et al. (2018) randomized 110 clusters of vacant lots in Philadelphia to greening, trash cleanup, or no intervention and followed nearby adults. Relative to no intervention, greening reduced reported feelings of depression and worthlessness; the broader poor-mental-health measure did not reach conventional statistical significance. The treatment included maintenance as well as physical change, so it does not isolate beauty from upkeep. The practical opening is substantial: test which sustained changes

to neglected places improve residents' lives, while following access and displacement separately.

27.4 — Enoughness in Housing, Food, Work, and Status

Because these systems interact, repair should identify the binding constraint instead of assuming every shortage has the same remedy. A household may need income, or it may be trapped by transport, care hours, benefit timing, housing location, or a debt obligation that makes nominal resources unusable. Enoughness improves when the binding constraint is removed without creating a new one elsewhere.

27.4.1 — Essentials That Fit Together

Housing stability includes tenure, repair rights and usable access; eviction threat, unsafe conditions or broken transport can make a roof insufficient. Dependable nourishment means a household can count on usable, nutritious and culturally appropriate food without hunger organizing the day or help requiring a public display of deprivation (WFP, 2025; Sen, 1999). Work with margin leaves time for life beyond the job and enough stability to use it. Sustainable care funds the schedules, transport, coordination, respite and income support that prevent another necessary task from becoming hidden exhaustion.

An arrangement can contain every required service and still leave a gap between them. Suppose a home-care visit ends at four and a family member can return at five. Increasing either person's competence does not cover that hour. The missing element is a compatible handover. A small change to timing may therefore make an otherwise well-resourced arrangement usable.

The best place to discover such a gap is the ordinary day it is meant to support. The household can describe when each commitment begins and ends. Providers can then identify which timing is fixed and which can be adjusted. This joins distinct

services around the life they are intended to make possible, while preserving the responsibilities of the people providing each one.

A successful fit is an achievement in its own right. No new building may be required, and the total hours supplied may remain unchanged. What has improved is the connection between them. Enoughness includes that relational work because isolated resources become usable life only when their interfaces hold.

27.4.2 — Margin for Different Lives

Shared room appears when a household can repair the car without missing rent, refuse unsafe hours without losing food, or take a child to care without risking dismissal. Different lives remain possible because each has usable margin (Mullainathan and Shafir, 2013; Shah et al., 2012). Section 1.4.1 examines the status pressures that can make this margin harder to sustain.

27.5 — The Hidden Tax on Attention Begins to Fall

Assess the attention returned when institutions remove forced vigilance and avoidable demands (Mani et al., 2013; Moynihan et al., 2015; World Bank, 2018). The time-accounting method in Section 27.1.1 separates the hours recovered from the residual cognitive load. Chapter 9's proof-budget approach calls for deleting unnecessary demands before digitizing them. Necessary forms must remain accessible, explainable, and correctable so the burden does not simply migrate to another channel.

27.5.1 — Less Crisis Monitoring

People need periods when vigilance can safely relax because institutions, reserves and trusted communication carry more of the detection burden (Haushofer and Fehr, 2014; Mullainathan and Shafir, 2013).

The operational question is how much monitoring a household must perform to remain safe. Compare portal checks, repeat notices, hours chasing status, missed-notice penalties, emergency calls, and unresolved complaints before and after redesign, then record which institution now performs the work. The object of review is restored system reliability, not a vigilance score attached to the person who had reason to worry.

27.5.2 — More Attention for Relationships and Learning

Unpredictable work, unpaid care and avoidable administrative demands can weaken the relationships households depend on during crisis (Addati et al., 2018; ILO, 2022).

Learning also protects democracy and labor. A person who can learn new skills, understand public decisions, read history, practice art, or follow civic debate is less trapped by the existing menu. Hidden burden narrows not only current comfort but future adaptability.

Attention must remain under the person’s authority. Because attention has public consequences, institutions and platforms will be tempted to manage it from above. The fair objective is narrower: reduce forced vigilance, manipulation, and institutional waste, then leave people freer to choose how they use their attention without having to justify those choices to a public system.

27.6 — Childhood With Institutional Trust

Children remain persons under protection, not proof objects for the Law, recruitment symbols, or exhibits in civilizational branding. This chapter can name material conditions children should no longer inherit, while safeguarding detail returns to the legal, clinical, educational, and ordinary-life domains whenever specialist authority is required. A child experiences institutions before understanding their theories. School meals, reliable teaching, protected play, and accessible care influence participation, development, and whether the future feels open. The evidence does not authorize one universal childhood design, but it does establish that nutrition, learning conditions, play, and mental health are not peripheral to generational stability (WFP, 2025; World Bank, 2018; Yogman et al., 2018; UNICEF, 2021).

27.6.1 — School With Reliable Meals and Care

Hunger and household emergency can consume the attention and energy children need to learn.

The scale of school-meal systems shows why reliable meals are infrastructure rather than episodic charity. The World Food Programme reports that at least 466 million children receive school meals globally, that 107 countries have a school-meals policy, and that annual investment reached roughly US$84 billion in 2024. Those totals do not establish adequate coverage, nutrition, procurement, or access everywhere, but they demonstrate that meals can be nationally owned systems whose quality, labor, supply chains, and reach are open to public improvement (WFP, 2025).

27.6.2 — Youth With an Open Future

An open-future test asks whether a young adult can still change course after a setback. Can they leave a bad job, change training,

move for opportunity, seek care, form a household, or pause education without one decision triggering default or years of penalty? Debt policy protects future room when ordinary course corrections remain survivable rather than becoming irreversible financial events (Sen, 1999; Nussbaum, 2011).

Students and families should be able to follow a complaint about unsafe or inaccessible services through to the resulting correction.

27.6.3 — Play and Beauty Protected

Play and beauty belong to childhood's rightful abundance. Children need time that is not optimized, spaces that are safe and accessible, beauty that is shared, and play that is not repeatedly interrupted by adult panic or institutional failure. Serious policy protects this freedom instead of treating it as whatever remains after serious work.

Protecting play requires material design. Safe streets, accessible parks, caregiver time, clean air, stable housing, cultural belonging, and freedom from violence determine whether children can test imagination, body, relationship, risk, and place (Yogman et al., 2018; UNICEF, 2021).

Intergenerational calm is truthful rather than evasive. Children should learn about hardship, climate risk, conflict, and injustice from institutions willing to repair what they can and explain what remains. Calm is not denial. It is the absence of needless dread added by preventable public failure.

27.7 — Local Life Inside a Wider Peace Order

Local belonging must not become moral enclosure. Migration, trade, climate shock, refugee duty, supply chains, and shared ecological risk mean every place is connected to burdens beyond its borders. A fair local life carries its own place with care while refusing to pretend that only insiders matter. Local life and wider interdependence are not opposites. Research on commons, coproduction, neighborhoods, and social infrastructure shows that durable cooperation depends on institutions people can recognize, relationships they can use, and rules that fit the scale of the problem. The peace task is to connect those local capacities to wider systems without erasing local authorship or exporting cost beyond sight (Ostrom, 1990; Ostrom, 1996; Sampson, 2012; Klinenberg, 2018).

27.7.1 — Local Trust, Production, and Care

Local trust needs recognizable institutions and people who can be reached when systems fail (Ostrom, 1996; Small, 2009).

Local production and care provide resilience without promising isolation. Nearby food networks, repair skills, health access, elder support, childcare, trades, energy systems, and civic knowledge reduce fragility when distant systems fail. Localism becomes unfair when it romanticizes poverty, excludes newcomers, or denies global dependence; the proper measure is whether local capacity widens security while remaining open to reciprocal exchange.

A successful local repair can become a reference case for other communities when the record shows the conditions, costs, maintenance, and local choices that made it work. The transferable value lies in the lesson, not in copying the place itself (Putnam, 2000; Klinenberg, 2018).

27.7.2 — Global Interdependence Without Erasure

Apply Chapter 18's authority and parity tests when distant standards enter local life. Polycentric governance supports coordination across levels without making one center the author of every place (Ostrom, 1990; OECD, 2024c).

Regional cooperation becomes clearer when responsibility is assigned at the scale where it can actually be carried. Local institutions can govern place-specific adaptation, while regional and higher-level bodies coordinate shared watersheds, grids, evacuation routes, finance, and technical capacity that no single jurisdiction can provide alone (IPCC, 2023).

Protection, adaptation and relocation plans should remain revisable as hazards, infrastructure, technology, and population change. Funding rules should not force a community to declare permanent protection, adaptation, or relocation simply to qualify for support when the safest path may change over time (IPCC, 2023).

27.8 — What Ordinary Abundance Actually Looks Like

Housing, nutrition, care, maintained assets, public trust, and contingency capacity cannot fully substitute for one another. Together they can reduce emergencies that consume time, health, attention, and future choice (WHO, 2018; WFP, 2025; OECD, 2026d; ISO, 2024a). Ordinary abundance is therefore a pattern of reliable access rather than a claim that every household will feel happy. The next examples test how everyday systems make that access visible at the point of use.

27.8.1 — Enough Repair Capacity

A practical repair test starts at the moment a neighborhood water outage is reported. The authority must know who can isolate the fault, who pays for safe temporary supply, how clinics and disabled residents receive assistance, whether private renters can obtain remedies, and when permanent service will be restored. A published repair time that starts only after a contractor accepts the job hides the hours residents have already lost. An accountable service measures time from first credible report to restored use, separates temporary supply from completed repair, and reports the distribution of delay.

Beauty and rest need no productivity justification. Their shared availability still depends on paid care, creative work, cleaning, building and maintenance (Gehl, 2010; Klinenberg, 2018; Twohig-Bennett and Jones, 2018).

27.8.2 — Room for an Unplanned Good

Ordinary abundance is also recognizable when access can be used without a special occasion. A public square with comfortable places to sit allows a person to pause without buying anything. A library with usable hours allows curiosity to continue after the

question first arises. A well-maintained route lets someone make a modest journey without first organizing a private rescue plan.

The value of these places includes the opportunity to be present without a transaction. A person can watch the afternoon change, read something unexpected, or meet someone without having to justify the visit through an immediate purchase. The supporting work remains real. Someone opens the building and cares for the space. Dependable provision makes the unplanned encounter possible.

A useful design question is how much preparation an ordinary visit requires. Clear opening information, a reachable entrance, and basic amenities can turn a nominally public place into a place people actually use. The result is not identical use by everyone. It is a wider range of people able to decide for themselves whether to come, stay, return, or leave.

This is a visible form of prosperity that reaches beyond emergency relief. A community has made room for experiences whose value need not be known before they occur. Peace supports discovery at this scale too: an ordinary day can contain something that was neither demanded by survival nor planned as an investment.

Chapter 28 — What It Takes to Build and Maintain

A peace architecture earns credibility only when it can be built. Diagnosis, enoughness, and worker dignity become actionable only when the proposal understands the incentives, capacities, contracts, routines, and regional dependencies that hold the old order together. Chapter 28 follows material peace from moral attraction into accountable construction.

Buildability is revealed wherever burden can be transferred. Reform can move burden onto underpaid staff, unpaid volunteers, households, local governments, host communities, subcontractors, or future maintainers while the central design still appears clean. The chapter follows implementation incidence through four questions: who gains, who pays first, who can refuse, and what looks successful only because exclusion sits outside the frame. The diagnosis should also distinguish obsolete or underfunded arrangements from those sustained by exploitation or habit.

Conversion centers workers and regions while distinguishing them from the systems that employ them. Defense, fossil energy, legacy bureaucracy, compliance, security, debt collection, and platform moderation all contain people who carry families, tax bases, and skill traditions through the work available. A fair transition finances the bridge before it closes the road and treats worker knowledge as implementation evidence (ILO, 2015; Causa et al., 2024).

People responsible for delivery must be able to challenge an infeasible timetable even after legislators, funders or boards endorse the proposal (Pressman and Wildavsky, 1973; Damschroder et al., 2022).

Maintenance enters the design before launch because early success can hide a future operating deficit. A pilot may rely on unusually committed people, temporary grants, borrowed facilities, forgiving vendors, founder labor, or exceptions that will

not survive routine operation. Those supports are part of the intervention. A design sustained only by exceptional subsidy demonstrates a protected experiment rather than ordinary buildability (Proctor et al., 2011; Fixsen et al., 2005).

A readiness record should report changing estimates, failed assumptions, and delayed savings, together with traceable corrections and repairs (van der Bles et al., 2020; Argyris and Schön, 1978). It should also show what another team must learn to reproduce the demonstrated service. If a crucial judgment still lives only in the founder’s head, the pilot has not yet shown that ordinary staffing can carry it. Following that dependency turns a general claim of readiness into a specific question about what has, and has not, become transferable.

Coalitions gain credibility through early results people can inspect: a queue that shortens without exhausting staff, a worker bridge paid before closure, a funded backlog, a usable form, or an error acknowledged and repaired. Credibility depends on burdens reduced and honestly reported.

28.1 — Why the Present Order Resists Change

A response matrix gives each form of resistance the intervention it actually requires. Predatory gain calls for constraint; dependent livelihood for a funded bridge; legitimate risk for bounded protection; professional caution for evidence; prestige defense for a different honor structure; missing capability for investment. This differentiation protects people without romanticizing power. Apply Chapter 19's protected-interest, authority, record, review-clock, and later-oversight requirements when a buildability proposal depends on sensitive information. Lawful secrecy doctrine, rights, remedies and institutional procedures require legal and administrative authority; material analysis identifies the resulting costs and burden shifts (Pierson, 2000; North, 1990).

28.1.1 — Money Incentives

Money incentives must be described where decisions are made. A ministry may announce reform while contractors bill through complexity; a firm may praise simplicity while revenue depends on premium access; a region may support repair while payroll and public revenue depend on the spending being reduced. Buildability follows the network of invoices, jobs, commissions, tax bases, contracts, and maintenance obligations rather than treating money as an abstraction.

The sharper material question is which earnings would remain legitimate after hidden burden is removed. A fair system still pays engineers, nurses, drivers, auditors, teachers, builders, translators, cybersecurity staff, and maintenance crews for useful work. It should not reward the preservation of unnecessary confusion, delay, desperation, or preventable scarcity. Distinguishing productive earning from burden-dependent earning is one of the chapter's central instruments (Stigler, 1971; North, 1990).

28.1.2 — Resistance Without Cartoon Villains

A useful resistance map asks what the actor gains, what burden they fear, what legitimate interest must be protected, what hidden advantage must be relinquished, and what conversion path exists (Lipsky, 1980; Pritchett et al., 2013).

The resistance map should also identify people who bear consequences without equal authority. Frontline workers and households may know where transition costs will appear first, so their evidence belongs in the map before commitments harden. Including that evidence early can distinguish resistance to reform from warnings that the proposed sequence would create an avoidable operational failure.

Convenience incentives are easy to miss because they are rarely announced. A confusing form can be convenient for an office that avoids hard judgment, a fragmented service for agencies that avoid shared responsibility, and a private workaround for people with status. The cost falls on those with less time, money, mobility, language access, social knowledge, or ability to complain (Pressman and Wildavsky, 1973; Lipsky, 1980).

28.2 — Verified Savings and the First Peace Dividend

Section 5.7.1's net-dividend calculation must be applied with a stated baseline and time horizon (Knight et al., 1996; Dunne et al., 2005; GAO, 2020). An apparent saving is not free cash until transition costs and continuing obligations are accounted for. Alternative uses need comparison before a surplus is publicly claimed. The first peace dividend is credible only when its burden reductions can be verified against a feasible counterfactual.

28.2.1 — Low-Risk Reinvestment

Choose early savings whose baseline, alternative explanations and burden shifts the public can inspect (GAO, 2020; Deaton and Cartwright, 2018).

Reinvestment should first reach the people and systems that carried the old burden. Staff, maintenance, and worker protection should come before branding or celebration.

Candidates for low-risk early reinvestment include maintenance backlogs, help desks, translation, accessibility, care continuity, worker transition funds, cyber hygiene and emergency reserves. Select destinations where need is established, capture risks can be checked and actual relief can be observed (ILO, 2015; Causa et al., 2024).

28.2.2 — Public Tracking of Reallocation

Use the public tracking record in Section 5.7.2 to follow avoided cost through deductions, reserves, allocation, delivery, and later review.

Before any remainder enters a public destination, each deduction needs an owner, evidence status, date, and overlap rule. Tracking then follows custody: where the amount sits, who allocates it, which obligation has claimed it, and what event

would require its reallocation or withdrawal. This keeps netting from becoming an untraceable promise.

A verified saving can support several purposes when it is divided rather than promised in full more than once. In a fictional allocation, a verified net cash saving of $100,000 could fund $60,000 of maintenance and $40,000 of debt relief. Promising the full $100,000 to each would create $200,000 of commitments against a single $100,000 source. Record the source amount, each commitment, and actual payments separately. A later revision to the saving must trigger a corresponding funding decision, rather than leave recipients carrying promises whose source has disappeared (GAO, 2020).

Keep released cash separate from additional capacity, lower risk, avoided repairs and trust, using Section 1.6.2's dimensional and overlap rules. Noncash gains do not automatically supply money for reallocation.

28.2.3 — Transition Costs Netted First

Deduct the funded worker, regional and implementation obligations in Sections 5.7 and 20.3 before claiming usable savings.

Residual risk belongs in the budget. A security system may shrink without disappearing. A public office may need old and new processes in parallel. A digital tool may need fallback support, and a climate transition may need temporary redundancy. Funding those conditions keeps the new system from becoming brittle.

Stage allocations and reserve uncommitted funds until conversion costs are observed; correct overestimated savings without withdrawing promised support from people carrying the transition.

28.3 — Conversion, Jobs, and the Non-Abandonment Rule

Displacement research shows that earnings losses can persist after a closure even when aggregate change is beneficial (Jacobson et al., 1993). The result matters for defense conversion and every other sector facing planned redesign. Section 20.3's worker and regional protections should apply before the plan is fixed, including worker authorship, funded support, and participation rights. A transition cannot count as successful if its public gain depends on leaving identifiable workers or communities without workable alternatives (ILO, 2015; Causa et al., 2024; Ansell and Gash, 2008).

28.3.1 — Worker Voice and Training in Implementation

Worker voice requires access to information, paid review time, independent representation, anti-retaliation, bargaining over conversion where lawful, appeal routes and public reasons for disagreements. Operational knowledge is implementation evidence; participation does not give any one organization a veto over the public purpose (ILO, 2015; Ansell and Gash, 2008).

Connect training to a credible regional project and procurement pipeline. Disclose uncertain demand, relocation, lost seniority and wage or bargaining risk before presenting training as security (ILO, 2015; Causa et al., 2024; OECD, 2026f).

Transition finance should also protect local credit, planning capacity, and community ownership so that conversion does not become a new route for outside extraction.

Major launch claims, public honors and donor celebrations must wait for a published worker-support and regional-continuity plan, enforceable protections and enough information for affected people to contest the plan. After launch, repeated safety reports, schedule failures, training gaps or supplier breakdowns should

trigger a documented response, a named owner and notice to workers explaining what changed or why a request was declined.

28.3.2 — Connecting Training to the First Order

Training and production need a meeting point. Consider a conversion proposal that offers experienced metalworkers a course for a new civilian product. Completing the course is useful, but a viable pathway also needs a buyer, a suitable workshop, and work scheduled when the trained people are available. The transition team can examine those commitments together before presenting the course as a destination.

A trial order can reveal what a qualification alone will not. Workers may discover that a familiar process needs a different finish or that the new product requires a test the old facility cannot yet perform. Those findings can guide a limited equipment purchase and a further period of practice. The order becomes a learning bridge between existing competence and a market that can actually use it.

Timing matters on both sides. A buyer who needs delivery before the conversion work is complete cannot provide the first dependable order. A training program that ends long before production begins may leave people waiting through another unsupported interval. A coordinated plan places recruitment, practice, equipment readiness, and the first delivery on a common calendar.

Worker knowledge improves this design because experienced people can identify which skills transfer and which assumptions are unrealistic. Their contribution is not merely assent to a plan. It changes the plan’s technical content. When that knowledge reaches procurement and training decisions early enough, conversion can preserve the depth of an existing workforce while giving it a credible new use.

28.4 — Legacy, Prestige, and Responsible Authorship

Recognition can support repair while distorting its purpose; assess whether recognized actors accept correction, relinquish control and leave others more agency (Bénabou and Tirole, 2006; Ariely et al., 2009; Pfeffer and Salancik, 1978; Reich, 2018). Prestige is a quiet subsidy. It can reward excellence and service, but it can also compensate people for standing above systems others must endure. A society may praise the donor who repairs a school more than the maintenance worker who kept it safe for twenty years, or reward crisis command more than prevention. Buildability must see these status payments because institutions defend the stories that make their leaders indispensable.

28.4.1 — Prestige Reattached to Repair

Honor verified repair and protected workers before spectacle.

Institutional authorship begins by identifying what an organization controls, what it helped create, and what it has left unrepaired. Contribution requires changed practice, funded remedy, protected criticism, and bounded future control; endorsement alone is insufficient. Continuing authority depends on accepting review and correction, not rewriting the past through sponsorship (Moore, 1995; Reich, 2018).

Institutional authorship is not a demand that existing institutions dissolve into a new brand. It is an invitation to take responsibility for what they can actually build. A university may contribute measurement literacy, a city may contribute service design, a labor body may contribute conversion knowledge, a firm may contribute technical capacity, and a public office may contribute lawful authority. Authorship is distributed according to competence and burden, not absorbed into a single central name.

Apply Section 6.5’s donor firewall and continuity tests when recognition, restricted funding or renewal rights could bend

institutional judgment (Pfeffer and Salancik, 1978; Froelich, 1999; Reich, 2018).

28.4.2 — Elite Exit From Capture

Hidden advantage often feels ordinary to those who possess it. Preferential access, private influence, insider timing, exemptions, donor proximity, and reputational shielding become background conditions. A workable exit makes those advantages visible and creates concrete routes for relinquishment without turning repair into theatrical self-humiliation.

Concrete exit routes include disclosing conflicts, relinquishing control, funding independent capacity, accepting procurement rules, protecting workers and critics from retaliation, and ending privileged channels (Pfeffer and Salancik, 1978; Stigler, 1971).

Peer expectations can also make elite exit harder. Powerful actors may defend hidden advantage not only for money but also for fear of losing status among their peers. A public culture that honors relinquished advantage can make exit less lonely. New prestige can therefore be attached to giving up hidden routes, not merely to funding visible projects.

28.5 — Coalitions, Early Wins, and Public Believability

A broad coalition needs credible early results without mistaking one successful pilot for a settled national answer. Participants may support the same project for different reasons, and the design should state those differences honestly. An early win earns confidence when affected people experience relief and the relevant public duties are funded. Evaluation must also identify the limits of what a pilot can establish before the model is expanded. Failure-ready design gives participants a way to revise or stop a program without concealing what went wrong.

28.5.1 — Early Wins With Real Relief

Coalitions across interests are strongest when they are honest about difference. A labor group may care first about wages, a city about service reliability, a firm about stable infrastructure, a donor about legacy, a household about access, an agency about capacity, and a researcher about evidence. These motives need not be identical to cooperate around burden relief.

A fair coalition must make authority and limits explicit (Damschroder et al., 2022; Proctor et al., 2011). Who decides, funds, audits, refuses, speaks for affected people, receives data, and can stop the effort? Which interests are public and which remain protected?

Early wins are easier to assess when a before-and-after comparison shows how everyday tasks changed. A parent stops making three trips for one service, a worker keeps income during conversion, a town receives repair funding before closure, or an office stops requesting documents it already has.

A before-and-after improvement can also occur without the proposed intervention. Finkelstein and colleagues (2020) randomized 800 hospitalized patients with complex needs to

Camden’s care-transition program or usual care. Readmission within 180 days was about 62.3 percent in the program group and 61.7 percent in the comparison group; the trial did not establish a reduction. Looking only at the program group’s decline after an unusually high-use period would have suggested a substantial gain. The comparison group improved similarly. For an early pilot, compare people selected under the same eligibility rule and time period, then ask which improvement exceeds the change that would otherwise have occurred. This helps direct further effort toward a mechanism that adds value.

28.5.2 — What Early Pilots Can Establish

A pilot’s setting is part of its evidence. Allcott (2015) examined 111 randomized evaluations of an energy-conservation program involving 8.6 million U.S. households. Predictions from the first ten sites overstated effects in the next 101. Early adopters and initially targeted households were unusually responsive. Random assignment within a site did not make the sites themselves representative. A replication program can use that finding constructively: record why a place volunteered, distinguish its users from the intended later population, and include ordinary delivery settings rather than only enthusiastic early partners. This tests whether a useful local result can survive the conditions under which the institution would normally operate.

Every pilot record should separate four decisions: whether the local mechanism operated, whether the benefit justified its burden, which conditions are required for transfer, and which authority may permit another use. The pilot does not by itself authorize rollout; its public value is a bounded account of what changed, for whom, at what cost and with what failure route (Deaton and Cartwright, 2018; Vivalt, 2020).

Important effects can occur outside the participant list. Egger et al. (2022) studied externally financed cash transfers to more than 10,500 poor households across 653 Kenyan villages. Randomized variation in where transfers were concentrated helped identify spillovers to nearby nonrecipients and firms. The study found wider economic gains with little price inflation and estimated a local transfer multiplier of about 2.5. That is not a national tax-financed multiplier or evidence that a program pays for itself. It demonstrates why an evaluation should follow spending and capacity into neighboring markets. A local pilot can otherwise miss both benefits reaching nonparticipants and costs displaced onto them.

Record anticipatory changes and shared staffing that could affect untreated units; then report which conclusions the design supports.

Staggered adoption raises a separate question from transferability: what did the reform change within the observed sites? Callaway and Sant’Anna (2021) provide a way to estimate effects by adoption group and time rather than compressing every change into one comparison. Appropriate never-treated or not-yet-treated units supply the counterfactual under a defended parallel-trends assumption, potentially conditional on observed characteristics. Earlier adopters need not be valid controls for later ones when effects evolve. Similar pre-reform trends are informative but do not establish the unobserved comparison.

A pilot intended for wider learning should leave a reproducible implementation packet: intervention description, staffing model, training, costs, dependencies, data definitions, accessibility and privacy requirements, adverse events, correction history, and the conditions that stopped or changed the work. Replication becomes a test of the recorded mechanism rather than an attempt to imitate the visible surface.

28.5.3 — Failure-Ready Pilots

Before launch, fund participant and worker protection, repair and specified data deletion under the pause conditions in Section 28.7.1. Feedback must be able to change the program (Proctor et al., 2011; Damschroder et al., 2022).

A stopping rule needs an interpretation rule. A missed target may reflect a weak mechanism, poor implementation, an inappropriate setting, insufficient time, inadequate measurement, or benefit limited to a narrower group. Predeclared review separates delivery failure from idea failure, preserves contrary evidence, narrows claims, and prevents a new population from inheriting unresolved uncertainty (Proctor et al., 2011; Damschroder et al., 2022).

Send the pilot's cost categories, support requirements, burden carriers, and failure flags to the material constraints packet in 28.7.1, including what was stopped, corrected, or redesigned after a serious objection.

28.6 — How Public Confidence Is Earned

Public confidence should be treated as a relationship supported by performance and correction, not permission for unrestricted expansion (Tyler, 2006; OECD, 2026d; van der Bles et al., 2020). An institution earns trust through what people can observe and challenge. Success on one project does not prove that its next proposal will work. Confidence should therefore remain revisable when evidence or experience changes.

28.6.1 — Usefulness, Uptake, and Their Limits

Practical improvements deserve recognition and further testing. They do not, by themselves, establish the Law's separate scientific claim about a latent boundary condition on unified conscious trajectories.

A practical study can estimate what an intervention changed under its stated comparison. Explaining why it worked is a further question; observed improvement alone does not identify the causal pathway. When results differ from expectations, revise the explanation and report what changed for those affected.

Public intervention claims should identify the study's question, setting, comparison, estimate, uncertainty and scope, retaining any condition necessary to interpret the finding. Record burden shifts, the decision supported and who must correct the public account (Deaton and Cartwright, 2018; Vivalt, 2020).

Uptake may reflect practical value, institutional pressure, funding, fashion or imitation. Public enthusiasm may also arise from hope, anger, belonging, novelty, celebrity, fear or relief. Investigate and replicate practical results before inferring validity from adoption or excitement. As participation grows, preserve refusal rights, independent scrutiny, donor boundaries and correction routes; enthusiasm must not silence critics or

determine expansion on its own (List, 2022; OECD, 2026d; van der Bles et al., 2020).

28.6.2 — Confidence Through Correction

Public confidence has more than one dimension. For the 2025 survey reported in 2026, the OECD gives an administrative-service satisfaction average of 68 percent among recent users and a national-government trust average of 40 percent for the wider respondent population. Both are unweighted averages of weighted country estimates, not percentages calculated by pooling all respondents. The questions and respondent bases differ; this is not a before-and-after comparison. The contrast motivates inquiry into how service experience relates to wider confidence, without by itself establishing a causal link (OECD, 2026d, Sections 1.3.1 and 3.1.2).

Use the correction record in Section 24.6 to connect a public acknowledgment with verified remedy for affected people. Publication alone does not establish that access was restored, exposure reduced or compensation delivered.

28.7 — From Buildability to Sequencing

Implementation frameworks distinguish readiness, adoption, fidelity, cost, sustainability, and adaptation because performance on one dimension cannot stand for all the others (Proctor et al., 2011; Damschroder et al., 2022; GAO, 2020). A workable project must specify what conditions are required before launch. It also needs a plan for monitoring whether those conditions survive ordinary operation. Sequencing should depend on evidence at each gate rather than a date chosen for narrative convenience.

28.7.1 — Material Constraints That Travel Forward

This volume supplies a material constraints packet rather than a transition calendar. For each proposed module, the packet names the protected floor, sustaining staff and suppliers, maintenance and host-community funding, lawful data route, residual risk, stop authority, remedy, and shutdown reserve. Transition authorities decide when and whether those conditions justify action.

The packet must be usable at a gate. Each condition receives an owner, evidence status, deadline, funding source, adverse threshold, and disposition if unmet. A missing safeguard is not converted into optimism; it delays, narrows, redesigns, or stops the module through the transition process and lawful public authority (ILO, 2015; Causa et al., 2024).

A material constraints packet should include a short interface agreement for every service that depends on several authorities. The agreement names the legally responsible provider, the funding owner, the supplier or workforce on which delivery depends, the minimum data needed for the handoff, the accepted service threshold, the alternative route during failure, and the person or office empowered to escalate an unresolved case. It distinguishes a transmitted request from an accepted obligation and a purchased asset from usable capability. Health, water,

transport, finance, and local government need not become one agency; they need compatible duties and a tested way to recognize one another’s failures. The agreement travels with the project when contractors, administrators, or funding sources change. Its purpose is to prevent the individual household from becoming the integration mechanism.

The handoff should also identify interactions among constraints. Worker transition can fail when housing is unavailable, a maintenance reserve can fail when procurement depends on one vendor, and a data safeguard can fail when staffing is too thin to operate it. Sequencing should therefore test which conditions depend on one another and which single failure could make several safeguards collapse at once.

28.7.2 — Gates, Pause, and Revision

Consider a fictional municipality deciding whether to retain two neighborhood service offices or consolidate their back-office work at a shared hub. Every figure, timetable, finding, and ruling in this case is illustrative. The example follows a proposal through independent appointment, contested evidence, appeal, revised authorization, and delivery. Its organizing principles are public reasons, relevant evidence, revision, and enforceable procedure (Daniels, 2000). The institutional arrangements below are a proposed application, not rules supplied by that source.

Authority comes first. Assume an existing legal framework in which a service director may redesign delivery within a council-approved appropriation, an independently appointed ombud commissions review, and a regional administrative appeals office may stay or set aside the director’s decisions. Its officers hold fixed terms under the regional appointment procedure, outside the project sponsor’s control. A conflicted officer is replaced from that office’s standing roster. Courts retain their ordinary

jurisdiction. The panel recommends; the director decides; the council appropriates; the appeals officer reviews; the operations manager delivers. None acquires another’s powers by endorsing the project.

The service receives 12,000 requests over 50 operating weeks annually. The existing arrangement is the recorded baseline. Option A retains both offices and repairs their workflows. Option B pools specialist processing at a hub while retaining telephone and paper access. Neither removes any of the nine existing jobs. Residents supporting A value nearby, familiar assistance. Managers supporting B expect fewer failed handoffs. Workers favor shared processing but challenge any saving obtained by transferring navigation and travel to applicants.

Before comparing estimates, the council adopts three project conditions. Ordinary assisted intake must be available within two business days; imminent filing deadlines require same-day assistance; users unable to use remote service must have an accessible journey of no more than 60 minutes round trip or a funded alternative. These are conditions for this hypothetical authorization, alongside existing eligibility, privacy, and other legal requirements. The council cannot waive a binding right by changing its project conditions. A cheaper building does not compensate for failure to provide required access.

The council funds up to $250,000 for one-time change and a three-year operating envelope of up to $1.20 million annually, with releases governed by the assumed budget law. Existing service does not depend on speculative savings. Estimates use a common price basis; price changes require an updated funding plan. Each option includes $720,000 for nine fully loaded posts at $80,000 each; $80,000 for inclusive access; $80,000 for systems and maintenance; and $80,000 for routine oversight and complaints. Premises cost $240,000 under A and $120,000 under

B. Annual totals are therefore $1.20 million and $1.08 million. B’s advertised recurring saving is $120,000.

One-time commitments remain separate. A requires $170,000: $60,000 for independent project review and appeal, $40,000 for repairs, $20,000 for paid training, and a $50,000 continuity reserve. B requires $250,000: the same review budget and reserve, plus $90,000 for conversion and $50,000 for training and up to twelve weeks of incremental parallel operation. The $60,000 review allocation comprises $18,000 for panel work, $12,000 for accessible hearings and administration, $10,000 for resident and worker advisers, $12,000 for appeals and reinspection, and $8,000 contingency. Routine oversight and project review perform different tasks. Unspent reserves are committed protection, not incurred expenditure. Current service continues from its ordinary budget during preparation; savings begin only when premises costs actually fall.

All numbered days below are business days. On day 0, the council publishes the scope, authority, conditions, and funding. Nominations open during days 1–4. On day 5, the ombud appoints three reviewers covering public finance, operations and accessibility, and administrative process, using published qualifications. The ombud records the reasons for selection and checks financial interests, recent vendor employment, and prior responsibility for the proposal. An affected person’s interest in obtaining good service is not a disqualifying conflict. Residents and workers select their own paid advisers; advisers supply evidence without claiming to represent everyone.

Appointment objections may be filed with the regional appeals office during days 6–10. It assigns an unconflicted officer who rules by day 15 and can require replacement from the qualified reserve list. Substantive review waits for unresolved appointment challenges. All three panel members participate; two votes can

adopt a recommendation, with the dissent attached. Fees are fixed independently of the recommendation. Only the ombud may remove a member for an evidenced conflict or failure of duty, with written reasons and regional appeal. The sponsor cannot remove a critic or withhold earned review fees because the findings are inconvenient.

During days 16–30, assuming appointments are settled, the panel assembles an evidence schedule with protected annexes. Later appointment resolution shifts the schedule rather than compressing response time. The record includes abandoned requests, staff observations, actual return journeys, and credible alternatives to consolidation. Vendors must supply material records required by the tender. A withheld record leaves the associated performance claim unsupported; it does not establish misconduct by itself. The panel can require further evidence or recommend that the affected component wait. Private service histories remain protected while the assumptions and decision reasons remain public.

The roster receives its own test. Seven caseworkers each supply 28 productive case hours weekly. At a mean half-hour of total specialist casework per request, including routine follow-up, their nominal capacity is 392 requests against average demand of 240. With two caseworkers absent, it falls to 280. A peak of 360 requests requires 180 case hours. The five remaining caseworkers provide 140 hours and the two trained supervisors can temporarily provide another 40. That would use every available casework hour and leaves no operating margin for variation. For the fictional peak, one trained relief worker therefore supplies 20 further funded case hours, raising nominal capacity to 200 hours against 180 hours of expected demand. The extra $800 for that week draws against the later-specified 500-hour staffing reserve; repeated peaks need a recurring staffing appropriation rather than

indefinite reserve use. A roster drill must confirm that supervision, urgent cases, breaks, and other duties remain covered. If the funded margin cannot be maintained under plausible combined absences, the director pauses consolidation rather than shifting risk onto overtime or applicants.

At the day-35 hearing, each side presents its strongest alternative and can question the other’s evidence through the chair. The director answers unresolved factual questions in writing. Two reviewers recommend B subject to the service conditions. The third prefers A because the travel evidence is incomplete. On day 45, the director issues a provisional decision for B with reasons and the dissent. Accessible notice identifies the appeal route and the ten-day filing period. No office closes before that period ends or a timely appeal is resolved.

On day 50, an affected-user association appeals without a fee. Its adviser shows that the sponsor’s journey table counted an outbound bus but ignored the last usable return. Some people unable to use remote service would face a 110-minute round trip. This is a challenge to compliance with a predeclared condition, not a demand that all inconvenience disappear. The regional officer stays closure within two days while preserving existing service and reversible preparation. The sponsor receives seven days to respond. A separately commissioned access assessor checks the journey; both sides can correct or contest the record. On day 60, the officer sets aside the authorization for a material evidentiary error and requires a revised comparison. The appeal changes the decision rather than merely adding another report.

The director proposes B*: shared processing plus two neighborhood counters, each staffed for ten hours a week, spread across business days and including an evening session. The additional annual provision costs $100,000: $40,000 for 1,000 fully loaded counter hours at $40 an hour, and $60,000 for

accessible rooms, equipment, and travel. Counter hours are additional to the nine posts. Preparation and urgent mobile assistance are assigned within the separately funded inclusive-access service; its roster must demonstrate capacity rather than count the counter hours twice. When remote service is unusable and the next counter session is too late, that service supplies an accessible visit or funded transport. The room contracts include counter setup and an option for temporary expanded access.

The local access route also requires a capacity test. Suppose, solely for this fictional scenario, that 12 percent of the 240 weekly requests need assisted local intake: about 29 visits. At a mean half-hour of counter assistance per visit, the counters provide 20 staffed hours per week and nominal room for 40 visits. An illustrative peak of 36 visits would consume 18 hours, leaving little margin; uneven daily arrivals still matter even when the weekly total fits. Counter assistance is separate from the specialist casework hours above. The access assessor must test time slots, the two-business-day standard, urgent same-day needs, accessibility, and any journey requiring a funded alternative. The inclusive-access team has its own roster and budgeted duties, rather than unlimited presumed capacity. Failure of this test prevents authorization of B* at the stated scope; an additional funded route must be provided first.

B* therefore requires $1.18 million annually and the same $250,000 one-time envelope. For a common three-year operating period after transition, A commits $3.77 million and B* $3.79 million: $170,000 plus three years at $1.20 million, against $250,000 plus three years at $1.18 million. B* requires $20,000 more. Both totals include the identical $50,000 reserve; neither treats it as already spent. The $20,000 annual operating difference is not a three-year net saving. Actual expenditure, retained reserves, and the timing of payments are reported separately. A

wider economic appraisal would also consider discounting, residual value, and noncash effects. Holding those annual amounts fixed, the two funding totals would meet after four operating years. That sensitivity makes the chosen horizon part of the public reason for the decision rather than a date selected to favor one option.

In this fictional case the access assessor verifies both revised options against the stated travel, scheduling, staffing, and urgent-assistance conditions. Neither dominates. A preserves more unscheduled local presence and uses less initial funding. B* may reduce repeat contacts through pooled specialist processing. Some residents still prefer A; their lost convenience remains visible. The director favors B* because of that expected service improvement within the authorized budget, not because all burdens have vanished. The panel’s revised report retains the minority preference for A and explains which factual objection the redesign resolved.

Before approving the trial, the council adopts a five-percentage-point improvement in first-contact resolution over the estimated repaired-A counterfactual as the minimum additional gain sought. At unchanged demand, a sustained gain of that size would produce 600 additional first-contact resolutions annually. Over three years, the extra $20,000 commitment would be about $11.11 per additional resolution. This is an undiscounted incremental funding illustration, not a measured effect or a complete cost-effectiveness estimate. It gives the public a concrete question: would the gain justify the additional commitment and reduced drop-in convenience? A shortfall or an imprecise estimate reopens review; uncertainty is not silently converted into either success or failure.

The amended proposal receives an accessible hearing and a fresh ten-day appeal period. An objector challenges the reduced

drop-in hours. A regional officer unconnected with the original panel reviews legal authority, procedural fairness, material factual error, compliance with the conditions, and whether the reasons support the choice. The sponsor responds within seven days; both parties can address new evidence. Within twenty days of filing, the officer upholds the amended authorization because funded alternatives meet the access condition and the director has explained the remaining tradeoff. The ruling does not certify a uniquely best policy. No closure occurs before this decision. Court access remains separate, and a new material fact can reopen the administrative record without making every unchanged preference an endless rehearing.

The finance officer verifies the appropriations and reserve before contracts begin. The operations manager keeps current service running during parallel operation. Payment milestones require tested paper intake, accessibility, exportable records, and training. The continuity reserve has an explicit deployment plan: $10,000 for temporary rooms, $20,000 for up to 500 additional staff hours at $40, $10,000 for restoring systems, and $10,000 for notices and record reconciliation. The existing core-service budget continues. Together with the room option, this finances a four-week replacement route while the council decides whether to restore A or fund another compliant arrangement. Quotes and a deployment drill must support the envelope before old leases are released. Any staffing reserve hours used during the surge must be restored or replaced by funded capacity before that four-week fallback is certified. A larger identified requirement must be funded first; the reserve does not authorize abandonment after four weeks.

The panel fixes outcome definitions before launch. For this trial, first-contact resolution means verified delivery by day thirty after the initial request, following only one applicant contact and

requiring no further applicant contact during that same thirty-day period. Silence or administrative closure is not evidence of delivery; correctly reasoned ineligibility decisions are reported separately rather than counted as benefits delivered. The denominator includes all requests under the same intake definition. Each reporting cohort is fixed by its intake date and is assessed only after its thirty-day delivery-and-repeat-contact window has matured. Records also track abandonment, waiting, staff effort, and nonremote access. To estimate repaired A, assume the evidence schedule contains protected, auditable records from comparable units already operating the A workflow. Their staffing, calendar periods, case mix, and outcome definitions are stated before comparison. Those units inform a counterfactual estimate, not an observed local A outcome. Matching does not by itself establish causal identification. If the comparator fails these checks, the first-contact advantage remains unresolved; the director must commission a better comparison or revise the decision without claiming that A has been tested.

The release gate requires six consecutive weeks meeting the access conditions, with no unresolved preventable loss of a filing right or material privacy failure. In week four of this illustration, a vendor defect loses eight paper-intake acknowledgments. Staff use the funded manual route, identify every affected case, restore notice, and obtain available deadline protection from the responsible authority. Correction and reinspection finish at the end of week five. The director withholds the unearned payment milestone; weeks six through eleven supply the new six-week test. The delay therefore remains inside the twelve-week parallel-operation allowance. Any forecast overrun returns to the council before new obligations are incurred.

After an assessor appointed by the ombud verifies the clean period and funded fallback, the director authorizes release of the

old full-time premises. Neighborhood counters continue. The ninety-day review clock begins with authorization to release those premises, not with the first day of parallel operation. Annual renewal remains a separate decision. Each report identifies the intake cohorts whose thirty-day outcome windows are complete; newer requests remain pending, not successful by default. Each missed access condition receives immediate correction; waiting beyond two days for more than one in ten ordinary applicants over two consecutive weeks additionally triggers system review and suspends further consolidation. Urgent harm never waits for that threshold. Renewal tests the expected first-contact improvement alongside costs and access. An unresolved benefit claim permits neither automatic expansion nor the cancellation of essential service. The director must improve capacity, narrow the change, or activate the funded alternative, with a published decision and appeal route.

The completed record makes disagreement productive. A critic exposes a missed burden; an appeal changes the design; the revised choice survives an explicit cost comparison; and a delivery defect changes the timetable. The remaining research question is sharply defined: does shared processing return enough usable time, under ordinary staffing and protected local access, to justify its additional commitment? That is a question a municipality and an independent research team can investigate together.

Chapter 29 — Counting the Project's Real Costs

Chapter 29 applies the book's lifecycle costing to the project itself, including correction, contraction and safe closure.

The chapter's purpose is analytical accountability: lifecycle obligations, shutdown reserves and uncertainty must remain visible enough to challenge (GAO, 2020; ISO, 2018).

The costing objects remain distinct. A taxonomy names categories; a scenario varies assumptions; an estimate gives a bounded quantity with uncertainty; an operating budget authorizes spending; a funding plan identifies lawful sources; and an actual-cost record reports expenditure after the fact. The chapter develops categories, scenarios, and estimate discipline. It neither authorizes spending nor solicits funds.

Those benefiting from launch must not control its cost assumptions or presentation without independent challenge (GAO, 2020; COSO, 2017). A shared resource creates a further difficulty. An archive may use staff, rooms, and equipment already supporting other work. Calling those resources free hides their competing uses; charging their entire cost again exaggerates the new project's burden. The self-account must distinguish the added obligation from existing capacity and identify what other work would be displaced by sharing it.

People cannot become the project's hidden fuel. Staff are workers rather than proof of mission; volunteer generosity has limits; donors do not acquire ownership by giving; viewers do not become participants by watching; and host communities are never scenery for an institution's public story. Future maintainers enter the account before they are hired.

Six questions organize every cost entry. What is the direct cost? Which cost could be displaced? What maintenance remains after launch attention fades? What correction or appeal route exists when the estimate is wrong? What does safe exit require? Who has standing and enough information to challenge the answer?

Accountability begins when the final question has a named challenger and a real answer.

A public cost dashboard can mislead through visual compression, resembling a donor prospectus, implementation mandate, country ranking, legitimacy score, or cross-architecture dossier. Any public version must disclose scope, omissions, uncertainty, maintainers, correction, data minimization, and record-regime separation before polish implies completeness (Pfeffer and Salancik, 1978; Reich, 2018).

Record-regime separation protects people and institutions. Research data, donor records, staff files, participation records, civic records, public-power records, archives, and artificial intelligence logs need different purposes, custodians, access rules, retention periods, correction routes, and deletion duties. Treating them as one pool for convenient analysis can turn self-costing into surveillance architecture. A staff payroll file and a research dataset should not become mutually searchable merely because the same project funds both.

29.1 — Every Project Must Count Its Own Cost

A serious cost practice combines an institutional calendar with event-triggered review. Budgets should be reopened before a public artifact launches or a vendor contract renews, and promptly when a donor restriction changes, moderation demand spikes, a security incident occurs, or a host community becomes newly exposed. Burden often moves during growth and failure rather than at tidy fiscal intervals. Mission-driven underfunding can hide essential administrative and labor costs to preserve an attractive public story. Count these supports using the scope, assumptions, lifecycle and update disciplines already established for public programs (GAO, 2020; Ebrahim, 2003; Gregory and Howard, 2009).

29.1.1 — Self-Burden Principle

An illustrative lifecycle account can expose omissions without becoming a budget. In the same arbitrary cost units and over the same project horizon, a base case totals 100: setup 20, operations 35, maintenance 15, oversight and accessibility 10, insurance and host mitigation 10, and contingency and shutdown 10. A low case of 85 assumes setup falls to 15 and operations to 25, with every other category unchanged and the same service standard still met. A high case of 130 keeps setup at 20, raises operations to 50 and maintenance to 20, and raises contingency and shutdown to 20; the other two categories remain 10 each. These are sensitivity assumptions, not evidence that either price path is achievable. Quotes, staffing requirements, service demand, and exit obligations would have to justify a real budget (GAO, 2020; ISO, 2018).

Use the role-boundary and operating-dependence tests in Section 29.7.1.

29.1.2 — Staff Wage Visibility

Count all delivery and support roles, including work less visible to participants. The account should show pay bands, benefits, hours, employment status, and whether the lowest-paid role can refuse unsafe work without losing livelihood (ILO, 2022; Kalleberg, 2011).

Contractors and freelancers need the same seriousness. A clean vendor invoice can conceal unstable hours, weak benefits, low pay, uncompensated revision, or psychological strain inside a subcontractor chain. The account should ask not only what the project paid, but who performed the work, under what terms, and whether procurement converted mission risk into someone else's insecure labor (Gregory and Howard, 2009; Addati et al., 2018).

Every wage account needs a repair route. Underpayment, misclassification, inadequate insurance, or emotional overload call for budget correction, back pay where owed, safer staffing, changed contracts, reduced scope, or pause.

29.1.3 — Host-Community Cost

Cost host-community effects before a project acquires a site or announces plans (OECD, 2024c; Jacobs, 1961). Residents who do not participate in the project still need protection from traffic, publicity, surveillance, pressure, rising costs and symbolic use (Sampson, 2012; Ansell and Gash, 2008).

Shared facilities create a useful opportunity to examine both contribution and capacity. A proposed public institution might use a nearby community hall on two evenings each week rather than construct its own meeting space. The agreement can fund the additional cleaning and caretaker time, specify access to storage, and preserve the hall's existing bookings. The host gains a contribution to a facility it already values, while the new institution avoids duplicating a suitable asset.

The terms should follow actual use. A quiet meeting and a large public event may require different support. The host needs a way to revise the arrangement when activity changes. Participants also need to know whom to contact about access, maintenance, or a scheduling conflict. A clear agreement lets shared use remain a relationship between institutions rather than a recurring favor requested from an overstretched caretaker.

This is one positive form of a host-community account. It identifies the resource being shared and funds the additional work without presuming ownership of the surrounding place. A new institution can become a dependable neighbor by leaving existing community uses stronger and making its own contribution easy to understand.

29.1.4 — Future Maintainer Burden

Apply the lifecycle duties in Chapter 12 to every project asset (ISO, 2024a; NDSA, 2026). Test unit costs at the staffing threshold where growth would require another moderator, interpreter, or support shift, rather than assuming that average costs stay constant as the project grows. A translation remains a service obligation when later corrections require updating every language edition rather than the original alone (Swanson et al., 2010; ISO, 2019).

A handover becomes usable when a successor can perform a real task with it. For an archive, that might mean locating a source file, identifying the version currently released, correcting its description, and restoring access from the preserved copy. The outgoing team can observe the attempt and discover which piece of knowledge still exists only in someone’s memory.

This test changes what documentation needs to contain. A list of accounts and filenames may locate the materials but leave the reason for a decision unexplained. A short record of why one

version superseded another can save a successor from repeating the same investigation. Conversely, a long account of institutional history may be less useful than the one instruction needed to complete a routine repair. The task supplies a standard of relevance.

Handover also has a labor cost in the present. The outgoing person needs time to explain and the incoming person needs time to practice. Treating both periods as productive work lets continuity be purchased openly. The institution is not simply paying twice for one role. It is paying for knowledge to cross between people while the service continues.

The result is a more generous inheritance. A future maintainer receives something understandable enough to improve rather than a finished-looking object whose operation depends on a departed founder. The cost belongs to the project that created the obligation, and the capability belongs to the people who will carry it next.

29.2 — Civic Place and Facility Cost Interfaces

The same governance principle should not impose the same paperwork on a village clinic and a national infrastructure fund. A low-risk local repair may need one public decision record, an accessible complaint contact, and a periodic independent sample of transactions. A high-value or high-risk facility may warrant detailed options appraisal, external engineering assurance, protected procurement disclosure, lifecycle reserves, and formal conflict review. Scale the evidentiary effort to the likely harm, irreversibility, expenditure, privacy sensitivity, and concentration of power, while keeping a minimum right to challenge and inspect lawful spending. Consolidating related safeguards into one auditable record can reduce duplicate reporting and staff exhaustion without surrendering accountability (HM Treasury, 2026b; Proctor et al., 2011).

29.2.1 — Land, Construction, and Site Cost

Land and construction accounting includes the opportunity cost of place. A site consumes more than purchase and building funds: it changes what else can occupy the land, who may be priced out, which public improvements become necessary, and whether noise, security, traffic, waste, water demand, or housing pressure move to neighbors.

The site record must also document existing uses and cultural history, ecological conditions, water and energy connections, transit, zoning, accessibility, permitting, and eventual site recovery (ISO, 2017; ISO, 2024a; ISO, 2024b).

A City-facing staff account must distinguish ceremonial roles from the support work that makes public access possible.

Staffing cost includes wages, benefits, training, supervision, scheduling, continuing education, conflict support, debriefing, backup coverage, and turnover. If a setting asks staff to carry

public scrutiny, moral conflict, or crisis contact, the budget must treat that load as occupational reality rather than personal devotion (GAO, 2020; OECD, 2015b).

29.2.2 — Records, Archives, and Public Memory Cost

Budget the record protections in Section 21.3 and preservation duties in Section 29.5.1, including translation, breach response and public explanation. Decisions about civic records and participation remain with the responsible legal and administrative authorities (ISO, 2012; NDSA, 2026).

Correction rights require people, time, and authority. If a record misstates a person's role, misclassifies a community, exposes a private detail, or turns a draft into public truth, someone must be able to challenge it without being punished for delay.

Security and insurance reveal who carries risk. Underestimating safety leaves staff and visitors exposed; overbuilding security can militarize ordinary life around the site. A fair account distinguishes safety threats from reputational concerns, identifies what insurance can cover and training can reduce, and states which risks are unacceptable. Expansion should stop when safety would require intrusive conditions.

29.2.3 — Exit and Shutdown Cost

Use Section 29.7.3's obligation-based shutdown reserve from the entry decision.

Funding a credible exit reduces pressure to preserve an institution merely because shutting it down has become financially or administratively painful. Continued operation should remain justified by continuing purpose and cost (Swanson et al., 2010; ISO, 2019).

The plan should identify which facilities, utilities, access improvements, records, or other local dependencies will remain after closure and who will lawfully own or maintain them.

Otherwise an institution can disappear while quietly transferring its residual operating burden to a municipality, landlord, community organization, or neighbor that never agreed to inherit it.

Release closure reserves against verified obligations, then publish a reconciliation identifying completed commitments and any continuing custody or funding duties.

29.3 — Community, Spiritual, and Basic-Floor Cost Interfaces

Community and optional spiritual institutions create a distinct cost field when meaning, care, housing, debt repair, belonging, and sacred language intersect. This chapter does not adjudicate doctrine or ritual. It asks what material obligations arise when a setting offers shelter, support, community, or a basic life floor. Safeguards should follow the activities and people involved rather than the institution's chosen language.

29.3.1 — Community Facility and Operating Cost

A Fairism House must fund safeguards appropriate to its activities, including food and fire safety, privacy, pastoral supervision and secular visitor dignity; sacred language does not substitute for protection.

The operating account should separate fixed property cost, variable occupancy cost, and activity-triggered safeguards. A quiet meeting, meal program, overnight shelter, youth event, counseling route, and records archive create different staffing, insurance, privacy, food-safety, accessibility, and supervision duties. Meaning makes honest activity-level costing more important, not less (Ebrahim, 2003; Froelich, 1999).

Include periodic replacement of roofs, heating, kitchens, accessibility features, equipment and records systems in the facility's funding plan (ISO, 2017; Gregory and Howard, 2009).

29.3.2 — Basic-Floor Cost Categories

Cost the basic life floor explicitly: shelter, food, utilities, hygiene, transport, communication, documents, medication access, disability accommodation, child safety where relevant, emergency routing, privacy and sufficient staff. Debt relief cost has to include administration, verification, counseling

boundaries, fraud controls, appeal, tax and legal review, privacy, and the risk of repeating old dependence through new conditions. Apply Section 6.5's donor firewall and Section 7.5's no-chain relief terms to the basic life floor (Ebrahim, 2003; Reich, 2018; OECD, 2026b).

29.3.3 — Member and Nonmember Safeguards

Protect members' privacy, pastoral boundaries, freedom to exit and freedom from internal status pressure. A complaint route must be available when basic-floor support is informally tied to affiliation, publicity, data use, testimony, volunteer labor or future participation. Nonmembers must retain services, safety, neighborhood access and civic standing when they decline religious participation.

Provide an independent review route that can detect coercion, favoritism, spiritual pressure or retaliation and is usable by people outside the community.

Imagine a community facility that offers a public meal and an optional gathering afterward. A person can come for the meal and leave when it ends. The timetable includes enough staff to serve and clear the meal without relying on attendance at the later activity. The practical freedom to decline is built into how the day is organized.

The accounts can reflect the same separation. Meal preparation and the optional gathering each receive their direct costs. A declared allocation covers the space they share. This makes it possible to support the public service without quietly financing it through an expectation that recipients join another activity.

Members also benefit from that clarity. Someone who stays does so for the gathering itself, and someone who serves the meal knows what work they have agreed to perform. A welcoming

institution becomes easier to trust when its ordinary arrangements leave people room to participate in different ways.

29.4 — Platform, Moderation, Labor, and Intellectual Property

Platforms that coordinate contribution, review, moderation, art, code, translation, research, or civic labor depend on work that must be counted from the outset, even when the visible contribution is only a post or edit. Thousands of small acts can accumulate into substantial labor, exposure, intellectual-property, and care burdens while remaining invisible in the platform's accounts. Budget for review, coordination, attribution, accessibility, technical support, and public response alongside the visible contribution (Gray and Suri, 2019; ILO, 2011). This accounting does not predesign points, roles, gates, or rollout, which require separate authority. Volunteer limits, retaliation safeguards, time-bound roles, and data-access restrictions must be established before participation becomes a test of identity (Gregory and Howard, 2009; Kalleberg, 2011; see Section 29.7.1).

29.4.1 — Moderation and Safety Cost

Moderation cost includes more than software, policy, and queue length. Moderators absorb harassment, grief, threats, disturbing content, manipulation, confusion, and public anger. Fund staffing, escalation, training, trauma-aware support, breaks, review, appeals, evidence handling and security for this work.

Ticket counts can hide exposure intensity. A queue of routine disputes is not equivalent to repeated credible threats or graphic material, and staffing models should distinguish duration, severity, recurrence, and recovery time. High-exposure work should therefore be assigned deliberately rather than treated as casual volunteer availability (Roberts, 2019; Gillespie, 2018).

Platform cost grows with reliability promises. Uptime, backups, accessibility, localization, abuse reporting, identity controls,

privacy review, audit logs, portability, and disaster recovery require people and money.

A platform needs funded arrangements for failure. Outages, breaches, vendor loss, moderator absence, or sudden public surges should trigger limits on affected functions, human fallback, clear communication, and recovery measures. If harm spreads faster than those measures can contain it, the platform has grown beyond its support.

29.4.2 — Intellectual Property and Authorship Cost

Intellectual-property burden appears when collaboration becomes unclear. Contributors need plain notice of what they keep, license, and permit the project to publish; how attribution and derivatives work; how confidential material is handled; and what withdrawal can accomplish after publication. Ambiguity turns enthusiasm into appropriation risk.

Default rights can differ with jurisdiction, employment status, commissioned work, collective authorship, and the form of the contribution. The project should therefore preserve provenance, identify the governing agreement and contributor status, and obtain specialist review before a contribution is reused beyond the purpose for which permission was given (Borgman, 2015; Ebrahim, 2003; WIPO, 2024).

Rights arrangements should be presented in comparable terms. Public reuse, limited internal use, proprietary arrangements, employment assignments, contractor terms, archive deposits, and model-training permission create different rights and risks. The account must fund legal review, contributor explanation, versioned consent, attribution, and correction of mistaken ownership claims (WIPO, 2024).

29.4.3 — Youth and Vulnerable Contributor Safeguards

Adults retain responsibility when young people participate. Minors should not carry moderation, publicity, fundraising, research, or community growth for adult convenience. Privacy, safeguarding, accessible complaints, appropriate supervision, and a real ability to leave without social punishment are part of the operating cost (UN Committee on the Rights of the Child, 2021; UNICEF, 2021).

Vulnerable contributors need stronger boundaries. Public ambition can attract people in crisis, seeking belonging, living with unstable housing or employment, or hoping contribution will create rescue. Their participation must remain bounded, supported, and independent of any promise of personal salvation (UN Committee on the Rights of the Child, 2021; W3C, 2023).

Contribution scores require defined purpose, expiry, correction, nontransfer and nonmigration rules; they must never rank human worth. Tell contributors before entry which records, credits and obligations survive withdrawal, and require a separate, explicit purpose for retaining a score after its task ends.

Before any scored or role-based participation system launches, identify who will answer disputed attribution or payment after a contributor leaves, who can correct a record, and which protections survive account closure. If those obligations are unfunded, apparent efficiency is unpaid risk. Closure procedures should include a final notice, an export or deletion choice consistent with lawful retention duties, and a named contact for later corrections. For minors, the record should state what changes at legal adulthood so consent and custody are not frozen indefinitely.

29.5 — Public Archives, Newsrooms, Compute, and Storage

Archives, newsrooms, compute infrastructure, and artificial intelligence support need separate budgets and authorities. Each carries distinct obligations around custody, review, continuity, and correction. Grouping them under one digital label can conceal dependencies that fail on different timelines. The next analysis asks what has to be maintained or staffed so that a public artifact remains available and trustworthy.

29.5.1 — Archive Survivability Cost

Archive survivability requires planned migration, redundant storage, format monitoring, fixity checks, metadata, access controls, preservation staff, and a rule for records that should not be kept. The public may remember the beautiful archive; future maintainers remember the corrupted file, obsolete format, missing consent flag, unclear rights note, or server invoice.

The 2026 NDSA preservation framework adds environmental sustainability to its preservation guidance, sharpening the cost question. More replicas, longer retention, and frequent migration can reduce one risk while increasing electricity use, storage expense, vendor dependence, and privacy exposure. Preservation plans should therefore justify copy count, location, integrity-check frequency, format strategy, and retention rather than treating unlimited accumulation as the safest default (NDSA, 2026).

The archive needs to know where copies exist, who can access them, how they are encrypted, when they are audited, and how deletion or correction travels across replicas (Conway, 2010; Borgman, 2015).

One proposed project record would give each consequential claim a stable identifier and link it to the exact source version and

passage supporting it. A study identifier would connect reports that reuse the same evidence, so several summaries would not appear to be several independent studies. An estimate would retain its outcome definition and comparison; an original proposal would be marked as a proposal. Corrections would identify which claims they supersede. A researcher or software tool could then follow a claim to its basis without rebuilding the entire argument. This structure makes evidence inspectable; the validity of the claim still depends on that evidence and the reasoning connecting it.

Reuse can be designed into the archive. In research data stewardship, FAIR means Findable, Accessible, Interoperable, and Reusable. These principles emphasize persistent identifiers, descriptive metadata, provenance, and clear conditions of use for both people and computational tools (Wilkinson et al., 2016). Access may require authorization; reuse does not require publishing private records.

29.5.2 — Newsroom Independence Cost

A newsroom within the Law of Fairness ecosystem would need a protected editorial budget and authority to publish findings unfavorable to the project. Source protection, legal review, correction, and investigation require paid time that cannot depend on producing praise. Its charter should separate donor reporting from editorial approval and specify who can enforce that separation when a founder or funder objects (UNESCO, 2025; Pickard, 2020).

A hypothetical newsroom can reserve a defined part of its annual budget for inquiries whose results are not known when the work begins. An editor authorizes the investigation and assigns the reporting time. A supporter funds the capacity to investigate

rather than purchasing a favorable conclusion or a guaranteed number of celebratory articles.

Sometimes careful work establishes that an allegation is unsupported. The newsroom still paid for a useful inquiry. Its internal record can explain what was checked and why the story was not published, while preserving confidential sources. Funding that work makes accuracy a feasible outcome instead of making publication the only way to justify the expense.

The same budget must support correction after publication. A record remains useful when a newsroom can revisit it as new evidence appears. Independence is therefore partly an allocation of time across the life of a claim: time to investigate before speaking and time to correct after others have begun relying on the account.

29.5.3 — Data-Center Energy and Water Cost

For each AI task, cost hardware and cloud use alongside evaluation, data handling, model monitoring, human review and escalation, accessibility, safety and privacy controls, and retirement.

Compute cost depends on the task. Draft assistance, search, translation, accessibility, moderation triage, archival description, data extraction, public response, and internal routing create different data, review, energy, and error profiles (Nielsen and Ganter, 2022).

Compute accounting should attribute resource demand to the service that causes it. Baseline facility load, incremental training or inference, storage growth, redundancy, cooling, network traffic, and backup power should be separated where measurement permits so one public tool is not assigned the footprint of an entire site. Shared infrastructure can then be

allocated by a declared rule and revised as workloads change (IEA, 2025; IEA, 2026b; Smith et al., 2026).

Water and grid burdens are local even when digital services are global. Cooling demand and new electrical load may be manageable in one region and damaging in another, especially where heat, drought, high prices, or constrained infrastructure already narrow public options. Siting and workload decisions should use location-specific energy, water, grid, and community baselines rather than describing computation as an immaterial service (IEA, 2025; IEA, 2026b; Smith et al., 2026).

Scenario ranges belong in public cost records because digital load is changing quickly. Berkeley Lab's 2025 update estimates that data centers could account for about 11.8 percent of United States electricity use by 2030, with a scenario range of roughly 9.5 to 15.3 percent. The estimate is national and assumption-bound, but it demonstrates why workload, location, grid constraints, efficiency, water conditions, and uncertainty should be disclosed before digital expansion is described as marginal (Smith et al., 2026).

The final compute question is whether the tool improves the stated service enough to justify its resource demands and risks. Compare it with a smaller system, a nondigital route, and the existing process. Report electricity, local water use, staff time, and service outcomes separately unless a justified conversion is supplied. Redesign or retire a tool when a feasible alternative meets the objective with lower relevant burdens and no offsetting deterioration.

29.6 — Standards, Participation, and Civic Tooling

Costing a civic tool confers no political or legal authority; its presentation must not imply ownership of fairness, proof of the Law or authority over participation. Civic tools are public artifacts before they are institutions. Maps, seals, voting pages, and campaign graphics can circulate without their original context. Essential scope labels must therefore travel with the tool; detailed hostile-reuse testing belongs in Section 29.6.2. Apply the project's lifecycle, access and vendor-exit costing to civic tools (EAC, 2021; W3C, 2023; OECD, 2015b; OECD, 2026f; NIST, 2024).

29.6.1 — Funding Boundaries for Civic Tools

Apply Section 6.5's donor firewall to civic operations: funding buys no authority over doctrine, votes, candidate status, transition timing or data.

Verification requires independence from the actor seeking legitimacy. A project cannot design the standard, collect the evidence, award itself the mark, control appeals, and present the result as public authority. Separate functions, declared methods, conflict rules, audit trails, and withdrawal procedures make verification reviewable (ISO/IEC, 2022; Swanson et al., 2010).

Donors may fund civic infrastructure under strict terms, but they may not choose questions, shape thresholds, receive early data, control dashboards, influence moderation, select featured outcomes, or treat a contribution as a claim to greater civic standing.

29.6.2 — Standards and Verification Cost

A transparency standard can be captured through ordinary line items: drafting support, certification fees, audit vendors, training, platform defaults, documentation formats, and committee access.

The account should identify who pays for standard-making, who benefits from compliance complexity, and whether compliance costs exclude smaller institutions or poorer jurisdictions.

Assign responsibility for funding, accreditation, methodology, conflicts, appeals, revocation, and revision, with reviewable records and resources to act.

Public artifacts must be reviewed in the forms they are likely to travel. A support chart can be cropped into a claim of establishment, a country map into hierarchy, a certification into status, and turnout into moral authority. Review therefore includes foreseeable detached uses and hostile crops, not only the full artifact as its designer intended (W3C, 2023; NIST, 2024).

Public voting produces bounded information about participation, concern, preference, or disagreement. Its authority should remain tied to the question and decision route for which participation was requested; broader use requires a separate mandate and correction path.

29.6.3 — Public Voting Infrastructure Cost

Budget public voting's identity, privacy, multilingual and disability access, question design, manipulation review, dispute handling, continuity, audit, appeals and public explanation.

For project polls, participatory budgets, standards consultations, and other advisory votes, the design should also address authentication proportionate to purpose, coercion, human support, and moderation. It should state who is excluded, how participation bias may shape the result, and what the result cannot authorize (EAC, 2021; National Academies, 2018b).

The jurisdictional boundary prevents this material account from becoming an accidental platform of authority. The chapter may count funding, standards, voting tools, compliance, staffing, security, maintenance, and exit, but it may not operate

government, enact legal routes, define constitutional rights, sequence rollout, or determine transition gates. Material cost is its jurisdiction (W3C, 2023; NIST, 2024).

29.7 — Staff Wages, Volunteer Limits, and Insurance

International labor guidance distinguishes paid work from volunteer work and treats working-time design as a condition of health and work-life balance (ILO, 2011; ILO, 2022; Addati et al., 2018). A living wage requirement anchors the ecosystem in local reality. The wage floor should account for housing, food, transport, health costs, caregiving, taxes, inflation, schedule stability, benefits, and the cost of working in a role with public exposure or emotional load. Track hours, after-hours messages, emergency rotations, unresolved tickets, emotional load, recovery time, staffing gaps, and repeated crises. Records must never shame fatigue or reward unhealthy availability. Persistent strain requires staffing, reduced scope, rest, revised timelines, or pause rather than gratitude (Gregory and Howard, 2009; Addati et al., 2018).

29.7.1 — Volunteer Labor and Role Boundaries

Volunteer role boundaries prevent an agreed commitment from expanding without renewed consent. A role should have a defined task, time limit, expense policy, safety line, support contact, and exit without shame.

Operational dependence is a signal that volunteer support needs review. If continuity, safety, professional review, moderation, or public response would fail without one volunteer, the institution should reduce the dependency, fund a paid role, or pause the activity until adequate support is available. This is an operating standard; employment classification requires separate review under the applicable law (ILO, 2011; Gregory and Howard, 2009).

Core safeguarding, moderation, custody, professional review, security, finance, and regular public-response roles should

ordinarily be paid; limited volunteering must not become the unfunded operating model of an essential function.

Support can include training, peer supervision, confidential counseling access, debriefing, rotation, breaks, leave, escalation, workload limits, and clear refusal rights. It should be professional enough that workers are not forced to become one another's therapists or to disclose personal distress in order to receive ordinary protection (Addati et al., 2018; Kalleberg, 2011).

Culture matters because praise can become pressure. An institution that celebrates sacrifice too often teaches people to hide fatigue and guilt one another into staying. A fairness project should honor sustainable care, accurate workload reporting, good boundaries, timely handoff, and the courage to pause before mission becomes identity.

29.7.2 — Liability Categories

Liability categories make foreseeable risk legible before harm. The account should distinguish workplace injury, harassment, data breach, professional error, event safety, facility hazard, transport, safeguarding, defamation, intellectual-property dispute, donor restriction, procurement failure, and governance conflict.

Assign responsibility, prevention, incident response, repair and finance to each risk category. Insurance can transfer some financial exposure, but it cannot replace safeguarding, labor standards, privacy governance, cybersecurity, correction, or direct care after harm. Deductibles, exclusions, notification duties, and tail coverage also belong in the cost map (ISO, 2018; COSO, 2017).

Reopen legal and ethical risk review before work with minors, events, sensitive records, moderation, restricted gifts, civic tools

or investigative publication changes the project's exposure (Swanson et al., 2010; NIST, 2024).

29.7.3 — Shutdown Reserve

The shutdown reserve should be calculated from actual obligations rather than a symbolic percentage. A modest public archive, for example, may owe final staff wages, lawful transfer or deletion of personal records, domain and hosting continuity through notice, vendor termination fees, participant notification, and preservation of evidence needed for appeal. If those obligations total eight units, a two-unit reserve is not made adequate by calling it 20 percent of a ten-unit launch budget. An exit plan should estimate each obligation with uncertainty, name the custodian and funding source, and state which commitments survive closure. A public tool that cannot afford to end safely cannot claim to have priced its full life (ISO, 2019; Swanson et al., 2010).

A shutdown reserve includes legal dissolution as well as continuing obligations to workers, participants, and host communities.

Exit support must be role-specific. Staff may need severance, references, benefits transition, wage correction, and records; participants may need continuity, referrals, data options, and assurance that departure is not betrayal; host communities may need restoration, notice, and unresolved-claim routes.

Test the reserve annually: shutdown authority, final payroll, notices, record transfer or deletion, service continuity, vendor release, and surviving claims. A failed step becomes a present funding or contract correction, not a future apology.

29.8 — Mission Independence and Procurement

Extend Section 29.8.1's donor-loss test to essential vendors and assess safe replacement as well as reserve duration (ISO, 2019; Swanson et al., 2010). A reserve has value only under the rules that would apply during an actual funding dispute. Money requiring a departing sponsor's approval cannot protect an independent exit. The test should therefore examine both usable liquidity and the authority to deploy it when the relationship deteriorates.

29.8.1 — Donor Independence as an Operating Constraint

Funding agreements must apply Section 6.5.1's non-ownership rule to the meaning of the Law, scientific conclusions, chapter emphasis, criticism, optional spiritual doctrine and choice of harms to investigate.

Donor reports and visits should use aggregate outcomes and audited cost categories where appropriate, with consent and privacy preserved. Any access to identifiable records for an independent financial audit requires a separate lawful purpose and safeguards; it is not a donor entitlement to personal stories or unpublished complaints (Reich, 2018; Ebrahim, 2003).

At renewal, model the loss of the largest donor: which activities would fail, how long reserves would last, and what safe contraction or replacement funding would require.

Restricted-gift agreements should state what happens when the original purpose becomes unsafe, obsolete or inconsistent with applicable law, safeguarding duties or the project's governing rules (Open Contracting Partnership, 2023; GAO, 2020).

29.8.2 — Procurement, Vendor Exit, and Lifecycle Accountability

Apply Section 6.4's procurement disclosure and lifecycle-accountability requirements to the project's own purchases, including labor, accessibility, security, data and exit obligations.

Before award, the packet also addresses environmental conditions, interoperability, and supplier concentration. During performance it records delivery, incidents, remedies, renewal evidence, and continuing fit with purpose (OECD, 2015b; OECD, 2025d; OECD, 2026f).

Vendor exit begins as a continuity and security obligation. Archives, moderation tools, identity services, civic platforms, and model-assisted workflows can fail during migration if access rights, logs, backups, documentation, and incident responsibilities are unclear. The cost of leaving must therefore include protection, detection, response, recovery, and communication, not only data export.

Security begins with an asset and dependency map. Records, systems, vendors, access rights, backups, interfaces, and the people harmed by alteration, deletion, leakage, or lockout must be known before controls can be proportionate. The same map should identify which functions can be performed by another supplier and how long a safe replacement would take (NIST, 2024).

Budget the portability and replacement requirements in Section 17.6.4 before procurement.

Part IX — Red-Team and Final Synthesis

A book about material peace should close by making itself answerable. Its strongest critics should be able to inspect the floorboards, wiring, budgets, incentives, language, and hidden transfers through which a humane promise could become another burden. Peace research and capability theory make that discipline necessary because visible calm can coexist with avoidable deprivation, dependence, or voicelessness (Galtung, 1969; Sen, 1999).

Part IX opens as an adversarial hearing. Every major material-peace proposition is tested under scarce capacity, adverse evidence, strategic pressure, mixed motives, local difference, measurement error, weak delivery, and foreseeable misuse. The governing question is where the burden goes when an idea is funded, measured, quoted, scaled, or administered by institutions that can be captured, exhausted, mistaken, or afraid (Mellers et al., 2001; HM Treasury, 2023).

Red-team discipline gives criticism evidence, standing, and consequence. A useful challenge identifies the failure mechanism, the people who would carry it, the signals that would reveal it, and the authority able to narrow, pause, redesign, or stop the work before injury becomes routine. It turns skepticism into practical learning with enough power to change the result.

Cost is a coequal test of moral seriousness. Money, labor, land, water, energy, administrative skill, staff attention, legal authority, public trust, and transition time determine whether a proposal becomes durable care or merely moves its invoice. Complete appraisal includes operation, maintenance, opportunity cost, residual risk, distribution, uncertainty, and obligations that emerge only after launch (HM Treasury, 2026b; IMF, 2018).

Incentives are part of the material architecture. Prestige can arrive before usefulness, donors and vendors can acquire leverage, bureaucracies can defend their own survival, and

reformers can be consumed by the work they hoped to sustain. The project therefore applies the same capture, concentration, exit, and maintenance tests to itself that it applies to other institutions (OECD, 2017b; Stigler, 1971).

Measurement also stands trial. Dashboards, burden registers, appraisal tools, and public artifacts can expose costs that ordinary accounts omit, yet they can acquire authority beyond their design. Categories, weights, rankings, and fluent summaries become unauthorized governance when they decide eligibility or prestige beyond their lawful, evidentiary or review limits. No such measure may determine human worth (Merry, 2016; Saltelli et al., 2020).

The synthesis can be no broader than the propositions that survive this hearing. Chapter 30 tests the claims; Chapter 31 identifies the work still required. It closes as a discipline of burden honesty, not a declaration that history, conflict, or scarcity has been solved. Its authority boundaries, evidence ceilings, unresolved dissent, maintenance costs and exit routes must remain visible alongside its horizon.

Chapter 30 — Red-Team Tests and Objections

A serious account of material peace hears the case against itself at full strength. The most useful critic exposes where an admired proposal shifts cost, risk, labor, delay, uncertainty, or loss onto people the proposal does not see. Adversarial collaboration and open criticism become productive when the challenge can alter the design instead of merely decorating its legitimacy (Mellers et al., 2001; Popper, 1945).

Skepticism is treated here as a public service. Questions about scarcity, incentives, violence, administrative capacity, local authority, or measurement often come from people who have watched humane language conceal new costs. A serious project gives those people access to the record, protection from retaliation, and a fair response process (Mellers et al., 2001; HM Treasury, 2023).

Use the challenge protocol in Section 30.9 to test poor data, exhausted users, adaptive adversaries, funding pressure and refusal that is safe on paper but costly in practice. A favorable finding has weight only if the review tested a serious challenge to the proposal and could have changed the decision.

One question governs the chapter: when an idea is wrong, incomplete, premature, captured, exported, excerpted, or administered under pressure, who carries the burden before the error is acknowledged? That question keeps institutional legibility subordinate to the knowledge, rights, and agency of people living inside the system (Scott, 1998; Ostrom, 1990).

Cost and scarcity require a harder comparison: does the preferred repair still outperform a realistic alternative once staffing, maintenance, opportunity cost, and burdens already carried outside the budget are counted (HM Treasury, 2026b; IMF, 2018)?

Then remove founding enthusiasm from the model. Would the design remain sound once ordinary careers, budgets, vendors, and

constituencies take over (Jensen and Meckling, 1976; OECD, 2017b; Carpenter and Moss, 2013)?

Measurement and implementation fail differently. One asks whether an indicator clarifies a bounded decision; the other whether an institution can act on what it learns. Section 30.6 develops both (Bevan and Hood, 2006; Saltelli et al., 2020; Andrews et al., 2017).

The hearing becomes an operating discipline, not a closing ritual. Failure tests are fixed before success creates a constituency; unresolved dissent remains visible; and review has its own budget and independence. Review findings must still reach an authority able to narrow, delay, redesign, pause, or retire a proposal after prestige and funding arrive.

Different expert communities should recognize their own failure conditions in the review. Economists need opportunity cost, incidence, recurrent liability, and optimism-bias correction. Security specialists need deterrence, deception, inspection failure, escalation, and emergency reversal. Rights specialists need privacy, appeal, record limits, and hostile-crop review. Administrators need staffing, procurement, translation, supervision, maintenance, and service continuity.

30.1 — The Utopianism Objection

The utopianism objection tests whether an attractive peace horizon still permits correction when evidence challenges its assumptions. Coherence, urgency, prestige, scale, and moral aspiration can each become shields against scrutiny if a proposal treats its own stated aims as proof of success. A responsible design instead specifies what observation would warrant revision, which affected people can initiate review, and who has the authority to change course. This objection protects the possibility of improvement by refusing to make any preferred outcome immune from criticism.

30.1.1 — Paradise Risk

Paradise language becomes unsafe when grief, disability, slowness, scarcity, continuing conflict, or dissent are treated as defects in people instead of conditions an institution must meet. Architectural hope asks whether fewer harms are hidden, fewer choices are trapped, and more repair is funded without pretending the human condition has been solved.

The danger begins when a coherent design is treated as evidence that operators will remain benevolent, central planners will understand local knowledge, and dissent merely proves misunderstanding. Popper’s piecemeal correction and Scott’s account of high-modernist failure imply a stricter rule: ambition should arrive in contestable increments, with feedback, exit, and reversal strong enough to survive the plan’s most devoted admirers (Popper, 1945; Scott, 1998).

A premortem makes paradise risk operational. Before launch, advocates and independent critics assume the proposal has failed and reconstruct plausible chains such as capture, staffing collapse, local refusal, fiscal stress, vendor lock-in, distorted measurement, or quiet departure. Each chain needs a signal, owner, treatment,

and decision consequence, or the exercise performs humility without granting it authority.

30.1.2 — Institutional Drift Risk

Institutional drift often enters through the language of necessity. A safeguard is postponed for efficiency, a review clock extended for workload, a donor exception tolerated for survival, or a dashboard simplified for communication until the design no longer carries its own burden. Use Section 30.9's independent review and correction powers to detect and reverse that movement.

Formal rules may remain while budgets, interpretations, enforcement priorities, vendor dependencies and internal coalitions change around them (Mahoney and Thelen, 2010; Hirschman, 1970; OECD, 2017b).

Drift indicators should reveal operating change rather than perform vigilance. Rising exception rates, delayed appeals, falling reversal rates, vendor concentration, unexplained eligibility shifts, staff turnover, maintenance deferral, and a widening gap between stated purpose and budgeted work can trigger deeper review. None proves capture alone, but the pattern can make correction possible before public breakdown becomes the first undeniable evidence.

In Section 28.7.2's constructed case, a disputed return journey overturns the first authorization, the amended design costs more over the chosen horizon, and a delivery defect delays implementation while essential service continues. The objection docket should preserve what each challenge changed and the disagreement that remains.

30.2 — The Cost and Scarcity Objection

Evaluate a proposal against credible alternatives using its full fiscal, material, institutional, and opportunity costs, including contingent liabilities and optimism bias (HM Treasury, 2026b; IMF, 2018). Money cannot immediately produce experienced engineers, clinical workers, trusted local partners, grid capacity, or a functioning appeals system, so record the time and scarce skills needed to make delivery possible. Distinguish ecological and technical limits from policy choices, historical underinvestment, artificial bottlenecks, and claims of scarcity that protect existing waste. Ask which other worthy undertaking would be delayed and who bears that displacement, applying the burden-conflict sequence in Section 1.5.3. The objection is answered only when a realistic comparison makes financing, capacity, uncertainty, mitigation, and a usable exit visible.

30.2.1 — Administrative Capacity Limits

Apply Section 28.7.2's staffing and service tests to caseload, language access, appeals, records, training and worker support.

State-capability research warns that importing the appearance of best practice can produce form without function, while project research supports modular learning and reference-class comparison. Large initiatives are safer when they begin from the actual problem and scale after operations become reliable (Andrews et al., 2017; Fukuyama, 2013; Flyvbjerg and Gardner, 2023).

A capability roadmap is more useful than a readiness rank. It names the task, present and required capability, formation time, scarce dependencies, evidence plan, and fallback if the gap persists. Evaluation also needs mandate, competence, quality review, publication, and follow-up beyond a reporting

requirement. OECD standards treat evaluation capacity as an institution of use and learning (OECD, 2022; OECD, 2025b).

30.2.2 — Fiscal Sustainability Challenge

A fiscal stress test should state what changes the decision. Consider a hypothetical proposal with 100 cost units of verified, realizable one-time savings. Necessary conversion, service continuity, maintenance and oversight consume 75; a prudential reserve withholds another 10. Only 15 remain allocable. A proposed new service costs 20 now and requires 12 each year thereafter. Approving it would fail twice: the initial cost exceeds the available remainder, and a one-time saving does not fund recurring operations. The defensible disposition is to defer the commitment, secure recurring finance, or reduce scope without violating the protected service floor.

Now test an adverse case: a confirmed 15-unit increase in conversion cost consumes the remaining allocable capacity. The reserve is not automatically released to preserve the announcement. The responsible body records the changed estimate, protects existing obligations, and reopens the decision. These invented numbers demonstrate the review method; they are not a forecast or evidence that a particular reform will fail. Actual appraisal still needs domain evidence, realistic alternatives, distributional review, and authority to refuse (IMF, 2018; HM Treasury, 2026b).

Fiscal affordability alone does not establish deliverability. Even a smaller program priced within the initial fifteen-unit remainder could fail if the only qualified maintenance team is already supporting an essential water system. Test workforce and physical capacity, protected rights and access, and funded correction or orderly shutdown alongside the budget. A rejected proposal may become viable through phased staffing, renewed

finance or changed timing. The red team should state which condition failed and what evidence would justify reconsideration.

30.3 — The Security and Bad Actor Objection

A security objection must begin with a specific protected interest and a credible account of the actor, capability, pathway, and potential harm. Examine whether proposed safeguards would reduce that risk and whether less restrictive alternatives could do so with lower civilian burdens. Apply Chapter 19's tests of necessity, lawful authority, proportionality, and review to any exceptional power. State the residual risk openly, including what the measure could damage if the threat estimate proves mistaken.

30.3.1 — Predatory Actor Risk

Which design reduces the gains from predation, improves detection, protects innocent people, and leaves a reviewable trail?

Clearer information and credible commitment solve different problems. Powell (2006) shows in a bargaining model how rapid shifts in power can undermine an agreement because a party cannot commit to respecting terms once its position improves. An inspection may resolve uncertainty about today's capability without securing tomorrow's bargain. Consider a hypothetical agreement in which one side must make an irreversible concession first: accurate disclosure alone does not determine whether later promises remain credible. The design question is whether sequencing, reciprocal obligations, or enforceable commitments can address that specific incentive.

Predatory actors often exploit seams between institutions. One office may see a financial anomaly, another a cyber intrusion, another labor coercion, and another unusual shipping, while no lawful process joins the evidence. Information sharing should therefore be purpose-limited, threshold-based, logged, reviewed, and correctable. Test what the compact requires when asymmetry persists, verification fails, or one party withdraws; use the reciprocal-burden and breach rules in Chapter 18.

30.3.2 — Deterrence Need

Test whether a posture's capabilities, communication and command can deter aggression without being mistaken for preparation for attack. Use the bounded deterrence review in Section 19.4 (Schelling, 1960; Jervis, 1978; SIPRI, 2026b).

Any claimed peace dividend must preserve the residual security function required by its threat model. Apply the net-cost comparison in 20.1; SIPRI's expenditure data establish the scale of current spending, not the cost of a particular conversion (SIPRI, 2026a; SIPRI, 2026b).

Inspection design should always ask what the inspection cannot see. The unseen field may include intimidation, unreported harm, hidden beneficial ownership, off-record threats, technical vulnerabilities, or the quiet fear of people asked to testify. Review therefore needs confidential channels, independent corroboration, and a record of uncertainty.

30.3.3 — Emergency Reversal Risk

Emergency-reversal review should combine weak-signal and operational learning, defined risk owners and recovery duties, and tests for deliberate deception (Weick and Sutcliffe, 2007; HM Treasury, 2023; SIPRI, 2026b).

Before a crisis, a reversal clause should define observable triggers, decision authority, scope, duration, protected rights, logging and data handling, review cadence, renewal standards, automatic expiry, replenishment, and the service and funding required to return to ordinary rules. A system able to enter emergency mode but unable to leave it has built a ratchet rather than resilience.

30.4 — The Human Nature and Incentives Objection

Test which incentives and dependencies let private advantage persist beneath public language (Jensen and Meckling, 1976; OECD, 2017b; Carpenter and Moss, 2013).

30.4.1 — Rewards That Follow the Result

Use Section 28.1's incentive map to identify who benefits from launch, expansion, favorable measurement, silence and closure.

Does prestige arrive before the relief it is meant to recognize?

An incentive can be examined by following what happens just before recognition is awarded. Suppose a program celebrates the number of people enrolled while its actual purpose is to help them complete a difficult qualification. Enrollment deserves to be recorded, but it occurs before the sustained teaching has done its work. The review can ask whether staff have enough reason and resources to support the less visible interval between joining and completion.

Changing the reward does not require pretending that one final number captures the whole service. The institution can recognize a successful intake and separately recognize durable completion, including the work needed to support people facing different starting conditions. Each achievement then retains its own meaning. Staff can see which contribution is being honored instead of trying to infer the desired behavior from a single headline measure.

A constructive incentive review looks for these mismatches between the moment of reward and the moment of benefit. It asks what useful work takes place after public attention has moved on. Funding and recognition can then be attached to that continuing work, making the institution's ambition more consistent with the experience it hopes to create.

30.4.2 — Independence at the Point of Decision

Test donor and vendor dependence together: can the project refuse a funder's demand while retaining lawful access to its records, interfaces and infrastructure?

Self-protection review should distinguish bad-faith concealment from fear produced by impossible workload. Some offices hide because they are corrupt; others hide because they are overwhelmed. The remedies differ, but both require visibility, protected candor, and enough capacity to correct the underlying condition.

For each critical dependency, test the owner, concentration threshold, replacement route and public account of what happens if the relationship ends (Carpenter and Moss, 2013; OECD, 2017b; Hirschman, 1970).

Controls should follow the complete influence chain. Beneficial ownership, lobbying, revolving-door employment, subcontractors, advisory roles, data access, and intellectual-property dependence can alter a decision without appearing in the formal procurement record. High-risk choices need disclosure, independent review, reasoned selection, alternatives, and pattern analysis across repeated decisions; no single cooling-off rule or role separation can substitute for that view (Carpenter and Moss, 2013).

Independence is clearest when a consequential answer is inconvenient to the institution supporting the work. A reviewer can ask the project to walk through such a decision in advance. Who receives the finding? Who controls the next payment? Who can preserve the record? Which other team can continue an essential service while the disagreement is resolved?

The exercise should end in an executable arrangement. A separate review budget can be released on stated terms. Records can be held in a form the operating team is able to retrieve. A

replacement provider can receive the information needed to continue the service. These are distinct pieces of independence, and each can be demonstrated through an ordinary action rather than an assurance about good intentions.

The result is a more useful answer to the human-nature objection. People can remain ambitious and organizations can remain protective of their reputations, while the design gives evidence a route to influence the next decision. Institutional strength comes partly from arranging responsibilities so that doing the necessary work does not depend on everyone having the same motive.

30.5 — The Local Difference and Cultural Imposition Objection

Shared questions about who pays, works, bears risk, maintains services, and may refuse a proposal do not establish one institutional answer for every community. Local governance, ecological conditions, language, history, and Indigenous authority can alter what legitimate implementation requires (Ostrom, 1990; Rodrik, 2007; UN General Assembly, 2007). A water-service design, for example, may need different ownership, consent, and maintenance arrangements in a dense city, a rural district, and an Indigenous territory. Review who defines the problem, controls evidence and procurement, owns the assets, and can revise or reject the design (Scott, 1998; Sen, 1999). Carry that authority through record custody, provider replacement, continuing finance, maintenance, and exit, as examined in Sections 22.3 and 22.4.

30.5.1 — Minority and Indigenous Safeguards

Protection begins before consultation is reduced to a meeting. Representative institutions, language, land, resources, cultural continuity, consent, and authority to shape or refuse a project all matter. The United Nations declaration supplies a rights floor, while local-governance research explains how externally imposed legibility can erase knowledge needed for stewardship. A material-peace proposal that ignores these authorities can become a route of dispossession (UN General Assembly, 2007; Ostrom, 1990; Scott, 1998).

Free, prior, and informed consent exceeds notice, a listening session, or a comment period. It concerns representative authority, culturally appropriate process, adequate information and time, freedom from coercion, and the ability to give, condition, withhold, or withdraw consent where rights are

affected. The 2026 United Nations guidance treats FPIC as a continuing project-lifecycle obligation whose demands extend far beyond a communications technique (UN Working Group on Business and Human Rights, 2026).

A consent process remains incomplete when the people most likely to be overridden cannot object safely or refuse without losing essential support. Minority and Indigenous protections are therefore not additions to local authorship; they are among the conditions that make local authorship fair.

30.5.2 — Shared Functions and Locally Authored Forms

A shared purpose can be translated into different local forms. Consider two communities seeking dependable access to clean water. One may operate a compact network with a permanent maintenance crew. Another may depend on dispersed sources and locally organized upkeep. The common question concerns dependable access; the work that sustains it can have a different institutional shape.

Comparison becomes productive when each design explains how it performs the relevant function. Who recognizes a fault? How is repair organized? What knowledge is needed to maintain the source? A proposal can learn from another community’s answer without copying its ownership structure or assuming the same distances and resources. The transferable insight may be a method of diagnosis rather than a complete institution.

Local authorship also makes room for invention. People who use a system can identify possibilities that an outside standard did not anticipate. The reviewing partnership can examine whether the proposed variation achieves the shared function and how it will be maintained. Agreement then concerns the reason for the arrangement and the evidence of its performance, while the form remains responsive to the place.

This is a stronger basis for exchange than choosing between rigid uniformity and complete incomparability. Communities can share experience about a recognizable task and still retain authority over how the task becomes part of their lives. Difference becomes a source of practical knowledge rather than an obstacle that cooperation must erase.

30.6 — The Measurement and Implementation Objection

Different analyses of the same evidence can disagree without representing independent replications: Silberzahn and colleagues (2018) gave one soccer dataset to 29 analyst teams, of which 20 reported a statistically significant association and nine did not. Auspurg and Brüderl (2021) argued that divergent interpretations of the research question explained much of the spread and obtained a narrower range by clarifying the target parameter and admissible models. The remedy is to define what is being estimated, identify defensible comparisons, and disclose sensitivity to modeling choices and unmeasured variables. Reanalysis tests dependence on choices within the existing records, whereas replication tests whether a finding recurs in new samples. Human or AI-assisted review should preserve the common origin of reused evidence and turn disagreement into a precise request for better measurement or assumptions.

30.6.1 — False Precision Risk

An uncertain indicator becomes falsely precise when presentation supplies more confidence than design has earned. Decimal places, rankings, color bands, and composite scores can imply stable knowledge even when the construct, denominator, missingness, weights, or causal interpretation remains contested.

A simple sensitivity display can make this risk visible. If two plausible sets of weights reverse which option ranks first, show both results and identify the judgment producing the reversal. Readers can then debate that choice directly instead of mistaking one selected score for a settled answer (Saltelli et al., 2020).

An incomplete record can still yield a useful bound. Consider a fictional census of 1,000 approved repair orders due by the review date: 720 are verified complete, 180 incomplete, and 100

unresolved. Completion is 80 percent among the 900 resolved records. Across the full cohort, however, it lies between 72 and 82 percent, depending on the unresolved cases. A 75 percent standard is therefore not yet established or ruled out; verifying completion at the review date for at least thirty unresolved orders would establish it. These are logical bounds, not confidence intervals; they assume the cohort is complete and the verified statuses are correct. Sampling or recording error requires additional analysis. The next useful work is to resolve the missing cases and examine whether absence is concentrated in inaccessible routes, rather than quietly dropping them from the denominator.

30.6.2 — Dashboard Authority Drift

A dashboard acquires governing power when a descriptive measure becomes a target, the target becomes an eligibility rule, and the eligibility rule becomes a proxy for worth or readiness. The shift can occur gradually enough that no single decision appears to have authorized it.

An accurate prediction can still answer the wrong question. Obermeyer et al. (2019) examined a widely used health-management algorithm that predicted spending as a proxy for need. At the same risk score, Black patients were sicker than White patients because spending did not represent need equally across groups. Reformulating the prediction target substantially reduced the disparity in the study. The error was therefore not merely too little data or too few decimal places; it lay in the bridge from the measured outcome to the intended purpose. A burden dashboard needs the same audit: what does its target actually represent, whose experience is missing from that representation, and would changing the target improve the decision?

AI-generated summaries create a distinct layer of authority drift. Fluent text can compress uncertainty, merge incompatible sources, omit material qualifiers, or invent causal language the record does not support. The NIST generative-AI profile and current public-sector evaluation guidance emphasize governance, provenance, testing, monitoring, incident routes, and comparison with the status quo (Autio et al., 2024; HM Treasury and Evaluation Task Force, 2026). Any AI-assisted public summary in this project must remain source-traceable, labeled, human-reviewed, evaluated for intended and unintended effects, correctable, and replaceable.

Use Section 4.6.2's correction and retirement rules when a display exceeds its authorized purpose. Notify downstream users which decisions may no longer rely on it, and preserve why its status changed.

30.6.3 — Implementation Capacity

Implementation evidence supports different decisions. In a fictional benefits office, an approval authorizes payment within ten working days. Receipt and transfer logs show that approved cases carrying one document code never reached the payment queue. Replaying the code in a secure test environment reproduces the break. The office repairs the routing rule, completes the approved payments, and uses its authorized contingency arrangements for urgent support. Those records establish a delivery defect and a basis for correction without estimating every later household consequence.

Whether the correction also reduced debt or improved health is a different question requiring a credible comparison and an appropriate follow-up period. A diagnostic record should retain the timing of the break, the cases affected, the competing explanations examined, and the test showing that the repaired

route works. Necessary records remain protected. An institution can fulfill an established obligation while continuing to study the wider benefits of doing so; it need not delay a demonstrated repair until every causal question is settled.

Complex interventions should be evaluated at three different levels. Process evidence asks whether implementation actually occurred, including acceptance by staff, coverage, fidelity, unintended exclusions, and operating cost. Impact evidence asks whether outcomes changed relative to an appropriate counterfactual, including adverse effects on nonparticipants and neighboring services. System evidence asks whether success depends on capacity taken from another clinic, agency, town, or future budget, and whether the benefit survives a correlated shock. A pilot showing shorter queues but longer waits at the appeals desk has improved a metric without establishing net service improvement. The evaluation plan should predeclare thresholds for scaling, revision, and stopping, and use methods proportionate to the problem rather than automatically privileging a single design (Proctor et al., 2011; Skivington et al., 2021; HM Treasury, 2026a).

30.7 — The Misuse Objection

Misuse review must test how a detached passage or simplified claim could justify practices the full argument would reject (Section 29.6.2; Autio et al., 2024). Calling deprivation, unpaid care, or broken services "enoughness" does not satisfy Chapter 2's protected floor and plural-aspiration limits (Ostry et al., 2016; Sen, 1999; Nussbaum, 2011). Apply Section 7.5's refusal audit to identify any benefit or right a recipient would lose for rejecting additional program demands. Migration claims must separate measured arrival-related effects from preexisting housing shortages, underinvestment, and employer practices, then identify who gains and who finances additional capacity (World Bank, 2023; Hainmueller and Hopkins, 2014). Residents and newcomers alike must be able to contest the account without being reduced to rival human problems.

30.7.1 — Public Artifact Misuse

Keep source, date, version, accessible equivalents and a responsible correction contact with high-reach summaries; correct or withdraw defective artifacts.

Check whether an ethical floor has become an eligibility score, an illustrative conditional example has become an established causal effect, a restricted security record exposes an operational weakness, or a cropped finding has lost a condition necessary to its meaning. Examine search snippets, automated translations, and AI summaries as well as the full document. Correct or narrow the misleading statement or graphic itself; withhold it when its essential condition cannot be preserved.

30.7.2 — Explanations That Travel Intact

A useful public explanation should carry its decisive relationship in the sentence that readers are most likely to repeat. Consider a

teaching account with a reserve of twenty units and a proposed expenditure of five. Saying that twenty units are available may leave the reader unsure whether the expenditure has already been deducted. Saying that the reserve begins at twenty and would fall to fifteen after the proposed payment preserves the sequence in ordinary language.

The improvement comes from making the relationship easier to retain, not from attaching a long qualification afterward. The explanation names the starting amount, the action, and the resulting amount together. Someone translating it can test whether all three survived. Someone quoting it can see which words are necessary for the calculation to remain intelligible.

The same method works for a service claim. A statement can distinguish appointments offered from visits completed in its main wording, rather than asking a reader to discover the difference in a distant note. Plain language is precise when it gives each number the right object. The aim is a compact explanation that remains useful when carried into a classroom or a public conversation.

This makes communication part of the design itself. A proposal that can be explained faithfully gives more people the ability to examine it. The test is whether a reader can reconstruct the relevant choice and ask a sensible question about it, even without the vocabulary of the original report.

30.8 — What Survives the Red-Team Test

This internal review leaves a set of conditional proposals, not a record of independent validation. Their costs, assumptions, failure routes, and misuse risks remain open to challenge. A proposal earns wider use only when evidence from the relevant setting supports it and the people responsible can revise, restrict, or stop it.

30.8.1 — Keeping the Insight and Revising the Design

What survives is a hierarchy of warranted statements, not a list of favored policies: ethical floors independent of PTVC, material propositions supported by their own evidence, conditional applications requiring stated premises, design hypotheses requiring tests, and open questions that remain unresolved. The red team succeeds when each survivor returns to the narrowest status it can honestly bear (Sen, 1999; Knight et al., 1996; HM Treasury, 2026b).

A review can produce progress by locating the level at which an idea remains useful. Suppose a service trial finds that shared scheduling reduces repeated visits, but the particular software is too difficult for staff to maintain. The scheduling insight and the software decision can receive different judgments. A simpler implementation may preserve the benefit while changing the part that made it expensive to sustain.

That distinction gives revision a clear direction. The next team can retain the demonstrated relationship, identify the implementation feature that needs replacement, and specify the comparison the new version must face. The knowledge gained in the first attempt remains usable. Improvement becomes a sequence in which each attempt leaves the next one a more precise question.

30.8.2 — Testing the Connections

The whole architecture should also be allowed to fail a joint test. A proposed combination of stable housing, service access, transparent finance, international cooperation, and protected rights has not demonstrated a fairer order if it can operate only by exhausting frontline workers, shifting maintenance debt into the future, or giving one funder control over records and criticism. A favorable judgment requires demonstrated improvement against realistic alternatives under available budget and staff, with essential rights and lawful review preserved, and with adverse scenarios that the institutions can actually survive. When evidence is incomplete, the verdict remains conditional. When those conditions are demonstrably violated, the design fails the test.

Each survivor should be stated at the level on which it was tested. A principle may survive while one implementation fails; a local program may work while a universal claim does not; a cost estimate may remain useful after its political conclusion is rejected. This layered result preserves warranted knowledge, retires what failed, and makes the remainder precise about population, place, time, assumptions, and burden.

30.9 — Red-Team Rules for Future Builders

A red-team challenge matters only if its findings can alter a real decision. Reviewers need independent access to evidence, sufficient resources, a named decision owner, a response deadline, and a durable record of unresolved dissent (Mellers et al., 2001; UN General Assembly, 2007). The review mandate should allow a proposal's scope, timing, financing, language, or continuation to change in response to findings. As the 2026 Magenta Book emphasizes, evaluation for learning and accountability must be designed before implementation rather than added as retrospective reporting (HM Treasury, 2026a).

30.9.1 — Failure Tests for Claims

A failure test turns confidence into a precommitted consequence. Before adoption, the proposal names the result that would narrow, pause, redesign, or retire it, who holds authority for that decision, what evidence is admissible, and what support is owed to people affected by stopping.

Failure tests must identify observations that could invalidate the premise itself, rather than routing every adverse result into implementation troubleshooting (Mellers et al., 2001; HM Treasury, 2023; HM Treasury, 2026a; HM Treasury, 2026b).

Use Section 29.2's proportional review standard, preserving affected people's right to challenge and a deadline that lets findings change the decision (HM Treasury, 2026a).

Further research has value when it can improve a decision. Consider an invented comparison of two otherwise equivalent options that already meet required rights and service floors. Under an expected-cost rule, option A costs $80,000 or $140,000 with equal probability; option B costs $100,000 in either state. Without further information, B is cheaper than A's $110,000 expected cost. Perfect information obtained before commitment would

select A in the low-cost state and B in the high-cost state, for an expected cost of $90,000. Its gross value is therefore $10,000, an upper bound on the value of an imperfect study for this decision. A $12,000 study would not recover its cost on those grounds alone, even before delay is counted. Required oversight and benefits to other decisions need their own stated justification. Value-of-information analysis separates choosing with current evidence from deciding whether to acquire more (Claxton, 1999). It can direct scarce research capacity toward the uncertainty most likely to improve the choice.

30.9.2 — Capture Tests for Funding Models

At funding renewal and vendor transition, rerun Section 30.4's capture tests alongside the continuity tests below (OECD, 2017b; Reich, 2018).

Run Section 29.8's largest-funder loss and independently accessible reserve tests: do publication, participant protection and essential services survive an unfavorable finding?

30.9.3 — Anti-Imperial Tests for Global Proposals

Apply the local-authority review in Section 30.5 with Indigenous authority, minority rights and ecological context intact. Not proceeding must remain an available outcome (Mellers et al., 2001; Carpenter and Moss, 2013; UN Working Group on Business and Human Rights, 2026).

When did affected communities gain power to change the design, and which choices remained open?

Repeat review before launch, scale and renewal, after major incidents, and when funding or technology changes risk assumptions.

Test Sections 22.3–22.4's exit obligations after sunk costs accrue, when refusal may expose the host to debt, displaced workers, abandoned equipment, service interruption or unusable

records. The continuing right must survive that dependence (Hirschman, 1970; OECD, 2017b; UN Working Group on Business and Human Rights, 2026).

Chapter 31 — Final Synthesis and the Peace Horizon

The final chapter begins where the book has been moving: with a test, not a victory. A civilization can speak tenderly about peace while leaving debt in the household, fear in the budget, delay in the clinic, shame in the queue, exhaustion in the caregiver, and risk in the future. The book ends by keeping those burdens visible beneath every beautiful phrase (Galtung, 1969; Sen, 1999).

The chapter’s final instrument is a material-burden register: Who pays? Who waits? Who works? Who is exposed? Who benefits? Who can refuse? Who repairs? Who carries the remainder? These questions give readers a practical starting point for examining economics, law, administration, infrastructure, climate, labor, migration, defense, and data governance.

The synthesis asks power to explain and carry the consequences of its decisions, preserving public challenge rather than prescribing one universal program (Ostrom, 1990; Weick and Sutcliffe, 2007).

The red team has narrowed the book’s confidence, and that is a gain. A peace theory unable to face cost, scarcity, security, incentives, local difference, measurement, and misuse is not ready to travel. Worthy confidence knows where it must be limited and what evidence would make it change.

The peace horizon is a direction, not a guarantee. It points toward more repairable systems, less defensive life, more honest public finance, dignified sufficiency, and a world less organized around hidden cost. Answerability remains ahead of triumph because any future that calls itself fair must remain correctable by the people who carry its burdens.

The register helps general readers notice when a public promise hides its bill where someone weaker must pay it, when delay is called neutral although it consumes a life, or when celebrated efficiency depends on labor removed from view. That attention is not cynicism. It makes hidden burden available to public reason

and recognizes the tired household, not only the analyst, as a legitimate witness of system cost.

The register remains usable by readers who reject PTVC or disagree with the book's proposals; scrutiny requires no belief test. Readers can disagree about taxation, markets, defense or public administration while asking whether each burden is named and its bearer can contest it. Take one initially persuasive proposal and follow the obligation it creates to the person least able to avoid it. Then take an opposing proposal through the same inquiry. Agreement may remain impossible, but disagreement can become more exact: about a consequence, an alternative, a distribution or a limit of authority. The closing research proposals keep two inquiries apart: whether an institutional change improves service, and whether a phenomenal measure can support a test of terminal closure. In Section 31.6, the practical study tests removing duplicate proof requirements and assigning a named case owner, separately and together, measuring applicant time and verified delivery. The scientific collaboration begins by agreeing calibration, endpoint rules and predictions that could distinguish terminal closure from specified rivals. Neither result is assumed in the other.

31.1 — What the Hypothesis Contributes to Material Peace

The book's material-peace framework makes overlooked consequences visible by asking whose burdens and gains a proposal actually changes. PTVC remains a heuristic for directing attention to felt consequences within qualifying streams; it does not itself establish causal evidence or confer authority to act, as Section 1.6 explains. The distinction protects both inquiry and the people affected by decisions. Mature hope reports what improved, who is still waiting, and when retreat or redesign is warranted. A funded remedy must be available on its public terms without requiring anyone to affirm the underlying hypothesis (Sen, 2009; Popper, 1945).

31.1.1 — Material Burden as Public Evidence

Use Section 28.7.1's record and Section 29.5.1's provenance rules, with standing for affected people omitted from the sample and protected source access for reviewers. Apply Section 30.6.2's AI safeguards: output must never determine human worth. Decisions to set aside testimony or close a complaint require accountable human review (Autio et al., 2024; HM Treasury and Evaluation Task Force, 2026).

These records are not phenomenal ledgers and cannot infer the total inner life of any person. Indicator and model governance therefore belong inside the evidentiary claim rather than in a technical appendix (Merry, 2016; Saltelli et al., 2020; Autio et al., 2024).

31.1.2 — A Shared Task Across Different Starting Points

People can agree on a repair while approaching the deeper question of fairness from different directions. One person may find the Law's proposed whole-life perspective compelling.

Another may be interested chiefly in a particular service that needs improvement. A shared undertaking can begin with the part both can examine: the service to be provided and the work required to sustain it.

The resulting cooperation has an intellectual advantage. Different starting points bring different questions to the same proposal. A participant attentive to lived experience may notice a burden that an administrative description has missed. A participant familiar with public finance may identify the funding interval that would otherwise interrupt the remedy. Their contributions become complementary when the proposal makes room for both kinds of examination.

This is a practical form of openness. The philosophical inquiry can continue in its own right while people develop arrangements they have independent reasons to support. Success in the shared task creates a better setting for further conversation because disagreement about the larger theory need not consume the work already possible together.

The contribution of the series is therefore also an invitation to attend. A public account is connected to lives lived from within, and a promising idea becomes more valuable when people with different knowledge can help make it work. Peace grows through that shared construction as well as through the arguments that first made construction imaginable.

31.2 — The Material Questions That Test Peace

Material peace cannot be judged from one favorable average because payment, labor, delay, exposure, benefit, refusal, repair, and residual cost may fall on different people. Treat each dimension as a separate question before combining the evidence into a public decision. Appraisal can compare feasible options, capability analysis can clarify practical freedoms, and indicator scholarship can reveal what conventional metrics omit (HM Treasury, 2026b; Sen, 2009; Merry, 2016). The account must name the households, places, institutions, and future budgets responsible for each consequence. Disagreement becomes more useful when participants can identify exactly which burden, estimate, or distribution they contest.

31.2.1 — Who Pays

Record monetary payments alongside work, delay, risk, and future obligations.

Incidence follows behavioral and institutional pathways beyond the legal line item. A treasury payment may shift work to municipalities or households; a firm may pass a levy through prices, employment or investment. The final distribution should be reported with uncertainty because market power, labor conditions, regional alternatives, and timing shape the result (HM Treasury, 2026b). Incidence should be revisited after implementation because initial estimates can miss avoidance, substitution, queue formation, and local capacity effects that appear only when people adapt to the new rule.

Time costs also differ: the same delay can impose greater harm on a child, patient, migrant, worker or household without reserves than on someone with usable alternatives. Equal delay is not always equal burden.

31.2.2 — Who Works

Use Chapter 8's work standards and Section 29.7's labor account, including household work that compensates for service failure (ILO, 2015; Andrews et al., 2017; Graham and Thrift, 2007). Review should distinguish workload that can be improved through staffing or process redesign from distress caused by threat, harassment, or morally impossible instructions because those conditions require different remedies and different authority to refuse. Apply Section 21.3's data limits to labor records so revealing hidden work does not enable surveillance or retaliation.

31.2.3 — Who Can Refuse

Apply Section 7.5's refusal test to actual dependence, not merely the formal option to say no.

Refusal applies to participation, data sharing, relocation, employment, treatment, consultation, donation and public visibility. A usable route protects applicable rights and essential support; it does not necessarily erase ordinary contractual obligations or every cost of leaving. Where exit alone cannot protect people, voice, appeal and alternative access must provide a practical route to change the arrangement (Hirschman, 1970; UN General Assembly, 2007; Sen, 1999).

Collective refusal requires representative authority; apply the continuing consent requirements in Section 30.5.1.

A refusal route should state the alternative service, transition period, portability rights, financial support, and appeal process available after the person or institution says no (Hirschman, 1970).

31.2.4 — Who Bears the Remaining Material Cost

Record the burden remaining after claimed savings, transfers and repairs.

After mitigation, remaining burden should be named, justified, reviewed, and where possible compensated or reduced rather than treated as the invisible price of progress (Coase, 1960; HM Treasury, 2026b; Sen, 2009). Compensation plans should specify when payment reaches the bearer, since reimbursement after a temporary burden ends may arrive too late to prevent borrowing or loss of service.

When a national decision shifts costs, funding must follow the obligation. Give affected institutions information, standing and a forum able to order correction; schedule review while the transfer can still change.

For temporary burdens, specify the owner, funding, review date, and end or transfer condition. Distinguish unavoidable costs from unassigned work; assign responsibility or explicitly accept the burden. Apply Section 30.3.3’s reversal rules to emergency measures.

31.3 — From Hidden Costs to Ordinary Abundance

Ordinary abundance means reliable opportunities and services, not a claim that scarcity, grief, or disagreement have disappeared. Gains are incomplete if they merely transfer fiscal or ecological costs to people who cannot refuse them (Sen, 1999; Raworth, 2017). Report released resources separately from the time and skilled capacity needed to turn them into staffed and maintained services (Vinsel and Russell, 2020). Compare benefits across locations and groups so an improving average does not conceal people who still wait. Include participation people previously had to forgo and the choices that become possible, not only the goods eventually delivered.

31.3.1 — Saved Waste Reinvested

Reinvestment remains a lawful public choice among uses that communities may value differently. Use Section 5.7.2's source-to-delivery record to expose the cost and feasibility of that choice; a burden analysis cannot determine it automatically.

Apply Section 30.6's comparison standards to the reinvestment, including displaced uses and wider changes. Reallocation or clawback must correct underperformance without abruptly abandoning essential workers and users.

31.3.2 — Peace as Breathable Order

The breathable order pictured in Chapter 27 is a horizon of reliable systems and plural freedom, not an inner state institutions can promise or score. Aggregate gains cannot excuse deprivation below the protected floor (Sen, 1999; Nussbaum, 2011; Galtung, 1969).

Protective margin can create shared benefits without being a pure public good. A reserve bed or crew can become unavailable to others when deployed. Its value lies partly in preventing one

interruption from spreading. Assess the margin against a stated stress case: what happens if demand rises while a key worker or supplier is unavailable? Count activation time and replenishment as well as nominal capacity. A crew without equipment or deployment authority is not the same reserve as a staffed, practiced response. Compare maintained spare capacity with feasible alternatives and current needs; a low utilization rate alone establishes neither waste nor resilience.

Preserve why a successful repair worked. If later budget cuts remove the staffing or redundancy that made the improvement possible, the same failure can return under a different name.

People should remain free to decide how to use the time that reliable systems return.

31.4 — Power Accountable for Its Consequences

Does power carry the cost of what it commands, breaks, delays and leaves behind? Budgets, reasons, remedies, maintenance, renewal and decommissioning make that consequence test observable. The test complements rather than supplies lawful authority (IMF, 2018; OECD, 2017b; Graham and Thrift, 2007). Report the interval between recognition of harm and repair, including interim relief and continuity of essential services while the remedy is completed.

31.4.1 — Decision Costs and Funded Repair

Financial and operational responsibility should follow the decision rights that create burdens, with enough resources assigned to the actor best able to prevent or manage the risk. This does not assign every loss to one institution or ignore shared causation.

When one institution selects the technology, another signs the contract, a third operates the service, and a fourth pays for complaints, each may deny control over the harmful result. Before failure, governance should map decision rights, information access, financial exposure, and repair duties together (Jensen and Meckling, 1976).

Fund independent review, evidence preservation, interim protection, appeal and repair before complaints arrive. Complainants should not carry the entire proof burden when institutions control the records, expertise and decision access. Exhaustion, displacement, illness or dependence on the service can make that imbalance harder to contest (HM Treasury, 2026a).

Use the differentiated remedies in Section 21.2.3, adding correction and notice for false records and visible retraction for public falsehoods. Distinguish individual relief from collective or future repair: payment to one claimant may leave a contaminated

site, discriminatory rule, broken archive or procurement failure unresolved. Affected people need standing to describe what restoration requires.

31.4.2 — Future-Carrying Power

Include long-lived operating, ecological and retirement obligations in the account. Discounting can compare time-separated costs, but the chosen rate is a substantive assumption for long-lived infrastructure and intergenerational harm. The record should show how results change under plausible rates and which duties remain even when present-value calculations make distant burdens appear small (HM Treasury, 2026b).

Future people cannot consent to present decisions. Current institutions can preserve their options by avoiding irreversible commitments when evidence is weak, protecting common assets, maintaining fiscal capacity and establishing review independent of the original sponsor. Source data, assumptions, maintenance manuals, legal records, environmental baselines and reasons for rejected alternatives should survive leadership turnover. This does not represent every future preference; it leaves later reviewers functioning systems and evidence with which to challenge inherited decisions.

31.5 — The Peace Horizon After the Red Team

The components of material peace can improve at different rates: more reliable housing may coexist with weak trust, debt reform with an unjust labor transition, and climate protection with displacement. Honest evaluation reports those mixed outcomes and allows partial gains or local reversal to remain visible. Use the process, impact, and system distinctions in Section 30.6.3 to state what spending changed, for whom, and under which conditions (HM Treasury, 2026a; OECD, 2022; OECD, 2025b). Each evaluation should name the decision it can inform, its deadline, the accountable authority, and the resulting change. Maintain a public fiscal record through political turnover, explaining any revisions to estimates, policy, or financing rather than treating the next announcement as a fresh beginning (HM Treasury, 2023; HM Treasury, 2026b).

31.5.1 — More Repairable Systems

Repairability combines redundancy, local competence, spare parts, documentation, user voice, modularity and authority to isolate a failing component (Ostrom, 1990; Weick and Sutcliffe, 2007; Vinsel and Russell, 2020).

Extend Section 24.6.1's exercises through restoration: recover records, replace vendors, relieve staff, correct false public information and withdraw temporary powers. Record access lost and workarounds used, while protecting participants' privacy, refusal and dignity (Weick and Sutcliffe, 2007).

31.5.2 — An Inheritance That Remains Open

A durable peace leaves future people more than a working arrangement. It leaves them the ability to improve it. Consider a public building designed so that rooms can change use without reconstructing the whole structure. Its present purpose matters,

but the design also creates possibilities that its first users may never need. The capacity to adapt is part of what later residents inherit.

Institutions can carry a similar openness. A decision record that explains why an arrangement was chosen gives successors something to reconsider when circumstances change. Training that includes diagnosis prepares people for unfamiliar faults. A relationship among neighboring services can make assistance available when one locality encounters a new problem. These are different ways of preserving room for the next decision.

The horizon is therefore not a final design held permanently in place. It is a civilization increasingly able to maintain what serves people and revise what no longer does. Its achievements can be passed onward because the knowledge and practical means of renewal travel with them.

That future has an ordinary expression. A person can enter a home, a workplace, or a public service and find that someone has thought about what happens after the first promise. The system has a continuing life, and the people within it have room to shape theirs. Peace becomes something received, used, and improved rather than something announced once.

31.6 — The Work That Must Be Sequenced Next

The next stage requires an actionable handoff rather than another declaration of aspiration. A signed constraints packet should identify what can proceed, what remains uncertain, and which protections or obligations cannot be deferred. The receiving authority may sequence or decline particular actions, but it must explain that choice against the recorded alternatives. Neither urgency nor the language of momentum can erase unresolved dissent or continuing duties (HM Treasury, 2026b; ILO, 2015). The handoff succeeds when another institution can understand the limits, assign responsibility, and know when the decision must be revisited.

31.6.1 — Material Constraints for Sequencing

Carry Section 28.7.1's packet forward, distinguishing prelaunch conditions, capacity developed during bounded operation and obligations surviving pause or closure.

Distinguish assets from the recurring staffing, maintenance and supplies that make them usable. Test replenishment and substitution: a reserve without resupply, a specialist without a training pipeline or a platform without a replacement can fail despite an impressive initial stock (HM Treasury, 2026b; OECD, 2025b; Vinsel and Russell, 2020).

Use Section 29.7.3's obligation-based shutdown costing. Distinguish orderly completion from emergency termination and identify which worker, user, record and residual-liability duties survive both (HM Treasury, 2026b; NIST, 2024; Hirschman, 1970).

Use the independent review and stopping rules in Section 30.9. The sequencing plan must also name who convenes each decision, supplies evidence, can veto on rights grounds, pays for remedy, and communicates the outcome. Separate authority must

not leave an obligation without an owner (Carpenter and Moss, 2013; OECD, 2022; NIST, 2024).

The scientific track remains a distinct invitation. Laboratories studying valence can develop measurement, routing, and endpoint tests within the research program described in The Science of Fairness. Public-service studies can test whether particular designs return usable time and capability. These tracks share a commitment to precise questions and informative comparisons while answering different questions with different evidence. Their connection is an opportunity for collaboration across the series, not a requirement that one result settle them all.

The next step can be a focused research collaboration. One proposed study would test two workflow changes: removing duplicate proof requirements and assigning a named case owner across handoffs. Four conditions would compare neither change, each separately, and both together. Where lawful and ethically appropriate, operational units would be randomly allocated to the four conditions. Teams that share staff would constitute one assignment unit. Common intake rules would identify in-scope requests before either change operates; this study cohort is not a presumption that every applicant qualifies for the requested benefit. Transfers and abandonment would retain the original assignment. Every arm would preserve eligibility rights and essential assistance. Ethics and operational review would precede enrollment; a known harmful process would be repaired rather than retained solely to create a control.

The primary comparison would be the combined workflow against the existing compliant workflow, analyzed by original assignment. It would pair average active applicant time over thirty days with verified delivery of the requested service by day thirty across the full intake cohort. Denials would be audited and reported separately, not counted as services delivered. Both

results are needed: evidence of a worthwhile time reduction and evidence that delivery has not fallen beyond a prospectively justified margin. A nonsignificant delivery difference is insufficient. Users and specialists would help justify the margin without relaxing entitlement or safety. Component comparisons and the factorial interaction would have a declared multiplicity plan. The additive interaction is the change from neither to both minus the sum of changes from neither to each component alone, calculated separately for time and delivery. Its interval would be estimated directly, not inferred from one significant arm and one nonsignificant arm.

Time would be measured in a consented sample drawn across all arms and intake outcomes, with inclusion probabilities and nonresponse recorded. Time spent before withdrawal still counts. Administrative records would retain the full in-scope cohort under a lawful, minimized data plan. The analysis would state which assumptions support inference beyond respondents and show how plausible missing reports change the conclusion. Staff hours would remain a separate outcome. Errors, repeat contacts, long waits, access failures, and privacy incidents would be monitored alongside delivery. A ninety-day follow-up would examine further work required to obtain or deliver the service after the initial thirty-day window, including deferred tasks, and any subsequent loss of service. Data collection would continue after enrollment ends until every included follow-up window has matured.

A twelve-week baseline and sixteen-week implementation period could serve as an initial planning frame, not a validated sample-size prescription. Power calculations would address independent team count, within-team correlation, unequal volumes, attrition, and the probability of meeting both primary criteria. The protocol would specify estimands, assignment,

treatment fidelity, confidence intervals, stopping authority, and responses to adverse events. Inference would respect the assigned operational units rather than count applicants sharing a unit as independent replications. Movement of staff or knowledge between arms would be recorded. Publish the protocol, data dictionary, analysis code, and appropriately protected results, including the recurring support required under ordinary staffing (HM Treasury, 2026a; Proctor et al., 2011). A well-estimated interaction would describe how the assigned components combine on the specified scale; it would not by itself establish the psychological pathway behind that result.

31.6.2 — Answerability Before Triumph

The final discipline joins open criticism, adversarial testing, scientific vulnerability, and comparative repair. An open society keeps criticism institutionally effective; adversarial collaboration lets rivals shape tests before results; the PTVC hypothesis remains open to testing, refinement, or rejection; comparative justice asks which remediable burdens can be reduced without waiting for a perfect order. The horizon belongs to no victor. It belongs to making power explain itself, cost visible, refusal protected, useful systems maintained, and failed systems withdrawable (Popper, 1945; Mellers et al., 2001; Sen, 2009).

A concrete precedent exists for organizing disagreement into discovery. The Cogitate Consortium and collaborators (2025) brought proponents of integrated information theory and global neuronal workspace theory into a preregistered comparison. Across 256 participants, separate cohorts contributed brain measurements using three complementary recording methods. The project separated theory proponents from data collection and analysis and specified divergent predictions before testing. Its

relevance here is the organization of a demanding comparison, not evidence for PTVC.

For this series, an initial collaboration could produce one agreed document stating which observations would distinguish the terminal hypothesis from specified rivals and which measurement assumptions connect those observations to the claim. Calibration and endpoint rules would be fixed independently of whether a result approaches zero. Simulation could test whether the proposed analysis distinguishes the declared models under those assumptions; a successful simulation would qualify the method for further work, not establish the hypothesis. Fresh data would then test a preregistered contrast. If the current design leaves rivals observationally indistinguishable, the collaboration's next product would be the measurement or discriminating prediction needed to separate them. That is a concrete task for laboratories and theorists to build together.

An initial calibration exercise can have a checkable success criterion. In an illustrative model with equally spaced signed readings, subtracting each history's own mean forces its sum to zero, whether or not the original sum was zero. Multiplying every reading by the same positive constant, by contrast, preserves whether the sum is zero. The same artifact occurs with irregular observations if a history is centered on its own duration-weighted mean: the resulting duration-weighted sum is forced to zero. Researchers could send synthetic histories with known zero and nonzero totals through the same preregistered processing pipeline and test recovery under declared missingness and measurement error. This checks the processing method, not a measure's validity for conscious experience. Recovering planted differences would be a useful first research product before applying the instrument to a terminal claim.

Peace becomes credible in a household able to plan past the next bill, a worker whose safety is not traded for output, a town that maintains what it builds, and a public office able to admit and repair an error. A reliable address and a predictable schedule give people more room to keep promises to one another. The horizon is not perfection. It is ordinary life with more time, security, capability, and room to begin again.

Glossary

A

Access Capture: Influence acquired through privileged proximity, such as private briefings, early drafts or advisory access, without formal decision-making authority.

Accessibility: Spaces, routes, equipment and interfaces that people can actually use across differences in mobility, perception, language and cognition, with maintenance that preserves that access.

Accumulated signed balance: Within one qualifying stream and a fixed valid representation, the positive phenomenal magnitude accumulated so far minus the negative phenomenal magnitude accumulated so far at a chosen cut. Under exact PTVC, the signed remainder from that cut to the truth-level endpoint equals the negative of this accumulated balance.

Administrative burden: The time, effort, confusion, documentation, waiting, repeated explanation, and procedural strain required to obtain, keep, challenge, or correct a service, right, benefit, remedy, or ordinary option.

Administrative Capacity: The staff, skills, systems, resources and lawful authority needed to carry out an institution's responsibilities reliably. It includes service delivery, records, procurement, supervision, language access, correction, appeals and continuity. Tax administration is one application.

Aging and Care Events: Changes in health, mobility, memory, income, housing or support that alter a person's care requirements over time.

Aging and Long-Term Care: The systems of housing, clinical care, personal assistance, family support, accessibility, financing, and social connection that sustain people as needs change with age or disability. Home-based and facility-based

care are evaluated by safety, dignity, staffing, continuity, choice, respite, and accountability rather than by a single preferred setting.

Anti-capture: Institutional design that limits the ability of donors, vendors, lenders, firms, officials, criminal networks, or other concentrated interests to convert dependence or privileged access into unanswerable control.

Anti-Money-Laundering Burden: The documentation, screening, delay and exclusion imposed by systems intended to detect illicit financial flows, assessed alongside their protective contribution.

Antitrust Evidence and Competing Risks: The evidence and tradeoffs used to assess competition, including market definition, entry, buyer power, foreclosure, innovation, remedies and administrability.

Attention as Material Capacity: Attention is a scarce coordinating resource with an opportunity cost distinct from clock time. Pain, caregiving, language translation, fear of status loss, and repeated context switching can make equal periods impose unequal demands. Observe time and cognitive demand separately, and record conditions that impede recovery.

B

Balance: On the primary additive PTVC branch, equality of a qualifying stream's complete finite positive and negative phenomenal magnitudes, equivalently a terminal signed residual of zero. Balance is intrastream and terminal; it does not require neutrality at every moment, equal gross experience across lives, or cancellation across people.

Bearer: The truth-level subject or unit to which qualifying conscious experiences belong and within which the proposed accumulation is evaluated. The bearer is determined by the declared stream, ownership, and routing rules, not by convenience of observation or legal identity alone.

Below-Floor Work: Employment whose pay or conditions fail to sustain basic material stability, despite participation in paid work.

Beneficial Ownership and Financial Route: The identification of who ultimately controls or benefits from an entity, asset or transaction, beyond its formal owner or intermediary.

Benefit Delay Cost: Harm attributable to delayed support, including arrears, interrupted care and crisis costs borne by applicants, dependents and other services.

Biological death: The end of biological life. It may be associated with a stream endpoint in some cases, but PTVC does not define biological death as the truth-level endpoint by stipulation.

Broken Maintenance: Failure to inspect, service, repair or renew a system sufficiently to preserve its intended function. Causes may include underfunding, weak planning, skill shortages, disrupted supply, unclear responsibility or corruption. A system may continue operating while this backlog accumulates.

Burden transfer: The movement of cost, risk, delay, labor, or maintenance from one actor, place, budget, or time period to another. A visible saving is incomplete until the transferred burden is identified.

Burden Transfer Across Scales: The movement of costs between households, local services, regions, national budgets and future periods through connected systems.

C

Capability: What a person can realistically do or become given their body, environment, support, rights, resources, access, and social conditions.

Capital Flight Risk: Exposure to destabilizing movement of residents' funds abroad, distinguished from withdrawal by foreign investors or a fall in new inflows. These channels can interact. The material concern is how financial adjustment affects people and institutions with less ability to move or protect their resources.

Care Access as Peace Infrastructure: A health system's ability to sustain an affordable and usable route from first contact through treatment, follow-up and recovery.

Care and Social Infrastructure Finance: The capital, operating funds, skilled labor and maintenance needed to sustain childcare, eldercare, disability support, health and community services.

Care Interruption as Harm Multiplier: The additional harm caused when an interrupted care pathway also disrupts medication, work, caregiving, records or recovery.

Care Worker Wages: Compensation for paid care work, evaluated with staffing, hours, safety and progression because workforce stability affects continuity of care.

Caregiver Support: Training, respite, income protection and practical assistance that help unpaid and paid caregivers sustain care without exhaustion or unsafe conditions.

Change-Order Risk: Exposure to post-award cost or scope increases arising from genuine uncertainty, poor specifications, strategic underbidding or opaque renegotiation.

Childcare Access and Labor Access: The relationship between reliable childcare and a caregiver's practical ability to take work, training or other scheduled opportunities.

Civilian protection: The practical protection of noncombatants from direct violence and from preventable collapse in food, water, shelter, care, movement, communication, and other essential systems during conflict or crisis.

Climate adaptation: Changes in infrastructure, institutions, finance, land use, services, and behavior that reduce exposure or vulnerability to climate-related hazards while preserving dignity, local authorship, and future options.

Closure: Satisfaction of a declared terminal target. On the primary additive PTVC branch, closure means that the finite positive and negative phenomenal magnitudes within one qualifying stream are exactly equal; equivalently, its terminal signed residual is zero.

Community Repair Cost: The resources required to restore safety, access and social confidence after harm affects people beyond the direct parties.

Competing Claims on Land: Overlapping claims involving title, customary use, Indigenous rights, housing, livelihoods, infrastructure, ecology and displacement that require contextual assessment.

Confidence-Building Steps: Limited, verifiable actions that reduce uncertainty or fear sufficiently to make further cooperation feasible.

Conscious stream: A temporally connected sequence or field of conscious experience treated as one unit under declared identity and routing criteria. It is not automatically identical to a biological organism, legal person, biography, memory narrative, or data record.

Consciousness: Subjective experience: the presence of states that are experienced by a bearer. The series treats the presence, structure, boundaries, and measurement of consciousness as empirical and conceptual questions rather than equating consciousness with any single report, behavior, neural signal, computational output, or biological marker.

Continuity: The ability of an essential service, right, relationship, payment, record, or care route to remain usable through ordinary disruption and to recover promptly after failure.

Continuity of Support: The preservation of coordinated help across transitions between crisis stabilization, clinical care, housing, addiction treatment, correctional release, community services, and ordinary life. It depends on reliable handoffs in medication, appointments, transport, insurance, housing, records, and trusted human contact.

Contribution, Pay, and Material Recognition: The relationship among socially useful work, compensation, and the material recognition required for people to sustain that work with dignity. Contribution includes paid and unpaid forms of labor, while compensation and recognition should reflect the conditions, skill, responsibility, risk, and continuity that shared systems depend on.

Controls and Freedom Risks: Risks to access, rights and ordinary activity created by emergency restrictions on capital, withdrawals, prices, foreign exchange or transactions.

Corruption drag: The wider cost created when entrusted power is bent toward private advantage or hidden access. It includes not only diverted money but delay, uncertainty, distrust, defensive duplication, weaker maintenance, and lost participation.

Corruption Drag Measures: Indicators of friction associated with hidden power, such as procurement anomalies, unresolved complaints and delay. They prompt investigation; no single indicator establishes corruption.

Corruption-Resistant Capacity: Administrative capacity designed to prevent capture while preserving the lawful discretion needed to resolve legitimate exceptions.

Credit Access and Exclusion: The availability and terms of useful credit, including barriers arising from discrimination, collateral, information gaps or weak local institutions.

Credit as Future Capacity: The use of purchasing power now against a future repayment claim to finance a household, enterprise or public investment.

Credit Data and Privacy: Limits governing the collection, use, sharing, correction and retention of information used to assess creditworthiness.

Creditor and Public Burden: The allocation of restructuring costs among creditors, taxpayers, workers, suppliers, future borrowers and public services.

Crisis response: A response to acute distress or danger that provides appropriate care and protection without making punishment the routine route into assistance.

Currency Trust: Confidence that money will remain sufficiently usable and stable to support saving, pricing, payment and contracting across time.

Cybersecurity and Tamper Resistance: Protection against unauthorized alteration, disclosure, theft, forgery or destruction of records, with the capacity to detect and correct compromise.

D

Dashboards as Bounded Attention Tools: Displays that help identify changes requiring attention in a system, without treating a signal alone as a diagnosis or a verdict.

Data Minimization in Service Access: Collecting only the data necessary for a defined service or decision purpose, with appropriate access, retention, correction and deletion rules.

Debt: An obligation against future income, assets or other resources, evaluated by its terms, productive use and effect on the borrower's later options.

Debt and Climate Inheritance: The obligations and environmental conditions passed to future generations, assessed alongside any durable capacity or protection created by present investment.

Debt and Credential Pressure: The interaction between educational opportunity, borrowing obligations and later occupational or household choices, including completion and earnings uncertainty.

Debt and Family Stress: The strain that repayment pressure, collection and financial uncertainty can place on household relationships, sleep and decision-making.

Debt Collection Humiliation: Loss of privacy, dignity or social standing caused by collection practices, distinct from the amount lawfully owed.

Debt Data Privacy: Control over the accuracy, circulation and continued use of debt information after it enters screening, collection and other decision systems.

Debt Relief and Household Stability: The relationship between reduced debt obligations and a household's ability to sustain essentials, refuse unsafe terms and plan beyond immediate payments.

Debt Service as Policy Narrowing: Repayment claims on future revenue can restrict responses to new public needs. Assess which functions become vulnerable under interest-rate, revenue, currency, or rollover stress, and whether the financed capacity remains reliable enough to justify its claim on later budgets.

Debt-Service Burden: The resources committed to interest and principal payments, considered alongside the benefits financed and the obligations remaining under stress.

Defense and Fossil Transition Income: Income continuity for workers and regions affected by defense conversion or fossil-fuel transition while viable alternative livelihoods are established.

Defensive duplication: Extra backups, checks, inventories, legal layers, security measures, or parallel systems created because a shared route is not trusted. Some duplication is prudent resilience; some becomes avoidable overhead.

Defensive Spending as Social Structure: Persistent reliance on private or parallel protective arrangements when shared services or commitments cannot be relied upon.

Deficits as Tool, Not Shame: Evaluation of budget deficits by their cause, duration and use, distinguishing cyclical

stabilization, emergencies, durable investment and recurring operating gaps.

Delay as Corruption Cost: Delay imposed or prolonged through bribery, favoritism or gatekeeping, distinguished from delay caused by capacity constraints or necessary review.

Dependency Risk: Exposure of essential work to the withdrawal or refusal of funding from a donor on whom the institution relies.

Digital Convenience: Time or effort saved through usable digital services, evaluated alongside the availability of assistance and viable alternatives.

Disability and Language Access: The accommodations and communication routes needed for people with differing disabilities and languages to use intake, services, correction and appeal.

Disability support: Tools, assistance and environmental or procedural adjustments that make participation possible for people with disabilities.

Documentation Burden: The work required to discover, obtain, prepare and submit the evidence a process demands, including navigation of unclear or fragmented requirements.

Domestic Opacity and Foreign Fear: Uncertainty across borders produced when a state conceals decisions or capabilities with military, financial, technological or ecological consequences for others.

Donor Exit Plan: An agreed route for continuing, transferring, narrowing or responsibly closing funded work if support ends or must be refused.

Donor firewall: A set of limits separating financial support from control over doctrine, evidence, appointments, standards,

public explanation, or transition decisions. The relevant question is the authority and dependency created by the funding arrangement, not the donor's stated intent.

E

Eldercare and Disability Support: Care and assistance that preserve safety, function, participation and choice as needs change with age, disability or illness.

Emergency Procurement Overreadiness: Emergency purchasing capacity or commitments that exceed a defensible risk requirement or persist without reassessment after conditions change.

Endpoint: The boundary at which a qualifying conscious stream ceases for purposes of the truth-level target. An endpoint must be defined by the stream model rather than equated automatically with biological death, loss of contact, device shutdown, or the last observed report.

Energy Access: Reliable, affordable and usable energy service, supported by the generation, networks, storage, maintenance and response capacity needed under local conditions.

Enoughness: A practical orientation toward sufficiency: having and using enough material, temporal, relational, and experiential room for a dignified life without treating endless accumulation as the measure of success. Enoughness is compatible with ambition, joy, generosity, and growth where they expand rather than consume human freedom.

Enoughness Without Imposed Scarcity: Sufficiency achieved through usable access and choice, without prescribing deprivation or one approved way of living.

Essential systems: The infrastructures and services whose failure quickly threatens life, health, agency, or basic

participation, including food, water, energy, housing, health care, sanitation, mobility, communication, payments, and critical public records.

Ethical floor: The minimum safeguards governing the project: present suffering receives care; consent and refusal remain meaningful; harmful testing is prohibited; dignity and rights do not depend on PTVC; and scientific, legal, institutional, and spiritual claims remain within their proper authority.

Evasion and Trust Collapse: The loss of willingness to contribute when people perceive shared obligations as unequally enforced; distinguish illegal evasion from lawful avoidance, confusion and inability to pay.

Exact closure: On a declared PTVC branch, the truth-level condition that a qualifying stream's terminal signed residual is exactly zero under the specified bearer, representation, domain, endpoint, and validity conditions. Exact closure is distinct from a near-zero estimate, a population mean of zero, or an uncertainty interval that merely contains zero.

Exit: A usable ability to leave an institution, contract, program, service, relationship, technology, or funding arrangement while retaining applicable rights and access to essential support, with protections for privacy and future opportunity. It does not necessarily erase ordinary contractual obligations or every cost of leaving.

Exit With Dignity: Departure from a program or institution without retaliation, humiliation, disproportionate penalties or loss of unrelated essential support.

Extractive Growth: Expansion in measured output, revenue or valuation achieved partly by shifting unaccounted costs onto workers, households, ecosystems or future periods.

F

Failure-Cost Disclosure: A statement of the supported costs and risks of inaction, separating observed consequences from scenarios and identifying evidence that could change the decision.

Fairness: A family of questions about how benefits, burdens, opportunities, rules, reasons, and remedies are distributed and justified. In this series, ethical and institutional fairness remains distinct from PTVC's scientific terminal-closure hypothesis; neither reduces to the other.

Family Stress and Material Systems: Household strain arising partly from wages, schedules, housing, care, transport, debt or service failures, rather than only from interpersonal causes.

Fast Relief as Cost Prevention: Timely assistance that prevents a defined escalation of harm, evaluated against a credible comparison and the full cost of delivery.

Financial Rent: Income arising from control of financial access beyond the value of the service and risk-bearing provided.

Fiscal Fear Rhetoric: Appeals to fiscal anxiety used to narrow policy choices without comparable scrutiny of alternatives and their consequences.

Food, Utilities, and Basic Stability: Reliable access to nutrition, water, power and other household utilities needed to sustain health, care, hygiene and usable shelter.

Fraud and Abuse Safeguards: Proportionate, correctable measures against misuse that distinguish intentional fraud from error, misunderstanding and changed circumstances.

Fraud Control Without Cruelty: Protection of shared resources without treating poverty, unusual circumstances or irregular records as proof of dishonesty.

G

Gross phenomenal magnitude: The sum of a stream's positive and negative phenomenal magnitudes before cancellation under the declared additive representation. Equal terminal balances can coexist with different gross magnitudes, as the hypothetical histories in Section 1.6.1 illustrate.

H

Health Delay Cost: Harm attributable to postponed diagnosis, treatment or support, including delays before contact, between referral and treatment, during care and in correcting errors.

Health Workforce Burden: The workload, staffing pressure, emotional strain and administrative friction carried by health workers, including their effects on service continuity and patient care.

Hidden burden: A cost carried in time, stress, risk, unpaid labor, waiting, maintenance, exposure, privacy, or lost options that is omitted from the visible price, budget, success claim, or public narrative.

Homelessness: The loss of shelter together with the safe storage, rest, privacy, address and routines needed to exercise ordinary options.

Household Panic as Public Evidence: Household distress treated as a signal for investigating possible material or institutional causes, rather than as proof of any single cause.

Housing and Neighborhood Stability: Continuity in home and neighborhood relationships that supports children's routines, schooling, health care and sense of safety.

Housing Rent Pressure: The degree to which housing payments constrain a household's ability to meet other essentials, absorb shocks and plan.

Housing Supply and Anti-Speculation: The interaction of construction, credit, land policy and investment behavior in housing availability, price and ownership; restraint of speculation does not itself create missing homes.

Housing, Food, Care, and Time: Four interdependent components of material stability. Housing anchors safety, privacy, storage, address, and continuity; food sustains health; care preserves function and recovery; and discretionary time makes navigation, relationships, rest, and planning possible. Failure in one domain can shift cost into the others.

I

Illness and Disability Events: Changes in health or function that alter care, accommodation, income, transport or housing needs, including sudden, progressive, fluctuating and lifelong conditions.

Incidence: The distribution of a policy, market, institutional, or infrastructure burden across the people and systems that ultimately carry its money, time, risk, labor, delay, privacy loss, exposure, or lost options.

Income Bridges: Temporary income support spanning the interval between a disrupted livelihood and a viable replacement.

Independent warrant: A reason for a policy or practice that does not depend on PTVC being true. Rights, care, public health, safety, capability, productivity, and institutional evidence can independently justify material repair.

Inflation as Household Pressure: The effect of rising prices on a household's usable resources, reflecting its spending mix and income changes as well as the headline inflation rate.

Infrastructure Inequality: Unequal access to reliable transport, communication, water, energy, public facilities and maintenance across places or populations.

Innovation That Lowers Burden: An innovation evaluated by the costs and constraints it removes, together with any burdens it introduces, rather than by novelty alone.

Institutional Delay Cost: Harm and additional work caused by institutional delay, attributed to its carriers and causes while distinguishing necessary review from avoidable friction.

Institutional Redundancy Under Mistrust: Parallel records, audits or procedures created because institutions cannot safely rely on shared information or one another's work.

Institutional Responsibility: Accountability for the contribution an institution's acts, omissions or service failures make to preventable harm.

Intergenerational Cost of Corruption: The later costs of corrupt decisions, including weakened assets, future repair bills and reduced opportunities inherited by people absent from the original decision.

Intrastream: Evaluated within one conscious stream at a time. For PTVC, positive or negative experience in one stream cannot be used to balance another stream.

J

Job Loss and Debt Events: The mismatch between sudden income loss, continuing household obligations and the timing of available assistance.

Journalistic Reconstruction Burden: The investigative work required to reconstruct public decisions from fragmented records, interviews, ownership information and disputed claims.

K

Knowledge Access: Usable access to the information, research, instructions and records needed for informed action and public judgment.

L

Labor Market Power: The ability of employers or other labor-market actors to influence pay and conditions when workers' practical alternatives are constrained.

Land as Shared Condition: Land considered as the location required for housing, work, food, infrastructure, ecological function and cultural life, alongside its role as property.

Land Rent: Returns attributable to control of a site and its location advantages, distinguished from returns to buildings, improvements and other produced capital.

Last-Minute Work Burden: The disruption imposed by short-notice changes to shifts, hours or availability, including canceled work and effects on care, transport and recovery.

Law of Fairness: The series' public name for the proposed Per-Stream Terminal Valence Closure hypothesis and the wider interdisciplinary inquiry built around it. In scientific use, it remains a candidate hypothesis rather than an established law of nature.

Learning Outcomes and Their Limits: Indicators of learning and educational progression, interpreted with context, selection and incentive effects rather than treated as complete measures of a student or school.

Ledger awareness: Conscious knowledge or tracking of a stream's accumulated phenomenal balance. PTVC does not require such awareness, and the hypothesis does not establish

that a person, institution, pastoral figure, or artificial system can read an actual personal ledger.

Legal and Compliance Overreadiness: Legal or compliance activity beyond what a defensible protective, evidentiary or coordinating purpose requires.

Lifecycle cost: The full cost of an asset, policy, contract, or system across design, construction, operation, staffing, maintenance, failure, correction, replacement, transition, and retirement rather than only its initial price.

M

Maintenance as Civilization Work: The continuing work required to preserve the function and reliability of shared physical and digital assets after their initial delivery.

Maintenance Debt Measures: Indicators of postponed inspection, repair, renewal or support, including the consequences and priority of the accumulated backlog.

Maintenance Failure: See Broken Maintenance.

Maintenance Labor Dignity: Recognition of maintenance workers' expertise, fair conditions and role in detecting recurrent failures and practical improvements.

Maintenance Skill: The knowledge and practical ability to inspect, diagnose, clean, calibrate, repair, replace and adapt physical or digital systems.

Market Fear and Capital Flight: Risk-driven financial adjustment that can include residents moving funds abroad, foreign investors withdrawing, or new inflows falling, with possible effects on currency, credit, imports, employment and household prices.

Material breathability: The usable margin that lets people plan, care, work, rest, recover, refuse harmful terms, and survive ordinary shocks without every essential need becoming a crisis.

Material incidence: The method of following a decision or disruption through the concrete people and systems that carry its money, time, health, labor, risk, delay, attention, and lost options.

Material peace: Peace understood partly through the reliable material conditions of ordinary life: safety, access, continuity, repair, enoughness, and reduced fear-driven overhead. It complements rather than replaces justice, voice, culture, and freedom from direct violence.

Maximal qualifying conscious stream: A qualifying conscious stream extended as far as the adopted identity and routing criteria permit, so the analyzed stream is not an arbitrary fragment of a larger routed stream.

Meaningful Work Without Coercion: Work valued for its purpose or contribution without treating that value as permission to override fair terms, safety, recovery, voice, or freedom to leave. A mission does not erase labor costs or make participation a condition of unrelated essential support.

Medication and Treatment Continuity: The preservation of clinically appropriate medicines, therapies, records, and follow-up across changes in insurer, prescriber, pharmacy, employer, residence, hospital, school, correctional setting, or disaster context. Continuity requires a current reviewable record, responsibility for handoff, and a route for timely correction when administrative change threatens care.

Mental Health as Material Stability: Mental health cannot be reduced to income, housing, or public services. Distress and

mental illness have biological, psychological, relational, developmental, cultural, and social dimensions, and clinical assessment belongs to qualified professionals. Material systems matter because they can intensify risk, interrupt care, or create conditions in which recovery is harder to sustain.

Metric Correction and Retirement: A governance rule requiring every operational metric to have review dates, complaint routes, misuse triggers, privacy limits, decision-relevance tests, correction procedures, and named retirement authority. These controls prevent a provisional measure from becoming permanent solely through contracts, software, spreadsheets, or institutional habit.

Municipal Austerity Pressure: Pressure to reduce municipal services or maintenance as debt obligations, fiscal rules and limited revenue constrain available choices.

N

Negative balance: A negative signed difference in which negative phenomenal magnitude exceeds positive phenomenal magnitude over the specified interval or accumulated portion of a stream.

Neighborhood Repair: Restoration of usable streets, public spaces and local facilities that support safe access and everyday participation.

Neutral origin: The independently justified zero point of the adopted valence representation, separating positive from negative phenomenal valence. The origin must be fixed or calibrated independently of the desired terminal result because shifting it can change cumulative closure. It is an experience-scale reference point, not terminal neutrality or a claim that the final conscious moment is neutral.

No adequate estimate: A prespecified inference status used when the target or estimand is undefined or cannot be identified or bounded under the available data and maintained assumptions, including because ethically permissible evidence is insufficient. It is distinct from a valid but unresolved analysis and from evidence rendered unusable by a failed protocol, consent, endpoint, measurement, provenance, computation, or participant-protection gate. The reason and affected estimand should be recorded.

Non-abandonment rule: The principle that a transition should not obtain its apparent success by leaving workers, regions, users, debtors, migrants, or communities to absorb the adjustment without a credible path to continuity and repair.

Nontransferable: Confined to the same qualifying conscious stream: one person's phenomenal valence cannot be used to balance another person's experience.

O

Opportunity Cost of Corruption: The value of the best feasible alternative use of money, time, skills, land or institutional attention consumed by corruption.

Opportunity Cost of Suspicion: The value of the best feasible alternative use of time and attention spent on repeated checking and reconstruction, such as work, care or investment.

Option Restoration: Expansion of a person's usable capabilities through changes that enable safer refusal, recovery, access, privacy and planning.

Ordinary abundance: A rich but maintainable everyday life built from dependable housing, food, care, energy, mobility, public space, time, beauty, learning, and other usable

capabilities rather than from permanent emergency or endless defensive consumption.

Overreadiness: Defensive capacity that exceeds a supportable risk model or persists after conditions change, tying up labor, money, attention, land, or institutional flexibility beyond what residual protection requires.

Overwork and Health Cost: Health, safety, care and public costs associated with excessive hours, inadequate rest or sustained workload beyond capacity.

Ownership: The prospectively specified rule assigning each qualifying phenomenal contribution to its bearer without omission, double counting, or cross-stream transfer. Ownership answers whose experience a contribution is; routing governs continuity through time and transitions.

P

Parental Time and Care: Time available for the relationships, practical care and support through which caregivers contribute to children's development and learning.

Parity and Capacity Matching: Allocation of commitments and exposure according to each party's practical capacity, while maintaining equal dignity and mutually intelligible obligations.

Partial, Full, and Structured Relief: Changes to debt obligations ranging from deferral and revised terms to principal reduction or discharge, under the distinct rules applicable to each borrower and debt type.

Peace dividend: Usable capacity demonstrably released by reduced conflict risk, corruption drag, defensive duplication, or overreadiness, after residual security, conversion, maintenance, worker, and transition costs are counted.

Pension and Workforce Liabilities: Future commitments to workers arising from employment, including pension obligations that extend beyond the current budget cycle.

Per-Stream Terminal Valence Closure (PTVC): The hypothesis that, within one separately routed maximal qualifying conscious stream, finite positive and negative phenomenal magnitudes are exactly equal at its true endpoint under a fixed representation calibrated to a neutral origin. The claim is intrastream and nontransferable; Section 1.6.1 states the proposal and explains the distinctions used in this volume.

Phenomenal magnitude: The represented amount of positive or negative felt experience assigned to a stream under a declared valence representation. Positive and negative gross magnitudes remain distinct from their signed net difference.

Phenomenal measure: A measure assigned to conscious experience for representing phenomenal valence across a stream. It is conceptually distinct from clock time, event count, memory strength, neural activity, or report frequency.

Phenomenal valence: See Valence.

Political Dynasty Risk: Structural advantages that enable public leadership to circulate within narrow family networks through inherited wealth, recognition, access and connections.

Positive and negative phenomenal magnitudes: On the primary additive branch, the nonnegative complete-stream totals of the positive and negative parts of the signed phenomenal measure. PTVC proposes that these two finite quantities are exactly equal within each qualifying stream.

Positive balance: A positive signed difference in which positive phenomenal magnitude exceeds negative phenomenal

magnitude over the specified interval or accumulated portion of a stream.

Precarity: A condition in which basic security depends on unstable income, housing, care, legal status, schedules, credit, or other fragile arrangements, so relatively small shocks can sharply narrow a person's usable options.

Preventive care: Care intended to prevent or reduce later illness and deterioration. Its benefits can include health and avoided suffering as well as changes in expenditure; the fiscal result depends on the service, population, and time horizon.

Prices and Moral Judgment: The distinction between a price signal and an ethical assessment: prices can reflect several market and institutional causes without establishing an exchange’s fairness.

Pricing Power Over Essentials: Supplier influence over the terms of necessary goods or services when buyers have limited ability to postpone purchase or use alternatives.

Private Security Overreadiness: Private protective expenditure exceeding a defensible risk requirement. The existence of private security or backup provision alone does not establish excess.

Productive growth: Expansion that leaves behind durable real capability such as housing, care, energy, skill, tools, infrastructure, knowledge, resilience, or maintainable production capacity.

Productive Wealth: Resources organized into enterprises, tools, assets or services that produce useful, durable capacity.

Prosperity as Repair Capacity: Prosperity assessed partly by the ability to detect damage, restore function and support recovery, rather than only by current output or wealth.

Prosperity Beyond GDP Alone: Assessment of prosperity using national accounts alongside evidence on distribution, well-being, ecological conditions and durable capability. National accounts record production, income, consumption, investment and balance sheets; they also support productivity and fiscal analysis.

Protected Security Domains: Information whose disclosure could create specific security or life-safety risks, subject to protected oversight rather than exemption from accountability.

Provenance: The documented origin and history of a source, record, artifact, or derived output, including the custody, transformations, permissions, versions, and corrections relevant to its interpretation and use.

PTVC: See Per-Stream Terminal Valence Closure (PTVC).

Public Capacity and Reserves: Institutional capability and buffers retained to sustain essential functions and respond to shocks before they become emergencies.

Public Land Stewardship: Management of public land with attention to present use, ecological and cultural value, and future public options rather than sale proceeds alone.

Public Services Under Suspicion: Service processes whose verification demands impose repeated burdens on eligible users, assessed against the actual integrity risks addressed.

Public Systems as Peace Infrastructure: Reliable shared services that reduce the private protective expenditure and work required to sustain ordinary life.

Public Tracking and Revision: A record connecting a verified burden reduction to released capacity and its eventual use, with correction when the claimed transfer fails.

Public-Sector Redesign Support: Staff time, training and continuity arrangements needed to implement changes to public processes, technology and service delivery.

Q

Qualifying conscious stream: A conscious stream that satisfies the declared scope, identity, routing, inclusion, and validity conditions for the target being considered. These criteria must be fixed independently of whether the stream appears to satisfy closure.

R

Reciprocal transparency: A transparency arrangement in which parties disclose comparable, relevant information under mutual rules, protected boundaries, verification, correction, and review rather than requiring one side to accept asymmetric exposure.

Redaction and Privacy Review: Assessment and removal or protection of sensitive information before disclosure, preserving public accountability while limiting foreseeable harm.

Reduced Litigation and Dispute Cost: Potential savings when clear records narrow factual disagreement. Their extent depends on the dispute and evidence; transparency does not guarantee lower litigation costs.

Reduced Panic as a Shared Benefit: The wider benefit that may follow when more reliable systems reduce avoidable household insecurity. Effects can extend to workplaces, schools, and public services. Such spillovers do not by themselves make every stabilizing service a public good in the technical economic sense.

Reference class: The prospectively declared domain over which a claim and its quantifier range, specified by eligible streams or cases together with the representation, endpoint regime, population, setting, and measurement architecture. It is not automatically the enrolled sample or analyzed cases; outcome-driven exclusions define a new claim.

Regional Cost Differences: Geographic variation in the cost and accessibility of essentials, relevant to how much usable security a nominal wage provides.

Regional Repair Priority: Priority for places with accumulated disinvestment, industrial loss, infrastructure decline or other burdens that weaken usable local opportunity.

Regional Transition Capacity: The local institutions, skills, services, infrastructure and finance needed to sustain a region through industrial or technological change.

Released capacity: Resources recovered when evidence supports reducing a defensive layer while funding residual protection, transition and the intended new use.

Reliability Metrics Without Ranking People: Measures of the dependable performance of a service, network, or institution, such as uptime, response time, continuity, failure frequency, recovery, or access. Their object is the system, not the worth of the people who depend on it, and the denominator, service standard, and failure condition must be explicit.

Rent Capture Versus Useful Return: Useful return rewards value created or risk responsibly carried. Rent capture derives income from control over scarcity, dependency, exclusion, or rules that prevent alternatives. The boundary is not always visible in an accounting statement because the same payment can contain both elements.

Rent, Debt, and Medical Delay: A household feedback loop in which high housing costs compress disposable income, debt carries unmet costs forward, and delayed medical care can turn money and time scarcity into greater health and financial risk. The interaction matters because households experience these systems jointly even when institutions administer them separately.

Repair as Peace Practice: Repair that can reduce conflict generated by unreliable shared systems, without assuming that improved infrastructure alone resolves antagonism.

Repair capacity: The people, authority, records, money, tools, reserves, and institutional memory needed to detect damage, restore function, support those harmed, correct causes, and prevent recurrence.

Repair Priority: A rule for ordering maintenance and remediation by the human and systemic consequences of delay, not only by asset condition or accounting category. The priority record links backlog to safety, health, school continuity, work, access, emergency response, household cost, and public trust.

Representation: A mathematical or conceptual encoding of phenomenal valence, time, state, or route that preserves the distinctions required by the stated research target.

Resilience: The capacity to preserve essential function, adapt under stress, and recover after disruption without transferring intolerable burden to people with less reserve.

Response Time: The elapsed time between a request, incident, or identified need and the relevant institutional response. Because delay can consume money, health, work stability, safety, options, and trust before formal losses are recorded,

response time is a substantive burden measure as well as an operational metric.

Restricted gift: Funding limited to a defined permitted use. In the proposed donor framework, acceptance must also preserve independence over scientific conclusions, doctrine, rights, public authority, personnel, records, and mission.

Restructuring: A negotiated or governed change to debt terms, such as maturity, interest, payment schedule, principal, or creditor coordination, intended to restore a viable future rather than preserve an impossible claim unchanged.

Retraining That Leads Somewhere: Training connected to a credible path into existing employment, advancement, self-employment or another useful role.

Risk Compression: The concentration of consequences when a household lacks the reserve to absorb an ordinary setback without cascading losses.

Route: A proposed process or construction through which a target might be reached or its consequences generated. A route has its own assumptions and evidence and must remain distinct from the target it is intended to explain or realize.

Routing: The prospectively specified treatment of continuation, pause, restoration, fission, fusion, copying, replacement, and unresolved continuity, including how earlier and later phenomenal contributions are assigned to streams.

S

Scarcity: A constraint on access or capacity. Physical scarcity arises from real material limits; administered scarcity arises from allocation or procedural design; manufactured scarcity is deliberately preserved or intensified to create leverage, rent, or exclusion.

Scarcity by Design: Supply or access constraints produced by institutional choices rather than unavoidable limits; distinguish protective rules from restrictions that preserve exclusion or rent.

School Resource Inequality: Unequal access to the staff, facilities, materials and support that shape students' actual learning conditions.

Security dilemma: A strategic interaction in which one actor's defensive preparation is interpreted as threat by another, prompting reciprocal measures that can increase insecurity even when neither side initially seeks escalation.

Security Overbuild: Protective measures that exceed a defensible risk requirement or remain after their justification has weakened.

Service capacity: An institution's practical ability to turn a service entitlement into timely, competent assistance at the point of need.

Signed remainder: The net signed phenomenal contribution from a chosen cut to the truth-level endpoint under the declared representation. Under exact PTVC, it equals the negative of the accumulated balance at that cut. It constrains the future net remainder, not either gross positive or gross negative total separately and not any particular event, timing, route, or mechanism.

Sovereign Bargaining Weakness: Reduced negotiating room when urgent financing needs, external exposure or fragmented creditors make refusal unusually costly.

Speculation Versus Capacity: The distinction between risk-bearing or price-discovery functions and speculative gains that create burdens without corresponding useful capacity.

Stability With Adaptation: Reliable essentials that allow supported, reversible learning and change without requiring participants to absorb unmanageable disruption.

Staffing and Training: The human capacity required to operate, interpret, explain, audit, secure, and correct public systems. Adequate staffing and role-specific training are infrastructure for transparency, service quality, privacy, continuity, and institutional learning.

Status Anxiety: Pressure to secure dignity or acceptance through visible achievement, position or comparison with others.

Status Pressure and Defensive Consumption: Spending partly intended to protect social acceptance or opportunity where access is influenced by visible signals of position.

Stolen Funds: Public or entrusted money unlawfully appropriated or diverted, including embezzled grants and payments obtained through false invoices. Bribery is an illicit exchange and may involve stolen funds, but a privately financed bribe is not automatically a matching theft from the public budget. Record the transfer and any public loss separately.

Strategic Fear as Economic Actor: The influence of perceived strategic threats on investment, inventories, insurance, procurement, research and contracting.

System-Level Burden Data: Evidence about institutional performance and burden, such as queues, service delays and correction, with the system rather than personal worth as the object of assessment.

T

Target: The truth-level property a hypothesis asserts. For the primary additive PTVC branch, the target is a terminal signed residual of exactly zero within each qualifying stream.

Tax as Shared Capacity: Taxation considered as financing for functions requiring collective provision, assessed by incidence, administrative burden, accountability and public return.

Technical Education Pathways: Educational and training routes that build practical technical capacity, including apprenticeships, trade schools, community colleges, professional institutes, certification programs, work-based learning, union training, adult retraining, and technical high schools. Their value is assessed by skill formation, mobility, completion, labor demand, working conditions, and public capacity.

The Law: See Law of Fairness.

Time Horizons and Daily Security: The relationship between reliable essentials and the period over which a person can realistically plan and accept delayed returns.

Time Scarcity: Insufficient discretionary time after paid work, care, travel, service navigation and other demands, including time costs intensified by low income.

Time Sovereignty and Dignity: Sufficient practical control over time to sustain rest, care, relationships and chosen activity alongside unavoidable obligations.

Timing, Conversion, and Double Counting: Checks that a claimed saving is available at the stated time, can be converted into its intended use and has not already been counted elsewhere.

Transaction costs: Resources used to arrange and sustain an exchange, including search, negotiation, verification, and enforcement.

Transit as Time Return: Improved access to destinations through reliable transport, assessed by travel time, predictability and cost rather than vehicle or route totals alone.

Transition Costs First: Recognition and funding of the obligations created by a change before treating gross reductions as usable savings.

Translation, Format, and Accessibility Costs: The resources needed to make public information usable across language, literacy, disability and format requirements.

Trauma and Decision Narrowing: A possible narrowing of perceived or usable options following trauma. This is a proposed mechanism to investigate, not a diagnosis made from prose.

Trauma Repetition: Repeated recounting of harm across service encounters. Safe, lawful handoffs can reduce repetition while preserving the person's control over sensitive information.

True endpoint: The actual endpoint of the qualifying conscious stream. It is distinct from a protocol endpoint or last observation and is not automatically identical to biological death.

Trust Destruction as Economic Loss: Economic costs that can follow a loss of warranted confidence, including shorter commitments, service avoidance and additional checking.

Trust Indicators and Their Limits: Measures of confidence in specified institutions or services, interpreted with their

population, context and object rather than as one universal level of trust.

Trust Repair as Economic Repair: The reduction of avoidable economic friction that can follow restored confidence in a system's competence, fairness and correction. Potential gains include less duplicate verification, easier service use and simpler contracting. They must be assessed against the cost of rebuilding trust and the safeguards still required.

U

Unpaid and Underpaid Care: Care work whose demands lack adequate compensation, rest or support, including burdens shifted to households when public or paid provision is insufficient.

Up-Front Cost Versus Downstream Harm: Comparison of visible initial expenditure with later harms and expenses caused or avoided, including costs carried outside the initiating budget.

Usable access: Access that can actually be exercised under real conditions of cost, distance, disability, language, time, safety, privacy, and administrative burden rather than merely existing on paper.

V

Valence: The positive, negative, or neutral evaluative character of conscious experience. Valence is distinct from arousal, reward, preference, behavior, social approval, and moral worth.

Verified savings: A demonstrated reduction in specified costs against a declared baseline and horizon. Gross savings, net usable fiscal capacity, noncash gains, and completed reinvestment are distinct statuses. A realized peace dividend

additionally requires evidence that the released capacity reached its stated use without hidden displacement of burden.

W

Wage and Price Dynamics: Interactions among pay, prices, productivity, input costs, demand, margins and supply; wage changes are one possible influence rather than a complete explanation of inflation.

Waiting as Dignity Cost: The loss of agency or dignity a person may experience while dependent on an institution's unresolved decision, distinct from elapsed time alone.

Waiting as Time Loss: Time consumed by waiting and associated work, distinguished from calendar delay and from deadlines after which a need becomes harder or impossible to remedy.

Warranted trust: Confidence grounded in clear rules, competent performance, correction, audit, and consequences for abuse. It differs from blind trust because it remains open to verification and challenge.

Worker Voice in Transition: Workers' participation in transition decisions, drawing on their knowledge of operations, safety, skills and the practical effects of proposed changes.

Y

Youth Debt and Early Constraint: Debt-related constraints on early adult income, mobility and opportunity before a stable reserve or bargaining position has formed.

Youth Future Confidence: A grounded expectation that future stability, opportunity and repair make preparation for adult life worthwhile.

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About the Author

Stephen Takowsky has spent his life studying complex systems: how rules shape them, how people cooperate and compete within them, and how hidden constraints can change what becomes possible. His path has crossed finance, competitive strategy, professional esports, and the leadership of global online communities, ultimately leading to the Law of Fairness project, a decades-long effort to investigate whether fairness itself may reflect a deeper structure of conscious life.

Competing as "MrGameTheory," Takowsky reached number one on the Civilization Revolution leaderboard and in the international Civilization IV and Civilization V leagues while pursuing a career in finance at Morgan Stanley in Beverly Hills. In 2016, aXiomatic, an ownership group led by Peter Guber and Ted Leonsis and backed by investors including Magic Johnson, Steve Case, and Tony Robbins, acquired a controlling interest in Team Liquid. Following the acquisition, Team Liquid selected Takowsky to captain its newly created Civilization team, later recognized by Guinness World Records as the first Civilization esports team. Across thousands of games, Civilization became an unconventional education in strategy: understand the whole system, think far beyond the next move, anticipate intelligent opposition, and keep searching when the obvious paths fail.

Strategy then became community leadership. Takowsky spent seven years at 505 Games, serving as Senior Digital Content & Engagement Manager and leading global communities for Terraria, which has surpassed 70 million copies sold, as well as Ghostrunner, WUCHANG: Fallen Feathers, Eiyuden Chronicle: Hundred Heroes, and other titles. He also founded the United Servers of Discord Alliance, bringing together leaders from major communities built around Roblox, Pokémon, Star Wars, League of Legends, Grand Theft Auto, Warframe, Animal Crossing, and other properties. Earlier, he served as administrator and producer

for TF2Outpost, a major Steam digital-goods marketplace. Across these very different environments, he kept returning to the same insight: better structure can expand what people are capable of building together.

Throughout those years, one question kept following him: could the apparent unfairness of life conceal a deeper structure? That question became the Law of Fairness. Takowsky began developing the idea more than twenty years ago while studying Religious Studies and Philosophy/Law and Society at the University of California, Riverside, and gave it its first public form in his 2009 book, *Of Grandeur*. In the years that followed, he sought out academics and serious thinkers across many disciplines, asking for insights, criticism, and sources that could help separate fact from speculation. What began as a philosophical intuition gradually became a research program built to face mathematics, measurement, competing explanations, and evidence.

The ten-book Law of Fairness series follows that question outward, from consciousness and scientific testing into ordinary life, meaning, and civilization. If the Law is true, then there may be no “free happiness” in the sense of positive experience remaining unresolved by the lifetime balance the hypothesis predicts. That possibility raises a profound practical question: what if humanity has been pursuing the wrong model of utopia? Rather than organizing life around the endless pursuit of pleasure, the series explores whether greater self-moderation, less needless suffering, and better-designed institutions could help remove some of the forces that keep people and societies trapped in conflict, corruption, and instability.

Takowsky’s fifty-year horizon is designed so that building a better future does not require tearing apart the present. It gives institutions and communities time to test reforms, build

infrastructure, prepare future leaders, improve technology, and correct mistakes before they spread. Just as importantly, it gives today's generation the chance to be remembered for beginning something larger than itself: laying the foundations for a more transparent, peaceful, and prosperous world it may never live to see.

The project's future is collaborative. Takowsky welcomes researchers, universities, laboratories, funders, technologists, students, critics, writers, artists, community builders, and volunteers willing to bring rigor, imagination, and courage to one of humanity's oldest questions. The next breakthrough may come from a theorem, an instrument, a dataset, a counterexample, or an experiment no one has yet imagined. *The Law of Fairness* began as one person's question. Its next chapter belongs to everyone willing to help discover what is true and build what that knowledge makes possible.

Explore the project and its official communities at LawOfFairness.com. Research, scholarly, media, funding, volunteer, and collaboration inquiries: team@lawoffairness.com.

The Peace of Fairness

Could lasting peace turn the costs of fear into the foundations of humanity's greatest future?

Humanity pays for conflict long before a shot is fired. Fear can commit enormous parts of the future to permanent readiness and force societies to spend for dangers they hope never arrive. The Peace of Fairness asks what could return when safer relationships allow some of that burden to fall without weakening the protection people still need.

The book follows that peace dividend into the systems that let ordinary life plan ahead. Greater stability can release capital, talent, and productive capacity now tied up in permanent readiness, allowing societies to direct more of their strength toward discovery, growth, and the next generation. Peace is not the end of the story, but the beginning of what humanity could build with more of its future back.



LawofFairness.com